



OPERATING MANUAL		
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SECTION A: PREFACE

The principal objective of an operating manual is to describe relevant operating procedures, instructions and process safety information in an orderly manner for use by operating personnel for safe and efficient operation of a plant facility. These operating procedures and instructions shall be up-to-date reflecting changes in plant hardware and operating practices carried out from time to time.

The Crude and Vacuum Distillation Unit-II was commissioned under the Visakh Refinery Expansion Project-I (VREP-I) to enhance the crude refining capacity of VR by 3.0 MMTPA. CDU-II is equipped with the latest technology. Its design provides for energy conservation; operational flexibility and maximization of product recoveries.

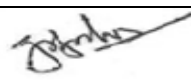
The original edition of operating manual of Crude Distillation Unit-II (CDU-II) was prepared prior to the commissioning of the unit in the year 1985 by M/s EIL. It was later updated in July 2008, based on the standard operating manual and process package provided by EIL. Plant Standing Instructions (PSI) issued from time to time based on operating experience and learning are available separately in field room.

Primary purpose of this revised operating manual is to integrate all the scattered operating procedures and instructions into a single operating manual while simultaneously fulfilling the requirements under Process Safety Management System (PSM) of Visakh Refinery. PSM-PR-04, which is based on OSHA-1910.119 standard, specifies the methodology and the format to be followed and the contents to be included in the preparation of an operating manual. Some of the new subjects that are incorporated in the manual due to PSM format are:

- Operating Limits and Consequence Deviations
- Upset Conditions and Stabilization
- Avoiding Deviations and Plant Upsets
- Temporary Operations
- Process Safety Information
- Special or Unique Hazards

Efforts have been made to include the relevant information in a concise, step-by-step, easy-to-read format so that they are within the comprehension of the readers. The users of this manual are encouraged to suggest ideas for further refinement and highlight typographical errors if any, to improve the overall quality of the manual.

Operating procedures & conditions given in this manual are indicative. These should be treated as general guide only for routine start-up and operation of the unit. The actual operating parameters and procedures may require minor modifications/changes from those contained in this manual as more experience is gained in operation of the Plant. For detailed specifications and operating procedures of specific equipment, corresponding Vendor's operating manuals/instructions need to be referred to.

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	Designation	DGM- Operations



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4	Feed and Product Characteristics	38	0	31-03-2012
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7	Description of critical control schemes	127	0	31-03-2012
8	Description of Distributed control System (DCS)	136	0	31-03-2012
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24	Sampling requirement and Sampling Procedures	326	0	31-03-2012
25	List of Plant Equipment	338	0	31-03-2012
26	Plant Chemicals a. Withdrawal management b. Max Storage allowable in the Plant c. Storage precautions d. Loading procedures e. Empty container disposal f. Handling Precautions g. Description of Chemical dosing system.	347	0	31-03-2012
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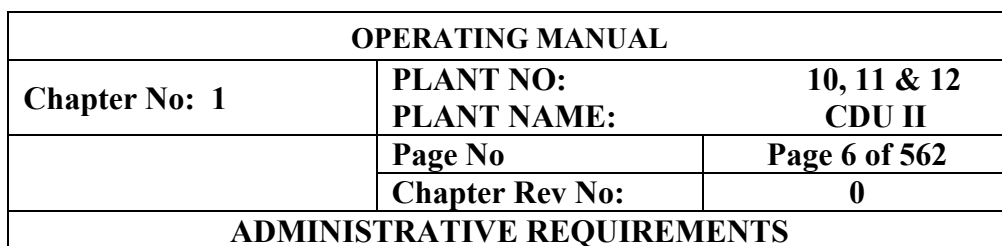
CHAPTER No:	TITLE	FROM PAGE NO	LATEST REV NO.	REV DATE
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29	Environmental Management a. Effluent Generation and Control b. Plant Emissions c. Solid Waste	382	0	31-03-2012
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31	Safe Work Practices a. Work Permit Procedures. b. Confined Space Entry procedure c. Procedure for Opening Process equipment and Piping d. Lockout/ Tag out Procedures e. Electrical isolation procedure f. Procedure for entry, presence, access and exit control in the Plant	404	0	31-03-2012
32	Standard Operating Procedures of Process Equipment.	444	0	31-03-2012
33	Chemical/HC spillage Handling Procedure	478	0	31-03-2012
34	List of Annexures : a) Unit master blind list b) Individual equipment blind list c) List of Vendor manuals d) Start – up & Shutdown check lists e) LEL detectors status f) Instrument air fail to open control valves g) DCP cylinders, First aid fire hose	492	0	31-03-2012



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35	Description of Utility systems	540	0	31-03-2012
36	Instrumentation Tags Description	549	0	31-03-2012

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	Name	P N VARA PRASAD
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This is to certify that this Operating Manual is current and accurate.

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SECTION D: DOCUMENT CONTROL

1. The administrative sections (Chapter 1 of PSM/GL/4.1) are approved by Division Head- Operations.
2. The original operating manual in file and tab format is maintained with the Division Head.
3. Three hard bound copies of the manual are issued as “Controlled Copy” to the respective plants- one for plant Manager, one for DCS and one for Field room. Controlled copy stamping is done on the following pages: “Title Page”, “Table of Contents” and First Page of every chapter.
4. Uncontrolled hard bound copies are made available to the plant personnel, Section Head, “Disaster Control Room” (formerly “Central Control Centre”), Refinery Engineering Documentation, Technical Department , HOD-Operations & YSF as training copies. The training copies are marked as “Training Copy”
5. In case of any doubt regarding the latest revision, the Original Copy is the reference document for confirmation.
6. All obsolete sections/chapters are removed by the Respective Division Heads. Revisions & additions are managed by way of “Plant Standing Instructions” which are annually integrated with the manual.

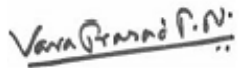
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SECTION E : REVISION OF THE OPERATING MANUAL

- This Operating Manual is revised for the following :
 - Change in Operating practice in any part of the Plant short and long duration.
 - Implementation of changes in Hardware and/or software systems of the Plant which have impact on procedure.
 - Change in Chemicals.
 - Changes in Safety systems.
- The revision of the Operating Manual is done in two stages :
 - Managing changes in the Operating Manual within one year cycle.
 - Updating Operating Manual annually.
- The revisions are issued as 'Standing Instructions'. The list of Standing Instructions is maintained in Section I of Chapter 1-Administrative Requirements of the Manual'.
- The Standing Instructions are backward integrated into the Operating Manual once in a year.
- The chapters which get revised at the time of revising operating manual, the Revision number of the Chapter which is revised is increased by "1". The Chapters which are not revised retain the same Revision Number.

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II. SECTION F : LIST OF ABBREVIATIONS

ABBREVIATION	EXPANSION
ATF	Aviation Turbine Fuel
ATP	Additional Tank age Project
BARC	Bhabha Atomic Research Centre
BA	Breathing Apparatus
BCW	Bearing Cooling Water
BFW	Boiler Feed Water
CCR	Continuous Catalytic Reformer
CAS	Chemical Abstracts Service
CBD	Closed Blow Down
CISF	Central Industrial Security Force
CPP	Captive Power Plant
CPWD	Central Public Works Department
DAF	Dissolved Air Flootation
DCP	Dry Chemical Powder
DOB	Daily Order Book
DMP	De-Mineralization Plant
DRN	Disposal Requirement Notice
DCN	Design Change Note
EPM	Environmental Procedures Manual
ESA	External Safety Audit
EHS	Environment Health Safety
ETP	Effluent Treatment Plant
ELCB	Earth Leakage Circuit Breaker
EMS	Environmental Management System
EDMS	Engineering Document Management System
E&P	Economics & Planning
FCCU	Fluid Catalytic Cracking Unit
HLPH	High Lift Pump House
HSD	High Speed Diesel
IA	Instrument air



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ABBREVIATION	EXPANSION
ISA	Internal Safety Audit
IWL	Inspection Work List
IFO	Internal Fuel Oil
JBO	Jute Batching Oil
KOD	Knock Out Drum
LLPH	Low Lift Pump House
LSHS	Low Sulfur Heavy Stock
MOC	Management of Change
MSIHC Rules	Manufacture, Storage , Import of Hazardous Chemical Rules
MSDS	Material Safety Data Sheet
MEROX	Mercaptan Oxidation
MES	Mechanical Engineering Services
MS	Motor Spirit
NDT	Non-Destructive Test
NHT	Naphtha Hydro-Treater
NRV	Non return Valve
OISD	Oil Industry Safety Directorate
OCP	Operational Control Procedure
OSTT	Off Shore Tanker Terminal
PDI	Plant Daily Instructions
P&ID	Piping & Instrumentation Diagram
PFD	Process Flow Diagram
PLC	Programmable Logic Control
PPM	Parts Per Million
PSI	Process Safety Information
PS&E	Process Safety & Environment
PSV	Pressure Safety Valve
PHA	Process Hazard Analysis
PAD	Process Analysis & Design
PPE	Personnel Protective Equipment
PMS	Project Management System



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ABBREVIATION	EXPANSION
PSMS	Process safety management system
PA	Paging Announcement
PSSR	Pre- Start Up Safety Review
PESO	Petroleum & Explosives Safety Organization
PIR	Project Initiation Request
QAP	Quality Assurance Plan
QRA	Quantitative Risk Analysis
RCA	Root Cause Analysis
RCW	Recirculating Cooling Water
RCR	Ramsbottom Carbon Residue
ROV	Remote Operated Valve
RED	Refinery Engineering Documentation
STEL	Short Term Exposure Limit
SSA	Surprise Safety Audit
SRU	Sulfur Recovery Unit
SWP	Safe Work Practice
SMPV	Static & Mobile Pressure Vessels
SKO	Superior Kerosene Oil
SAC	Strong Acidic Cations
SBA	Strong Basic Anions
SRN	Straight Run Naphtha
SR	Short Residue
T&I	Turnaround & Inspection
TLV	Threshold Lower Value
TSV	Thermal Safety Valve
TC	Turnaround Cycle
TOB	Turnover Book
TBP	True Boiling Point
UEL	Upper Explosive Limit
VRCFP	Visakh Refinery Clean Fuels Project
VGO	Vacuum Gas Oil
VBU	Visbreaker Unit



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ABBREVIATION	EXPANSION
VREP	Visakh Refinery Expansion Project
VD	Vacuum Diesel
YSF	Yard Shift Foreman

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SECTION G: LIST OF COPY HOLDERS

List of the Controlled Copy holders are as given below:

S.NO	COPY TYPE	DESIGNATION OF THE COPY HOLDER
1	Original	Division Head-Production Block
2	Controlled Copy	Unit Manager, CDU II Unit DCS, Field room
3	Training Copy (Hard Copy)	HOD-Operations
		HOD-F&S
		Technical Services
		Refinery Engineering Documentation.
		All Plant Personnel
		YSF

Note:

1. "Controlled Copy" means that the Plant Division Head will monitor it for its status, incorporate changes as & when required, ensure its applicability and accessibility.
2. Training copy will be available in soft as well as hard copies.

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SECTION H: RECORD OF REVISIONS TO THE MANUAL

1. The revisions to the Operating Manual are made through issue of Standing Instruction.
2. The Standing Instructions are issued either to revise the existing operating procedure in the Operating Manual in part or as an addendum.
3. The Standing Instructions are issued by the Respective Division Head-Operations.
4. The Standing Instructions for respective plants are filed in a separate File with tab separators and kept as records.
5. List of Standing Instructions issued are recorded and maintained in Chapter 1, Section I of the respective operating manual.
6. The Standing Instructions which describes recurring operational activity are identified among the standing instructions issued and incorporated in the corresponding chapter of the Operating Manual.
7. The revision of the Operating Manual is carried out annually to ensure the operating procedures are current and accurate.
8. The record of the Standing Instructions issued is retained long term in Operations Department.

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	Name	P N VARA PRASAD
	Designation	DIVISION HEAD-PRODUCTION BLOCK



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SECTION I: PART-I: STANDING INSTRUCTIONS - IN USE

SL. NO	STANDING INSTRUCTION NO.	STANDING INSTRUCTION TITLE	DATE OF ISSUE	
1.	ADM/OPRN/PRODN/ SI/003	Equipment draining	Aug 2000	Standing Instructions Incorporated in Operations Manual Chapter-28
2.	ADM/OPRN/PRODN/ SI/008	CDU-II startup Check-List	March 2000	Standing Instructions Incorporated in Operations Manual Chapter-34
3.	ADM/OPRN/ PROD /SI/07	CDU-II Desalter Online De-sludging	Feb' 2002	Standing Instructions Incorporated in Operations Manual Chapter-16
4.	OPRN/PROD/SI/011	Process Units Effluent Monitoring	Nov' 2001	Standing Instructions incorporated in Operations Manual Chapter-28
5.	ADM/OPRN/ PROD /SI/012	VREP I/ VREP- II –OWS System	Aug' 2001	Procedure incorporated in Operations manual Chapter-28
6.	ADM/OPRN/ PROD /SI/016	Refinery Fuel Gas System Management and Control	Feb' 2005	Standing Instructions incorporated in Operations Manual Chapter-15
7.	ADM/OPNRN/OM&S/ SI/017	11-E-40A/B commissioning procedure	May'2005	Standing Instructions incorporated in Operations Manual Chapter-.16
8.	OPRN/ADMN/SI/20	Standing Instructions on feed tank change over	Dec' 2005	Incorporated in Operations Manual Chapter-15
9.	ADM/OPRN/PRODN/ SI/23	Empty oil & chemical drums collection for washing in CDU block	March' 2006	Standing Instructions incorporated in Operations Manual Chapter-26.
10.	ADM/OPRN/PROD/SI /30	Standing Instruction for Improving Aesthetics of MOI Control Room	May' 2010	Standing Instructions incorporated in Operations Manual Chapter-8
11.	ADM/OPRN/PRODN/ SI/32	procedure for monitoring online ER probes, PIN matrixes and corrosion coupons for high acid crudes in CDU-II	Dec'2010	Standing Instructions incorporated in Operations Manual Chapter-.34
12.	ADM/OPRN/PRODN/ SI/34	Procedure for commissioning of PFD in CDU-II	April' 2011	Standing Instructions incorporated in Operations Manual Chapter-.16

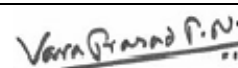


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13.	ADM/OPRN/PRODN/ SI/035	Avoid congealing of Heavy oil R/D lines from CDU-2	May 2011	Standing Instructions incorporated in Operations Manual Chapter-.15
14.	ADM/OPRN/PRODN/ SI/036	APH water washing procedure	July 2011	Standing Instructions incorporated in Operations Manual Chapter-.06
15.	ADM/OPRN/PRODN/ SI/037	To apprise Merox DCS shift-In-charge in case of fluctuation in sour water flow.	Aug 2011	Standing Instructions incorporated in Operations Manual Chapter-15
16.	ADM/OPRN/PRODN/ SI/038	Procedure for fuel oil gun cleaning	Sept'2011	Standing Instructions incorporated in Operations Manual Chapter-15
17.	ADM/OPRN/PRODN/ SI/042	Car seals Management	Dec 2011	Standing Instructions incorporated in Operations Manual Chapter-15

PART-II: RECORD OF STANDING INSTRUCTIONS CANCELLED

SL. NO	STANDING INSTRUCTION NO.	STANDING INSTRUCTION TITLE	DATE	RE V.	EXPIRED/ INCORPORATED
1.	ADM/OPRN/PROD N/SI/002	Effluent Monitoring	Aug 2000	0	EXPIRED
2.	ADM/OPRN/PROD N/SI/013	Work Permit System in the Units	Sep 2000	0	Invalid. New work permit system Procedure incorporated in Operations Manual Chapter-31
3.	ADM/OPRN/PROD N/SI/014	Safety Job Card (Red Job Card System)	Oct 2000	0	Invalid. New work request Procedure incorporated in Operations Manual Chapter-31

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INTRODUCTION

The Crude and Vacuum Distillation Unit-II was commissioned under the VISAKH Refinery Expansion Project-I (VREP-I) to enhance the crude refining capacity by 3.0 MMTPA.

The facilities put up under VREP-I are listed below:

- i) Grass-root Crude and Vacuum Unit of 3.0 MMTPA capacity designed to process Basrah crude.
- ii) Grass-root Fluidized Catalytic Cracking Unit of 0.6 MMTPA
- iii) Bitumen Blowing Unit which was later revamped to 225 TMTA unit in VREP-II with a new Biturox reactor
- iv) VREP-I BCW system.
- v) Capacity augmentation of the existing utilities and offsite facilities to meet the additional requirements.

CDU / VDU-II have a design capacity to process 3.0 MMTPA of crude oil. The crude processing capacity was enhanced to 3.2 MMTPA by installing Pre-Flash Drum in 1996. The design feed stocks for the unit are Basrah and Bombay High (BH). In addition, two check cases have been considered for the unit design, namely Kuwait and Heavy Arabian Mix crudes. The unit was designed based on 330 on-stream days per annum. The CDU is designed to produce Liquefied Petroleum Gas (LPG), Straight Run Naphtha, (SRN), Heavy Naphtha (HN), Superior Kerosene Oil (SKO), High Speed Diesel (HSD), and Reduced Crude Oil (RCO). The unit is also designed for special product Aviation Turbine Fuel (ATF)

The CDU also comprises the Naphtha Stabilizer section and the SRN Caustic and Water Wash sections.

The VDU is designed to process RCO from CDU and to produce Light Vacuum Gas Oil (LVGO), Heavy Vacuum Gas Oil (HVGO), Slop cut and Short Residue (SR). CDU / VDU are designed to operate in conjunction and independent operation of either of these units is not considered.

The crude oil is pumped from the off-site storage tanks to the Crude Distillation Unit. The various stages of operation it undergoes are as follows:

- Crude Pre-heating in process heat exchangers and desalting of crude in Desalter.
- Again preheating the desalted crude in process heat exchangers.



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- Heating of pre-heated crude oil in Atmospheric Fired Heaters.
- Fractionation in Atmospheric Distillation Column.
- Stabilization of Straight run naphtha in Stabilizer unit & Caustic - Water wash Treatment.
- Products steam stripping (Heavy Naphtha, Kero and Diesel) and routing to designated Product tanks or treatment facilities (in case of special products like ATF, Bitumen etc.).
- Heating of Atmospheric Column bottoms (Reduced Crude Oil) in Vacuum Heater.
- Fractionation of RCO in Vacuum Distillation Column.
- Routing of products (VGO, Slop Cut, and Vacuum Residue) to designated tanks/units.

The products from CDU can be routed as follows:-

- 1) Stabilizer off Gas to FCCU-II
- 2) LPG to the Amine Treating Unit (ATU)
- 3) Stabilized Naphtha to
 - a. Stabilized Naphtha storage tanks
 - b. MS tanks
 - c. 6" T/o downstream of 11-E-19 for giving hook-up to VRCFP.
- 4) Heavy Naphtha to
 - a. ATP diesel line
 - b. Heavy naphtha intermediate tank
 - c. Stabilized naphtha rundown line
 - d. Storage
 - e. Slop
 - f. 4" line hook-up given to route HN to VRCFP (NHT-CCR).
- 5) Kerosene to
 - a. Storage
 - b. Diesel Header
 - c. ATP Diesel Header
 - d. FO Blend
 - e. MEROX when on ATF regulation
 - f. Slop.



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- 6) DSL to
 - a) Sour diesel storage tanks
 - b) ATP diesel header
 - c) FO blend to HFO header
 - d) FO blend to RFO header
 - e) LDO blend header
 - f) To FCCU-2
 - g) To DHDS upstream of 11-E-23
 - h) To CDU-3 for flushing oil.
 - i) Slop

The VDU products are routed as follows:-

1. Hot well oil to TK 17.
2. VGO to
 - (a) FCCU-I/FCCU-II as hot feed
 - (b) VGO storage tanks
 - (c) Slop
 - (d) LDO header.
3. Slop Cut to
 - (a) Vacuum furnace along with RCO (As recycle stream)
 - (b) As product rundown, second part gets mixed with SR product up stream to 12-E-01 A/B/C. (provision also there to mix with SR down-stream of 12-E-01A/B/C).
 - (c) To CDU-I cooler box.
 - (d) To FCCU-II via recycle control valve.
4. SR to
 - a) VBU storage
 - b) Direct VBU feed.
 - c) To HFO line
 - d) To RFO line
 - e) To BBU feed
 - f) Slop header
 - g) 10" startup line/circulation line back to crude inlet line to Preheat train – I.
 - h) LDO header.

The unit is designed for a turndown capacity of 50%.



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DESIGN BASIS		

DESIGN BASIS

The Crude Distillation Column has been designed to process 3 MMTPA. Design of all the equipments other than crude column was based on Basrah crude. Crude column design was based on Kirkuk crude for both Kerosene (SKO) and Aviation Turbine Fuel (ATF) operations.

The unit was designed to process the crudes of API 31.3 ° to 36 ° API with marginal shortfalls in throughput. All the exchangers are specified to process Basrah crude only.

Modifications, under the “BH Conversion Project”, have been done in the plant to process 3 MMTPA of Bombay High crude also.

3.1.1 ELECTRICAL DESALTER:

a) Design feed:

Crude	Basrah	Bombay high
Kg/ hr	367647	367647
Sp. Gravity @ 15 °C	0.848	0.8284
API @16 °C	33.6	39.2
Total Sulfur, wt %	1.9	0.15
Wax content wt%	6.0	14.7
RVP @ 100 °F, psi	7-9	5.5
H2S content	Nil	Nil
Viscosity @ 20 °C	9.0 cst	-
Viscosity @ 37.8 °C	5.9 cst	-
Pour point °C	-15	+30
Conradson carbon residue, wt%	4.3	-
Characterization factor	11.9	-
Water & sediments, vol %	0.15	0.05
Salt content, (ptb)	3.0	5.0
Vanadium, ppm	18.0	4.24
Nickel, ppm	5.4	6.2
Iron, ppm	1.0	7.14
TBP distillation,	wt%	wt%
IBP-150 °C	19	24.5
150- 250 °C	16.7	19.6
250-370 °C	19.8	23.2
370 °C plus	45.5	32.7

b) Design product:

Salt content: 5 mg/l or 5% of the salt content of the raw crude whichever is greater.



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3.1.2 DISTILLATION UNIT:

a) Design feed:

The design feed for the unit is BASRAH crude oil with the characteristics indicated in the table 1.

b) Design products:

i. Product battery limit conditions:

➤ Atmos section:

Product	TBP cut range °C	Temperature °C	Pressure, Kg/cm ² A
LPG	C3-C4	40	8.0
Stabilized naphtha	C5-130	43	5.0
Heavy Naphtha	130-160	43	5.0
Kerosene/ATF	160-270/160-230	43	5.0
Diesel	270-380/230-380	43	5.0
RCO	380+	343	14.4

➤ Vac section:

Product	TBP cut range °C	Temperature °C	Pressure, Kg/cm ² A
Light vacuum gas oil	380-400	70/213	5.0
Heavy vacuum gas oil	400-530	70/240	5.0
Slop distillate	530-550	90	9.0
Vacuum residue	550+	90	9.0
Vacuum residue(bitumen unit feed)	550+	250	9.0



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3.2 Equipment Design Basis:

3.2.1 ELECTRIC DESALTER :

Design inlet chloride as NaCl	85.601 mg/l
Design outlet chloride as NaCl	5 mg/l
Required process water	5 vol, %
Insoluble water in desalted crude	0.2 vol. %
Oil content of effluent brine	100 ppm Max.
Required pH	7.0-8.5
Operating temperature	120-130 °C.

3.2.2 ATMOSPHERIC DISTILLATION COLUMN:

The crude distillation column has been designed to handle KIRKUK crude at capacity of 3 million tons/yr. the crude distillation column has been processed various other crudes like Kirkuk, Kuwait and 50:50 light heavy Arabian crudes with the same heat removal at the circulation reflux exchangers as for designed for Basrah crude.

3.2.2.1 Pressure:

The atmospheric column reflux drum operating pressure was set to 2.6 Kg/cm² abs. in order to obtain total condensation of the over head product at 45 °C. Accordingly the flash zone pressure has been fixed at 3.2 Kg/cm².

3.2.2.2 Over flash and bottom stripping steam:

Over flash (6 vol % on crude) and bottom stripping steam rate (28 kg/m³ of reduced crude) have been fixed to produce reduced crude containing not more than 10 volume per cent of gas oil boiling below 380 °C.

3.2.2.3 Heat removal:

The location and amounts of heat removal by the various circulating refluxes are selected to balance the tower loading and also to make it possible to recover heat in reboiler. The heat removal from the column is as below:

Diesel CR	5.0 MM K.cal/hr
Kerosene CR	11.0 MM K.cal/hr
Top pump around	7.4 MM K.cal/hr

50 °C temperature drop is taken for circulating reflux.



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3.2.2.4 Top Reflux:

The design reflux is selected to give an overhead temperature which prevents condensation of water at the top of the tower.

3.2.3 NAPHTHA STABILISER:

The stabilizer has been designed to make a naphtha bottom product of RVP 10 max. and top overhead product of LPG contains not more than 1 mol. %. Mol. % C5.

3.2.4 NAPHTHA CAUSTIC/ WATER WASH SYSTEM:

The caustic wash system is designed to remove all hydrogen sulphide in naphtha and reduce the mercaptan content to 10 ppm. A circulation rate of 25 % of naphtha is taken for caustic and water circulation. The caustic hold has been fixed to give a batch time of 6 days.

3.2.5 VACUUM DISTILLATION COLUMN:

3.2.5.1 Number of stages:

A single stage dry vacuum distillation system is provided for FCC feed preparation.

3.2.5.2 Flash zone temperature:

The flash zone temperature is set at 395 °C to achieve the desired vaporization at the pressure in the flash zone.

3.2.5.3 Tower pressure:

The operating pressure is selected such that there is no requirement of steam to achieve the desired vaporization and the tower diameter is minimized. A pressure of 24 mm Hg at flash zone ensures that the ejector system suction pressure will be 5 mm Hg. Abs.

3.2.5.4 Column internals:

Packed column has been provided for achieving low pressure drop. Glitsch grid has been provided in the wash zone. Chimney trays are provided for the draw off of the side stream products. Demister pads are provided above the wash zone to prevent carryover of asphaltenes and at the top of the tower (to minimize carryover of hydrocarbons into the ejectors system).



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3.2.5.5 Recycle:

The vacuum column is designed with a recycle rate equal to 6 v% of the tower feed in order to ensure satisfactory product quality.

3.2.5.6 Pump Around:

Pump around locations and duties are chosen to balance the column internal loading while maximizing the crude preheat.

3.2.5.7 Bottom Quench:

The tower bottom temperature is kept at 350 °C to reduce possible cracking during hold up in the tower. The quenching is achieved by returning a quench streams to the tower at a temperature of 250 °C after heat exchange between vacuum residue and crude.

3.3 Material Balance (design case):

3.3.1.1 Atmospheric Distillation column: (Basrah Crude)

i.) SKO operation:

Product	TBP cut Range °C	Sp. Gr @ 15 °C	Vol % cut	Wt % cut	kg/h	m ³ /h @ 15 °C	MTPA
Gas + Lt. Naphtha	IBP-130	0.6671	19.6	15.65	57537	84.975	469502
Hy. Naphtha	130-160	0.77229	5.6	5.1	18750	24.278	153000
Kerosene	160-270	0.8104	19.17	18.32	67353	83.111	549600
Diesel	270-380	0.85275	17.87	17.97	66066	77.474	539099
RCO	380+	0.96478	37.76	42.96	157941	163.707	1288799
Crude		0.848	100	100	367647	433.545	3000000



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ii.) ATF operation:

Product	TBP cut Range °C	Sp. Gr @ 15°C	Vol % cut	Wt % cut	kg/h	m ³ /h @ 15 °C	MTPA
Gas + Lt. Naphtha	IBP-130	0.6671	19.6	15.65	57537	84.975	469502
Hy. Naphtha	130-160	0.77229	5.6	5.1	18750	24.278	153000
ATF	160-230	0.78863	12.57	11.69	42978	54.497	350700
Diesel	230-380	0.85251	24.47	24.6	90441	106.088	737999
RCO	380+	0.96478	37.76	42.96	157941	163.707	1288799
Crude		0.848	100	100	367647	433.545	3000000

3.3.1.2 Naphtha Stabilizer Material Balance for Basrah crude:

Products	TBP cut Range (°C)	Sp. Gr. @ 15 °C	kg/h	m ³ /h	MTPA
LPG	C ₃ – C ₄	0.5545	8343	15.046	68079
Stabilised Naphtha	C ₅ – 130	0.6836	49194	71.963	401423
Light Naphtha	IBP-130	0.6771	57537	87.009	469502

3.3.1.3 Vacuum Distillation Column Material Balance for Basrah crude:

Product	TBP cut Range °C	Sp. Gr @ 15°C	Vol % cut on RCO feed	Wt % cut on RCO feed	kg/h	m ³ /h @ 15°C	MTPA
LVGO	380-400	0.891	7.84		11435	12.834	93309
HVGO	400-530	0.9205	41.84	39.92	63050	68.495	514489
Slop Distillate	530-550	0.961	5.96	5.94	9382	9.763	76555
Vac. Residue	550+	1.02	44.36	46.90	74074	72.622	604447
RCO feed	380+	0.96478	100	100	157941	163.707	1288799



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3.3.2.1 Crude Distillation Column Material Balance for Bombay High crude:

Product	TBP cut Range °C	Sp. Gr @ 15°C	Vol % cut	Wt % cut	kg/h	m ³ /h @ 15°C	MTPA
Gas + Lt. Naphtha	IBP-110	0.690	17.3	14.1	51838	75.13	423000
Hy. Naphtha	110-140	0.747	6.9	6.42	23603	31.60	192600
Kerosene*	140-240	0.791	19.6	19.28	70882	89.61	578400
	140-270	0.803	26.2	25.68	94412	117.57	770400
Diesel	240-380	0.852	27.2	27.7	101838	119.53	831000
	270-380	0.855	20.6	21.3	78308	91.59	639000
RCO	380+	0.889	29	32.5	119486	134.40	975000
Crude		0.828	100	100	367647	444.02	3000000

*ATF cannot be produced from BH crude due to high Aromatics in it

3.3.2.2 Naphtha Stabilizer Material Balance for Bombay High crude:

Products	TBP cut Range (°C)	Sp. Gr. @ 15 °C	kg/h	m ³ /h	MTPA
Fuel gas			484		3950
LPG**	C ₃ – C ₄	0.549	7610	13.86	62100
Stabilised Naphtha	C ₅ – 130	0.726	43744	60.25	356950
Light Naphtha	IBP-130	0.690	51838	75.13	423000

** LPG quantity corresponds to 94% recovery based on 2.2% by wt. LPG on crude



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3.3.2.3 Vacuum Distillation Column Material Balance for Bombay High crude:

Product	TBP cut Range °C	Sp. Gr @ 15°C	Vol % cut on RCO feed	Wt % cut on RCO feed	kg/h	m ³ /h @ 15°C	MTPA
LVGO	380-400	0.861	13.36	12.91	15420	17.91	125830
HVGO	400-510	0.878	64.68	63.7	76108	86.68	621040
Slop Distillate	510-550	0.927	6.74	7.01	8378	9.04	68360
Vac. Residue	550+	0.960	15.22	16.39	19580	20.4	159770
RCO feed	380+	0.889	100	100	119486	134.4	975000

The performance of the above design has further been checked even for the following cases. The material balance has been tabulated given as under.

a) Crude Distillation Column Material Balance for Basrah crude SKO operation without Heavy Naphtha production

Product	TBP cut Range °C	Sp. Gr @ 15°C	Vol % cut	Wt % cut	kg/h	m ³ /h @ 15°C	MTPA
Gas + Lt. Naphtha	IBP-140	0.68476	21.61	17.45	64154	93.689	523500
Kerosene	140-270	0.80553	22.76	21.62	79435	98.675	648600
Diesel	270-380	0.85275	17.87	17.97	66066	77.474	539100
RCO	380+	0.96478	37.76	42.96	157941	163.707	1288800
Crude		0.848	100	100	367647	433.545	3000000



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b) Crude Distillation Column Material Balance for Basrah crude ATF operation without Heavy Naphtha production

Product	TBP cut Range °C	Sp. Gr @ 15°C	Vol % cut	Wt % cut	kg/h	m ³ /h @ 15 °C	MTPA
Gas + Lt. Naphtha	IBP-140	0.68476	21.61	17.45	64154	93.689	523500
ATF	140-240	0.78922	17.89	16.65	61213	77.562	499500
Diesel	240-380	0.85546	22.74	22.941	84338	98.588	688200
RCO	380+	0.96478	37.76	42.96	157941	163.707	1288800
Crude		0.848	100	100	367647	433.545	3000000

3.3.3.1 Crude Distillation Column Material Balance for Kuwait crude SKO operation:

Product	TBP cut Range °C	Sp. Gr @ 15°C	Vol % cut	Wt % cut	kg/h	m ³ /h @ 15°C	MTPA
Gas + Lt. Naphtha	IBP-130	0.6615	15.5	11.8	43382	65.581	354000
Hy. Naphtha	130-160	0.7603	4.8	4.2	15441	20.309	126000
Kerosene	160-270	0.8031	18.5	17.1	62868	78.282	513000
Diesel	270-380	0.8742	16.7	16.4	60294	68.97	492000
RCO	380+	0.9785	44.5	50.5	185662	189.741	1515000
Crude		0.869	100	100	367647	422.883	3000000



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3.3.3.2 Vacuum Distillation Column Material Balance for Kuwait crude:

Product	TBP cut Range °C	Sp. Gr @ 15°C	Vol % cut on RCO feed	Wt % cut on RCO feed	kg/h	m ³ /h @ 15°C	MTPA
LVGO	380-400	0.893	9	5.74	10657	11.934	89961
HVGO	400-530	0.922	35	34.85	64703	70.177	527978
Slop Distillate	530-550	0.99	8	7.32	13590	13.727	110898
Vac. Residue	550+	1.0299	48	52.09	96711	93.903	78916
RCO feed	380+	0.9785	100	100	185661	189.741	1515000

3.3.4.1 Crude Distillation Column Material Balance for Kirkuk crude SKO operation:

Product	TBP cut Range °C	Sp. Gr @ 15°C	Vol % cut	Wt % cut	kg/h	m ³ /h @ 15°C	MTPA
Gas + Lt. Naphtha	IBP-130	0.6855	19.7	16	58824	85.812	480000
Hy. Naphtha	130-160	0.776	6.2	5.7	20956	27.005	171000
Kerosene	160-270	0.8022	21.6	20.53	75478	94.089	615900
Diesel	270-380	0.8605	18.9	19.27	70846	82.331	578100
RCO	380+	0.967	33.6	38.5	141544	146.374	1155000
Crude		0.869	100	100	367647	435.611	3000000



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3.3.4.2 Vacuum Distillation Column Material Balance for Kirkuk crude:

Product	TBP cut Range °C	Sp. Gr @ 15°C	Vol % cut on RCO feed	Wt % cut on RCO feed	kg/h	m ³ /h @ 15°C	MTPA
LVGO	380-400	0.904	9.5	7.8	11040	12.212	90090
HVGO	400-530	0.9173	42.4	40.52	57354	62.525	468006
Slop Distillate	530-550	0.943	5.3	4.93	6978	7.4	56941
Vac. Residue	550+	1.03	42.8	46.75	66172	64.245	539963
RCO feed	380+	0.967	100	100	141544	146.374	1155000

3.3.5.1 Crude Distillation Column Material Balance for 50:50 Light: Heavy Arabian crude SKO operation:

Product	TBP cut Range °C	Sp. Gr @ 15°C	Vol % cut	Wt % cut	kg/h	m ³ /h @ 15°C	MTPA
Gas + Lt. Naphtha	IBP-130	0.6704	15.8	12.17	44743	66.74	365100
Hy. Naphtha	130-160	0.7567	4.9	4.26	15662	20.7	127800
Kerosene	160-270	0.7997	18.1	16.63	61140	76.45	498900
Diesel	270-380	0.8645	17.9	17.78	65368	75.61	533402
RCO	380+	0.9881	43.3	49.16	180735	182.91	1474798
Crude		0.87035	100	100	367648	422.41	3000000



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3.3.5.2 Vacuum Distillation Column Material Balance for 50:50 Light: Heavy Arabian crude:

Product	TBP cut Range °C	Sp. Gr @ 15°C	Vol % cut on RCO feed	Wt % cut on RCO feed	kg/h	m ³ /h @ 15°C	MTPA
LVGO	380-400	0.921	7.41	6.91	12489	13.56	101909
HVGO	400-530	0.942	39.39	37.56	67884	72.06	553934
Slop Distillate	530-550	0.95	4.99	4.8	8675	9.13	70790
Vac. Residue	550+	1.039	48.21	50.73	91687	88.19	748165
RCO feed	380+	0.9881	100	100	180735	182.94	1474798

3.4 FEED/ PRODUCT BATTERY LIMIT CONDITION

FEED STOCK: CRUDE (@ 1.0 kg/cm² & 30 °C)

PRODUCT	Pressure (kg/cm ²)	Temp (°C)
LPG	8.0	40
Stabilized Naphtha	5.0	43
Heavy Naphtha	5.0	43
Kerosene/ATF	5.0	43
Diesel	5.0	45
RCO	14.4	343
LVGO	5.0	*213/70
HVGO	5.0	*240/70
(Slop/RFO/HFO)	9.0	90
Short Residue(BBU feed)	9.0	250
Short Residue	9.0	90
Hot well Oil	5.0	40



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3.5 UTILITIES CONDITIONS AT UNIT BATTERY LIMIT:

Utilities And Their Specifications:

3.5.1 LP Steam:

	Minimum	Normal	Maximum	Mech. Design
Pressure (Kg/cm ²)	3.5	4.0	5.0	6.5
Temperature (°C)	Saturated	150	170	190

3.5.2 MP steam:

	Minimum	Normal	Maximum	Mech. Design
Pressure (Kg/cm ²)	10.0	11.0	12.0	13.5
Temperature (°C)	Saturated	250	280	300

3.5.3 HP steam:

	Minimum	Normal	Maximum	Mech. Design
Pressure (Kg/cm ²)	34	36		
Temperature (°C)	Saturated	370	390	

3.5.4 Instrument Air:

	Minimum	Normal	Maximum	Mech. Design
Pressure (Kg/cm ²)	5.0	6.0	7.0	9.5
Dew Point (°C) at 1.0 Kg/cm ²	-40°C	-40°C	-40°C	-40°C
Oil Content (ppm)	0.0	0.0	0.0	0.0
Temperature (°C)	30	40	45	65

3.5.5 Plant air:

	Minimum	Normal	Maximum	Mech. Design
Pressure (Kg/cm ²)	4.0	5.0	6.0	9.5
Dew Point (°C)	No Free Moisture	-	No Free Moisture	-
Oil Content (ppm)	0.0	0.0	0.0	0.0
Temperature (°C)	30	40	45	65



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3.5.6 Raw Water:

Turbidity (ppm)	15
M. Alkalinity as CaCO ₃ (ppm)	50-192
Ca Hardness as CaCO ₃ (ppm)	30-150
Total Hardness as CaCO ₃ (ppm)	22-300
Silica as SiO ₂ (ppm)	50 (max)
Chlorides as Cl ((ppm)	30-200
Sulfates as SO ₄ (ppm)	80-
Iron as Fe (ppm)	0.25
TDS as CaCO ₃ (ppm)	700
Total Suspended Solids (ppm)	23
pH	7.0-9.0
Conductivity at 25 ⁰ C micro mho/cm	approx 5.0
Pressure Kg/cm ²	Operating =3.5, Mech. Design = 7.0
Temperature ⁰ C	Operating =32, Mech. Design = 65

3.5.7 Cooling water:

	Unboosted	Boosted	Mech. Design
Pressure Kg/cm ²	2.0	3.5	
Temperature ⁰ C	33	44	65

3.5.8 Boiler feed Water (MP):

	Supply	Mech. Design
Pressure Kg/cm ²		5.2
Temperature ⁰ C	120	150

3.5.9 DM feed water:

Turbidity (ppm)	Nil
Hardness as CaCO ₃ (ppm)	Nil
Silica as SiO ₂ (ppm)	0.05
Chlorides as NaCl ((ppm)	0.05
Iron as Fe (ppm)	Nil
Conductivity at 20 ⁰ C micro mho/cm	1.0 (max)



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pH	6.5-8.0
Pressure (Kg/cm ²) at grade	3.0
Temperature °C	Ambient
Mech. Design Pressure (Kg/cm ²)	7.0
Mech. Design Temperature (°C)	65

3.5.10 LP condensate:

	Maximum	Mech. Design
Pressure (Kg/cm ²)	4.5	6.5
Temperature °C	147	170
Oil Content	15	15
Conductivity micro mho/cm	1.0	1.0

3.5.11 Fuel oil at unit Battery Limit:

	Minimum	Normal	Maximum	Mech. Design
Pressure (Kg/cm ²)	8.0	10.0	11.0	18.0
Temperature °C	110	130	200	200

	Fuel Oil	LSHS
Specific Gravity @ 15°C	0.959	0.9756
Viscosity, cst at 82 °C	100	39.4
Viscosity, cst at 100 °C	45	23.6
Sulfur content (wt.%)	4.5	0.7
Ash Content (ppm)	-	0.1 (max)
Sediment (ppm)	-	0.25(max)
Flash Point °C	>93	>93
Pour Point °C	+30	+51
Heating Value (Kcal/kg gross)	10,200	10,200
Net H/C ratio	9480	9480

Normally LSHS only will be used. However FO will be used for short duration when LSHS is not available.



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3.5.12 Fuel gas at unit Battery limit:

	Minimum	Normal	Maximum	Mech. Design
Pressure (Kg/cm ²)	3.5	4.0	4.5	7.0
Temperature °C	30	40-50	60	70

3.5.13 Electricity supply to unit:

	Volts	Phases	Cycles
Lighting	230	1	5C CPS
Emergency power for instruments	110	1	50
For interlocks	110 DC		

3.6 UTILITY CONSUMPTION:

Utility consumption rate:-

Utility	Consumption
L.P Steam (Kg/hr)	1800
M.P Steam (Kg/hr)	17260-10000
H.P Steam (Kg/hr)	*21100
Cooling Sea Water (m3/hr)	3257 (*3606)
D.M Water (m3/hr)	27*
Service Water (m3/hr)	17.4*
Fuel Oil (T/hr)	6.54
Fuel Gas (T/hr)	5.45

1. (*) When FCC is down
2. * intermittent Generation

3.7 CHEMICAL CONSUMPTION

Chemical	Chemical name	Average consumption(w.r.t crude feed rate)
Neutralizer	Ammonia	2 PPM *
Filmer	EC1021A	1.4 PPM
Demulsifier	EC2040A	5 PPM
caustic	----	5 PPM max.

* 1 PPM for ATMOS neutralizer and 1 ppm for VAC neutralizer.



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3.8 EQUIPMENTS DESIGN CONSIDERATION:

3.8.1 Roto-dynamics Machinery:

All roto-dynamic machines are over designed to 120% of limiting design flow. All the pumps are motor driven and following pumps however will have a turbine driven spare: crude charger, crude booster (presently for PFD), atmos col. Reflux and KERO CR.

3.8.2 Heat Exchangers:

All the coolers, condensers and other heat exchangers are over designed to 110% of limiting design on flow and duty. Condensers must have 20% spare philosophy, i.e., 20% overdesign on flow and duty (ex. Trim condenser (2+1) each of 60% duty. There are no spares available for other exchangers. For air fin coolers, extent of cooling can be maximized up to 42°C. Preheat exchangers are for obtaining maximum possible preheat and each exchanger has block and bypass valves. Stacked exchangers are not more than two or three.

3.8.3 Heaters:

The combination burner firing as well as individual burner firing facility is there. Either FO or FG can take the full load if required. Turn down ratio of heater is 50%. Turndown ratio of burners for oil firing is 1:3 and for gas firing is 1:5. Atomizing steam is MP steam at pressure of 11.5 Kg/cm². Stack height has to be maximum of 60 meters and the diameter has to be such that flue gas exit velocity shall be more than 20meters/sec at turndown condition. Soot blowers operating with electric motors for 11-F-01 and for 12-F-01 inst. Air operated with pneumatic and retractable soot blowers. Both heaters are provided with air pre-heater.

3.8.4 Instruments:

All the instruments are under Centralized (Distributed Digital Control) Automatic Computerized control and pneumatically controlled. There are no local controls on instruments. All instruments have power supply of 110V, 50Hz. Safety valves have 100% spares and they are all provided with block valves and bypass valves. All control valves must have isolation and bypass valves. All field junction box have to be explosion proof. All the pressure gauges and dial thermometers should have 6" diameter.



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3.8.5 Atmospheric Column:

The material of construction is Carbon steel with SS410 cladding up to light diesel and Monel clad is present up to tray#4 from top. Trays are SS410 except top four trays which are made up of Monel. The trays are valve trays except the chimney trays at draw-off. Overhead condenser shell is Monel clad 3mm over CS. Tubes are of titanium (and before 2010 T&I tubes are made of copper and nickel (70:30)). Channel section is of Monel clad 3mm thick. Overhead drum has to be 100% cement lined.

3.8.6 Vacuum Column:

The column has structured packing and the material of construction is CS with SS410 clad up to 250°C limit during T&I material is upgraded to SS316 2.5 Mo (min) metallurgy. The top section and the overhead vapor line are of only CS. Surface condenser is floating head type and its shell is made up of CS + Monel clad 3mm. Tubes are made up of Cu & Ni and upgraded to titanium tubes. Channel section also has 3 mm Monel clad and the drum needs 100% cement lining. The associated lines to drum are of CS. Surface condensers are floating head type and only 12-E-07A having back flushing facility.

3.8.7 Stabilizer:

The stabilizer also has valve trays and internals are of SS410. Overhead vapor line is made of CS. Condensers shell is of CS with Monel lining and tubes are of Cu & Ni in ratio (70:30). Channel section is having Monel clad of 3 mm. Drum has 100% cement lining and re-boiling is provided by KERO CR stream



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FEED AND PRODUCT CHARACTERISTICS

4.1 FEED SPECIFICATIONS AND INLET BATTERY LIMIT CONDITIONS

4.1.1 Feed Characteristics:

- The Crude Distillation Column has been designed to process 3 MMTPA of Basrah crude for both Kerosene (SKO) and Aviation Turbine Fuel (ATF) operations. Modifications, under the “BH Conversion Project”, have been done in the plant to process 3 MMTPA of Bombay High crude also.

Property	Basrah	Bombay high
SP.GR	0.848	0.8284
API @ 16 °C	35.4	39.2
RVP@38 °C psi	5.8	5.5
Pour point , °C	-15	+30
Wax Content % Wt	6.0	14.7
Total Sulfur % Wt	1.9	0.15
Salt content (ptb)	3.0	5.0
Viscosity	9.0 cst @ 20°C	2.876 KV @ 40 °C
	5.9 cst @ 37.8°C	2.404 KV @ 50 °C
Asphaltenes (wt %)	1.0	0.05
Total light ends (wt%)	5.02	2.62

4.1.2 Feed at battery limit Conditions:

Feed Stocks	Pressure	Temperature	Source
Crude oil	1.0 Kg/cm ²	Ambient (30°C)	Storage tank

4.1.3 TBP Distillation:

Temperature, °C	Basrah (wt%)	BH (wt%)
IBP-150	19.0	24.5
150-250	16.7	19.6
250-370	19.8	23.2
370 Plus	45.5	32.7



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4.1.4 Metal content weight ppm:

Metal	Basrah	Bombay high
Vanadium	18.0	4.24
Iron	1.0	7.14
Nickel	5.4	6.2
Sodium	-	-

4.2 PRODUCT SPECIFICATIONS AND OUTLET BATTERY LIMIT CONDITIONS:

4.2.1 Products TBP at ranges for crude and vacuum unit are as follows:

Product	PG Case ($^{\circ}\text{C}$)	BH Case ($^{\circ}\text{C}$)
Overheads	IBP-130	IBP-110
Heavy Naphtha	130-160	110-140
ATF/KERO	160-230/160-270	140-240 /140-270
Diesel	230-380/270-380	240-380/270-380
LVGO	380-400	IBP-410
HVGO	400-530	400-510
Slop	530-550	510-550
Short Residue (VR)	550+	550+

4.2.2 Liquefied Petroleum Gas (LPG) (As per IS-4576 standards)

Property	Basrah	Bombay High
Yield tones/annum	68075	62100
Sp. Gravity at 15/4 $^{\circ}\text{C}$	0.554	0.549
Vapor Pressure at 65 $^{\circ}\text{C}$	16.87 Kg/Cm ²	16.88
Sulphur wt%	<0.003	-
H ₂ S Wt%	0.45*	-
Dryness	no free entrained water	no free entrained water
Volatility: evaporation	2	2
Weathering	95% Vaporization by volume at 760 mm HG pressure	95% Vaporization by volume at 760 mm HG pressure

* LPG for further treatment.



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4.2.3 Stabilized Naphtha (SRN):

Property	Basrah	Bombay High
Rate tones/annum	401423	356950
Pour point °C	0.69	-
Sulfur mercaptans, PPM	10	27
Sp.Gravity@40 ⁰ C	NA	0.726
RVP (max) Psi	10	7.0
E.O.N clear	53.4	
Paraffins vol %	77.8	60.9
Naphthenes, vol %	16.9	29.2
Aromatics, vol %	5.3	9.9
ASTM-distillation	Temperature	
IBP	-	49
10%	51.5	70
30%	65.5	83
50%	84	94
70 %	96	105
90%	111.5	120
FBP	118	138

4.2.4 Heavy Naphtha (HN) Product:

Property	Basrah	Bombay High
Rate tones/annum	153000	192600
Sp.Gravity@15 ⁰ C	0.7723	0.747
ASTM D-86(Vol. %)	Temperature ⁰ C	
IBP	118	126
10%	136	138
30%	141	143
50%	144	149
70%	148	153
90%	156	161
FBP	177	177
Viscosity 20°C CST	NA	NA
Sulfur wt%	0.016	48 ppm
Paraffins vol %	65.8	52.7



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Naphthenes, vol %	19.5	30.0
Aromatics, vol %	14.7	17.2
Flash point °C	20	20

4.2.5 Kerosene Product:

Property	Basrah	Bombay high
Flash point °C	45	48
Smoke point mm.	25	
Rate tones/annum	549600	770400
Sp.Gravity@15°C	0.8104	0.803
Sulfur Wt %	0.158	0.25
Viscosity Cst @ 20° C	2.2	1.32
Viscosity Cst @37.8 °C	1.5	1.30
Diesel index	68.7	33
ASTM D-86(VOL %)	Temperature °C	
IBP	148	143
10%	175	177
30%	192.2	195
50%	206	208
70%	221	222
90%	251	241
FBP	291.9	260

4.2.6 ATF product:

Property	Basrah	Bombay high
Flash point °C	43	
Smoke point mm.	25	
Rate tones/annum	350700	578400
Sp.Gravity@15°C	0.7886	0.791
Sulfur Wt %	0.098	0.013
Viscosity Cst @ 20° C	1.6	0.93
Viscosity Cst @37.8 °C	1.2	0.95
Freezing point, °C	-50	-47
Diesel index	71	
ASTM D-86(VOL %)	Temperature °C	



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IBP	150	110
10%	170	149
30%	182	161
50%	190	171
70%	199	183
90%	218	202
FBP	249.6	240

4.2.7 Diesel (DSL):

Property	Basrah	Bombay high
Flash point $^{\circ}\text{C}$	115	
Smoke point mm.	NA	
Rate tones/annum	539099	639000
Sp.Gravity@15 $^{\circ}\text{C}$	0.8527	0.855
Sulfur Wt %	1.6	0.129
Viscosity Cst	8.7	
Viscosity Cst @37.8 $^{\circ}\text{C}$	5.2	
N ₂	NA	52 mg/l
ASTM D-86(VOL %)	Temperature $^{\circ}\text{C}$	
IBP	236.9	241
10%	272	269
30%	293	284
50%	307	300
70%	323	322
90%	<366	366

4.2.8 Diesel (DSL): (during ATF regulation)

Property	Basrah	Bombay high
Flash point $^{\circ}\text{C}$	94	
Smoke point mm.	NA	
Rate tones/annum	737999	831000
Sp.Gravity@15 $^{\circ}\text{C}$	0.8525	0.852
Sulfur Wt %	1.32	0.129
Viscosity Cst @ 20 $^{\circ}\text{C}$	NA	3.74
Viscosity Cst @37.8 $^{\circ}\text{C}$	4.0	3.2



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ASTM D-86(VOL %)	Temperature °C	
IBP	213.5	175
10%	246	224
30%	270.5	246
50%	289.5	264
70%	311.5	289
90%	<366	348

4.2.9 Light Vacuum Gas Oil (LVGO):

Property	Basrah	Bombay high
Flash point °C	79	
Rate tones/yr	93309	125830
Sp.Gravity@15°C	0.891	0.861
Sulfur Wt %	2.45	0.174
Viscosity Cst @ 20 °C	34	17.3
Viscosity Cst @ 37.8 °C	16	12.5
Metal Content	NA	
96% distillate	400	467

4.2.10 FCCU FEED : (LVGO + HVGO)

Property	Basrah	Bombay high
Pour point °C	+48	+30
Sulfur Wt %	3.5	0.19
Rate Kg/hr	50980	
Sp.Gravity@15°C	0.933	0.90
Viscosity Cp	1.12@246°C	NA
	0.77@314.2°C	NA
Metal Content	NA	
ASTM D-86 (VOL %)	Temperature °C	
5%	410.1	379
10%	422.0	395
30%	444.7	425
50%	462.0	442



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70%	480.0	467
90%	511.4	512
95%	529.2	538

4.2.12 Slop Distillate:

Property	Basrah	Bombay high
Flash point $^{\circ}\text{C}$	NA	
Pour point $^{\circ}\text{C}$	+55	+34
Rate Kg/hr	4831	9950
Sp.Gravity@15 $^{\circ}\text{C}$	0.961	0.93
Sulfur Wt %	6.0	0.35
Viscosity Cp	0.9@378.7 $^{\circ}\text{C}$	NA
Metal Content	NA	
ASTM D-86(VOL %)	Temperature $^{\circ}\text{C}$	
5%	469.1	425
10%	486.7	448
30%	511.3	505
50%	525.3	549
70%	546.4	-
90%	573.8	-
95%	599.0	-

4.2.13 Vacuum Residue (VR/SR):

Property	Basrah	Bombay high
Pour point $^{\circ}\text{C}$	+60	NA
Rate Kg/hr	91931	
Sp.Gravity@15 $^{\circ}\text{C}$	10.56	0.99
Sulfur Wt %	5.50	0.68
Viscosity Cp	3.0@382 $^{\circ}\text{C}$	NA
	37.2@220 $^{\circ}\text{C}$	NA
Metal Content	NA	
ASTM D-86(VOL %)	Temperature $^{\circ}\text{C}$	
5%	513.1	464
10%	526.5	520
30%	574.8	601



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50%	616.2	641
70%	661.2	702
90%	711.5	794
95%	713.7	819

4.3 PRODUCT'S BATTERY LIMIT CONDITIONS:

Products	Pressure Kg/cm ²	Temperature °C	Destination
Stabilizer off Gas	8.1	35	FCCU-2/FG header
LPG	16.0	40	MEROX
Light Naphtha	5.5	40	storage tank
Heavy Naphtha	5.5	40	into Diesel
Kerosene/ATF	5.5	44	storage tank
Diesel(DSL)	5.5	44	storage tank
VGO	7.0	70/140	Storage tank/FCCU
Vacuum Residue	9.0	90/250/160	Storage tank/BBU/VBU/FO



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CRUDE CHEMISTRY

Crude oil is one of the two major fossil fuels on earth, the other being coal. It is the major and a cost effective energy source today; though efforts are on to discover other means. Crude oils vary widely in appearance and consistency from country to country and from field to field. However, all crude oils consist essentially of hydrocarbons.

Hydrocarbons:

Organic compounds of carbon and hydrogen are known collectively by the name hydrocarbons. As carbon has a valency of four and hydrogen is monovalent, it can normally be expected that carbon should form only one tetra-hydride by combining with four atoms of hydrogen. Such a compound known as methane or CH_4 does exist, but as carbon can also combine with itself and can also leave some of its valencies unsatisfied by getting involved in unsaturated bonds or linkages, the number of hydrocarbons is truly myriad.

Saturated and Unsaturated Hydrocarbon Compounds:

In any compound made up of carbon and hydrogen the carbon atoms behave as though they had four arms and the hydrogen atoms behave as though each had only one arm. Each arm of the carbon atom must always be occupied, that is, it must be holding something, either a hydrogen atom or another carbon atom. When all the carbon arms or bonds are used to hold other atoms, the compound is said to be “saturated”. Similarly, a compound which does not have all the carbon arms or bonds taken up by other atoms is said to be “unsaturated”. There are millions of different ways in which carbon and hydrogen atoms can be connected together to form hydrocarbon molecules. To help describe these in a systematic way, Science has classified hydrocarbons into various families depending on their molecular structure. In petroleum chemistry, hydrocarbons are classified primarily into four groups

Type of Hydrocarbon Group:

Type of Hydrocarbon	Group
Paraffins	Saturates
Naphthenes	Cyclic saturates
Olefins	Unsaturated
Aromatics	Unsaturated



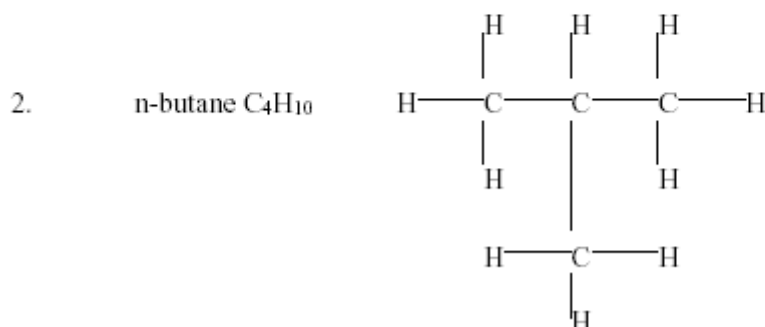
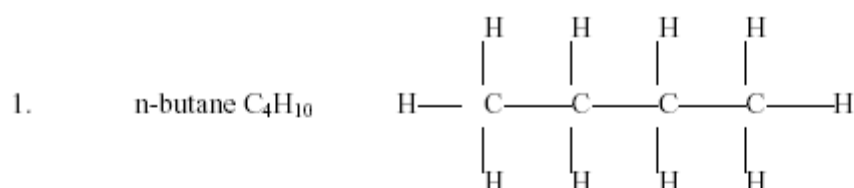
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Paraffinic Family:

The first family of hydrocarbons is Paraffins. They are saturated hydrocarbons with the general formula C_nH_{2n+2} .

Normal (Straight Chain) Paraffins & Isomers:

There are two ways in which carbon and hydrogen can be combined in butane



In the normal butane, the chain is straight where as in the iso-butane, the chain is branched, although both compounds have the same number of carbon and hydrogen atoms. For heavier hydrocarbons there can be more isomers.

Properties of Paraffins:

- 1) Good natural stability. However, high reactivity in presence of oxygen or under the influence of heat.
- 2) Low effect of temperature on viscosity. Highly paraffinic lubricants have high viscosity index.
- 3) At a sufficiently high molecular weight they form waxy solids. Paraffinic crudes are good sources of waxes.
- 4) Paraffinic hydrocarbons in the gasoline range burn too readily and lead to the 'knocking' phenomenon. They are poor components in gasoline blends.
- 5) In lubricants they lead to high pour points.
- 6) As motor oil components they tend to form hard carbon deposits,



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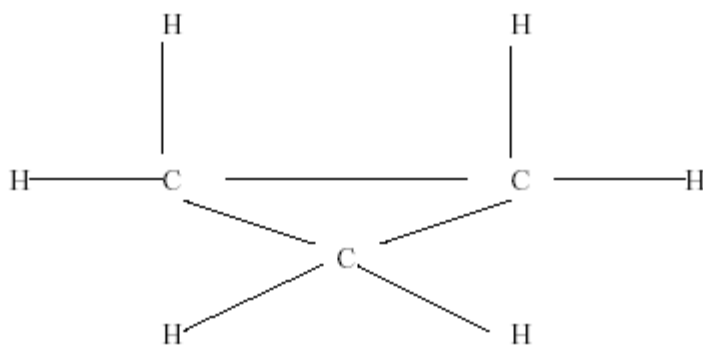
Properties of Iso-Paraffins:

- 1) For the same carbon number (number of carbon atoms in the hydrocarbon molecule), iso-paraffins have lower boiling points than paraffins.
- 2) They make better components in gasoline blends; they have better (higher) octane rating.

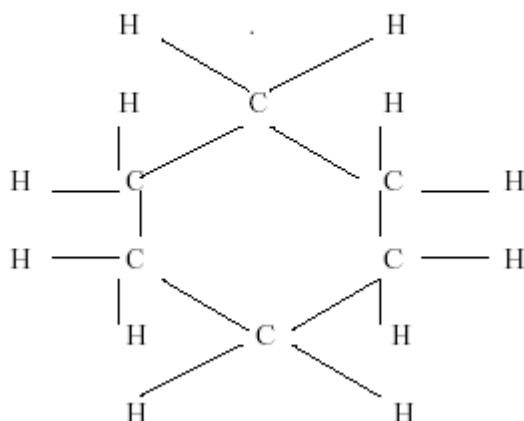
Naphthenic Family:

Naphthenic hydrocarbons have fewer hydrogen atoms per molecule than paraffins. This is because they have a closed or ring structure. Naphthene molecules with one ring have the general formula C_nH_{2n} . They are also known as cyclo-paraffins. While rings can be small (3, 4 carbon atoms) or large (above 6) many naphthenes in petroleum have 5 or 6 membered rings.

C_3H_6 Cyclopropane



C_6H_{12}





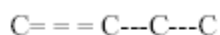
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Properties of Naphthenes:

- 1) Naphthenes in motor oils form soft fluffy carbon deposits
- 2) Viscosity is affected more by temperature change. Viscosity index is lower for naphthenic lubricants for paraffinic ones.
- 3) Naphthenic lubricants have low pour points.
- 4) Naphthenes in gasoline improve its octane rating e.g. n-heptane has 0 octane and methyl cyclo-hexane, 78 octane number.

Olefin Family:

Olefinic are unsaturated hydrocarbons with the general formula C_nH_{2n} . While olefins as such are not normally found in natural crude oil, they are produced by cracking reactions. The simplest member of this family is ethylene. Like paraffins, the higher members of the olefinic family can exist in straight chain (normal) or branch (iso) structure. The location of the double bond can vary, leading to different isomeric compounds.



butylene-1



butylene - 2

Properties of Olefins:

- 1) Olefins are highly reactive. Thus their presence in gasoline or lubricating oils leads to interaction with oxygen to form sludge, gum and varnish.
- 2) In gasoline the presence of some olefins does improve octane rating (anti -knock properties). However, anti-oxidants will have to be added to suppress oxidation tendencies.

Aromatic Family:

Aromatics are unsaturated ring type hydrocarbons of a special chemical category. In these structures, alternating double and single bonds having a property known as resonance confer some stability and other special characteristics.

Aromatic streams from a refinery normally contain benzene or its derivatives, condensed aromatic hydrocarbons like naphthalene or their derivatives.

Properties of Aromatics:

- 1) In view of the property of electronic resonance, benzene hydrocarbons are quite stable.
- 2) High octane values render aromatics excellent blended components.
- 3) Have high solvency power. They make good commercial solvents.
- 4) They are poor viscosity index components in lubricating oils.
- 5) Aromatics in kerosene produce smoky flames (low smoke point).



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Crude Oil:

Various systems of classification have been attempted since the early production of crude oil from the last century. Based on the nature of hydrocarbons present in Crude Oils, they are broadly classified into

Paraffin Based Crude Oils: These consist mainly of paraffinic hydrocarbons and little / no Asphaltenes matter. They usually give good yields of paraffinic wax, high grade SKO and high grade lubricating oils.

Asphaltene Based Crude Oils: They contain little / no paraffin wax but Asphaltene material is usually present in large proportions. They consist of mostly Naphthenes. Lube Oils of these crude oils are more sensitive to temperature. These crude oils give high quality Gasoline.

Mixed Base Crude Oils: These crude oils exhibit considerable overlapping of the both types described above. A majority of the crude oils are of this type.

Recent classifications are based on their API gravity (calculated from specific gravity) and sulphur content. Generally the higher the API gravity (or lighter the crude) the more distillate products it contains and the higher is its value.

$$O_{API} = \frac{141.5}{Sp.Gr @ 15.6^{\circ}C} - 131.5$$

Sulphur is a significant factor in the crude cost as it is an impurity. The sulphur content in the petroleum products is restricted by product specifications. High sulphur crudes also have to be processed after all, but the investment and operational costs are high. Sulphur in crude occurs in different forms like free Sulphur, Hydrogen Sulphide, Mercaptans, etc.

Nitrogen is also present in crude oils in very minute quantities in the form of Indoles, Pyridines, Quinolines, etc. Nitrogen compounds create problems while processing and to the stability of the products. Catalyst deactivation or poisoning, gum formation are some of the offshoots of Nitrogen.

Pour point is also important factor to the extent of handling the crude oil. Crude oils with high pour point require special handling facilities such as heat tracing and tank heating coils. Sometimes pour depressing additives are also used. Pour point is an indication of-wax content in crude oil.



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Low Sulfur crudes:

Crude	Origin	API gravity	Sulfur (Wt %)
Tapis	Malaysia	47.6	0.05
Skua	Australia	40.6	0.09
Bombay High	India	39.2	0.15
Qua Iboe	Nigeria	36.1	0.26
Ratna	India	35.2	0.26
Ravva	India	35.13	<0.1
Labuan	Malaysia	33.2	0.09
Mirri Light	Malaysia	29.19	0.23

High Sulphur crudes:

Crude	Origin	API gravity	Sulfur (Wt %)
Kuwait	Kuwait	31.2	2.53
Dubai	UAE	31.05	2.0
Kirkuk	Iran	35.6	1.99
Basrah	Iraq	33.6	1.9
Umm Shalf	UAE	35.77	1.39
Upper Zakum	UAE	33.9	1.05

RANGE OF HYDROCARBONS IN TYPICAL CRUDE

Component	No. of carbon atoms in the component
Gas	1-4
Virgin Naphtha / Straight Run Gasoline	4-12
Distillates (kerosene – Diesel)	10-18
Gas Oil	14-22
Lube Oil Distillate	16-26
Residuum (Reduced Crude, Heavy Fuel, Asphalt etc.)	26+



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RANGE OF PRODUCTS MADE FROM CRUDE OIL

Propylene
 Liquefied Petroleum Gas - LPG
 Gasoline (Petrol)/SRN
 Naphtha
 Aviation Turbine Fuel (ATF)
 Mineral Turpentine Oil (MTO)
 Kerosene
 Diesel Oils
 Jute Batching Oil
 Lube Oils
 LSHS / Fuel Oils
 Asphalt
 Sulphur

The Crude Distillation Unit of Visakh Refinery Expansion Project consists of Atmospheric and vacuum Distillation Sections. The unit was originally de-signed to process 3.0MMPTA of Basrah crude. It is also capable of processing other feed stocks like Kuwait (31.3 API), Kirkuk (36 API) and 50:50 light / heavy Arabian mix crudes. Besides the unit is also capable of processing Bombay High crude (39.2 API) at 90% design capacity without any change in unit configuration or equipments and for capacities higher than 90%, the following constraints are encountered.

- Low preheat and consequent overloading of Atmospheric Furnace.
- High product rundown temperatures
- Stabilizer Re-boiler duty.

The main equipment of the unit include an electrostatic Desalter, an Atmospheric Distillation Column (ADU), strippers, furnaces, Vacuum Distillation Column (VDU), ejectors, pumps, exchangers, etc. The plant produces straight run products as well as Heavy Vacuum Gas Oil (HVGO) the feed stock for FCC units and Vacuum Residue (VR), the feed stock for Bitumen Blowing Unit (BBU).

From 1991 with the non-availability of Basrah crude, the unit processed mostly BH crude with reduced throughputs due to the above mentioned constraints. To overcome these constraints, the unit was modified in two stages in 1991 and 1993 to process BH crude (or similar crudes) at 100% capacity.



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The unit process can be classified into two sections as given under.

5.1 ATMOSPHERIC SECTION

Crude oil from the off-site tanks is supplied as feed to the unit. In the Atmospheric Section, crude is preheated in a series of preheat- exchangers. The sensible heat of various products and circulating refluxes (CRs) is used for this purpose. After the crude attains a preheat of 125 °C, it is sent into a Desalter for removal of water, salts, metals, sludge and any other impurity. This is achieved by controlled addition of water to the crude to form an emulsion and the separation of the emulsified water from the crude under an electric field. The water thus injected into the crude extracts the salts in the crude. After desalting, the crude is boosted to a higher pressure and is split into two streams. One stream is preheated by the Atmospheric Distillation Column products and the other by the Vacuum Distillation Column products and refluxes .After passing through the preheat exchangers, crude from both the streams is combined and sent into a Pre-Flash Drum (PFD), which facilitates the removal of lighter products from the preheated crude. This works on the principle of flashing. The bottoms of the PFD are then boosted again in a turbine driven pump and sent into an exchanger for further preheat gain in the form of sensible heat of Circulating Oil from FCCU-II. There is also a provision to route crude to the PFD downstream of the Circulating Oil exchanger. The outlet of this exchanger is routed to the Atmospheric Furnace for further heat gain. Once the crude attains the required coil outlet temperature, which is dependent on the crude being processed, it is sent into the flash zone of the Atmospheric Distillation Column.

Crude when it enters the Column is partly converted into vapor phase, which travels up in the Column (enriching section). The balance crude oil which is in the liquid form travels to the bottom section (stripping section) of the column. Stripping steam is introduced at the bottom of the column to strip off the lighter fractions in the bottom product. This also helps in the vaporization of hydrocarbons by lowering of partial pressure† (*Dalton's law of pressures*). The vapors in the enriching section are separated into four fractions namely the overhead fractions and three side draw-offs. The overhead fraction is condensed totally in the overhead condensers and is collected in the reflux drum. To control the top temperature of the column, a part of the above condensed vapors, known as Unstabilized Naphtha, is sent to the column as top reflux. The balance is used as a feed to Naphtha Stabilizer. The stripping steam used in the Atmospheric column gets condensed along with the overhead vapors and is accumulated in the boot of the reflux drum. This condensate, known as Sour water, can either be used for



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Desalter wash water injection, or can be routed to the Sour Water Stripping Unit (SWSU) for treatment.

Heavy Naphtha, Kerosene / ATF and Diesel are the three side draw-offs. They are further steam stripped† (*importance of steam stripping*) in strippers to meet the product specifications. Three Circulating Refluxes (CR's) Top Pump Around, Middle Pump Around (Kerosene CR) and Bottom Pump Around (Diesel CR) are also drawn separately; their sensible heat is removed and then they are returned to the column to maintain the temperature profile inside the column. Unstabilized Naphtha, which is a mixture of Naphtha and LPG, from the overhead reflux drum, is sent to the Stabilizer. The process in the Stabilizer is a simple two-component distillation with a reboiler and overhead condensers. Unstabilized Naphtha is preheated in an exchanger, which extracts the sensible heat from Stabilized Naphtha, and sent into the Stabilizer. The heat input to maintain the bottom temperature of the Stabilizer is provided by a reboiler, which uses Middle Pump Around (Kero CR) as the heating medium. The overhead vapors from the Stabilizer are condensed in the overhead condensers and collected in the reflux drum. To maintain the top temperature, a part of the above condensed liquid i.e., LPG is sent as reflux to the Stabilizer. The balance LPG is routed to MEROX for further treatment. The Stabilized Naphtha or Straight Run Naphtha (SRN) from the Stabilizer bottom is routed to the Caustic Wash Section for the removal of mercaptans and H₂S present in it by washing it with caustic solution. The outlet of the Caustic Wash Section is then routed to the Water Wash Section where the caustic treated

SRN is washed with water to remove any caustic carry-over in SRN. The outlet of this section is routed to storage tank.

The bottom product of the Atmospheric Distillation Column, known as the Reduced Crude Oil is sent to the Vacuum Distillation Column for further processing.

5.2 VACUUM SECTION

The Reduced Crude Oil from the Atmospheric Distillation Column is mixed with Slop-Cut Distillate Recycle, heated and partially vaporized in the Vacuum Furnace and is introduced into the flash zone of the Vacuum Distillation Column. The liquid portion of the feed drops into the bottom section of the Column and is withdrawn as Short residue. The vaporized portion rises up in the tower and is fractionated into three side stream products. The Short Residue is partially routed back to the Vacuum Distillation Column bottom as quench after transferring some of its sensible heat to the crude oil. The balance can be routed either as feed to the Bitumen Blowing Unit, or the Vis-Breaker Unit or can be sent to Fuel Oil storage



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tanks. Slop-Cut Distillate is withdrawn as the first side draw-off from the Wash Zone of the Vacuum Distillation Column. The vapors rising from the Wash Zone pass through the demister pad to ensure the removal of entrained Asphaltenes and metals to achieve the quality of FCCU feed. A slip stream of Slop Distillate, known as Slop-cut Recycle, can be routed to the Reduced Crude Oil stream which is then routed to the Vacuum Furnace.

The Slop-Cut Distillate is mixed with the Short Residue and sent to the Fuel Oil pool. In case the Short Residue is being regulated for Bitumen directly from the column, then there is a provision to route the Slop Distillate to a Cooler Box in CDU-I. The hydrocarbon vapor from the Wash Zone is condensed in the HVGO and LVGO sections. HVGO is drawn off as the second side stream along with the Internal Reflux and the Circulating Reflux. Internal Reflux (Wash Oil) is a hot HVGO stream from the pump discharge routed to the Column below the HVGO bed to avoid coke formation in the Wash Zone. Circulating Reflux is used to preheat the crude, generate steam and then returned to the Vacuum Column at the top of the HVGO section. ***The product can be routed either as hot feed to FCCU-II or to the storage tank.*** LVGO is the third side stream drawn along with the Circulating Reflux and Internal Reflux to the HVGO packing. Circulating Reflux is used to preheat the crude and is cooled further before returning to the Vacuum Column at the top of the LVGO section.

Vacuum is maintained by a three-stage ejector system, with three ejectors in each stage, and surface condensers. The Vacuum Column overhead vapor flows through the 1st stage ejectors. The discharge from the 1st stage ejectors goes to the primary condenser. The non-condensable in this condenser are drawn by the 2nd stage ejectors, whose outlet is again routed to a secondary condenser. The non-condensable of the secondary condenser are drawn by the 3rd stage ejectors. The discharge of the 3rd stage ejectors goes to an after-condenser and non-condensable vapors are routed to atmos Heater (hot well off gas burners) or vented to the Atmosphere through the Hotwell Drum. The condensate from all the condensers is routed to the Hotwell Drum through dip legs. Water and oil carry over in the condensate are separated in the Hotwell Drum. Water can be used for Desalter wash water injection or can be routed to the Sour Water Stripping Unit. Oil which is carried over along with the tower overheads and enters the Hotwell is pumped to the Hotwell Oil intermediate storage tank or the slop oil tank.

5.3 CHEMICAL INJECTIONS

i. De-emulsifier

De-emulsifier solution is added to the crude at the feed pump suction to break the water-crude emulsion. De-emulsifiers are surface activating agents and act on the interfacial surface tension of crude and water emulsion.



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ii. Caustic

Caustic solution is injected either at the feed pump suction or at the suction of the feed booster pump. Caustic is injected to remove the hydrolysable salts, which if not removed, can get converted into HCl and cause corrosion in the system.

iii. Neutralizer

Neutralizer solution is injected to the overhead vapour line and top reflux line of Atmospheric Column and the Vacuum Column overhead system. This helps in maintaining a stable pH at the column overhead area, which is very essential for a good Corrosion control.

iv. Corrosion Inhibitor

Corrosion Inhibitor solution is injected into the overhead vapour line and the Top Reflux line of the Atmospheric Distillation Column and in the overhead vapour line of the Vacuum Distillation Column. This helps in preventing the contact of corrosive water and acids with the shell of the column. It acts by forming a film on the surface of the shell in the overhead area. This film is impervious to the acids formed.

5.4 DISTILLATION

The Chemical Engineering operation used in the Crude / Vacuum Distillation Unit to process crude oil (or range of hydrocarbons) into a range of products is Distillation. Distillation technique is employed to separate various product cuts from the crude petroleum in this primary distillation unit.

Distillation is one of the many separation processes employed in chemical industry. It is a physical process (not necessarily involving chemical reactions) where separation is achieved using differences in their boiling points or, in other words, difference in volatility. The application of this technique ranges from the simplest binary distillation to the most complex distillations like azeotropic or extractive distillations.

5.4.1 PRINCIPLE

If solution of two components with different boiling points is allowed to flash in a vessel, the liquid and vapour portions separate and after sufficient time attain equilibrium. The vapors will be richer in lighter components and the liquid residue therefore leaner. Suppose the vapours are condensed and flashed again, the resulting vapours will be richer in the lighter components. By repeating the procedure, we will reach a stage when vapours will be full of the lighter components (ideally). Similarly, by repeatedly heating and flashing the liquid portion, we will eventually end up with a liquid which is hundred per cent the heavier component. The same principle is used in a distillation column with an integration of the above process and each step where the heavier component in the vapours is condensed and



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the lighter component in the liquid is vaporized known as an equilibrium stage. The total number of theoretical stages will be decided by the extent of separation relative volatility of the components involved.

In a typical distillation operation, the feed is introduced into a vertical cascade of stages. Vapour rising in the section above the feed (called enriching / rectifying section) is washed with liquid to remove the less volatile components. The wash liquid is provided by condensing the vapour from the top (rich in more volatile component) called reflux; a portion of the condensate is removed as distillate. In the section below the feed (called stripping section), the liquid is stripped of the volatile component by vapours rising from bottom. The vapours are generated by supplying the necessary energy at the bottom through a reboiler or furnace. The liquid rich in less volatile component is removed from the bottom.

5.4.2 MULTICOMPONENT SYSTEM

Through a binary system is ideal to design and operate, many of the separations encountered in the industry are not so; they involve more than two components. The principles of binary solutions are generally applicable to such distillations but nevertheless some special consideration and techniques are needed more volatile components are designated as light and the less volatile, heavy. Suppose a solution has to be separated by distillation, the majority component among the bottom product components compared to the other lighter components is called the light key component. Similarly, the one among the distillate components which is present in considerable amount when compared to the other heavier components is the heavy key component. With this key component as the basis, now the problem of multi-component distillation is treated in much the same way as binary distillation. The difficulty of separation, as measured by the number of trays for a given reflux ratio, is fixed by the key component concentrations in the products. It is, therefore, important to establish the key components in a multi-component distillation.

The distillations involved in petroleum industry are further complicated by aspects like withdrawal of side streams (apart from top and bottom products), circulating refluxes, stripping steam, etc. Here, the separation is achieved not directly on the basis of components but by boiling ranges. Each product comprises multiple components and its end use requirements specify certain properties such as boiling range, flash point, specific gravity, viscosity, etc., rather than component purity. The design of such systems is very complex and cannot be accomplished by totally theoretical methods; pilot plant studies coupled with past experience generally yield satisfactory results. Many software packages are available in the



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present day market to evaluate the performance of the existing columns and also to design new columns.

5.4.3 DISTILLATION TOWERS

Fractionating towers and related equipment are mechanical devices for repeatedly establishing equilibrium between ascending vapour and descending liquid and repeated separation of the two phases. Hence, a means of attaining a large interface for contact and affecting a complete separation of the two phases must be incorporated in any successful design. The choice of contacting device in a column depends on

- Operating pressure and pressure drop.
- Turn-down ratio
- Nature of the solution (foaming tendency, presence of solids, etc.)
- Number of side streams

The most generally used in the industry are tray towers in which the liquid and vapour are contacted in steps or trays or plates.

Types of trays:

- Trays with down-comers (Bubble -cap, Sieve, Valve)
- Trays without down-comers (Duel flow, baffle)
- Multi down-comer trays
- Collection or chimney trays

In tray towers, the liquid from the stage above flows across each tray and through a down-comer to the tray below. The gas passes upward through opening in the tray, then bubbles through the liquid, disengages and passes on to the next tray above. The depth of liquid on the tray required for gas contacting is maintained by an overflow weir. Each tray of the column is a stage. The number of equilibrium stages (theoretical trays) determines the number of actual trays. Tray spacing is usually decided on the basis of adequate insurance against flooding & entrainment and on expediency in construction, maintenance & cost. It varies from 300 - 900 mm depending on the diameter and service of the tower. Column diameter or cross sectional area is determined on the basis of gas/liquid volumes to be handled. Packed columns are preferred to tray towers under the following circumstances:

- for columns of less than 2 ft. diameter
- for acids and other corrosive materials
- for foaming liquids
- for thermally sensitive liquids which require low liquid hold-up
- for lower pressure drop or for vacuum operation
- for greater mass transfer efficiency



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The design of packed towers involves the determination of HETP (Height Equivalent to Theoretical Plate). Packings can be either random or structured. The various types of packings available are

Generation	Type of packing
First	Raschig, Lessing, Cross Partion Rings, Berl saddles
Second	Pall Rings, Hypak, Intalox Metal Tower packing(IMTP), Cascade Mini Rings (CMR), Nutter rings
Third	Gempak, Mellapak, Intalox (structured)

Care should be exercised to distribute feed and reflux streams uniformly throughout the cross-section of the packing to avoid channeling. The withdrawals will be from collector or chimney trays.



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GENERAL

CDU-II is an integrated unit which has both CDU/VDU and BBU. The CDU/VDU unit process can be divided into 7 sections as given under.

- 1) Crude Distillation Unit
- 2) Naphtha Stabilizer Unit
- 3) Caustic / Water Wash Section
- 4) Vacuum Distillation Unit
- 5) Tempered Water / DM Water System
- 6) Steam Generation
- 7) Chemical Injection Facilities

6.1 CRUDE DISTILLATION UNIT

The Crude Distillation Unit is further classified into the following sub sections

- Feed supply
- Pre-Heat Train-I
- Desalter
- Pre-Heat Train-II
- Pre-Flash Drum
- Atmospheric Heater
- Atmospheric Distillation Column/Main Column
- Product Strippers
- Product Coolers and Routing to Storage Tanks

6.1.1 FEED SUPPLY

Crude is supplied to the unit as feed from the crude tanks. Crude enters the unit in a 10" header line at a pressure of 1-2 kg/cm² or 6-8 kg/ cm² in the case of Refinery tanks or ATP tanks respectively. 11-PG-101 is provided to indicate pressure of the crude from the storage tanks.



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This crude is fed to a crude charge pump 11-P-01A/B to raise the feed crude pressure to 24 kg/cm² g. The rated capacity of the pumps is 482 m³/h. However, the pump can be operated at a design limit of 520 m³/h. Both the pumps are centrifugal type. While one pump (01A) is motor driven, the other (01B) is turbine driven one. Normally 01A is used and 01B is kept as a standby. The discharges of 01A and 01B have PG's 11-PG-103 and 11-PG-102 respectively. The discharge pressure of the pumps is indicated on the DCS panel as 11-PI-101. There is a low pressure switch 11-PSL-101 on the pump discharge which sounds alarm on the annunciator panel in the DCS room to indicate low discharge pressure. Turbine auto cut-in facility was provided in case the discharge pressure of the discharge pressure of the motor driven pump comes below 18 kg/cm² g. However this facility has been defunct.

There are provisions for injecting Demulsifier and Caustic at the pump suction with individual isolation valves. A 4" service water connection and a 4" flushing oil connection are provided for use during shut-down. There is also a 6" provision to route the feed pump suction or RCO pump discharge to slop via the unit slop header. The crude charged to the unit is recorded and integrated in 11-FR/FQ-104. From the pump discharge header, the crude is sent into the Preheat Train-I. Crude pre-heating is done in two modes – BH mode and PG mode depending on the type of the crude being processed. Though BH mode operation is more common, PG mode is also used in special cases. For this reason, both the cases are discussed in the following sections.

6.1.2 PREHEAT TRAIN-I (PHT-I)

This preheat train is totally upstream of the Desalter. It has 7 exchangers from 11-E-01 to 11-E-07. This preheat train helps in heating the crude to the desalting temperature which helps in improving the crude oil's fluidity by decreasing its viscosity and also helps in thorough mixing of desalting water. Various product streams are used to preheat the crude in this train.

1.BH Mode: The unit is required to be run in Bombay High (BH) mode whenever low sulphur crude is processed. Low sulphur crudes have low tar content and lighter ends are more. Hence, these crudes require lesser preheat.

2.PG Mode: The unit is operated in Persian Gulf mode for processing high sulphur crude. High sulphur crudes have high tar content and the percentage of lighter components is lesser. Hence, the pre-heat required for such crudes is more.

The following table given on the next page gives operating temperatures for both BH and PG cases. All those exchangers (listed on the table) can be bypassed and isolated



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individually on the shell and tube side. There are TI's on all the shell and tube-side outlets of the above exchangers and the condition can be monitored from the DCS panel. The crude thus preheated to 125-135°C goes to the Desalter.

Facility to Place 11-E-7 before /After Desalter

11-E-07 can be placed before as well as after the Desalter. Normally it is placed in the upstream of the Desalter. For better efficiency of the Desalter, its pressure has to be maintained between 10.0 -11.0 kg/cm² at a temperature ranging from 120 to 130 °C. If the temperature is high and pressure is low, crude will vaporize, which is undesirable for Desalter operation. This can occur only when preheat at the upstream of Desalter is high and hence the above facility was proposed. The switching over of 11-E-07 to the downstream of desalter will help in reducing vaporization in desalter and thereby would lead to a gain in total crude preheat.

Heat exchanger (shell side and tube side fluids)	BH mode operation		PG mode operation	
	Crude inlet temperature, °C	Crude outlet temperature, °C	Crude inlet temperature, °C	Crude outlet temperature, °C
11-E-01 (crude/ HN)	30	37.5	30	35
11-E-02 (Crude/ KERO)	37.5	52.5	35	44
11-E-03 (crude/diesel)	52.5	66	44	60
11-E-04A/B (crude/TPA)	66	101	60	95
11-E-05 (Crude/ KERO)	101	117.6	95	111.5
11-E-06 (crude/diesel)	117.6	128.3	111.5	118
11-E-07 * (crude/ LVGO)	128.3	135.1	118	125

* There is a provision to place 11-E-07 either upstream or down-stream of Desalter.



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6.1.3 DESALTER

Crude oil brings along with it salts of Sodium, Magnesium etc., metals like Arsenic, Vanadium etc., and sludge. Although these are present only in small amounts, their presence can result in serious problems in down-stream equipment viz., heat exchangers heaters and columns.

Specific effects resulting from salts in the crude are:

- At high temperatures, hydrolysable salts in the crude like $MgCl_2$ hydrolyse to Hydrochloric acid which causes severe corrosion in Crude Distillation Column and piping.
- Salts of Sodium and Calcium act as a catalyst for coke formation in furnace and heat exchanger tubes which cause plugging and reduced heat transfer rates in the exchangers. Excessive coke formation results in escalation of hot spots on heater tubes which can have serious and disastrous consequences for heater tubes.
- Salts and solids, concentrated in the residuum of distillation towers, result in high ash content and degradation of the product
- The presence of metals like Arsenic and Nickel act as poison to catalysts that may be used in the down-stream process units.

Thus removal of the impurities in the crude oil is very essential to avoid their deleterious effects. Desalting of the crude is a very effective method to remove these impurities.

The advantages of crude Desalting are given below:

- A Desalter is a shock absorber for smoothing out abnormalities due to slugs of water during feed tank switches.
- Slop oil injection to the crude can be achieved.

At a modest cost Crude + Desalter wash water mixture enters the Desalter which employs electrostatic elements to coalesce and separate water from oil. Wash water is used for reducing the salt content in crude oil so that the salts are within acceptable and specified limits in the desalted crude. Application of strong electric field hastens the process of coalescing and therefore settling of all unwanted material. When the crude-water mixture is subjected to a high-potential electric field, the tiny water droplets get distributed between the electrodes forming dipoles. This makes it possible for the tiny particles to coalesce and form bigger particles of sufficient weight to settle down. The force of attraction between two droplets must be of sufficient magnitude to break through the oil film. Thus in effect, oil-water mixture separates into two phases in Desalter. While the oil phase floats on top and overflows, water with its dissolved salts, metals, mud and iron oxides settle down at the



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bottom. At a level below the electrodes an interface is formed between hydrocarbons and water. This inter phase level is controlled by regulating the effluent water withdrawal rate from the vessel.

Caustic injection in crude, upstream of Desalter, is done to neutralise acids present in crude and convert them into salts. These salts are then removed by wash water in the Desalter. Caustic injection down-stream of Desalter is provided to neutralise any other acid traces formed at the Desalter operating conditions. Brine is associated with crude both as a fine suspension of droplets and more permanent emulsion. To break these light emulsions Demulsifier chemical is added to the crude line at the Battery Limit. This ensures better functioning of the Desalter.

Exact details of the Desalter can be obtained in the vendor's manual.

The description of desalting process in CDU-II is dealt in three subsections

- Desalter (11-V-02) Description
- Desalter Wash Water System
- Desalter Operation

Desalter (11-V-02) Description:

This is a single stage Desalter unit with the following specifications:

Parameter	Specifications
Electric power requirement	20 Kv
Pressure drop across the mixed valve (kg/cm ²)	0.7 - 1.0
Operating temperature °C	120-130 ° C
Operating pressure (kg/cm ²)	10.5
Salt content of fresh crude (max. lb/1000bbl)	20
Salt content of desalted crude, max.	10
Water content of the product crude (%wt.)	0.1 BS&W
Effluent water oil content (% vol ppm)	100

Crude and Desalter wash water enter the vessel (11-V-02) at the bottom. Desalted crude leaves from the top of the vessel. There is a provision to bypass the Desalter if required through a 12" line. Crude outlet from top of the vessel is connected to the crude booster pumps (11-P-02 A/B) suction. Drain connections to CBD / OWS are also provided on the Desalter for maintenance activities.



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Since Desalter is a liquid filled vessel, the Desalter pressure control (11-PIC-105) is achieved by manipulating 11-PV-105 provided on crude line at the inlet of Preheat Train-I. 11-PSV-101 (set pressure 12.5 kg/cm² g.) is provided to protect the Desalter from over-pressurization due to blocked outlet. Discharge of the Pressure Safety Valve is connected to the flash zone of Atmospheric Column. The Pressure Safety Valve is located close to the Atmospheric Distillation Column to reduce the discharge pipe length where two phase flow may occur after PSV discharge. The Desalter approach temperature of the crude-wash water mixture downstream of the mix valve is indicated on the DCS panel by 11-TI-101.

The crude-wash-water mixture is subjected to intense electric field created by two grid type electrodes located in the Desalter. The power to these electrodes is supplied by 3 transformers mounted on the Desalter. Panel mounted push button are provided to **control** the power supply. There is a provision of 3 ammeters and 3 voltmeters in the field to indicate the condition of the electric field. There is also an indication of the voltage on the DCS panel as V1901A&B and VC1901. The higher the amperage, the more conducting is the electric field in the Desalter. There is a trip provision on the power supply to the Desalter which gets activated at high amperage.

Two types of instruments are provided for measuring and controlling the inter phase level

- 1) 11-LIC-101 which is a Displacer type instrument
- 2) Agar ID and OW (Interface Detector and Oil-Water) probe.

The Agar Probe System has three probes on the Desalter which measure the water concentration in the zone they are present. The details of the Agar Probe System are given in the table below.

Probe #	Location	Significance
LC1101A	At 45 ° angle	Indicates the interphase level
LI1101B	Below the lower electrode	Indicates the water concentration in the crude layer. Also known as the emulsion level indicator
LI1101C	Bottom	Indicates the sludge level

Either of the two LIC's can be used for controlling the inter-phase level. This is achieved by changing the software switch LC1101SS on the DCS panel. The LIC controls the inter phase level by operating the control valve LCV-1101 (on the brine line in the SWSU), which controls the flow of effluent water in a 4" line from the Desalter. LIC-1101A has software LAL and LAH alarms. A low level switch 11-LSL-102 is mounted on the top of the Desalter drum to protect the system. Actuation of 11-LSL-102 switches off the power



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supply to Desalter locally. There is a provision of 5 try-cocks in the lower half of the Desalter for physical verification of the interphase level.

The material that collects at Desalter bottom is a thick sludge that is corrosive and often has a tendency to choke piping. This sludge material needs to be cleaned periodically. There is a provision to carry out desludging operation with the unit online. Desalter wash-water pumps (11-P-12 A/B) are used for this purpose. A 3" line (Desludging connection) is provided from the pump discharge. There are distributors / Spargers inside the Desalter for effective desludging. Desludging is a batch process. It loosens up the muck collected at the bottom and facilitates easy exit from outlet. Care should be taken during this operation as sudden jerk can upset oil-water interphase leading to water and salt carryover downstream of the Desalter.

The degree of mixing between the crude and the wash-water can be varied by changing the pressure drop (DP) across the mix-valve 11-DPV-101. Sample points are provided on Crude oil inlet to Desalter, sludge water, from vessels itself and crude oil outlet from Desalter. Hot samples are cooled in a sample cooler. Sampling cock lines have been provided with a 3" LP steam flushing line to clean the line of any crude after taking samples. Sample cocks, called try cocks, are provided on Desalter at various elevations of the vessel. These are used to check interphase level physically against that indicated by LIC-1101A. There is also a bypass provision for the Desalter for operational flexibility. There is a 2" LP steam provision to serve as a steam out point.

Desalter Wash Water System:

Stripped water from Sour Water Stripping Unit, Atmos Sour water, Hotwell water, MAB condensate from FCCU-II (*if the chloride content is less than 240 .ppm*), Service water and also DM water can be used as wash-water for desalting. This water is first collected in Desalter Water Vessel (11-V-04) whose level is maintained by the level controller 11-LIC-103. A low level alarm 11-LAL-103 is provided to indicate the low level in the water drum.

Desalter water pumps 11-P-12 A/B delivers water from 11-V-04 to 11-E-18 where LP steam is used to preheat the water to 125 °C. The flow of the wash water is controlled by the FCV 11-FIC-102. There is a 3" water injection provision with a check valve and isolation upstream of a manually operated mix-valve to supply the preheated water to the Desalter. There is also a provision to route the stripped water from SWSU directly to the Desalter in a 4" line at a temperature of 110 to 120 °C.



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The degree of mixing between the crude and the wash water can be varied by changing the pressure drop (DP) across the mix-valve. Normally a DP of 0.5 kg/cm² gives efficient mixing. The crude approach temperature to the Desalter can be controlled manually by operating the bypass of one of the exchangers between 11-E-01 and 11-E-07.

Desalter Operation

Some of the Parameters that are to be closely monitored to realize good Desalter operation are:

1) Wash water

Dilution is the primary purpose of wash water injection. Acids like HCl and cyanide lose their corrosive potency and salts can be washed out in solution with sufficient quantity of water. Most Desalters are designed to use 4-6 volume percent water based on the crude charge.

Wash water injection provision provided just upstream of Desalter mix valve is to be used in crudes which have severe tendency to emulsify. Wash water can also be injected before the crude charge pump. Injection at this point results in maximum contact and also prevents the sediments from settling in the exchanger tubes and fouling them. But care should be taken such that the intense shearing agitation in the preheat train does not create so tight an emulsion that cannot be resolved in the Desalter. The severe shearing effect due to the crude pump impellers should also be considered here. The quality of water is a very important aspect.

It is recommended that ammonia-free and caustic-free wash water having a pH of 6.5 is used in the desalter for effective removal of chloride salts, scaling salts, sediments and water in the crude. Lower Desalter wash water pH of 6.5 reduces the amount to hydrogen sulphide in the effluent water, lowers the filterable solids, i.e., iron sulphides in the desalted crude, reduces the quantity of oil in the effluent water and keeps the magnesium and calcium carbonates in the water phase. These salts become insoluble in water with increasing pH and deposit at the bottom of the Desalter as sludge. Both ammonia and caustic will accelerate the precipitation of sludge in the Desalter.

The use of un-stripped sour water recycle to the Desalter can lead to operational problems of the Desalter. The most common problems are listed below

a) It stabilizes the crude emulsions in the Desalter resulting in water carry over from Desalter which can upset the overhead corrosion control. The following disadvantages result when water is carried over from Desalter.



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- Water carried over from the Desalter results in increased pressure drop in heat exchanger train and the heater.
- Steam occupies 7 to 10 times the volume compared to hydrocarbon. This causes fractionators tray vapour loading.
- Energy consumption increases due to vaporization of the entrained Desalter wash water.
- Higher quantities of water in the overhead will raise the dew point temperature of the water in the overhead that may result in the condensation of water inside the tower. This can result in an aggressive level of corrosion attack on the top tray.

b) Ammonia present in the sour water recycle causes high pH which has several disadvantages.

1. The filterable solids and metals from the crude oil will not be removed.
2. Higher amounts of filterable solids (less than 20 microns) have been found in the desalted crude compared to the raw crude. These solids are found to have the same elemental makeup as the deposits found in the heat exchangers and heaters.

2) Settling time

The settling time of the water droplets is very critical for the removal of water from the oil. Settling time is directly proportional to the viscosity of the fluid and inversely proportional to the square of the droplet size and difference in density of the drop and the fluid.

3) Temperature

The water separation is accelerated by the rise in temperature of the oil as the viscosity of settling medium – oil – decreases with temperature, but water carryover in desalted crude increases with temperature due to the greater solubility of water in hydrocarbons at higher temperatures. Hence care should be taken that the design operating temperature is not exceeded for most of the time

4) Desalting Chemicals

There are two types of emulsions. One is a continuous phase of oil with some emulsified water and the other has water in the continuous phase with hydrocarbons suspended in it. Generally, solid particles are coated with heavy oil or wax, which inhibits water contact. Wetting agents or emulsifiers, strip the oil and aid in water contact. But emulsifiers being semi-polar in nature induce polarity on water droplets and also change the surface tension, making it difficult for the droplets to coalesce. Demulsifiers either weaken the water film or



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reduce / change polarity of the droplet. This helps in resolving the emulsified oil. Caustic is used to control the pH in the Desalter.

5) Alternating Electric Fields

The thin film of emulsifier surrounding the water globule is ruptured by the sharp force of the alternating electric field and the water so released coalesces. This mechanism involves the vibration of all the water droplets within the electric field. The same effect is also achieved by the collision of water globules with each other.

6) Mixing valve

This is used to produce maximum contact of wash water with the crude oil and is critical for optimising the Desalter performance. The pressure differential (dP) across the mix valve controls the degree of emulsification. Increase in the mix valve dP causes a proportional increase in salt removal. But the emulsions formed under very high dP are so tight that the water droplets will not coalesce, resulting in water and salt carryover, defeating the very purpose of Desalter.

7) Pressure

The purpose of pressure is to keep the system from boiling. The agitation resulting from boiling would result in severe carryover of BS&W and salts. The operating pressure of Desalter is determined by adding 1.021 to 1.361 kg/cm² to the vapour pressure of the crude oil at the operating temperature.

8) Level

A steady interface is to be maintained between the oil emulsion and water phase in the Desalter. An emulsion usually of oil in water exists below this interface. This interface supports particulates, solids encapsulated by oil and emulsified with the water. The operating level of this interface should be maintained below the lower electrode in order to avoid an amperage overload, which could shut-down the electrical system. Also high interface level can result in not only emulsion carry over to the distillation tower upsetting its conditions, but also can accelerate the fouling of exchangers. At the same time, too low interface level can result in oil carry-under, upsetting the downstream Effluent Treatment Plants.

9) Agar Probe System



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Agar instruments function by the principle of energy absorption. All the materials absorb at different rates. AGAR technology uses this fundamental physical property to differentiate between two materials based on the rate at which the energy is absorbed. One common example of two such materials is oil and water. Water absorbs energy at a much higher rate than hydrocarbons. When an AGAR ID and OW (Interface Detector and Oil-Water) antenna is surrounded by water, the energy from that antenna is absorbed very quickly. This puts a high load on the attached transmitter, and the transmitter then sends a high signal back to the PS (Power Supply / Signal). However, when oil is present at the antenna, the rate at which energy is absorbed from the antenna is much lower, resulting in a lower load and signal sent to the PS. When calibrated properly for oil and water, an AGAR ID and OW will use this transmitter load to determine the concentration of oil. The signal from the transmitter to the PS is then converted to a 4-20 mA current that is proportional to the amount of water (water concentration by volume) in the area around the antenna from 0% (all oil) to 100% (all water).

This system is very effective since it achieves control not by measuring the imaginary interface level but by measuring the actual water content at different elevations in the system. There are 3 agar probes on the Desalter at the locations as shown **in the figure**. AGAR ID and OW, LC-1101A located at 45° inclination indicates the interface level of crude and water in that location. AGAR ID and OW, LI-1101B located horizontally just below the central line of the Desalter vessel (just below the power grid), indicates the water concentration in the crude layer at that location. During normal operation, it reads 0 to 1%, indicating that the water concentration in that zone is 0 or 1%. If it shows a value greater than 0, it indicates that the emulsion level is rising in the Desalter.

LI-1101C located at the bottom of the Desalter vessel indicates the concentration of water at that location. During normal operation without any sediment, it reads 100%, indicating the water concentration to be 100% at that level. However, an indication below 100 indicates the accumulation of sludge in the Desalter and the necessity for online desludging. The desalted crude from the Desalter is taken from the top of the vessel to the suction of the Crude Booster Pump 11-PM-02A. The standby pump 11-PT-02B is used for PFD service. However, this pump can be used when in Crude Booster mode, when PFD is out of service. The Booster Pumps give a discharge pressure of 33.6 kg/cm² g. The discharge pressure of 02A and 02B are indicated in the field by 11-PG-113 and 11-PG-114 respectively. There is a provision for caustic injection at the suction of the Booster Pump. The crude thus boosted is sent to the Preheat Train-II for further preheating.

6.1.4 PREHEAT TRAIN-II



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The crude from the Desalter is split into 2 streams. One stream goes to the Atmos-side exchangers, whereas the other goes to the Vac-side exchangers. The flow to the Atmos-side exchangers is indicated and recorded by 11-FI/FR-105. The flow to the vac side exchangers is controlled, indicated and recorded by split flow control valve 11-FRC-101.

Since, the pre-heat requirement of both the types of crude is different, so, the sequence of exchangers in preheat train II is slightly different for the two operating modes. This motive of the alignment is to add more pre-heat to the high sulphur crudes. The sequence of the exchangers in the path of the crude flow in the BH and PG modes is given as follows.

BH Mode

Atmos-side Exchangers

Heat exchanger(shell side/ tube side fluid)	Crude inlet temperature, °C	Crude outlet temperature, °C
11-E-08 (crude/ diesel)	131.1	140.6
11-E-11 (Kero CR / crude)	140.6	169.3
11-E-10 (crude/ Kero)	169.3	172.2
11-E-09 (Kero CR / crude)	172.2	180.9
11-E-12 (Diesel/ Crude)	180.9	204.1
11-E-14 (Diesel CR / crude)	204.1	236.6
11-E-13 (diesel/ crude)	236.6	252.2
11-E-15 A/B (diesel CR/ crude)	252.2	278.8
11-E-16 (SR/ crude)	278.8	281.1

Note:-Now a days, the sequence of exchangers that is being followed in BH mode of operation is 11E8→11E11→11E10→11E9→11E12→11E13→11E14→11E15→11E16

Vac-side Exchangers

Heat exchanger(shell side/ tube side fluid)	Crude inlet temperature, °C	Crude outlet temperature, °C
12-E-01ABC (Kero CR / crude)	131.1	172.1
12-E-02 (crude / HVGO)	172.1	193.9
12-E-03 (SR/ crude)	193.9	204.7
12-E-04 (crude/ HVGO)	204.7	233.4
12-E-05 A/B (HVGO CR/ crude)	233.4	258.8
12-E-06A/B (crude/ SR) *	258.8	265.5



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PG Mode

Atmos-side Exchangers

Heat exchanger(shell side/ tube side fluid)	Crude inlet temperature, °C	Crude outlet temperature, °C
11-E-08 (crude/ diesel)	120	126
11-E-09 (Kero CR / crude)	126	161.5
11-E-10 (crude/ Kero)	161.5	168
11-E-11 (Kero CR / crude)	168	204.5
11-E-12 (Diesel/ Crude)	204.5	218
11-E-13 (diesel/ crude)	218	235
11-E-14 (Diesel CR / crude)	235	251.5
11-E-15 A/B (diesel CR/ crude)	251.5	278
11-E-16 (SR/ crude)	278	290

Vac-side Exchangers

Heat exchanger(shell side/ tube side fluid)	Crude inlet temperature, °C	Crude outlet temperature, °C
12-E-01ABC (Kero CR / crude)	120	164
12-E-02 (crude / HVGO)	164	181
12-E-03 (SR/ crude)	181	213.5
12-E-04 (crude/ HVGO)	213.5	241.5
12-E-05 A/B (HVGO CR/ crude)	241.5	271.5
12-E-06A/B (crude/ SR) *	271.5	290

*RCO can be routed to 12-E-06 A/B on its tube side.

The crude thus pre-heated goes to the PFD and 11-E-40 A/B for flashing and further preheating respectively.

6.1.5 PRE-FLASH DRUM (11-V-10)

Pre-Flash Drum was introduced to enhance the crude throughput of the unit by 0.2 MMTPA. This is achieved by the removal of lighter components in the crude by flashing it and thereby reducing the load on 11-F-01. A maximum of 20 – 25% by volume of the total crude entering the PFD can be achieved depending on the type of crude being processed.

The temperature and pressure of the crude after the pre-heat train is 250 °C+ and 20 kg/cm² g. respectively. When this crude is allowed to flash i.e., its pressure is suddenly decreased, the lighter fractions in the crude whose vapour pressure at that temperature equals the



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system pressure will vaporize. The operating pressure of the PFD varies with the crude being processed.

Crude enters the PFD approximately in the middle of the vessel. There is a provision to record the crude inlet pressure and temperature on DCS panel as P1901R and T1901R respectively. Crude that flashes in the PFD exits in the form of vapours from the top of the PFD and enters the 12th tray of the Atmos Column. A demister pad is provided on the vapour line to knock off any liquid droplets entrained in the vapour. The amount of flashing depends on the pressure in the PFD, which is controlled by PR1902. The crude level in the PFD is controlled by LR1902. Level gauge glasses are also provided on the shell of the PFD for physical verification of the level. There is also a high-level and low level switch provision. The un-flashed heavier crude goes to the PFD turbine 11-PT-02B from where it is boosted to 25 kg/cm² g. and sent to the PFD manifold through 11-E-40 A/B. The crude outlet flow from the PFD and its temperature are indicated on the DCS panel by F1902R and T1902R (provided on the outlet of crude from PFD) respectively. The PFD has two pressure safety valves (120-PSV-1201 A&B) set at 25.5 kg/cm² g. The discharge of the PSV is connected to the flash zone of the Atmospheric Distillation Column (down stream of Desalter RV).

There is a provision to bypass crude flow to PFD in case of emergency, from the DCS panel. This is done by operating the ROV switch on the auxiliary panel. There is an interlock on this ROV operation. When ROV is opened, the LCV-1902 gets closed. This interlock is provided to prevent the crude entry into the PFD when it is bypassed. A switch is provided on the auxiliary panel to trip the turbine (11-P-02B) from the DCS panel. Operating this switch on the auxiliary panel will close the steam shut off valve on the steam inlet to the turbine, thereby cutting off steam to the turbine. There is a provision to reset the SDV manually in the field, by which the SDV can be reset to open position. The outlet* of the PFD is routed to 11-E-40 A/B for further heat gain. Sensible heat from Circulating Oil of FCCU-II is utilized to preheat the crude cover heat further in this exchanger. The outlet of 11-E-40 A/B, goes to the Atmospheric Furnace for further heat gain.

**There is a provision to place 11-E-40 A/B either upstream or downstream of the PFD. This provision is made at the PFD manifold.*

11-E-40 A/B (Crude/Circ. Oil)

Atmospheric Furnace (11-F-01) was limiting the throughput due to low preheat. In spite of repeated cleaning of the preheat exchangers, the feed preheat was low and hence high load



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on 11-F-01. Hence to reduce the load on the furnace, the above scheme has been implemented.

11-E-40 A/B can be placed before PFD or after PFD. But it usually it is placed on the downstream of PFD because 11-E-16 & 12-E-06 A/B outlet stream temperature are sufficient enough for flashing out the light ends of the crude in PFD. This exchanger can be placed in service even after bypassing PFD. This scheme has experienced preheat pick up of up to 15 °C when feed was 500 m³/hr. Thus 11-E-40 A/B has helped in recovering the heat lost In PFD due to flashing, thereby sustaining higher throughput.

6.1.6 ATMOSPHERIC HEATER:

Refer P&ID No.

The major equipments of this section are Atmospheric Heater, Air Pre-heater, ID Fan, FD Fan and Steam Decoking Pot. The description of the Atmos Heater is divided into the following sub sections:

- 1) Process System
- 2) Fuel System
- 3) Air Preheating System
- 4) Trip and Interlock System
- 5) Steam Air Decoking

6.1.6.1 Process System:

The crude from the PFD enters the Atmospheric Furnace, where it is heated further so that it reaches the required flashing temperature. This is a vertical cylindrical type heater with absorbed design heat duty of 40.226 MM Kcal/h for crude oil and 0.9 MM Kcal/h for superheated steam same was revamped during 2010 T&I and heat duty was raised to 45.68 MM Kcal/h and 0.92 MM Kcal/hr for superheated steam (total 46.6 MM Kcal/h). The crude coming to the heater gets split into 4 streams and enters the 4 crude passes of the furnace under the flow recorders and controllers 11-FRC-301/302/303/304. Low flow alarms 11-FAL-301/302/303/304 are provided on all the passes to protect the heater pass tubes in case of low flow through each pass. Steam connections are provided downstream of the FCV's for purging the respective coils in case of emergency. The isolation valve for the purging steam line is provided at a safe distance from the heater.

The feed enters the convection zone of the heater first. It is located at the top of the furnace above the radiation zone. The purpose of this zone is to increase the thermal efficiency of the furnace by extracting heat from the flue gas leaving the radiation zone. This zone consists of 12 rows of tubes with 8 tubes per row. These 12 rows are arranged in 3 bundles



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with 4 rows per bundle. While the top 2 rows of this zone are finned type the bottom two are bare tubes. All the others are studded type. The effective length of each tube in this zone is 9058mm. The material of construction of the tubes is 5% Chromium + 0.5% Molybdenum. Additional convection zone was provided during 2010 T&I to increase the heater efficiency and heat duty. The additional convection zone consists of 8 rows of tubes with 8 tubes per row. In these 8 rows, top 4 rows of this zone are 24 SPP studded tubes, below it 2 rows of 12 SPP studded tubes and the bottom two are bare tubes.

Soot in the flue gas gets deposited on the tubes in the convection zone. This reduces the heat transfer efficiency of the tubes which can be noticed by increasing flue gas temperature and decreasing heat pick up in the convection zone. Hence a provision for the external cleaning of convection tubes is made in the form of soot-blowers. They are retractable motor-operated soot-blowers. They are arranged in 2 rows with 4 blowers in each row and additional 1 row was provided in 2010 T&I for new convection zone. Procurement of soot blowers for this zone is in progress. Soot blowers can be operated from the grade level at the bottom of the furnace. Pressure gauges are provided on the inlet and outlet of each pass in the convection zone. The pressure drop across each pass is a measure of coke formation in the tubes. Temperature indicators (11-TI-302, 307, 312, and 317) are provided on outlet of the convection zone to measure the temperature gain in this zone.

The convection zone has a single row (8 tubes) of steam coil in which saturated MP steam (@10.5 kg/cm² g.) generated in the unit gets superheated (~ to 350 °C). This superheated steam is used as stripping steam in Atmospheric Distillation Column and Product Side Strippers. The material of construction of these tubes is carbon steel. Pressure gauges, 11-PG-323 & 11-PG-324 and temperature gauges 11-TG-301 & 11-TG-324 are provided on the inlet and outlet of the coil respectively. Temperature indicator 11-TI-324 is provided to indicate the superheated steam outlet temperature on the DCS panel. The superheated steam coil is provided with 2 safety valves 11-PSV-301 & 11-PSV-302. One PSV is kept bypassed in lock-open position, while the other is kept bypassed in lock-close position. The vent in the steam line is routed to the Atmosphere through a silencer. The temperature of the superheated steam is controlled by the 11-TRC-303, by injecting BFW through 11-TV-303. Temperature can also be controlled by allowing more superheated steam into the LP steam header (by operating the MP steam to LP steam globe valve at the grade level near the 12-F-01 FD fan).

The coils come out from the bottom of the convection zone, and enter the radiation zone. Radiation zone is the combustion chamber which is the cylindrical casing of the furnace, lined with refractory materials and ceramic fibres. In this zone, heat is primarily transferred



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by radiation from the flame and hot combustion products. The radiation zone Coils have vertical tubes arranged concentric to the casing. Each pass has 22 tubes each of 19090mm (weld to weld). These tubes are 6" NB Sch 40 type. Each pass has provision to monitor skin temperatures at three different levels in the furnace. They are used to monitor the condition in the furnace and also give an indication of hotspots in the furnace. There is also a provision of temperature indicators in the firebox near the floor level.

After gaining heat in the radiation zone, the outlets of the four passes combine and enter the Atmospheric Distillation Column through the 24" transfer line. The outlet temperature of each pass is measured by 11-TI-306, 311, 316, and 321. The common outlet transfer line temperature is measured by the 11-TI-322 and is controlled and recorded by the COT (coil outlet temperature) 11-TRC-301. The control is achieved by varying the quantity of fuel to the furnace. There is an alarm for high transfer line temperature as 11-TAH-133.

6.1.6.2 Fuel System

11-F-01 is a dual fired furnace i.e., either fuel oil or fuel gas or both can be used. The atmospheric heater has a total of 12 burners. Of late, 3 burners (No. 2, No. 5 & No. 8) have been dedicated to utilize off gases from the vacuum distillation column's hot well drum as the burning fuel. During 2010 T&I all the burners were replaced with 20 new burners ZEECO make (16 are combined firing and 4 are hot well off gas burners).

a) Fuel Gas System

Fuel gas is supplied to the unit from the Battery Limit in an 8" header. This is further branched into a 6" header to the Atmospheric heater. This FG line is steam traced to avoid condensation of heavier components, as carry over of liquid droplets of Hydrocarbon to the burner must be avoided.

FG to main burners passes through a mass flow meter (F1315) and shutdown valve 11-SDV-303. This SDV is connected to interlock logic. 11-FR/FQ-308 indicates FG flow in DCS room. It is provided with FAL and FAH. A local PG and a TG are provided to indicate pressure and temperature at field. 11-PI-308 indicates FG pressure in the DCS room. A low pressure alarm 11-PAL-303 is also provided. Fuel gas pressure low trip is set at 0.2 kg/cm²g.

In case the fuel gas pressure goes below the trip value, only 11-SDV-303 will get closed. If fuel gas tip pressure falls below the set value, chances of flame failure and subsequent accumulation of un-burnt hydrocarbons in the firebox is possible. This can lead to the possibility of explosion or back fire in the heater. Hence the provision of "**FG pressure**



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low” trip was provided. There is a provision to cascade the fuel gas pressure to the 11-F-01 COT, 11-TRC-301 through a selector switch on the auxiliary panel on the DCS panel. A 2” FG tapping upstream of 11-SDV-303 has been branched off for pilot burners. The pilot gas pressure is normally adjusted manually and is maintained at a pressure of 0.7 kg/cm² g. In case of low pilot gas pressure, 11-PAL-302 is provided to actuate an alarm. Low pilot gas pressure will alert the operator when pilot gas pressure falls.

b) Fuel Oil System

Fuel oil is supplied to the unit from the Battery Limit in a 3" header. This is further branched into a 3" header to the Atmospheric heater. FO line is steam traced to maintain temperature and avoid congealing. Mass flow recorder and integrator 11-FR/FQ-305 are provided on main FO supply line and 11-FR/FQ-306 is provided on the main FO return line from heater. Since this is a closed circuit through which FO circulation is maintained, the net consumption of fuel oil is measured as the difference between FI-305 and FI-306.

Shutdown valves 11-SDV-301 A/B are provided on the FO supply and return headers respectively. Local PG's and TG's are provided on the supply line to show pressure and temperature of FO supply. 11-PRC-301 indicates the pressure of fuel oil on the DCS panel. Pressure is maintained by 11-PRC-301, which regulates 11-PV-301 on the fuel oil supply line. There is a provision to cascade the fuel oil pressure 11-PRC-301 to the 11-F-01 COT, 11-TRC-301 through selector switch, on the auxiliary panel.

A low-pressure trip alarm has been provided on supply line. Actuation of this alarm shuts 11-SDV-301 A/B and cuts off only the fuel oil firing in the Furnace. Since FO is normally a thick heavy liquid, it needs to be always maintained in circulating state. If it is left stagnant and unused in burners and piping, it can get congealed despite the fact that tracing steam of the FO circuit is on. Circulation in heater area (FO piping forming a closed circuit across all passes called fuel oil ring) is maintained even when no fuel oil burner is in use. A ratio of 2:1 FO supply to return is normally maintained to obtain a good control on firing and prevent congealing of FO system. FO is drawn by individual burners through ¾" lines from header and balance quantity is sent to the return line. When there is no need of FO firing in the heater, circulation can be maintained. Purge steam connections are provided on each oil burner. FO burners are to be kept steam purged when idle.

When FO is fired, it is atomised or sprayed as a fine mist for realising complete combustion. The spraying of FO is done by de-superheated MP steam in FO burners. Atomising steam is supplied to heater through a 4" header. The differential pressure



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controller 11-DPIC-301 controls the atomising steam pressure, taking pressure signal from FO supply and MP steam simultaneously. Atomising steam pressure is maintained about 2.0 kg/cm² above the FO pressure. Atomising steam flow is recorded by 11-FR-307. Local PG and TG are also provided on this line. 2" flushing oil connection is provided on FO supply line up stream CBD/OWS drain is provided on FO return line. These provisions are to flush the line within Battery Limit during heater shut down. When furnace operates on combination fuel-either Fuel Gas operates on PIC and Fuel Oil on PIC/TIC cascade or Fuel Oil operates on PIC and Fuel Gas on PIC/TIC cascade mode. Selector switch is used to select only one fuel for COT control by cascading.

c) Off Gas System

In earlier mode of operations, the off gases from the vacuum column hot well drum used to be released to atmosphere through a vent. Later, an 8" provision was given for serving 3 burners on atmospheric heater, during 2010 T&I number of burners were increased to 4 ZEECO burners.

6.1.6.3 Air Preheating System

11-F-01 is balanced draft furnace. Both the convection and radiation sections are used for heating crude. The combustion chamber houses the radiation section of tubes. The convection section provided at the top of radiation section serves to increase the thermal efficiency of the furnace by utilizing further heat from the flue gas. Tubes are arranged vertically in the radiation zone and horizontally in the convection zone.

The following are the major parts in the Air Preheating system of the furnace

a) Forced Draft Fan

This is a centrifugal type fan. It supplies the air required for combustion in the balanced and forced draft operation of the furnace. The inlet of this fan is provided with variable guide vanes to regulate the flow of air. The guide vanes are journaled in the fan shaft vicinity with a spherical pivot in a hub ring with cylindrical drilling and at the outside with cylindrical pivots. Sealing discs below and above the bearings keep the dirt away.

The guide vanes have a regulator which is actuated by a swivel-stem actuator. The actuator is steered pneumatically and is designed in such a way that, in case of air failure, the guide vane regulator will move to open position by means of an installed spring. There is also a provision to operate the inlet guide vane regulator manually by means of a hand wheel which is located at the bottom of the actuator. The duct connection on the suction and



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discharge sides of the fan is effected via a soft material compensator, which prevents the transfer of external forces, like the forces due to thermal expansion, to the fan. The discharge side of the fan has a multi-vane damper in between the fan casing and soft material compensator, which isolates the discharge duct of the fan. The fan shaft with the impeller and guide vanes is seated in two oil-lubricated sliding bearings. The fixed bearing is located on the motor-side and the mobile bearing is situated opposite to it.

These bearings are lubricated by means of rotating lubrication rings and have intermediary chamber. The oil level is seen in the sight glass and the oil operating temperature should not exceed 80 °C and brief peak temperatures up to 90 °C are sealed off by a labyrinth seal. The fans are driven by polyphase induction motors via couplings, which are directly coupled to the fan. The fans are dynamically balanced. This warrants a running performance which is free of any vibration. In case of ID fans, an uneven caking on the impeller will create unsteadiness of run.

b) Induced Draft Fan

This is a centrifugal type fan. It controls the flue gas flow from the furnace by the variable inlet guide vane mechanism same as mentioned above. There is a provision to indicate the wide open position of the suction vanes in the DCS room on the auxiliary panel.

c) Recuperative type Air Preheater

This is an assembly of rectangular cast tubes that have fins on either side at the hot end and fins only on the flue side at the cold end. The tube lengths, size and pitch will vary as per requirements in individual cases. Using the tubes as building block, air preheater of any size can be made to suit heat duty and pressure drop. The entire tube assembly is built inside a steel frame made by beams and fully insulated casings. Flue gas and air terminal connections are made of rectangular flanges formed from rolled steel sections.

These flanges form an integral part of the air preheater frame and are sturdy and are capable of carrying considerable external loads. The entire air preheater assembly forms part of the ducting system. Because of the orientation of flat surfaces it is essential that flue gas flow is always in the vertical direction.

The flue gas will be on single pass, vertically down and air can have a number of passes depending upon allowable pressure drops. The pressure drop allowable is decided on a case-to-case basis. Generally it is 50-100 mm WC on the air side and slightly lower side on the flue-gas side. The tube is made of two half sections, cast independently and then bolted together. To prevent air leakage in the longitudinal direction, two grooves have been



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provided on the flanges on the either side. Asbestos ropes are placed in the grooves before bolting. The flanges provide the necessary gap for the flue gas passage. The flanges also have peripheral grooves on all four sides to accommodate rope to ensure air tightness between adjacent tubes.

d) Drop-Out Doors

Drop-Out Doors are provided to supply combustion air in case of Natural Draft operation or in case of emergency. The drop-out doors are double-flap isolator type, actuated pneumatically by double-acting power cylinders and 4-way solenoid valves. The following provisions are also made to operate them in case of Instrument air failure as extra safety devices.

- Weight loading to open the DOD's by the force of gravity.
- An air accumulator tank of sufficient capacity to operate the DOD's.

Explosion proof limit switches are provided to indicate fully open and fully close positions by means of indication lamps provided on the auxiliary panel in the control room. Switches are provided in the control room to open and close each drop out door. The open or close position of the DOD's is indicated by the lamps provided on the auxiliary panel. These are actuated by the limit switches.

e) Stack Damper:

Stack Damper is provided to prevent the flue gases from escaping directly without heat exchange in the Air-Preheater. It also helps in the direct escape of flue gases when the furnace is in Forced Draft or Natural Draft operation. The stack damper operates either full open or full close. During 2010 T&I the stack damper was replaced with full shut off damper with controlled damper operation. The new stack Damper is a multi-Louver isolation and control damper with pneumatic plus manual control. The damper is provided with counter weight to the **FAIL SAFE OPEN** position.

The damper is designed as FAIL OPEN position and it shall attain FAIL OPEN position on failure of:

1. Supply air failure with supply air pressure switch set at 2.5 Kg/Cm² falling.
2. Electric Supply to the Control Panel by auto operation of solenoid valve.
3. The Signal failure i.e. signals pressure falling below 0.2 Kg/Cm².

The damper shall open with decrease in signal air pressure and accordingly close with increase in signal air pressure. It is provided with Pneumatic linear actuators for operation



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and control of the damper. The damper is also provided with Winch and cable for manual operation of the stack damper from grade. In auto operation, disconnect the winch by either removing shackles from winch arm at damper level or by disengaging the worm wheel and shaft provided on the winch machine by rotating the hand wheel in anti-clockwise direction at grade level. The worm shaft can be locked by the lever below hand wheel in clockwise direction.

Explosion proof limit switches are provided to indicate fully open and fully close positions by means of indication lamps provided on the auxiliary panel in the control room. Switches are provided in the control room to open and close the Stack Damper. The open or close position of the Stack Damper is indicated by the lamps provided on the auxiliary panel. These are actuated by the limit switches. There is a provision to open the Stack Damper manually from the field by winch operation in case of emergencies. Air is required for the combustion of fuels in a furnace. It is supplied by the FD fan (11-FM-01). The air flow is regulated by adjusting the variable inlet guide vane mechanism to maintain proper dP across the furnace and for maintaining the required excess air in the flue gas. About 15 to 20% excess air in case of fuel oil and 10 to 15% in case of fuel gas is found to give satisfactory performance of the furnace. An oxygen analyzer (AR-1801) is also provided at the outlet of the flue gas to monitor the excess air regularly.

Air pre-heating system helps in recovering the sensible heat from the flue gas further, after the furnace convection zone, which is utilized to preheat the combustion air. This increases the fuel economy and also the heater efficiency approximately by 10 %. There are separate Air Preheating systems (APH) for Atmos and Vacuum furnaces.

Forced Draft fan (11-FM-01) draws atmospheric air and forces it through the APH. Induced Draft fan (11-FM-02) draws the flue gas through the APH, and returns it to the stack above the stack damper after recovering heat from it. Care should be taken to maintain the return temperature of flue gas above its dew point (typically 175 °C) to avoid condensation which would otherwise result in acid corrosion. The design of the furnaces in CDU-II gives immense flexibility in their operation. There are 3 modes of operation as given under.

2.5% O₂ on AR-1801 equals to 10-15% of excess air

a) Balanced Draft operation

In this mode of operation, the FD and ID fans both function simultaneously and help in recovering the heat from the flue gas. Flue gas is drawn by the ID fan through a duct connection that is taken from the castable below the Stack Damper. This duct is lined with refractory. Pressure and temperature gauges are provided to give local indication of the pressure and temperature of flue gas entering the air pre-heater. PI and TI are also provided



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to give the indication in the control room. A winch operated shut-off blade hand controller is provided in the duct to isolate the flue gas duct. Hot flue gas enters the APH at the top and exchanges the heat with combustion air. Flue gas exiting the APH is routed to the ID fan suction. This fan draws the flue gas and conveys them back to the stack above the stack damper through a duct. 11-TAL-803 at the ID fan inlet provision is there to give an alarm when the flue gas temperature falls below the dew point temperature to avoid corrosion. In case of low temperature, the air flow through the APH can be adjusted so that the temperature can be increased. In case there is an abnormal rise in temperature of flue gas leaving the APH, TAH will give an alarm. 11-PAH-808 is also provided to give an alarm in case of high furnace box pressure in the arch zone.

Combustion air under pressure from the FD fan is ducted through the bottom of the APH. PG and TG give the local indication of the pressure and temperature of the cold air entering the APH and a PI gives the cold air pressure indication in the control room. A PAL (11-PAL-806) is provided on the discharge of the FD fan to indicate malfunction of FD fan controls or tripping of the fan. After exchange of heat with the flue gas, hot air is sent into the hot air distribution duct running around the circumference of the heater. This duct is provided with a TG and PG to give a local indication of the temperature and pressure of the hot air. There is also a TI and a PI to give an indication in the control room. A TAH (11-TAH-801) is provided to give a high temperature alarm. This hot air duct branches into the plenum chamber of the furnace, where the burners are mounted. Hot air for combustion enters the burners and is used for firing with corresponding saving in fuel.

b) Forced Draft

In this mode of operation, the FD fan of the furnace will be running and the ID fan is stopped and the stack damper is kept open. Flue gas escapes to the Atmosphere directly without preheating the combustion air. This is done mainly when isolation of the APH or ID fan is required to carry out maintenance activities.

c) Natural Draft

There is also a provision to operate the furnaces in natural draft, wherein there is no requirement of FD and ID fans. There is a provision of 5 Drop Out Doors (DOD's) on the combustion air duct of 11-F-01. To operate the furnace in Natural Draft, DOD's and stack damper are opened. Atmospheric air goes inside the furnace by the action of the draft in the furnace and aids in combustion. The flue gas escapes directly through the Stack Damper without preheating the combustion air.

Standing Instruction on APH water washing (SI: 36):



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1. STANDING INSTRUCTIONS:

- 1.1. APH water washing is to be carried out after meeting all the requirements specified under S.no 2.
- 1.2. Follow the procedure specified under S.no 3 and also additional precautions mentioned in the PDI by the unit manager.

2. Requirements for APH water washing :

- 2.1. Technical Clearance
- 2.2. PDI instructions from unit manager
- 2.3. Clearance from ETP
- 2.4. Clearance from power plant for service water usage
- 2.5. Sufficient caustic inventory in the unit
- 2.6. LP steam availability for APH hot water wash
- 2.7. Utility air hose arrangement up to APH bottom man way for purging.
- 2.8. Clearance from YSF
- 2.9. Maintenance help for opening and closing of APH bottom & top man ways
- 2.10. Universal PH indicator for checking effluent water PH in field

3. Procedure to be followed during APH water washing:

1. APH water washing to be carried out based on flue gas side pressure drop and glass APH approach/APH outlet temperature of flue gas. Technical will organize for a pressure drop survey and recommend APH cleaning if required. It may be noted that carrying out APH water wash without necessity may expose the metal components more frequently to thermal shocks and exposure to lower PH water, which may cause corrosion.
2. After obtaining Technical recommendation, take clearance from YSF, ETP-I and PP-II for APH water washing (one day in advance).
3. After ensuring all the items under S.no 6 are met, Heater operation to be changed to forced draft, for that wide open heater Stack damper at the rate of 5% increment, observe heater fire box, if required adjust fires.
4. Start reducing ID fan loading slowly to Zero Percent and stop ID fan.
5. Before start of water wash, APH to be completely taken out of service and isolated so that there is no flue gas or air ingress into the APH. Close ID fan suction damper.
6. Flue gas ingress during APH water washing can make the water more acidic (due to Sox in flue gas) which will result in excessive corrosion of the APH components.
7. Check the condition of APH drain line & unplug the same (if necessary).
8. Open APH bottom man way to prevent water flow along with solid particles in to ID fan duct during water washing (flue gas will not come out during opening of bottom man way,



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as draught is from bottom to top, same thing can be checked by placing small piece cotton at the APH bottom man way, the cotton piece will be sucked in to the APH due to the draught present).

9. The APH bottom man way can be used for purging of APH with utility air for displacing the flue gas contained in the APH. Barricading red tape to be tied to the APH bottom man way, so that no person should enter in to APH.
10. Arrange utility air hose connection up to APH bottom man way.
11. After ensuring that the heater arch pressures are not close to trip value, start APH air purging with Utility air at slow rate, from bottom man way to top, for about 4 hrs. During this process around 80% of the flue gases contained in the APH will be displaced through stack.
12. After 4hrs of air purging, stop air purging, close APH Flue gas inlet damper, open APH top man way and start air purging with Utility air at slow rate, from bottom man way to top man way for about 4 hrs.
13. During air purging, it is to be ensured that no person is exposed to the vented air, which may be high in Sox content.
14. APH to be cooled up to 80-90 °C before start of water washing. After reaching APH temperature to 80-90 °C, take clearance from YSF, ETP-I and power plant for APH water washing.
15. Start APH water washing with hot water at 80-90 °C from the top section (Removal of deposits is more effective at higher temperature). LP Steam is to be introduced along with wash water for achieving high temperature. The globe valves available on steam lines shall be used to control steam flow to wash water.
16. Immediately after start of water wash, start Caustic injection into APH O/L water to neutralize the OWS effluent water (to maintain a PH of 7.0 – 8.0) for protecting biological treatment step at ETP.
17. During APH water wash, effluent water sample to be physically checked for color and two samples(one immediately after APH outlet and other after caustic injection) to be sent to lab as follows
 - o Immediately after start of water wash
 - o At the end of water wash (after noting down field result)
18. In between during the progress of water washing, APH effluent water PH to be checked in field every 2 hours.
19. After 10-12hrs, the APH hot water washing to be shifted from top to bottom section and APH effluent water PH to be checked in field every 2 hours.
20. APH water wash is to be continued till 2 consecutive sets of water samples are reported to have PH in the range of 6.5-7.0. After confirmation in field the last sample of APH effluent



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water to be sent to lab for testing final PH. (Estimated duration of water washing is typically 24-48hrs, depending on the size of the APH).

21. After stopping water wash, APH is to be purged with Utility air from APH bottom to top, to dry out the surface of glass tubes and finned plates before allowing flue gas entry.
22. After completion of air purging, close APH bottom & top man ways.
23. After ensuring the APH is dry, open APH flue gas inlet and ID fan suction dampers and start ID fan at 2%(minimum) loading.
24. Slowly load ID fan and ensure steady and slow raise GAPH Inlet Temp (It should not cross design temperature) and simultaneously divert combustion Air to APH.
25. While taking APH into service the APH air bypass damper to be adjusted in such a way that the tube skin temperature in the cast APH section should be at least 10 °C more than the dew point of flue gases.(dew point of flue gas is 160 °C).

6.1.6.4 Trip and Interlock System:

Trip Values for 11-F-01

Trip	Unit	Value
FD fan discharge pressure (low)	mm Aq.	+15
ID fan suction pressure (High)	mm Aq.	-45
Furnace pressure (high)	mm Aq.	+2.95
Fuel oil pressure (low)	Kg/cm ² g.	+2.77
Fuel gas pressure (low)	Kg/cm ² g.	0.20
Furnace pass flow (low)	m ³ /h	42

The following is the description of the consequences arising from the above conditions *given that all the trips are in auto interlock mode.*

FD fan discharge pressure (low): The activation of this trip alarm (Annunciator alarm) may be due to the tripping of FD fan or due to the faulty indication of the combustion air discharge pressure. This results in the following

- a) DOD's gets opened
- b) Stack Damper gets opened and ID fan trips

ID fan suction pressure (high): The activation of this trip alarm (Annunciator alarm) may be due to the tripping of ID fan or due to the faulty indication of the flue gas suction pressure. This results in the following

- a) Stack Damper gets opened

Furnace pressure (high): The activation of this trip alarm (Annunciator alarm) results in the following



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a) Stack Damper gets opened and ID fan trips

b) DOD's get opened and FD fan trips

If this alarm does not get reset within 30 seconds of its activation, the fuels to the furnace get cut off (SDV's get closed)

Fuel oil pressure (low): The activation of this trip alarm (Annunciator alarm) results in the following

a) Fuel oil SDV's (supply & return) close if the pressure low alarm does not get reset within 9 seconds.

Fuel gas pressure (low): The activation of this trip alarm (Annunciator alarm) results in the following

a) Fuel gas SDV closes immediately.

Furnace pass flow (low): The activation of this trip alarm (Annunciator alarm) results in the following

a) Fuels getting cut-off to the furnace (SDV's get closed) immediately

b) Stack Damper opens, ID fan trips, FD fan also trips and DOD's open

The Operator Interface in the DCS room is provided with a set of switches, push buttons and indications to enable the smooth and safe operation of the plant. The Annunciator panel has a provision of alarms which alert the operator when unsafe operating conditions arise. There is a provision to operate (open / close) the DOD's and Stack Damper from the Annunciator panel. There is also a provision of lamp indications to indicate the open /close conditions of the DOD's and Stack Damper (their activation results only when the limit switch corresponding to the open / close position is engaged as the case may be). There is a software alarm provision to indicate the limit switch (open / close) activation on the DCS panel. There is also a provision of Emergency Shut-Down push buttons to ensure the safety of the personnel and also the equipment.

The following trip switches are provided on 11-F-01

DOD – SD Check Trip Switch: This trip, when in auto mode, opens stack damper under the following conditions:

a) Furnace pressure high (Annunciator alarm)

b) FD fan trips / FD fan discharge pressure low (Annunciator alarm)

c) ID fan trips



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- d) Both fuel oil and fuel gas pressure low (Annunciator alarm)
- e) Pass flow low-low
- f) Activation of the DOD-SD ESD

FD Fan Trip Switch: This trip, when in auto mode, opens the DOD's under the following conditions

- a) Furnace pressure high (Annunciator alarm)
- b) **FD fan trips** / FD fan discharge pressure low (Annunciator alarm)
- c) FD fan motor amps low (Annunciator alarm)
- d) Any of the DOD's is not closed completely (close limit switch deactivation)
- e) Pass flow low-low
- f) Activation of the DOD-SD ESD

ID Fan Trip Switch: This trip, when in auto mode, opens the SD under the following conditions

- a) Furnace pressure high (Annunciator alarm)
- b) **FD fan trips** / FD fan discharge pressure low (Annunciator alarm)
- c) ID fan motor amps low / ID fan suction pressure high (Annunciator alarm)
- d) Both fuel oil and fuel gas pressure low (Annunciator alarm)
- e) Stack Damper is not closed completely (close limit switch deactivation)
- f) Pass flow low-low
- g) Activation of the DOD-SD ESD

Fuel Oil Trip Switch: This trip, when in auto mode, closes the fuel oil supply and return SDV's under the following conditions

- a) Fuel oil pressure low (Annunciator alarm)
- b) Signal from any of the three timers (details of the timers are given below)
- c) Activation of the Fuel-ESD
- d) Pass flow low-low

Fuel Gas Trip Switch: This trip, when in auto mode, closes the fuel gas SDV under the following conditions

- a) Fuel gas pressure low (Annunciator alarm)
- b) Signal from any of the three timers (details of the timers are given below)
- c) Activation of the Fuel-ESD
- d) Pass flow low-low



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Timers: Three timers are provided in the furnace trip circuit, which will start counting when their respective alarms are activated. If the alarm is not brought back to normal value within 30 seconds (as counted by the timer), the timer relay will activate the fuel oil and fuel gas trips. If the alarm is brought back to normal value within 30 seconds (as counted by the timer), the timer will get reset i.e., the timer indication will become zero.

a) **Furnace Pressure High Timer:** It gets activated when furnace pressure high alarm comes on the annunciator panel and gets reset when the alarm is normalized. If the alarm condition is not reset, the fuels to the furnace get cut-off (given the fuel trips are in auto mode)

b) **FD Fan Trip Timer:** It gets activated when the FD fan trips. This should result in ID fan tripping and DOD's and Stack Damper opening within 30 seconds or else fuels get cut-off. Gets reset when FD fan is running and all the DODs close or the FD fan trip is bypassed.

c) **ID Fan Trip Timer:** It gets activated when the ID fan trips. This trip should result in the opening of Stack Damper within 30 seconds or else fuels get cut-off. Gets reset when ID fan is running and the stack damper is completely closed or the ID fan trip is bypassed

Note: There is a provision to set the timer to any time duration between 0-60 seconds

Interlock system: Interlocks are provided to ensure safe operation of the equipment. They ensure that corrective action is taken automatically whenever unsafe operating conditions arise due to process upsets, mal-operation of Instruments or equipment etc. But the interlocks can be made ineffective / inactive by bypassing the trip switches provided on the DCS panel.

Interlocks on furnace operation: The following interlocks ensure the safe operation of furnace. The following description is valid only when the trips described above are in **auto-interlock** mode.

Note: All the references to the 'open' and 'close' positions of DOD's and Stack Damper mean that the open or close limit switches are engaged and the close indication is active on the Annunciator panel

- Drop out doors opening or closing
- Stack damper opening or closing
- FD fan starting or stopping
- ID fan starting or stopping



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- e) Fuel SDV's opening or closing
- f) Low Pass Flow

Interlock 1: DOD Operation

- i) When the furnace is in natural draft mode, all the DOD's have to be in fully open position. If any one DOD leaves the fully open position (either by operation or by instrumentation malfunction), then the fuel to the furnace will cut off.
- ii) When the furnace is in forced draft mode, DOD's are in fully closed position and if any of the DOD's is not in fully closed position, FD fan is tripped. In case the FD fan is tripped, the DOD's open automatically.
- iii) When the furnace is in balanced draft mode, DOD's are in fully closed position. They get opened automatically if the
 - FD fan trips
 - Arch pressure high gets activated
 - ID fan trips and Stack Dampers gets opened

Interlock 2: Stack Damper Operation

- When the furnace is in natural draft mode, the SD has to be in fully open position. If the SD leaves this position, the fuel to the furnace is cut off.
- When the furnace is in forced draft mode, the SD has to be in fully open position. If the SD leaves this position, the fuel to the furnace is cut off.
- When the furnace is in the balanced draft mode, the SD has to be in fully closed position. If it leaves this position, the ID fan is tripped. The SD gets opened if
 - ID fan gets tripped
 - Arch pressure high gets activated
 - FD fan trips /any DOD leaves fully closed position
 - Fuel oil / fuel gas SDV's both get closed (command)
 - Activation of DOD-SD ESD

Interlock 3: FD Fan Operation

- When the furnace is in natural draft mode, **the FD cannot run** as long as the DODs are in open position.
- When the furnace is in forced draft mode, FD fan gets tripped if any DOD leaves fully closed position or the furnace pressure is too high.
- When the furnace is in the balanced draft mode, the FD gets tripped if any DOD leaves fully closed position.
- FD fan cannot run if furnace pressure high alarm is active



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Interlock 4: ID Fan Operation

- i) When the furnace is in natural draft mode, the ID cannot run if FD fan is not running.
- ii) When the furnace is in forced draft mode, only FD fan runs and ID fan is stop condition.
- iii) When the furnace is in the balanced draft mode, the ID fan gets tripped if
 - SD is not fully closed
 - any DOD leaves fully closed position (FD trips, hence SD opens)
 - FD fan is tripped
 - Furnace pressure is high

Interlock 5: Fuel SDV's Operation

The fuel SDV's get closed by operating the ESD on the auxiliary panel. Individual ESD's are also provided for fuel oil and fuel gas. The fuel SDV's get opened only after the individual "SDV Reset" push button on the auxiliary panel is operated. The following are the interlocks on the Fuel SDVs

- i) When the furnace is in natural draft mode, fuel is cut off if any DOD or stack damper leaves the fully open position.
- ii) When the furnace is in forced draft mode, fuel is cut off if stack damper leaves fully open position / FD fan trips.
- iii) When the furnace is in the balanced draft mode, the fuel is cut off in case any of the three timers gets activated.

Interlock 6: Low Flow in the Furnace Passes

This interlock shuts down the fuel SDV's to the furnace whenever the furnace pass flow in any coil is low. The trip value set for the low pass flow is 42 m³/hr.

6.1.6.5 COKE FORMATION ON HEATER TUBES

Coke formation on heater tube surfaces can be attributed to various factors. Here are the most prominent ones:

❖ Heater Charge Properties

- Sodium Content – causes rapid fouling
- Asphaltene Content – increases fouling
- Calcium Content
- Flow Improvers- potential increase in fouling
- Crude Properties –API & viscosity

Operating Parameters

- High Heater Outlet Temperatures
- Process Velocity



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- ✓ Low Mass Velocity, Increases Film Temperature
- ✓ Loss of Velocity Steam
- ✓ Low Cold Oil Velocity (target 6 ft/sec minimum)
- Uneven Heat Distribution - “hot spots or cold spots in firebox”
- Residence Time Above Cracking Threshold
- Low Flow (Turndown) or Poor Flow Distribution
- Feed Interruptions

❖ Other Issues

- Changes Made During Heater Revamp:
- ✓ Change to Low NOx Burners
- ✓ Fuel Heating Value Changes
- ✓ Heavier API Crude Feed
- ✓ Addition of Air Preheat
- ✓ Change Size, Number, and/or Metallurgy
- Fuel Contaminants- Tip Plugging

METHODS OF CLEANING HEATER TUBES OR DECOKING

Steam-Air Decoking (Steam & Air Burn)

- Performed when heater is off-line
- Heater box temperature is reduced/cooled

Mechanical (Pigging)

- Performed when heater is off-line
- Heater box temperature is cooled

On-Line Spalling

- Performed while heater is in service
- 1 pass is taken out of service while other passes remain on-line

DECOKING OPTION	TYPES OF HEAVY OIL HEATERS			
	CRUDE HEATER	VACUUM HEATER	VISBREAKER HEATER	DELAYED COKER
STEAM-AIR DECOKING	YES	YES	YES	YES
MECHANICAL PIGGING	YES	YES	YES	YES



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SPALLING	NA	NA	YES	YES
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Two of the most important decoking techniques are steam-air decoking and mechanical pigging. They have been discussed in the following section:

1. Steam Air Decoking

When petroleum fractions are heated to high temperatures, a fraction of it gets decomposed unavoidably over a period and accumulates itself as coke deposits inside the tube walls. This coke layer results in increased pressure drop and increased tube metal skin temperature, which may result in hot spots.

Steam Air Decoking (SAD†) of atmospheric heater tubes is done to remove coke deposit from inside heater tubes with the help of steam and air. Removal of coke results in clean heater tube internals and improves heater performance by better heat transfer to process, fluid. SAD also achieves low pressure drop through heater tubes and reduces chances of hot spot on heater tubes. Need for steam air decoking of heater tubes is indicated by increased pressure drop and harder firing.

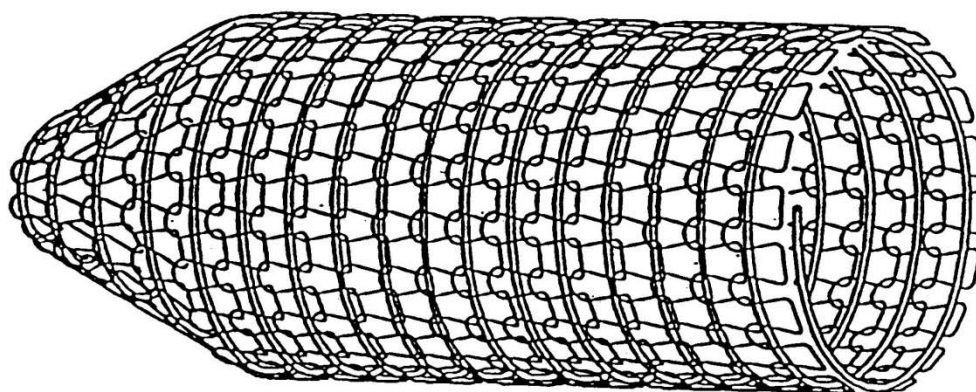
This dedicated SAD arrangement comprises decoking drum (11-V-05), piping and instrumentation for plant air, service water and MP steam. MP steam connection to each pass flow is provided to dislodge carbon deposit from inside the heater tubes. Local and DCS panel mounted flow indications (11-FE-310 to 313) are provided on each steam connection. Plant air is required to ignite the remaining coke fill deposit clinging on the inner wall of tubes and achieve final cleaning. Local and DCS panel mounted flow indications 11-FE-309 are provided on the main 4" plant air header. 11-FE-309 indicates total plant air consumption during decoking operation.

Service water quench provision is given on 11-V-05 to quench the contents before letting them to atmosphere. To carry out SAD, pass flow inlet and outlet of the furnace 11-F-01 are isolated from process network and connected by means of swing elbows with the decoking network. Heater pass flow outlets are connected to the decoking pot 11-V-05. While MP steam is introduced in the tubes it is fired from outside. Thermal shock caused by the flame, cracks the coke scales inside the tubes and flowing steam dislodges them. These coke particles are carried to 11-V-05 after being quenched in the decoking pot by service water. When no more coke is removable as indicated by relatively clear colour of the effluent, air along with steam is introduced into pass flow to blow out the coke inside tube while firing is on in the heater. Oxygen burns coke at high temperature. Burning of the coke is indicated by increased tube metal temp.



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PIGGING



- Coke is removed from the heater tube by pumping a metal studded foam or plastic “pig” with water and air. The metal studded “pig” rotates such that it scrapes the coke off the inside of the heater tube.
- Different size and abrasiveness “pigs” are used in the decoking process. “Pigs” are slightly larger than the inside diameter of the heater tube.
- Usually “pigs” are pumped through heater several times forward and backward until overall differential pressure across the tube (inlet to outlet) is restored to its original “unfouled” condition.
- Typical decoking time is 18 to 24 hours per heater depending on setup time.

6.1.7 ATMOSPHERIC DISTILLATION COLUMN:

Crude oil after final heating in Atmospheric Heater is fed to the Atmospheric Distillation Column. This Column has 40 valve type trays of SS410S material except the 37th to 40th tray which are of SS316 material with monel lining. The column has a stripping section at the bottom.

Description of entire column has been taken up zone wise.

1 Flash zone



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2. Bottom section
3. Middle section
4. Overhead section

6.1.7.1 Flash Zone

Heated and partly vaporized crude oil enters the flash zone under the 7th tray. Hydrocarbon vapours flash in this section and get liberated. Non-flashed liquid moves down which is predominantly the bottom product called Reduced Crude Oil (RCO). Certain degree of over flashing of crude is desirable for proper stabilization of RCO, and fractionation of gas oil components.

Over-flashing is achieved by setting up COT at slightly higher temperature than what is actually required. The required temperature of flash zone is 357 °C while that of feed is 360 °C. This over-flashed material mostly condenses at the 7th tray. The condensed liquid is withdrawn from 7th tray and put back on the 6th tray. Overflash liquid travels down from 7th tray to 1st tray. It strips off heavier components coming up from RCO stock collected at column bottom which otherwise could move up and cause discolouration of Heavy Diesel stream. Flow of overflashed liquid could be increased either by increasing the COT and condensing more material on 7th tray or by reducing the Heavy Diesel draw-off rate. However the second option will lead to less diesel yield and higher energy consumption without any advantage. Too large flow of over flash liquid may result in the drop of bottom temperature and lighter bottom product, i.e., RCO.

Over flash flow is indicated by 11- FI/ FR-402. This is a 6" line with a U-loop. 11-FE-402 is mounted in the liquid seal. This seal provides adequate liquid build up on upstream of FE and ensures unflickering, steady flow through orifice. In addition, it provides some back pressure which is required to prevent flashing just downstream of flow orifice due to pressure drop.

MP steam is introduced in the column through 11-FRC-401, below tray 1 for stripping of RCO. Stripping steam helps in removing lighter components from the heavier products by reducing their partial pressures and vaporizing them without requiring additional heat.

Minimum 22 kg/h stripping steam per m3/h of RCO is to be used for effective stripping.

Hydrocarbon vapours liberated by flashing move upward along with steam in the column for further mass transfer at the trays in the upper section. The Desalter pressure relief valve and the Pre-flash Drum pressure relief valve discharges are connected to the flash zone.

6.1.7.2 Bottom Section



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Reduced Crude Oil (RCO) product is collected at the bottom of the column. The column bottom level is indicated and controlled by 11-LRC-401. LRC-1401 goes to the ratio block of vacuum heater. Manipulating the RCO flow to vacuum heater can do column bottom level control. LRC is provided with software high / low level alarms. In Addition, 11-LAL/LAH-401 are provided in DCS panel. 11-TR-401 shows product RCO temperature. RCO is pumped out from the bottom of the column at by RCO pumps 11-P-10 A/B to any of the following destinations.

- 10" line to Vacuum Furnace (12-F-01) as RCO feed. During normal operation, the RCO flow (FX2100) is regulated by the column bottom level control. This total RCO flow in turn gives the set value to the vacuum furnace pass flow controllers.
- Swing elbow has been provided at the inlet and outlet of the heater passes, therefore steam and air decoking can be done either way.
- There is a provision to route RCO in an 8" line to 12-E-06 A/B (i.e., VR lines) for utilization of these exchangers during light crude processing. After exchanging the heat in 12-E-06 A/B, RCO goes to the vacuum furnace.
- To SR manifold during start-up. This is a start-up line.

6.1.7.3 Middle Section

Middle section of the column has product withdrawal and circulating reflux network. In order to maximize heat recovery and balance the column loading for maintaining proper temperature profile across the column, three circulating refluxes (CR) are considered viz., Top Pump Around, Kerosene CR and Heavy Diesel CR. These circulating refluxes are drawn from their draw-off **seal** boxes and are routed to preheat trains for recovery before entering back to the column again.

a) Top Pump Around (TPA)

TPA is drawn from the 37th tray by the pump 11-P-09 A/B and is cooled by routing it through 11-E-04 A/B. It enters the Atmospheric Column on the 39th tray. 11-TI-411 and 11-TI-414 indicate its draw-off and return temperature respectively. TPA flow is indicated and controlled by 11-FRC-406. The return temperature is to be maintained above 100 °C to prevent the condensation of water vapour which may result in acid corrosion of the column top.

b) Middle Pump Around (Kerosene Circulating Reflux)



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Kerosene CR is drawn from 20th tray by the pump 11-P-08 A/B and is cooled by routing it through 11-E-09 and 11-E-25 in parallel and then through 11-E-11 (BH case). It is then boosted by 11-P-08 C/D. In BH operation mode, it flows through 12-E-01 A/B/C and then enters the Atmospheric Column on 22nd tray. 11-TI-403 and 11-TI-413 indicate its draw-off and return temperature respectively. Kero CR flow is indicated and controlled by 11-FRC-405. There is a pressure indication 11-PI-432 at the 20th tray in the Kerosene zone. There is a provision to route hot Kerosene to the unit flushing oil header from the discharge of 11-P-08 A/B.

c) Bottom Pump Around (Heavy Diesel (HD) Circulating Reflux)

HD is drawn from the 12th tray by pumps 11-P-07 C/D and is cooled by passing it through 11-E-15 A/B and 11-E-13. The seal flushing facility to these CR pumps is provided from the outlet of 11-E-23, Diesel product cooler. The CR return enters the Atmospheric Column on the 14th tray. 11-TI-402 and 11-TI-412 indicate its draw-off and return temperature respectively. HD CR flow is indicated and controlled by 11-FRC-404.

Product Draw-Off

Heavy Naphtha (HN), Kerosene, and HD products flow by gravity from the 28th, 20th and 12th tray respectively to strippers 11-C-02, 03 and 04 under respective level control of strippers (viz., 11-LIC-404, 403, 402 respectively). This draw-off from the draw-off boxes includes the respective CR for Kerosene and Diesel cases. Vapour return lines from HN, Kerosene, and HD strippers back to the fractionator column are provided just two tray above the draw-off for HN and two trays above the draw-off for Kerosene, and HD. 11-TI-403, 404, and 405 indicate the draw-off temperatures of HD, Kerosene and HN respectively from the column in the in DCS panel. 11-TI-407, 408, 409 indicate the vapour return temperature from HD, Kerosene and HN strippers respectively. 11-TI-206 indicates the temperature at the 30th tray. An Elevation of 3m for HN and Kerosene and 4m for HD, from draw off nozzle to each level control valve of the stripper has been provided to exert back pressure necessary to prevent flashing just downstream of control valve. This also prevents two phase flow in draw off piping.

6.1.7.4 Over Head Section

The overhead vapours of Atmospheric Column pass through the overhead condensers 11-E-17 A to H (in 4 banks) and are totally condensed. The condensate gets collected in the Overhead Naphtha Accumulator (reflux drum) 11-V-01. Any of this overhead condenser banks can be isolated for maintenance. There is also a provision of service water connection at the inlet of each condenser to occasionally wash away the deposits of ammonium salts in the tubes.



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Top pressure of Atmospheric Column is maintained by 11-PRC-409 A/B which Manipulates 11-PV-409A on outgoing uncondensed gases from 11-V-01 and 11-PV-409B on incoming fuel gas. Since isolation valve is provided in overhead line between Atmospheric Column and overhead Naphtha accumulator, 11-PSV-401 A/B/C (set at 4.8 kg/cm² g.) are provided at the top of column. Snuffing steam provision is given on the PSV outlet header to quench the vapours in case of emergency or as a safety precaution to be opened during thunder storms.

Condensed Hydrocarbons are allowed to settle in reflux drum where steam condensate (water) settles in vessel boot and then flows to the Sour Water Stripper Unit on its pressure. 11-TI-410 indicates the temperature of the reflux drum in DCS panel. 11-PSV-403 on 11-V-01 has also been provided for the safety of the Atmospheric Column top PSVs. Uncondensed gases from 11-V-01 are routed to flare through 11-PV-409A and FX- 801 indicates the mass flow rate of the flare gases.

Water-Naphtha interface controller 11-LDIC-406 controls level of water in the boot and operates 11-LV-406 on 3" sour water line. LIC-1406 has software high / low level alarm (11-LAH/LAL-406) in DCS panel. 11-FI/FR-450 shows sour water flow in DCS panel when Atmos sour water is routed to 11-V-04.

Accumulated hydrocarbon in 11-V-01 is pumped back to Atmospheric Distillation Column as top reflux on the 40th tray by 11-P-06 A/B. Reflux flow is controlled by 11-FRC-403 which is cascaded to the Atmospheric Column top temperature controller 11-TRC-403.

Excess quantity of Naphtha in Reflux drum is pumped by 11-P-06 A/B to the Naphtha Stabiliser as feed through a 6" line. The flow of Unstabilised Naphtha to the Stabiliser is controlled by 11-FRC-503 which is cascaded to the Naphtha Accumulator (11-V-01) level controller 11-LRC-405. LIC-1405 has software high / low level alarms (11-LAH/LAL-405).

There is a ¾" provision for injection of Neutraliser and Corrosion Inhibitor on the Atmospheric Distillation Column overhead vapour line and the reflux line to maintain the desired pH.

Product Strippers

There are three side strippers for stripping out side draw-off products from Atmospheric Column viz., Heavy Naphtha Stripper, Kerosene Stripper, and Heavy Diesel Stripper. The



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flash point and the IBP of the distillate streams are controlled adjusting the stripping steam flow in each of the strippers.

i) Heavy Naphtha Stripper (11-C-02)

Six valve type trays (SS410S) are provided in HN stripper of 1200 mm diameter. HN is admitted on 6th tray under level control 11-LIC-404. The HN stripper is provided with software low level alarm 11-LAL-404. Minimum 1.5m elevation is provided between 11-LV-404 and feed in nozzle to provide back pressure and prevent flashing in the piping.

Superheated MP steam from the 11-F-01 is used for stripping. Steam flow is regulated by 11-FRC-409 which manipulates 11-FV-409 on the steam line. MP steam reduces the partial pressure of hydrocarbon components inside the Stripper and helps them to get vaporised at a lower temperature. Vapours move up the stripper column. Mass transfer between the down coming Heavy Naphtha liquid from the 6th tray to bottom and uprising steam-hydrocarbon vapours takes place on each tray. Stripped HN vapours enter the Atmospheric Distillation Column at the 30th tray. Finally stripped HN from the bottom of the stripper is drawn by Heavy Naphtha pumps (11-P-05 A/B) and sent to the product cooling section.

Minimum 16 kg/hr stripping steam per m³/h of product is to be used for effective stripping

A temperature difference of maximum 12 °C between the stripper inlet and outlet temperatures ensures proper stripping.

Temperatures of the incoming and outgoing HN from the stripper are indicated by 11-TI-404 and 11-TI-409 respectively.

HN stream from 11-P-05 A/B discharge is cooled down in 11-E-01 by crude and further by cooling water in 11-E-26. 11-TI-104 indicates final rundown temperature of HN.

HN can be routed as follows to

- Heavy Diesel (sweet / sour) storage tanks, **PG HN rundown** (blending stream) and Stabilised Naphtha rundown, through 11-FRC-103 in a 3" line
- To Stabilised Naphtha rundown and ATP Diesel through 11-FRC-107 in a 3" line
- To slops in a 3" line
- To NHT-CCR in a 3" and a 4" (via 6" SRN) line as hot feed.



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ii) Kerosene Stripper (11-C-03)

Six valve trays (SS410S) are provided in the Kero Stripper of 2000 mm diameter. Kerosene is admitted on 6th tray under level control 11-LIC-403. The Kerosene stripper is provided with software low level alarm 11-LAL-403. Minimum 1.5m elevation is provided between the inlet nozzle and 11-LV-403 to provide back pressure and prevent flashing in piping.

Superheated MP steam from 11-F-01 is used for stripping. Steam flow is regulated by 11-FRC-408 on the steam line. MP steam reduces partial pressure of Hydrocarbon Components inside stripper and helps them vaporise at lower temperature. Vapours move up the stripper column. Mass transfer between the down coming Kerosene liquid from the 6th tray to bottom and uprising stream hydrocarbon vapours takes place on each tray.

Stripped Kerosene vapours enter the Atmospheric Column at the 22nd tray. Finally stripped Kerosene is drawn by Kerosene product pumps (11-P-04 A/B) and sent to the product cooling section.

Minimum 16 kg/hr stripping steam per m³/h of product is to be used for effective stripping

Temperatures of the incoming and outgoing Kero from the stripper are indicated by 11-TI-403 and 11-TI-408 respectively.

A temperature difference of maximum 12 °C between the stripper inlet and outlet temperatures ensures proper stripping.

Kerosene stream from 11-P-04 A/B discharge is cooled down in 11-E-10, 05 & 02 by crude and further by cooling water in 11-E-24 & 24A. 11-TI-203 indicates final rundown temperature of Kerosene.

Kerosene can be routed as follows to

- MEROX unit either as Kerosene / ATF through 11-FRC-203 in a 6" line (there is a 4" branching to receive ATF from CDU-I from this line at the battery limit)
- Diesel storage tanks (sweet / sour) through 11-FRC-204 in a 6" line
- LDO pool with the flow indication 11-FQ-205 in a 3" line (there is a 4" branching to route Kerosene to VBU from this line)
- FO pool with the flow indication 11-FQ-206 in a 3" line. (RFO and HFO lines)
- Slops in a 4" header.

iii) Heavy Diesel (HD) Stripper (11-C-04)

Four valve trays (SS410S) are provided in the HD Stripper of 2200 mm diameter. HD to be stripped is admitted on 4th tray under level control 11-LIC-402. The HD stripper is



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provided with software low level alarm 11-LAL-402. Minimum 1.5m elevation is provided between the inlet nozzle and 11-LV-402 to provide back pressure and prevent flashing in piping.

Superheated MP steam 11-F-01 is used for stripping. Steam flow is regulated by 11-FRC-407 on the steam line. MP steam reduces partial pressure of Hydrocarbon Components inside stripper and helps them vaporise at lower temperature. Vapours move up the stripper column. Mass transfer between the down coming HD liquid from the 4th tray to bottom and uprising stream hydrocarbon vapours takes place on each tray.

Stripped HD vapours enter the Atmospheric Column at the 14th tray. Finally stripped HD is drawn by HD product pumps (11-P-03 A/B/C) and sent to the product cooling section.

Minimum 16 kg/hr stripping steam per m³/h of product is to be used for effective stripping

Temperatures of the incoming and outgoing HD from the stripper are indicated by 11-TI-402 and 11-TI-407 respectively.

A temperature difference of maximum 12 °C between the stripper inlet and outlet temperatures ensures proper stripping.

HD stream from 11-P-03 A/B/C discharge is cooled down in 11-E-14, 12, 08, 06 & 03 by crude and further by cooling water in 11-E-23 & 23A. 11-TI-202 indicates final rundown temperature of HD.

HD can be routed as follows to

- DHDS as hot feed in a 10" line through 11-FRC-3501 from the up-stream of 11-E-03
- Diesel storage tanks (sweet / sour) in a 6" line through 11-FRC-202
- LDO pool in a 4" line under the flow indication 11-FI/FQ-207
- A slip stream to flushing oil system from the upstream of 11-E-23 in a 2" line
- A slip stream as seal flushing for 11-P-07 C/D from the downstream of 11-E-23
- Slops in a 6" line from the down-stream of 11-FV-202
- FCCU-II as hot diesel from the upstream of 11-E-23

6.2 NAPHTHA STABILISER

Unstabilised Naphtha obtained in Atmospheric Column overhead reflux drum 11-V-01 contains lighter ends like C₃ and C₄ which vaporize at normal Atmospheric conditions. This



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Naphtha if stored as such in storage tanks will release lot of Hydrocarbon vapours and can create unsafe conditions and pressurization of the storage tank. To avoid this problem the lighter components of Naphtha are removed in a column. This process is called Naphtha stabilization.

Naphtha stabilization is carried out in Naphtha Stabiliser (11-C-05) where C₃ and C₄ hydrocarbons are removed from Naphtha. The Stabiliser is a distillation column which has 30 valve type trays (SS410S). It is provided with a PSV (11-PSV-501) set at 14.0 kg / cm² g. The PSV outlet is routed to the flare header.

Unstabilized Naphtha from the Top Reflux Pump (11-P-06 A/B) discharge is first heated up in Stabiliser feed / bottom exchanger (11-E-19 A/B) by exchanging heat with the outgoing stabilized Naphtha product. There is a provision to route the CDU-I Unstabilised Naphtha to 11-C-05, the flow of which is indicated by F1505. Feed enters the column on the 17th tray under the flow control 11-FRC-503 which is normally cascaded with 11-LIC-405 of 11-V-01. 11-TI-501 indicates temperature pick up from 11-E-19 A/B before entering the column.

Overhead vapours from Stabiliser (11-C-05) containing C₃ and C₄ components come out from column top in a 12" overhead line. This line is routed through Stabiliser overhead condensers 11-E-20 A/B/C/D. The condensed liquid, LPG, is collected in the reflux drum (11-V-03) and consists of C₃ and C₄ components.

The Stabiliser overhead pressure is maintained by pressure controller 11-PRC-501 A/B. 11-PIC-501 acts as a split controller on 11-PV-501A mounted on the condensers bypass line and 11-PV-501B mounted on the off-gas line from 11-V-03 to FCCU-II (14-V-11).

In case of decrease of Stabiliser top pressure below the set value, 11-PRC-501 opens 11-PV-501A to allow hot vapours directly into 11-V-03, bypassing the condensers. If the pressure in 11-C-05 increases above the set value, then 11-PV-501B would open to release the excess pressure to the FCCU-II sweet fuel gas distribution network. There is a provision to route the Off-gas through 11-PV-501B directly to the flare also.

LPG pumps 11-P-11A/B function as both LPG product and reflux pumps. Reflux flow which is controlled by 11-FRC-501 can be cascaded with the column top temperature (30th tray) indicator and controller 11-TI-510. A 2" minimum flow line (spill-back) from the discharge header of 11-P-11 A/B is to the pump suction is also provided.

11-LIC-502 controls LPG product flow to maintain reflux drum level and it is cascaded to 11-FRC-502 on the LPG product flow line. LPG is sent to the Amine Treating Unit in MEROX for the removal of H₂S and Mercaptans. 11-V-03 also has level high / low alarms in the DCS panel (11-LAH/LAL-502).



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A slip stream of LPG can also be sent to the LPG vaporiser of FCCU-II in a 2" line. Sour water is collected in the boot of 11-V-03. The interface level of water and LPG is indicated by DL-1503. High water level in boot may result in water carryover with LPG and it will affect the Amine Treating Unit at MEROX. Hence it is drained at a controlled rate to OWS periodically.

A technician should always be present during the draining of the water

11-PSV-502 set at 14.0 kg/cm² g. is provided on the Stabiliser reflux drum whose discharge is route to flare header. It prevents vessel from getting over pressurized in case of external fire.

A 2" service water line connection is provided on 11-V-03 to fill the vessel and wash the Stabiliser column with water during shut down.

A thermo siphon Stabiliser reboiler 11-E-25 is provided at Stabiliser bottom to supply the necessary heat for boiling the Unstabilised Naphtha. Kerosene CR from the discharge of 11-P-08 A/B is used as heating medium. 11-TR-403 & 11-TI-507 indicate the Kerosene CR supply and return temperature. The bottom temperature is indicated and controlled by 11-TRC-501. The control is achieved by adjusting the flow of Kerosene CR through 11-FV-504 which can be cascaded to the Stabiliser bottom temperature as indicated by 11-TRC-501. 11-TI-506 & 505 indicate the stabiliser bottom reboiler shell-side (Naphtha-side) inlet and outlet temperatures respectively.

Stabilised Naphtha gets collected at the bottom of the Stabiliser and the bottom level is controlled by 11-LIC-501. There is also an indication of bottom level high and low alarms as 11-LAH/LAL-501 in the DCS panel. This Stabilised Naphtha goes under the pressure of the Stabiliser to 11-E-19 A/B where it exchanges heat with the feed and then it gets cooled further in the salt water cooler 11-E-21. There are SRN hot feed provisions from the upstream of 11-E-21.

- A 4" line to FCCU 2 (for FCCU 2 start up).
- An 8" line to NHT-CCR.

After 11-E-21, SRN goes to the Caustic & Water-wash system.

6.3 NAPHTHA CAUSTIC & WATER-WASH SYSTEM

The purpose of this section is to bring down the H₂S and Mercaptans in the Stabilised Naphtha below 10 ppm. The main equipment in this unit section are:



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- a) Naphtha caustic wash drum (10-V-01)
- b) Naphtha water wash drum (10-V-02)
- c) Caustic circulation pumps (10-P-01 A/B)
- d) Water make-up pumps (10-P-03 A/B)

Naphtha from 11-E-21 flows through a 6" line to the caustic wash drum (10-V-01). Upstream to the 10-V-01, a vortex mixer 10-X-01 has been provided. Caustic solution is injected into Naphtha upstream of this mixer. The mixer helps in the efficient mixing of caustic solution with Naphtha with minimum pressure drop across it. Thorough mixing of caustic solution and Naphtha enables the transfer of H₂S / Mercaptans from the Hydrocarbon phase to the caustic solution phase. Sufficient residence time is given in 10-V-01 for the separation of the two phases: Hydrocarbon phase on top and caustic solution in water phase at the bottom. The circulation rate is measured by 10-FI-101. The concentration of circulating caustic diminishes gradually. This is replaced by fresh charge. Make-up caustic solution is supplied from MEROX unit to the suction of Caustic circulation pumps (10-P-01 A/B) which is diluted to a concentration of 5 Boumi. The spent caustic is drained from the pump suction line to the Chemical Sewer at MEROX. Naphtha goes out from the top of 10-V-01 to 10-V-02 for water wash. There is a continuous water make-up (measured by 10-FI-103) to this drum from 10-P-03 A/B, from the service water line. Equivalent amount of water is drained from the wash drum under the level control 10-LIC-101. High level and low level alarms 10-LAH/LAL-101 have been provided for 10-V-02.

After Caustic and Water Wash treatment, SRN can be routed to:

1. SRN storage tank in a 6" line, under the flow indication 11-FR/FQ-201. This flow is controlled by the Stabiliser bottom level controller 11-LIC-501. Down-stream of 11-FV-501, there is a provision to route the CDU-I Naphtha to the SRN storage tank
2. MS pool under the flow indication FR0104 in a 6" line. There is a selector switch to cascade the Stabiliser bottom level controller on the SRN to MS pool also by operating the software switch LI1501S1 in the DCS panel.
3. Slops in a 4" line.

6.4 VACUUM DISTILLATION UNIT

The Vacuum Distillation Unit is further classified into the following sub sections

1. Vacuum Heater
2. Vacuum Distillation Column (fractionation)
3. Product cooling and routing to storage tanks



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6.4.1 VACUUM HEATER (12-F-01)

The major equipment of this section are Vacuum Furnace, Air Pre-heater, ID Fan, FD Fan and Steam Decoking Pot.

The description of the Atmospheric Heater is divided into the following sub sections:

- 1) Process System
- 2) Fuel System
- 3) Air Preheating System
- 4) Trip and Interlock System
- 5) Steam Air Decoking

6.4.1.1 Process system

RCO from the 11-C-01 enters the vacuum furnace, where it is heated further so that it reaches the required flashing temperature. This is a cabin type fired heater with a design heat duty of 13.28 MM Kcal/h, which was increased to 15.24 MM Kcal/h. The RCO coming to the heater gets split into 4 streams and enters the 4 passes of the furnace under the flow recorders and controllers 12-FRC-101/102/103/104. Low flow alarms 12-FAL-101/102/103/104 are provided on all the passes to protect the heater pass tubes in case of low flow through each pass.

Steam connections are provided downstream of the FCV's for purging the respective coils in case of emergency. The isolation valve for the purging steam line is provided at a safe distance from the heater.

RCO enters the convection zone of the heater first. It is located at the top of the furnace above the radiation zone. This zone consists of 6 rows of tubes with 4 tubes per row. While the top 4 rows of this zone are studded type, the bottom two rows are bare tubes. The material of construction of the tubes is 9% Chromium+1% Molybdenum. The tubes are 6" NB Sch. 40 type, during 2010 T&I additional convection zone was provided. This zone consists of 8 rows of tubes with 4 tubes per row. In the additional zone, top 4 rows are SPP16 studded tubes, next two rows are SPP9 studded tubes and bottom two rows are bare tubes.

The convection section has a single row of 8 retractable pneumatic-operated soot-blowers for the external cleaning of the convection tubes. During 2010 T&I soot blower provision was given for additional convection zone (8 nos) and presently 4 number of soot blowers from old setup (alternate) were shifted to additional convection zone. This was done to



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carryout soot blowing of the new tubes installed during T&I. New soot blowers were under procurement and will be fixed in the setup. They are operated at the convection zone platform of the furnace. Pressure gauges are provided on the inlet and outlet of each pass in the convection zone to indicate the pressure drop across each pass. Temperature indicators 12-TI-302, 307, 312 & 317 are provided on the outlet of the convection zone to measure the temperature gain in this zone.

The coils come out from the bottom of the convection zone, and enter the radiation zone, which is the combustion chamber. The radiation zone has 20 tubes per each pass. Each pass has 34 tubes. These tubes are 4" NB, 6" NB & 8" NB Sch. 40 type. Each pass has provision to monitor skin temperatures at three different levels in the furnace. They are used to monitor the condition in the furnace and also give an indication of hotspots in the furnace. There is also a provision of temperature indicators in the firebox 12-TI-125, 126, 127 & 128 near the floor level.

After gaining heat in the radiation zone, the outlets of the four passes combine and enter the vacuum distillation column through the 52" transfer line. The outlet temperature of each pass is measured by 12-TI-106, 111, 116, and 121. The common outlet transfer line temperature is measured by 12-TI-122 and is controlled and recorded by 12-TRC-133 (coil outlet temperature). The control is achieved by varying the quantity of fuel to the furnace. There is also an alarm for high transfer line temperature as 12-TAH-133.

6.4.1.2 Fuel System

12-F-01 is a dual fired furnace i.e., either fuel oil or fuel gas or both can be used. It has 12 burners and were upgraded to 16 numbers during 2010 T&I.

i) Fuel Gas System

Fuel gas is supplied to the unit from the Battery Limit in an 8" header. This is further branched into a 3" header to the Vacuum Heater. This FG line is steam traced to avoid condensation of heavier components, as carry over of liquid droplets of Hydrocarbon to the burner must be avoided.

FG to main burners passes through a mass flow meter F2406 and a shutdown valve 12-SDV-105. This SDV is connected to interlock logic. 12-FR/FQ-107 records and integrates the FG flow to 12-F-01. It is provided with FAL and FAH. Local PG and TG are provided to indicate pressure and temperature at field.

12-PI-308 indicates FG pressure on the DCS panel. A low pressure alarm 12-PAL-108 is also provided. Fuel gas pressure low trip is set at 0.2 kg/cm² g. In case the fuel gas pressure is low, only 12-SDV-105 will get closed. If fuel gas tip pressure falls below the set value, chances of flame failure and subsequent accumulation of un-burnt hydrocarbons in the



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firebox is possible. This can lead to the possibility of explosion or back fire in the heater. Hence the provision of FG pressure low trip. There is a provision to cascade the fuel gas pressure to the 12-F-01 COT, 12-TRC-133 through a selector switch on the auxiliary panel in the DCS room.

A 2" FG tapping upstream of 12-SDV-105 has been branched off for pilot burners. The pilot gas pressure is normally adjusted manually and is maintained at a pressure of 0.7 kg/cm² g. In case of low pilot gas pressure, 12-PAL-107 is provided to actuate an alarm. Low pilot gas pressure will alert the operator when pilot gas pressure falls.

ii) Fuel Oil System

Fuel oil is supplied to the unit from the Battery Limit in a 3" header. This is further branched into a 2" header to the Vacuum heater. FO line is steam traced to maintain temperature and avoid congealing. Flow recorder and integrator 12-FR/FQ-105 is provided on main FO supply line and 12-FR/FQ-106 is provided on the main FO return line from heater. Since this is a closed circuit through which FO circulation is maintained, the net consumption of fuel oil is measured as the difference between FI-105 and FI-106. Shutdown valves 12-SDV-102 A/B are provided on the FO supply and return headers respectively. Local PG's and TG's are provided on the supply line to show pressure and temperature of FO supply. 12-PRC-101 indicates the Pressure of fuel oil on the DCS panel. Pressure is maintained by 12-PRC-101, which regulates 12-PV-101 on the fuel oil supply line. There is a provision to cascade the fuel oil pressure 12-PRC-101 to the 12-F-01 COT, 12-TRC-133 through selector switch, on the auxiliary panel.

A low-pressure trip alarm has been provided on supply line. Actuation of this alarm shuts 12-SDV-102 A/B and cuts off only the fuel oil firing to the Furnace. Since FO is normally a thick heavy liquid, it needs to be always maintained in circulating state. If it is left stagnant and unused in burners and piping, it can get congealed despite the fact that tracing steam of the FO circuit is on. Circulation in heater area (FO piping forming a closed circuit across all passes called fuel oil ring) is maintained even when no fuel oil burner is in use. A ratio of 2:1 FO supply to return is normally maintained to obtain a good control on firing and prevent congealing of FO system. FO is drawn by individual burners through ¾" lines from header and balance quantity is sent to the return line. When there is no need of FO firing in the heater, the circulation can be maintained. Purge steam connections are provided on each oil burner. FO burners are to be kept steam purged when idle. When FO is fired, it is atomised or sprayed as a fine mist for realising complete combustion. The spraying of FO is done by de-superheated MP steam in FO burners.



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Atomising steam is supplied to heater through a 4" header. The differential pressure controller 12-DPIC-103 controls the atomising steam pressure, taking pressure signal from FO supply and MP steam simultaneously. Atomising steam pressure is maintained about 2.0 kg/cm² above the FO pressure. Atomising steam flow is recorded by 12-FR-108. Local PG and TG are also provided on this line. 2" flushing oil connection is provided on FO supply line up stream. CBD/OWS drain is provided on FO return line. These provisions are to flush the line within Battery Limit after heater shut down. When furnace operates on combination fuel-either Fuel Gas operates on PIC and Fuel Oil on PIC/TIC cascade or Fuel Oil operates on PIC and Fuel Gas on PIC/TIC cascade mode. Selector switch is used to select only one fuel for COT control by cascading.

6.4.1.3 Air Preheating System

12-F-01 is balanced draught furnace. It is a cabin-type heater. Both the convection and radiation sections are used for heating crude. The combustion chamber houses the radiation section of tubes. The convection section provided at the top of radiation section serves to increase the thermal efficiency of the furnace by utilizing further heat from the flue gas. Tubes are arranged horizontally both in the radiation and the convection zones. The following are the major parts in the Air Preheating system of the furnace

a) Forced Draft Fan

This is a centrifugal type fan. It supplies the air required for combustion in the balanced and forced draft operation of the furnace. The inlet of this fan is provided with variable guide vanes to regulate the flow of air. The guide vanes are journalled in the fan shaft vicinity with a spherical pivot in a hub ring with cylindrical drilling and at the outside with cylindrical pivots. Sealing disc suction and discharge sides of the fan is effected via a soft material compensator, which prevents the transfer of external forces, like the forces due to thermal expansion, to the fan. The discharge side of the fan has a multi-vane damper in between the fan casing and soft material compensator, which isolates the discharge duct of the fan. The fan shaft with the impeller and guide vanes is seated in two oil-lubricated sliding bearings. The fixed bearing is located on the motor-side and the mobile bearing is situated opposite to it.

These bearings are lubricated by means of rotating lubrication rings and have Intermediate chamber. The oil level is seen in the sight glass and the oil operating temperature should not exceed 80 °C and brief peak temperatures up to 90 °C are sealed off by a labyrinth seal. The



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fans are driven by polyphase induction motors via couplings, which are directly coupled to the fan. The fans are dynamically balanced. This warrants a running performance which is free of any vibration. In case of ID fans, an uneven caking on the impeller will create unsteadiness of run.

b) Induced Draft Fan

This is a centrifugal type fan. It controls the flue gas flow from the furnace by the variable inlet guide vane mechanism same as mentioned above. There is a provision to indicate the wide open position of the suction vanes in the DCS room on the auxiliary panel.

c) Recuperative type Air Preheater

This is an assembly of rectangular cast tubes that have fins on either side at the hot end and fins only on the flue side at the cold end. The tube lengths, size and pitch will vary as per requirements in individual cases. Using the tubes as building block, air preheater of any size can be made to suit heat duty and pressure drop. The entire tube assembly is built inside a steel frame made by beams and fully insulated casings. Flue gas and air terminal connections are made of rectangular flanges formed from rolled steel sections.

These flanges form an integral part of the air preheater frame and are sturdy and are capable of carrying considerable external loads. The entire air preheater assembly forms part of the ducting system. Because of the orientation of flat surfaces it is essential that flue gas flow is always in the vertical direction.

The flue gas will be on single pass, vertically down and air can have a number of passes depending upon allowable pressure drops. The pressure drop allowable is decided on a case-to-case basis. Generally it is 50-100 mm WC on the air side and slightly lower side on the flue-gas side. The tube is made of two half sections, cast independently and then bolted together. To prevent air leakage in the longitudinal direction, two grooves have been provided on the flanges on the either side. Asbestos ropes are placed in the grooves before bolting. The flanges provide the necessary gap for the flue gas passage. The flanges also have peripheral grooves on all four sides to accommodate asbestos rope to ensure air tightness between adjacent tubes.

d) Drop-Out Doors

Drop-Out Doors are provided to supply combustion air in case of Natural Draft operation or in case of emergency. The drop-out doors are double-flap isolator type, actuated pneumatically by double-acting power cylinders and 4-way solenoid valves. The following



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provisions are also made to operate them in case of Instrument air failure as extra safety devices.

o Weight loading to open the DODs by the force of gravity.

o An air accumulator tank of sufficient capacity to operate the DODs.

Explosion proof limit switches are provided to indicate fully open and fully close positions by means of indication lamps provided on the auxiliary panel in the control room. Switches are provided in the control room to open and close each drop out door. The open or close position of the DOD's is indicated by the lamps provided on the auxiliary panel. These are actuated by the limit switches.

e) Stack Damper

Stack Damper is provided to prevent the flue gases from escaping directly without heat exchange in the Air-Preheater. It also helps in the direct escape of flue gases when the furnace is in Forced Draft or Natural Draft operation. The stack damper operates either full open or full close. During 2010 T&I, the stack damper was replaced with full shut off damper with controlled damper operation. The new stack Damper is a multi-Louver isolation and control damper with pneumatic plus manual control. The damper is provided with counter weight to the **FAIL SAFE OPEN** position.

The damper is designed as FAIL OPEN position and it shall attain FAIL OPEN position on failure of:

1. Supply air failure with supply air pressure switch set at 2.5 Kg/Cm² falling.
2. Electric Supply to the Control Panel by auto operation of solenoid valve.
3. The Signal failure i.e. signals pressure falling below 0.2 Kg/Cm².

The damper shall open with decrease in signal air pressure and accordingly close with increase in signal air pressure. It is provided with Pneumatic linear actuators for operation and control of the damper. The damper is also provided with Winch and cable for manual operation of the stack damper from grade. In auto operation, disconnect the winch by either removing shackles from winch arm at damper level or by disengaging the worm wheel and shaft provided on the winch machine by rotating the hand wheel in anti-clockwise direction at grade level. The worm shaft can be locked by the lever below hand wheel in clockwise direction.

Explosion proof limit switches are provided to indicate fully open and fully close positions by means of indication lamps provided on the auxiliary panel in the control room. Switches are provided in the control room to open and close the Stack Damper.



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The open or close position of the Stack Damper is indicated by the lamps provided on the auxiliary panel. These are actuated by the limit switches. There is a provision to open the Stack Damper manually from the field by winch operation in case of emergencies.

Air is required for the combustion of fuels in a furnace. It is supplied by the FD fan (12-FM-01). The air flow is regulated by adjusting the variable inlet guide vane mechanism to maintain proper dP across the furnace and for maintaining the required excess air in the flue gas. About 15 to 20% excess air in case of fuel oil and 10 to 15% in case of fuel gas is found to give satisfactory performance of the furnace. An oxygen analyzer (AR-2501) is also provided at the outlet of the flue gas to monitor the excess air regularly.

2.5% O₂ on AR-2501 equals to 10-15% of excess air

Air preheating system helps in recovering the sensible heat from the flue gas further, after the furnace convection zone, which is utilized to preheat the combustion air. This increases the fuel economy and also the heater efficiency approximately by 10 %. There are separate Air Preheating systems (APH) for Atmospheric and Vacuum Furnaces. Forced Draft fan (12-FM-01) draws Atmospheric air and forces it through the APH. Induced Draft fan (12-FM-02) draws the flue gas through the APH, and returns it to the stack above the stack damper after recovering heat from it. Care should be taken to maintain the return temperature of flue gas above its dew point (typically 175 °C) to avoid condensation which otherwise would result in acid corrosion.

The design of the furnaces in CDU-II gives immense flexibility in their operation. There are 3 modes of operation as given under.

a) Balanced Draft operation

In this mode of operation, the FD and ID fans both function simultaneously and help in recovering the heat from the flue gas. Flue gas is drawn by the ID fan through a duct connection that is taken from the castable below the Stack Damper. This duct is lined with refractory. Pressure and temperature gauges are provided to give local indication of the pressure and temperature of flue gas entering the air pre-heater. PI and TI are also provided to give the indication in the control room. A winch operated shut-off blade hand controller is provided in the duct to isolate the flue gas duct. Hot flue gas enters the APH at the top and exchanges the heat with combustion air. Flue gas exiting the APH is routed to the ID fan suction. This fan draws the flue gas and conveys them back to the stack above the stack damper through a duct. 12-TAL-503 (at the inlet of ID fan) is provided to give an alarm when the flue gas temperature falls below the dew point temperature to avoid corrosion. In case of low temperature, the air flow through the APH can be adjusted so that the



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temperature can be raised. In case there is an abnormal rise in temperature of flue gas leaving the APH, TAH will give an alarm. 12-PAH-507B gives an alarm in case of high furnace box pressure in the arch zone.

Combustion air under pressure from the FD fan is ducted through the bottom of the APH. PG and TG give the local indication of the pressure and temperature of the cold air entering the APH and a PI gives the cold air pressure indication in the control room. 12-PAL-506 is provided on the discharge of the FD fan to give an indication in case of low discharge pressure of the FD fan. After exchange of heat with the flue gas, hot air is sent into the hot air distribution duct running west of the heater. This duct is provided with a TG and PG to give a local indication of the temperature and pressure of the hot air. There is also a TI and a PI to give an indication in the control room. A TAH (12-TAH-501) is provided to give a high temperature alarm. This hot air duct branches into the plenum chamber of the furnace, where the burners are mounted.

b) Forced Draft

In this mode of operation, the FD fan of the furnace will be running and the ID fan is stopped and the stack damper is kept open. Flue gas escapes to the Atmosphere directly without preheating the combustion air. This is done mainly when isolation of the APH or ID fan is required to carry out maintenance activities.

c) Natural Draft

There is also a provision to operate the furnaces in natural draft, wherein there is no requirement of FD and ID fans. There is a provision of 2 Drop out Doors (DOD's) on the combustion air duct of 12-F-01.

To operate the furnace in Natural Draft, DOD's and stack damper are opened. Atmospheric air goes inside the furnace by the action of the draft in the furnace and aids in combustion. The flue gas escapes directly through the Stack Damper without preheating the combustion air.

6.4.1.4 Trip and Interlock System

Trip values of 12-F-01:

Trip	Unit	Value
FD fan discharge pressure (low)	mm Aq.	+15
ID fan suction pressure (High)	mm Aq.	-45
Furnace pressure (high)	mm Aq.	+2.95
Fuel oil pressure (low)	Kg/cm ² g.	+2.77



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Fuel gas pressure (low)	Kg/cm ² g.	0.20
Furnace pass flow (low)	m ³ /h	14

The following trips are provided on 12-F-01

DOD – SD Check Trip: This trip opens stack damper under the following conditions

- Furnace pressure high (Annunciator alarm)
- FD fan trips / FD fan discharge pressure low (Annunciator alarm)
- ID fan trips
- Both fuel oil and fuel gas pressure low (Annunciator alarm)
- Pass flow low-low

FD Fan Trip: This trip opens the DOD's under the following conditions

- Furnace pressure high (Annunciator alarm)
- FD fan trips** / FD fan discharge pressure low (Annunciator alarm)
- FD fan motor amps low (Annunciator alarm)
- Any of the DOD's is not closed completely
- Pass flow low-low
- ESD is activated.

ID Fan Trip: This trip opens the SD under the following conditions

- Furnace pressure high (Annunciator alarm)
- FD fan trips** / FD fan discharge pressure low (Annunciator alarm)
- ID fan motor amps low (Annunciator alarm)
- Both fuel oil and fuel gas pressure low (Annunciator alarm)
- Pass flow low-low
- ESD is activated
- Stack damper opens when in Balanced Draft

Fuel Oil Trip: This trip closes the fuel oil supply and return SDV's under the following conditions:

- Fuel oil pressure low (Annunciator alarm)
- Signal from any of the three timers (details of the timers are given below)
- ESD is activated
- Pass flow low-low

Fuel Gas Trip: This trip closes the fuel gas SDV under the following conditions

- Fuel gas pressure low (Annunciator alarm)
- Signal from any of the three timers (details of the timers are given below)



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- c) ESD is activated
- d) Pass flow low-low

Timers: Three timers are provided in the furnace trip circuit, which will start counting when their respective alarms are activated. If the alarm is not brought back to normal value within 30 seconds (as counted by the timer), the timer relay will activate the fuel oil and fuel gas trips. If the alarm is brought back to normal value within 30 seconds (as counted by the timer), the timer will

- a. **Furnace Pressure High Timer:** It gets activated when furnace pressure high alarm comes on the annunciator panel and gets reset when the alarm is normalized.
- b. **FD Fan Trip Timer:** It gets activated when the FD fan is not running, and gets reset when FD fan is running or the FD fan trip is bypassed.
- c. **ID Fan Trip Timer:** It gets activated when the ID fan is not running, and gets reset when ID fan is running or the ID fan trip is bypassed.

Interlock system: Interlocks are provided to ensure safe operation of the equipment. They ensure that corrective action is taken automatically whenever unsafe operating conditions arise due to process upsets, mal-operation etc. But the interlocks can be made ineffective / inactive by bypassing the trip switches provided on the DCS panel.

Interlocks on furnace operation: The following interlocks ensure the safe operation of furnace. The following description is valid only when the trips described above are in **auto-interlock** mode.

- a) Drop out doors opening or closing
- b) Stack damper opening or closing
- c) FD fan starting or stopping
- d) ID fan starting or stopping
- e) Fuel SDV's opening or closing
- f) Low Pass Flow

Interlock 1: DOD Operation

- i) When the furnace is in natural draft mode, all the DOD's have to be in fully open position. If any one DOD leaves the fully open position (either by operation or by instrumentation malfunction), then the fuel to the furnace is cut off.



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ii) When the furnace is in forced draft mode, DOD's are in fully closed position and if any of the DOD's is not in fully closed position, FD fan is tripped. As get reset i.e., the timer indication will become zero.

In case the FD fan is tripped, all the DOD's will open automatically.

iii) When the furnace is in balanced draft mode, DOD's are in fully closed position. They get opened automatically if the

- FD fan trips
- Arch pressure high gets activated

Interlock 2: SD Operation

- When the furnace is in natural draft mode, the SD has to be in fully open position. If the SD leaves this position, the fuel to the furnace is cut off.
- When the furnace is in forced draft mode, the SD has to be in fully open position. If the SD leaves this position, the fuel to the furnace is cut off.
- When the furnace is in the balanced draft mode, the SD has to be in fully closed position. If it leaves this position, the ID fan is tripped. The SD gets opened if
- ID fan gets tripped
- Arch pressure high gets activated
- FD fan trips /any DOD leaves fully closed position

Interlock 3: FD Fan Operation

- When the furnace is in natural draft mode, **the FD cannot run** as long as the DOD's are in open position.
- When the furnace is in forced draft mode, FD fan gets tripped if any DOD leaves fully closed position or the furnace pressure is too high.
- When the furnace is in the balanced draft mode, the FD gets tripped if any DOD leaves fully closed position.

Interlock 4: ID Fan Operation

- When the furnace is in natural draft mode, the ID cannot run if FD fan is not running.
- When the furnace is in forced draft mode, the ID is in stopped condition. No interlock.
- When the furnace is in the balanced draft mode, the ID fan gets tripped if
- SD is not fully closed
- any DOD leaves fully closed position
- FD fan is tripped
- Furnace pressure is high



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Interlock 5: Fuel SDV's Operation

The fuel SDV's get closed by operating the ESD on the auxiliary panel. Individual ESD's are also provided for fuel oil and fuel gas. The fuel SDV's get opened only after the individual "SDV Reset" push button on the auxiliary panel is operated. The following are the interlocks on the Fuel SDV's

- When the furnace is in natural draft mode, fuel is cut off if any DOD or stack damper leaves the fully open position.
- When the furnace is in forced draft mode, fuel is cut off if any DOD leaves fully open position / FD fan trips
- When the furnace is in the balanced draft mode, the fuel is cut off in case any of the three timers gets activated.

Interlock 6: Low Flow in the Furnace Passes

This interlock shuts down the fuel SDV's to the furnace whenever the furnace pass flow in any coil is low. The trip value set for the low pass flow for 12-F- 01 is 14 m³/h.

6.4.1.5 Steam Air Decoking

Steam air decoking (SAD) of vacuum heater tubes is done to remove coke deposit from inside heater tubes with the help of steam and air. Removal of coke results in clean heater tube internals and improves heater performance by better heat transfer to process fluid.

SAD† also achieves low pressure drop through heater tubes and reduces chances of hot spot on heater tubes. Need for steam air coking of heater tubes is indicated by increased pressure drop and harder firing.

This dedicated SAD arrangement comprises decoking pot (11-V-05) (which is common for 11-F-01 and 12-F-01), piping and instrumentation for plant air, service water and MP Steam. MP steam connection to each pass flow is provided to dislodge carbon deposit from inside the heater tube. Local and DCS room mounted flow indications (12-FE-111 to 113) are provided on each steam connection plant air is required to ignite the remaining coke film deposit cleaning inner wall of tubes and achieve final cleaning for each 4" size plant air connections. 12-FE-110 indicates the total plant air consumption.

Service water quench provision is given on 11-V-05 as well as on decoking line to quench the contents before letting out to atmosphere. To carry out SAD, pass flow inlet and outlet of the furnace 12-F-01 are isolated from process network and connected by means of swing elbows with the decoking network.



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Heater pass flow outlets are connected to the decoking pot (11-V-05). While MP steam is introduced in the tubes it is fired from outside, thermal shock caused by the flame cracks the coke scales inside tube and flowing steam dislodges them. These coke particles are carried to 11-V-05 after being quenched in the line as well as in the decoking pot by service water. This operation is called spalling. When no more coke is removable by spalling as indicated by relatively clear colour of effluent, air along with steam is introduced into pass flow to burn out the coke inside tube while firing is on in the heater, oxygen burns coke at high temperature, burning of the smoke is indicated by increased tube metal temperature and same should not be allowed to go beyond 625 °C Number of passes selected for spalling and coke burning is largely dependent on limitations posed by steam availability and piping network. Sudden release of coke during spalling may result in choking of the piping handling effluent or may cause hot spots of the tube during burning. SAD of only one pass at a time should be done if limitations in steam and SAD piping are experienced.

6.4.2 VACUUM DISTILLATION COLUMN

The vacuum column (12-C-01) has three sections of different diameters. Top section is of 5000 mm diameter. Middle section is of 6600 mm diameter and bottom section is of 5000 mm diameter. As the Vacuum Column operates under vacuum, the vapour velocities are high. Sections where the vapour load is low are smaller in diameter for cost saving. The upper portion of the column has 3 packed beds of SS410S material. During 2010 T&I the LVGO packing (Existing : 25 M3 M-PaK1.5+7.5M3 MPak2.0+ 17M3 Hy-Pak 1.5) and HVGO packing (52M3 Hy-Pak 3.0 + 52M3 M-Pak 3.0) were replaced with IMTP-40 and IMTP-70 respectively with SS316 2.5% Mo metallurgy. There are 3 side draw-off trays (chimney type) for slop cut, HVGO and LVGO, one each below the three packed beds. The bottom portion has 3 disc and doughnut type heat transfer trays. Short residue is drawn as bottom product.

Description of Vacuum Column has been taken up zone wise starting from bottom.

6.4.2.1 Vacuum Column Bottom Section:

The partially vaporised RCO feed stock coming from the Vacuum Heater enters the column in the flash zone below the slop draw-off chimney tray. The vaporised portion rises up in the tower and is fractionated into 3 side stream products. The liquid portion of the feed drops into the bottom section of the tower and is withdrawn as Short residue (VR). 12-TR-203 and 12-PR-202 indicate the Column flash zone temperature and vacuum respectively on the DCS panel.



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The Column bottom level is maintained by 12-LIC-202's action on 12-LV-202 on VR rundown line. LI-2201 also indicates level on DCS panel. Both LIC and LI have software high level and low level alarms 12-LAH/LAL-201. 12-TI-201 indicates temperature of the column bottom on the DCS panel. LAH-2108 and LAL-2109 are also provided for column bottom on the DCS panel.

A tangential vapour horn is provided for flash zone inlet, which minimises entrainment of heavier hydrocarbon liquid droplets. There is a 3 inch LP steam line provision to column bottom below tray 1 for steaming out purpose.

Vents of all pumps taking suction from Vacuum Column are connected back to Vacuum Column above flash zone through a common vacuum line. With this arrangement all pumps connected with Vacuum Column can be vented to the column.

Temperature at the bottom section is normally quenched to about 350 °C at a vacuum of - 735 mm Hg (g.). RCO is fed to the flash zone at 390 °C.

6.4.2.2 Short Residue Draw-off

Short residue (VR) is drawn by VR + Quench pumps (12-P-01 A/B) from the Vacuum Column (12-C-01) bottom. VR pump discharge can be routed to following destinations:

- To Vacuum Heater (12-F-01) through 8" start-up line which joins the RCO feed line to 12-F-01. This is used during start-up only.
- A slip stream of the pump discharge can also be routed to the pumps suction in a 2" line as spill-back.
- The pump discharge from 12-P-01 A/B exchanges heat with crude in 11-E-16, 12-E-06 A/B and 12-E-03. The outlet of 12-E-03 at a temperature of 250 °C can be routed as follows.
 - It is partly sent to the 12-C-01 bottom in a 6" line for quenching and maintaining the bottom temperature at 350 °C to prevent cracking of SR and lead to other problems like deterioration of vacuum, plugging or suction strainers of 12-P-01 A/B leading to loss of suction. The quench flow is recorded and controlled by 12-FRC-204.
 - It can be partly sent as feed to the Bitumen Blowing Unit in a 3" line.
 - The rest of it can either be sent to 12-E-01 A/B/C and then to 12-E-09 A/B/C/D or can be routed directly to the 12-E-09 A/B/C/D where it is cooled. TIC-2103 located on VR product rundown header to tanks controls TV-2103 on the tempered water line to 12-E-09 A/B/C/D.

- Downstream of the coolers, SR can be routed to



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- IFO pool partially in a 6" line.
- Storage via the three-way FCV 12-FRC-405 through an 8" RFO line, or to the HFO pool by adjusting 12-FV-405. The HFO flow is measured by 12-FR-406.
- LDO pool partially, in a 4" line, measured by 12-FI-407.
- VBU as hot feed from the upstream of 12-E-09 A/B/C/D or to VBU storage tanks in 10" line. (TIC-2420 on rundown line maintains VR temperature by bypassing a certain amount of hot VR to the rundown line)
- BBU storage tanks via HFO line (at unit limit tie-in).

6.4.2.3 Vacuum Slop Draw-off:

The zone immediately above the flash zone is known as Wash Zone. It consists of a packed bed (Hipack type). Above this there is a 150 mm thick SS410 Glitsch Grid packing to remove the entrained asphaltenes. Spray nozzles are provided above the packing for proper distribution of the internal reflux. The spray nozzles were replaced with Lechler's spray nozzles (designed by Shell) were fitted (28nos). The Wash zone also has a demister pad above the spray nozzles of the wash zone packing. The vapours rising from the wash zone pass through a demister pad provided to trap the entrained droplets of Heavy hydrocarbons which could otherwise adversely affect the HVGO/LVGO quality. Slop-cut is drawn by slop distillate and recycle pumps (12-P-02 A/B). The draw-off temperature is indicated by 12-TI-204. The level on the chimney tray is regulated by 12-LI-203 by operating 12-LV-203 on the slop-cut rundown line to the VR line. Slop + Recycle pump (12-P-02 A/B) discharge has following destinations.

One part of slop distillate goes to the furnace under flow control 12-FRC-109 through a 3" line. The purpose of the recycle stream to vacuum furnace is to maximise HVGO recovery from the slop cut. The slop-cut recycle from slop pump is mixed with RCO from Atmospheric column before getting heated in vacuum furnace. Additional provision was given during 2010 T&I for routing the slop-cut to FCCU-II from the downstream of slop-cut recycle control valve.

A slip stream of the pump discharge can also be routed to the pumps suction in a 2" line as spill-back.

As product rundown, a part of Slop Distillate pump discharge gets mixed VR product up stream to 12-E-01 A/B/C. 12-LV-203 on slop line maintains slop level on the chimney tray of slop section.

The product stream can also be routed to the storage tanks through the CDU-I cooler box.



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6.4.2.4 Heavy Vacuum Gas Oil (HVGO) Draw-off:

The majority of rising hydrocarbon vapours from slop zone wash section are condensed in HVGO section by circulating reflux to yield the side draw-off product. HVGO product, Internal Reflux (IR) + Circulating Reflux (CR) is withdrawn as the second side stream. 12-TI-205 indicates the draw-off temperature. HVGO draw-off from the Column is routed to HVGO product + CR pumps (12-P-03 A/B). There is a provision to route SR to these pump suction (during flushing of the Vacuum Column bottom). The HVGO discharge from the pumps is routed as

a) **HVGO Internal Reflux:** One part goes as Internal Reflux (IR) for packing washing of Wash zone of vacuum column without any heat exchange. HVGO IR (Wash Oil) is regulated by 12-FRC-202 in such a way that proper washing of the packing is always achieved for all throughputs. One distributor is provided for proper distribution of HVGO IR over the entire cross section area of the column packing. Strainers (12-X-01 A/B) are provided to arrest the carryover of foreign materials back into column. Pressure gauges across the filters indicate the pressure drop across the filters and its rise would indicate the need to change over of the filter in line. 12-TI-206 shows HVGO IR temperature.

b) **HVGO Circulating Reflux:** HVGO CR + product from 12-P-03 A/B discharge is split into two streams. One stream goes through 12-E-05 A/B and 12-E-02 and the other through 12-E-04. Here, it exchanges its sensible heat with the crude. The two streams are then combined and routed to the MP steam generators 12-E-10/10A parallelly. The stream coming out of the steam generators is routed partly as HVGO CR back to column at through the CR strainer 12-X-02 A/B. This flow is controlled and recorded by 12-FRC-203. The CR return temperature is indicated by 12-TI-208. The remaining volume is either routed as hot feed to FCCU-II or to product rundown line or both at the same time.

HVGO Products:

The HVGO can be routed to the following:

- To the 6" Hot feed line, under the flow controller 12-FRC-402. The temperature of this stream is indicated by TX-2402.
- To 12-E-12 A/B where it gets cooled down to 80 °C and goes to the product storage tanks in a 4" line under the control of 12-LIC-204, which controls the HVGO level in the Column. The product can also be routed to LDO pool, whose flow is recorded by 12-FR/FQ-408.



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- c) A slip stream of HVGO is routed as quench to 13-PV-402 under the flow controller 13-FRC-404.
- d) To slops in a 6" line

6.4.2.5 Light Vacuum Gas Oil Draw-off (LVGO):

The rising uncondensed hydrocarbon vapours from HVGO zone are condensed in LVGO section by circulating reflux to yield the side draw product LVGO product + Internal Reflux (IR) + Circulating Reflux (CR) is withdrawn through 10" line from the third chimney tray. 12-TI-207 indicates draw-off temperature on the DCS panel. LVGO draw-off from the column is routed to LVGO product + CR + IR pumps (12-P-04 A/B), whose discharge is routed as

- a) **LVGO Internal Reflux:** This stream joins the HVGO CR stream and goes as a reflux over the HVGO packing without any heat exchange. The flow of this stream is controlled by 12-FRC-201. This stream joins the cold HVGO CR stream upstream of the strainers 12-X-02 A/B.
- b) **LVGO Circulating Reflux:** This stream splits further into two streams. One stream goes 11-E-07 to exchange its sensible heat with the crude and then goes to 11-E-22 for cooling. The other stream goes to 11-E-22A directly for cooling. The outlet of 22 & 22A combine and go through the LVGO CR strainer 12-X-03 A/B to the top of the LVGO packing as top reflux. The CR flow is controlled by 12-FRC-205 and the return temperature is indicated by 12-TI-210.
- c) **LVGO Product:** A part of 12-P-04 A/B discharge stream gets cooled in 12-E-11 and can be routed to HVGO storage tanks or FCCU-II. The flow of this stream is controlled by 12-LRC-205 and is recorded / integrated by 12-FR/FQ-403. The product can also be routed to Diesel or LDO pool controlled by 12-FRC-404.

LVGO product is routed as follows:

- As hot feed to FCCU-II along with hot HVGO through 12-LV-205
- To VGO storage tanks along with HVGO through 12-LV-205
- Either to Diesel/LDO/SR or HVGO (at battery limit) storage tanks through 12-FV-404 in a 3" line. There is a provision to cascade the level controller on the LDO FCV
- To flushing oil system in a 2" line
- To slops along with HVGO.

6.4.2.6 Vacuum Column Overhead System:



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Steam and small amount of light hydrocarbons produced as a result of cracking, pass out of the vacuum column top through a 36" size line. The uncondensed vapours flow through a demister pad provided in the column top vapour space to set of steam ejectors. Demister pad prevents carry over of liquid droplets to ejectors and condenser system. The Vacuum Column top temperature is indicated by 12-TI-209. The overhead pressure is indicated and controlled by 12-PRC-206 on DCS panel. 12-PSV-202/ 202B set at 3.5 kg/cm g. are mounted on top of the Vacuum Column to protect the column from over pressurisation. Column top vacuum is normally operated at -751 mm Hg (g.).

Vacuum is created and maintained in the column by 3-stage ejector system with condensers.

The 1st stage ejectors (12-J-01 A/B/C), 2nd stage ejectors (12-J-02 A/B/C) and the 3rd stage ejectors (12-J-03 A/B/C) are designed *for a capacity factor of 1/7, 2/7 and 4/7 totalling 150% of normal capacity*. If required each ejector element can be isolated by cutting off steam and isolating suction inlet valves.

The vapours from the Vacuum Column top are sent to the primary condenser 12-E-07A. The non-condensable from the primary condenser are drawn by the 2nd stage ejectors, which is again routed to the secondary condenser (12-E-07B). The non-condensable from the secondary condenser are drawn by the 3rd stage ejectors and routed to the after condenser (12-E-07D). The non-condensables from this stage are routed to the hotwell drum through a dip leg. The non-condensables from here enters to knock out pot and then are either routed to 11-F-01 Hot well off gas burners 17,18,19 and 20 or it has to be routed to atmosphere (vent the non-condensable into atmosphere). The condensate from all the three condensers drops to the Hotwell drum (12-V-01) through the barometric legs. Cooling water is supplied to the primary condenser in an 18" header. The water outlet from the primary condenser is bifurcated into two parts. One part feeds the cooling water to the secondary condenser and the next part feeds the after-condenser. There is a provision to back flush the primary condenser. A draining provision is provided in the up-stream of the after-condenser. 12-PRC-206 is mounted on the non-condensable line from 12-E-07D, controls the Vacuum Column overhead pressure by routing a part of the non-condensable vapours (before letting them into the Hotwell drum) to the inlet of the 1st stage ejectors. A 3" fuel gas line is provided near the inlet of the 1st stage ejectors, for backing-in of fuel gas during the start-up and shut-down to maintain the column in positive pressure.

MP steam to all ejectors is supplied in parallel by 6" header. Ejector steam consumption is indicated by 12-FR-207. Strainers are provided on the MP steam line to arrest line scales



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etc. from reaching into ejectors and adversely effecting performance of overhead system. Ejector steam pressure is controlled by 12-PRC-207. This should be maintained constant as far as possible for smooth operation of the Vacuum Column. Corrosion Inhibitor and Neutraliser injection facility into overhead vapours have been provided both on Vacuum column top overhead.

Hotwell (12-V-01) is a horizontal vessel provided to separate out condensed hydrocarbons from vacuum column overhead stream and provide effective sealing of the system operating under vacuum from Atmosphere. Hotwell is provided with 3 compartments created by two internal baffles of 400 mm and 800 mm height. The process lines called barometric legs carrying the condensate from all ejector condensers are kept dipped into water in the left side of the compartment (left of 400 mm baffle). Dip legs are provided so that no air ingress is possible. An overflow line with a U-loop seal and valve (called as siphoning loop) is provided on the water side of the Hotwell Drum. This siphoning loop ensures that a minimum water level in the water compartment of the drum is maintained, so that water doesn't rise back into the barometric legs, thereby bringing down the vacuum. At the crest of this siphon U-loop, a 1" vent having an opening downwards is provided to break the siphoning effect, and hence called, the siphon breaker.

The water-hydrocarbon interphase is formed in the middle compartment from where hydrocarbon overflows to its higher baffle height (800 mm) compartment (slop oil compartment) and water remains in the middle sour water compartment. The sour water from the middle compartment is pumped out by sour water pumps (12-P-06 A/B) to the Sour Water Stripping unit.

12-LDIC-201 controls the Hotwell interface level acting on 12-LV-201 which is mounted on the sour water line. 12-FI/FR-650 shows sour water flow on DCS panel when Hotwell sour water is routed to 11-V-04. 12-LT-201 is mounted in the middle section of the hotwell to monitor the oil-water interphase. Condensed and separated hydrocarbon components (Hotwell oil) are collected in oil compartment. Hotwell oil is pumped out by Hotwell oil pump (12-P-05 A/B) to Hotwell oil storage tank or to slop. Hotwell oil pumps are interlocked with Hotwell oil high level alarm (12-LAH-205) and low oil level alarm (12-LAL-205) for auto cut-in and cut-off respectively. These high and low level switches automatically start and stop the Hotwell oil pumps. This flow is recorded by 12-FI/FR-206. Vapours from 12-V-01 are routed to the hot well knock-out pot where entrained liquid is knocked off and routed to OWS periodically. Vapours from the knock-out pot are either routed to 11-F-01 Hot well off gas burners 17,18,19 and 20 or it has to be routed to atmosphere (vent the non-condensable into atmosphere). Steam connection is provided on



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this vent line to Atmosphere for dilution of hydrocarbon vapours which are vented. Hot well off gas flame arresters (2 no's) are provided on the off gas line to 11-F-01.

6.4.2.7 Vacuum Pumps Priming

The pumps in the Vacuum section that take suction from the Vacuum Column are served by a vacuum vent system. The vents from the various Vacuum section pumps are connected to a header which enters below the first tray of the Vacuum Column. The vent system is necessary so that vacuum can be pulled in a pump casing before the suction valve is opened. This facility ensures proper priming of pumps operating under vacuum. The following pumps are served by the system.

- 11-P-10 A/B RCO pumps
- 12-P-01 A/B Short Residue pumps
- 12-P-02 A/B Slop Distillate and Recycle pumps
- 12-P-03 A/B HVGO pumps
- 12-P-04 A/B LVGO pumps

6.5 MP STEAM GENERATION

MP steam is generated from BFW in the unit by recovering heat from HVGO CR + Product stream. BFW is received from off-sites through a 4" header taken from which a 2" tapping is for steam generation. Steam is generated in kettle type steam generators 12-E-10 / 10A by exchanging heat with HVGO. The level of the BFW in the steam gen is controlled by 12-LIC-301/302 by operating 12-LV-301/302 on the BFW lines to 12-E-10/ 10A respectively. The temperature of HVGO from the steam generators is controlled by 12-TRC-302/303 by operating the three-way control valves 12-TV-302/303 respectively on 12-E-10/10A respectively. Saturated MP-steam is generated in this system at 10.5 kg/cm² g. pressure and 190.7 °C. Two PSV's (set at 13.5 kg/cm² g.) have been provided on each steam generator to protect the steam generators from over-pressurisation in the event of external fire. A 2" vent line is provided to vent out the steam from the PSV's to atmosphere.

In order to depressurise and blow down steam safely, 12-V-03 (steam blow down drum) has been considered. Blow down from steam generators 12-E-10 and 12-E-10A enters the blow down drum (12-V-03) and gets quenched by service water and then routed to OWS. This drum has a vent at safe height. Its drain connection has water seal provision.

6.6 TEMPERED WATER AND DM WATER SYSTEM

DM water at 60 °C is used in a closed circuit to cool Short Residue and Bitumen product streams. This is used as these products should not be cooled below their pour points. Salt water or service water are also not used to prevent scale formation in the exchanger tubes at



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high operating temperatures, which would otherwise lead to severe loss in heat transfer efficiency.

This system consists of 12-V-01 (tempered water drum), 12-P-07 A/B (tempered water pumps), 12-E-08 A/B (tempered water coolers), 12-E-09 A/B/C/D (Short Residue coolers) and 13-E-02 A/B/C (Bitumen product trim coolers). 12-V-02 is a vertical cylindrical vessel of Height $\square$ Diameter, 5400 $\square$ 1800 mm. The vessel is elevated 6000 mm above the ground, so as to provide the necessary NPSHR to the tempered water pumps and avoid cavitation. It is provided with a level indicator (12-LI-401), low level alarm (12-LAL-401), high level alarm (12-LAH-401) and a temperature gauge (12-TG-401). DM water makeup line has been provided to make up for evaporation loss. For heating the tempered water during the start-up, a 1½" LP steam line is provided to admit open steam into the tempered water drum. Tempered water at 60 °C drawn by tempered water pump is routed to 12-E-09 A/B/C/D and 13-E-02 A/B/C located in the plant. The hot tempered water is maintained by adjusting the bypass globe valve across the cooler 12-E-08 A/B. The outlet temperature is indicated by 12-TI-405.

Tempered water is also used as bearing cooling water for 12-P-01 A/B (Short Residue pumps)

6.7 CHEMICAL INJECTION FACILITIES

Crude oil contains salts such as chlorides of Magnesium and Calcium. These salts if not removed, hydrolyse in the system to form Hydrochloric acid. There is also a possibility of formation of acidic Hydrogen Sulphide (formed from dissociation of heavy sulphur compounds present in crude) which is then dissolved in the crude. Both of these concentrate in the overhead system and form acid solutions, which are corrosive. Measures must be taken to overcome their harmful effects.

The overhead system including condensers and reflux drum is made of carbon steel. In order to protect this section, caustic solution, ammonia solution and corrosion inhibitors are added at various points. The purpose of injecting caustic at the outlet of Desalter is to achieve better mixing of these chemicals with crude and neutralise the acids/salts mainly HCl and H₂S as soon as they formed (at a temperature of 120°C and above). The reaction products i.e. Sodium and Ammonium salts go along with reduced crude (RCO). The balance acids and acid gases if any will go up to the overhead system where ammonia is injected in the overhead vapour line for neutralisation. Amount of ammonia should be controlled in such a way that pH of reflux drum water remains at around 6.5 Injection of caustic at the outlet of Desalter should be maintained in such a way that the salt formation should be low in the overhead circuit which might form scales in the overhead condensers tubes.



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A slightly acidic condition of the overhead system is desirable to keep ammonium salts in solution, which if precipitate, would foul and plug the condensers. Corrosion against slightly acidic (pH 6.5) condition is minimized by adding corrosion inhibitors in the overhead vapour line. The inhibitor is also added in reflux line. Top section of the column is also benefited from the injection of corrosion inhibitors mainly in the reflux line. These inhibitors are high boiling point compounds and can perform satisfactorily at higher column top temperature also. The amount of inhibitor injected depends upon the type of inhibitor used and is generally specified by vendor. Adjustment is made by operating personnel depending upon iron contents in the reflux drum water. Inhibitors are filling organic compounds which cover entire metal surface of the system with a thin film. This prevents contact of corrosive water with metal surface.

Demulsifier is added to the crude to break the water-crude emulsion. Water-crude emulsion behaves like a single phase and does not get separated easily. Demulsifier helps the process and ultimately in Desalter vessel water is separated out. Demulsifiers are surface activating agents and acts on interface surface tension of crude and water emulsion.

6.7.1 CAUSTIC INJECTION SYSTEM

Caustic is received at unit battery limit either from MEROX or from Power Plant-I. There are two caustic solution vessels in the unit 11-V-07 A/B. Concentrated solution is received in these vessels from unit battery limit and then diluted to 5 Beo. Each vessel is provided with service water connection for dilution and 2" plant air connection through spargers for agitation and mixing. Vent, level gauge and overflow lines are also provided on each tank. There is also a provision of sample point on each drum. Caustic solution is pumped by 11-P-13 A/B/C to the following destinations

1. Crude charge pump suction
2. Downstream of Desalter through a vortex mixer 11-X-01.

The injection pumps are metering pumps which give a maximum discharge pressure of 15 kg/cm² g. Caustic injection rates can be varied from 0 to 75 lph, by varying the stroke of the pump. The rate of injection can be measured by using the graduated cylinder provided on the pumps

6.7.2 DEMULSIFIER INJECTION SYSTEM

The chemicals that are used as DMF come under the trademark names EC2040A and Embreak. De-emulsifier chemical is received in drums and is pumped into the De-



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emulsifier vessel using a pneumatic pump. This is further diluted with water in the De-emulsifier vessel in the ratio of 1:3. De-emulsifier injection pumps (11-P-16 A/B) takes suction from the de-emulsifier vessel and is injected to the suction header of the crude charge pumps 11-P-01 A/B. The injection pumps are metering pumps which give a maximum discharge pressure of 15 kg/cm² g. De-emulsifier injection rates can be varied from 0 to 16 lph, by varying the stroke of the pump. The rate of injection can be measured by using the graduated drawdown cylinder provided on the pumps.

6.7.3 NEUTRALISER INJECTION SYSTEM

A battery of liquid Ammonia cylinders is provided for meeting Neutraliser requirement. Ammonia gas is drawn from cylinders at reduced pressure into Ammonia solution vessel (11-V-06 A/B). A **rotameter 11-FI-601** indicates Ammonia gas consumption. The cylinders are connected to a common manifold. A pressure gauge (11-PG-601) and a PSV (11-PSV-601) are provided on the Ammonia collection common manifold. Ammonia solution vessel is a vertical vessel with a water seal on its vent to prevent escaping of Ammonia while preparing solution. Water seal will blow off during excess pressure build up in the vessel indicating the saturation of Ammonia in the solution. The Ammonia solution prepared is injected by ammonia solution dosing pumps 11-P-14 A/B/C to following destinations:

- To Atmospheric Column Top (Vapour line) through rotameter 11-FI-602
- To Atmospheric Column Top Reflux line through rotameter 11-FI-603

Ammonia (10% soln) dosing rate is normally 10 lph for both the sections. The injection pumps are metering pumps which give a maximum discharge pressure of 15 kg/cm² g. Ammonia injection rates can be varied from 0 to 20 lph, by varying the stroke of the pump. The rate of injection can be measured by using the graduated cylinder provided on the pumps

The neutralizer to the Vacuum Column overhead lines is supplied by 12-P-08 A/B, at rates varying from 0 to 20 lph to the Vacuum Column overhead vapour line.

6.7.4 CORROSION INHIBITOR INJECTION SYSTEM

Corrosion Inhibitor is received in drums and is pumped into the corrosion inhibitor vessels. NALCO 5186 or equivalent filming amine solution (1% conc.) is prepared by diluting it in Kerosene in 1:3 ratio in the Corrosion Inhibitor solution vessels. The solution is pumped by 11-P-15 A/B to the following destinations

- Atmospheric Column overhead
- Vacuum Column overhead system



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The injection pumps are metering pumps which give a maximum discharge pressure of 15 kg/cm² g. Corrosion inhibitor injection rates can be varied from **0 to 10 lph**, by varying the stroke of the pump. The rate of injection can be measured by using the graduated cylinder provided on the pump



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7.1 ATMOSPHERIC SECTION:

It is important that the operation of crude distillation unit be conducted to produce products of desired quality. At the same time appropriate controls should be exercised on certain parameters to prolong the life of the equipment. The following discussion gives guidelines about affect of the variables and measures to be taken to achieve desired results.

7.1.1 Desalter Operating Variables:

Only operating experience with desalter can determine optimum operating conditions. No two crudes behave alike at the same desalting conditions. But, all are affected similarly by change in desalting conditions.

i) *Water Injection and Pressure Drop:*

Water injection should be started only after the crude temperature reaches specified level and power is switched on to the grid. Initially the injection rate should be limited between 2 to 6% of crude flow rate and point of injection should be just ahead of the emulsifying valve.

The pressure drop across emulsifying valve should be adjusted to give the required degree of desalting. The higher the pressure drop the more efficient the contact between the salt in the crude and the injection water. Too high pressure drop will result in excessive emulsification and poor separation of oil water, resulting in carry-over of water in the desalted crude. A pressure drop between 0.2- 1.0 kg/cm² is normally sufficient and the value has to be decided based on oil content in effluent and desalter inlet and outlet salt content.

Additional emulsification can be obtained by injection of water before the feed pump suction (11-PM-01A/B). Injection at this point results in maximum contact and also prevents the sediments from settling in the exchanger tubes and fouling them. But care should be taken such that the intense shearing agitation in the preheat train does not create so tight an emulsion that cannot be resolved in the Desalter. The severe shearing effect due to the crude pump impellers should also be considered here. The quality of water is a very important aspect.

Optimum water injection rate and pressure drop across mixing valve should be established to get the desired desalting of crude. Once this is done the conditions should be maintained steady and should be varied only for changes in feed rate and feed qualities.

ii) *Oil/ Water Interface level:*



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The Oil water interface level should be kept below the centre line of the vessel. Incorrect operation of the interface level controller can result in more water in desalted crude due to less hold up time available for oil (high interface level) and more oil carry-over in brine-water due to less hold up time available for water (low water level). Also too high an interface level may put watery mixture up between the electrodes and cause them to short out.

iii) *Desalter Vessel Pressure:*

The pressure in the vessel should be maintained at about 10 to 11 kg/cm²g. A low pressure will cause vaporization of crude and high pressure will result in chances of lifting of the safety valve on the desalter. Sudden variation in the operating pressure may cause hunting of the desalter level control valve with consequent fluctuation of the interface level. If there is vaporization in desalter, it results in hazardous condition, erratic operation and loss of desalting efficiency.

iv) *Desalter Temperature:*

Temperature is another important variable which affects oil water separation in Desalter. Most crude oils have an optimum operating temperature range of 120 to 130°C. Lower the temperature higher the viscosities of the oil which slows down the separation rate. As conductivity of crude increases with temperature, operating temperature beyond the range will lead to drop in grid voltage and high amperage which imposes limitation on good separation. Excessive amperage will eventually cause the circuit breaker to open removing the grid voltage and rendering the electrical system inoperable until the thermal relay is closed. Moreover very high temperature may lead to vaporization of crude in the desalter.

v) *Demulsifier injection:*

Stable emulsions can also be broken by use of demulsifying chemicals. The amount of chemicals required depends on the nature of the emulsion, type of crude and other operating conditions like residence time, temperatures etc. Tests should be made to ascertain the required chemicals injection rate for optimum operation of desalting unit.

vi) *Voltage and Amperage:*

The electrical panel houses pilot lights, a voltmeter and ammeter. The voltmeter gives the voltage across the primary circuit of the transformer. The ammeter gives the current flow. These meters give an indication of the performance of the grids inside the desalter. In case, if



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crude/water emulsion is too tightly bound or if the interface level is too high there will be increase in the amperage and the voltage will drop. Take corrective action to break the emulsion or reduce the interface level.

For more details about the operation of desalter the vendors operating and maintenance manual should be referred.

7.1.2 Heater Outlet Temperature and Column Pressure:

The quantity of crude oil vaporized during its passage through the heater depends on transfer temperature and pressure at the flash zone of the column. In order to achieve proper recovery of distillates little over flash is maintained, by keeping the transfer temperature slightly higher. This flow is about 6% volume of crude flow rate. This also indicates the presence of liquid levels in the trays down below the diesel draw off for avoiding dry operations of these trays (nos. 7 to 11). Maintaining high over flash rate will result in more consumption of energy. Heater outlet temperature is controlled by 11-TRC-301. lower temperature will not give desired distillate recovery, bottom product RCO will be lighter and all side draw offs will also be proportionate to lighter. Higher than normal temperature enhances cracking possibility and at the same time specification of every product may not be met.

The pressure in the column is maintained by split range pressure controller 11-PRC-409 A/B. A low pressure aids in greater vaporization. All products will be heavier and there will be gas loss from reflux drum. Higher than normal pressure will result in reverse effects. Efforts should be made to operate the column at the designed pressure of 0.6 kg/cm²g at the reflux drum. Adjust cooling water flow in the cooler 11-E-17 A/B/C/D/E/F/G/H in such a way that there is total condensation and both the control valves (11-PV-409 A/B) of split range controller remain shut.

7.1.3 Crude Column Top Temperature:

The column top temperature is controlled by regulating amount of overhead reflux through 11-FRC-403. Top temperature is continuously recorded by the recorder. Lowering of top temperature will reduce FBP of naphtha and flash point of Heavy Naphtha. Too low a temperature will start steam condensation at the top section of the column (as the dew point will be reached), which may likely to increase rate of corrosion at the top. Raising the temperature will increase FBP of overhead naphtha and IBP (flash) of Heavy Naphtha as well as the product and CR draw off temperatures. In some cases, the temperature may go to such a value that pumps design limit might be violated.



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Increasing the top temperature also increases the naphtha yield. For lighter crudes where naphtha yield is high, increasing top temperature will increase naphtha yield further and overhead drum 11-V-01 level may increase.

7.1.4 Pump around Flows (Circulating Reflux):

The pump around/ circulating reflux serve mainly in withdrawal of heat from the column and to reduce the vapor liquid traffic in the appropriate section of the distillation column. There are three circulating flows- Top Pump Around (TPA), KERO circulating reflux and diesel circulating reflux. These flows are respectively controlled by flow meters 11-FRC-404/405/408. The return temperature is maintained by operating the respective exchangers. Return temperature are indicated by 11-TI -414(TPA), 11-TI-413 (KERO CR) and 11-TI-412(Diesel CR). A high TPA flow will result in decrease of overhead reflux and affect the quality of light naphtha. The overhead condenser duty will come down as there will be correspondingly less O/H product. Similarly a high KERO CR flow will tend to lower the plate temperature of heavy naphtha and kerosene draw off resulting in lighter product in these trays. The gap between naphtha and kerosene will decrease.

Likewise a high diesel CR will tend to lower the draw off temperature of kerosene and diesel. Increase of circulating reflux will result in higher crude preheat temperatures by greater recovery in heat exchanger train.

7.1.5 Product Withdrawal Temperature:

The withdrawal temperature of a product from the column influences the end point of the product. This is determined by quantity of the product withdrawn from the stripper. An increase in withdrawal rate of the side stream increases the withdrawal temperatures and the end point of all side stream lower down the column unless withdrawal rate lower down the column are reduced correspondingly. For example, if kerosene withdrawal rate is increased, the internal reflux in the trays below the draw off tray will be reduced which will lead to flow of heavier vapors above the tray. This increases the end point of kerosene. If diesel withdrawal rate is not reduced to maintain its plate temperature, its initial boiling point (flash) will go up.

Similar reverse action takes place when withdrawal temperature is lowered by reducing the quantity of withdrawal.



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7.1.6 Stripping Steam:

i) *Atmospheric tower stripping steam:*

At 10.5 kg/cm²g pressure and 350 °C, superheated steam is used to strip lighter fraction from the reduced crude in the lower part of the crude tower. Design steam rate is about 4585 kg/hr for Basrah crude. This is assumed to be optimum rate for economical stripping and should not be varied much. Lowering the rate below the optimum may leave some lighter component in the reduced crude and is undesirable. Exceeding the design rate might cause entrainment of reduced crude into the diesel because of excessive vapor velocity and also will overload the over head condenser system. The flow of steam controlled by 11-FRC-401

ii) *Stripping steam in Heavy Naphtha, Kerosene and Diesel strippers:*

The initial boiling points and flash points of heavy Naphtha, Kerosene and Diesel products are controlled to some extent by varying the stripping steam rate to stripper 11-C-02/03/05 respectively. Steam at 10.5 kg/cm²g and 350 °C used for these strippers. Steam flows are indicated by 11-FI409/11-FI-408/ 11-FI-407 and regulated by respective control valves.

It is advisable not to exceed the steam flow rate of its designed value viz. Heavy naphtha stripper 444 kg/hr. kerosene stripper 1496 kg/hr and diesel stripper 1906 kg/hr, as this will tend to lift some of the high boiling materials. In case if the desired flash point could not be reached by designing rate of stripping steam, the draw off temperature at the product just above it to be increased to enhance its flash.

7.1.7 Corrosion Control of Overhead System of Distillation Column:

Hydrochloric acid formed from the hydrolysis of salt present in the crude and hydrogen sulphide formed dissolved in the crude (formed from the dissociation of heavy sulphur compounds present in crude), goes to the overhead system. Both form acid solutions which are very corrosive. Measures must be taken to overcome their effects.

The overhead system including condensers and reflux drum are made of carbon steel. Only to protect this section caustic ammonia solution and corrosion inhibitors are added at the following points:

Caustic injection:

1. Suction of crude booster pump i.e. ahead of desalted crude preheat trains.
2. After the preheat trains before booster pump 11-PM-02A/B suction



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Ammonia injection:

1. Suction of crude booster pump ahead of desalted crude preheat train.
2. Column top reflux line.
3. Crude column overhead vapor line.

Corrosion inhibitor injection

1. Column top reflux line.
2. Crude column overhead vapor line.

The idea of injecting caustic and ammonia at the outlet of Desalter is for better mixing of these chemicals with crude and neutralizes the acids/acid salts mainly HCl and H₂S as soon as it is formed (120 °C and above). Chance of H₂S formation at this temperature is remote. The reaction product is sodium and ammonium salts goes along with the reduced crude. The balance acids and acids gases if any will go up to the overhead system where ammonia or ammonium solution is injected either along with reflux or in the overhead vapor line for neutralization. Amount of ammonia should be controlled in such a way that pH of reflux drum sour water remains around 6.0 to 6.5. Injection of caustic and ammonia at the outlet of Desalter should be maintained in such a way that the salt formation should be low in the overhead which might scale up the overhead condensers tubes.

A slightly acidic condition of the overhead system is desirable to keep ammonium salts in solution, which if precipitates would foul and plug the condensers.

Corrosion against slightly acidic condition is minimized by adding corrosion inhibitors in the overhead line. The inhibitor is also added in reflux line. The amount of inhibitor injected depends upon the type of inhibitor used and generally specified by vendor. However, slight adjustment is made by operating personnel depending upon the content in the reflux drum water. These inhibitors are filming organic compounds (amines) which covers entire metal surface of the system with a thin film. This prevents contact of corrosive water with metal.

Top section of the column is also benefited from the injection of inhibitors mainly in the reflux line/ these inhibitors are high boiling compounds and can perform satisfactorily at higher tower temperatures.

7.2 STABILIZER TEMPERATURE AND PRESSURE:

Stabilizer removes the majority of butane and lighter hydrocarbons from the naphtha stream. These are recovered as overhead LPG product. High top temperature will make overhead product heavier, even pentanes may be carried into LPG, making it off-spec. Lower



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temperature will reduce LPG, make. Vapor pressure of LPG may go beyond the specified limit if top temperature is too low.

Bottom temperature if too low will result in higher than allowable vapor pressure (RVP) of naphtha and at the same time it will reduce LPG make.

Low pressure in the column will cause higher amount of hydrocarbon compounds (propane and butanes) to escape into fuel gas system. This has got similar affect as that of higher temperature in the column. The RVP of naphtha has to be controlled by the re-boiling and the LPG spec has to be obtained by varying the top pressure and reflux.

The stabilizer pressure has to be maintained at constant value and there should not be drastic changes in the parameters affecting the stabilizer pressure. It has to be ensured that gas generation does not fluctuate drastically because that will yield in disturbance of fuel gas header pressure and may result in flaring and vaporization.

7.3 VACUUM SECTION:

The following variables of the vacuum column influence the quality of the products and should be controlled to meet the product specifications.

7.3.1 Transfer Line Temperature:

The transfer line temperature is controlled by 12-TRC-133 on the outlet of the vacuum heater, and should be adjusted to maintain the flash zone temperature around 395 °C, this temperature determines the degree of vaporization and the level of heat in the liquid vapor mixture, entering the vacuum tower for fabrication. This temperature will be varied depending on the quantities of the desired distillates. But the furnace outlet temperature should not be allowed to go beyond the designed limit of 415 °C after which the degree of cracking increases rapidly. Detrimental effect of cracking or coke deposition on heater tubes, transfer line and bottom line sections. It also increases quantity of non-condensable going to the overhead system. The cracking can also have a detrimental effect in the curing qualities of asphalt. Too low a transfer temperature will result in lower yields of vacuum distillates and also the vacuum residue from the column becomes lighter.

7.3.2 Vacuum Column Pressure:

The column top pressure is controlled by 12-PRC-306 which recycles some non-condensables to the ejector 12-J-01 A/B/C inlet line. The top pressure should be maintained around 7-9 mm Hg absolute. Increase in pressure will result in reduced yield of vacuum



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distillates and may lead to cracking of the feed. Reduction in top pressure further may result in carryover of LVGO into the overhead system. The lower the pressure, the higher is the feed vaporization and higher is the distillates yield.

7.3.3 flash zone pressure and temperature:

It is of paramount importance to maintain a high vacuum and temperature at the flash zone with in the prescribed limit to obtain maximum yield distillates. The designed flash zone pressure is 24 mm Hg and temperature 395 °C. Fluctuation of vacuum will affect the product quality adversely besides producing mechanical stress on column internals. Increasing the flash zone temperature will result in greater yield distillates but cracking possibility is enhanced. If for any reason, vacuum starts falling sharply, firing in the heater should be reduced to bring down temperature of feed.

7.3.4 LVGO system:

The LVGO system is a combination of LVGO product, LVGO circulating reflux and internal reflux for HVGO packing. The circulating reflux is sprayed over the LVGO packing through a distributor. This stream is taken from the column and a part of LVGO after exchanging its sensible heat with crude in 11-E-07 and cooled in coolers 11-E-22/22A in parallel, it is then returned at a temperature of 65 °C to the column (12-C-01) under flow controller 12-FRC-205 to maintain column to temperature of 80 °C, Because of crude throughput maximization, it was observed that the Vac. Column temperature was frequently crossing the desired temperature ultimately resulting in under control, LVGO CR from 11-E-07 outlet is modified accordingly and routed to 12-E-12 A (top cooler), the outlet of 12-E-12 A routed to 11-E-22/22A in parallel. Increasing the reflux will reduce the top temperature which will simply increase the energy consumption. Reduction in reflux rate will increase the top temperature that will overload the ejector and increase the slop production thereby losing LVGO yield.

The internal reflux to HVGO packings from LVGO draw off tray is maintained by diverting a part of the LVGO from 12-PM-04 A/B pump discharge. This flow is regulated by 12-FRC-201. The LVGO product withdrawal rate is regulated by 12-LIC-205, which controls the LVGO level. LVGO rundown flow is indicated by F2403R. LVGO draw off temperature is 213 °C. the draw off temperature as well as LVGO product rate can be varied by increasing or decreasing either LVGO internal reflux or HVGO circulating reflux. Increase in reflux means reduction of LVGO draw off temperature and LVGO product flow rate and vice-versa.

A software switch is provided fro LVGO system when it is being routed to either HVGO/LDO. When the switch is kept in position 1, LVGO level will be controlled by its



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own LVC (LI2205). When the switch is kept in position 2, LVGO level i.e. LI2205 will be cascaded with FR2404 (i.e. LVGO to LDO/diesel).

7.3.5 HVGO Draw off:

The HVGO system is a combination of Wash liquid for wash zone packing, circulating reflux (HVGO packing) and HVGO product. Heat transfer requirement is met through HVGO pump around flow. A higher than the design pump around rate will result in overloading of packing in HVGO. A pump around flow rate lower than the requirement will result in carry-over of heavier ends to LVGO section.

If the withdrawal rate is lower than the design it will result in lower draw off temperature, reducing recovery of HVGO. Higher rate of withdrawal will result in increased tray temperature. This may lead to carry-over of asphaltenes to HVGO. The wash liquid is given to the wash zone packing for avoiding choking of the packing area because of heavy asphaltenes. The choking of the packing zone would result in higher differential pressure across this zone and this will adversely affect the column performance. About 45 m³/hr of HVGO is normally supplied as wash oil, the flow of which is regulated by 12-FRC-202. An increase in wash oil flow will reduce the carryover of asphaltenes in HVGO stream. More the desired quantity of this stream may adversely affect the quality/yield of Vacuum residue. Less quantity of Wash Oil may result in the carryover of heavy asphaltenes into HVGO stream.

7.3.6 Slop Distillate & Recycle:

The object to provide slop recycle to furnace flow is to get desired over flash so as to ensure proper recovery of distillates. A higher recycle rate will unnecessarily increase the energy consumption. Recycle + slop distillates are draw from the chimney tray below the wash zone packing. The draw-off rate of slop distillate product is regulated by 12-LIC-203 which controls the level in the draw off tray. The recycle stream flow rate is controlled by 12-FRC-102. Recycle rate at normal throughput is about 12.0 m³/hr.

7.3.7 Quench:

Quench flow is a slip of vacuum residue at 250 °C obtained after exchanger 12-E-03 and the flow is regulated by 12FRC-204. Its temperature is indicated by 12-TI-202. The purpose of providing quench is to prevent, coke formation at the bottom of the column by quickly cooling Vacuum Residue from 395 °C to 350 °C. A lower temperature than this, i.e. higher quench is not wanted because of disproportionate increase of energy loss, whereas lower quench flow may lead to coke formation due to cracking.



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DESCRIPTION OF DCS		

NOTE: Manual copy not yet provided by Instrumentation.

Standing Instruction for Improving Aesthetics of MOI Control Room : SI 30

1.0 Objective:

DCS Stations of CDU-I/II/III, FCCU-I/II, MEROX, PP-II,TPH,DHDS,DHDS-SRU are in MOI Control room. To improve and maintain MOI Control room Aesthetics, Standing Instructions have been developed.

2.0 Background:

VIP's frequently visit the MOI Control room where DCS stations of CDU-I/II/III,FCCU-I/II,MEROX,PP-II,TPH,DHDS,DHDS-SRU exist. The MOI control room needs to be clean and maintained properly. The standing Instructions have been prepared to improve and maintain the aesthetics of the MOI Control room.

3.0 Responsibility:

The overall responsibility to implement these guidelines rests with the unit Shift-in-charges at DCS stations of CDU-I/II/III, FCCU-I/II,MEROX,DHDS,DHDS-SRU,TPH,DCS –Instrumentation group. Section Head of respective locations to ensure compliance of the guidelines.

4.0 A) The following guidelines to be followed by Operations for improving the aesthetics of MOI Control room:

- a) Eatables are not allowed inside the MOI Control room.
- b) Footwear are not allowed inside the MOI Control room.
- c) One month old TOB's only are to be kept in MOI Control room. Old documents, files etc are not allowed inside.
- d) 2 chairs are to be kept at each DCS Station.
- e) The following Registers only need to be kept in MOI Control room at DCS Stations.
 - i) DCS TOB
 - ii) Summary book
 - iii) C/V Parameter data bank
 - iv) Shift Event review register



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- v) Stack Analyzer register
- vi) Trip interlock bypass register
- vii) DCS Alarms setting register
- viii) Tracking register for ensuring all control valves remain in 'AUTO' or 'Cascade'
- ix) Permits entry register

f) Important telephone nos, Trips data, Standing Instructions etc need to be kept in a file properly.

B) The following guidelines are to be followed by Maintenance-Instrumentation for improving the aesthetics of MOI Control room:

- a) Instrumentation group to ensure proper cleanliness at DCS Panels and in the entire MOI control room.
- b) While laying new cables/new fittings inside MOI control room, it should be in proper way without affecting the aesthetics of the room.
- c) For any new addition of items, review is to be carried out and permission to be taken from Division Head –Production block.

All unused/old items belonging to Instrument group are to be removed from MOI after replacement with new items.



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DESCRIPTION OF ADVANCED PROCESS CONTROLS		

DESCRIPTION OF ADVANCED PROCESS CONTROLS

In a process plant the basic operating parameters measured in the field are flow, temperature, pressure, level etc. Plant operators depend on these measured parameters for achieving control on plant performance. However, qualitative control cannot be directly achieved to the fullest potential by the operator as there is no online indication for product quality. Operator has to necessarily rely on feedback from quality control laboratory and individual operating experience. Operators with different levels of experience operate the plant differently, at times resulting in a possible loss of optimum plant performance.

Safety of plant personnel and equipment is paramount in a running plant. Yet the plant must not run under 'over safe condition' which results in the loss of certain achievable benefits. In a plant, various constraints appear for short durations and practically it is not possible for the operators to keep track of each and every constraint and take action on a real time basis. APC takes care of such a situation by taking small actions at a time and keeping strict control on various constraints to ensure they are not violated while optimizing the plant performance.

With the advent of high powered computers it has become possible to make available, critical calculated parameters using empirical equations, scientific relations and basic plant parameters on a real time basis. Calculated parameters such as Reflux ratio, Dew point margin etc., and inferential like SRN RVP, HVGO 95%, HD Flash etc., can be used directly as controlled variables for improving the plant performance.

Advanced Process Controllers maximize or minimize a given variable for attaining the defined benefits in the plant. Variables to be maximized and minimized are identified and limits are set for each variable. A limit is the maximum or minimum value of a variable beyond which it becomes unacceptable for ensuring favorable unit conditions. Predictions are made for each variable and the controller downloads the desired move to the DCS. APC control encompasses the benefits of feed maximization, better and sustained product quality, reduced energy consumption, safety and less of manual intervention from DCS operating personnel.

9.1 PROFIT CONTROLLER APPLICATIONS:

Profit controller improves production performance while promoting operational flexibility. The generic structure of Profit controller supports its application to solve the following problems in process plants:



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9.1.1 Feed maximization:

Many times process constraints can be better managed to result in higher production rates. DCS operators typically do not have time to closely monitor the symptoms leading to a constraint. Profit controller being a multivariable controller, monitors the process frequently and makes small adjustments to compensate for any changes in feed rate due to a constraint. Extent of increase in feed rate depends on how conservatively the process was originally operated.

9.1.2 Real time quality control:

Profit controller includes Profit Sensor Pro, a comprehensive easy-to-use data analysis and inferential model development application. It simplifies the task of identifying and developing relationships among process variables offering online capability that allows for real-time quality control (Lab inferential). This allows tighter product quality control without the long delays associated with lab analyses.

Accuracy of lab inferential is effectively maintained through the use of online model updation with periodic laboratory feedback information. Improvement in quality is derived from improved process stability, fewer process upsets and more consistent control across operator shifts.

9.1.3 Optimization:

Profit controller provides a perfect framework for implementation of desired optimization objectives. The built in model calculates the desired target conditions of the process and the controller moves the process to the optimal resting conditions while still controlling the process within the specified constraints. Lab inferential of products quality when included as a controlled variable in the process model allows for optimization objectives, such as feed maximization or product value optimization to occur.

9.2 APC NOMENCLATURE:

9.2.1 Controlled Variable (CV):

The process variables that must be maintained at some value or within certain limits, and should be optimized are called as CVs. CV limits are defined by the operator and can be a measured or calculated value. RMPCT controller tries to control and maintain the CVs within the operator specified limits (HI & LO).



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9.2.2 Manipulated Variable (MV):

The “handles” that are adjusted to keep the CVs within limits (or at set points), and to optimize the process. The controller always honors MV limits specified by the operator. Manipulated variables can be a regulatory controller’s set point (SP) or output (OP). With the help of MVs, RMPCT controller maintains the CV within limits.

9.2.3 Disturbance Variable (DV):

These are measured variables not under the control of RMPCT controller but have an effect on the controller’s CVs. By predicting the future effects of the DVs on the CVs, the controller can take action to prevent CV excursions before they occur. The DV’s provide information for feed-forward control. RMPCT Controller only takes the PV of the DVs and takes necessary action on the MVs so that the CVs are within limits.

Controller status (displayed on RMPCT graphics):

STATUS	DESCRIPTION
INACTIVE	Controller point is inactive. RMPCT is not tracking the CV’s, it is not executing calculations and it is not controlling the MVs.
CONTROL OK	Controller is running (without optimization)
INITIALIZING	Controller is aligning itself with the current process conditions and running integrity checks on points.
BLANK (Nothing displayed)	Controller is receiving input from the process and making prediction calculations, but there is no controller output.
SHEDDING CONNECTIONS	Controller has been shut off and is shedding control to the configured regulatory PID loops.
WAIT FOR EXECUTION	Controller is waiting for background calculations to complete. Background calculations are lower priority calculations.
HANDLING CONSTRAINTS	The controller is indicating that it has encountered competing control objectives and that bringing all variables within the constraints cannot be managed at the current execution or cannot be managed within the prescribed horizon. Caution: Competing constraints are usually temporary conditions, provided that operating conditions have not changed since the controller was tuned.
OPTIMIZING	Controller is running and the control objectives at the current execution have been met. Controller is now attempting to optimize the process based upon economic objectives.



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Understanding CV summary (displayed on RMPCT graphics)

In the controller graphic, when the **CV summary** is clicked on the graphics, certain CV related fields appear whose significance is as follows:

Field	Field type	Default Access	Description	User notes
CV #	Read Only	No Access	The index of the variable	
CV Description	Read Only	Operator	Description that characterizes the variable's function or location	Click the field to display the CV detail display.
Status	Editable	Operator	Status of this variable GOOD, DROP, PRED, CRIT, INIT, WDUP or INAC.	Selecting the status allows you to drop a CV under control or take dropped CV under control. The controlled can automatically drop a CV. Dropping a CV in DROP status means the CV will not be used once the condition causing DROP state clears.
Value	Read Only	No Access	The value of the CV at the current execution. This can be the measured process value or a predicted value	Dash displays when the source value is bad or unavailable for longer than the number of bad values allowed. Values display in color: Cyan= Displayed value within the limits Yellow= Displayed value near the limits Red = Displayed value violating the limits
Future	Read Only	No Access	Displays the future predicted value after three measurement cycles plus delay	Values display in color: Cyan= Displayed value within the limits Yellow= Displayed value near the limits Red = Displayed value violating the limits
SS value	-	-	Predicted steady state value.	
Lo limit	Editable	Operator	Lowest end of operating range.	Green up arrow displays when CV is ramped up. ◆◇-Solid or hollow yellow diamonds display when hard or soft limit constraint is reached for a CV.
High limit	Editable	Operator	Highest end of operating range.	Green down arrow displays when CV is ramped down.



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				◆◇-Solid or hollow yellow diamonds display when hard or soft limit constraint is reached for a CV.
Set point	Editable	Operator	The operating field of the CV	Click in set point column to make entry. Setting same Hi and low limits also establishes a set point. Value is grey when set point is in use.

Understanding MV summary (displayed on RMPCT graphics)

In the controller graphic, when the **MV summary** is clicked on the graphics, certain MV related fields appear whose significance is as follows:

Field	Field type	Default Access	Description	User notes
MV #	Read Only	No Access	The index of the variable	
MV Description	Read Only	OPER	Description that characterizes variable's function or location	Click the field to call the MV detail display.
Status	Read Only	No Access	Status ON, FFWD, HIGH, LOW, SERV, INIT, INAC.	Whether the MV is dropped or used as the feed forward value is set on the MV process tuning display or on the MV detail display
Value	Read Only	No Access	Value of the MV at current execution.	Value shown is the OP or SP when status is ON. Otherwise it is the PV.
Move	Read Only	No Access	The size and direction of change made to the immediate downstream controller at the last execution	
Future	Read Only	No Access	Displays the future predicted value after three measurement cycles plus delay	Values display in color: Cyan=Displayed value within limit Yellow=Displayed value near limit Red=Displayed value violating limit
SS value	Read Only	No Access	Predicted steady state value.	
Lo limit	Editable	OPER	Lowest end of operating	▲Arrow displays when MV is



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			range.	ramped up. ◆◇-Solid or hollow yellow diamonds display when hard or soft limit constraint is reached for a MV.
High limit	Editable	OPER	Highest end of operating range.	▼Arrow displays when CV is ramped up.
Mode	Editable	OPER	Control Mode: Either RMPCT or Operator	In operator mode, profit controller stops outputting changes to the downstream controller. Field is locked if the MV is critical. Critical or non critical is set on MV tuning display or MV detail display.
Critical	Read Only	No Access	This column displays a "C" if the value is critical	

Understanding DV summary (displayed on RMPCT graphics)

In the controller graphic, when the **DV summary** is clicked on the graphics, certain DV related fields appear whose significance is as follows:

Field	Field type	Default Access	Description	User notes
DV #	Read Only	No Access	The index of the variable	
DV Description	Read Only	OPER	Description that characterizes variable's function or location	Click the field to call the MV detail display.
RMPCT Status	Editable	OPER	Status GOOD, CRITIC, DROP or INAC	A critical DV has -C on its suffix. The controller shuts down (OFF) when a critical DV goes bad. Selecting the status allows to drop a DV in use or put in use a dropped DV. Critical DV's cannot be dropped. Non-critical DV's can be dropped if they go bad



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9.3 CONTROL OBJECTIVES AND STRATEGIES

The advanced process control applications are designed to reduce variations of key unit operating variables through model based predictive control, which includes feed forward disturbance rejection. The prime objective of increasing overall unit profitability is achieved by utilizing RMPCT in conjunction with supporting calculations and intermediate regulatory controls. Supporting calculations are provided to supplement existing process measurements. The intermediate regulatory controls provide a more stable foundation for implementation of the multivariable control applications.

9.3.1 General Control Objectives:

- Provide stable operation of entire unit
- Maximize unit throughput
- Maintain product quality
- Increase run length of equipments by maintaining its guiding parameters within limits
- Minimize energy & utility consumption (Fuel gas, Fuel oil and steam)
- Maximize the yield of valuable products
- Avoid extra safe operations and operate close to the safer limits.

9.3.2 Overall Control Strategy:

Based on the level of interaction among the variables and their respective settling time, four different RMPCT controllers have been configured for the entire CDU-II unit. The variables for the RMPCT controllers have been chosen to achieve the control and LP optimization objectives as mentioned in the respective sections. The four controllers are as follows:

1. Atmospheric Section Controller- ADU2
2. Vacuum Section Controller- VDU2
3. Naphtha Stabilizer Controller- NSU2

9.3.2.1 **ADU2 Control Strategies:**

Atmospheric section controller has a total of 30 CVs, 21 MVs and 1 DV. The variables for the Atmospheric section controller have been chosen to achieve the following optimization strategies:



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- **Atmos Heater Pass balancing:**

The regulatory level control of the individual passes in cascade with the master flow controller is broken and the individual pass flows are manipulated through the RMPCT controller for achieving pass outlet temps balancing even while maximizing thruput.

- **Throughput Maximization:**

Total feed will be maximized subject to Heater Skin temperatures and desalter pressure control valve opening. Ex-PFD crude flow FR1804.PV will be used as the tag for maximization.

- **SRN Maximization:**

Un-stabilized Naphtha flow to Stabilizer will be maximized, subject to the SRN 95%.

- **Heavy Naphtha Maximization:**

Heavy Naphtha flow will be maximized, subject to Kero / ATF Flash point.

- **Kero / ATF Maximization:**

Kerosene will be maximized, subject to draw temp and Kero/ATF FBP (not a direct CV).

- **HD Maximization:**

The HD rundown will be maximized, subject to 95% Point and minimum over flash flow.

9.3.2.2 VDU2 Control Strategies:

Vacuum section controller has a total of 18 CVs, 9 MVs and 1 DV. VDU2 controller has the following optimization strategies:

- **Vacuum Heater Pass balancing:**

The regulatory level control of the individual passes in cascade with the Atmos column bottom level controller is broken and the individual pass flows are manipulated through the RMPCT controller for achieving pass outlet temps balancing while maintaining atmos column bottom level.

- **Vacuum column top temperature Minimization:**

Vacuum column top temperature will be minimized by regulating LVGO CR flow (FR2205).

- **HVGO Maximization**

HVGO will be maximized subject to HVGO 95% by regulating HVGO IR & CR flow (FR2202 / 3).



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9.3.3.3 NSU2 Control Strategies:

The NSU2 controller has a total of 3 CVs, 3 MVs and 1 DV. NSU2 controller optimization strategies include the following:

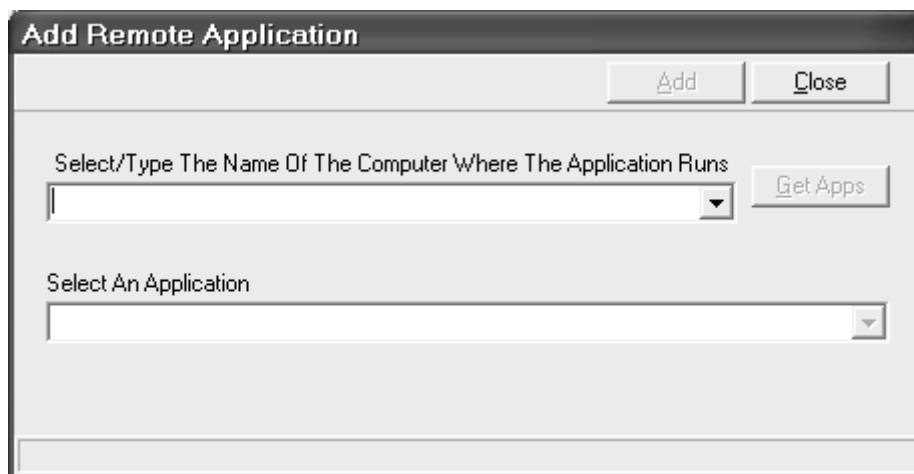
- **LPG Maximization:**

LPG will be maximized subjected to its C5 specification based on field weathering and Stabilizer top PCT while maintaining control on SRN RVP.

9.4 APC START UP, SHUTDOWN & HANDLING PROCEDURE:

9.4.1 Putting the APC Controller ON:

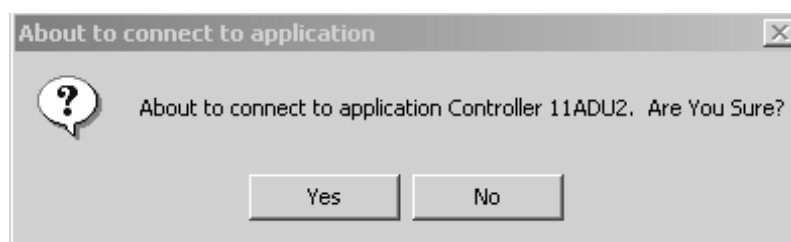
- Switch on the power to the Profit Viewer machine beside DCS. To login use User name “**administrator**” and password for this user is “**password**” and press **Enter**.
- In the Profit Viewer machine open the ‘**Profit Viewer**’ Application, i.e. go to **Start>Programs>Honeywell HiSpec Solution>Profit Viewer** (One can also use the Shortcut of the same available in the ‘**Profit**’ folder on the Desktop)
- Ensure that the Controller Application is present in that window and its status is “**ACTIVE**” (Controller status is there at the right end of the Profit Viewer Screen).
- If the Controller Application is not present, then click the drop down button in the Program Applications & select **Add Profit Viewer Remote Application**. You will see a window like this:





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- Click on the drop down button 'Select / Type The Name of the Computer where the Application Runs' and select the 'CDU2APCSERVER'.
- Then select the Controller (e.g. ADU2, VDU2 & NSU2) & then click **Add**, repeat this for rest of the controllers to be able to view all in the Profit Viewer Window.
- After the Controller Applications are added in the Profit Viewer machine, select one of the controller (say ADU2) & click '**View**', you will see a face plate like this:



- Click **Yes** to view this Application .

If the Controller Application is present but its status is displayed as '**Inactive**' then contact APC Engineer. Controller can be placed in service only after it is made '**Active**' by the APC engineer.

- Click **WARM**. This will change colour of **WARM** button to yellow & a face plate will come. Click **Yes** to put the controller in **WARM** mode.

Why to put in WARM mode?

In WARM mode controller works like a simulated controller and the limits of all MVs & CVs can be checked and set right before putting the controller 'ON'. In Warm mode, move size and steady state values can also be checked. Then each MV, CV, DV (one by one) can be placed in RMPCT mode or DROP mode as desired by the operator.

In Warm mode, it is recommended to check whether controller is giving proper 'Moves' or not and the future / steady state value is desirable or not.

- After one minute the colour will change to cyan & the status appears as "**WARM**" disclosing the SS Value (Steady State Value) and Moves.
- Set carefully HI & LO Limits for each CV & MV and then observe the MV Moves.
- Put the MVs (which show permissible moves) in **RMPC** mode from **OPR** mode (in the extreme right hand side column). When MV1 is put on RMPC mode, following face plate will appear:





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Click **Yes**; you will see the RMPC mode getting selected on the right hand side against MV1.

Note: At least one MV is required to be in **RMPC** mode before putting the total controller in **ON** Mode. Otherwise the controller gets put off automatically.

- i. Put the controller 'ON' (after carefully examining that all CVs & MVs high and low limits are set properly, the 'MOVE' size as indicated by the controller against each MV is acceptable and taking the CVs & MVs online).
- j. Confirm that for all MVs, which have been taken into RMPCT, the status has become ON in Profit Viewer and have gone to SPC mode in DCS.



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SHED MODES (required mode for placing in RMPCT):

- ADU2 – MV SHED MODEs:

No	Tag Number	Descriptions	SHED MODE
1.	FR1101.SV	PREHEAT TRAIN SPLIT FLOW	AUTO
2.	FR1301.SV	11F01 PASS-A FLOW	AUTO
3.	FR1302.SV	11F01 PASS-B FLOW	AUTO
4.	FR1303.SV	11F01 PASS-C FLOW	AUTO
5.	FR1304.SV	11F01 PASS-D FLOW	AUTO
6.	TR1301.SV	11F01 COT	AUTO
7.	TR1403.SV	ATMOS COLUMN TOP TEMP	AUTO
8.	PR1409.SV	ATMOS COLUMN TOP PRESSURE	AUTO
9.	FR1406.SV	TOP PUMP AROUND FLOW	AUTO
10.	FR1405.SV	KERO CIRCULATING REFLUX FLOW	AUTO
11.	FR1404.SV	DIESEL CIRCULATING REFLUX FLOW	AUTO
12.	FR1103.SV	HN PRODUCT RUNDOWN TO DIESEL	AUTO
13.	FR1107.SV	HN PRODUCT RUNDOWN TO SRN	AUTO
14.	FR1203.SV	KERO PRODUCT RUNDOWN	AUTO
15.	FR1202.SV	HEAVY DIESEL PRODUCT RUNDOWN	AUTO
16.	FR3501.SV	HEAVY DIESEL TO DHDS	AUTO
17.	FI1409.SV	HN STRIPPING STEAM	AUTO
18.	FR1408.SV	KERO STRIPPING STEAM	AUTO
19.	FR1407.SV	HD STRIPPING STEAM	AUTO
20.	FR1401.SV	ATMOS COLUMN BOTTOM STRIPPING	AUTO
21.	FR1204.SV	KERO TO DIESEL RUNDOWN	AUTO



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VDU2 – MV SHED MODES:

No	Tag Number	Descriptions	SHED MODE
1.	FR2101.SV	12F01 PASS-A FLOW	AUTO
2.	FR2102.SV	12F01 PASS-B FLOW	AUTO
3.	FR2103.SV	12F01 PASS-C FLOW	AUTO
4.	FR2104.SV	12F01 PASS-D FLOW	AUTO
5.	TR2133.SV	12F01 COT	AUTO
6.	FR2201.SV	LVGO IR FLOW	AUTO
7.	FR2202.SV	WASH OIL FLOW	AUTO
8.	FR2205.SV	LVGO CIRCULATING REFLUX FLOW	AUTO
9.	FR2203.SV	HVGO CIRCULATING REFLUX FLOW	AUTO

NSU2 – MV SHED MODES:

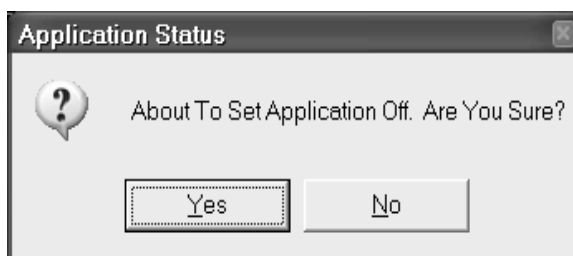
No	Tag Number	Descriptions	SHED MODE
1.	PR1501.SV	STABILISER TOP PRESSURE	AUTO
2.	FR1501.SV	STABILISER REFLUX FLOW	AUTO
3.	TR1501.SV	STABILISER 3RD TRAY TEMP	AUTO

Note: Before taking any MV on RMPCT mode, at the DCS level the same need to be in the correct (SHED) mode of operation (either Manual / Auto / Cascade mode). If any MV is not in its pre-defined shed mode while taking in RMPCT mode, the MV's status in the controller graphics would appear as 'INIT' (initialization).

9.4.2 Putting the APC Controller OFF: Any controller can be put 'OFF' from the 'Profit Viewer Graphics' as well as from DCS control panel.

- From Profit Viewer Graphics:

a) Click on the **OFF** tab (at top row). A face plate will appear:





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b) Click “Yes” to switch the controller **OFF**. It will take few seconds to get switched off.

- *From DCS (using Master Switch):*

In case of any communication failure between APC server and client, a master switch is configured in the DCS to put OFF individual APC controllers as well as all 3 together. Under normal circumstances, it is always recommended to switch off APC controllers from APC client PC. However, this master switch can be used to put OFF all the three controllers at a time (like in case of emergency or any process upset when Operator wants to take control in his hands).

- 1) The Master switch schematic name in DCS is **GR0293**.
- 2) To switch OFF the controllers, go to the above schematic page & click on fields as required by the DCS supervisor.

DCS schematic page showing the master switch is given below.



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APC SCHEMATIC IN DCS GRAPHIC PAGE GR0293:

HPCL	CDU – II	RMPCT
CONTROLLER	DESCRIPTION	SWITCH
ADU2	ATMOS SECTION CONTROLLER	 Click here
VDU2	VACUUM SECTION CONTROLLER	 Click here
NSU2	NAPH STABILISER CONTROLLER	 Click here
		MASTER SWITCH
TURN OFF ALL THREE CONTROLLERS		 
CRUDE SWITCH		
	NEW LO LIMIT	NEW HI LIMIT
11F01 FEED	440	450
11F01 COT	360	362
	START / STOP	
	RUN TIME (MIN)	
	0	

9.4.3 Crude Switch Program:

HPCL CDU – II handles Crude Switch once in 2-3 days. During this, operator has to reduce the feed and then switch tanks. Whenever crude switch is taking place from Low Sulfur type to High Sulfur type and vice versa, main operating parameters which gets changed are the feed rate and 11F1 COT. In order to facilitate automatic change / shift in these important operating parameters, an APC level program has been developed. This Crude Switch utility (program name is CR_SWITCH) will help the DCS Panel operators in changing the Feed and 11F1 COT in a timely and smooth manner. DCS graphic (GR0293) is prepared to help panel operators to change the limits according to requirement.



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- Handling the crude switch program:

Procedure:

- First check the new limits (Low and High) one wants the Feed and 11F1 COT set values to go in the DCS graphic page **GR0293**. If required, change the limits, by clicking on the targets provided.
- Prior to feed tank switch, to reduce the feed rate, start the program exactly 60 minutes ahead of tank switch. Program will reduce the feed rate gradually within that period keeping atmospheric column operation smooth.
- COT will start changing after a time lag of 10 minutes (after the crude switch program was made active) and will be changed gradually to the new limit in the next 50 minutes by the program.
- Once the plant is stabilized at the lower feed rate, feed rate shall be raised upto the potential available, either through APC in the normal way or by Crude Switch program once again by setting proper target values as explained above
- If the panel operator wants to intervene in-between (in case of any process upset during the crude switch program run) he may do so by stopping the program.
- Keep wide limits for all MVs in controllers, as controller has to take action accordingly to maintain column profile as much as possible.
- Specially, pass flows need to be made wide in controller limits all the time, as 11F1 pass flows control the Feed and 12F1 pass flows are controlling Atmospheric column bottom level.
- Don't change any of the limits of Feed and 11F1 COT when this program is active. If any of these limits are changed in between, program will take it as an OFF request and set itself off.
- Don't set the program active for less than 30 M3/hr feed change, as program will take 60 minutes irrespective of the amount of change. This level of change can very well be done from controller limits directly.



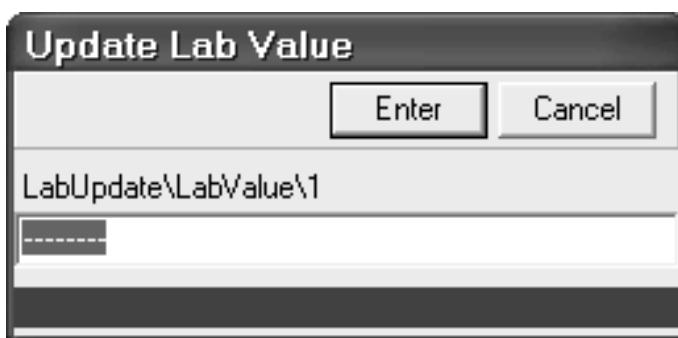
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UNDERSTANDING MANUAL LAB-UPDATE:

Lab-Update has to be done for validating the inferential. This can improve the quality of future predictions.

9.5.1 How to do Lab-Update?

- In the Profit Viewer machine open the Profit Viewer Application and then open the “Lab update”, i.e. go to **Start>Programs>Honeywell HiSpec Solution>Lab Update** (One can use the Shortcut of the same available in the Desktop in the Profit folder).
- Type “**CDU2INF**” in the “Connect to Application” field & click **Connect**.
- Enter the Date and time of actual lab sampling for each individual inferential.
- Click the corresponding cell under **Lab Value** and a face plate will appear. Enter the lab reported value correctly & click Enter.



- After few seconds, it updates and returns with **Average at Sampling Time** (inferential predicted value) and **Calculated Bias**, press Enter.
- After this again a Face plate will come & will show the **Delta Bias**. Click Enter to update the bias.

Notes:

- For lab bias updation it is important to enter exact sample collection time (Do not round off) and correct lab value.



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- If Operator feels LAB value is not correct then better not to enter. Wrong data entry / unreliable values will eventually drift the calculation from the actual and future predictions will deteriorate affecting the controller's performance. If by mistake wrong value was entered, do not update the bias.
- vii. For continued better prediction it is advisable to do the lab update in the order of time. For example, Nov 28th 1100 hrs sample should be entered first, Nov 28th 2300 hrs sample second, Nov 29th 1100 hrs sample third etc.

9.5.2 Automatic Lab-Update:

Timely lab updation helps in improving the accuracy level of inferential predictions which are used as Controlled variables in APC for meeting the desired objectives. To improve the inferential predictions, automatic lab updation feature has been enabled through an APC level program. Every 5 minutes, CDU-II APC server checks for any new value being entered in LIMS by QC Lab crew and automatically updates the corresponding inferential.

The normally expected values (low and high) for each of the product property have been defined and values in violation of these limits will not be updated. Similarly non-numeric values will not be considered for updation.

9.6 PROCESS PARAMETERS TRENDING:

- i. **Trending & plotting-** There are various parameters related to MV/CV/DV/Controller being historized in the Embedded PHD archive files. To plot a particular MV/CV/DV/Controller parameter one has to open Process Trend Application (Start>Programs>Honeywell Uniformance Desktop> Process Trend).
- ii. For trending in the client machine it will ask for, Server name. The server name in the client machine is IP address "**10.6.0.92**".
- iii. To add the Tag or plot definitions, press F12 or go to Plot> add/modify Tag or plot definition. Now importing of the Tags for trending can be done from tag explorer or by manual entry. Maximum eight parameters can be plotted in a file, the file can be saved with the ".plt" extension.
- iv. Once a ".plt" file is saved, all its settings (Tag name, HI, LO limits Plot format) will remain as it is, which can be used at any moment.
- v. To change the plot definition or to add / remove tags, press F12 or by right click and select appropriate option. The plots characteristics can be customized by pressing F9 key or by right click. Hair line can also be introduced by pressing F11 key or by right click.



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9.7 OPERATING INSTRUCTIONS:

9.7.1 General:

- Before making a controller ON check whether the controller is Active or not.
- Check HI / LO Limits of all MVs & CVs. It is recommended to put MV limits relaxed and CV limits as per specifications & requirements.
- It is always recommended to make the controller WARM before making it ON.
- For taking any MV in RMPCT mode, ensure that it is in desired DCS (Shed) mode of operation (Auto / Manual / Cascade). When any of the four controllers is put OFF by an operator, all the MVs in that controller will go back to their normal DCS mode (Shed Mode) of operation.
- At least one MV has to be in RMPC mode to make the controller ON. If not the controller will automatically get switched OFF and all the MVs in that controller will go back to their normal DCS mode (Shed Mode).
- Value of any parameter on the RMPCT graphic page in CYAN color means that parameter is within the low & high limit (Normal).
- Value of any parameter on the RMPCT graphic page in YELLOW color means that parameter is touching the low or high limit including its soft/ hard limits.
- Value of any parameter on the RMPCT graphic page in RED color means that parameter is violating the set low or high limit.
- In the event of failure of any instrument or requirement for checking any instrument:
 - ❖ Check whether that instrument is contributing (directly or indirectly) to any MV / CV / DV in any of the controller (ADU2 / VDU2 / NSU2). This can be checked from the 'Gain Delay' graphics page or MV / CV/ DV tags list.
 - ❖ Check whether that instrument is contributing (directly or indirectly) to any calculated or inferential variable. This can be checked from the list of input tags for calculated & inferential variables (chapter: 6.0).
 - ❖ Accordingly, all the related MV, CV, DV, Inferential & Calculated tag need to be taken out from RMPCT mode. After restoration of normalcy of the instrument, the concerned variables can be placed in RMPCT mode again.
- If any of the CV including calculated tag & inferential tags and DV value goes bad (No value; -----) the controller drops that CV / DV automatically. If any of the source tag becomes bad, the calculated / inferential value also goes bad and gets dropped automatically. However if any of the CV or DV shows erratic / random values, then it



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does not get dropped automatically and it is advisable to drop that CV / DV manually immediately.

- When a MV is dropped from RMPC mode, operator has to decide whether he wants to put that MV in Service mode or in Feed forward mode. Feed forward mode is useful for prediction purpose provided the indication is healthy, whereas keeping in Service mode will not give any information of that MV to the controller for any calculation. However for safety reasons the default mode for "When MV In Manual" option has been configured as "DROP" to avoid unnecessary moves (when the MV is checked by Instrumentation without putting in drop mode) based on FFWD.
- During communication failure or to put off all three APC controllers at a time use the Master Switch configured in the DCS. Operator can call the schematic **GR0293** to access the master switch.
- When any of the three APC controllers is Switched OFF due to any reason, an alarm will come on the DCS panel saying the respective APC controller is switched OFF.
- Never put the APC controllers ON from the DCS Schematic page. This is to avoid possible upset / disturbance in the unit. Instead operator must check the MV / CV limits on the profit viewer before putting controller ON.
- Periodically check for automatic lab updation of inferential. In case of discrepancy, inform Technical-APC group.
- If any of the CV has status as WDUP, it means that the CV cannot be controlled within the set limits as its MV's set low / high values are limiting.
- If any of the MV shows status as HIGH or LOW, check its output at the DCS level for locks if any & take proper actions to bring its output into the floating range.
- When any MV is taken into APC, it will go to **SPC** mode in DCS. There will not be any mode change for CVs & DVs.
- If the communication between APC machine and DCS gets broken, all MVs in the RMPCT controllers will go back to their Shed Mode.



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9.7.2 Controller Specific:

- *ADU2 Controller:*
 - a) For taking feed maximization and pass temp balancing strategies under APC control, ensure that FR1804 is in manual and pass flows are in Auto. Under APC control, furnace pass flows (in SPC mode at DCS) will be manipulated for achieving these objectives. When APC controller is switched off, pass flows will go to AUTO (shed) mode and FR1804 will remain in manual. DCS supervisor to ensure that FR1804 is taken back into Auto with pass flows in Cascade mode (normal regulatory control).
 - b) Product flows in each rundown line are considered as individual MV. Make sure that Product lines, which are not in operation are kept out of APC (i.e., MVs are made OFF). e.g., when only Kero rundown, FR1203, is in line, then keep the other MV FR1204 (Kero to Diesel) OFF from APC Controller and set proper limits for FR1203. In general, if there are more than one r/d MV for the same product, ensure that the one that can be manipulated is in APC. Keep the other MV in Auto.
 - c) In the controller, along with product inferential like 95% points, Draw Temperatures are also considered as CVs. Whenever inferential is predicting better, make limits for draw temperature relaxed, so that these CVs does not have any conflict with the inferential limits.
 - d) If 95% inferential value is showing bad or wrong value, then make that CV off and make draw temperature limits tight. One can control the product quality with draw temperature in such a situation.
 - e) Dew point inferential is based on empirical equation and hence it is not always perfect. Keep limits considering 4-5 degree positive offset.
 - f) If any of the skin (or Arch) temperatures are showing bad or wrong values, the maximum skin (or Arch) temp CV shall be kept dropped and inform APC group. The particular tag will be dropped from the calculation and the CV can be taken back inline. The maximum skin (or Arch) temperature will be calculated based on remaining indications. Make sure, that the tag is taken back inline by APC group when the indication is attended and value returned to normal range.



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- *VDU2Controller:*

- For taking pass temp balancing strategy under APC control, ensure that LR1401 is in manual and pass flows are in Auto. Under APC control, furnace pass flows (in SPC mode at DCS) will be manipulated for achieving this objective. When APC controller is switched off, pass flows will go to AUTO (shed) mode and LR1401 will remain in manual. DCS supervisor to ensure that LR1401 is taken back into Auto with pass flows in Cascade mode (normal regulatory control) and Atmospheric column bottom level under control.
- In the controller, along with product inferential like 95% points, Draw Temperatures are also considered as CVs. Whenever inferential is predicting better, make limits for draw temperature relaxed, so that these CVs does not have any conflict with the inferential limits.
- If 95% inferential value is showing bad or wrong value, then make that CV off and make draw temperature limits tight. One can control the product quality with draw temperature in such a situation.
- If any of the skin (or Arch) temperatures are showing bad or wrong values, the maximum skin (or Arch) temp CV shall be kept dropped and inform APC group. The particular tag will be dropped from the calculation and the CV can be taken back inline. The maximum skin (or Arch) temperature will be calculated based on remaining indications. Make sure, that the tag is taken back inline by APC group when the indication is attended and value returned to normal range.
- As vacuum column is normally faced with instrumentation problems because of viscous fluid, get the maintenance and instrument rectification on time for better and reliable APC

- *NSU2 Controller:*

- In the column, both Top Temperature and bottom temperature are controlled. For column stability and LPG yield / quality, keep the bottom temperature relaxed and control top parameters tight.

9.8 APC MONITORING & MANAGEMENT:

9.8.1 Role of DCS supervisor :

- Strictly follow the APC startup, shutdown and operating instructions given in Section 7.0 & 8.0.



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2. When APC is **ON**, the RMPCT controllers will be downloading the set point for MVs to DCS in order to keep the CVs within limits. DCS supervisor needs to monitor the condition of all variables in all the controllers and functionality of RMPCT. Following the color coding (Cyan for normal; Yellow for operation on limits & red for operation beyond limits) will make it easier for real time monitoring.
3. Check for periodic automatic lab updates of inferentials and if it is not done, do it manually.
4. The profit viewer computer kept at the DCS is directly used for controlling the plant operation. From system security point of view following points need to be ensured:
 - The computer should not be used for purposes other than APC.
 - No other software should be downloaded on to this computer.
 - Floppy, CD and USB drives should not be used.

Any disturbance / virus attack in this computer would make the APC unavailable.
5. To ensure proper flow of information with respect to functioning of RMPCT controller on round the clock basis, relevant observations need to be mentioned in shift TOB.

Role of APC group:

1. Technical APC group shall oversee the service factor of all APC controllers and circulate performance reports for individual RMPCT controllers on a daily and monthly basis and carry out trouble shooting as and when required.

NOTE: Assets related to APC like server, Profit viewer computer and associated accessories will be monitored and managed by Technical-APC group.



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PRE-COMMISSIONING ACTIVITIES

There is a large amount of preparatory work which should be performed by the operating crew before starting of the unit after completion of mechanical jobs. A planned check of the unit will not only set the foundation of a smooth start-up but will also provide a firm basis for acquainting operators with the equipment. Start-up is a critical period and the operators must know the operation of equipments.

Some of the pre-commissioning works can be carried out simultaneously along with construction. But care in the organization of this work is necessary so that it will not interfere with construction work. It is most important to plan schedule and record with check-lists and test schedules all the preliminary operation and to co-ordinate the construction program.

10.1 INSPECTION:

Inspection of the following items is primarily contractors' responsibility. Nevertheless, it is important for operating crew to participate in the inspection to the fullest extent since for many of them this may be the only opportunity before start-up to examine the internals of some of the equipment.

10.1.1 Inspection of Vessels and Columns:

Inspection of the interior of the vessels, columns, heaters and other equipments that are not normally accessible during operation will be made to ensure that they are completed clean and correctly installed.

10.1.2 Piping and Accessories:

Piping and accessories will be checked against drawings and specifications. Piping support and hangers will be inspected to ensure that all anchorages are firm. Valves will be checked for proper packing and mounting. Spring supports are to be checked for the cold settings and later for hot settings while the plant is in operation.

10.1.3 Instruments:

Instruments will be checked, starting from the controller and proceeding logically through the control loop. Cascade control system will be checked from the impulse point of primary loop. Operating crew should check proper mounting of control valves.



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10.1.4 Relief Valves:

Relief valves will be set in the shop and mounted before the system pressure test. Block valves ahead and after relief valves will be checked for lock open or lock close position as per P&ID. Relief valves will be checked against specifications. Rupture discs should not be mounted during tightness test.

10.2 PREPARATION OF UNIT:

- Check the unit for completion of mechanical work against P&ID.
- Remove all construction debris lying around in the unit and clean up the area.
- Install blinds as per master blind list.
- Safety valves should be kept blinded during flushing and re-installed afterwards. These should be shop tested and set at the stipulated values.
- Ensure that underground sewerage system is in working condition. Clear plugging, if any. Check by flushing with water.
- Check that communication between units, control room, offsites and utilities are complete and in working condition.
- Ensure that the required lube oil, grease and other consumable are available in the unit.

10.3 COMMISSIONING OF UTILITIES:

Commissioning of various utilities in the DC unit is described below. Prior to commissioning of each utility system, the utility section should be informed about it and the load which is expected.

10.3.1 Steam Network:

Network shall be blown through completely from battery limit with a strong steam flow in order to clean the lines. The following steps are recommended.

- Check network, all equipment will be disconnected to avoid entry of flushed material.
- Drain all the low points. If necessary open steam trap inlet flanges.
- Open slowly battery limit valve and let the temperature rise in the header, slowly and steadily.
- Check support of fixed points and expansion loops.



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- When line is hot blow it through completely with a strong steam flow.
- Close battery limit valve and prepare another network. When the blowing is satisfactory, reconnect all equipment and remount steam traps. Recharge header as above.

Note: The following precautions are to be taken while blowing/commissioning steam header:

- To drain the low points of the lines before and during heating period in order to avoid water accumulation, which leads to hammering.
- To open drain/vent during cooling period to prevent vacuum formation
- To isolate the instruments, remove orifice plates and control valves; to re-install the orifice plates and control valves blowing is over.

10.3.2 Cooling Water and Service Water:

Network shall be cleaned from battery limit with a strong water flow. All equipment will be disconnected at the inlet and reconnected when lines are cleaned. Control valves and orifice plates will be removed and re-installed after the lines become clean. When system has been flushed charge the lines to the operating pressure.

The following precautions to be taken:

- To open vents at high points in order to expel air from equipment and piping
- To open the battery limit valve, slowly and steadily.

10.3.3 Boiler Feed Water:

Flanges are to be opened at the inlet of the connected equipment and at any convenient low point. Network shall be first cleaned with raw water from battery limit with a strong water flow. The net work shall then be drained and rinsed with boiler feed water. Open flanges will be made up and header charged after the network is clean. Usual precautions for instruments like isolation, removal etc. are to be taken as mentioned above in case of steam network.

10.3.4 Instrument and Plant Air:

Network shall be blown through completely from battery limit with strong flow of air in order to clean and dry the lines. All joints and connection shall be checked for tightness with soap solution. Header and branch lines will be blown through with a high flow rate of air. During all tests, the instruments and control valve shall be carefully isolated from the system.

10.3.5 Fuel Gas and Fuel Oil Networks:



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Networks shall be blown through from battery limit with a strong steam flow for fuel oil and a strong air flow for fuel gas in order to clean the lines. During this operation, orifice plates and control valves shall be removed. Special care shall be taken to prevent water from entering the furnace. The fuel gas headers will be commissioned before firing the Coker furnace.

10.4 WATER FLUSHING OF PROCESS LINES:

To clean scales and foreign matters etc. from inside, lines and equipment are flushed with water wherever possible. Temporary water connections should be provided at convenient locations in the system for carrying out water flushing. The following points should be remembered during water flushing.

- Low point drains and high point vents should be flushed.
- All instruments connection should be isolated, orifice plates removed. Control valves isolated and by-passed. In case there is no bypass, remove control valve and flush the line. The valve will be installed after clean water starts coming out and further flushing may be continued.
- If there is any heat exchanger in the line flushing should be done upto and around the exchanger using by-pass line. It should be ensured that dirty water from initial flushing does not get into the exchanger. Wherever by-passes are not available, the flanged joints at the inlet of heat exchanger should be first opened and the line flushed till clear water starts coming out. Then reconnect flange and flush through the exchanger.
- At each opening of the flanged joints, a thin metallic sheet should be inserted to prevent dirty water from entering the equipment or piping.
- The flow of water should preferably be from top to bottom for flushing of heat exchanger coolers. The bottom flange of the equipment should be opened to permit proper flushing.
- The flushing should be carried out with maximum possible flow of water till clear water starts coming out
- Vertical lines which are long and rather big (say over 100 mm dia.) should preferably be flushed from top to bottom. This will ensure better flushing. Filling the lines and releasing from bottom is also helpful. The rundown lines can also be flushed conveniently to the respective tanks.
- It should be ensured in all flushing operation that design pressure of lines and equipment are never exceeded. After flushing of lines and equipment, water should be thoroughly drained from all low points. Lines and equipment containing pockets of



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water should not be left idle for a long time. It is preferable to dry these lines and equipment with air after water flushing.

- For flushing of stainless steel lines and equipments DM water shall be used.

10.5 FUNCTIONING TEST:

All rotary equipment (including dosing pumps) will undergo functional test to check their performance.

10.5.1 Motors:

Each motor should be checked and started to ensure that it has the correct direction of rotation. The motor speed should be checked with tachometer to ensure that RPM is correct. The manufacturer's lubrication schedule should be used to ensure that all lubrication points have been serviced. After a short run each bearing should be felt to ensure that it is free and not overheated.

10.5.2 Pumps:

Generally the pump casing is opened and checked to ensure that it contains no foreign material. Pump casing need not be opened up in case the pump had been stored carefully and the blinds covering suction and discharge nozzles were not removed in storage or in transit. Cleanliness of suction line and installation of line mesh strainer should be ensured. Pumps and motor will be aligned and then tried on water. Temporary connections may have to made, if required. When running a pump designed for hydrocarbons on water the discharge valve may have to be throttled so that the rated amperage is not exceeded. Ensure that the vents and drains of the pumps are clear. Pressure gauge tapping will be flushed and filled with sealing fluid wherever necessary.

10.5.3 Turbines:

Turbines should be de-coupled and run using steam as per manufacturer's instructions.

10.5.4 Compressors:

It is important to make sure that the inside of piping (especially suction piping) around the compressor has been cleaned. The compressor cylinder should have no stress of piping. The lube oil lines and compressor suction lines downstream of the suction filter should be acid



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cleaned and pickled. Vendor's instructions may be followed in this regard. A fine mesh strainer (about 100 mesh) should be inserted in the suction line. Lube oil should be used as per manufacturer's recommendation. The compressor should be run for a few hours following the manufacturer's instructions regarding media and mode of trial operation. A close watch should be kept on critical parameters like bearing temperature, discharge pressure and temperature, lube oil temperature and pressure, etc. When the compressor is found to function satisfactorily, it should be connected to the rest of the system.

Note: Vendor's instructions shall be followed as far as the testing medium for pumps and compressors is concerned. In general, casing design pressures and rated motor amperes shall not be exceeded.

10.6 CALIBRATION OF INSTRUMENTS:

The following guidelines may be adopted for checking and calibration of all instruments.

10.6.1 Orifice Plates:

Before each orifice plate is installed the orifice taps should be blown clear. The orifice plate should be callipered to check, if the correct size orifice plate is installed. The plate should then be installed after checking for the correct direction.

10.6.2 Differential pressure Transmitters and Receivers:

Ordinarily these should be calibrated locally against a manometer. The calibration should be checked at the receiver which may be board or locally mounted recorder or indicator.

10.6.3 Pressure Transmitters and Receivers:

This should be checked in place. The calibration of the receiver should be checked at the same time.

10.6.4 Alarms:

All alarms auto start and cut off systems should be checked by simulating the conditions. Check the sequence of start-up/ shut down interlocks by simulation.



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10.7 FURNACE DRYING:

The furnace refractory must be thoroughly dried out so that it does not crack when the Heater is brought into operation. The drying should be done by gradual heating of the refractory so that no spalling or cracking takes place due to sudden vaporization of moisture from the refractory. The drying procedure for heater refractory is given separately.

10.8 TIGHTNESS TEST:

The purpose of this test is to check the tightness of flanges, joints, manholes etc. (except pumps and control instruments). This operation can be integrated with steam purging activity aimed at expelling air prior to introducing hydrocarbon into the unit.

- Drains at low points will be opened and after draining is over, these will be closed. Vents will be opened; pressure gauges will be installed on each circuit.
- Steam is progressively admitted where connections are available. Circuits which do not have direct admission of steam will be supplied through hoses.
- The temperature of the whole installation is increased slowly and free expansion of lines is checked. The condensed water is drained while the temperature of the circuit rises.
- When temperature is steady, vents are progressively closed in order to get the desired pressure by keeping a vent slightly opened. A steam make-up is maintained. All joints will be checked for leaks. If leaks are detected system will be depressurized and leaks will be attended and the system retested.
- For vacuum system perform vacuum test if tightness test is satisfactory.

10.9 CHARGING OF CHEMICALS:

Small quantity of the corrosion inhibitor, ammonia & demulsifier are required for use in the CDU/VDU unit (no chemical is used in BBU). It is important that these chemicals should be available for use from the time the unit is put on stream.

Corrosion inhibitor is received in drums and is used directly from the drums. Proceed as follows:

- Receive two full drums of corrosion inhibitor.
- Open the two bungs on each drum and install vent connection on one opening and valve connection on the other in each drum.
- Load the two drums on the rack.
- Hook up the outlet of drums one each to the suction of the corrosion inhibitor pumps respectively with hoses.
- Check for leaks and keep ready for starting the pump.
- Similarly keep ready ammonia & demulsifier dosing system for use.



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PREPARATORY OPERATIONS & ACTIVITIES FOR COMMISSIONING

Start up and normal operating procedures are described in this section. Start up and shutdown are the most critical periods in operation. It is then that the hazardous possibilities for fire and explosion are greatest.

The hazards encountered most frequently in start up and shut down of units are accidental mixing of air and hydrocarbons and contacting of water with hot oil. Other hazards primarily associated with startup are pressure, vacuum, thermal and mechanical shocks. These can result in fires, explosions, destructive pressure surges and other damages to unit as well as injury to personnel.

Fires occur when oxygen and fuel vapor or mists are mixed in flammable proportions and come in contact with an ignition. They may run out of control or touch off devastating explosion. Pressure surge from unplanned mixing of water and hot oil may cause damage of equipment and / or loss of valuable production. This may result in expensive, costly down time on process unit. Fires usually follow if the explosion bursts lines or vessels.

Preparation for start - up begins with a complete review of the start up procedure by the operating crew. Activities of CDU/VDU should be coordinated with OMS and other units and utilities section. Start up of the unit involves the following consecutive phases:

- Preparation of the unit.
- Removal of air from the unit by steaming.
- Tightness testing under steam pressure.

11.1 PREPARATION:

Prior to actual commissioning of the plant it should be established that all preparatory works have been successfully completed and all equipment are ready to function. Ensure that:

- Blinds are installed as per master blind list. Each removal and insertion of a blind should be noted and installed by the operator in charge.
- All vessels, piping, equipment are pressure tested, flushed and ready for service.
- All rotating equipment such as pumps motors etc. have undergone functional test successfully.
- All instruments have been checked and calibrated. Control should be on manual.



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- All safety valves are in position after setting and testing. Isolating valves will be left in lock open position. Spare valves should be kept isolated.
- All utility headers are charged.
- Flare, closed blow down, sewer and flushing oil systems are in operable condition.
- All related units are informed of the start - up plan.
- All pre-commissioning activities are completed.
- Fuel oil and fuel gas blinds are removed and both headers charged.
- Refractory dry out for heater is carried out.
- Tracing steam to the lines in opened.

Tightness and vacuum test will form part of pre-commissioning activities for the first start up. For subsequent start ups the tightness test and vacuum test can be done in conjunction with the step of elimination of air.

11.2 STEAM PURGING AND TIGHTNESS TESTING:

For this purpose, Unit is divided into following sections.

CDU/VDU

1. Desalter and crude preheat system and PFD.
2. Atmospheric column, top, middle and bottom pump around systems and product circuits up to battery limit
3. Stabilizer and stabilizer reflux drum
4. Naphtha caustic wash
5. Vac column for vacuum test.
6. Vac products and Vac reflux systems.

Steam connection is given at appropriate places to remove air from equipment and pipes. Water to coolers and condensers is isolated. Low point drains of pipes and equipment are opened to remove condensate. High point vents are opened for air removal. Steaming is done till O₂ content of the system reduces to less than 1%. For leak testing vents and drains are throttled, pressure is built up to 1kg/cm²g for atmos column and for stabilizer pressure to be build up to 5 kg/cm²g and all product systems pressure to be build up to 7 kg/cm²g and system checked for any leaks. Leaks are attended after depressurizing.

In vacuum section, vacuum is pulled in the vacuum column and the system checked for holding the vacuum. Air will be removed from vacuum section during vacuum pulling. Equipment and lines will be included during this activity.



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Low point drains of pipes and equipment are opened to remove condensate. High point vents are opened for air removal. Steaming is done till O₂ content of the system reduces to less than 1%. For leak testing vents and drains are throttled, pressure is built up to 5 kg/cm²g for product system and system is checked for any leaks. Leaks are attended after depressurizing.



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NORMAL START -UP PROCEDURE

12.1 Introduction

Start up and normal operating procedures are described in this section. Start up and shutdown are the most critical periods in operation. It is then that the hazardous possibilities for fire and explosion are the greatest.

The hazards encountered most frequently in start up and shut down of units are accidental mixing of air and hydrocarbons and contacting of water with hot oil. Other hazards primarily associated with start up are pressure, vacuum, thermal and mechanical shocks. These can result in fires, explosions, destructive pressure surges and other damages to unit as well as injury to personnel.

Fires occur when oxygen and fuel vapor or mists are mixed in flammable proportions and come in contact with a source of ignition. They may run out of control or touch off devastating explosion. Pressure surge from unplanned mixing of water and hot oil may cause damage of equipment and loss of valuable production. Extensive costly down-time on the process unit may result. Fires usually follow if the explosion bursts lines or vessels.

Preparation for start-up begins with a complete review of the start-up procedure by the operating crew. Activities of CDU / VDU and BBU should be coordinated with OMS, other units and utilities section. Start-up of the unit involves the following consecutive phases:

- Preparation of the unit
- Removal of air from the unit by steaming
- Tightness testing under steam pressure
- Backing in fuel gas
- Cold oil circulation and removal of water
- Hot oil circulation
- Bringing the unit on stream

12.2 Summary of Start-up

Normally both CDU and VDU will be running. However in case of some problem in VDU, CDU above can be kept running for some time by diverting RCO to fuel oil tanks.



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Two types of start-up are envisaged.

- Start-up of both CDU and VDU simultaneously
- Start-up of only CDU

Start-up of the unit will be done by taking crude oil in Atmospheric section and gas oil in vacuum section.

12.2.1. Preparation

Prior to actual commissioning of the plant it should be established that all the preparatory works have been successfully completed and all equipment are ready to function. Ensure that:

1. Blinds are removed as per the master blind list. Each removal and insertion of a blind should be noted and installed by the operator in-charge.
2. All vessels, piping and equipment are pressure tested, flushed and ready for service.
3. All rotating equipment such as pumps motors etc., have undergone functional test successfully.
4. All instruments have been checked and calibrated. Control should be on manual.
5. All safety valves are in position, after setting and testing Isolating valves will be left in lock open position.
6. All utility headers are charged.
7. Flare, closed blow-down (CBD), sewer and flushing oil systems are in operable condition.
8. All related units are informed of the start-up plan.
9. All pre-commissioning activities are completed.
10. Fuel oil and fuel gas blinds are removed and both headers charged.
11. Refractory dry-out for heater is carried out.
12. Tracing steam to the lines is opened.

Tightness and vacuum test will form part of pre-commissioning activities for the first start-up. For subsequent start-ups the tightness test and vacuum test can be done in conjunction with the step of elimination of air.

12.2.2. Steam Purging and Tightness Testing

For this purpose, the unit is divided into following sections:

- i. Desalter and crude preheat system.
- ii. Atmospheric column, top, middle and bottom pump around systems and product circuits up to the battery limit.



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- iii. Stabilizer and stabilizer reflux drum.
- iv. Naphtha and caustic wash.

Steam connections are given at appropriate places to remove air from equipment and pipes. Water to coolers and condensers is isolated. Low point drains of pipes and equipment are opened to remove condensate. High point vents are opened for air removal. Steaming is done till O₂ content of the system reduces to less than 1%. For leak testing, vents and drains are throttled, pressure is built up to 1 kg/cm² g and the system is checked for leaks. Leaks are attended on, after depressurizing.

In vacuum section, vacuum is pulled in the vacuum column and the system is checked for holding the vacuum. Air will be removed from vacuum section during vacuum pulling. Equipment and connected lines will be included during this activity.

12.2.3. Backing in Fuel Gas

Steam is cut off slowly and fuel gas is backed in, for which, provision has been made in the design. Avoid pulling in vacuum during this period. This may cause entry of air in the system and damage of vessels which are not designed for vacuum. It is advisable to back in fuel gas section wise. After all the sections have been floated on fuel gas, water is drained from low point drains, and draining should be recorded. In vacuum section, after system tightness is proven, maximum vacuum is pulled. This vacuum will be broken by fuel gas admission (0.5 Kg/cm² g. pressure). Positive pressure in the system will facilitate draining of residual water. ***Removal of water from the system is an important step for smooth start-up.***

12.2.4. Cold oil Circulation

Atmospheric Unit: For Atmospheric unit, crude oil is used during cold oil circulation. The purpose of cold oil circulation is to try out pumps and control system and for better water removal. Crude oil is pumped at about 50% of design rate by crude charge pumps 11-P-01 A/B. Crude oil passes through crude preheat system. Two trains of desalted crude preheat system, Atmospheric furnace and into Atmospheric distillation column 11-C-01. Reduced crude oil pump 11-P-10A/B takes suction from column bottom. The discharge of this pump bypasses vacuum heater 12-F-01 and flows through the Short Residue circuit to the slop header circuit through start-up line and then to the crude tanks. The cold circulation is then established. During cold oil circulation one of the crude oil tanks will remain floating with the system. After establishing cold oil circulation for an hour, stop the pumps and circulation



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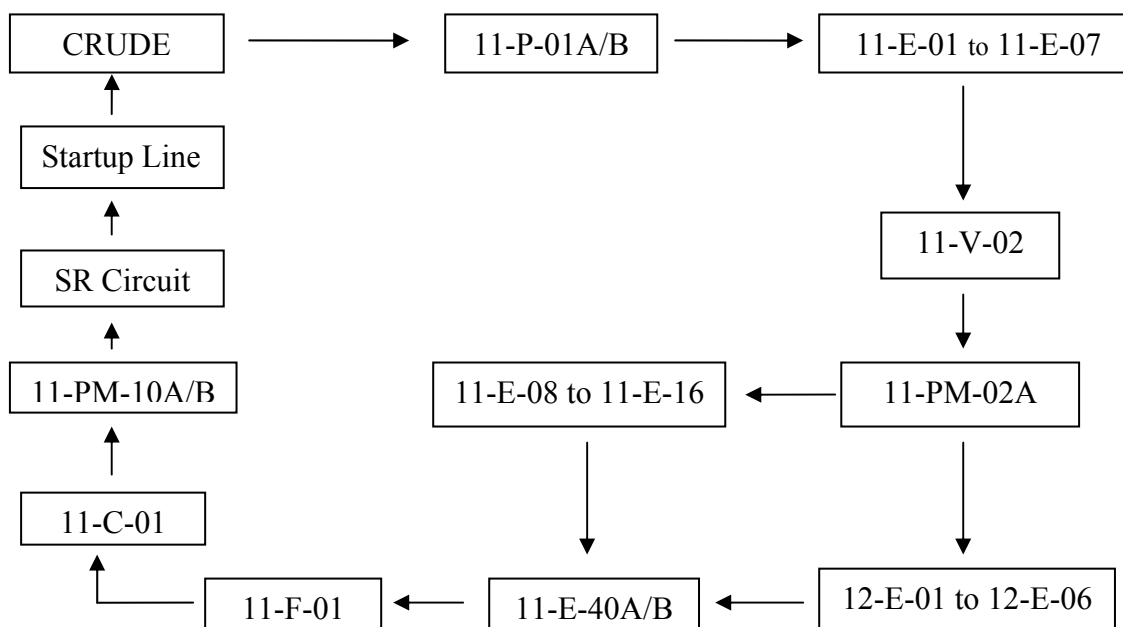
and allow water to settle at the lowest points. Drain water from all low points and restart cold circulation. Repeat the procedure till all the water in the system is removed.

Vacuum Unit: In vacuum unit, gas oil is used for cold circulation. Gas oil is charged in the vacuum column through flushing oil connection provided for the purpose. This oil is put in the vacuum heaters. LVGO system, HVGO system, slop distillate and quench system and these systems are circulated individually.

After establishing cold oil circulation for an hour, stop the pumps and circulation and allow water to settle at the lowest points. Drain water from all low points and restart cold circulation. Repeat the procedure till all the water in the system is removed.

Systematically, the cold oil circulation for both CDU & VDU can be done as per the following sequence:

- Line up 11-P-01A/B suction and fill Desalter at a slow rate, using feed pump turbine. Open 1½" relief valve bypass and displace fuel gas to fractionators (11-C-01). Control Desalter pressure manually by 11-PV-105 at 7 kg/cm² then close unit limit feed valve.
- Line up the unit as per the following circuit

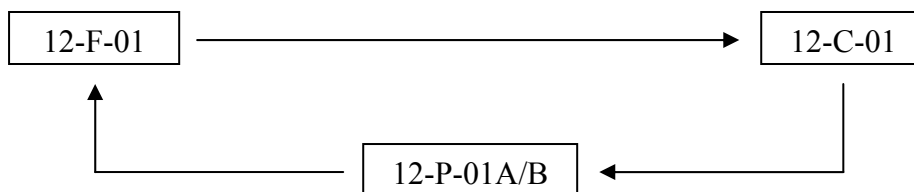




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Take flushing oil to 11-C-01 bottom through flushing oil tie-in and bring up level.
Start 11-PM-10A/B, 12-LV-202 bypass open and fill SR exchanger up to feed pump suction.

- c) Isolate Desalter. Switch control level switches to startup mode. Start feed pump turbine and control Desalter downstream pressure at 9.0 kg/cm^2 by operating 11-PV-105 manually. Keep heater pass flows wide open and fill all exchangers. Add flushing oil at 11-C-01 bottom to make up level
- d) Maintain column (11-C-01) pressure at 1.0 kg/cm^2 and release displaced gas to flare. Shut off fuel gas and commission surface condenser to check any pressure buildup in 12-C-01. Open hot well vent.
- e) When the system is filled with flushing oil, start cold oil circulation by running 11-PM-10A/B.
- f) By circulating cold oil, water in the system will settle at low points. Stop circulation after one hour, allow water to settle, drain water from all low points. Restart circulation and repeat this operation until no water drains out.
- g) Commission tempered water system by taking DM water to 12-V-02.
- h) Take flushing oil to 12-C-01 top, through flushing oil tie-in at 12-P-01 discharge manifold under 12-FRC-205. After getting LVGO level, start reflux to HVGO packing at $30 \text{ m}^3/\text{hr}$. through FRC-201 and build up HVGO level. Similarly, build up level in slop cut, running 12-PM-3A/B and routing through 12-FRC-202
- i) Take flushing oil to 12-F-01 passes and build up 12-C-01 bottom level and establish internal circulation as shown below.



- j) Circulate cold oil through all circuits i.e. bottom, slop cut, HVGO, LVGO, for about an hour. Drain water after settling and repeat until systems are completely water free.



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- k) Operate all control valve bypasses and exchanger bypasses to remove water from these points

12.2.5. Heater Firing and Hot Oil Circulation

Atmospheric Unit: When cold-oil circulation has been well established and water in the system has been thoroughly drained out, heater firing will be done. About 50% of normal flow will be maintained through heater coils. Transfer temperature is raised to 120 °C at a rate of 20 °C/h, (MAX). This temperature is maintained for about four hours to remove residual water from the system. The transfer temperature will be further raised to 250 °C at 30 °C /h. Top temperature of the column will gradually rise. When it reaches more than 100 °C, steam will escape from the column and condense in overhead Naphtha accumulator. Drain condensate at regular intervals. As top temperature rises further, column pressure is slowly raised to its normal value. Higher pressure will help in better condensation of vapors. As level starts building up in overhead Naphtha accumulator 11-V-01, refluxing is started to maintain desired top temperature. Complete hot bolting at 250 °C.

System temperature will be further raised to 365/385 (BH/AM). Top temperature and pressure are maintained. Admit stripping steam to crude column at a transfer temperature of about 300 °C. When Naphtha production increase, which is indicated by rising oil level in crude column overhead Naphtha accumulator, it is routed to Stabilizer column 11-C- 05.

Naphtha is diverted directly to Stabilizer column and build up level slowly at the column bottom. When levels appear in the crude column side stream strippers, maintain these levels at about 50% and line up BPA (Bottom Pump Around –DCR), MPA (Middle Pump Around – KCR) and TPA (Top Pump Around-TPA) reflux circuits. Water is drained before starting BPA, MPA and TPA pumps. Flow and temperature of top pump around, middle pump around and bottom pump around are adjusted to maintain temperature profile in the column. Line up all product circuits up to battery limit. Start respective product pumps when level appears in the strippers. Route all products to slop header. Once sufficient level (70%) is built up at Stabilizer column bottom, Stabilizer column will be brought on stream by gradually cutting in the heating medium i.e., KCR in re-boiler. As the temperature of the Stabilizer column rises, pressure will increase and level will appear in the Stabilizer reflux drum. Stabilized Naphtha will be routed to storage tanks. When level appears in the Stabilizer reflux drum refluxing will be started. When LPG quality is on-spec, it will be sent to the Amine Treating Unit. Reduced Crude Oil at the outlet of Atmospheric column bottom will be routed



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to fuel oil tanks till vacuum section is ready to receive RCO. When the normal transfer temperature of 365/385 (BH/AM) has been attained, make necessary adjustments and normalize the operating conditions.

Vacuum Unit: Once cold oil circulation has been established and water in the system is drained out burners are lit in the vacuum heater 12-F-01. Transfer line temperature is slowly raised to 90 °C at 30 °C /h. Hold this temperature for 2 hours. Activate all the circuits in the system using start-up lines. Transfer line temperature is then raised to 120 °C and held for 4 hours. During this period, water in the system will be removed as steam in the overhead system. After removing water completely from the system, transfer line temperature is raised to 230 °C at the rate of 30 °C /h. Any level build up on trays is pumped to slop. When RCO from Atmospheric unit is available, it is lined up to the vacuum unit. Establish once through flow in the unit by gradually displacing the flushing oil in the unit. Transfer line temperature is raised to 300 °C at the rate of 30 °C/h. Hot bolting is done by holding the transfer line temperature at 300 °C. After hot bolting is completed, increase the transfer line temperature to 360 °C @ 30 °C/h. Light fractions will be collected on the upper trays as transfer line temperature is increased. Pump around and internal refluxes should be maintained continuously Tray levels are maintained by adjusting the product routing to slop. At this stage recycle stream from column bottom to heater inlet is commissioned. Gradually vacuum is pulled. Transfer temperature is then gradually increased to 395 / 402 °C (BH/AM) at 30 °C/h. Quench rates, draw-off rates etc., are adjusted and steady conditions are maintained. When products are on spec. they are routed to the respective tanks.

Systematically, hot oil circulation for both CDU/VDU can be as per the following plan:

- Fire both heaters and bring up coil outlet temperature at 30°C/hr rate.
- When transfer line temperature reaches 120°C hold firing rate and maintain circulation in both heaters for about 4 hrs, to remove final traces of water.
- Closely follow 12-C-01 pressure, commissioning 3rd stage ejector if required to contain pressure build up
- Admit small amount of steam into 11-F-01 superheating coil and vent through silencer. Outlet temperature should not exceed 350°C.
- Slowly raise furnace COTs to 220°C/hr rate
- Check all the equipments and pumps during this period.
- When 11-C-01 top temperature crosses 100°C, steam will escape from column and condensate collects in reflux drum keep interface controlling service.
- Hot bolt at 200-250°C and cap off all bleeders.



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12.3 DETAILED PROCEDURE

12.3.1 ATMOS SECTION

Elimination of Air

Air from various equipment, piping etc. is eliminated by steaming. Steaming of various Sections of the unit to expel air can be carried out simultaneously or in a convenient Sequence. The following basic things should be taken care of during steaming of the equipments / systems:

1. Cooling water to the condensers and product coolers to be isolated and water should be drained out from the condensers, coolers etc.
 2. Keep water side vents and drains of condensers / coolers opened
 3. Keep all pumps isolated
 4. Keep instruments like PT, FT etc., isolated
 5. Power supply to the Desalter is cut off
- i) Admit steam into the system by opening the valve on 3" steam line joining the discharge header of crude charge pump 11-P-01 A/B.
 - ii) Prior to this activity, isolate Desalter and introduce LP steam. Keep watch on Desalter pressure which should be maintained around 0.5 kg/cm² g. If necessary, throttle steam. Allow steam to vent from Desalter top. Purge Desalter safety valve inlet line. Air will be purged from both inlet and outlet lines of safety valve. MP steam should not enter Desalter.
 - iii) Open steam slowly into the system and allow piping and equipment to gradually warm up. Drain condensate frequently from low points.
 - iv) Keep booster pump suction and discharge valves closed
 - v) Line up overhead condensers 11-E-17 A-H, and overhead Naphtha accumulator 11-V-01 with the column. Ensure cooling water to Condensers is isolated and water is drained out.
 - vi) Open vents of overhead Naphtha accumulator and crude column.
 - vii) Line up strippers, product circuits and circulating reflux circuits up to the respective pumps. Keep pump suction and discharge valves closed.
 - viii) Commission fuel gas header to furnace, after removing the battery limit blind. Fire the heater at a small rate. Heater is fired to avoid condensation in tubes and subsequent hammering if steam is introduced in cold tubes. ***Please refer vendor Operating Manual for heater start up procedures.*** Maintain heater Arch temperature at about 250°C. Pilot



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burners will be lighted initially. Main burners can be lighted afterwards, if required. Keep open the vents at crude column top and overhead Naphtha accumulator.

- ix) Slowly open steam into all passes of the heater coils through emergency steam purging connections. Also open steam to the crude column bottom via stripping steam line.
- x) Drain condensate from low points and allow system to warm up. Regulate the steam flow to heat up the system gradually. Continue purging till steam comes out from the top of the crude column and overhead Naphtha accumulator.
- xi) Back up steam up to the Booster Pump discharge. Wedge open discharge valve upstream flange to release air and condensate. Drain condensate from exchangers and lines from all low points. Steam the bypass lines of the exchangers.
- xii) Purge product and pump around circuits by backing up steam from crude column. Steam the product circuit up to the battery limit and vent through sample points or any convenient high and low point drains. It may be necessary to augment steam supply at the discharge of product pump by connecting temporary hoses to vent or drain points in the lines. B/L flange may be wedge opened for good steaming. Box up the flange under slight steam pressure.
- xiii) Steam out RCO and VR circuit up to isolation valve at the B/L. Eliminate air through all exchanger vent points and drain out condensate from all exchanger drain points.
- xiv) Back up steam into the top reflux line and vent from the bleeder valve upstream of the Stabilizer feed control valve 11-FV-501.
- xv) Vigorously steam for about 2 hours and then shut off crude column vent. Allow steam to vent from overhead Naphtha accumulator and other points as mentioned above.
- xvi) Maintain 1.0 kg/cm² g. pressure in column flash zone (11-PI-401) by regulating the quantity of steam being introduced into the system.
- xvii) Line up the Stabilizer overhead condenser 11-E-20 A to D and reflux drum 11-V-03 to Stabilizer column. See that vents in the cooling water side of condenser are kept open. Open vent valves of Stabilizer column and its reflux drum.
- xviii) Remove blind on the utility steam connection at the Stabilizer column bottom and admit steam slowly. Drain condensate from low points and allow the system to warm up.
- xix) When steam comes out from Stabilizer top, vigorously steam for about half an hour. Then shut off this vent and allow steam to come out of reflux drum vent.
- xx) Maintain Stabilizer column pressure at 1.0 kg/cm² g. (11-PIC-501) by regulating the quantity of steam. Open the pressure control valve 11-PV-501 and globe valve on its bypass line. Steam for about half an hour then shut off the control valve and the globe valve.
- xxi) Purge feed line by backing up steam from the Stabilizer column.
- xxii) Purge Stabilizer bottom outlet circuit up to the battery limit along with caustic wash system. Vigorous steam venting from various points of the unit as described above will



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be continued for about 4 hours. It is estimated that oxygen will be eliminated at the end of this period. Samples may be checked from different location to check that O₂ content is less than 0.5%.

- xxiii) During the period of steaming, systems should be checked for leaks. Attend the leaks by depressurizing the system. Retest and purge. When sufficient steam comes out from all the vents and drains, reduce steam inlet to maintain positive pressure.

Backing in Fuel Gas

Before admitting fuel gas into the system, all vents and drains are to be closed. Admit fuel gas through fuel gas make-up control valve. Steam entering the system is throttled slowly. Rate of fuel gas backing is adjusted in such a way that operation of the refinery gas system is not adversely affected. During gas backing, ensure that vacuum is not formed in the system due to condensation of steam. Furnace fires are cut off. Cooling water is commissioned to the overhead condensers and coolers. Maintain the pressure in the system around 0.5 – 1.0 kg/cm² g. As the system cools, condensate will accumulate. The drain at each low point must be opened and the condensate be drained. Never leave open drains unattended. Each drain which is opened and checked should be listed in the start-up check list. A log should be kept for the draining activities showing time of check and absence of condensate. Each drain must be closed as soon as gas issues from it. When the draining is complete the system is ready to take crude oil.

Steps in backing fuel gas into the systems are given below:

- Reduce steam to the various points in the crude column and furnace coils. Close all drains and vents except any one vent at a convenient point to vent steam to maintain system pressure.
- Close all the vents one by one. Adjust steam inlet rate to maintain system pressure about 0.5 kg/cm² g. Cap off all vents properly to avoid leakage of gas through these points. Throttle the drain valves to allow only condensate to flow to keep the system hot. Keep watch on the system pressure. Adjust steam inlet if necessary.
- Open the pressure control valve 11-PCV-409B very slowly to admit fuel gas into the system. Close the vent valve which was kept open at a convenient point to maintain the steam pressure. Slowly reduce and finally shut off steam inlet to the system. Watch system pressure which should not be allowed to cross 1.0 kg/cm² g. Fuel gas inlet is to be regulated such that other running units are not affected by this operation. Ensure that all the vents and drains are fully closed.
- Cut off furnace fires when steam to the unit is shut off.



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- v) Commission water to the overhead condensers and coolers after venting air from water side. Open suction and discharge valves of pumps.
- vi) As the system cools, drain condensate that collects at all low points in the system. Always shut off drains after gas starts issuing from it.
- vii) Reduce steaming in Stabilizer section and close all vent and drain valves and admit fuel gas into the Stabilizer by gradually opening the control valve 11-PV- 501.
- viii) As Stabilizer pressure tends to increase, cut off steam and maintain Stabilizer pressure around 1.0 kg/cm² g. Check for leaks, if any.
- ix) Purge flare header with fuel gas and open B/L isolation valve. This flare header should now be lined up to various equipments.
- x) As the system cools, the steam in the systems will condense and collect at low points in the unit. Repeat draining operation every half an hour till no more condensate is drained. Most of the problems in unit start-up can be avoided if condensate draining is done thoroughly at this stage. When all condensate has been drained and the unit is under fuel gas pressure of about 1.0 kg/cm² g, crude oil can be taken into the unit.
- xi) Take all instruments on line which were kept isolated during steaming.

Receiving crude in the Unit

The crude tank that will feed the unit will be prepared in advance by thorough draining of water. Sample will be analyzed and dips will taken. It will then be lined up to the unit.

- a. Open suction valves of crude charge pumps 11-P-01 A/B and crude booster pumps 11-P-02 A/B.
- b. Slowly charge crude from the crude tanks. Crude will begin to flow by gravity into the line and displaces air. This air is vented from the crude pump vents. *In subsequent start-ups this operation may not be required as off-site's line up to the battery limit will remain full.*
- c. Make sure that electrical supply to Desalter is cut-off. Open the discharge value of crude pump and allow the crude to flow to the Desalter. Alternatively, Desalter can be kept filled with Crude oil and by-passed.
- d. Release displaced fuel gas into the flare system from overhead Naphtha accumulator 11-V-01, and do not exceed column pressure beyond 1.0 kg/cm² g.
- e. When gravity flow of crude oil stops start crude charge pump 11-P-01 A/B. Continue charging crude oil at a slow rate. Preferably start turbine for low flow rates.

Note: Before warming up the stand-by pump of any hot pump through the 1" warm-up line across NRV of stand-by pump, it is to be ensured that suction valve of the stand-by pump is kept open. This is to avoid pressurization of the pump and its suction line to the discharge pressure of running pump.



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Establishing cold oil circulation

Circulating cold oil will carry liquid water in the system to low points in the unit. Stop circulation after about an hour, allow water to settle. Drain water from all low points. All plugged drains must be cleared. When draining water, be careful to prevent unnecessary loss of oil to the sewer. Restart circulation once draining from all low points is over. This operation of circulation, stopping, settling and draining will be repeated till no further water separates out. The steps involved to establish cold oil circulation are detailed below:

- With crude charge pump 11-P-01 A/B running, carefully operate the Desalter 11- V-02 pressure controller PIC-1105 manually and build up a pressure of 8.0 to 9.0 kg/cm² g. in the Desalter. Keep the bypass of booster pump open for continuing circulation. Crude oil flow rate should be kept at the minimum, limited by FG venting capacity from overhead Naphtha accumulator to flare.
- Start Booster Pump 11-P-02 A/B and close the bypass valve gradually. Regulate the coil flow through FIC-1301 to 1304 of 11-F-01 such that crude flow to unit records about 225 m³/h (about 50% of normal throughput). Also divide the flow equally through two crude preheat exchanger trains by operating FRC – 1101.
- Stabilize the Desalter pressure at 9.5 kg/cm² g and put PIC-1105 on auto. Observe the performance. Crude oil will gradually displace fuel gas and build up level in crude column.
- Keep watch on crude column level (**LI-1401**)
- Maintain column pressure at 1.0 kg/cm² g by operating PIC-1409. Displaced gas will be released to flare.
- When sufficient level is built up in the column bottom, start RCO pump 11-P-10 A/B and route the crude to Crude tank through VR circuit to slop for about 1 hour to displace free water. Check that there is no flow of crude oil to vacuum furnace.
- Keep the crude feed tanks floating with crude charge pump suction.
- Vent the exchangers in RCO circuit to CBD, so that these become full of crude. Activate exchanger by passes; Bleed residual fuel gas from the RCO circuit.
- Stop all pumps every one hour and allow settling for one hour. Carry out water draining from all the low points in the system including column and exchangers. Restart circulation. Repeat this operation till no further water separates out. A check for water content of circulating crude oil is useful information.

Firing the Heater

When cold circulation has been well established and water in the system has been thoroughly drained out, heater firing will be done. Ensure the following before firing the heater:



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1. Cooling water flow is established in all overhead condensers and product coolers.
2. Tempered water system is activated.
3. Tracing steam is commissioned.
4. Set level controllers (LIC-1404, 1403, 1402) of strippers 11-C-02/03/04 at 50% level (auto mode). Check that valves remain opened as there is no level in the strippers.
5. All safety isolation valves are left lock opened.

Raising Temperature to 120°C

- a) Charge atomizing steam header to the heater. Commission fuel oil supply and return lines and establish circulation. Fire the burners one by one. The rate of increase of transfer line temperature will be restricted to 20 °C per hour. When the transfer line temperature reaches 120 °C at heater outlet hold firing rate to maintain this temperature for four hours.
- b) Lower column pressure to 0.5 kg/cm² g. or less to facilitate water removal as water-vapor to the column overhead. Keep watch on the column pressure. The excess pressure will be released to flare.
- c) Monitor all temperature and pressure reading of column and heater.
- d) While raising the temperature it will be observed that crude passing through preheat trains will be gaining heat from the RCO. Check for presence of water in product and circulating pumps' suction and drain out.
- e) Watch performance of RCO pump as it may tend to lose suction with rise in crude oil temperature. If required, increase column pressure to 1.0 Kg/cm²
- f) Maintain temperature of crude at the outlet of Vac. Residue cooler 12-E-09 A/B/C/D about 40°C to avoid cavitation due to hot RCO going back to the crude charge pump (11-P-01 A/B) suction.
- g) Maintain crude column level by matching RCO pump discharge rate with crude intake to the unit. Flow is to be maintained at about 50% of the normal throughput.
- h) At the end of 4 hours, carry out test for water content in circulating oil. Water content equal to or less than that of tank sample is indication of good water removal. A value of about 0.2 wt% of water is often obtained.

Raising Temperature to 250°C

- i) After holding the temperature at 120°C for 4 hours, transfer temperature will be further raised at a rate of 30°C/h to 250°C and held at this temperature.
- ii) Closely watch all instrument readings and check their performance.



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- iii) Hold crude column at about 50% level. Manipulate, if necessary, the RCO pump discharge rate.
- iv) Top temperature (11-TI-403) of the column will rise gradually. When it crosses 100°C, steam will escape from column and condense in the overhead Naphtha accumulator.
- v) When level appears in the accumulator 11-V-01, check and drain it out. Local checks for levels are to be made.
- vi) As top temperature rises further, slowly raise column pressure a little at a time to its normal value of (BH/AM) 2.5/2.4 kg/cm² g. at accumulator. After steadying out, put pressure controller 11-PIC-409 on auto to hold this pressure. Higher pressure will help in condensation of vapors which would otherwise escape.
- vii) Check for appearance of oil level in crude column overhead Naphtha accumulator. Start refluxing when oil level builds up. Care will be taken so that water does not go in the reflux stream. Refluxing will be started at a small rate. No overhead product will be withdrawn at this stage.
- viii) Start hot bolting in the transfer line, RCO circuit, column bottom manholes and other flanges in hot service where temperature exceeds 200°C.
- ix) Cap off all drains that were used to drain water.

Raising Temperature to 365°C

- i) When hot bolting of the portion where temperature reaches 200°C is over, start raising the transfer temperature at 30°C/h and continue hot bolting in other areas where temperature touches 200°C.
- ii) Watch water and oil levels in overhead Naphtha accumulator 11-V-01. Commission water and oil level controllers 11-LDIC-406 and 11-LIC-406 and put them on auto to hold about 50% level respectively. Watch the instrument performance.
- iii) Regulate top refluxing to maintain the column top temperature (11-TI-403) of (BH/AM) say 118/110°C.
- iv) When Naphtha make increases as indicated by rising oil level in crude column overhead Naphtha accumulator, route this product to Stabilizer column.
- v) Maintain about 50% oil level in overhead Naphtha accumulator by regulating the quantity of Naphtha withdrawn.
- vi) In case RCO product temperature at the outlet of 12-E-09 A/B/C/D goes beyond 45°C, Receiving slop crude tank may experience high temperature which should be taken care of.



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- vii) When column top temperature and pressure reaches its normal value, line up Bottom (HSD CR), middle (KERO CR) and top pump around circuit. Always drain water before activating the circuits. Start refluxing at minimum flow rate after draining water from pump casing drain.
- viii) When level appears in side strippers, drain water from pump suction and line up product discharge (heavy Naphtha, kerosene, & HD) to slop header. Heating medium to Stabilizer reboiler 11-E-25 should not be commissioned which will be commissioned at a later stage when continuous feed to Stabilizer will be available. The control valve 11-TV-504 will remain 100% open on manual. When sufficient level is built up in the side strippers (about 40%) start withdrawing off-spec. products to slop header which is lined up to crude storage tank.
- ix) Check performance of level controllers of Heavy Naphtha stripper 11-LIC-404, Kerosene stripper 11-LIC-403, and HD stripper 11-LIC-402.
- x) Activate stripping steam header by draining Condensate from drain points provided near crude column and strippers.
- xi) When crude column bottom temperature reaches 300°C, admit stripping steam step by step about 250 kg/h (11-FRC-401) in each step. With introduction of stripping steam, amount of vapour flowing to the upper section will go up.
- xii) Divert RCO to vacuum furnace, provided vacuum section is ready to receive RCO. Otherwise divert to slop. Divert other side products to Fuel oil tanks / slop tanks.
- xiii) Watch column top pressure and temperature. Adjust circulating refluxes and top reflux if necessary. Raise the circulating reflux flow to suit conditions.
- xiv) When transfer temperature reaches (BH/AM) 365 / 385°C put TIC-1409 on auto and steady out the conditions. Introduce stripping steam to side strippers. Ensure that condensate is drained before allowing steam flow to the strippers.
- xv) Adjust cooling water, if necessary, to all product coolers to maintain run-down temperature of Heavy Naphtha, Kerosene and HD around 45°C.

Bringing up Naphtha Stabilizer System

- i) Start taking Naphtha from crude column overhead Naphtha accumulator by maintaining its level (11-LIC-405) at about 50%
- ii) Take about 75% level (11-LT-501) in Stabilizer bottom through 11-FIC-503.
- iii) Commission 11-**PIC-501** on auto setting at about 10.0 kg/cm² g. Initially the control valve 11-**PV-501** will remain fully closed.
- iv) Keep 11-FIC-502 (LPG product to ATU) manually closed.



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- v) Divert KERO CR through Stabilizer reboiler 11-E-25 partially by adjusting to rise stabilize bottom temperature gradually. Restrict rise in temperature to about 25 °C/h.
- vi) Stabilizer pressure will tend to rise along with the bottom being heated by KERO CR. **PIC-1501** should be manually operated to gradually open 11-PV-501 towards fuel gas system to maintain the pressure of the column. Condensable accumulated in condensers should be periodically released to reflux drum 11-V-03. When normal pressure (**8.0 kg/cm² g. at 11-PIC501**) is reached normal working of the column will be established. During the process of heating up, Stabilizer bottom level will fail if rate of heating is too fast. Divert column bottom to caustic wash unit through feed / Stabilizer bottom exchanger 11-E-19 A/B.
- vii) Raise gradually Stabilizer tray temperature by adjusting 11-TV-501. When temperature stabilizes at around 180 °C put 11-TIC-501 on auto. Sudden increase of temperature and pressure will destabilize the column making heavy fraction to go to the top. Then it will take longer time to stabilize. So increase or decrease of temperature or pressure of Stabilizer shall be done gradually.
- viii) When level appears in reflux drum, drain water from boot and pumps 11-P-11 A/B. Start refluxing to the column keeping 11-FRC-501 on manual control. Maintain top temperature about 60 / 70 °C (BH/AM). When reflux drum pressure reaches around 10.0 kg/cm² g., open 11-PV-501 manually put 11PIC-501 on auto.
- ix) Stabilizer column will remain on total reflux. Check sample of LPG for weathering test. If it is found to be on-spec, route this product slowly to Amine Treating Unit under flow control 11-FIC-502 cascaded with reflux drum level control 11-LIC-501. Any abrupt operation in temperature, flow or pressure is likely to make LPG off-spec.

Normalizing the Operating Conditions

When the normal transfer line temperature 365/385 °C (BH/AM) has been attained maintain temperature and make further adjustments.

- i. Adjust heater firing to maintain 365/385°C (BH/AM) at heater outlet.
- ii. Check for normal product run down temperatures.
- iii. When conditions have become steady take all remaining controls on auto one-by one. Particular care should be taken in case of heaters control. Safety shutdown system for low fuel gas pressure (which was on bypass initially) should be made operative.
- iv. Take product samples. Make adjustment on operating parameters to bring the products on-spec.



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- v. When products are on-spec., route them to respective tank / down-stream units.
- vi. When vacuum unit is ready to receive RCO, RCO will be diverted to vacuum furnace inlet.
- vii. Check locally temperature, pressure, flow and level of different equipment and streams. Also check for normal functioning of pumps, heaters and other equipment.

Commissioning of Desalter

Desalter will be brought into service at this stage before commissioning vacuum section.

- i. Stabilize Desalter 11-V-02 pressure at about 10.5 kg/cm² g. by PIC-1105. Control the Desalter temperature at about 120°C. Check oil water interface level through the try lines and check for any presence of vapor.
- ii. Switch on the power supply to Desalter. High voltage and low amperage should be indicated by voltmeter and ammeter respectively.
- iii. Line up for water injection at Desalter inlet and start injection at about 4% (by volume) of crude throughput. Also line up effluent water circuit.
- iv. Start caustic injection pump 11-PM-13 A/B/C and inject caustic solution into suction of crude charge pump 11-P-01 A/B and crude booster pump 11-P-02 A/B to maintain the effluent brine pH at about 10.0
- v. Start Demulsifier pump 11-P-16 A/B and inject Demulsifier at crude charge pump suction at the rate of about 1.5 to 2 ppm on crude charged.
- vi. Commission LP steam and water in the wash water heater 11-E-18. Commission level controller **LIC-1102** and route Desalter water to waste water treatment plant.
- vii. Take samples of crude before and after Desalter and readjust operating condition for getting following performance.
 - i. Salt content as NaCl at the outlet should be 3.0 mg/1 or 5% of salt content of raw crude whichever is greater.
 - ii. The insoluble water content in desalted crude should be less than 0.15% (Max) by volume and the effluent brine should have oil content less than 100 **ppm (Max)**.

12.3.2 Vacuum Section

While start-up in Atmospheric section is in progress, vacuum section should be made ready systematically as per the following sequence:

1. Elimination of air and vacuum test
2. Fuel Gas backing in and water draining
3. Cold circulation with gas oil



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4. Hot circulation
5. Feed cut-in
6. Bringing up the unit
7. Stabilization

It is to be ensured that RCO does not enter the vacuum section while it is still not ready. Blinding at the appropriate places may be necessary if valves are suspected to be passing. It becomes mandatory if maintenance work is planned in the vacuum column while Atmospheric section is on-stream.

Elimination of air

Air will be eliminated from the system by pulling vacuum.

- i. Commission water to ejector condensers 12-E-07 A/B/C.
- ii. Ensure normal level in hot well with vent open to Atmosphere from hot well catch pot 12V03. Ensure that hot well seal compartment is filled with water and the seal is maintained.
- iii. Slowly start pulling vacuum in the system. Commission one of the third stage ejectors, to be followed by second and first stage. Vacuum pulling should be done slowly. 12-PRC-206 may be used for this.
- iv. Air will be gradually drawn out from the system including furnace coil and other connected piping and equipment.
- v. Watch pressure in the system. Gradually lower the pressure to the maximum possible extent. Operate control valves and exchanger bypass to subject all equipment and pipes to vacuum for pulling out air.
- vi. When there is no further lowering of pressure, block off the system. Isolate at ejector inlet and outlet and simultaneously shut off steam. Isolate 12-PV-206.
- vii. Observe the rate of fall of vacuum. Initially there may be rapid drop during the process of blocking off. If the rate of fall of vacuum does not exceed ***0.05 kg/cm² g per hour (approximately 40 mm Hg per hour)***, the system is assumed fairly tight. Otherwise thorough leak checking has to be carried out by pressurizing the system to about 1 kg/cm² g. with air and applying soap solution on every flange / joint.

Backing in Gas

1. Ensure all vents are closed and properly capped.
2. Slowly admit fuel gas through fuel gas back up line and break the vacuum. Adjust gas backing such that fuel gas system is not disturbed during this operation. Pressure surges during fuel gas backing in should be avoided.



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3. Isolate the liquid seal at the hot well during fuel gas backing. Valves in hot well vent, drain and overflow line are to be closed.
4. Maintain the pressure in the system at 0.5 kg/cm² g. by regulating the fuel gas inlet valve or by opening the vent to Atmosphere on the hot well.
5. Open the low point drain one by one and drain out condensate completely. Check list is to be prepared to ensure that all drains are made free of condensate. A log should be kept for drain check showing the time of check, absence of condensate and initials of the person who made the final check on the drain. Each drain must be closed as soon as gas issues from it.
6. Back up gas from column to the heater up to feed inlet flow control valves and then to the short residue pump discharge through start up line.
7. Back up gas from column to all pump around reflux and internal reflux circuits.
8. Back up gas from column to all the product run-down lines through internal and circulating reflux lines from the column.
9. Drain out water from the following points:
 - ✓ Preheat Exchangers
 - ✓ Transfer line
 - ✓ Pump around and reflux lines
 - ✓ All product circuits up to B/L
 - ✓ Quench line through control valve bleeders
 - ✓ All pumps
 - ✓ All exchangers and coolers
10. Repeat draining operation every half an hour till no more condensate is drained. Maintain the whole system at 0.5 kg/cm² g. fuel gas pressure. Ensure all vents and drain points are closed and capped off.

Cold oil circulation

Commission cooling water to surface condensers 12-E-07 A/B/C (three-stage ejector condensers).

- i. Commission tempered water system if not commissioned already by taking DM water to 12-V-02. Establish the system by running 12-P-07 A/B.
- ii. Admit flushing oil into the column through the 4" gas oil (cutter) line through the connection provided for this purpose.
- iii. Build up level in vacuum column up to 70%. Settle and drain water.
- iv. Start the pump 12-P-01 A/B and keep gas oil flow of about 25 m³/h through each coil of 12-F-01 keeping FRC-101/102/103/104 on manual.



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- v. Maintain the column bottom level at by making up, if required, with gas oil. Stop circulation after 1 hour. Allow water to settle and drain out from all low points. Repeat the operation till no water appears through the drains.
- vi. Route about 15 m³/h of start up gas oil to the top of the column through the 4" start-up line connected to LVGO CR return header. Regulate the flow by 12-FRC- 205.
- vii. When level appears in LVGO draw-off tray, start refluxing in the HVGO packing at a rate of around 20 m³/h through 12-FRC-201. Through LVGO CR circuit, maintain a flow of around 50 m³/h by controlling 12-FRC-205
- viii. Similarly activate HVGO pump and reflux the wash zone packing. Control the flow @ 20 m³/h by keeping 12-FRC-202 in manual control. Put around 200 m³/h through HVGO CR circuit to maintain the minimum pump capacity though 12- FRC-203.
- ix. When level appears in the slop distillate draw-off tray, start 12-P-02 A/B after draining water. Recycle about 20 m³/h of slop distillate to the furnace inlet though 12-FRC-109. Raise the furnace feed flow appropriately to accommodate this quantity.
- x. Maintain column and tray levels around 50%. Now circulation is established through all circuits except quench line.
- xi. Operate all control valve bypasses and exchanger bypasses to remove water from these points.
- xii. Stop all pumps every one hour and allow a settling time of one hour. Drain water from all low points in the system. Restart circulation. Repeat this operation till no further water issue out of the drains.
- xiii. In case level tends to increase due to too much flushing oil drawn into the system, some of it can be routed to slop.
- xiv. Drain-out water from the quench line through all low points. Divert RCO from VR circuit to Vacuum column through quench line c/v for about 10 minutes to flush out water / condensate from the quench line c/v and bypass circuit. Care should be taken to close the quench line c/v and bypass immediately after flushing to avoid vacuum column build-up.
- xv. During the circulation care should be taken to maintain the system pressure at 0.5 kg/cm² g. by FG back-up. Release the excess gas to Atmosphere from the hot well in case pressure goes beyond 1 kg/cm² g.

Hot Oil Circulation

When cold oil circulation has been well established and water in the system has been thoroughly drained out, the system is ready for hot oil circulation. Utmost care should be taken in de-watering the system completely to prevent water shot and damage to the column internals.



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After ensuring that atomizing, snuffing and emergency steam headers to the vacuum furnace are commissioned, hot oil circulation is established as follows

- Establish fuel oil circulation through the Fuel Oil circuit of 12-F-01.
- Restart pump 12-P-01 A/B and establish flow through each furnace coil of 12-F-01 at about 25 m³/h. Put all the four flow controllers 12-FRC- 101/102/103/104 on auto control and observe the performance.
- Continue tray to tray circulation including pump around as mentioned earlier.
- Maintain column bottom level around 50%
- Fire the furnace as per the normal procedure. Light 1 or 2 burners initially and increase the number of burners as the temperature is raised gradually taking care that the burners are equally spread in the fire box. When steady flame condition is established, increase transfer line temperature to 120°C at 30°C/h rate with the help of **TRC-2133**. Hold the temperature for about 4 hours. Ensure that entire column is heated up to about 110°C. This will ensure that the water accumulated in the trays will be removed as steam in the overhead system.
- After holding transfer temperature at 120°C and removing water completely from the entire system, transfer line temperature is to be raised further. Any oil level build-up in the hot well and trays may be pumped out to slop.
- Before raising temperature beyond 120°C shut off 4" start-up line connection to LVGO circulating reflux. Also stop flow of internal reflux from tray to tray (FRC-201 / 202). Stop LVGO, HVGO and slop distillate pumps as and when they lose suction.
- Continue gas oil circulation in the bottom section of column.

Cutting in Feed

When the system temperature is 120°C it is assumed to be dry. Raise the transfer temperature to 220°C at the rate of 30°C/h. Reduce the column level to minimum operating level, confirm with second level transmitter and route excess gas oil to slop.

- Maintain the system pressure around 0.5 kg/cm² g. (make-up fuel gas if necessary or release the pressure through hot-well vent at controlled rate)
- Hold the temperature until hot-bolting is completed in the column, transfer lines, exchangers etc.
- Divert RCO slowly from 11-P-01 A/B discharge to 12-F-01. Line up 12-P-01 A/B discharge to VR circuit which in-turn is lined up to the slop tank. Change level control of crude column bottom to vac. Furnace flows once these are steady. Vacuum column bottom level controller should be connected to 12-LV-202 and put on auto control.



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- iv. Close 12-P-01 A/B to 12-F-01 and RCO to VR circuit. Monitor 12-F-01 firing. As RCO temperature is 300°C firing in 12-F-01 requires to be reduced to keep the temperature around 300°C.
- v. Hold the temperature at 300°C and carryout hot bolting of flanges, heater coil plugs, manholes, etc. Activate quench oil circuit and maintain about 10 m³/h flow. With diversion of RCO to 12-F-01, once-through flow will be established and gas oil will be displaced by RCO to slop tank. After displacing gas-oil, flow can be re-routed to the appropriate tank. Normal capacity of 12-P-01 A/B is about 133 m³/h.
- vi. Maintain system pressure at 0.5 kg/cm² g. by backing-up fuel gas if necessary.
- vii. During this period ensure the following:
 - Tempered water circulation is established
 - All coolers are commissioned
 - All product tanks are lined up
 - Slop tank has enough ullage to cater to the requirements of unit start-up.
- viii. The hot well water side has to be maintained at 50 % water level. If sufficient water is not available, fill up with fresh water through utility connection. Ensure that the seal compartment drain line is blinded down stream of block valve to prevent mal-operation of drain valve and losing of seal level.
- ix. After completing hot bolting at 300 °C, bring the unit on stream by raising transfer temperature at the rate of 30 °C/h.
- x. Light fractions will be collected in the upper trays as transfer temperature is increased. Tray pump-arounds and internal refluxes should be commissioned. Adjust product routing to slop to hold tray levels. Watch run-down temperature while routing products to tank.
- xi. Commission slop-recycle stream from the discharge of 12-P-02 A/B to RCO feed line of the heater. Adjust flow to about 6-7 m³/h through 12-FRC-109.
- xii. Shut off FG to 12-C-01 overhead, if open.
- xiii. Adjust quench flow to keep 12-C-01 bottom temperature at 350°C.
- xiv. Commission steam to third-stage ejectors 12-J-03 A/B/C after thorough draining of condensate from the steam header. Before that ensure that hot well catch pot vent is opened to Atmosphere.
- xv. Commission 12-PRC-07 and maintain the steam pressure at desired level.
- xvi. Pull vacuum by increasing the steam flow rate gradually. Care should be taken so that the pumps connected to the system are not affected. Activate pressure controller 12-PRC-206 manually to avoid pressure system **surge** in the column
- xvii. Activate sour water pump at hot well bottom and route water to sour water stripper.



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- xviii. Anticipate lifting of light oil from trays into hot well when vacuum is pulled. Pump out the oil from hot well through slop oil pump 12-P-05 A/B to slop. Check the operation of interface level controller 12-LDIC-201.
- xix. Once the system is normalized i.e., all pumps are working satisfactorily, commission the 2nd stage ejector 12-J-02 A/B/C and increase the system vacuum gradually. Watch the performance of the pumps.
- xx. Once system is stabilized with 2nd ejectors on operation, commission 1st stage ejector 12-J-01 A/B/C gradually. Pull vacuum to about 7mm of Hg absolute pressure at the top of the column. Put pressure controller 12-PIC-206 on auto and set the top pressure in such a way that flash zone pressure reads 24 mm Hg absolute.
- xxi. Raise the transfer line temperature to 395 – 400°C as required based on the crude in slow intervals. Adjust quench rate, draw off rates etc., on pro-rata basis and steady out conditions.
- xxii. Send samples of LVGO, HVGO and Short residue. Route them to respective tanks when they are on-spec.

Raising to Normal Throughput

At this stage normal operating conditions are established for both Atmospheric section as well as vacuum section with 50% of normal capacity of the plant. All products are routed to their respective storage tanks / down-stream plants.

Feed rate of the plant is to be raised to its normal capacity in steps as outlined below:

- Raise the feed to Atmospheric furnaces by 50 – 60 m³/h.
- Steady out transfer temperature at (BH/AM) 365/385°C. Check all furnace controls for proper functioning.
- Adjust draw-off of Heavy Naphtha, Kerosene, LVGO and HVGO to maintain respective draw off temperatures.
- Make adjustment to the stripping steam in the main column bottom and side strippers.
- Raise circulating refluxes proportionately to maintain column temperature profile.
- Adjust RCO draw off to maintain main column bottom level and adjust draw off of LVGO, HVGO and Slop Distillate in the Vacuum Column. Raise the circulating refluxes of vacuum column proportionately.
- Steady out all parameters including vacuum furnace controls.
- Check samples of all products. If any product goes off spec. divert it to slop tank and make adjustment as detailed under operating variables to bring the product to specification.



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- Raise the throughput further only when all products are on-spec. Raise the throughput to maximum by steps with the procedure outlined above. Establish operating conditions in each step.
- Keep watch on the running equipment, heaters etc. Look for leaks and any abnormalities.
- Take hourly log readings and report any abnormal conditions to the supervisor immediately.
- Keep watch on ejector performance. Adjust water flow to the condensers, and product coolers to achieve required temperature, pressure, vacuum etc.

Refer to section which shows the normal operating conditions and make necessary adjustment on operating parameters.

Chemical Injection to Atmospheric Section

Start caustic injection pumps 11-P-13 A/B/C and inject caustic (about 5% solution in water) at the suction of crude charge pumps 11-P-01 A/B and crude booster pumps 11-PM-02 A/B. Rate of caustic injection should be limited to a maximum to avoid caustic embrittlement of downstream equipment and process lines at higher temperatures. Line up ammonia cylinders and inject ammonia gas in to the crude column overhead vapor line. Adjust the rate of injection of ammonia to such an extent that the crude column overhead reflux drum water pH is around 6.5 ± 0.2 . Start corrosion inhibitor dosing simultaneously and adjust the rate of injection as per requirement.

12.3.3 CDU/VDU Startup Procedure after Short Shutdown

a) Pre startup checks

- i) Check all the pump couplings, direction of rotation, lube oil levels and BCW flow to all pumps.
- ii) Check all the utilities and inform the respective units/sections about their consumption
- iii) Check then proper functioning of fire fighting equipment in unit
- iv) Check all the line-ups

b) Startup

The startup procedure is similar to that discussed in feed cut in and stabilization section with slight modifications. The modifications are

- i) The 11-F-01 COT can be raised at the rate of 120°C/hr
- ii) Hot bolting is not required in the vacuum section.



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During T&I normally many agencies from various departments like projects, maintenance electrical, instrumentation, rotary, onsite, off-site, civil, Technical (PAD, Inspection, MES) and minor projects carry out their respective jobs in parallel such that the down time of the unit / equipment is optimally used for completing various planned jobs in time.

Normally towards the end of the T&I under pressure of time to meet planned schedules, situations do occur with chances of improper completion of jobs as per standard practice. If these gaps are not corrected through a proper system of checking, it induces unintended and unknown amount of risk into the startup process which could lead to major accidents or costly down time.

Thus need exists to do a comprehensive check from various angles by respective disciplines, when the unit is getting ready for startup. In the recent past, there were instances in so many refineries, in which the unit had to be immediately shut down due to the need to change improper gaskets which remained in position without proper checking.

Start-up clearance procedure to be followed to ensure safe, smooth and incident free startup of the Unit after planned Turnaround & Inspection (T&I). The start-up clearance procedure is intended to ensure that the plant equipment is properly closed as per checklist and restored to normalcy for a safe start-up and operation after T&I. In other words, it ensures that no additional risk is induced into the whole system due to improper work during T&I. This start up clearance is a prerequisite for unit / equipment start-up and shall be completed before introducing any hydrocarbon.

13.1 DEFINITION OF ‘START-UP CLEARANCE’:

“Startup clearance” is a formal means of assurance by various functional disciplines about the satisfactory and proper completion of their planned respective functional activities during the T & I. This clearance serves as assurance of non-addition of any safety risk and also authorization for the commencement of startup activity. This clearance is normally issued after thorough checking for proper completion of all planned activities as per work procedure.



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Steps for unit start-up post T&I consists of the following:

1. Reviewing and assessment of T&I jobs completion status.
2. Seeking 'start-up clearance' from each section of concerned dept.
3. Receiving the "start-up clearance" and 'exception job's list' (if any)
4. Obtaining approval from HOD (Maint. & Opns.) For start-up, pending 'exception jobs'.
5. Obtaining 'final start-up clearance' from division head - operations.
6. Proceeding for start-up.

1. Reviewing and assessment of T&I jobs completion status:

The unit Manager (Operations) shall take assessment of the readiness of the unit for start-up and intimate concerned sections of Maint. / Tech. / Projects seeking **"Start-up clearance"**. Keeping in mind the nature of jobs carried out in the unit during the period of shut down, he would originate-

- a. **Letter of Request for "Start-up Clearance"** for obtaining full-fledged start up clearance from all concerned sections (4 days prior to start-up date).
 - b. **"Waiver for Start-up Clearance"**, if any, for obtaining 'Waiver' for some sections whose activities were minor / nil during the S/D. The waiver shall be adjudged and approved by Division Head - Operations. Letter to be initiated 4 days prior to start-up date (parallel to case- a.)
2. Seeking 'start-up clearance' from each section of concerned dept.

All the concerned sections (Maintenance / Tech. / Projects) should be notified 96 hours in advance by UAT Leader (Manager - Operations), indicating clearly the date and time of the proposed commencement of start-up activities and indicating the time by which this 'clearance' should reach him back.

3. Receiving the "start-up clearance" and 'exception job's list' (if any)

After receiving the intimation about startup plan the concerned Section Head (Maint. / Tech. / Projects) should immediately assign the respective UAT member (or alternate designated person) for thorough & systematic checking (through check list) of relevant activities carried out by them. They shall ensure internally that proper and detailed system of checking is put in place for thorough checking prior to issuance of 'Start up clearance'



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4. Obtaining approval from HOD (Maint. & Opns.) For startup, pending 'exception jobs' (if any).

The 'Startup Clearance' from individual sections of Maint. / Tech. / Projects to be taken. Each of the respective section head has to develop detailed / elaborate internal check list (which need not be sent to operations) using which a detailed checking should precede the issuance of 'Start up clearance'. Developing and implementing the required 'internal checking system' is the responsibility of respective functional section heads of Maint. / Tech. / Projects. It is expected that the 'Start up clearance' shall reach the Unit Area Team leader (Opns) within 24 ~ 36 hours of receipt of intimation (at least 48 hours before commencement of startup).

5. Obtaining 'final start-up clearance' from division head – operation

The final 'Startup Clearance' to be taken from the Division Head-Operations. After obtaining clearance only actual field activities for startup are to be started.

6. Proceeding for start-up :

Check the unit for completion of mechanical work against P&ID with new changes.

1. Remove all construction debris lying around in the unit and clean up the area.
2. Install blinds as per master blind list. Each blind removal and insertion of blind should be entered in the blind register.
3. Safety valves should be kept blinded during flushing and re-installed afterwards. These should be shop tested and set at the stipulated values.
4. Ensure that underground sewerage system is in working condition. Clear plugging, if any. Check by flushing with water.
5. Check that communication between units, control room, Offsite and utilities are complete and in working condition.
6. Ensure that the required lube oil, grease and other consumable are available in the unit.
7. The proceedings for startup activities are as below
 - a. COMMISSIONING OF UTILITIES
 - b. WATER FLUSHING OF PROCESS LINES
 - c. FUNCTIONING TEST
 - d. CALIBRATION OF INSTRUMENTS
 - e. FURNACE DRYING
 - f. TIGHTNESS TEST
 - g. CHARGING OF CHEMICALS



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The explanations of above activities are mentioned in chapter number 10. Please refer chapter number 10.

8. After these preliminary activities are completed the startup procedure is same as the normal startup procedure. The steps to be followed as below
 - a. Removal of air from the unit by steaming
 - b. Tightness testing under steam purge
 - c. Backing in fuel gas
 - d. Cold oil circulation and removal of water.
 - e. Hot oil circulation.
 - f. Bringing the unit on stream.

The explanations of above activities are mentioned in chapter number 12. Please refer chapter number 12.

Note: Before going for the start up the startup check list to be filled by respective person as mentioned in the document. The startup check list document is available in chapter number 34.



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All the equipment used in the unit has some operating limits. The limits of any equipment, line, etc should always be respected. The limiting parameter for all equipments will be different and the values should always be kept in mind to follow them. The deviation will result in severe consequences like unit upsets as well as equipment damage. This may also result in unsafe working conditions. The values can be obtained from the design package of the unit or the soft copy present in form MS- Excel sheets.

14.1 PUMPS:

The pumps have their operating parameters provided by the vendor. Every machine is designed to work at some optimum condition and also the maximum deviation it can handle from its optimum value. For pumps, the operating parameters are maximum and minimum flow, suction pressure and discharge pressure, differential head and net positive suction head.

For a pump, the vendor provides the value of normal flow as well as maximum and minimum flow. If the pump flow is within the limits of maximum and minimum flow, then it will be healthy. If the pump flow is beyond its maximum limit, it means that the pump is overloaded and it is pumping liquid more than what it has been designed. So there is a chance of the pump tripping on overload. In such conditions, pump amperes has to be continuously monitored and compared with full load current. The pump flow has to be more than the minimum flow so otherwise the pump starts losing suction and will suffer more vibrations. This damages the pump bearings, seal as well as can harm the pump impellers.

The pump suction pressure also has to be more than its minimum suction pressure. If the pump doesn't get suction at the desired pressure, it will also damage the pump bearings, as well as the mechanical seal and if the problem is not attended and the pump is kept running, the impellers will also get damaged. The pump discharge pressure also cannot be violated as it will overload the pump in trying to increase the discharge pressure.

The pump differential head and NPSH are required to know the exact pumping capacity of the pump and if the pump is not able to meet them, it is an indication that something is wrong with pumping capacity.



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14.2 HEATERS:

Operating temperatures and pressures of the heater is the temperature and pressure at which the process liquid enters and leaves the heater. If the inlet temperature of the process liquid in the heater is low, the energy required to raise it to required temperature will be very high. Sometimes it will not even be possible for the operator to bring the COT to the desired value due to limiting parameters like arch and skin temperatures. The operating pressure also has to be between the limits because high pressure will develop chances of tube rupture and leaks. Low inlet pressure means the velocity of process fluids in the line will be slow and that will lead to higher cracking and coking of heater tubes. Outlet pressure also need to be maintained at a desired value and drop in the outlet pressure when inlet pressure is normal, is an indication of heater tubes getting coked up.

Heater TURNDOWN value is 50%. It means that if the heater feed is less than 50% of its operating value, then it cannot be kept in service and has to be shutdown. This is for meeting minimum flow requirements of pass flows. The fuel gas pressure has to be maintained between its maximum and minimum values. If it goes below its minimum value then the gas flow will experience back pressure and the supply will get stopped creating unsafe conditions. If the pressure is very high, it will damage the gas rings and the tip size will also get enlarged or the opening will get eroded. The same is for fuel oil because very high steam pressure will erode the oil gun tips. Very less fuel oil pressure will congeal the fuel oil lines and thus will stop the fuel oil flow completely.

Fuel oil viscosity should also meet the desired specifications because a very high fuel oil viscosity again will stop the flow of oil in the lines. Atomizing steam pressure also has to be maintained because a high steam pressure will erode the tip of the gun and a very low steam pressure means that atomizing will not take place. There has to be a minimum amount of difference between the pressure of oil and steam so that it can atomize fuel oil. The fuel oil will not ignite until it is in atomized form. So the atomizing steam has to be properly used for best results in the heater.

14.3 HEAT EXCHANGERS:

Heat exchangers are used as preheat exchangers, coolers and condensers in the unit. Even though the usage may be different, operating parameters for all of them are mostly same. They have inlet and outlet temperatures which need to be maintained. Every preheat exchanger needs to maintain the outlet temperature because the outlet of the exchanger will be the inlet stream to some other equipment. If the outlet doesn't meet the operating value i.e.



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the deviation is very high, the temperature of the stream will be very less or very high for the next equipment. If the temperature of the inlet is very less, it becomes very difficult to attain the outlet temperature because the requirement of temperature raise will exceed the duty that exchanger can provide. Also if the temperatures are very high at the inlet and outlet, they may result in channel section leaks, and also gasket may give up resulting into emergencies.

The pressure on the shell side and tube sides are also monitored as the operating procedures. If the pressure is too high on either side, it may result into leaks and thus creating unnecessary emergencies and equipment damage. A very less pressure on either side means sufficient flow will not be established and the heat transfer will not be effective. Thus the operating pressure should be maintained as mentioned in the design manual to get maximum output from every exchanger under safe limits.

14.4 COLUMNS:

The column limits on the conditions like the column DP and the product draw-off temperatures. The parameters like DP are indicators for column internal condition like vapour load and flooding. High column DP is indicative that the column is flooding or the vapour load in the column is very high. For the total height of 11-C-01, the column DP should not exceed 0.40 kg/cm². More column DP will result in the colour of products going off spec as the heavier ends will contaminate the lighter products by rising up in the column. Also a very high vapour velocity may erode the internal surfaces of the column and also cause damage to the trays.

Column draw-off temperature for every products draw-off is specified in the design package. The draw-off temperatures affect the qualities of the products. A very high draw-off temperature of a product will make it heavier. The product may become off on distillation. Also if the temperature at a particular tray is high, the CR pumps design suction temperature may be violated and the pump may get damaged. A very low draw-off temperature means the product will become lighter and the IBP and the flash of it may go down.

14.5 CDU-II ALARM VALUES:

TAG NO.	DESCRIPTION	SETTING
LL1401	Atmos column bottom level low	40
LH1401	Atmos column bottom level high	70
LL2201	Vac bottom low switch	40
LH2201	Vac. Column bottom high	70



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LL2203	Vac. Column HVGO level low	30
LL2204	Vac. Column LVGO level low	30
LL1501	Stabilizer bottom level low	30
LH1501	Stabiliser bottom level high	75
LL1902	PFD level low	30
LH1902	PFD level high	80
LL2202	Vac. Slopcut level low	30
LL2401	Tempered water low	30
LH2401	Tempered water high	80
LL11P11A	LPG pump seal level switch A	-
LL11P11B	LPG pump seal level switch B	-
PH1807	11F01 ID Fan Suction Pr. Indication	-60
PH1807A	11F01IDFan Suc. Pr.high	-40
PH2507	ID fan suction Pr.high	-60
PH2507A	ID fan suction Pr.high	-40
PL1806	11F01FDFan dis.Pr.Low	15(18)
PL2506	FD fan Discharge Pr.Low	8(14)
PR1808	11F01 Pr. Control	+1.0
PR1809	11F01Pr.Control(switch)	+2.95
PR2508	12F1 Pr.Control	+1.0
PO2509	12F1 Arch Pr trip	+2.95
11-PAL-301	11F1 fuel oil pressure low	2.77
11-PAL-303	11F1 fuel gas pressure low	0.2
12-PAL-106	12F1 fuel oil pressure low	2.77
12-PAL-108	12F1 fuel gas pressure low	0.2
11FR301	Crude to Pass A	42
11FR302	Crude to Pass B	42
11FR303	Crude to Pass C	42
11FR304	Crude to Pass D	42
12FR101	RCO to Pass A	14
12FR102	RCO to Pass B	14
12FR103	RCO to Pass C	14
12FR104	RCO to Pass D	14



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15.1 OPERATING VARIABLES OF ATMOSPHERIC SECTION:

It is important that operation of CDU be conducted to produce products of desired quality. At the same time appropriate controls should be exercised on certain parameters to prolong the life of the equipment. The following discussion will give the guidelines about affect of the variables and measures to be taken to achieve the desired result.

15.1.1 Desalter Operating Variables:

Only operating experience with desalter can determine optimum operating conditions. No two crudes behave alike at the same desalting conditions, but all are affected similarly by change in desalting conditions.

a) *Water Injection and Pressure Drop:*

Water injection should be started only after crude reaches a specified level and power is switched on to the grid. Initially the injection rate should be limited between 2-6% of crude flow rate and point of injection should be just ahead of the emulsifying valve. Pressure drop across the emulsifying valve should be 0.3-1.0 kg/cm². Higher pressure drop ensures more efficient contact between the salt in the crude and the injection water. Too high pressure drop will result in excessive emulsification and poor separation of oil and water, resulting in carry-over of water in desalted crude. Injection of water can also be done before feed pump suction. Injection at feed pump suction point results in maximum contact and also prevents the sediments from settling in the exchanger tubes and fouling them. But care should be taken such that the intense shearing agitation in the preheat train does not create so tight an emulsion that cannot be resolved in the Desalter. The severe shearing effect due to the crude pump impellers should also be considered here. The quality of water is a very important aspect.

b) *Oil water Interface:*

The oil water interface level should be kept below the centre line of the vessel. Incorrect operation of the interface level controller can result in more water in the desalted crude due to less hold up time available for oil (high interface level) and more oil carryover in brine due to less hold up time available for water (low water level). Also too high interface level may put watery mixture up between the electrodes and cause them to short out.



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c) *Desalter Vessel Pressure:*

The pressure in the vessel should be maintained about 10.5 kg/cm². A low pressure will cause vaporization of the crude and high pressure will result in lifting of the safety valve on desalter.

d) *Desalter Temperature:*

Temperature is another important variable which affects oil water separation in desalter. Most crudes have an optimum operating temperature range of 120-130 °C. Lower the temperature higher the viscosities of the oil which slows down the separation rate. As conductivity of crude increases with temperature, operating temperature beyond the range will lead to drop in grid voltage and high amperage which imposes limitation on good separation. Excessive amperage will eventually cause the circuit breaker to open, removing the grid voltage and rendering the electrical system inoperable until the thermal delay is closed. Moreover very high temperature may lead to vaporization of crude in desalter.

e) *Demulsifier Injection:*

Stable emulsions can also be broken by use of demulsifying agents. The amount of chemicals required depends on the nature of emulsion, type of crude and other operating conditions like residence time, temperature, etc. tests should be made to ascertain the required chemical injection rate for optimum operation of desalting unit.

f) *Voltage and Amperage:*

The electrical panel houses 3 numbers of voltmeters and ammeters. The voltmeter gives the voltage across the primary circuit of the transformer. The ammeter gives the current flow. These meters give an indication of the performance of the grids inside the desalter. Incase, if crude/water emulsion is too tightly bound or if the interface level is too high the amperage will increase and voltage will drop. Take corrective action to break the emulsion or reduce the interface level.

15.1.2 Heater Outlet Temperature and Column Pressure:

The quantity of crude oil vaporized during its passage through the heater depends on transfer temperature and pressure at the flash zone of the column. In order to achieve proper recovery of distillates little over flash is maintained, by keeping the transfer temperature slightly



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higher. This flow is about 6% volume of crude charge. This also indicates the presence of liquid levels in the trays down below the diesel draw off for avoiding dry operations of these trays (nos. 7 to 11). Maintaining high overhead flash rate will result in more consumption of energy. Heater outlet temperature is controlled by 11-TRC-301. Lower temperature will not give desired distillate recovery, bottom product RCO will be lighter and all side draw offs will also be proportionate to lighter. Higher than normal temperature enhances cracking possibility and at the same time specification of every product may not be met.

The pressure in the column is maintained by split range pressure controller 11-PRC-409 A/B. A low pressure aids in greater vaporization. All products will be heavier and there will be gas loss from reflux drum. Higher than normal pressure will have reverse effects. Efforts should be made to operate the column at the designed pressure of 0.6 kg/cm²g at the reflux drum. Adjust cooling water flow in the cooler 11-E-17 A/B/C/D/E/F/G/H in such a way that there is total condensation and both the control valves (11-PV-409 A/B) of split range controller remain shut.

15.1.3 Crude Column Top Temperature:

The column top temperature is controlled by regulating amount of overhead reflux through 11-FRC-403. Top temperature is continuously recorded by the recorder. Lowering of top temperature will reduce FBP of naphtha and flash point of heavy naphtha. Too low a temperature will start stem condensation at the top section of the column which may likely to increase rate of corrosion at the top. Raising the temperature will increase FBP of overhead naphtha and IBP (flash) of Heavy Naphtha.

15.1.4 Pump Around Flows:

The pump around/ circulating reflux serve mainly in withdrawal of heat from the column and to reduce the vapor liquid traffic in the appropriate section of the distillation column. There are three circulating flows- Top Pump Around (TPA), KERO circulating reflux and diesel circulating reflux. These flows are respectively controlled by flow meters. 11-FRC-404/405/408. The return temperature is maintained by operating the respective exchangers. Return temperature are indicated by 11-TI -414(TPA). 11-TI-413 (KERO CR) and 11-TI-412(Diesel CR). A high TPA flow will result in decrease requirement of overhead reflux and affect the quality of light naphtha. The overhead condenser duty will come down as there will be correspondingly less O/H product. Similarly a high Kero CR flow will tend to lower the plate temperature of heavy naphtha and kerosene draw off resulting in lighter product in these trays. The gap between naphtha and kerosene will decrease.



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Likewise a high diesel CR will tend to lower the draw off temperature of kerosene and diesel. Increase of circulating reflux will result in higher crude preheat temperatures by greater recovery in heat exchanger train.

15.1.5 Product Withdrawal Temperatures:

The withdrawal temperature of a product from the column influences the end product of the product. This is determined by quantity of the product withdrawn from the stripper. An increase in withdrawal rate of the side stream increases the withdrawal temperatures and the end point of all side stream lower down the column unless withdrawal rate lower down the column are reduced correspondingly. For example, if kerosene withdrawal rate is increased, the internal reflux in the trays below the draw off tray will be reduced which will lead to flow of heavier vapors above the tray. This increases the end point of kerosene. If diesel withdrawal rate is not reduced to maintain its plate temperature, its initial boiling point (flash) will go up.

Similar reverse action takes place when withdrawal temperature is lowered by reducing the quantity of withdrawal.

15.1.6 Stripping Steam:

a. Atmospheric Column Stripping Steam:

At 10.5 kg/cm²g pressure and 350 °C, superheated steam is used to strip lighter fraction from the reduced crude in the lower part of the crude tower. Design steam rate is about 4585 kg/hr for Basrah crude. This is assumed to be optimum rate for economical stripping and should not be varied much. Lowering the rate below the optimum may leave some lighter component in the reduced crude and is undesirable. Exceeding the design rate might cause entrainment of reduced crude into the diesel because of excessive vapor velocity and also will overload the over head condenser system. The flow of steam controlled by 11-FRC-401.

b. Stripping steam in heavy naphtha, kerosene and diesel stripper:

The initial boiling points and flash points of heavy Naphtha, Kerosene and Diesel products are controlled to some extent by varying the stripping steam rate to stripper 11-C-02/03/04 respectively. Steam at 10.5 kg/cm²g and 350 °C used for these strippers. Steam flows are indicated by 11-FI409/11-FI-408/ 11-FI-407 and regulated by respective control valves.



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It is advisable not to exceed the steam flow rate of its designed value viz. Heavy naphtha stripper -444 kg/hr. kerosene stripper-1496 kg/hr and diesel stripper-1906 kg/hr, as this will tend to lift some of the high boiling materials. In case if the desired flash point could not be reached by designing rate of stripping steam, the draw off temperature at the product just above it to be increased to enhance its flash.

15.1.7 Corrosion Control of Overhead system of Distillation Column:

Hydrochloric acid formed from the hydrolysis of salt present in the crude and hydrogen sulphide formed dissolved in the crude (formed from the dissociation of heavy sulphur compounds present in crude), goes to the overhead system. Both form acid solutions which are very corrosive. Measures must be taken to overcome their effects.

The overhead system including condensers and reflux drum are made of carbon steel. Only to protect this section caustic ammonia solution and corrosion inhibitors are added at the following points:

Caustic injection:

3. Suction of crude booster pump i.e. ahead of desalted crude preheat trains.
4. After the preheat trains before booster pump 11-PM-02A/B suction

Ammonia injection:

4. Suction of crude booster pump ahead of desalted crude preheat train
5. Column top reflux line.
6. Crude column overhead vapor line.

Corrosion inhibitor injection

3. Column top reflux line.
4. Crude column overhead vapor line.

The idea of injecting caustic and ammonia at the outlet of Desalter is for better mixing of these chemicals with crude and neutralizes the acids/acid salts mainly HCl and H₂S as soon as it is formed (120 °C and above). Chance of H₂S formation at this temperature is remote. The reaction product is sodium and ammonium salts goes along with the educed crude. The balance acids and acids gases if any will go up to the overhead system where ammonia or ammonium solution is injected either along with reflux or in the overhead vapor line fro neutralization. Amount of ammonia should be controlled in such a way that Ph of reflux drum sour water remains around 6.0 to 6.5. Injection of caustic and ammonia at the



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outlet of Desalter should be maintained in such a way that the salt formation should be low in the overhead which might scale up the overhead condensers tubes.

A slightly acidic condition of the overhead system is desirable to keep ammonium salts in solution, which if precipitates would foul and plug the condensers.

Corrosion against slightly acidic condition is minimized by adding corrosion inhibitors in the overhead line. The inhibitor is also added in reflux line. The amount of inhibitor injected depends upon the type of inhibitor used, and generally specified by vendor. However, slight adjustment is made by operating personnel depending upon the content in the reflux drum water. These inhibitors are filming organic compounds (amines) which covers entire metal surface of the system with a thin film. This prevents contact of corrosive water with metal.

Top section of the column is also benefited from the injection of inhibitors mainly in the reflux line/ these inhibitors are high boiling compounds and can perform satisfactorily at upper tower temperatures.

15.1.8 Stabilizer Temperature and Pressure:

Stabilizer removes the majority of butane and lighter hydrocarbons from the naphtha stream. These are recovered as overhead LPG product. High top temperature will make overhead product heavier, even pentanes may be carried into LPG, making it off-spec. Lower temperature will reduce LPG, make. Vapor pressure of LPG may go beyond the specified limit if top temperature is too low.

Bottom temperature if too low will result in higher than allowable vapor pressure (RVP) of naphtha and at the same time it will reduce LPG make. Low pressure in the column will cause higher amount of hydrocarbons (propane and butanes) to escape into fuel gas system. This has got similar effect as that of higher temperature in the column.

15.2 OPERATING VARIABLES OF VACUUM SECTION:

The following variables of the vacuum column influence the quality of the products and should be controlled to meet the product specification.



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15.2.1 Transfer Line Temperature:

The transfer line temperature is controlled by 12-TRC-133 on the outlet of the vacuum heater, and should be adjusted to maintain the flash zone temperature around 395 °C. this temperature determines the degree of vaporization and the level of heat in the liquid vapor mixture, entering the vacuum tower for fabrication. This temperature will be varied depending on the quantities of the desired distillates. But the furnace outlet temperature should not be allowed to go beyond the designed limit of 415 °C after which the degree of cracking increases rapidly. Detrimental effect of cracking or coke deposition on heater tubes, transfer line and bottom line sections. It also increases quantity of non-condensable going to the overhead system. The cracking can also have a detrimental effect in the curing qualities of asphalt. Too low a transfer temperature will result in lower yields of vacuum distillates and flash point of distillate cut may be lower.

15.2.2 Vacuum Column Pressure:

The column top pressure is controlled by 12-PRC-306 which recycles some non-condensables to the ejector 12-J-01 A/B/C inlet line. The top pressure should be maintained around 7-9 mm Hg absolute. Increase in pressure will result in reduced yield of vacuum distillates and may lead to cracking of the feed. Reduction in top pressure further may result in carryover of LVGO into the ejector system.

15.2.3 Flash zone pressure and temperature:

It is of paramount importance to maintain a high vacuum and temperature at the flash zone with in the prescribed limit to obtain maximum yield distillates. The designed flash zone pressure is 24 mmHg and temperature 395 °C. Fluctuation of vacuum will affect the product quality adversely besides producing mechanical stress on column internals. Increasing the flash zone temperature will result in greater yield distillates but cracking possibility is enhanced. If for any reason, vacuum starts falling sharply, firing in the heater should be reduced to bring down temperature of feed.

15.2.4 LVGO system:

The LVGO system is a combination of LVGO product, LVGO circulating reflux and internal reflux for HVGO packing. The circulating reflux is sprayed over the LVGO packing through a distributor. This stream is taken from the column and a part of LVGO after exchanging its



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sensible heat with crude in 11-E-07 and cooled in coolers 11-E-22/22A in parallel, it is then returned at a temperature of 65 °C to the column (12-C-01) under flow controller 12-FRC-205 to maintain column to temperature of 80 °C. Because of crude throughput maximization, it was observed that the Vac. Column temperature was frequently crossing the desired temperature ultimately resulting in under control, LVGO CR from 11-E-07 outlet is modified accordingly and routed to 12-E-12 A (top cooler), the outlet of 12-E-12 A routed to 11-E-22/22A in parallel. Increasing the reflux will reduce the top temperature which will simply increase the energy consumption. Reduction in reflux rate will increase the top temperature that will overload the ejector and increase the slop production thereby losing LVGO yield.

The internal reflux to HVGO packing's from LVGO draw off tray is maintained by diverting a part of the LVGO from 12-PM-04 A/B pump discharge. This flow is regulated by 12-FRC-201. The LVGO product withdrawal rate is regulated by 12-LIC-205, which controls the LVGO level. LVGO rundown flow is indicated by F2403R. LVGO draw off temperature is 213 °C. the draw off temperature as well as LVGO product rate can be varied by increasing or decreasing either LVGO internal reflux or HVGO circulating reflux. Increase in reflux means reduction of LVGO draw off temperature and LVGO product flow rate and vice-versa.

A software switch is provided fro LVGO system when it is being routed to either HVGO/LDO. When the switch is kept in position 1, LVGO level will be controlled by its own LVC (LI2205). When the switch is kept in position 2, LVGO level i.e. LI2205 will be cascaded with FR2404 (i.e. LVGO to LDO/diesel).

15.2.5 HVGO system:

The HVGO system is a combination of Wash liquid for wash zone packing, circulating reflux (HVGO packing) and HVGO product. There wash liquid is given to the wash zone packing for avoiding chocking of the packing area because of heavy asphaltenes. The chocking of the packing zone would result in higher differential pressure across this zone and this will adversely affect the column performance. About 80 m³/hr of HVGO is normally supplied as wash oil, the flow of which is regulated by 12-FRC-202. An increase in wash oil flow will reduce the carry over of asphaltenes in HVGO stream. More the desired quantity of this stream may adversely affect the quality/yield of Vacuum residue. Less quantity of Wash Oil may result in the carry over of heavy asphaltenes into HVGO stream.



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15.2.6 Slop Distillate and Recycle:

The object to provide slop recycle to furnace flow is to get desired over flash so as to ensure proper recovery of distillates. A higher recycle rate will unnecessarily increase the energy consumption. Recycle + slop distillates are draw from the chimney tray below the wash zone packing. The draw-off rate of slop distillate product is regulated by 12-LIC-203 which controls the level in the draw off tray. The recycle stream flow rate is controlled by 12-FRC-102. Recycle rate at normal throughput is about 12.0 M3/hr.

15.2.8 Quench:

Quench flow is a slip of vacuum residue at 250 °C obtained after exchanger 12-E-03 and the flow is regulated by 12FRC-204. Its temperature is indicated by 12-TI-202. The purpose of providing quench is to prevent, coke formation at the bottom of the column by quickly cooling Vacuum Residue from 395 °C to 350 °C. A lower temperature than this, i.e. higher quench is not wanted because of disproportionate increase of energy loss, whereas lower quench flow may lead to coke formation due to cracking.

15.3 FEED TANK SWITCH OVER IN CDU.

15.3.1 LOW SULPHUR TO HIGH SULPHUR

Feed tank changeover Standing Instructions: SI20

Following procedural steps are to be adopted before change-over of feed tank:

- Reduce feed rate to 440 m³/hr
- Suspend water injection to desalter.
- Desalter try-cock levels to be set between 70-75% and to be confirmed with try-cock levels (2&3). In case of any Desalter grid voltage / amperes fluctuations, level can reduce to suit. But it should be informed to the unit manager.
- Wait for an hour after tank switch over to make changes in feed rate.
- Adjust CRs and yields accordingly.
- Inform FCCU /MEROX/DHDS/YSF.

Preflash Drum

- Slowly decrease the vaporization watching product colors in general and diesel color in particular.
- Increase the flow through the furnace keeping a watch on PFD level.



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➤ Atmospheric Section

- Increase the Furnace COT gradually as per technical data.
- Make changes in reflux rates to attain design column pressure and vapor line temperature.
- Keep watch on bottom level and adjust Vac feed accordingly.
- Stop additional reflux and product pumps.

➤ Vacuum section

- Increase Furnace COT as per technical data.
- Adjust refluxes to maintain column temperature profile.
- Reduce SR pump spillback to the minimum and wide open the discharge valve.
- Adjust slop cut pump spillback also.
- Stop additional LVGO pump.
- Place additional tempered water pump if required.
- Take into service additional SR/Tempered water coolers.
- Change SR quench flow rate to suit the column bottom temperature
- Route Slop cut to SR/HVGO via CDU 1 cooler box to decrease load on SR/Tempered water coolers.

➤ Chemical injection

- Introduce caustic solution injection
- Adjust Atmos and Vac neutralizer rates to attain designed pH values of sour waters.

➤ Product routing

- Route Diesel R/D to sour DSL storage/DHDS in coordination with OM&S/DHDS
- Route SR to FO/VBU storage or BBU in coordination with OM&S.

➤ Sampling

- Send product & crude (both FC & DC) samples after four hours after tank change over
- Samples to be tested for Sulphur also as per OM&S requirement.



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15.3.2 HIGH SULPHUR TO LOW SULPHUR

- Follow **standing instructions** for Feed tank changeover **SI20**.
- Pre-flash Drum
 - a. Slowly increase the vaporization watching product colors in general and diesel color in particular.
 - b. Decrease the flow through the furnace keeping a watch on PFD level.
- Atmospheric Section
 - a. Decrease the Furnace COT gradually as per technical data.
 - b. Make changes in reflux rates to attain design column pressure and vapor line temperature.
 - c. Keep watch on bottom level and adjust Vac feed accordingly.
 - d. Start additional reflux pump and product pumps.
- Vacuum section
 - 1) Decrease Furnace COT as per technical data.
 - 2) Adjust refluxes to maintain Column temperature profile.
 - 3) Increase SR pump spillback to the maximum and adjust the discharge valve.
 - 4) Adjust slop cut pump spillback also.
 - 5) Stop additional tempered water pump if required.
 - 6) Take out additional SR/Tempered water coolers service.
 - 7) Change SR quench flow rate to suit the column bottom temperature.
 - 8) Route Slop cut to SR to maintain SR level.
- Chemical injection
 - 1) Caustic solution injection.
 - 2) Adjust Atmos and Vac neutralizer rates to attain designed pH values of sour waters.



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➤ Product routing

- 1) Route Diesel R/D to sour DSL storage/DHDS in coordination with OM&S/DHDS.
- 2) Route SR to FO/VBU storage or BBU in coordination with OM&S.

➤ Sampling

- 1) Send product & crude (both FC & DC) samples after four hours after tank change over.
- 2) Samples to be tested for Sulphur also as per OM&S requirement.

15.4 Standing Instruction on Heater fuel oil gun cleaning (SI 38):

15.4.1 Fuel Oil burners are required to be cleaned / repaired, if

- a) Fuel oil fires to burners pressure is holding on higher side even for lesser flow
- b) Fire pattern is not good – Sooty, flickering and impingement in spite of adjustment.
- c) Fuel Oil dripping from the burner.
- d) Coke formation on the fuel oil burners.
- e) Leaks on the Fuel oil or steam hoses of the oil burners.

15.4.2 REQUIREMENTS FOR OIL GUN DROPPING:

- a. Asbestos Gloves
- b. Face shield / Goggles.
- c. Bench Vice with necessary arrangements.
- d. Proper sized pipe wrenches.
- e. Two number of lead gaskets
- f. DCS shift in charge clearance(after stack damper opening)

15.4.3 STANDING INSTRUCTIONS:

- o Inform DCS shift in charge before going to start oil gun dropping activity and ensure that stack damper is in open condition, during oil gun dropping & placing back the oil fire.
- o Remove the oil fire and ensure that gas fire and pilot are there in the particular burner. Adjust The firing in other burners.



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- Open the purge steam valve fully to purge the oil gun. (Ensure only dry steam is admitted)
- When all signs of flame disappear, close the purge steam & atomizing steam valve.
- Open burner drain to drain the oil present inside the burner assembly.
- Ensure that no oil accumulation inside burner by viewing through burner view Ports.
- Do not allow oil gun more time without steam purging when gas fire is in-side as the oil burner tip will get damaged.
- If oil accumulation is not there, drop the oil gun by wearing face shield & asbestos Gloves, after ensuring that the steam valves (Atomizing & purge steam) are not passing. After removing the gun, fix the sealing plate.
- If oil accumulation is there then remove all fires and close air registers for cooling before dropping.
- After dropping the gun soak it in the soaking pit for 1 hour (after cooling).
- After soaking, fix the oil gun in the bench vice and open the gun parts one by one, clean them and compare them with the new spare parts.
- Replace the worn out or damaged parts of oil gun (Oil tip, atomiser etc).
- After cleaning the oil gun, care should be taken to reassemble the oil gun in the reverse order of dismantling.
- Assemble the parts of the burner slowly by taking care of the threaded nipples and gaskets.
- ***Over tightening and wrong threading can damage the parts.***
 - After assembling the parts check for leaks if any by using steam.
 - Fix the oil gun with new lead gaskets & Inform DCS shift in charge before light up of the cleaned burner.
 - Purge the gun with steam for testing leaks, if any.
 - If no leaks are present, then slowly open the fuel oil block valves after ensuring gas fire & Atomising steam is present in the particular burner.
 - Carry out fine adjustments, so that proper flame is there without any impingement.
 - Update the burner spares register.



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15.5 Standing Instruction to apprise Merox DCS shift in charge in case of sudden fluctuation in unit sour water flow to SWSU(SI 37):

15.5.1. OBJECTIVE:

To appraise Merox DCS shift in charge in case of sudden fluctuation in unit sour water flow to SWSU (say $\geq 2\text{m}^3/\text{hr}$), so that corrective actions can be taken and the following can be avoided.

- Disturbances in unit & stripper conditions, leads to poor stripping operation.
- Carryovers of sour water with oil into Acid Gas flare KOD (280V05) due to sudden rise in level of surge drum.
- Sudden rise in Acid Gas flare KOD level, resulting in spillage of oil around the flare, with consequent fires in the flare vicinity.

15.5.2.BACKGROUND:

- Sour water from FCCU-I, FCCU-II and VBU goes into the sour water Surge Drum(16V-05) in SWSU-I.
- Rich Sour water surge drum (16V-05) is a horizontal vessel provided with baffle plates which divide the drum into three compartments.
- Sour water enters the middle compartment, where phase separation takes place. Water, being heavier, underflows to the third compartment. Oil overflows from the top of the baffle to the first compartment. From first compartment oil drains to OWS continuously through siphon loop arrangement.
- The drum is connected to the Acid Gas Flare header so as to vent any disengaged
- Hydrocarbon vapors coming along with the sour water. The drum is provided with PSV-1106 and PSV-1107 which discharge to the Acid Gas Flare header.
- From the third compartment, Stripper Feed pumps 16P-06 A/B transfer the sour water from 16V-05 to the I Stripper Column (16C-04) through preheat exchanger 16E-05A/B/C under flow control FIC-II03.
- Sour waters from CDU-I, VDU-I, CDU-II, VDU-II, CDU-III and VDU-III goes in to the Lean Sour water Surge Drum (16V-12) in SWSU-2. This drum works identical to 16V05.
- The drum is connected to the Acid Gas Flare header so as to vent any disengaged
- Hydrocarbon vapors coming along with the sour water. The drum is provided with PSV-1201 and PSV-1202 which discharge to the Acid Gas Flare header.
- From the third compartment, lean sour water is pumped by 16P-15 A/B to II Stripper Column (16C-05) through a preheating system.



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11. There is an option available in Merox to route sour waters from FCCU-1, FCCU-2 & VBU to lean oil surge drum (16V-12) in SWSU-2.
12. There is an additional option available for FCCU-1/ FCCU-2 to route sour water to MS block SWSU.

15.5.3. STANDING INSTRUCTIONS:

1. Always maintain steady sour water flow to SWSU and apprise Merox DCS shift in charge (4454), when sour water flow fluctuation is greater than 2m³/hr.
2. During unit normal operation, check the sour water samples for any oil carryover at every 2 hours frequency, in case any abnormality is noticed, inform Merox DCS shift in charge and discontinue sour water routing to SWSU immediately. Resume water intake after reconfirming there is no oil carry over.
3. During plant upsets and during boot LT's failure, check the sour water samples for any oil carry over at 15 minutes frequency and in case any abnormality is noticed, inform Merox DCS shift in charge and discontinue sour water routing to SWSU immediately.
4. Resume after reconfirming no oil carryover is there.
5. The above instruction is equally applicable in case of sour water routing from FCCU-I /II to MS block, in that case information should be given to MS block DCS shift in charge (4504).

15.6 Standing Instruction on Car seal Management (SI42):

15.6.1. OBJECTIVE:

- To ensure car seals on the inlet and outlet isolation valves in open position for the PSV's in service.
- To ensure car seals on all the First Aid Fire Hose reels, TSV's, Sour water drain to OWS valves, Critical Utility lines at Battery limits as per the identified list, chemical system drain valves and other important valves as per process requirement.
- To ensure the car seals are maintained on the all the specified valves and their condition is monitored on a regular basis.



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15.6.2 **BACKGROUND:**

- In the past some shift incidents/upsets were experienced due to inadvertent isolation of some critical valves in the unit. To avoid such incident/upsets, car seals to be provided at the following equipment isolation valves.
- **PSV's:** The pressure safety valve (PSV) is a type of valve used to limit the pressure in a system or vessel which can build up by a process upset, instrument or equipment failure or fire. Inadvertent isolation of a PSV will defeat its purpose and therefore periodic checks are to be carried out. Upstream and downstream isolation valves are to be car sealed in open position to prevent such inadvertent closure.
- **Utilities:** Utilities such as Bearing Cooling water supply & return, DM water, Instrument Air, Fuel oil supply & return, Fuel Gas and Nitrogen form an integral part of the Unit Operation. Continuous supply of these utilities is very important and failure of any one of these utilities can lead to partial or total shutdown of the Unit. Inadvertent isolation of these utilities can be prevented by car sealing these utility lines valves in open condition at the Unit Battery limits.
- **FAFHR:** First aid fire hose reels are provided in the Unit to aid in firefighting at times of emergencies. Usage of fire water for other activities like lines flushing and floor cleaning will lead to overloading of ETP's as well as hampering the treatment of the effluents at ETP. Car seals are provided to ensure usage of FAFHR only for the firefighting purposes.
- **Chemical system drain Valves:** Chemicals often require different treatment steps than oily water streams before final disposal, therefore Chemical should not be drained to OWS. Car seals to be fixed on chemical system drain valves to avoid chemicals draining to OWS.
- **Sour water system drain Valves:** Sour water draining to OWS will increase the load on ETP's, therefore Sour should not be drained to OWS. Car seals to be fixed on Sour water system drain valves to avoid Sour water draining to OWS.
- **Other Specified valves as per PDI:** There are some critical equipment or lines in the Unit for which the valves to be either to remain open or in close condition as per the existing process conditions. Car seals are to be provided to valves on such lines or equipment as per PDI instructions.



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15.6.3. STANDING INSTRUCTIONS:

1. Car seals to be provided and periodic checks to be carried out as mentioned below:

Car Seals provision to	Monitoring duration	Standing Instructions
PSV'in Service	Monthly DRJ	Provide car seals to the inlet and outlet valves in open condition.
Utility lines	Weekly DRJ	Car Seals to the following utility lines with valves at Battery limits in open conditions a. Instrument Air b. BCW Supply c. BCW Return d. IFO Supply e. IFO Return f. Fuel Gas Supply g. Nitrogen
FAFHR's	Monthly DRJ	Car Seals to all the FAFHR lines with valves in closed conditions.
Chemical system & Sour water drain Valves	Weekly DRJ	Car Seals to all the chemical system & Sour water drain valves to OWS in closed conditions.
Other Specified valves as per PDI	Weekly DRJ	Car seals to only those specified valves either in open condition or in closed condition as per the requirements advised through PDI.

2. Car seals can be broken in case of any process requirement by intimating concerned shift supervisor, same to be entered in car seal register and in FIELD/DCS TOB.
3. After meeting the process requirement the broken car seals to be restored to their normal position as early as possible after taking clearance from concerned shift supervisor, same to be entered in car seal register and in FIELD/DCS TOB.

15.7 Standing Instruction on Refinery Fuel gas system Management and control (SI 16):

1. Maintain constant pressure in VBU fractionator (11PR-1501). Other variables affecting the gas generation rate and under operational control(viz. Feed rate, Heater COT, Stabilizer re boiling etc.) should not be varied drastically.



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2. Sudden variation in feed quality, cooling water pressure/ Temperature also affects the Fuel gas generation rate in the unit. As those variations are not under the operational control, immediately information to be given to YSF when such variations are observed.
3. During startup consume minimum quantity of F.G. Inform to F.G. controlling unit in advance.
4. During normal run of the plant maintain constant consumption of fuel gas.
5. In case of any change in FG consumption, prior permission from FG controlling unit needs to be obtained in coordination with YSF.
6. Always follow the instructions from FG controlling unit for increase/decrease in FG consumption.
7. Record the summation value of total gas released from unit to flare in TOB in every shift.

15.8 Standing Instructions to avoid congealing of heavy oil rundown lines from the unit. SI 35

4. OBJECTIVE:

To avoid congealing of heavy oil rundown lines from the unit.

5. BACKGROUND:

SR obtained from vac. Column bottom is routed to various specified rundowns like RFO, HFO, VBU feed tanks and as BBU hot feed. Due to various planner requirements, different specs of SR having different viscosity grades V-10, V-30 and V50 grade bitumen and are routed to designated tanks. SR being a heavy product, during the change over of SR from one tank to another, it is required to flush the previous lines with cutter after changeover in case of RFO and HFO tanks. This prevents congealing of the lines. Heat tracing and steam tracing is provided along the length of the line from the unit to tank farms to prevent congealing.

Feed to BBU is obtained from CDU-I/II/III VAC bottom short residue (SR). Bitumen product from BBU is being routed to RFO, HFO and VBU tanks during ullage problem or due to off-spec bitumen. As the product being heavier, line has to be flushed with cutter after change over. This prevents congealing of the lines.

6. SCOPE:

This “Standing Instruction” is generated for routing of the bitumen product due to off spec. or ullage problem. In such cases, the product has to be routed to VBU tank line in CDU-II.



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7. **RESPONSIBILITY:**

The overall responsibility to implement these guidelines rests with the unit shift-in charges (Field & DCS) and technicians.

8. **STANDING INSTRUCTIONS:**

A. The following are different cases for routing of Bitumen product to RFO, HFO, VBU tank lines:

- i. **BBU start up:** During BBU start-up, initially cutter will be taken and displaced with the SR from CDU's. During the process, air to reactor is introduced and off spec product is being routed to HFO or VBU tank. In that process, initially the product having viscosity of SR is being routed to rundown and after a period of time the viscosity increases and product (spec. near to the V30 or V50 grade) still flows through the HFO or VBU line. The off spec bitumen having higher viscosity and pen should have higher EHT temperature than that of SR rundown in off-sites.
- ii. **BBU feed change over:** whenever Feed to reactor changes, the rundown will be routed to VBU tank or HFO line till the product is on spec. The product routed till the time is highly viscous and proper heat tracing till the tank is to be ensured. After completion of routing the line to be flushed with cutter.
- iii. **Production of V30 grade:** During production of V30 grade, viscosity has to be more than 2500 cst and if the product doesn't meet the requirement it has to be routed to HFO or VBU tank line. During this time, the product obtained will be highly viscous and hence the product has to be routed at a minimum temp of 140 °C to prevent congealing.
- iv. **Direct bitumen from vac column:** whenever high sulfur crudes like IM, UM are processed, direct bitumen is obtained from vac column bottom. During that period SR obtained has high viscosity and till the product is on spec. (if pen is above 100 m) SR has to be routed to RFO, HFO or VBU tanks. During these conditions the heat tracing has to be maintained above 140 °C and as per the technical advice the line temperatures are to maintain around 90 °C. So chances of congealing of line are more when there is sudden stoppage of pump (or due to suction loss) or slower rates of flow.



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As per Technical,

- The minimum temperature to be maintained for the SR which has to be routed to RFO or HFO is 90-95 deg C. (line design 120 deg C)
- The minimum temperature to be maintained for the bitumen which has to be routed to Tank 35 or Tk34 via new or old rundown line is 140-150 deg C. (line design 170 deg C)
- The minimum temperature to be maintained for routing SR or bitumen to VBU tank is 90- 160 deg C (line design 266 deg C)

The above said temperatures are the operating temperatures of the lines which are to be maintained in the offsite area. Hence, when ever the bitumen is being routed to RFO or HFO the heat tracing temperature is not sufficient in case of sudden stoppage of Bitumen product or slower rates of flow and may cause plugging. During the above activities, the off spec. bitumen product has to be routed via VBU line to respective tanks only.

Note: while processing high sulfur crudes like IM, UM or bituminous crudes (bituminous crudes list to be provided by technical) even though vac column conditions are not adjusted for direct bitumen, it has to be ensured that the SR to be routed to VBU tanks only as the product SR may have bitumen product qualities.

- Due to potential hazard, Bitumen cannot be routed to RFO or HFO lines as the design temperature of the lines are 120 deg C and the EHT for RFO and HFO lines are 90 -110 deg C and for bitumen, minimum temperature has to be maintained is 140 deg C.
- During processing of bituminous crudes in CDU and even though vac column conditions are not adjusted for direct bitumen, it has to be ensured that the SR to be routed to VBU tanks only as the product SR may have bitumen product qualities. Flushing of RFO and HFO to be done after changeover of the rundown. Considering 1000 m length of the line then total HFO/RFO line content to be displaced is around 50m³. At a cutter flow rate of 50 m³/hr, it has to be flushed for one hour. If cutter rate at 25 m³/hr means at least 2 hrs lines has to be flushed.
- The bitumen product due to off spec. or ullage problem, product has to be routed to VBU tank line, as its line temperature is maintained in the bitumen range with MP steam tracing.
- Old and new bitumen lines cannot be flushed with cutter as the lines are floating on the bitumen tanks and will off spec. the bitumen product.



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9. THE FOLLOWING ACTION ARE TO BE FOLLOWED BY CDU-II PERSONNEL:

- i.Route Bitumen from BBU or vac column bottom through designated bitumen rundown lines only.
- ii.Do not route bitumen product via HFO/RFO lines, as the line temperatures are limiting.
- iii.Route off-spec bitumen from BBU or SR of bituminous crudes from vac. bottom to VBU tanks only.
- iv.After tank switch, flushing of HFO or RFO are to be done to avoid congealing, as advised in the above instructions.
- v.Bitumen cannot be flushed with cutter as the lines are floating on the rundown tanks.
Due to planner requirement, if product is to be routed against the above instructions, permission has to be taken from the division head operations.



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The major equipments of the units are as follows:

- 16.1 Crude Charge Pump
- 16.2 Crude Booster Pump
- 16.3 Desalter
- 16.4 Pre-Flash Drum
- 16.5 Atmospheric Heater
- 16.6 Atmospheric Column
- 16.7 Naphtha stabilizer
- 16.8 RCO pumps
- 16.9 Vacuum Heater
- 16.10 Vacuum Column
- 16.11 Ejectors
- 16.12 HVGO pumps
- 16.13 SR pumps
- 16.14 Condensate recovery unit

16.1 CRUDE CHARGE PUMP 11-PM- 01A/B:

It is a motor and turbine driven centrifugal pump. The pumping temperature is 30°C and viscosity at pumping temperature should be 7.5 cst. Vapor pressure at pumping temperature has to be 0.5 kg/cm² absolute. Density of the fluid has to be 840 kg/m³. It has a design pressure of 31.7 kg/cm²A and temperature of 65°C. The casing and the impellor are made of carbon steel. It has a mechanical seal to protect out flow of liquid from the casing.

Operating Conditions:

Normal flow rate	482 m ³ /hr
Minimum flow rate	219 m ³ /hr
Suction Pressure	2.0 kg/cm ² A
Discharge Pressure	24.45 kg/cm ² A
Differential Head	367.3 meters
NPSH available	6 meters



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Pump Change over Procedure: (motor to turbine)

- First take clearance from CPP and also from PP-II for consumption of HP steam.
- Ensure all the utilities like BCW, seal steam and self coolant are open. Suction valve is open and discharge valve is in closed position. Casing has to be thoroughly drained.
- Check the pump basement and remove any foreign materials present.
- Drain the condensate from turbine and vent some steam to get dry steam.
- Check the pump freeness and ensure all the vents and bleeders are capped up.
- Check the auxiliary oil system:
 - ✓ Lube oil level
 - ✓ Lube oil pressure
 - ✓ Trip lever pressure
 - ✓ Cooling water system is commissioned
 - ✓ Check DP gauge of filters.
- Engage trip lever and start rolling of turbine.
- Place AOP (auxiliary oil pump) in AUTO mode.
- Increase the RPM of turbine and slowly close motor discharge valve against the feed discharge pressure and adjust the discharge pressure to the value required by DCS supervisor by increasing the turbine RPM (speed) and simultaneously closing the running pump discharge valve. This activity has to be done very slowly, ensuring there are no jerks in feed pump discharge pressure. Continuous watch has to be kept on the pump discharge pressure.
- Switch over seal flushing to 11-PM-02A from motor to turbine.
- When the running pump discharge is fully closed and amps become low, confirm the flow with DCS supervisor. Check the healthiness of turbine and with his consent stop the idle running pump. Check for leaks after changeover of the pumps.

Pump Change over Procedure: (turbine to motor)

- First take clearance from CPP for placing an HT motor and also from PP-II for reduction of HP steam.
- Ensure all the utilities like BCW, seal steam and self coolant are open. Suction valve is open and discharge valve is in closed position. Casing has to be thoroughly drained.
- Check the pump basement and remove any foreign materials present.
- Check the pump freeness.
- Ensure all drains and vents are capped up.
- Inform the field person to start the motor.



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- Start the pump and adjust the discharge pressure to the value required by DCS supervisor by opening the discharge and simultaneously reduce the turbine RPM while watching the feed discharge pressure. This activity has to be done very slowly, ensuring there are no jerks in feed pump discharge pressure. Continuous watch has to be kept on the pump discharge pressure.
- Switchover seal flushing to motor.
- Check the motor amperage and healthiness of pumps and then stop the turbine.

16.2 CRUDE BOOSTER PUMP 11PM 02A/B:

It is a motor as well as turbine driven centrifugal pump. The pumping temperature is 120°C and viscosity at pumping temperature should be 1.85 cst. Vapor pressure at pumping temperature has to be 9.5 kg/cm² absolute. Density of the fluid has to be 789 kg/m³. It has a design pressure of 41.0 kg/cm²A and temperature of 145 °C. The casing and the impellor both are made of carbon steel. It has a mechanical seal to protect out flow of liquid from the casing.

Operating conditions:

Normal flow rate	468 m ³ /hr
Maximum flow rate	515 m ³ /hr
Minimum flow rate	234 m ³ /hr
Suction Pressure	11.8 kg/cm ² A
Discharge Pressure	34.6 kg/cm ² A
Differential Head	238.4 meters
NPSH available	6 meters

Pump Change over Procedure: (motor to turbine) when PFD not in service.

- First take clearance from CPP and also from PP-II for consumption of HP steam.
- Ensure all the utilities like BCW, seal steam and self coolant are open. Suction valve is open and discharge valve is in closed position. Casing has to be thoroughly drained.
- Check the pump basement and remove any foreign materials present.
- Drain the condensate from turbine and vent some steam to get dry steam.
- Check the pump freeness and ensure all the vents and bleeders are capped up.
- Engage trip lever and start rolling of turbine.



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- Reduce Desalter pressure to 09.50 Kg\cm².
- Increase the RPM of turbine and slowly close motor discharge valve against the booster discharge pressure and adjust the discharge pressure to the value required by DCS supervisor by increasing the turbine RPM (speed) and simultaneously closing the running pump discharge valve. This activity has to be done very slowly so that the booster discharge pressure and flow, as well as the Desalter pressure do not fluctuate. Continuous watch has to be kept on the pump discharge pressure.
- When the running pump discharge is fully closed and amps become low, confirm the flow with DCS supervisor and with his consent stop the idle running pump. Check for leaks after changeover of the pumps.
- Keep the stand by booster pump in warm up condition.
- Place necessary trips online after the activity.

16.3 DESALTER (11-V-02):

Desalter handles crude and water together. The desalter receives thoroughly mixed crude oil and water.

The operating conditions of desalter are as follows:

Normal Flow rate	367647 kg/hr
Pressure	9.0 – 11 kg/cm ² A
Temperature	120-130 °C
Material of Construction	Carbon Steel
Corrosion Allowance	3 mm
Insulation	100 mm (hot)
Design Pressure	13.5 kg/ cm ² A
Oil Content in Water	100 ppm (max)
Inlet chlorides as NaCl	85.601 mg/l
Outlet chlorides as NaCl	5.0 mg/l

Desalting is the accepted industry term for the electrostatic process for removing contaminants such as salts, solids and water from crude oil at a refinery.

Crude oil brings along with it salts, particularly those of Sodium, Magnesium etc. metals like Arsenic, Vanadium etc. In addition crude oil contains solids such as finely divided sand



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particles, clay, drilling mud and rust and scale. Most of the contaminants are present in crude as it is produced from oil wells and accumulated during the transportation by tankers. Although these are present only in small amount, their presence can result in serious problems in down stream equipment viz, heat exchangers, heaters and columns. Hence the need for their removal is important before processing.

When crude oil enters a refinery, it typically contains a small amount of water, approximately 0.1 to 0.5 % by volume. The water in the crude oil contains water-soluble salts, and the crude oil contains insoluble particulate matter. To lower the level of impurities, water must be added first to the crude.

A mix valve mixes the wash water in to crude by dispersing water in to extremely small droplets. This intimate mixing of water and oil causes fresh water to contact brine droplets and various water soluble impurities in the oil. The water and oil mixture is then piped in to the Desalter vessel where the mixture is metered out at a low velocity in to an electrical field. The electrical field causes the oil and water to separate. The droplets of wash water (now combined with droplets of brine) form large drops that are separated from the crude oil by high voltage electricity and force of gravity. This process is called electrostatic coalescence. Application of strong electric field hastens the process of coalescing and therefore settling of all unwanted material. When the mixture of crude and water is subjected high potential electric field, the tiny water droplets get distributed between the electrodes forming dipoles. This makes it possible for the tiny particles to coalesce and form bigger particles of sufficient weight to settle down. The force of attraction between two droplets must be of sufficient magnitude to break through the oil film.

Desalter employs electrostatic elements operating at 415 volts to coalesce and separate water from oil for reducing the salts contents in crude oil so that it is within acceptable and specified limits in desalted crude. When oil water mixture enters the high voltage electric field, it separates into two phases in desalter. The oil phase floats on top and overflows, while water with its dissolved salts, metals, mud and iron oxides settles down at the bottom.

At a level below the electrodes an interface is formed between Hydrocarbons and water. Regulating desalted water withdrawal from the vessel controls this inter-phase level.

If the wash water is not added, the total population of water droplets in crude oil would not be sufficient for much coalescence (combining) of water droplets to occur. The addition of wash water to the crude oil increases the total water volume in crude and permits removal of contaminants.



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Caustic injection upstream of desalter in crude is done to neutralize acids present in crude and convert them into salts. These salts are then removed by desalter water in desalter. Caustic injection down stream of desalter is provided to neutralize any other acid traces formed at desalter operating conditions.

Brine is associated with crude both as a fine suspension of droplets and more permanent emulsion. To break these light emulsions demulsifier is added. This improves the effectiveness of desalter. Demulsifier is injected into crude upstream of desalter. Provision also exists for injection of demulsifier into the crude line at battery limit.

Caustic solution and demulsifier can also be added into the crude before the first exchanger in the preheat train-I.

Stripped water from Sour water stripper (SWSU) and MAB condensate from FCCU-II will be used as desalting water. Also, sour water from Atmos. overhead drum (11V01) and hot well water drum (12V01) are recycled to wash water drum. This helps in reducing the load on SWSU. Provision is there to make up fresh water using service water or DM water.

Following is a general description of crude desalting system of CDU given under three subsections viz. desalter description, desalter water system & desalter operation. Major equipments of this section are desalter, desalter water pumps and desalter water vessel.

16.3.1 Standing instructions for Desalter Online Desludging: SI 10

The Desalter is located in 1st stage PHT-1 after successive preheat exchangers i.e., (11-E-01, 11-E-02, 11-E-03, 11-E-04A/B, 11-E-05, 11-E-06 & 11-E-07) utilizing relatively low temperature fluids such as HN, Kero, diesel, TPA, Kero, diesel and LVGO respectively. The outlet temperature of this preheat train is between 120-130°C. Service water/DM water/stripped water (sour water from MEROX) is used for desalting which will be stored in vessel (11V04). The water will be preheated to 90°C in 11E18 using LP steam before injecting to desalter through 11PM 12A/B pumps.

In normal desalting process, water will be mixed with crude u/s of mix valve upstream of the desalter (provision is there for injecting a part of wash water into crude line suction also upstream of 11P01). On mixing, the undesirable salts present in the crude get dissolved in the wash water and hence get separated from crude. The electrostatic field applied in the desalter



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helps in breaking the oil water emulsion and thus the two liquid phases separate out. The hot brine from the desalter is routed to sour water stripper unit (merox for heat recovery and subsequently transfer to ETPs (effluent treatment plant).

1.0 Important checks during desludging.

1. Desludging operation has to be carried out once in three days for 1/2hour.
2. The important watch should be on interphase level that is to be maintained at about 40 to 45%.
3. Interphase level should not be dropped during process of desludging.
4. Continuous checking & confirmation through trycocks is a must.
5. Desalter LT /Agar probe must be checked by instrumentation before carrying out desludging operation.

2.0 Steps to be followed during desludging.

1. Inform all concerned Units and YSF before Desludging Operation.
2. Water injection to desalter via mix valve to be suspended by closing the field block valve located west of the desalter.
3. Resume wash water through a 3" line u/s of wash water isolation valve (with 4 lateral tie ins) and which is going to desalter (at bottom shell) directly with a check valve and a block valve located at north of desalter,
4. The water rate can be controlled from MOI (FI1102).
5. Route effluent through normal effluent line up. The wash water entry to desalter is from the bottom where the sludge and emulsion that is present separates out of the desalter through the normal effluent water outlet.
6. Maintain all process conditions of desalter normal, including power, pressure, temperature & interphase and desludging to be continued until the sludge is removed from the desalter by time to time check of trycock from 1 to 5 nos.

Once the sludge is removed and effluent water is clear the online desludging process can be stopped by closing the block valve in the field and desalter normal wash water injection can be resumed. The routing of clear effluent water should be restored as earlier. When routinely done once in three days it is supposed to take about half an hour for desludging.



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16.4 PRE-FLASH DRUM:

Crude enters the PFD approximately in the middle of the vessel. There is a provision to record the crude inlet pressure and temperature on DCS panel as P1901R and T1901R respectively. Crude that flashes in the PFD exits in the form of vapours from the top of the PFD and enters the 12th tray of the Atmos Column. A demister pad is provided on the vapour line to knock off any liquid droplets entrained in the vapour. The amount of flashing depends on the pressure in the PFD, which is controlled by PR1902. The crude level in the PFD is controlled by LR1902. Level gauge glasses are also provided on the shell of the PFD for physical verification of the level. There is also a high-level and low level switch provision. The un-flashed heavier crude goes to the PFD turbine 11-PT-02B from where it is boosted to 25 kg/cm² g. and sent to the PFD manifold through 11-E-40 A/B. The crude outlet flow from the PFD and its temperature are indicated on the DCS panel by F1902R and T1902R (provided on the outlet of crude from PFD) respectively. The PFD has two pressure safety valves (120-PSV-1201 A&B) set at 25.5 kg/cm² g. The discharge of the PSV is connected to the flash zone of the Atmospheric Distillation Column (down stream of Desalter RV).

There is a provision to bypass crude flow to PFD in case of emergency, from the DCS panel. This is done by operating the ROV switch on the auxiliary panel. There is an interlock on this ROV operation. When ROV is opened, the LCV-1902 gets closed. This interlock is provided to prevent the crude entry into the PFD when it is bypassed. A switch is provided on the auxiliary panel to trip the turbine (11-P-02B) from the DCS panel. Operating this switch on the auxiliary panel will close the steam shut off valve on the steam inlet to the turbine, thereby cutting off steam to the turbine. There is a provision to reset the SDV manually in the field, by which the SDV can be reset to open position

Standing instructions for commissioning of PFD : SI 34

10. OBJECTIVE:

To have safe commissioning of PFD without effecting the unit operation.

11. SCOPE:

This “Standing Instruction” is generated for commissioning of PFD in a smooth manner and to reduce the risk factor in CDU-II.



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12. RESPONSIBILITY:

The overall responsibility to implement these guidelines rests with the unit shift-in-charges (Field & DCS) and technicians.

13. STANDING INSTRUCTIONS:

The following are the changes to be made before commissioning the PFD.

- Feed rate to be reduced to 440 m³/hr and adjust the parameters accordingly.
- Drain the cold stock from the PFD to suction and discharge of 11-PT-02B turbine to CBD.
- Trip check the 11-PT-02B turbine steam SDV.
- Trip check the ROV with PDF level control valve (LR1902) and close the ROV after ensuring the bypass valve in wide open position.
- Stroke check the PFD pressure control valve (PR1902).
- Ensure PFD ROV in closed position and PFD bypass valve in wide open position.
- Close the block valve of level control valve (LR1902) and stroke check the control valve.
- Bypass 11-E-40A/B in co-ordination with FCCU-2.

After completion of the above activities build up PFD level initially by opening turbine warm up line up to ~10-15%. After that PFD drum filling to be done by opening the level control valve block valve and by opening the level control valve by 5 %.

Note: 1. Ensure the PFD inlet pressure P1901R is maintained less than (or around) 17 kg/cm².g

2. Take clearance from TPH, to route KERO to DSL as a precaution against discoloration during commissioning of PFD.

3. Take clearance from Power plant for consumption of HP steam around 7 tons for placing PFD turbine (11-PT-02B).

The steps are to be followed for placing the PFD into service.

- Drain the crude from the PFD to suction and discharge lines at pump to CBD.
- Keep a close watch on the CBD drum temperature and level. Ensure level is under control and CBD pump AUTO cut-in is working.(ensure CBD drum temperature is maintained below 80 °C and continuous supply of cooling water is provided to the coil of the CBD drum to ensure temperature under control).



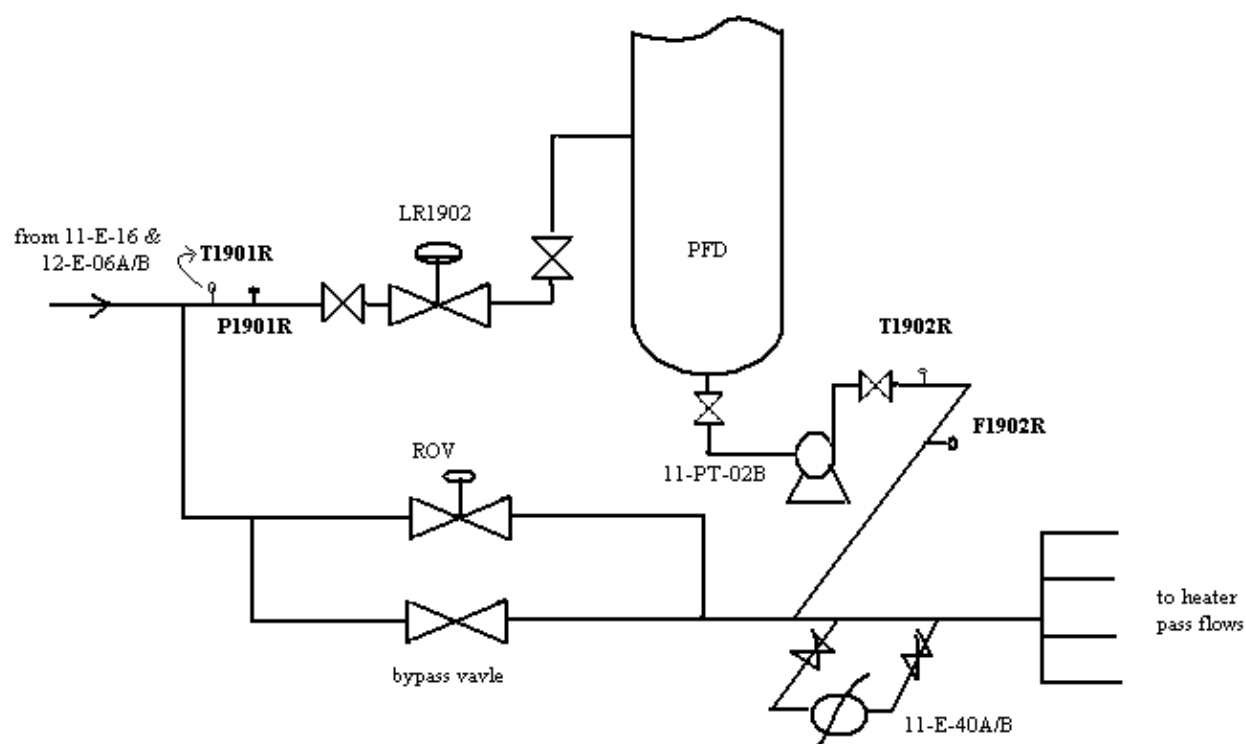
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3. After attaining temperature above 150 °C at PFD draw off temperature (T1902R) and PFD level around 50% and pressure above 5 kg/cm², start rolling of turbine 11-PT-02B.
4. When F1902R starts sensing the flow, Increase the speed of the turbine and simultaneously reduce the PFD bypass valve. Ensure no starvation of the flow towards the heater pass flows.
5. If the flow from turbine not developing then
 - i.) Increase the speed of turbine to get the turbine to heater (F1902R) flow.
 - ii.) Ensure that the PFD pressure is above 5 Kg/cm²g and if not, increase PFD level / hot feed to PFD to increase the pressure. Open the PFD LCV further to reduce the back pressure on the turbine (the flow through turbine will get established / increase.)
 - iii.) Reduce the PFD inlet pressure (P1901R) by reducing the booster pump discharge.
 - iv.) If still flow is not developing then stop the turbine, as there is a chance of seal failure due to vapor lock. (If turbine was stopped due to no flow then again the pump content has to be drained and again it has to be tried as mentioned earlier).
6. After ensuring the flow from F1902R, slowly close the PFD bypass valve and speed up the turbine in steps.
7. While closing the PFD bypass valve ensure PFD inlet pressure P1901R should not increase drastically. (Maintain less or around 18 kg/cm²). If the pressure is crossing 23 Kg/cm²g, immediately open the PFD level control valve (LR1902) by keeping a close watch on the PFD level. If the pressure doesn't come down then open the PFD bypass valve and stop the PFD turbine and check the system for any abnormalities. This is done to avoid the popping of the downstream Relief valves present on the exchangers (11-E-16, 12-E-01 to 05A/B with a set valve of 35.2 Kg/cm²g).
8. Slowly increase the speed of the PFD turbine in steps (PFD outlet flow = F1902R) and simultaneously reduce the PFD bypass valve and finally close it after full flow is ensured from the turbine.
9. After attaining the flow from PFD to heater the feed selectors switch has to be changed over to PFD mode (FR1804S1 position 1- controls 11-F-01 pass flows with respect to F1902R and position 2- controls 11-F-01 pass flows with respect to F1104R).
10. Route turbine outlet steam to system (to LP header) after ensuring dry steam from the turbine.



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Schematic of PFD:





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16.4.1 Standing instructions on 11E40A/B commissioning procedure : SI 17

1. **Objective :** To have safe commissioning of 11E40A/B without effecting unit operations.
2. **Background :** 11E40 A/B (crude /Circulating oil) exchanger was introduced to recover sensible heat from FCCU-II circulating oil and to increase the preheat of the crude in CDU II to give heat relief to heater. Inlet of 11E40A/B is taken from 14PM06A/B (circulating oil pumps) discharge and outlet is joined to 14E07(MP steam generator) upstream. A bypass facility at 14E07 and isolation facility at unit limit is provided for this scheme.
- 3 **Scope:** Commissioning of 11E40A/B is a crucial operation in view of light oil/ cold stock entry to column and reactor especially after cold startup of the unit. standing instructions are generated based on the past experiences.
4. **Responsibility:** FCCU-II field/DCS officers/Operations technicians.
- 5 **Commissioning of 11E40A/B when CDU II online**

1. Keep Crude side is in bypassed condition at CDU-II(it should be floating).
2. Displace condensate in the circuit /drain at low points.
 - CBD/OWS near 11E40A/B bypass valve.
 - Unit limit circulating oil inlet and outlet drains.
 - CBD/OWS on 11E40 inlet and outlet at CDU-II.
3. Displace cutter with feed in the 11E40A/B circuit to the column .
4. Circulate feed in the 11E40A/B circuit for at least 5hrs with 11E40A/B bypass in close condition and 1hr in open condition.
5. Check for feed in the circuit at the following locations.
 - 11E40A/B circulating outlet to CBD/OWS at CDU-II
 - FR406 control valve downstream drain to CBD/OWS.
 - Circulating oil to 11E40A/B drains to OWS at unit limit.
 - Circulating oil pumps suction to CBD/OWS.
 - 11E40A/B bypass loop.
6. Flush with feed @ 35M3/Hr for 4hrs to eliminate lighter oil in the circuit to main column and pump out column tray 12 stock to slop.



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7. During hot oil circulation (MC flash zone temp of 170- 180DegC), take feed to tray 12 and flush out for 4 hrs through CDU-II circuit.
8. Check periodically drain points at the loops mentioned in item No 5.
9. Monitor circulating oil supply and return temperatures.
10. After ensuring flow thorough flushing, open circulating oil bypass valve to about 8turns.

6 Commissioning of 11E40A/B during FCCU-II on line and CDU-II startup in progress

In this case FCCU-II unit is on line and CDU-II is under shutdown is considered. Whenever CDU-II is under shutdown, circulating oil from FCCU-II to CDU-II bypass will be kept open condition and inlet will be isolated at unit limit , outlet will be in floating condition. Once the CDU-II comes on line and stabilized, follow the below procedure for re-commissioning of 11E40A/B with circulating oil.

1. Coordinate with CDU-II and ensure the unit is stabilized and crude is through 11E40A/B.
2. Keep open unit limit inlet block v/v by 2turns and wait for 2hrs. this is to ensure complete filling of the loop. Ensure FR405 flow is not reduced.
3. Inform CDU-II to drain inlet and outlet to CBD till the outlet temperature on circulating oil reached 100degC.
4. After ensuring hot liquid in the circulating oil inlet and outlet loop, keep wide open 11E40A/B inlet valve at unit limit in slow steps.

7 Activity during placing of CDU-II Pre-flash Drum

The circulating oil temperature may fall down due to low temperature crude entry into 11E40A/B during commissioning of CDU-II PFD.

1. Prior to commissioning of CDU-II PFD , bypass 11E40 circulating oil at FCCU-I
2. Once the PFD commissioning is completed and stabilized, ensure circulating oil 11E40A/B outlet temperature is at normal condition.
3. Close the 11E40A/B bypass in slow steps.

16.5 ATMOSPHERIC HEATER:

Crude enters at the top of the convection zone or the fired heater. From the bottom of the convection zone, the coils are routed to the radiation zone. From radiation zone crude comes out of the heater and enters atmospheric column 11-C-01. Crude oil is heated from 280 °C to



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375 °C and 260 °C to 367 °C in PG and BH case respectively before entering 11-C-01 for fractionation.

The heater 11-F-01 is a cylindrical, cabin type fired heater having four passes. The radiant section of the heater is provided with 22 tubes each of 19090mm (weld to weld). These tubes are 6" NB Sch 40 type. The convection section has 12 rows of tubes with 8 tubes per row. These 12 rows are arranged in 3 bundles with 4 rows per bundle. While the top 2 rows of this zone are finned type the bottom two are bare tubes. All the others are studded type. The effective length of each tube in this zone is 9058mm. The material of construction of the tubes is 5% Chromium + 0.5% Molybdenum. Additional convection zone was provided during 2010 T&I to increase the heater efficiency and heat duty. The additional convection zone consists of 8 rows of tubes with 8 tubes per row. In these 8 rows, top 4 rows of this zone are 24 SPP studded tubes, below it 2 rows of 12 SPP studded tubes and the bottom two are bare tubes. The crude passes are numbered as pass A to pass D. Skin thermocouples have been provided on heater pass tubes to know the metal temperatures at convection zone outlet, middle of the radiation zone and outlet of the radiation zone.

Flow in each pass is regulated by individual pass flow controllers namely 11-FRC-301/302/303/304. These controllers get input signal from respective flow transmitters (FT-1301 to FT-1304) and set point signal from a ratio controller. Controllers generate output signal and manipulates respective flow valves (FR1301 to FR1304). And Low flow alarms 11-FAL-301/302/303/304 for each pass respectively.

A ratio controller or pass balancer maintains ratio of crude flow in a particular pass to total crude flow. The ratio controller/pass balancer functions in such a way that the weighted average temperature at the outlet of each pass is maintained almost the same. The pass balancer receives software input signal from FR1804 (feed selector switch. Total crude flow or PFD bottom flow), 11-TRC-301 (Crude oil Temperature exit Atmospheric heater), 11- FI/FR-402 (Crude over flash flow), 11-TI-306, 311, 316, and 321 (Outlet temperature of individual passes) and FR1301 to FR1304 (Current values of all crude heater pass flows). The output from the pass balancer regulates individual crude flow through the individual passes by manipulating the Flow control valves (11-FRC-301/302/303/304) located on the individual passes.

The distribution of crude through each pass should be adjusted in such a way that the heat duties and hence COT (Coil outlet temperature) of all the passes are more or less same.



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Depending on heat duty variation, pass flow will vary. In equality of flow through each pass to the extent of 10% can be tolerated while operating on manual mode or with pass balancer.

All the 4 passes join together at the outlet of radiation zone. 11-TI-302, 307, 312, and 317 shows the intermediate heating of crude in the convection zone.

To avoid repetitive description, features of only pass-A have been taken up. For other passes identical arrangements exist.

11-FRC-301 controls and indicates crude flow through pass-A. Through the transmitter FT - 1301 from the same element, one flow low-low alarm 11-FAL-301 has been provided in DCS. Actuation of this alarm will trip fuel supply to all burners of the heater excluding pilots. A local PI indicates field pressure and DCS bound 11-TI-302 indicates temperature of crude at radiation zone inlet. 11-TI-306 indicates temperature in DCS of crude of pass-A at heater outlet. Software alarm TAH and TAL on the COT has been provided to alert the operator against inadequate flow or inadequate firing in that particular pass. Finally, heated crude enters a common manifold of 24" size called transfer line before entering atmospheric column.

- The common outlet transfer line temperature is measured by the 11-TI-322 and is controlled and recorded by the COT (coil outlet temperature) 11-TRC-301, which regulates quantity of fuel to the furnace. Typical COT values are 375 ° C (IM) and 365 ° C (BH).
- Pressure drop across the heater coil is a measure of internal tube condition and increase in pressure drop indicates tube fouling due to coke formation in the tubes.
- Emergency coil MP steam connections are given in each pass, down stream of pass flow control valves to displace crude from the coil to the column during an emergency or after normal shut down operation.
- Soot blowers, using MP steam are provided in convection zone of the heater. Need of soot blowing will be indicated by poor heat pick-up in convection zone and increase in flue gas temperature. The soot blowers can be operated from grade level also.
- Emergency steam, soot blower steam and decoking steam connections are taken from MP steam header.
- Snuffing steam (furnace purging steam) is connected to convection zone and heater box in order to extinguish fire by steam blanketing. LP steam is used for this purpose.
- To facilitate safe approach to emergency steam/snuffing steam during an emergency scenario isolation valve on steam line needs to be provided at a safe distance of at least 15 meters from heater.



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Fuel System:

11-F-01 is a balanced draught furnace (and can be operated in natural draught at 70% efficiency). Both the convection and radiation sections are used for heating crude. The combustion chamber houses the radiation section of tubes. The convection section provided at the top of radiation section serves to increase the thermal efficiency of the furnace by utilizing further heat from the flue gas. Tubes in radiation and convection zone are arranged horizontally.

11-F-01 is a dual fired furnace i.e., either fuel oil or fuel gas or both can be used. The atmospheric heater has a total of 12 burners. Of late, 3 burners (No. 2, No. 5 & No. 8) have been dedicated to utilize off gases from the vacuum distillation column's hot well drum as the burning fuel. During 2010 T&I, all the burners were replaced with 20 new burners ZEECO make (16 are combined firing and 4 are hot well off gas burners (number 16 to 20)).

a) Fuel Gas System:

Fuel gas is supplied to the unit from the Battery Limit in an 8" header. This is further branched into a 6" header to the Atmospheric heater. This FG line is steam traced to avoid condensation of heavier components, as carry over of liquid droplets of Hydrocarbon to the burner must be avoided.

FG to main burners passes through a mass flow meter (F1315) and shutdown valve 11-SDV-303. This SDV is connected to interlock logic. 11-FR/FQ-308 indicates FG flow in DCS room. It is provided with FAL and FAH. A local PG and a TG are provided to indicate pressure and temperature at field. 11-PI-308 indicates FG pressure in the DCS room. A low pressure alarm 11-PAL-303 is also provided. Fuel gas pressure low trip is set at 0.2 kg/cm²g.

In case the fuel gas pressure goes below the trip value, only 11-SDV-303 will get closed. If fuel gas tip pressure falls below the set value, chances of flame failure and subsequent accumulation of un-burnt hydrocarbons in the firebox is possible. This can lead to the possibility of explosion or back fire in the heater. Hence the provision of "**FG pressure low**" trip was provided. There is a provision to cascade the fuel gas pressure to the 11-F-01 COT, 11-TRC-301 through a selector switch on the auxiliary panel on the DCS panel. A 2" FG tapping upstream of 11-SDV-303 has been branched off for pilot burners. The pilot gas pressure is normally adjusted manually and is maintained at a pressure of 0.7 kg/cm² g. In



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case of low pilot gas pressure, 11-PAL-302 is provided to actuate an alarm. Low pilot gas pressure will alert the operator when pilot gas pressure falls.

b) Fuel Oil System

Fuel oil is supplied to the unit from the Battery Limit in a 3" header. This is further branched into a 3" header to the Atmospheric heater. FO line is steam traced to maintain temperature and avoid congealing. Mass flow recorder and integrator 11-FR/FQ-305 are provided on main FO supply line and 11-FR/FQ-306 is provided on the main FO return line from heater. Since this is a closed circuit through which FO circulation is maintained, the net consumption of fuel oil is measured as the difference between FI-305 and FI-306.

Shutdown valves 11-SDV-301 A/B are provided on the FO supply and return headers respectively. Local PG's and TG's are provided on the supply line to show pressure and temperature of FO supply. 11-PRC-301 indicates the pressure of fuel oil on the DCS panel. Pressure is maintained by 11-PRC-301, which regulates 11-PV-301 on the fuel oil supply line. There is a provision to cascade the fuel oil pressure 11-PRC-301 to the 11-F-01 COT, 11-TRC-301 through selector switch, on the auxiliary panel.

A low-pressure trip alarm has been provided on supply line. Actuation of this alarm shuts 11-SDV-301 A/B and cuts off only the fuel oil firing in the Furnace. Since FO is normally a thick heavy liquid, it needs to be always maintained in circulating state. If it is left stagnant and unused in burners and piping, it can get congealed despite the fact that tracing steam of the FO circuit is on. Circulation in heater area (FO piping forming a closed circuit across all passes called fuel oil ring) is maintained even when no fuel oil burner is in use. A ratio of 2:1 FO supply to return is normally maintained to obtain a good control on firing and prevent congealing of FO system. FO is drawn by individual burners through ¾" lines from header and balance quantity is sent to the return line. When there is no need of FO firing in the heater, circulation can be maintained. Purge steam connections are provided on each oil burner. FO burners are to be kept steam purged when idle.

When FO is fired, it is atomised or sprayed as a fine mist for realising complete combustion. The spraying of FO is done by de-superheated MP steam in FO burners. Atomising steam is supplied to heater through a 4" header. The differential pressure controller 11-DPIC-301 controls the atomising steam pressure, taking pressure signal from FO supply and MP steam simultaneously. Atomising steam pressure is maintained about 2.0 kg/cm² above the FO pressure. Atomising steam flow is recorded by 11-FR-307. Local PG



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and TG are also provided on this line. 2" flushing oil connection is provided on FO supply line up stream CBD/OWS drain is provided on FO return line. These provisions are to flush the line within Battery Limit during heater shut down. When furnace operates on combination fuel-either Fuel Gas operates on PIC and Fuel Oil on PIC/TIC cascade or Fuel Oil operates on PIC and Fuel Gas on PIC/TIC cascade mode. Selector switch is used to select only one fuel for COT control by cascading.

c) Off Gas System

Vapors from Hot well drum (12-V-01) are routed to the Atmospheric heater burner through flame arresters. Off gases from the hot well are burnt in Atmos heater. An SDV (ZS-2901) is provided on this line, which will close in the event of Atmospheric heater (11-F-01) trip. Provision for routing the off gases from 12-V-01 to atmosphere exists to route gases to atmosphere in the event routing of off gases to Atmos. furnace is not possible. An SDV (ZS-2900) is provided on the vent line, which can be opened from DCS. Steam connection is provided on this vent line to atmosphere for dilution of hydrocarbon vapors, which are vented. Three burners are provided in Atmos. heater for burning off gas. Fuel gas connection also is provided for these burners. During 2010 T&I number of burners were increased to 4 ZEECO burners.

When furnace operates on combination fuel, either fuel gas operates on auto mode and Fuel Oil on cascade or fuel oil operates on auto mode and Fuel Gas on cascade mode. Selector switch is used to select only one fuel for COT control by cascading.

Heater operating conditions:

Operating Temperature (°C)	270 (IN)	380(OUT)
Operating Pressure (kg/cm ² -a)	17.5	4.5

Heater Section:

Design Pressure	30 kg/cm ² -a
Material of Construction	5 Cr + 0.5 Mo (A335 Gr P5)
Desired efficiency	88 % (LHV)
Stack height	71412 mm above grade

Note: For further details, please refer to furnace improvements of CDU-II crude heater, Volume-I, page 21.



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16.6 ATMOSPHERIC COLUMN:

Crude oil after final heating in atmospheric heater is fed to the atmospheric column at 375°C (IM) and 367°C (BH). This column has 40 valve type trays including baffle trays and three chimney trays for side stream withdrawal. The column has a stripping section at the bottom.

Description of entire column has been taken up zone wise.

- Flash zone
- Bottom section
- Middle section
- Over head section

16.6.1 Flash Zone:

Heated and partly vaporized crude oil enters the flash zone. Hydrocarbon vapors flash in this section and get liberated. Non flashed liquid moves down, which is largely the bottom product called RCO (Reduced Crude oil). Certain degree of over flashing of crude is desirable for proper stabilization of RCO and fractionation of gas oil (diesel) components.

Over flashing is achieved by setting up COT at slightly higher temperature than actually required. Required temperature of flash zone is 357 °C while that of feed is 360 °C. This over flashed material mostly condenses at 7th tray. The condensed liquid is withdrawn from 7th tray and put back on 6th tray. Over flash liquid travels down from 7th tray to 1st tray. It strips off heavier component coming up from RCO stock collected at column bottom which otherwise could move up and cause coloration of heavy diesel stream. Flow of over flashed liquid could be increased either by increasing COT and condensing more material on 7th tray or by reducing heavy diesel draw off. However the second option will lead to less diesel yield and higher energy consumption without any advantage. Too large flow of over flash liquid may result in drop in bottom temperature and lighter bottom product RCO.

Over flash flow is indicated by 11- FI/ FR-402. This is a 6" line with a U-loop. 11-FE-402 is mounted in the liquid seal. This seal provides adequate liquid build up on upstream of FE and ensures no flickering, steady flow through orifice. In addition, it provides some back pressure which is required to prevent flashing just downstream of flow orifice due to pressure drop.



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MP steam is introduced in the column through 11-FRC-401, below tray 1 for stripping of RCO. Stripping steam helps in removing lighter components from the heavier products by reducing their partial pressures and vaporizing them without requiring additional heat. Hydrocarbon vapors liberated by flashing move upward along with steam in the column for further mass transfer at trays in upper section.

Pressure relief valve discharge from Desalter and PFD are routed to the flash zone.

16.6.2 Bottom Section:

Reduced Crude Oil (RCO) product is collected at the bottom of the column. The column bottom level is indicated and controlled by 11-LRC-401. LRC-1401 goes to the ratio block of vacuum heater. Manipulating the RCO flow to vacuum heater can do column bottom level control. LRC is provided with software high / low level alarms. In Addition, 11-LAL/LAH-401 are provided in DCS panel. 11-TR-401 shows product RCO temperature.

RCO at a temperature of 340 to 370°C is pumped out from the bottom of the column at by RCO pumps 11-P-10 A/B to any of the following destinations.

- 10" line to Vacuum Furnace (12-F-01) as RCO feed. During normal operation, the RCO flow (FX2100) is regulated by the column bottom level control. This total RCO flow in-turn gives the set value to the vacuum furnace pass flow controllers.
- Swing elbow has been provided at the inlet and outlet of the heater passes, therefore steam and air decoking can be done either way.
- There is a provision to route RCO in an 8" line to 12-E-06 A/B (i.e., VR lines) for utilization of these exchangers during light crude processing. After exchanging the heat in 12-E-06 A/B, RCO goes to the vacuum furnace.
- To SR manifold during start-up. This is a start-up line.

16.6.3 Middle Section:

Middle section of the column has product withdrawal and circulating reflux network. In order to maximize heat recovery and balance the column loading for maintaining proper temperature profile across the column, three circulating refluxes (CR) are considered viz., Top Pump Around, Kerosene CR and Heavy Diesel CR. These circulating refluxes are drawn



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from their draw-off **seal** boxes and are routed to preheat trains for recovery before entering back to the column again

Duty controllers are provided on CR circuits to control CR flow rates to column. These duty controllers take corrective action based on actual CR duty and desired CR duty. For a particular type of crude and crude throughput, the CR under reference will have certain duty. This will be governed by total crude flow and specific heat of CR and is called CR duty. Actual CR duty is also computed by duty controller based on real time measurement of a) temperature difference between CR draw off and CR return stream b) CR flow rate and c) specific heat of CR. While total crude flow, CR temperature difference and CR flow are measured by various instruments, specific heat of CR is fixed by operator in software for computation purpose and no on line measurement for this parameter is available. Actual and desired CR duty is calculated in the duty controller as under:

Actual CR duty = Measured CR flow * CR temperature difference * Sp. heat of CR stream.

Desired CR duty= (Desired CR duty/desired total crude flow)* Actual crude flow.

The CR draw off and the CR return stream to the atmospheric column temperature difference is achieved by routing the respective streams through the preheat trains and exchanging heat with crude oil in the preheat exchangers.

Inputs that are to be manually provided by operator are i) Sp. heat CR stream and ii) the ratio (Desired CR duty/desired total crude flow) for each crude. Desired CR duty should be estimated on prorated feed basis to CDU. This is typical to all such duty controllers on CR lines. Desired CR duty is compared with actual CR duty and flow of CR is varied to achieve desired CR duty.

a) Top Pump Around (TPA)

TPA is drawn from the 37th tray by the pump 11-P-09 A/B and is cooled by routing it through 11-E-04 A/B. It enters the Atmospheric Column on the 39th tray. 11-TI-411 and 11-TI-414 indicate its draw-off and return temperature respectively. TPA flow is indicated and controlled by 11-FRC-406. The return temperature is to be maintained above 100 °C to prevent the condensation of water vapour which may result in acid corrosion of the column top.



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b) Kerosene Circulating Reflux:

Kerosene CR is drawn from 20th tray by the pump 11-P-08 A/B and is cooled by routing it through 11-E-09 and 11-E-25 in parallel and then through 11-E-11 (BH case). It is then boosted by 11-P-08 C/D. In BH operation mode, it flows through 12-E-01 A/B/C and then enters the Atmospheric Column on 22nd tray. 11-TI-403 and 11-TI-413 indicate its draw-off and return temperature respectively. KERO CR flow is indicated and controlled by 11-FRC-405. There is a pressure indication 11-PI-432 at the 20th tray in the Kerosene zone. There is a provision to route hot Kerosene to the unit flushing oil header from the discharge of 11-P-08 A/B

c) Diesel Circulating Reflux:

HD is drawn from the 12th tray by pumps 11-P-07 C/D and is cooled by passing it through 11-E-15 A/B and 11-E-13. The seal flushing facility to these CR pumps is provided from the outlet of 11-E-23, Diesel product cooler. The CR return enters the Atmospheric Column on the 14th tray. 11-TI-402 and 11-TI-412 indicate its draw-off and return temperature respectively. HD CR flow is indicated and controlled by 11-FRC-404.

d) Product draw-off:

Heavy Naphtha (HN), Kerosene, and HD products flow by gravity from the 28th, 20th and 12th tray respectively to strippers 11-C-02, 03 and 04 under respective level control of strippers (viz., 11-LIC-404, 403, 402 respectively). This draw-off from the draw-off boxes includes the respective CR for Kerosene and Diesel cases. Vapour return lines from HN, Kerosene, and Diesel strippers back to the fractionator column are provided just two tray above the draw-off for HN and two trays above the draw-off for Kerosene, and HD. 11-TI-403, 404, and 405 indicate the draw-off temperatures of HD, Kerosene and HN respectively from the column in the DCS panel. 11-TI-407, 408, 409 indicate the vapour return temperature from HD, Kerosene and HN strippers respectively. 11-TI-206 indicates the temperature at the 30th tray. An Elevation of 3m for HN and Kerosene and 4m for HD, from draw off nozzle to each level control valve of the stripper has been provided to exert back pressure necessary to prevent flashing just down stream of control valve. This also prevents two phase flow in draw off piping.

16.6.4 Over Head Section:

The overhead vapours of Atmospheric Column pass through the overhead condensers 11-E-17 A to H (in 4 banks) and are totally condensed. The condensate gets collected in the



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Overhead Naphtha Accumulator (reflux drum) 11-V-01. Any of this overhead condenser banks can be isolated for maintenance. There is also a provision of service water connection at the inlet of each condenser to occasionally wash away the deposits of ammonium salts in the tubes.

Top pressure of Atmospheric Column is maintained by 11-PRC-409 A/B which manipulates 11-PV-409A on outgoing uncondensed gases from 11-V-01 and 11-PV-409B on incoming fuel gas. Since isolation valve is provided in overhead line between Atmospheric Column and overhead Naphtha accumulator, 11-PSV-401 A/B/C (set at 4.8 kg/cm² g.) are provided at the top of column. Snuffing steam provision is given on the PSV outlet header to quench the vapours in case of emergency or as a safety precaution to be opened during thunder storms.

Condensed Hydrocarbons are allowed to settle in reflux drum where steam condensate (water) settles in vessel boot and then flows to the Sour Water Stripper Unit on its pressure. 11-TI-410 indicates the temperature of the reflux drum in DCS panel. 11-PSV-403 on 11-V-01 has also been provided for the safety of the Atmospheric Column top PSVs. Uncondensed gases from 11-V-01 are routed to flare through 11-PV-409A and FX- 801 indicates the mass flow rate of the flare gases.

Water-Naphtha interface controller 11-LDIC-406 controls level of water in the boot and operates 11-LV-406 on 3" sour water line. LIC-1406 has software high / low level alarm (11-LAH/LAL-406) in DCS panel. 11-FI/FR-450 shows sour water flow in DCS panel when Atmos sour water is routed to 11-V-04.

Accumulated hydrocarbon in 11-V-01 is pumped back to Atmospheric Distillation Column as top reflux on the 40th tray by 11-P-06 A/B. Reflux flow is controlled by 11-FRC-403 which is cascaded to the Atmospheric Column top temperature controller 11-TRC-403.

Excess quantity of Naphtha in Reflux drum is pumped by 11-P-06 A/B to the Naphtha Stabiliser as feed through a 6" line. The flow of Un-stabilized Naphtha to the Stabiliser is controlled by 11-FRC-503 which is cascaded to the Naphtha Accumulator (11-V-01) level controller 11-LRC-405. LIC-1405 has software high / low level alarms (11-LAH/LAL-405).

There is a ¾" provision for injection of Neutraliser and Corrosion Inhibitor on the Atmospheric Distillation Column overhead vapour line and the reflux line to maintain the desired pH.



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16.7 NAPHTHA STABILIZER:

Unstabilised Naphtha obtained in Atmospheric Column overhead reflux drum 11-V-01 contains lighter ends like C_3 and C_4 which vaporize at normal Atmospheric conditions. This Naphtha if stored as such in storage tanks will release lot of Hydrocarbon vapours and can create unsafe conditions and pressurization of the storage tank. To avoid this problem the lighter components of Naphtha are removed in a column. This process is called Naphtha stabilization.

Naphtha stabilization is carried out in Naphtha Stabiliser (11-C-05) where C_3 and C_4 hydrocarbons are removed from Naphtha. The Stabiliser is a distillation column which has 30 valve type trays (SS410S). It is provided with a PSV (11-PSV-501) set at $14.0 \text{ kg / cm}^2 \text{ g}$. The PSV outlet is routed to the flare header.

Unstabilized Naphtha from the Top Reflux Pump (11-P-06 A/B) discharge is first heated up in Stabiliser feed / bottom exchanger (11-E-19 A/B) by exchanging heat with the outgoing stabilized Naphtha product. There is a provision to route the CDU-I Unstabilised Naphtha to 11-C-05, the flow of which is indicated by F1505. Feed enters the column on the 17th tray under the flow control 11-FRC-503 which is normally cascaded with 11-LIC-405 of 11-V-01. 11-TI-501 indicates temperature pick up from 11-E-19 A/B before entering the column.

Overhead vapours from Stabiliser (11-C-05) containing C_3 and C_4 components come out from column top in a 12" overhead line. This line is routed through Stabiliser overhead condensers 11-E-20 A/B/C/D. The condensed liquid, LPG, is collected in the reflux drum (11-V-03) and consists of C_3 and C_4 components.

The Stabiliser overhead pressure is maintained by pressure controller 11-PRC-501 A/B. 11-PIC-501 acts as a split controller on 11-PV-501A mounted on the condensers bypass line and 11-PV-501B mounted on the off-gas line from 11-V-03 to FCCU-II (14-V-11).

In case of decrease of Stabiliser top pressure below the set value, 11-PRC-501 opens 11-PV-501A to allow hot vapours directly into 11-V-03, bypassing the condensers. If the pressure in 11-C-05 increases above the set value, then 11-PV-501B would open to release the excess pressure to the FCCU-II sweet fuel gas distribution network. There is a provision to route the Off-gas through 11-PV-501B directly to the flare also.



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LPG pumps 11-P-11A/B function as both LPG product and reflux pumps. Reflux flow which is controlled by 11-FRC-501 can be cascaded with the column top temperature (30th tray) indicator and controller 11-TI-510. A 2" minimum flow line (spill-back) from the discharge header of 11-P-11 A/B is to the pump suction is also provided.

11-LIC-502 controls LPG product flow to maintain reflux drum level and it is cascaded to 11-FRC-502 on the LPG product flow line. LPG is sent to the Amine Treating Unit in MEROX for the removal of H₂S and Mercaptans. 11-V-03 also has level high / low alarms in the DCS panel (11-LAH/LAL-502).

A slip stream of LPG can also be sent to the LPG vaporiser of FCCU-II in a 2" line. Sour water is collected in the boot of 11-V-03. The interface level of water and LPG is indicated by DL-1503. High water level in boot may result in water carryover with LPG and it will affect the Amine Treating Unit at MEROX. Hence it is drained at a controlled rate to OWS periodically.

11-PSV-502 set at 14.0 kg/cm² g. is provided on the Stabiliser reflux drum whose discharge is route to flare header. It prevents vessel from getting over pressurized in case of external fire.

A 2" service water line connection is provided on 11-V-03 to fill the vessel and wash the Stabiliser column with water during shut down.

A thermo siphon Stabiliser reboiler 11-E-25 is provided at Stabiliser bottom to supply the necessary heat for boiling the Unstabilised Naphtha. Kerosene CR from the discharge of 11-P-08 A/B is used as heating medium. 11-TR-403 & 11-TI-507 indicate the Kerosene CR supply and return temperature. The bottom temperature is indicated and controlled by 11-TRC-501. The control is achieved by adjusting the flow of Kerosene CR through 11-FV-504 which can be cascaded to the Stabiliser bottom temperature as indicated by 11-TRC-501. 11-TI-506 & 505 indicate the stabiliser bottom reboiler shell-side (Naphtha-side) inlet and outlet temperatures respectively.

Stabilised Naphtha gets collected at the bottom of the Stabiliser and the bottom level is controlled by 11-LIC-501 down stream of caustic and water wash sections. There is also an indication of bottom level high and low alarms as 11-LAH/LAL-501 in the DCS panel. This Stabilised Naphtha goes under the pressure of the Stabiliser to 11-E-19 A/B where it exchanges heat with the feed and then it gets cooled further in the salt water cooler 11-E-21. Naphtha from 11-E-21 flows through a 6" line to the caustic and water wash drums (10-V-01 and 10-V-02).



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16.8 RCO PUMPS:

It is a motor driven centrifugal pump. The pumping temperature is 343°C and viscosity at pumping temperature should be 1.25 cst. Vapor pressure at pumping temperature has to be 3.76 kg/cm² absolute. Density of the fluid has to be 780 kg/m³. It has a design pressure of 15.3 kg/cm²A and temperature of 380°C. The casing and the impellor are made of 11-13% Cr-steel. It has a mechanical seal to protect out flow of liquid from the casing.

Operating Conditions:

Normal flow rate	200 m ³ /hr
Maximum flow rate	232 m ³ /hr
Minimum flow rate	83 m ³ /hr
Suction Pressure	4.1 kg/cm ² A
Discharge Pressure	12 kg/cm ² A
Differential Head	101.3 meters
NPSH available	3.0 meters

Pump Change over Procedure:

- Ensure all the utilities like BCW, seal steam and self coolant are open, suction valve is open and discharge valve is closed. Drain the standby pump casing for removal of any entrapped water.
- Ensure the discharge is fully closed and warm-up is open.
- Close warm up after ensuring the readiness of pump.
- Start the pump and adjust the discharge pressure to the value required by DCS supervisor by opening the discharge and simultaneously closing the running pump discharge valve. This activity has to be done very slowly so that the flow does not fluctuate heavily. During this activity, amps of both the pumps have to be checked and ensure that it doesn't go beyond FLC for either of the pump.
- When the running pump discharge is fully closed and amps become low, confirm the flow with DCS supervisor and with his consent stop the idle running pump. Check for leaks after changeover of the pumps.
- Keep the stand by pump in warm up condition.



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16.9 VACUUM HEATER:

Major equipments of this section are Vacuum heater, air pre-heater, ID fan, FD fan and steam decoking pot (Common for Atmospheric and vacuum heater), separate APH including ID and FD fans have been provided for the crude and vacuum heaters.

RCO enters at the top of the convection zone of the fired heater. Part of the slop distillate produced from vacuum column is also routed along with RCO for the purpose of providing over flashing. From the bottom of the convection zone the coils are routed to the radiation zone. From radiation zone, RCO comes out of the heater and enters Vacuum column 12-C-01. Reduced Crude oil is heated 405°C (IM)/ 398°C (BH) before entering the 12-C-01 for fractionation.

The heater 12-F-01 is a single cell, cabin type fired heater having four parallel passes. The radiant section of the heater is provided with 20 bare tube/pass of 9% Chromium+1% Molybdenum grade 4" NB Sch. 40 size except the last two tubes of each pass which are of 8" NB Sc 40 and 6" NB Sch 40 from outlet. The convection section has 4 tubes per row and consists of 6 rows of tubes with 4 tubes per row. While the top 4 rows of this zone are studded type, the bottom two rows are bare tubes. The material of construction of the tubes is 9% Chromium+1% Molybdenum. The tubes are 6" NB Sch. 40 type, during 2010 T&I additional convection zone was provided. This zone consists of 8 rows of tubes with 4 tubes per row. In the additional zone, top 4 rows are SPP16 studded tubes, next two rows are SPP9 studded tubes and bottom two rows are bare tubes. The RCO passes are numbered as pass A to pass D. Skin thermocouples have been provided on heater pass tubes to know the metal temperatures at convection zone outlet, middle of the radiation zone and outlet of the radiation zone.

Individual pass flow controllers namely 12-FRC-101/102/103/104 regulate flow in each pass. These controllers get input signal from respective flow transmitters (FT2101 to FT2104) and set point signal from a ratio controller. Controllers generate output signal and manipulates respective flow valves (FV -2101 to FV-2104).

The ratio of RCO in a particular pass to the total crude flow is maintained by a ratio controller or passes balancer. The ratio controller/pass balancer functions such a way that the weighted average temperature at the outlet of each pass is maintained almost the same, Ratio controller receives software input signal from 11-LRC-401 (Through a selector switch) which is responsible for Atmospheric column bottom level, Pass balancer gets input from each pass



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flow (12-FRC-101/102/103/104) and each coil outlet temperature (12-TI-106, 111, 116, and 121) and then controls individual pass flow in order to maintain heat balance of all the passes. Pass balancer is an advanced control feature conceived for better furnace control.

The distribution of RCO through each pass should be adjusted in such a way that the heat duties and hence COT (Coil out let temperature) of all the passes are more or less same. Depending on heat duty variation pass flow will vary. In equality of flow through each pass to the extent of 10% can be tolerated, while operating on manual mode or without pass balancer. 12-FAL-101/102/103/104 are provided on all the passes to protect the heater pass tubes in case of low flow through each pass.

All the four passes join together at the outlet of radiation zone. DCS indication of temperature for each pass is provided (12-TI-106, 111, 116, and 121) and there is also an alarm for high transfer line temperature as 12-TAH-133. Temperature indicators 12-TI-302, 307, 312 & 317 are provided on the outlet of the convection zone to measure the temperature gain in this zone.

To avoid repetitive description features of only pass-A have been taken up. For other passes identical arrangements exists.

FIC-2101 controls and indicates RCO flow through pass-A, through transmitter FT-2101 has been provided. 12-FAL-101 actuation of this alarm will trip fuel supply to all burners of the heater, excluding pilots. A local PI indicates field pressure and DCS bound 12-TI-302 indicates temperature of RCO at radiation zone inlet. 12-TI-106 indicates temperature in DCS and a local PI shows pressure of RCO of pass-A at heater outlet, software alarm TAH and TAL on the COT has been provided to alert the operator against inadequate flow or inadequate firing in that particular pass. Prolonged high temperature may lead to coking up of that particular pass, Finally heated RCO enters a common manifold 52" size called transfer line before entering Vacuum column.

The common outlet transfer line temperature is measured by 12-TI-122 and is controlled and recorded by 12-TRC-133 (coil outlet temperature), which regulates quantity of fuel to the furnace.

Pressure drop across the heater coil is a measure of internal tube condition and increase in pressure drop indicates tube fouling due to coke formation in the tubes. Emergency coil MP steam connections are given in each pass, down stream of pass flow control valves to displace RCO from the coil to the column during an emergency or after normal shut down operation.



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Soot blowers, using MP steam are provided in convection zone of the heater. Need of soot blowing will be indicated by poor heat pick-up in convection zone and increase in flue gas temperature. The soot blowers can be operated from grade level also.

Emergency steam, soot blower steam and decoking steam connections are taken from 8"MP steam header.

Snuffing steam (furnace purging steam) is connected to convection zone and heater box in order to extinguish fire by steam blanketing. LP steam is used for this purpose.

To facilitate safe approach to emergency team/snuffing steam during an emergency scenario isolation valve on steam line needs to be provided at a safe distance of at least 15 meters from heater.

ii) Fuel System:

12-F-01 is a balanced draught furnace (and can be operated in natural draught at 70% efficiency). Both the convection and radiation sections are used for heating crude. The combustion chamber houses the radiation section of tubes. The convection section provided at the top of radiation section serves to increase the thermal efficiency of the furnace by utilizing further heat from the flue gas. Tubes in radiation and convection zone are arranged horizontally.

12-F-01 is a dual fired furnace i.e., either fuel oil or fuel gas or both can be used. The vacuum heater has a total of 12 burners. During 2010 T&I, all the burners were replaced with 16 new burners ZEECO make.

a) Fuel Gas System:

Fuel gas is supplied to the unit from the Battery Limit in an 8" header. This is further branched into a 3" header to the Vacuum Heater. This FG line is steam traced to avoid condensation of heavier components, as carry over of liquid droplets of Hydrocarbon to the burner must be avoided.

FG to main burners passes through a mass flow meter F2406 and a shutdown valve 12-SDV-105. This SDV is connected to interlock logic. 12-FR/FQ-107 records and integrates



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the FG flow to 12-F-01. It is provided with FAL and FAH. Local PG and TG are provided to indicate pressure and temperature at field.

12-PI-308 indicates FG pressure on the DCS panel. A low pressure alarm 12-PAL-108 is also provided. Fuel gas pressure low trip is set at 0.2 kg/cm² g. In case the fuel gas pressure is low, only 12-SDV-105 will get closed. If fuel gas tip pressure falls below the set value, chances of flame failure and subsequent accumulation of un-burnt hydrocarbons in the firebox is possible. This can lead to the possibility of explosion or back fire in the heater. Hence the provision of FG pressure low trip. There is a provision to cascade the fuel gas pressure to the 12-F-01 COT, 12-TRC-133 through a selector switch on the auxiliary panel in the DCS room.

A 2" FG tapping upstream of 12-SDV-105 has been branched off for pilot burners. The pilot gas pressure is normally adjusted manually and is maintained at a pressure of 0.7 kg/cm² g. In case of low pilot gas pressure, 12-PAL-107 is provided to actuate an alarm. Low pilot gas pressure will alert the operator when pilot gas pressure falls.

b) Fuel Oil System:

Fuel oil is supplied to the unit from the Battery Limit in a 3" header. This is further branched into a 2" header to the Vacuum heater. FO line is steam traced to maintain temperature and avoid congealing. Flow recorder and integrator 12-FR/FQ-105 is provided on main FO supply line and 12-FR/FQ-106 is provided on the main FO return line from heater. Since this is a closed circuit through which FO circulation is maintained, the net consumption of fuel oil is measured as the difference between FI-105 and FI-106. Shutdown valves 12-SDV-102 A/B are provided on the FO supply and return headers respectively. Local PG's and TG's are provided on the supply line to show pressure and temperature of FO supply. 12-PRC-101 indicates the Pressure of fuel oil on the DCS panel. Pressure is maintained by 12-PRC-101, which regulates 12-PV-101 on the fuel oil supply line. There is a provision to cascade the fuel oil pressure 12-PRC-101 to the 12-F-01 COT, 12-TRC-133 through selector switch, on the auxiliary panel.

A low-pressure trip alarm has been provided on supply line. Actuation of this alarm shuts 12-SDV-102 A/B and cuts off only the fuel oil firing to the Furnace. Since FO is normally a thick heavy liquid, it needs to be always maintained in circulating state. If it is left stagnant and unused in burners and piping, it can get congealed despite the fact that tracing steam of the FO circuit is on. Circulation in heater area (FO piping forming a closed circuit



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across all passes called fuel oil ring) is maintained even when no fuel oil burner is in use. A ratio of 2:1 FO supply to return is normally maintained to obtain a good control on firing and prevent congealing of FO system. FO is drawn by individual burners through ¾” lines from header and balance quantity is sent to the return line. When there is no need of FO firing in the heater, the circulation can be maintained. Purge steam connections are provided on each oil burner. FO burners are to be kept steam purged when idle. When FO is fired, it is atomised or sprayed as a fine mist for realising complete combustion. The spraying of FO is done by de-superheated MP steam in FO burners.

Atomising steam is supplied to heater through a 4” header. The differential pressure controller 12-DPIC-103 controls the atomising steam pressure, taking pressure signal from FO supply and MP steam simultaneously. Atomising steam pressure is maintained about 2.0 kg/cm² above the FO pressure. Atomising steam flow is recorded by 12-FR-108. Local PG and TG are also provided on this line. 2” flushing oil connection is provided on FO supply line up stream. CBD/OWS drain is provided on FO return line. These provisions are to flush the line within Battery Limit after heater shut down. When furnace operates on combination fuel-either Fuel Gas operates on PIC and Fuel Oil on PIC/TIC cascade or Fuel Oil operates on PIC and Fuel Gas on PIC/TIC cascade mode. Selector switch is used to select only one fuel for COT control by cascading.

Heater operating conditions:

Operating Temperature (°C)	340 (IN)	405(OUT)
Operating Pressure (kg/cm ² -a)	3.57	0.07

Heater Section:

Design Pressure	18 kg/cm ² g
Material of Construction	9 Cr + 1 Mo (A335 Gr P9)
Desired efficiency	88% (LHV)
Minimum stack height	60 meters.

Note: For further data, please refer to furnace improvements of CDU-II vacuum heater, Volume-I, page 20.



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16.10 VACUUM COLUMN:

The vacuum column (12-C-01) has three sections of different diameters. Top section is of 5000 mm diameter. Middle section is of 6600 mm diameter and bottom section is of 5000 mm diameter. The column has 3 packed sections. It is provided with 3 total draw-off trays for LVGO, HVGO and Slop cut. Vacuum Residue is drawn as bottom product. Chimney trays are provided for all side draw off products. The vacuum column is designed to operate in dry vacuum mode. Vacuum indicators are provided at top and flash zone of the column.

Description of Vacuum column has been taken up zone wise starting from bottom.

16.10.1 Vacuum Column Bottom Section:

The partly vaporized RCO feed stock coming from the vacuum heater enters the column in the flash zone below the slop draw off chimney tray. The vaporized portion rises up in the tower and is fractionated into 3 side stream products. The liquid portion of the feed drops into the bottom section of the tower and is withdrawn as Vacuum Residue (VR). Vacuum Residue is also called as Short Residue (SR). Column bottom has 3 disc and donut type trays.

The Column bottom level is maintained by 12-LIC-202's action on 12-LV-202 on VR rundown line. LI-2201 also indicates level on DCS panel. Both LIC and LI have software high level and low level alarms 12-LAH/LAL-201. 12-TI-201 indicates temperature of the column bottom on the DCS panel. LAH-2108 and LAL-2109 are also provided for column bottom on the DCS panel.

A tangential vapour horn is provided for flash zone inlet, which minimises entrainment of heavier hydrocarbon liquid droplets. There is a 3 inch LP steam line provision to column bottom below tray 1 for steaming out purpose.

Vents of all pumps taking suction from Vacuum Column are connected back to Vacuum Column above flash zone through a common vacuum line. With this arrangement all pumps connected with Vacuum Column can be vented to the column.

Temperature at the bottom section is normally quenched to about 350 °C at a vacuum of -735 mm Hg (g.). RCO is fed to the flash zone at 390 °C.



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16.10.2 Vacuum Residue Draw-off:

Vacuum Residue (VR) is drawn by VR + Quench pumps (12-P-01A/B SR pumps) at 350 °C from the column (12-C-01) bottom. VR pump discharge can be routed to following destinations:

- a) To Vacuum Heater (12-F-01) through 8" start-up line which joins the RCO feed line to 12-F-01. This is used during start-up only.
- b) A slip stream of the pump discharge can also be routed to the pumps suction in a 2" line as spill-back.
- c) The pump discharge from 12-P-01 A/B exchanges heat with crude in 11-E-16, 12-E-06 A/B and 12-E-03. The outlet of 12-E-03 at a temperature of 250 °C can be routed as follows.
 - i. It is partly sent to the 12-C-01 bottom in a 6" line for quenching and maintaining the bottom temperature at 350 °C to prevent cracking of SR and lead to other problems like deterioration of vacuum, plugging or suction strainers of 12-P-01 A/B leading to loss of suction. The quench flow is recorded and controlled by 12-FRC-204.
 - ii. It can be partly sent as feed to the Bitumen Blowing Unit in a 3" line.
 - iii. The rest of it can either be sent to 12-E-01 A/B/C and then to 12-E-09 A/B/C/D or can be routed directly to the 12-E-09 A/B/C/D where it is cooled. TIC-2103 located on VR product rundown header to tanks controls TV-2103 on the tempered water line to 12-E-09 A/B/C/D.
- d) Down stream of the coolers, SR can be routed to
 - i. IFO pool partially in a 6" line.
 - ii. Storage via the three-way FCV 12-FRC-405 through an 8" RFO line, or to the HFO pool by adjusting 12-FV-405. The HFO flow is measured by 12-FR-406.
 - iii. LDO pool partially, in a 4" line, measured by 12-FI-407.
 - iv. VBU as hot feed from the upstream of 12-E-09 A/B/C/D or to VBU storage tanks in 10" line. (TIC-2420 on rundown line maintains VR temperature by bypassing a certain amount of hot VR to the rundown line)
 - v. BBU storage tanks via HFO line (at unit limit tie-in).

16.10.3 Vacuum Slop-Cut Draw Off:

Immediately above flash zone, a wash zone consisting of one section packed bed is provided. Slop section is a bed with a demister pad above the zone packing. The vapors rising from the



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wash zone pass through a demister pad provided above the wash section to trap entrained droplets of heavy hydrocarbons, which could otherwise adversely affect HVGO/LVGO quality.

Slop is drawn by slop distillate pumps (12-P-02 A/B) at 350 ° C. This is first side draw off from the bottom.

The draw-off temperature is indicated by 12-TI-204. The level on the chimney tray is regulated by 12-LI-203 by operating 12-LV-203 on the slop-cut rundown line to the VR line. Slop + Recycle pump (12-P-02 A/B) discharge has following destinations.

- As recycle stream to vacuum furnace along with RCO. One part of slop distillate goes to the furnace under flow control 12-FRC-109 through a 3" line. The purpose of maintaining slop recycle is to provide necessary over flash in the vacuum column. The slop recycle from slop pump is mixed with RCO from atmospheric column before getting heated in vacuum furnace. However if any premature coking is observed, the slop recycle may be reduced accordingly. Additional provision was given during 2010 T&I for routing the slop-cut to FCCU-II from the downstream of slop-cut recycle control valve.
- As product rundown, a part of Slop Distillate pump discharge gets mixed VR product up stream to 12-E-01 A/B/C. 12-LV-203 on slop line maintains slop level on the chimney tray of slop section

16.10.4 Heavy Vacuum Gas Oil Draw off:

The majority of rising hydrocarbon vapors from slop zone wash section is condensed in HVGO section by circulating reflux to yield the side draw product. HVGO product internal reflux (IR) + circulating reflux (CR) is withdrawn as a second side steam.

12-TI-205 indicates draw off temperature. HVGO after draw off is pumped by HVGO product +CR pump (12-P-03 A/B) at 314 ° C and splits in to two streams.

One part goes as internal reflux for packing washing of slop section of vacuum column without any heat exchange. HVGO IR is regulated by 12-FRC-202 in such a way that proper washing of the packing is always achieved for all throughputs. Height of packing is fixed and dependent on heat withdrawal requirement by slop quench stream. One distributor is provided for proper distribution of HVGO IR over entire cross section area of the column



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packing. Strainers (12-X-01 A/B) are provided to arrest carry over of foreign materials back into column. Pressure gauges across the filters indicate the pressure drop across the filters and its rise would indicate the need to change over of the filter in line. 12-TI-206 shows HVGO IR temperature.

The other part is routed to preheat exchanger trains as HVGO CR + product.

HVGO CR + product from 12-P-03 A/B discharge is split into two streams. One stream goes through 12-E-05 A/B and 12-E-02 and the other through 12-E-04. Here, it exchanges its sensible heat with the crude. The two streams are then combined and routed to the MP steam generators 12-E-10/10A parallel. The stream coming out of the steam generators is routed partly as HVGO CR back to column at through the CR strainer 12-X-02 A/B. This flow is controlled and recorded by 12-FRC-203. The CR return temperature is indicated by 12-TI-208. The remaining volume is either routed as hot feed to FCCU-II or to product rundown line or both at the same time.

16.10.5 Light Vacuum Gas Oil Draw off (LVGO):

The rising uncondensed hydrocarbon vapors from HVGO zone packing are condensed in LVGO section by circulating reflux to yield the side draw product. LVGO product+ internal reflux (IR) + circulating reflux (CR) is withdrawn through 10" line from the third chimney tray. 12-TI-207 indicates draw-off temperature on the DCS panel. LVGO draw-off from the column is routed to LVGO product + CR + IR pumps (12-P-04 A/B), whose discharge is routed as

LVGO IR stream: The packed bed in HVGO zone below the LVGO zone is washed by LVGO internal reflux stream. This stream enters the column without any heat exchange. LVGO IR is regulated by 12-FRC-201 in such a way that proper washing of packing is achieved for all throughputs. One distributor is provided for proper distribution of LVGO IR stream over entire cross sectional area of the column packing. Strainers (12-X-02 A/B) are provided on LVGO IR return to arrest carry over of any foreign materials back into column.

LVGO Circulating Reflux: This stream splits further into two streams. One stream goes 11-E-07 to exchange its sensible heat with the crude and then goes to 11-E-22 for cooling. The other stream goes to 11-E-22A directly for cooling. The outlet of 22 & 22A combine and go through the LVGO CR strainer 12-X-03 A/B to the top of the LVGO packing as top reflux.



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The CR flow is controlled by 12-FRC-205 and the return temperature is indicated by 12-TI-210.

LVGO Product: A part of 12-P-04 A/B discharge stream gets cooled in 12-E-11 and can be routed to HVGO storage tanks or FCCU-II. The flow of this stream is controlled by 12-LRC-205 and is recorded / integrated by 12-FR/FQ-403. The product can also be routed to Diesel or LDO pool controlled by 12-FRC-404.

LVGO product is routed as follows:

- 1 As hot feed to FCCU-II along with hot HVGO through 12-LV-205
- 2 To VGO storage tanks along with HVGO through 12-LV-205
- 3 Either to Diesel/LDO/SR or HVGO (at battery limit) storage tanks through 12-FV-404 in a 3" line. There is a provision to cascade the level controller on the LDO FCV
- 4 To flushing oil system in a 2" line
- 5 To slops along with HVGO.

16.11 Ejectors:

Fluid handled: Composition

Component	Kg/hr	MW	Moles/hr	° API
Non condensable	840	33.7	24.93	-
Oil vapor (condensable)	500	250	2.0	30.31

Vapor pressure data of condensable oil vapors:

Temperature, °C	Vapor pressure, mm Hg abs
40	0.011
50	0.025
60	0.056
70	0.120
80	0.23



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Operating conditions:

	Temperature, °C		Pressure	
	Maximum	Normal	Maximum	Normal
Inlet	95	80	-	5.0 mm Hg abs
Outlet	-	-	-	1.1 kg/cm ² .a

Mechanical design:

Design temperature	340 °C
Design pressure kg/cm ² .abs	14 /full vac.

Material of construction:

Ejector nozzle	SS
Ejector steam chest	steel
Suction chamber and diffuser	cast iron and steel plate
Steam condenser tubes	70:30 Cu-Ni
Valves	steel bodies

Description:

In Vacuum column overhead system, three stage ejectors with condensers are provided to maintain the desired vacuum. MP Steam is used as a motive fluid in ejectors. Vacuum column overhead vapor line (36") is routed to first stage ejectors (12-J-01 A/B/C). First stage ejectors outlets are joining a common header and which is routed to shell side of 1st stage inter condensers (12-E-07A). Uncondensed vapors from inter condensers (12-E-07A) are joining a common header and then routed to second stage ejectors (12-J-02 A/B/C). Second stage ejectors outlets are joining a common header, which is routed to shell side of second stage inter condenser (12-E-07B). Uncondensed vapors from inter condensers (12-E-07B) are routed to third stage ejectors (12-J-03 A/B/C). Third stage Ejectors (12-J-03 A/B/C) outlets are joining a common header which is routed to surface condenser (12-E-07D). Condensate streams from surface condensers (12-E-07A, 12-E-07B and 12-E-07D) are routed through barometric legs to Receiver Vessel called hot well drum (12-V-01). These barometric legs are dipped in water the ejector condensate receiver (12-V-01) for sealing purposes. Uncondensed vapors from third stage ejectors (12-J-03 A/B/C) are routed to the Hot Well Drum through a dip leg. 12-PRC-206 is mounted on the non-condensable line from 12-E-07D, controls the Vacuum Column overhead pressure by routing a part of the non-condensable vapours (before letting them into the Hotwell drum) to the inlet of the 1st stage



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ejectors. A 3" fuel gas line is provided near the inlet of the 1st stage ejectors, for backing-in of fuel gas during the start-up and shut-down to maintain the column in positive pressure.

The 1st stage ejectors (12-J-01 A/B/C), 2nd stage ejectors (12-J-02 A/B/C) and the 3rd stage ejectors (12-J-03 A/B/C) are designed *for a capacity factor of 1/7, 2/7 and 4/7 totalling 150% of normal capacity*. If required each ejector element can be isolated by cutting off steam and isolating suction inlet valves.

Cooling water is supplied to the primary condenser in an 18" header. The water outlet from the primary condenser is bifurcated into two parts. One part feeds the cooling water to the secondary condenser and the next part feeds the after-condenser. There is a provision to back flush the primary condenser. A draining provision is provided in the up-stream of the after-condenser.

MP steam to all ejectors is supplied in parallel by 6" header. Ejector steam consumption is indicated by 12-FR-207. Strainers are provided on the MP steam line to arrest line scales etc. from reaching into ejectors and adversely effecting performance of overhead system. Ejector steam pressure is controlled by 12-PRC-207. This should be maintained constant as far as possible for smooth operation of the Vacuum Column. Corrosion Inhibitor and Neutraliser injection facility into overhead vapours have been provided both on Vacuum column top overhead.

Operation procedure:

1. Commissioning:

- Ensure all the mechanical jobs are completed on the ejectors with all accessories.
- Check all the blinds have been removed or not.
- Check Hot well is properly filled of water or not. If not fill it up and commission Hot well overflow loop.
- Charge steam header to the ejector. Drain condensate from the low point drains. Ensure steam is clear before it is charged to the ejectors.
- Open cooling water to all the surface condensers.
- Before opening steam to the ejectors, open the strainer bleeder and purge the strainer.
- Open all the isolating valves of the ejectors to be placed in service on the first and subsequent stages of the ejectors.
- Open upstream and downstream valves of the pressure controller for controlling vacuum.



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- i. Open steam slowly to the ejectors to the last stage. Wait till the vacuum stabilises. Next open steam to the preceding stage ejectors and so on until the required vacuum is obtained.

Note: when an ejector in a stage is in service, the other ejectors in that stage should be positively isolated, if they are not in service to avoid circulation of the non-condensables in that stage. If any of the ejector is out of service, is to be placed in service, the following is the sequence of steps to be followed.

- ✓ Open the ejector outlet valve.
 - ✓ Open steam to the ejector
 - ✓ Open hydrocarbon inlet valve to the ejector.
- If an ejector in service has to be taken out, then the alternative ejector has to be first placed in service.
- j. When the hot well level starts increasing commission hot well oil and water pumps. When the hot well oil rate becomes steady put the oil pump on AUTO.

2. De-commissioning:

- a) Close the hydrocarbon inlet valve to ejectors.
- b) Close steam to the ejector
- c) Close the outlet valve to the ejector.

The above sequence of operation is for shut down of an ejector. And for total shut down of all the ejectors, the above sequence of operation should be carryout from the last stage to first stage and followed by

- ✓ Opening make up water to the hot well drum
- ✓ Closing cooling water to surface condensers
- ❖ For total shut down of all ejectors, take out steam from first stage ejectors to last stage ejector in that order and isolate steam header to ejectors.
- ❖ Open make up water to hot well and maintain level
- ❖ Isolate all ejectors upstream and downstream valves
- ❖ Close cooling water to all surface condensers.

Trouble shooting:

If vacuum starts falling, the reason may be:

- Insufficient inlet steam pressure
- Inlet water temperature to the condensers higher then normal temperature
- Air leaks in the tail pipe of inter condensers
- Flooding of the inter condensers by excessive water flow (direct cooling water condensers).



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- Starving of any inter stage condensers by insufficient water flow.
- Plugging of water distribution system in the condensers.
- Plugging of the tail pipe.
- Plugging of the steam nozzle and jets due to pipe scale.
- Steam leak at nozzle throat.

The defects mentioned are to be located and rectified for proper operation.

16.12 HVGO PUMPS:

It is a motor driven centrifugal pump. The pumping temperature is 315°C and viscosity at pumping temperature should be 0.7 cst. Vapor pressure at pumping temperature has to be 0.023 kg/cm² absolute. Density of the fluid has to be 725 kg/m³. It has a design pressure of 17 kg/cm²A and temperature of 340°C. The casing and the impellor are made of 11-13% Cr-steel. It has a mechanical seal to protect out flow of liquid from the casing.

Operating Conditions:

Normal flow rate.	333 m ³ /hr
Maximum flow rate.	400 m ³ /hr
Minimum flow rate.	211 m ³ /hr
Suction Pressure	1.78 kg/cm ² A
Discharge Pressure	12.22 kg/ cm ² A
Differential Head	144 meters
NPSH available	>6 meters

Pump Change over Procedure:

- First take clearance from CPP as it is a HT pump.
- Ensure all the utilities like BCW (supply and return), seal steam are open, suction valve is open and discharge valve is closed. Drain the standby pump casing for removal of any entrapped water.
- Ensure the discharge is fully closed and warm-up is open.
- Close warm up after ensuring the readiness of pump.
- Start the pump and adjust the discharge pressure to the value required by DCS supervisor by opening the discharge and simultaneously closing the running pump discharge valve. This activity has to be done very slowly so that the flow does not fluctuate heavily. During this



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activity, amps of both the pumps have to be checked and ensure that it doesn't go beyond FLC for either of the pump.

- When the running pump discharge is fully closed and amps become low, confirm the flow with DCS supervisor and with his consent stop the idle running pump. Check for leaks after changeover of the pumps.
- Keep the stand by pump in warm up condition.

16.13 SR PUMPS:

It is a motor driven centrifugal pump. The pumping temperature is 350°C and viscosity at pumping temperature should be 3 cst. Vapor pressure at pumping temperature has to be 0.0316 kg/cm² absolute. Density of the fluid has to be 840 kg/m³. It has a design pressure of 29.7 kg/cm²A and temperature of 415°C. The casing and the impellor are made of 11-13% Cr-steel. It has a mechanical seal to protect out flow of liquid from the casing.

Operating Conditions:

Normal flow rate.	133 m ³ /hr
Maximum flow rate.	174 m ³ /hr
Minimum flow rate.	80 m ³ /hr
Suction Pressure	0.33 kg/cm ² A
Discharge Pressure	21.3 kg/ cm ² A
Differential Head	249.6 meters
NPSH available	3.5 meters

Pump Change over Procedure:

- First take clearance from CPP as it is a HT pump.
- Ensure all the utilities like tempered water, seal steam are open, suction valve is open and discharge valve is closed. Drain the standby pump casing for removal of any entrapped water.
- Ensure the discharge is fully closed and warm-up is open.
- Close warm up after ensuring the readiness of pump.
- Start the pump and adjust the discharge pressure to the value required by DCS supervisor by opening the discharge and simultaneously closing the running pump discharge valve. This activity has to be done very slowly so that the flow does not fluctuate heavily. During this



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activity, amps of both the pumps have to be checked and ensure that it doesn't go beyond FLC for either of the pump.

- When the running pump discharge is fully closed and amps become low, confirm the flow with DCS supervisor and with his consent stop the idle running pump. Check for leaks after changeover of the pumps.
- Keep the stand by pump in warm up condition.

16.16 Condensate recovery unit: (CRS)

The condensate recovery unit was commissioned in the 2010 T&I. The main aim is to recover the condensate obtained from the steam traps of various steam tracings. The steam trap outlets are joined together and same were routed to flash vessel located at CRS skid (at south of BBU compressors). In the flash vessel the steam and condensate were separated and condensate was routed through bucket trap to the collector drum and when the collector drum reaches 5 kg.cm² pressure the content will be routed to condensate tanks located at FCCU-II. The top uncondensed steam from flash vessel is routed to seal steam header with a check valve.

Total no. of steam traps connected to CRS: **233**

List of critical steam traps connected to CRS:

1. Steam tracing of Bitumen product take off from reactor- North of 13PM-03A
2. Steam tracing of bitumen R/D at pump discharge- North of 13E-03A
3. Steam tracing of CDU-II SR to BBU C/V loop- East of BBU off gas KOD
4. Steam tracing of BBU Long circulation loop-South of 13E-01A/B
5. Steam tracing of SR to HFO, RFO, IFO, LDO, and start up circulation loop at SR manifold- North of Preheat train-II block v/v.
6. Steam tracing of SR to VBU R/D at SR manifold- North of 12PM-07B
7. Steam tracing of SR LCV manifold- West of SR LCV
8. Steam tracing of SR pump spill back- North of 12PM-02B
9. Steam tracing of CDU-III SR to BBU- North of 12PM-01B
10. Steam tracing of SR pump discharge- North of 12PM-01A
11. Steam tracing of SR to HFO, IFO, and LDO at B/L- South of BBU off gas KOD-2
12. Steam tracing of 12F-01 FO Supply & Return- North of pillar no. N26
13. Steam tracing of HVGO to slop & slop header- South of slop manifold



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14. Steam tracing of HVGO hot feed- West of 11PM-03C
15. Steam tracing of SR quench C/V loop- South of 11PM-04B
16. Steam tracing of HVGO R/D, RFO, and Bitumen R/D at B/L- North of 11F-01 MP steam manifold
17. Steam tracing of HW off gas flame arrestor loop- North of flame arrestor
18. Steam tracing of 11F-01 FO Supply & return- West of inst. air KOD
19. Steam tracing of SR to VBU at B/L- South of EBL

Flash Drum & Pump:

Flash drum pressure is holding at 2 kg/cm²g. Condensate from flash drum is pumped out by a Pressured Powered Pump for which MP steam acts as a motivator. Pump is Operating at a temp. 75 deg C & pressure 4.5 bar.

Condensate Drum:

Condensate drum at FCCU-II is commissioned which is having a LT, & LOW & HIGH level switches. FCCU-II CRS condensate header joins to 4" header before connecting to condensate drum. A TG & PG provided at the drum inlet condensate header.

PROCEDURE:

Pressured Powered Pump Package Unit:

1. Slowly open supply to provide pressure at the PP pump inlet valve.
2. Check that trap is operational. Open gate valves in the condensate inlet and discharge line.
3. Open valves ahead of the unit allowing condensate to enter the receiver and fill the PP pump body. Pump will discharge when full. PP pumps should cycle periodically cycle.
4. Set the motive pressure about 2 kg above total back pressure including frictional losses.
5. If the condensate to the pump is slow. The pump will not operate till it is full. The strokes will be intermittent.



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Do's:

1. Before attempting any maintenance, be sure that the unit is completely isolated and relieved of any internal pressure. Motive supply, exhaust, condensate inlet and discharge lines should all be closed.
2. Check whether condensate reaches pump.
3. Check the vent is open to atmosphere, check strainers weekly for chokes and clean.
4. Check motive pressure. It must be 2 bar above the back pressure.
5. When the available motive steam or air pressure exceeds 8.7 kg/cm², a pressure reducing arrangement is required to reduce pressure going to pump.
6. The discharge line should rise vertically up immediately after outlet DCV to cover the entire head and then it should discharge to the tank by a sloping line. This is necessary so that frictional losses are kept to minimum and realize maximum advantage of gravity.

Don'ts:

1. Do not close the vent under any circumstances
2. Do not allow the motive pressure to fluctuate
3. Do not install crooked delivery line.
4. The discharge line should never be smaller than the bore of the outlet valve, it should be properly sized to handle the quality of condensate.
5. Do not hammer the pump if it does not work. Go to the trouble shooting schedule.
6. Do not allow steam in the condensate to enter the pump body. Steam must be separated before hand by providing proper trap.

Trouble shooting of Pressured Powered Pump:

S no	Symptom	Cause	Check and cure
1.	Pump fails to operate on start up	a. Motive supply closed. b. Condensate inlet line closed or flow is inadequate c. Condensate discharge line closed. d. Motive pressure insufficient to overcome back pressure	1. Open valves to supply motive pressure to pump 2. Open condensate valve, check upstream strainers and clean. Wait till sufficient condensate flows to the drum and fill the pump. 3. Open all discharge lines to allow free discharge from pump to destination. 4. Check motive pressure and static back pressure. Adjust motive pressure to 1 to 2 bar more than total back pressure.



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2.	Supply line/ equipment flooded, but pump appears to cycle normally.	a. Pump under sized. b. Insufficient motive pressure to achieve rated capacity. c. Restriction in condensate inlet line d. Inlet or outlet check valve stuck open (debris)	a. Verify the rated capacity as per the TIS capacity table. Increase check valve size or install additional pump as required. b. Check motive pressure setting and maximum back pressure during operation. Compare with capacity table of TIS increase motive pressure as required to meet the load requirements. c. Clean the strainer and check that all valves are fully open. d. Isolate check valves and relieve line pressure. Clean the DCV's
----	--	--	---

Pressure reducing valve (DP143):

- i) Ensure that all connections are properly made and that all valves are closed.
- ii) Close all valves at reducing valve station, including valves on by pass line if fitted.
- iii) Check that adjustment screw is turned fully anti-clockwise until spring is slack.
- iv) Open small valve in pressure control line.
- v) Blow through the approach pipe work by removing the cap and screen from the strainer protecting the steam trap draining the upstream pipe work. Replace upon completion. Do not remove the screen from the main line strainer during this operation. Although this should remove most of the dirt which is present, it may be necessary to examine and clean the main line strainer at regular intervals.
- vi) Slowly open the upstream isolation valve until it is fully open.
- vii) Using a suitable spanner slowly turn adjustment screw in a clockwise direction until desired downstream pressure reading is obtained.
- viii) Holding the adjustment screw in position with the spanner tighten down the locknut to secure the setting of the adjustment spring, making sure that the "C" washer stays in position.
- ix) Slowly open the downstream valve until it is fully open.

Maintenance of the valve:

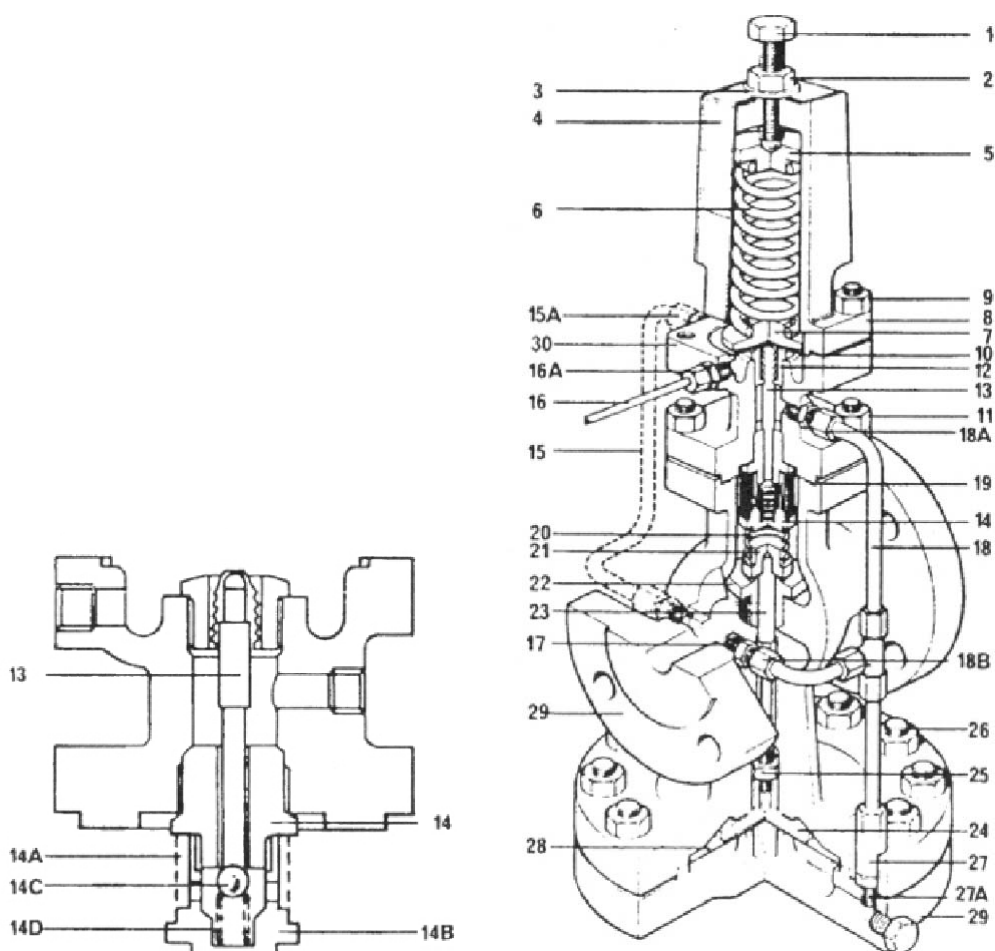
It is recommended that the valve is dismantled once in every twelve to eight months for a complete overhaul and ideally this is to be carried out with the valve removed from the line.



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The parts that may require replacing or refurbishing are listed below. Main valve (22) and valve head (21), pilot valve assembly (14), pilot diaphragms (10), main diaphragms (28). Diaphragms and cleaning:

If the valve is dismantled and either the main diaphragms or the pilot diaphragms are not renewed care must be taken not to turn the diaphragms over- refit them in exactly the same position as and when dismantled. The control orifice in the adapters (17) and (27a) and the interconnecting pipe assembly (18) as well as the pressure sensing pipe work (16) or (15) must be kept clear of dirt. Blow through with compressed air if necessary- do not use a drill on either of the control orifice, one of which contains a split pin, as enlargement of the orifices might upset the operation of the valve.



Pressure reducing valve (DP143)



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UPSET CONDITIONS & STABILIZATION

In any running unit, upsets are bound to happen. Sometimes they can be avoided and sometimes the conditions are unavoidable. The upsets can be of any type, nature. It can be a situation encountered earlier or a totally new situation. The dealing with upset conditions needs a blend of knowledge and experience for successful handling. Some unit upsets which have been observed earlier are discussed here. The probable reasons for upset and the stabilization methods are also discussed here.

17.1 FEED PUMP LOSING SUCTION:

- This can be confirmed by the feed pump suction pressure gauge. If it is low, immediately ask TPH to check the offsite booster pump and check line-ups and make corrections immediately.
- If the upset is not from offsite, then check the feed pump condition and immediately change the pump if any abnormality is found. Check DP of the running pump suction strainer for any plugging due to which the pump may be losing suction. Take clearance from CPP and PP-II before change over of the pump as this is an HT motor and stand by pump is turbine driven.

Two different upset conditions are observed based upon the status of PFD in the unit:

When PFD in service:

PFD level gives some cushion and unit can be sustained for some time and during that time reduce heater feed flow (for sustaining PFD level) and heater firing. If the feed flow does not increase to maintain PFD level and thereby feed to heater cannot be sustained at even lower feed rate, unit has to go for shut down.

When PFD not in service:

Keep a check on heater COT as it will shoot up very quickly. If it cannot be controlled by pressure of FO and FG then remove fires from the heater. If the feed has failed totally and there is no resumption immediately, unit has to go for emergency shut down

17.2 DESALTER PRESSURE FLUCTUATION:

1. Following control valves to be checked for faults/fluctuations:

- Desalter PCV
- FCV1101
- PFD LCV (If it is in AUTO mode, level to be checked for fluctuations)



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- iv) Wash water FCV to Desalter
 - v) Effluent water / Desalter Interface LCV (If it is in AUTO mode, level to be checked for fluctuations)
 - vi) Pass flow control valves (When PFD is not in service)
2. Feed pump discharge pressure to be checked for fluctuations
 3. Water carryover from desalter and probable loss of suction of booster pump
 4. Desalter PCV malfunctioning

In brief:

- This will happen if the feed pump discharge pressure is varying vigorously. Here the feed pump pressure has to be stabilized and if necessary, the desalter pressure control valve has to be operated manually for sometime to stabilize it.
- The same will also happen if there is any sudden fluctuation in desalting water inlet flow rate or the effluent flow rate. Also if there is any problem in the desalter LCV, the pressure will start swinging badly. In such cases, the fluctuation in water flow should be avoided as much as possible. The root cause has to be analyzed immediately and the problem to be sorted out. If there is a problem with level control valve, desalter water injection has to be removed after permission from Unit Manager/YSF.
- The desalter interface level also has to be kept at a stable value and any changes required have to be done very slowly. Any sudden change in set point will give a jerk to effluent water flow and thus result in a fluctuation in desalter pressure.
- If the Desalter mixing valve DP is high and proper separation is not there then carryover of water from Desalter to next preheat exchangers, Desalter pressure will start to swing vigorously.
- If the Desalter pressure control valve gets opened more on its own, the feed to the unit will suddenly increase. Also the Desalter pressure will shoot up very quickly. In this case, the pressure control valve has to be bypassed immediately and rectified on priority.
- In case the Desalter pressure transmitter is malfunctioning, the Desalter pressure will not be displayed correctly but the control valve will operate depending upon the value transmitter is showing. In that case immediately the control valve has to be taken in manual control and transmitter to be rectified on priority.
- If the PFD level control valve fluctuating badly, this will give back pressure on Desalter and pressure will start to fluctuate drastically. If this happens then that control valve has to be immediately taken in manual and rectified on priority.
- When PFD not in service and if the heater pass flows control valve gets wide opened or closed in one or more passes, it will give sudden jerks to Desalter pressure and the pressure



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will start to fluctuate drastically. If this happens then that control valve has to be immediately bypassed and rectified on priority.

- Also check the heater pass flows and the preheat train –II split control valve position. Any problem in the control valve like getting stuck suddenly at wide open or full close condition will also result in booster pump pressure and flow fluctuations
- If all the above said conditions are Ok, Desalter RV to be checked for passing. Passing in Desalter RV may lead to pressure fluctuation. Same can be confirmed by reducing atmos. Column Flash zone temperature but if the passing is very low, drop in temperature will be minimum to identify

17.3 BOOSTER PUMP SUCTION PRESSURE FLUCTUATION:

- This will happen when the desalter pressure is fluctuating. Since the booster takes direct suction from the desalter, to ensure steady booster suction and hence discharge pressure, it is very important to keep desalter pressure at a steady value.
- Check the feed booster pump condition and if any abnormality is found, immediately bypass PFD and change the turbine line up to booster line up and change the pump. Check the suction strainer of the pump for any plugging due to which the pump may be losing suction. Take clearance from CPP and PP-II before changeover of the pump as this is HT motor and other is turbine.
- Check the PFD level control valve position. Any problem in the control valve like getting stuck suddenly at wide open or full close condition will also result in booster pump pressure and flow fluctuations.
- Also check the heater pass flows (when PFD not in service) and the preheat train –II split control valve position. Any problem in the control valve like getting stuck suddenly at wide open or full close condition will also result in booster pump pressure and flow fluctuations.

17.4 FUEL OIL PRESSURE FLUCTUATIONS:

- This may happen due to upsets in FO supply pressure from IFO. In that case, the pressure will fluctuate heavily and accordingly control valves will respond. Take control valve in manual and keep the pressure at some steady value by operating control valve manually. Once the control valve becomes unsteady, it becomes very difficult for it to stabilize on its own in AUTO-CASCADE mode. Also the COT has to be monitored because FO fluctuations mean that COT is bound to fluctuate. Check the COT and vary FO control valve opening or gas consumption accordingly as per the situation. FO header pressure fluctuation from IFO



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can be confirmed as this will cause FO pressure fluctuations in both the heaters and can be ruled out if the fluctuations are observed in only one heater.

- The other reason can also be a problem in the FO pressure control valve. It may get stuck and will result in no response from control valve. The graphics will show it to be operating but actual control valve will not be operating in the field. In this case, first the reason of the pressure fluctuation has to be confirmed as non-operation of the valve and the same has to be rectified by maintenance.
- The problem will also arise if the transmitter is malfunctioning. In that case, take control valve on manual control and transmitter steam tracing needs to be checked for being effective. If the steam tracing is effective then the transmitter needs to be checked by maintenance.
- Passing of steam purge line to any of the FO guns will cause FO pressure fluctuations. Check for each purge to be tightly shut.
- COT fluctuation when FO is in CASCADE mode.
- SDV malfluctuating

17.5 FUEL GAS PRESSURE FLUCTUATIONS:

- This may happen when the fuel gas header pressure is varying very quickly. In that case, the pressure will fluctuate heavily and accordingly flow fluctuates and leads to heater COT to fluctuate. Take FG flow control valve in manual and keep the pressure at some steady value by operating control valve manually. Once the control valve becomes unsteady, it becomes very difficult for it to stabilize on its own. Also the COT has to be monitored because FG fluctuations mean that COT is bound to fluctuate. Check the COT and vary FG control valve opening or gas consumption accordingly as per the situation. (To be observed in both heaters, same as FO pressure fluctuations)
- The other reason can also be a problem in the FG control valve. It may get stuck and will result in no response from control valve. The graphics will show it to be operating but actual control valve will not be operating in the field. In this case, first the reason of the pressure fluctuation has to be confirmed as non-operation of the valve and the same has to be rectified by maintenance.
- The problem will also arise if the transmitter is malfunctioning. In that case, take control valve on manual control and transmitter steam tracing needs to be checked for being effective. If the steam tracing is effective then the transmitter needs to be checked by maintenance.
- Liquid carryover in FG, the best way to confirm is to see the FG fire. The flame will be smoky with fire flies.



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- Abnormal increase in flow and drop in pressure will be observed if a FG tip gets disengaged.
- SDV malfunctioning.

17.6 11-F-01 PASS FLOWS FLUCTUATIONS:

- This can be a result of booster or PFD turbine discharge pressure fluctuations. If the booster or PFD turbine flow is fluctuating, then the feed to heater pass flows will vary, resulting in heavy variations in the control valve opening to counter the effect. The variations will affect the flow through the pass flows. In this case (PFD not in service), the booster discharge has to be stabilized first because that is the root cause of the problem. After stabilizing the booster discharge pressure, if the control valves do not get stabilized, then operate them manually and stabilize the feed flow. In case of PFD in service, check the healthiness of turbine and proper steam consumption of turbine. Due to HP steam fluctuation turbine speed will vary and lead to pass flow fluctuations.
- The feed flows can also fluctuate if a pass flow transmitter fails. For example, if Pass A FT starts showing “0” value then the control valve will open further to normalize the value. But the actual flow is not zero in the tube. But on the opening of control valve, flow will start to increase (in case PFD in service, PFD level will decrease and level control valve will open) and thus the total feed going to the unit will start increasing, thus upsetting the unit. The only solution here is to watch the outlet temperature and analyze if the flow has really reduced or is it transmitter problem. In case of transmitter problem, the control valve has to be taken on manual immediately and output should be given around the value before upset. Then monitor the pass outlet temperature and accordingly operate the control valve. Transmitter has to be rectified on priority basis. Also check for the steam tracings of the transmitter lead lines to be effective.
- If PFD not in service and the pass flows are also fluctuate if there is a restriction of flow from upstream like too much closing of 11-FRC-101 (PHT II split valve). In this case, the control valves get wide opened but then also the sufficient feed flow cannot be established. Here, 11-FRC-101 needs to be opened further to establish sufficient flow in the heater.

17.7 ATMOSPHERIC COLUMN PRESSURE SHOOTING UP:

- The column pressure can shoot up if there is excess vaporization in the column. This can be seen and concluded from the column conditions. If the case is of excessive vaporization, the COT needs to be reduced. Sometimes crude layers may get formed and instead of a blend of crudes coming to a unit, the case may happen that the lighter component of the blend is



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coming to the column. Thus this lighter crude will vaporize to a larger extent increasing the column pressure. CR's to be increased to maximum possible extent.

- In some cases, it may happen that 11V01 may get full of liquid and start exerting back pressure on the vapor coming from column to 11V01. In this case, column pressure will shoot up very quickly. If the level is observed to be very high in gauge glass, it has to be reduced to approximately 50% immediately. Once the level comes down, column pressure will also get reduced. During such situations, column top temperature and COT can be temporarily reduced to reduce the yield of naphtha.
- Again if the boot level of 11V01 is full, water will come to HC section of the drum and this water will go back to column as reflux stream. This reflux stream will contain water and thus water will return to column by this path, causing the top pressure to shoot up. Here, the top temperature and pressure will come down for some time and then start shooting up.
- If the column bottom stripping steam control valve gets opened more and the steam flow starts going high, the column top pressure will rise. In this case, the problem has to be identified and then flow to be reduced to normalize column conditions.
- The column top pressure will also shoot up if there is a reduction in the cooling water pressure or if the cooling water fails totally. The situation has to be handled depending on the extent of water failure.
- The column pressure may also shoot up and stay at higher value if there is any vapor lock in the column. This can only be reduced by releasing the vapor lock from the column.
- Pressure also shoots-up if water is entering the column in high volumes. This happens when water is present in crude and proper separation of water from crude is not done at the desalter outlet. The carried over water will create upsets in the column.
- If the control valve of fuel gas make up to column gets wide open, then the pressure will start shooting and will result in high column pressure. In that situation the control valve upstream or downstream block valve needs to be pinched down and adjusted depending on the value of column pressure.
- If any of the CR flow fails then that will also result in shooting up of the column pressure. Immediately the CR flow has to be restored. If that is not possible, other CR to be maximized and feed rate may be needed to reduce after clearance from Manager/YSF.
- If the top reflux flow control valve gets more or wide opened, the pressure of the column will shoot up. The control valve in that case needs to be bypassed and rectified on priority.
- If the column bottom level is abnormally more, the column pressure will swing and hold at a high value. In that case, level to be cross checked by column DP value as well as by checking the levels in the gauge glass.
- If the column top pressure transmitter fails, the value shown by it will be wrong.



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- If column DP increases and causes flooding. This will lead to flaring from the column due to sudden release of lighter fractions.
- Water carryover from 12E10/10A
- If overhead condensers salt water valves drop seated during back flushing of condensers, column pressure will shoot up. Same can be confirmed by checking the individual salt water valves and same has to be reverted back.

17.8 11-V-01 LEVEL HIGH:

- Sudden increase in level at same operating conditions indicates naphtha yield has increased. In such cases, the lighter end content of the crude may have increased. So reduce the top temperature of the column but do not go below dew point. Both the reflux pumps are to be placed in service in this case to bring down the level. Reduction of COT may also help in this situation.
- In case where the stabilizer feed control valve is having a problem, 11-V-01 level will increase. If the control valve got stuck and not opening beyond some value the control valve has to be by passed and rectified on priority basis.
- The drum level can also go up if there is a problem with top reflux pumps. In that case, if sufficient flow is not obtained which is required to maintain the top temperature, the top temperature will shoot up and naphtha yield will rise. Also the pump will have to take more load for level reduction as reflux flow has come down.
- Level to be cross checked with gauge glass if increase is instantaneous

17.9 COLUMN BOTTOM LEVEL HIGH:

- The main reason of the column level going suddenly high will be failure of RCO pumps. In that case, try to place the standby immediately. If the standby pump is not starting immediately then feed rate has to be reduced till the time the pump is available. If the problem continues for a longer time then follow the procedure discussed in chapter 19.
- If the column level is slowly building up and the overflash is high, increase the diesel yield so that the overflash comes down to normal value and the level increasing is stopped. But while doing this, ensure that the diesel color is ok and not going off-spec.
- If vacuum heater pass flows control valves encounter a problem, 11-C-01 bottom level will start increasing. Then it becomes necessary to rectify the control valve or use bypass to provide sufficient flow.
- If either kerosene, diesel flows come down drastically or fail totally, the liquid content will get accumulated in the column bottom and raise the level.



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- If there is a sudden dip in 11F-01 COT due to FO or FG pressure fluctuation or some other reasons, bottom level will go up.

17.10 COLUMN BOTTOM LEVEL LOW:

- The column bottom level can come down if there is excess vaporization in the column. This can be seen and concluded from the column conditions. If the case is of excessive vaporization, the COT needs to be reduced. Sometimes crude layers may get formed and the instead of a blend of crudes coming to a unit, the case may happen that the lighter component of the blend is coming to the column. This lighter crude mixture will have lower diesel yield. Accordingly diesel flow has to be reduced and over flash to be normalized (which will reduce in this case). The liquid has to be allowed to go down in the column to maintain the level.
- Also if the column pressure comes down suddenly, vaporization will be enhanced and the contents rising up will be more. This will reduce diesel yield and thus the column level will also come down if the diesel flow rate is not reduced.

17.11 RCO PUMPS PROBLEMS:

- Any seal leaks in RCO pumps will lead to a sudden fluctuation of flow and result in vacuum heater tripping most of the times. Also due to RCO being at a very high temperature, the chances of fire are high by auto ignition. So RCO pump seal leaks create very severe situations in the unit. Immediately RCO pump has to be changed in case of any abnormality.
- Column level will start building up.
- Vacuum heater temperature will also start to shoot up if its feed reduces suddenly. Immediately fires need to be removed because high temperature will lead to coking and even chances of tube rupture may develop.
- Vacuum feed failure procedures have to be followed in case RCO pumps are not available for a longer period.

17.12 DIESEL-CR PUMPS LOOSING SUCTION:

- The Diesel section level can come down if there is excess vaporization in the column. This can be seen and concluded from the column conditions. If the case is of excessive vaporization, the COT needs to be reduced. Sometimes crude layers may get formed and the instead of a blend of crudes coming to a unit, the case may happen that the lighter component of the blend is coming to the column. This lighter crude mixture will have lower diesel yield. So, if the rundown of diesel is not reduced, the liquid level in the diesel section trays will



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come down and Diesel CR pump will start loosing suction. Diesel CR pump discharge can be pinched down to save pump from any damage. Accordingly diesel flow has to be reduced and over flash to be normalized (which will reduce in this case). The liquid has to be allowed to go down in the column to maintain the level.

17.13 COLUMN DP HIGH:

- Column DP high means either this will be due to excess vaporization in the column, or it is because column flooding. Actually both these causes cannot be segregated but the primary cause of high DP has to be known. In case of more vaporization, each section DP will generally be high. For each plate, maximum DP has to be approximately 0.01 kg/cm². If the vaporization is more, the DP in each section will be more than its design DP (no. of trays in section * 0.01 kg/cm²).
- But if there is flooding in a particular section, DP there will be very high but other sections will not be high. Flooding in one section may lead to flooding in the ones below it, for example, heavy flooding in KERO will result in fluctuations of Diesel yield and also increase in diesel yield. In this case, Kero product flow and Diesel product flow have to be increased. The reasons for column DP have to be carefully analyzed.
- If the column bottom level is abnormally high, the column DP will go up. In that case, the level has to be reduced by pumping out the content. The level should be brought down to approximately 50%.
- If either of the pressure transmitters of the column is not working properly, it will give wrong indication and hence mislead the person operating the DCS.
- If the feed rate is more than what is prescribed by TSD, the column may not be able to take that much load. It will result in increased DP of the column.

Whatever may be the cause of increase of DP, either increasing rundown or reducing COT will reduce column DP. But the counter effects of each have to be considered before taking any action. If problems like increased draw-off temperature or increased bottom level are encountered, counter action has to be taken.

17.14 COLUMN TOP TEMPERATURE FLUCTUATIONS:

- The column top temperature will go up if sufficient reflux flow is not present. The reflux is for controlling the temperature and if the temperature is not under control, it means either the reflux is not present or effective reflux is not present (reflux at proper temperature). Here the



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pumps need to be checked for proper pumping and need to be changed if the pumping is not sufficient or any fluctuations are there.

- The drum temperature also needs to be limited below the operating temperature of condenser outlet. High drum temperature means the reflux temperature is high and thus the heat transfer is less than the desired value. To reduce drum temperature over head condensers to be back flushed.
- Top reflux and all CR control valves also need to be checked in case of sudden rise or fall of column top temperature. In case of any control valve getting closed, the same has to be bypassed and attended by maintenance.

17.15 PRODUCT DRAW-OFF TEMPERATURES HIGH:

- If the product draw-off temperatures are high the rundown rates need to be reduced. Drawing more of a product increases the draw-off temperature. Draw-off has to be reduced for the product to cut down its temperature. If the draw-off rate starts increasing the column DP, it better to reduce COT because this is an indication of excess vaporization and carry over to higher trays.
- Also the higher draw-off temperature may be due to insufficient flow of CR. Or the case may be that CR return temperatures are high and thus they are not providing desired cooling effects. Here the CR pump need to be checked as well as exchangers for checking that bypasses are not open.

17.16 STABILIZER RE-BOILING FLUCTUATION:

- This can happen if the KERO-CR flow is fluctuating. KERO-CR is the heating medium in the re-boiler and any fluctuation in KERO-CR flow will work as change in flow of heating medium. The control valve will take care of that fluctuation but if it is too much then re-boiling temperature will start to vary. In this case stabilize the CR flow and then the temperature of re-boiling.
- If the response of KERO-CR control valve becomes erroneous or it gets stuck at a value, the re-boiling will fluctuate. In this case, the flow has to be controlled by bypass valve and the control valve has to be rectified on priority.
- If the level control valve of stabilizer opens suddenly and the level starts to fall down, the re-boiling will shoot up very quickly. If the converse happens, the re-boiling will reduce very fast.
- This can also happen if the feed to stabilizer valve gets wide opened or closed and thus reducing or increasing the re-boiling respectively.



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- If the temperature indicator thermocouple is faulty then it will show a wrong value, misleading the DCS operator.

17.17 STABILIZER TOP PRESSURE FLUCTUATIONS:

- This can happen if there is any problem with 11-PRC-501 A/B. If the FG control valve gets wide opened or full closed, the pressure will start to collapse or build-up respectively. The split control valves has to be immediately bypassed and system to be normalized. Rectification of the control valve has to be done on priority.
- If the stabilizer top reflux flow varies suddenly due to a problem either in pump or the control valve, the temperature of the stabilizer top section will vary and also the pressure. The root cause of the flow variation has to be accessed and then rectified. If the problem is with pump, change the pump to the standby. If the problem is with control valve, bypass it and establish the flow through bypass.
- If the LPG drum level is low, the LPG pump will not get suction and that will let the temperature and pressure of the system to shoot up. In this case, build-up the level by reducing rundown. If required the rundown can be closed by informing MEROX.
- The pressure also shoots-up in case of salt water failure or salt water pressure reduction. In this situation, inform to Manager/YSF and follow the instructions.
- The pressure also shots up if by some reason, 11-V-03 level becomes full. The gas flow will experience back pressure and result in increasing the stabilizer pressure. Immediately the level has to be brought down by increasing the rundown flow.
- If the top pressure transmitter is malfunctioning, it gives wrong indication misleading the DCS supervisor.
- Drum boot level filled up, water carryover in reflux.

17.18 12-F-01 PASS FLOWS FLUCTUATIONS:

- This can be a result of RCO pump discharge flow fluctuations. If the RCO discharge is fluctuating, then the feed to heater pass flows will vary, resulting in heavy variations in the control valve opening to counter the effect. The variations will affect the flow through the pass flows. In this case, the RCO discharge has to be stabilized first because that is the root cause of the problem. After stabilizing the RCO discharge flow, if the control valves do not get stabilized, then operate them manually and stabilize the feed flow.
- The feed flows can also fluctuate if a pass flow transmitter fails. For example, if Pass A FT starts showing “0” value then the control valve will open further to normalize the value. But the actual flow is not zero in the tube. But on the opening of control valve, flow will start to



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increase and thus the total feed going to the vacuum heater will start increasing, thus upsetting the heater conditions. The only solution here is to see the outlet temperature and analyze if the flow has really reduced or is it transmitter problem. In case of transmitter problem, the control valve has to be taken on manual immediately and control valve should be given around the value before upset. Then monitor the pass outlet temperature and accordingly operate the control valve. Transmitter has to be rectified on priority basis. Also check for the steam tracings of the transmitter lead lines to be effective.

17.19 VACUUM SECTION DRAW-OFF TEMPERATURES SHOOTING –UP:

- If the draw-off temperature of any particular product shoots-up, check for the circulating reflux of the product. The pump may not be giving sufficient pumping to maintain the draw-off temperature in the limits. Check pumps and in any problem is with the pump then change the pump.
- The CR flow may get interrupted incase the control valve gets affected like gets stuck or closed. In this case, control valve has to be bypassed and rectified on priority basis.
- If all the product draw-off temperatures are high, it shows that the vaporization is more than desired and heavier ends are getting carried to top. In this case it is better to reduce COT.

17.20 SUDDEN VACUUM DROP:

- The vacuum drop may be due to reduction of MP steam pressure or condensate carryover in the steam. The ejector steam is responsible for holding the vacuum and if the steam fails then the chances of maintaining vacuum are feeble. The steam supply has to be normalized immediately or the feed rate has to be reduced after clearance from MP/ YSF. In case of condensate carryover, proper draining has to be done to make steam free of condensate.
- If the salt water fails or the pressure drops, the holding of vacuum becomes very difficult. Here also the water supply has to resume otherwise feed rate has to be reduced after clearance from Manager/YSF.
- The vacuum may also get disturbed if the vacuum column bottom level is holding very high. In this case, the column level has to be brought down by pumping it out.
- If these corrections also are not able to give any improvements, we have to suspect leaks in the vacuum system and check for it. There are chances of blockage in dip legs also, which can affect the system vacuum.
- If the COT is high, it increases cracking in the system and result in the loss of vacuum. In such cases vacuum heater COT needs to be reduced quickly.



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- If the hotwell water level is holding low, the vacuum will start breaking. In that case the water level has to be restored to normal value and thus it will bring column to normal conditions.
- In case the hotwell gas to 11-F-01 SDV gets closed and the hotwell vent does not open, the vacuum will start getting upset. In that case, the SDV has to be reset immediately or the hotwell vent to be opened temporarily.
- If the hotwell water level transmitter fails and the thermo-siphon leg gets clogged, the vacuum will not hold and the column will get upset. Here gauge glass to be checked for level confirmation and hotwell bottom drain to be opened and level to be brought down to 60%.
- If any of the CR flow fails, vacuum will get upset. Immediately the flow has to be restored and if that is not possible, maximize other CRs and reduce feed if required after clearance from Manager/YSF.
- If the top pressure transmitter fails then the faulty indication will misguide the DCS operator.
- Pumps vents opening

17.21 VACUUM BOTTOM LEVEL HOLDING HIGH:

- If the SR pump is not taking proper suction due to strainer plugging or pump problems, level will start building up. In that case, the pump has to be changed and the problem has to be found out.
- If the rundown line up is wrong.
- If the bottom level transmitter fails, it misleads the DCS operator.
- If there is a sudden raise in RCO flow, the level in vacuum column will go up due to rise in column feed.
- Sudden increase in RCO flow or reduction in firing reduces the vacuum heater COT and thus the level at bottom starts building up and yields are lost. Cot has to be normalized immediately.

Drop in Vacuum will cause bottom, level to fill up.



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AVOIDING DEVIATIONS AND PLANT UPSETS

In a process unit, there are some upsets that cannot be avoided. But some incidents and upsets can be avoided by timely action and precautions. Proper checking in the field, following DRJs, and ensuring that all the systems and procedures are followed is the only way to avoid upsets. Some problems and upsets are discussed in the previous chapter and here we are discussing the possible methods of avoiding those upsets.

18.1 FEED PUMP SUCTION LOSS:

- In the case where feed pump is losing suction because of insufficient flow from offsite, there is nothing much that can be done to prevent it because the root cause of the upset is not from the unit. But the suction loss due to suction strainer plugging can be avoided by timely cleaning of suction strainer. The suction strainer cleaning of feed pump is a part of DRJ and it should be followed every time it falls under the schedule. If it is cleaned as per the schedule, then this situation can be avoided a number of times.

18.2 DESALTER PRESSURE FLUCTUATIONS:

- If the fluctuations are due to the feed pump discharge fluctuations, the same can be avoided by proper monitoring of feed pump. In most cases, close monitoring of such critical equipments will give us some indications before the problem becomes more critical. If any problem is found with the pump before unit gets upset, same has to be referred to rotary and advice to be taken for necessary actions.
- Water injection should not be given any sudden changes. If there is a need to increase or reduce the water injection rate to desalter, same should be done very slowly so that the interface level does not fluctuate.
- If the desalter effluent level control valve gets stuck or fails, there is nothing much that can be done to avoid such situations. The failure of control valve does not give any indication before the actual failure. So, the actions have to be taken after failure of valves and they need to be taken very quickly.

18.3 BOOSTER PUMP SUCTION PRESSURE FLUCTUATION:

- If the booster pressure is fluctuating due to desalter pressure fluctuations, the precautions and checks required for avoiding desalter pressure fluctuations have to be taken.



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- The suction loss due to suction strainer plugging can be avoided by cleaning of suction strainer. As the stand by pump being used for PFD the suction strainer cleaning of feed booster pump should be carried out when the PFD was not in service. If it is cleaned then this situation can be avoided.

18.4 FUEL OIL PRESSURE FLUCTUATIONS:

- Individual fuel oil lines to be flushed whenever any shutdown is there. And proper circulation to be established for FO system.
- Lead lines of the pressure transmitter need to have proper steam tracing. In case the steam tracing is not effective, the transmitter will show faulty reading and thus it will result in unit upset. So, during rounds the effectiveness of tracing need to be checked.
- Any operation of supply or return block valves have to be done very slowly as their response is very quick. It has to be done by a person with experience or under the supervision of a person with experience.

18.5 FUEL GAS PRESSURE FLUCTUATIONS:

- Lead lines of the pressure transmitter need to have proper steam tracing. In case the steam tracing is not effective, the transmitter will show faulty reading and thus it will result in unit upset. So, during rounds the effectiveness of tracing need to be checked.
- While placing fire in heater utmost care to be taken and Fuel gas to be opened slowly and it is ensured that pilots are on.

18.6 11F01 PASS FLOWS FLUCTUATIONS:

- For keeping the pass flows at a stable value, PFD turbine discharge or the booster discharge pressure and flow has to be kept at a steady value. So, the measures discussed for keeping the turbine or booster normal are required to keep the pass flows also steady. The healthiness of the pump ensures that pass flows do not get upset due to its feed variation.
- The DCS operator has to ensure that there should not be any high variation in the heater flows during feed switch. Care to be taken and step by step feed to heater to be increased or decreased.



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18.7 ATMOSPHERIC COLUMN PRESSURE SHOOTING UP:

- The column overhead drum 11V01 level should be closely monitored in case of any unit upset and only LT value should not be trusted. Gauge glass level should be checked after draining the gauge glass. Same should be done with the boot level also and ensure that both the levels are within normal operating limits (not very low and not very high).
- During rounds in the field, all the condensers 11-E-17A/B, C/D, E/F, G/H should be checked for any leaks, abnormality, valves drop seating should be observed in field so that the problems are identified and resolved before becoming cause for major upset.
- The split control valves (gas make-up to column and gas to flare) should be checked regularly so that any problem in them is identified quickly. Any problem in it will result in column pressure variations.
- Water carryover can be minimized during feed tank switch by taking out water injection prior to tank switch and settling of interface to be ensured.

18.8 12F01 PASS FLOWS FLUCTUATIONS:

- For keeping the pass flows at a stable value, the RCO pump discharge pressure and flow has to be kept at a steady value. The healthiness of the pump ensures that pass flows do not get upset due to its feed variation.
- The DCS operator has to ensure that there should not be drastic increase or decrease in Atmos bottom level and heater pass flows.

18.9 SUDDEN DROP IN VACUUM:

- Steam has to be ensured to ejectors and the steam has to be condensate free. Proper draining of condensate should be ensured for the MP steam line.
- Close monitoring of pumps will ensure that there is no vacuum drop due to failure of pumps. Running pump healthiness and standby pump readiness has to be ensured.
- Both flame arrestors strainers have to be cleaned periodically and ensure one flame arrester kept in service and timely drained so that there is no backpressure on vacuum column. The water level has to be sufficient so that the dip legs are immersed in the water. Hot-well off gas fires need to be checked to ensure that gas incineration continues uninterrupted.
- Vacuum column level should not be allowed to go very high. Parameters like flash zone temperature of vacuum column have to be monitored as indicators of vacuum bottom level.
- Checks for leaks have to be done thoroughly if other parameters correction is not able to improve the vacuum condition.



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NOTE: The problems caused by failures of instruments like control valves, transmitters, etc do not give any indication of their happening. So, there is no way we can avoid the upsets caused by their failure. But the time taken to suspect or conclude the failure of an instrument plays a very important role in the extent of upset it can cause. A robust approach in identification of problem will result in quick and comfortable trouble shooting.

Regular field round need to be effective. Lots of emergencies can be avoided by close observation in the field. Two most important things that need to be checked in the field are the healthiness of running equipment and readiness of standby equipment. The DCS supervisors can avoid many emergencies by timely assessment of unit condition changes and by checking trends of varying parameters. If any upset is timely handled, it solves majority of the problem compared to what happens if the action is not timely



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EMERGENCY PROCEDURES

When the safe routine operation is interrupted, emergency procedures are sometimes required to overcome the potential hazards. If continuation of unit operations on limited basis is impractical the unit must be shut down as safely as possible.

19.1 GENERAL

The emergency procedures attempt to overcome the hazard of a quick shut down. Emergency can result from equipment failure and from interruption of utilities. Certain features have been designed into the plant to minimize the likelihood of emergencies. These include spare pumps, exchangers etc. Operators should be thoroughly familiar with emergency procedures and understand the reasons for each work. Good judgment must be exercised as no written procedure can completely cover all details or problems that can arise in an emergency. Judgment is more likely to be exact if prior thought and planning have been made. As during emergency there is no time, emergency procedure must be learnt before hand.

The steps to be taken during various emergencies are given below.

It depends on common sense and prevailing condition of emergency. But normal encountering emergency, we have to follow the sequence of operation for the safe unit shutdown.

1. Inform YSF, MEROX, Power Plat, FCCU, DHDS and TPH.
2. Trip heater 11-F-01 and 12-F-01. Purge and isolate fuel oil and fuel gas with steam.
3. Take out feed by stopping feed pumps and PFD turbine and isolate West battery limit feed Valve.
4. Monitor 11-C-01 pressure, accordingly open PCV Bypass valve (if pressure goes up) and if pressure goes down open Fuel Gas make-up, close pressure control bypass.
5. Isolate stripping steam control valve and block valve and MP to LP steam.
6. Close BFW block valve to steam gens.
7. If power available use Top Reflux Pump for contain top temp and pump out level to slop.
8. If power available pump out levels and close run-down valves. Close pump discharge valves.
9. Close 12-C-01 ejectors steam and ensure slight positive pressure in Vac column (0.5 kg/cm²g) by operating Fuel Gas make-up and close Hot Well vent.
10. Close Hot Well Water and Atmos Water and Effluent water to SWSU and open the Hot Well Water and Atmos water side block valves to OWS.
11. Check all the equipments for any leak.



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19.2 POWER FAILURE

The unit will be brought down as indicated below during power failure.

ATMOSPHERIC SECTION:

- i. Shut off individual burners and flush out oil form burners with steam.
- ii. Shut off stripping steam to the main column and side strippers.
- iii. Close all pump discharge valves.
- iv. Hold pressure in the crude column and Stabilizer by operating PRC-1409 and PRC-1501.
If required, open PRC1409 by pass valve to control the atmospheric column pressure.
- v. Hold levels in columns, vessels by manually closing the respective control valves /isolation valves. When power is restored, establish circulation by running crude pump, **booster pump** and RCO pump as per normal practice.

VACUUM SECTION:

- i. Shut off ejector steam.
- ii. Close non-condensable return line to inlet of first stage ejectors.
- iii. Open fuel gas to inlet of first stage ejectors to bring up the system pressure to 0.5 kg/cm² g, if power is not resumed immediately or start-up of unit is delayed.
- iv. Close all individual burner valves and flush out fuel oil burner guns one by one.
- v. Close all pumps' discharge valves.

As soon as power is available and unit start-up is delayed, flush the system with flushing oil. Take fresh **gas oil** (cutter) and start the unit as per normal start-up procedure as described earlier.

19.3 LOSS OF STEAM

Effects on Atmospheric Section

This will affect the following:

- Stripping steam to crude column and strippers
- Atomizing steam
- Tracing Steam



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In the event of loss of steam, the unit will be brought down as follows:

- i. Cut off fire to the heater. FD/ID fans may be kept running
- ii. Shut off all burners and flush out all oil burners, if possible.
- iii. Keep fuel oil circulation on
- iv. Resort to circulation in the bottom section using VR Circuit and crude tank. Vacuum unit will be on short circulation through its furnace.
- v. Start close circulation by diverting 11-P-10 A/B discharge to 11-F-01 inlet through VR circuit and crude tank.
- vi. Stop all the other pumps.
- vii. Hold levels and pressures by taking all controls on manual.
- viii. Switch over to normal start-up circulation, once emptying out of the Atmospheric column is over.

If the duration of steam failure is long it is advisable to flush the lines and equipments with flushing oil. The procedure outlined under normal shut down procedures in **section -5** for flushing and emptying out shall be followed.

Effects on Vacuum Section

- Atomizing steam to burners
- Ejectors
- Tracing Steam

The following steps have to be taken for safe shut down of the unit.

- i) Close ejector steam valves.
- ii) Close non-condensable return to inlet of first stage ejectors.
- iii) Back in fuel gas and maintain pressure at about 0.5 kg/cm² g.
- iv) Shut down heater and isolate all burners. Purge out oil burners if steam permits. FD and ID fans may be kept running.
- v) Start circulation from 12-P-01 A/B to 12-F-01 inlet. Keep the coil flow at minimum level.
- vi) Start emptying out heavy stocks to slop by diverting a part of 12-P-01 discharge through VR circuit.
- vii) Pump out products to slop.
- viii) Cut in flushing oil to the bottom of vacuum column.
- ix) Unit can be started back as per normal procedure on resumption of steam supply.



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19.4 LOSS OF COOLING WATER

ATMOSPHERIC SECTION

If cooling water failure is total, the unit will be brought down immediately.

- All fires to the heater including pilot will be cut off.
- Burner guns will be steam purged.
- Other pumps can be run to empty out and cool the system as far as possible (if run down temperature permits).
- Stop stripping steam to crude column and strippers.
- Crude column refluxing should continue at a flow rate to cool down the column.
- Column pressure is to be watched. If necessary, open PR1409 control valve bypass valve to control the pressure.
- Crude column Circulating refluxing should continue to cool down the column. Stop Top reflux pump.
- Stop crude pump, crude booster, PFD booster if cooling water could not be resumed.
- Other pumps can be run to empty out and cool the system as far as possible (if run down temperature permits)
- If cooling water supply is not restored soon and if flushing oil from off-sites is available at sufficient pressure, RCO can be flushed out through VR circuit.

If cooling water failure is partial, it can be tackled by reducing throughput and throttling water to cooler and condensers proportionately. In such cases careful watch should be kept on column pressure and also on the pump where cooling water is used as coolant.

VACUUM SECTION

Loss of cooling water will result in failure of ejector condensers and other product coolers. The following actions are to be taken in such emergency

- Watch system pressure for any abrupt changes.
- Cut off steam to ejectors. Normally with barometric seal provided, Atmospheric air is not expected to enter the system. However, this is an added precaution.
- Shut down the heater and close all burner valves. Flush oil burners and purge the heater.
- Cut off feed and resort to circulation.
- Pump out heavy oils and take in flushing oil. Continue circulation.
- Unit can be started as per normal procedure on resumption of cooling water.
- Observe Tempered water system temperature .Open make up water if Temp is crossing 90 Deg. Over flow tem water drum if required to avoid hammering.



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- If SR R/d temp is limiting, Open Quench C/V to maintain minimum flow to SR pump.
- Pump out heavy oils and take in flushing oil.
- Cut off feed and resort to circulation.

In case of partial failure of cooling water, Unit can be kept on running with reduced throughput in conjunction with Atmospheric section by throttling cooling water to the condensers. Close watch to be kept on the pumps, which are running with cooling water as coolant.

19.5 LOSS OF INSTRUMENT AIR

Failure of instrument air will cause the control valves to go to fail-safe position. Pressure, temperature, Flow and level indications of various equipment and headers will be available in the control room. Emergency shutdown to be operated. Fuel Gas Make-up to Atmos column control valve block valve should be shut and flare release control valve bypass valve should be operated to control the pressure.

19.6 LOSS OF FEED

ATMOSPHERIC SECTION

When PFD online: Feed failure will be indicated audio-visually by low total flow alarm provided at the feed pump discharge. Immediate attention may be paid for locating such failure and restore the flow. PFD level may act as cushion to sustain minimum flow to heaters for some time during feed failure due to offsite crude booster pump failure.

If pass flows low flow alarms activates (when PFD online) this may occur due to improper functioning of 11-PT-02B, immediately turbine to be checked and PFD to be bypassed by operating PFD ROV and stopping PFD turbine.

When PFD not in service: Feed failure will be indicated audio-visually by low flow alarms provided in the crude heater pass flow controllers 11-FRC-1301 to 1304. Immediate attention may be paid for locating such failure and restore the flow.

In case feed flow could not be restored immediately, further reduction of flow will cause activation of low flow switches which will off fuel oil and fuel gas to the heater. Failure of crude supply to unit may occur due to improper functioning of off-sites Crude Pump Unit, Crude Charge Pump 11-PM-01 A/B and Booster Pump 11-PM-02A. If Crude Charge pump's



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discharge pressure tends to come down cut down unit throughput to restore normal pressure. Look for the cause of the trouble. The operational factors responsible for pump malfunctioning are low level in the tank, improper lining up, choking of strainers and maintenance of improper temperature and pressure conditions in the Desalter. Any operational mistake which is readily detected will be corrected at once. Standby pumps will be pressed into service for any trouble with the running pump. If feed supply cannot be restored immediately the unit will be brought to hot circulation by diverting 11-P-10 A/B discharge to furnace inlet through VR line. Temperature will be maintained around 200 °C. Products will be stopped. Water to coolers will be throttled. If the interruption is for a longer period the unit will be shut down as per normal procedure.

VACUUM SECTION:

Feed failure to vacuum section can be due to interruption in the Atmospheric section. Failure will be indicated by pass flow low alarm. Further reduction in flow will activate and trip fuel oil and fuel gas to the furnace. If the failure is due to Atmospheric column bottom pump, start the spare pump. If the flow cannot be restored immediately, the following steps are to be taken.

- i) Cut firing rate in the furnace in case FSL is not activated already.
- ii) Put the unit of circulation by diverting 12-P-01 A/B discharge to furnace inlet through start-up line.
- iii) Bring down the unit as per normal procedure.

19.7 HEATER TUBE FAILURE

ATMOSPHERIC HEATER 11-F-01:

Vacuum section will be put under circulation and will be shut down as per normal procedure. Atmospheric section will be brought down either by emergency procedure or by normal procedure depending upon the extent of damage. A small crack in the tube may eventually coke up and may not warrant and emergency shut down.

The following steps are to be taken in order to bring down the unit immediately in the event of any major failure:

1. Feed to the heater will be stopped.
2. Fires to be minimized in the heater. Once HC in the furnace is burnt, take out all fires including pilots.



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3. Snuffing steam will be opened in radiation and convection sections. Emergency coil steam will be opened in all the passes of the heater.
4. Product / pump around pumps will be stopped when they loose suction.
5. Top refluxing will be continued as long as possible.
6. Column pressure will be closely watched for any rise of pressure due to additional amount of steam put into the heater coil.
7. Stripping steam to crude column and stripper will be stopped.
8. RCO circuit will be flushed by taking flushing oil from offsite.

VACUUM HEATER 12-F-01:

Atmospheric section will be continued, diverting RCO through VR circuit and will be shut down with normal shut-down procedure depending upon RCO handling facilities. If the leak is small it does not warrant and emergency shut-down otherwise it has to be an emergency shutdown. The plant supervisor has to assess the extent of damage and take recourse to a normal shut-down or an emergency shut-down.

The following steps are indicated to bring down the unit on emergency in the event of a failure of heater tube:

- i. Fires to be minimized in the heater. Once HC in the furnace is burnt, take out all fires including pilots Snuffing steam will be opened to heater box.
- ii. Feed to the furnace will be cut off (RCO will be diverted through VR circuit).
- iii. Emergency steam to coil will be opened and content of heater coils will be displaced to column
- iv. Cut off ejector steam and close cooling water to ejector condensers and drain water from them. Vacuum should be broken with steam to avoid air entry through ruptured tube.
- v. Float the unit on fuel gas and allow gas to escape to furnace by sufficiently increasing steam to heater coils.
- vi. Empty out the tower by pumping from all trays and column bottom till pumps loose suction.
- vii. Take flushing oil into the column and dilute its content, pump out the oil from the column.
- viii. Drain out remaining oil from the column to CBD.
- ix. Steam out the unit as per normal procedure and hand over to maintenance for installing blinds as per master blind list.



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19.8 DCS FAILURE

We can view pages related to our unit on any of the YOKOGAWA control panels installed in the MOI and also closely monitor APC system. If further control of the unit is required from other unit control panel, then engineering keys for any control operation is required. In case all the Yokogawa Control panels have failed, then no option is left but to go for a unit shutdown. So at that time the trip bypass has to be kept in auto mode and press all the five ESD switches. And then ask the field crew to switch off all the pumps.

Upon power failure to DCS, there will be no control from control room and all the controllers and SDV's will go to their respective fail safe positions Depending on the duration of failure, either operation can be resumed or unit shutdown resorted to.

19.9 FEED BOOSTER PUMP

Consequence:

1. Desalter pressure shoots up and Desalter RV may operate if the pressure of 12.5 kg/cm²g is breached.

Actions:

1. Take the Desalter pressure controller to manual and shut down the valve completely.
2. Stop the feed charge pump.
3. Take steam clearance from power plant and start Feed turbine pump.

If both the pumps are not running then, prepare for an emergency shutdown

In Atmos section

1. Trip the heater, keep the fuel oil in circulation
2. Purge out the oil from burners.
3. Stop stripping steam to the column and strippers.
4. Stop all other pumps one by one except the RCO pump and keep the Atmos section in short circulation mode via SR circuit.
5. Maintain the pressure in the Atmos column by bypassing PR1409 to flare.
6. Close the discharge valve of all pumps and stop them.

In Vac section:

1. Put off steam supply to steam ejectors.
2. Isolate all the ejectors. Close non-condensable return line to the first stage ejectors.



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3. Open fuel gas make-up line to the Vac column, and there-by maintain positive pressure of 0.5 kg/cm²g.
4. Trip the heater and cut-off the fuel gas and fuel supply to it. Keep the fuel oil circuit in circulation mode.
5. Isolate all individual burners and flush out the oil using steam.
6. For the entire furnace, use purging and snuffing steam.
7. Keep the Vac section in short circulation mode.

19.10 PFD TURBINE TRIPS:

If PFD turbine trips, then Atmos heater will also trip because of low pass flow interlock. Try starting the turbine pump but if it is not starting then by-pass the PFD and start the unit.

19.11 RCO PUMP

Consequences:

1. Atmos column bottom level will increase.
2. Feed to the Vac column will stop.
3. Vac section downstream units will suffer.

Actions:

Reduce over flash rates and the amount of over flash. Instruct the field crew to immediately rush and start the stand-by pump or restart the pump after checking the condition. In the meanwhile, reduce the feed rate to the Atmos heater and increase the heater COT. IF the standby pump not available and the same pump is not starting then, trip the Vac heater. Put the fuel oil to Vac heater in circulation mode. Close all block valve of fuel oil and fuel gas. Put Vac section in short circulation mode and stop all pumps of the Vac side except the SR pump.

19.12 ATMOS COLUMN TOP REFLUX PUMP

Consequences:

1. Atmos column top pressure shoots up.
2. Since reflux is immediately affected, the top temperature also starts rising.
3. Atmos column overhead drum level would start rising alarmingly.
4. Stabilizer would loose feed.
5. Atmos section product quality would degrade.



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Actions:

1. Bypass PR1409 to release excess pressure to flare.
2. Take clearance from power plant and start the turbine pump.
3. If the problem sustains for a bit longer time then bring down heater COT and if required cut down feed rate.
4. Increase all CRs to maximum.
5. Reduce stripping steam flow rate to the main column bottom.

19.13 KERO CR PUMP

Consequences:

1. Temperature in the column will increase.
2. Product will go off spec.

Actions:

1. Start turbine pump (11-PT-08 B) by taking clearance from Power Plant.
2. Cut down the feed rate.
3. As temperature in the Atmos column is going to increase the Diesel, TPA CR to maximum and also increase the reflux flow rate.
4. Decrease the heater COT.
5. Heavy Naphtha flow will increase, so start drawing off more Heavy Naphtha.

19.14 DIESEL CR

Consequences:

1. Stripper level will go.
2. Diesel draw-off temperature will increase.
3. Over-flash will decrease.

Actions:

1. Decrease COT
2. Reduce the feed rate.
3. Try to place standby pump.

19.15 DIESEL PUMP

Consequences:

1. Over flash will increase.
2. Diesel slips to RCO and will go to Vac column.
3. Vac column top temperature will increase and chances of loosing vacuum are there.



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Actions:

1. Cut the heater feed rate.
2. Increase the COT.
3. Increase KERO yield so as to allow maximum vaporization.
4. Try to place standby pump.

19.16 SR PUMP

Consequences:

1. Vac column level will start building up.
2. Vacuum will start going down.
3. Bottom temperature will start increasing as there will be no quench flow.
4. There will be no BBU feed if SR is routed to BBU from our unit.

Actions:

1. SR pump failure may have caused due to cavitation, so try opening the vent valve.
2. Start extra HVGO pump to pump out SR from the column.
3. If BBU is taking feed from our unit then, reduce air and put BBU in short circulation mode.
4. Reduce feed rate.

19.17 HVGO PUMP

Consequences:

1. HVGO level in the Vac column will become high.
2. Wash zone pickings in the Vac column can get choked.
3. Gradually Vacuum column bottom level will start increasing.
4. Degradation in SR quality.
5. LVGO level will drop and the LVGO pump may lose suction.
6. HVGO feed supply to FCCU will be interrupted
7. Drop in column vacuum.
8. MP steam generation in 12-E-10/10A steam gens. will be affected.

Actions:

1. Trip the heater and drop fires in the burners individually, and flush oil with steam.
2. Take HVGO to FCCU feed control valve in manual and then close it.
3. Empty LVGO from the column.
4. Cut off MP steam supply to the ejectors and isolate them.
5. Put the Vac section in short circulation in short circulation mode.
6. Empty slop level in Vac column and then stop the slop pump.



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7. Stop all the Vac section product pumps except SR pump.
8. Maintain a minimum feed rate in the Atmos section and adjust the operating conditions accordingly.
9. If the BBU section is not receiving feed from CDU-2 then Bitumen section will not be affected at all, but if BBU is receiving feed from our unit then CDU-1 or CDU-3 may be asked to compensate for the loss in feed.
10. Maintain a minimum flow rate for the unit and adjust the operating conditions accordingly.

19.18 LVGO PUMP

Consequences:

1. Vac column top temperature will shoot up.
2. Vacuum will loose in the column.
3. Hot feed to FCCU will stop.
4. LVGO will drop down to HVGO.

Actions:

1. Take out feed to FCCU.
2. Decrease COT.
3. Decrease feed to the heater.

19.19 TEMPERED WATER PUMP

Consequences:

1. Cooling for SR and Bitumen product will get affected.
2. Bitumen and SR lines run-down temperatures would be high.
3. SR pumps would have to be stopped because tempered water is used as bearing coolant for 12-P-01/02.

Actions:

In case of tempered water failure the unit can be brought down as per the following plan:

1. Trip the heaters immediately, and keep the fuel oil system under circulation. Put off all fires and flush out the oil.
2. Stop SR pump immediately.
3. Empty the Vac column HVGO, LVGO and Slop-Cut levels.
4. Stop MP steam supply to ejectors and then isolate it.
5. Introduce Fuel Gas to the Vac column, so as to maintain a positive pressure of 0.5 kg/cm²g.
6. Stop all the product pumps and let the CR pumps run.



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7. Maintain close circulation in the Atmos section.
8. Go for normal start-up when the Tempered water service has been resumed to normal operation.
9. Empty the Vac column and Atmos column products to Slop.
10. Close the discharge for all the pumps.
11. Block stripping steam supply to the main column and the side strippers and close the block valves at the shell flange.

For Bitumen section:

1. If the failure is for short duration then cut off feed, air and boiler feed water to the Bitumen reactor, and then put BBU in circulation mode.
2. If the failure is for longer duration then dilute the system with cutter and then shut-down the unit.

19.20 LOSS OF BOILER FEED WATER

Loss of BFW in CDU/VBU will affect the HVGO rundown temperature and CR. Reduce the throughput and by pass the steam generator. After BFW sustains increase the feed rate as accordingly.

Loss of BFW will affect the BBU unit steam generators and the Reactor temperature control. If the failure is for longer duration, the unit is to be shutdown as per normal procedure.

19.21 LOSS OF BEARING COOLING WATER

In CDU, all the pumps will be affected and all are to be stopped. Check for the bearing house temperatures and if the temperatures are higher, then all the pumps are to be stopped. If the failure is for longer duration, unit to be taken to shut down.

In BBU, Operation of air compressor and product pumps will be affected. These are to be stopped accompanied by stoppage of feed, air and BFW to the Reactor. Establish circulation. If the failure is for longer duration, the system should be diluted with flushing oil and the unit should be shutdown. Circulation may be re-established when water is resumed. Then, the unit is started again as per normal procedure.



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BRIEF SUMMARY OF EMERGENCY PROCEDURE:

Case	Consequence	Action required
Power failure to unit only	All pumps, IN/FD fans of furnace will trip	Follow emergency procedure
Power failure to DCS only	There will be no control from control and all the controllers & SDV's will go to fail safe positions.	Follow emergency procedure
Power failure to unit and DCS	Same as above two	Follow emergency procedure
BFW failure	1. BFW supply to PRDS will stop. 2. BFW to steam generators will stop which leads to high R/d temp of VGO.	1. Stop the steam let down system. 2. Reduce the throughput and by pass the steam generator.
Failure of the crude charge pump and crude booster pump including stand by pump	1. There will be no feed to CDU furnace. (IF PFD not in service). 2. Feed to furnace will be there of PFD turbine is running.	1. Follow emergency procedure. 2. Take out fires in heater and stop turbine, and go to emergency shut down procedure.
Failure of the RCO pump including stand by pump	Crude column bottom level will increase & may cause vibration problem due to stripping steam	Stop stripping steam to column & follow the emergency shut down procedure for VDU section only.
Failure of the SR/VR pumps including stand by pumps	Vacuum column bottom level will increase & may cause will increase & may cause loss of vacuum	Follow the emergency shut down procedure for VDU section only.
Failure of the CR pumps including stand by pumps	Column operating condition in that particular CR zone will disturb leading off spec products	Route these products to slop header until the same CR pump will be normalized.
Failure of the Product pump including stand-by pump	Product will overflow to just below chimney tray leading to off spec product from this tray	Route this product to slop header until the same product pump will be normalized



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RE-STARTUP AFTER EMERGENCY SHUTDOWNS

The emergencies once normalized, the unit has to again go for start-up. In case of emergency shutdown, many times it happens that the conditions are normalized very soon and the heater COT, column pressure, etc don't get upset very much. In such cases, unit has to go for hot start-up due to constraints of meeting the throughput. The different emergencies will have different time and in some cases, the normalcy can be very fast. For example, for emergency like power failure, power supply can resume in 15 minutes only and we have go start the unit again. But in case of problems like critical pumps failure, heater tube failure, clearance for start-up may not come for hours or even days.

So, if the unit has to go for start-up after emergency situation, different procedures are to be followed in different cases. The actions to be taken for start-up after any particular emergency are discussed as below:

20.1 POWER FAILURE:

In case of the power failure, shut down the unit as mentioned in Chapter 19 under the shutdown in case of power failure. Whenever the power supply is resumed and clearance is available from MP/CMP/YSF for start-up, the system has to be flushed with flushing oil. However if the power failure is long, the system should be steam purged. Then again fresh cutter is taken and start-up is done as per normal start-up procedure.

20.2 STEAM FAILURE:

In case of failure of steam, the shut down procedure discussed in Chapter 19 has to be followed. Atmospheric column has to be emptied out by pumping the level and after that normal start-up circulation has to be started as discussed in the Chapter No. 12. But if the steam failure is for a long duration, it is advisable to flush the lines and equipments with flushing oil. The procedure outlined under normal shut down procedures in Chapter 21 for flushing and emptying out has to be followed. Once the clearance is there, proceed with start-up as per normal start-up procedure.

20.3 LOSS OF COOLING WATER:

In case of partial failure of cooling water, the unit can be sustained at a lower feed rate. But in case of total cooling water failure, the unit has to be taken for a shutdown.



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During partial cooling water failure, the flow through coolers and condensers can be throttled proportionately. In such cases careful watch has to be kept on column pressure and also on the pumps which use cooling water as coolants. The feed rate can be increased to normal value once the clearance has been given for desired usage of cooling water.

If the cooling water failure is total and not restored soon, and if flushing oil from off-sites is available at sufficient pressure, RCO can be flushed out through VR circuit. Alternatively, steam purging can be done to displace the heater stuff. After resumption of cooling water, start-up has to be done as per normal start-up procedure.

20.4 LOSS OF INSTRUMENT AIR:

The failure of instrument air will place all the control valves in fail safe mode. Cross check all the control valves to be in their fail safe positions. Flows, temperature, pressure and other unit parameters can be controlled by operating the bypass of the control valves. To facilitate better control of flows through the bypass line, the bypass valves provided are globe valves. Try to sustain the unit if the loss of instrument air is for a short time but go for emergency shutdown if the time is prolonged. After the instrument air supply is resumed, again go for start-up as per normal start-up procedure.

20.5 LOSS OF FEED:

In case of loss of feed, if the supply cannot be restored immediately then the unit has to be brought to hot circulation by diverting 11-PM-10 A/B discharge to atmospheric heater through VR line. Temperature will be maintained around 200°C. Products routing will be stopped and water to coolers will be throttled. If the interruption is for a longer period then the unit has to be shut down as per normal procedure. Again after solving the problem, unit has to be started as per normal start-up procedure.

Loss of feed to vacuum section can be due to interruption in atmospheric section. If the failure is due to RCO pumps 11-P-10 A/B, immediately start the spare pump and normalize the flow. If the flow cannot be restored immediately, the unit has to be taken on circulation by diverting 12-PM-01A/B discharge to furnace inlet through start-up line. Then proceed to shutdown as per normal procedure and after solving the problem go for start-up as per normal start-up procedure.



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20.6 HEATER TUBE FAILURE:

In case of heater tube failure in 11F01, the plant has to be taken for an emergency shutdown. The decision whether the shutdown has to be emergency or normal, rests in the hands of plant supervisor available at that time with consent of YSF/MP. The steps for emergency shutdown are discussed in Chapter 19. After the shutdown and completion of maintenance jobs, the unit has to be started again as per normal procedure.

If there is a tube leak in 12F01, there is no immediate need to shut down CDU. Atmospheric section will be continued, diverting RCO through VR circuit and will be shutdown by normal shut down procedure depending upon the RCO handling facilities. Whether the shut down has to be as per normal or emergency depends upon the extent of damage. After the arrest of leak, the start-up has to be done as per normal procedure.



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NORMAL SHUT DOWN PROCEDURE

21.1 PREPARATION

While shutting down the unit care should be taken, to avoid ingress of air into the system until all hydrocarbon vapors have been removed. All related units including utilities should be informed about the shutdown plan. It is to be ensured that a slop tank having sufficient ullage is lined up to the unit to receive the slops / off-spec material. Closed Blow down (CBD) should be emptied out and should be kept in a position to receive draining / flushing from the equipment. Flushing oil should be made available.

21.2 SUMMARY

The summary of shutdown procedure is as follows:

The throughput will be gradually reduced to about 50% of normal capacity by decommissioning PFD. Necessary adjustment will be done in reflux rates / withdrawal rates in both the sections to keep the product on specification. Number of burners will be reduced in the furnaces 11-F-01 and 12-F-01, if necessary. Chemical injection will be discontinued. Adjustment will be done in Stabilizer column operation to stop production of LPG, which will be released to the fuel gas system. Naphtha caustic wash will be bypassed. Desalter will be taken out of service. Transfer temperature of vacuum section will be gradually reduced to 350 °C. Products will be diverted to slop. Vacuum section will be put to close circulation and RCO from Atmospheric section will be diverted to fuel oil tanks / slop tanks through SR circuit. Vacuum section circulation will be continued and allowed to cool off on its own.

Atmospheric section will be running at 50% throughput with all products except light Naphtha on specification. 12-F-01 fire will be cut-off as there is no heat load. Cut-off steam to ejectors one by one. Bring the system to positive pressure by backing up fuel gas. Pump out the materials to slop when temperature reaches below 200 °C. Take flushing oil and flush out all heavy materials to slop. Bring down the temperature of Atmospheric section slowly to 300 °C. Divert product to slop when they become off-spec. Cut off stripping steam. Re-circulate RCO along with other product through slop header. Back up fuel gas when pressure tends to become low. Bring down the temperature to 200 °C.

Cut off fire in 11-F-01. Allow to cool down while on circulation. When system is sufficiently cooled empty out all equipment to slop. Flush out the heavy oil system with gas oil. Flush out



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fuel oil system. Steam out columns, exchangers etc., to remove hydrocarbon vapors so that maintenance work can be started.

The important steps in the shutdown procedure can be summarized as below:

1. Decommission PFD.
2. Reduce feed rate to 50%
3. Discontinue Chemical Injection facilities
4. Decommission Desalter
5. Shut down vacuum section
6. Discontinue LPG Production
7. Shutdown Stabilizer
8. Shutdown Atmospheric section

21.3 DETAILED PROCEDURE

The procedure indicated below will ensure a safe and smooth shutdown of the unit but is not mandatory. It is important that operators understand the purpose and effect of each activity. In particular, operators should make sure that their actions will not result in the creation of hazardous conditions either because actions taken were wrong or these were taken at wrong time.

Atmos section shut-down procedure:

A) Bypassing PFD & Reduction of Feed Rate

- i) Reduce the feed rate slowly to 450 m³/h and bypass / shutdown PFD by operating the ROV. Simultaneously, change the heater pass flow made to F1104R from F1902R by operating the software switch FR1804S1.
- ii) Inform FCCU-II regarding CDU-II shutdown and ask them to slowly withdraw circulating oil to 11-E-40 A/B.
- iii) Reduce throughput gradually to 50% of normal rate step by step. Cut down coil flow about 10% in each step. Operate Atmospheric furnace flow controllers (FRC-1301 to 1304) for this purpose.
- iv) Make adjustment on all operating variables in each step so that normal pressure and temperature conditions are maintained and the products are also on-spec, which allows them to be routed to the respective tanks.
- v) Shut down the vacuum section (**refer section E**).



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B) Discontinuing Chemical Injection Facilities

- i) Stop caustic injection pumps 11-P-13 A/B/C. Isolate the following:
 - ii) Caustic injection to suction of Crude charge pump
 - iii) Caustic injection to suction of Crude booster pump
 - iv) Stop Demulsifier pumps 11-P-16 A/B and isolate the valves near crude charge pump.
 - v) Stop liquid ammonia solution injection and isolate the following injection lines:
 - i) Ammonia injections to crude column overhead vapor line and reflux line.
 - ii) Ammonia injection to vacuum column overhead vapor line and down stream of ejector.
 - iii) Stop corrosion inhibitor injection after stopping the ammonia injection.
 - vi) Isolate the vessels including cylinder and drums, which contain chemicals.

Depressurize and drain pumps and lines including the calibration pots.

C) De-Commission the Desalter

(Refer *Vendors instruction manual*)

- i) Stop water to Desalter and then switch off power.
- ii) Shut off Desalter effluent water (brine) line after ensuring the water level is lowered to minimum.
- iii) Wide open the emulsifying valve (mix valve). The pressure control system of the Desalter should continue to work so that booster pump suction pressure is ensured.

D) Cut off LPG Production and Shut down Stabilizer

Control system of Stabilizer column shall be changed from total condensation to partial condensation operation. LPG downstream treatment unit will be informed about discontinuing LPG production.

- i) Increase reflux to Stabilizer and cut off LPG production. Operate **FRC-1501** (Reflux controller) and **FRC-1502** (LPG product) for this.
- ii) Reduce the column top pressure very gradually by operating **PRC- 1501**. Care must be taken to reduce the pressure very slowly to avoid lifting of heavy ends to the top and sudden release of heavy gas to the fuel gas system.
- iii) Bring the column to total reflux operation, by maintaining pressure such that only sufficient vapor condenses to maintain the reflux flow.
- iv) Reduce feed to Stabilizer.



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- v) By utilizing the available pressure, keep emptying out the Stabilizer bottom to storage tank and maintain Stabilizer bottom level. Pressure control valve **PV-1501** may be closed to aid the emptying out.
- vi) Reduce heating medium flow to reboiler 11-E-25
- vii) Take **TIC-1501** on manual and bypass KCR to exchanger 11-E-25 through **TV-1501**.
- viii) Pump out reflux drum (11-V-03) liquid to Stabilizer till pump loses suction.
- ix) Keep receiving feed to Stabilizer till Atmospheric column runs on total reflux and then cut off feed to the Stabilizer. Close the isolation valve in the Stabilizer feed line and empty out the Stabilizer bottom as explained below.
- x) Open **LV-1501** and empty out Stabilizer to storage tank by its own pressure bypassing the caustic wash system. Heat up bottom if necessary for this purpose. Care must be exercised to prevent gas blow by while emptying out the column.
- xi) Stabilizer column pressure, Naphtha flow and Stabilizer bottom level will give indication about emptying out of the column. After emptying out isolate the column by shutting off valve on the inlet and outlet streams and then depressurize slowly by releasing to flare through depressurizing valve.
- xii) After depressurizing, residual liquid in column, piping and exchangers will be drained to **CBD or OWS**.
- xiii) Open the LP steam connection provided at the Stabilizer column bottom and start steaming. Open vent at top and reflux drum. This step can be taken up only after shutdown of the rest of the unit so that cooling water can be isolated for effective steaming. Steam out for a period of about four hours, after which it will be discontinued.
- xiv) Allow the system to cool down by putting water and with vent valves full open.

E) Bringing Down Vacuum Section

This should be done first after throughput reduction:

- i) Start reducing 12-F-01 outlet temperature by operating **TRC-2133** at 30 °C per hour to 350 °C.
- ii) Divert products to slop when they go off-spec.
- iii) Start column to furnace circulation by diverting 12-**P-01 A/B** discharge to 12-F-01 through start-up line. Simultaneously divert Atmospheric column Bottom (11-P-10 A/B discharge) to SR circuit, which will be lined up to Fuel oil tanks / slop tanks.
- iv) Stop side stream withdrawal and increase internal refluxes to make the bottom lighter.
- v) Cut off fire to furnace 12-F-01 as there is no major heat removal from the system.

Purge out the furnace as defined in furnace shut-down procedures.



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- vi) When circulating RCO temperature is around 250 °C, stop ejectors one by one closing 12-J-01 A/B/C first and then the other stages. Close hot well liquid seal valve, vent valve and non-condensable return valve along with the last stage ejector. Care should be taken to prevent air entry into the system.
- vii) Bring the system pressure to around 0.5 kg/cm² by backing up with fuel gas.
- viii) Maintain bottom level by opening 12-**PM-01 A/B** discharge to SR circuit whenever necessary.
- ix) Allow the system to cool down to 200 °C, when the temperature reaches below 200 °C, empty out the column to slop / fuel oil tanks through SR circuit along with RCO.
- x) Line up flushing oil to 12-C-01 LVGO section and flush out all circuits by running circulating pumps.
- xi) When sufficient level is built up at the column bottom, start 12-**P-01 A/B** and circulate it through furnace coil.
- xii) When the system is thoroughly flushed, divert 12-**P-01 A/B** to slop. Pump out liquid from all trays to **the slops** by connecting the respective product lines with the slop header. Stop the pump when it loses suction.
- xiii) Blind the fuel gas line. Isolate all inlet and outlet lines at the battery limit.
- xiv) Reverse blinds on CBD lines from various equipments.
- xv) Isolate cooling water to condensers.
- xvi) Drain all the oil in the CBD. Ensure adequate ullage in CBD drum by pumping out CBD drum contents to slop.
- xvii) Open steam to column and open furnace coil emergency steam. Steam out all piping/exchangers etc. Continue steaming vigorously for about 8 hours to make the system hydrocarbon free.
- xviii) Once steaming is over, isolate steam and keep all vents and drains open for handing over to maintenance.

F) Reducing Heater Outlet Temperature (Atmospheric Section)

Atmospheric section is running with 50% of normal capacity with:

- RCO routed through SR circuit to Fuel oil tanks.
- Rest of the products are going to respective tanks.

- i) Start reducing heater outlet temperature by TRC-1301 @ 30 °C/h. All controls will be taken on manual.
- ii) As temperature of crude oil drops, there will be less of distillate products and column pressure will tend to fall. Admit fuel gas into the reflux drum by pressure controller PV-1409B if fuel gas is available from elsewhere.



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- iii) As product temperature starts dropping, products will go off-spec. Divert all off-spec distillate products to slop at the battery limit.
- iv) The overhead product yield will gradually come down. Stop withdrawing Naphtha product and resort to total refluxing. Liquid in reflux drum will be emptied out into crude column slowly
- v) Gradually reduce product withdrawal rates and pump around rates.
- vi) Stop pump around pumps one by one starting from TPA pump.
- vii) Stop product pumps when they lose suction.
- viii) With lowering of the heater outlet temperature, RCO yield will increase and viscosity will come down. Line up the **slops** receiving tanks to crude charge pump suction and establish close circulation of RCO through SR circuit. Isolate the crude feed tank.
- ix) When heater temperature drops to about 200 °C, cut off all fires. **Purge out burners and fire box as per details under furnace shut down procedure described in General Operating Manual.**
- x) Continue circulation till system gets cooled down sufficiently to about 100 °C at the column base. After this, stop crude charge pump 11-P-01 A/B. Route RCO to slop tank. Stop booster pump 11-P-02 A/B and RCO pump 11-P-10 A/B when they lose suction.
- xi) Inform MEROX to charge flushing oil header. Take flushing oil to the distillation column through RCO pump 11-P-10 A/B suction. Build up about 30% level at the base of the column.
- xii) Start circulation through SR circuit to crude charge pump and complete the circuit up to the column. Take flushing oil to feed pump suction and use pumps 11-P-01 A/B, **11-P-02 A/B** and 11-P-10 A/B for this purpose.
- xiii) Flush out heavy oil from exchangers/control valve bypasses by operating them one by one.
- xiv) Continue circulation till entire circuit is flushed out.
- xv) Check the consistency of the circulating material. When it is sufficiently light, stop circulation and pump out the material to slop tank. The flushing operation may have to be repeated more than once.
- xvi) Slowly open steam to the discharge of crude charge pumps 11-P-01 A/B and flush the lines and equipments. Isolate Desalter and drain the content to the CBD system. Admit LP steam and thoroughly steam by opening vent and try-cocks. Stop steaming when oil is fully emptied in Desalter, care should be taken steam should not enter CBD system. Keep the Desalter pressure around 0.3 kg/cm² during steaming.
- xvii) Steam the crude exchangers from the discharge of the crude booster pump 11-P-02 A/B and displace the material into the column. Open emergency steam to the four passes of the furnace 11-F-01 to expedite the displacement.



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- xxviii) Pump out the material collected in crude column to slop tank. Run RCO pump as and when necessary.
- xix) Steam out the discharge of RCO pumps and route the material into slop tank. Care should be taken to avoid steaming beyond battery limit.
- xx) After maximum quantity of material has been removed from the unit, drain all equipment and lines to CBD one by one. CBD drum level and temperature should be watched during this draining.
- xxi) Isolate fuel gas header from the reflux drum (11-V-01). Reverse the blind to close position) of the fuel gas header in the plant battery limit.
- xxii) Depressurize the crude column to flare. After depressurization, close all valves releasing to flare. Close cooling water at plant the battery limit and drain out water from condensers and product coolers.
- xxiii) Start steaming of crude column, preheat exchangers, furnace coils and coolers and condensers as detailed under the section of start-up procedures. Continue steaming till all hydrocarbon vapors are removed from the system. Vents at column top and reflux drum will be opened after ensuring clear condensate appears in the reflux drum.
- xxiv) Flush out fuel oil header by taking flushing oil at unit battery limit. Drain out oil to OWS. Steam out the header till hydrocarbon is eliminated.
- xxv) Shut off tracing steam to the lines.
- xxvi) Carry out insertion of blinds as per the master blind list for isolating the unit for maintenance.

21.4 SPECIAL SHUTDOWN PROCEDURES

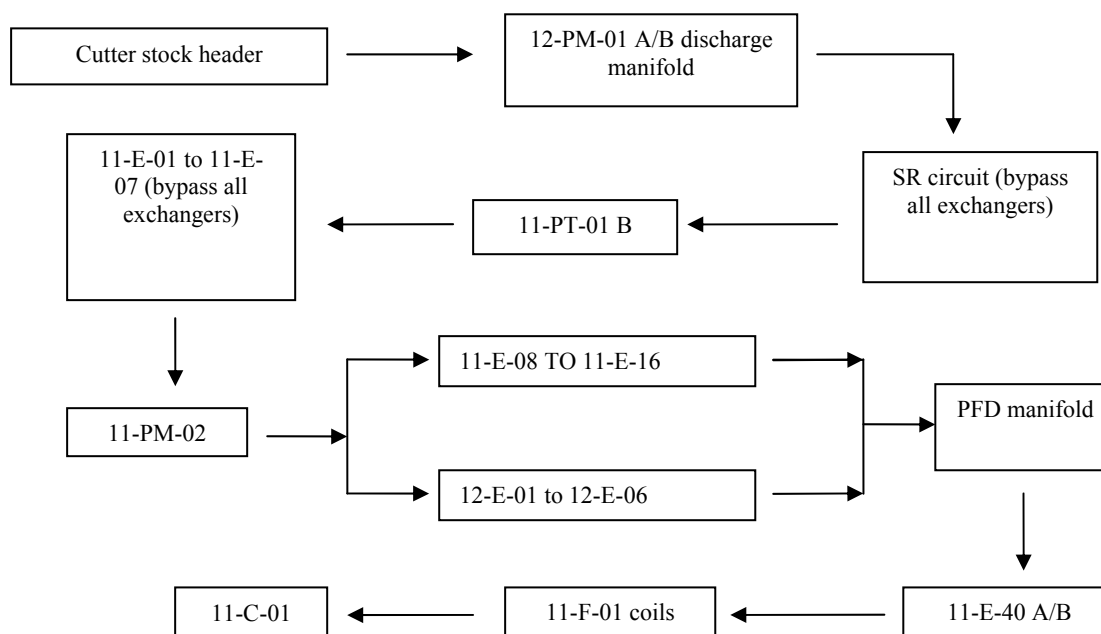
21.4.1 ATMOS SECTION CUTTER FLUSHING PROCEDURE

- 1) Ensure Desalter is in bypassed condition and keep Desalter R/V bypass valve in open position and empty out crude from Desalter to CBD. Keep close watch on Desalter top pressure to avoid creation of vacuum in Desalter.
- 2) Cutters flush and displace crude from exchangers (11-E-01 to 11-E-07) to CBD (inclusive of all tie-ins) using individual flushing oil tie-ins.
- 3) Bypass and isolate the crude side of exchangers of 11-E-08 to 11-E-16 (both BH and PG mode tie-ins) and cutter flush to CBD using individual flushing oil tie-ins.
- 4) Bypass and isolate both sides of the exchangers of 12-E-01 to 12-E-06 and cutter flush both sides to CBD using individual flushing oil tie-ins.
- 5) Take cutter to 12-PM-01 A/B discharge manifolds and flush SR circuit (11-E-16, 12-E-06 A/B, 12-E03, 12-E-01 A/B/C, 12-E-09 A/B/C/D [SR side]) to slop tank.



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- 6) Cutter flush both preheat trains (crude side) and 11-F-01 coils to 11-C-01 as shown in the circuit below:-



- 7) Pump out 11-C-01 bottom stock to slop tank again via SR circuit. Since, SR circuit is involved in both Atmos. and Vac. Side flushing operation, it is to be carried out alternatively. Continue the flushing cycles till we observe clear cutter stock at SR sample point.
- 8) After thorough flushing, depressurize individual equipment to CBD and prepare for steaming.
- 9) Remove blind on 11-C-01 vent & 11-V-03 vent boiler makers. Isolate 11-C-01 overhead condensers and stabilizer overhead condensers salt water side and remove boiler makers on salt water lines to drain salt water.
- 10) Close 11-PV-409B (fuel gas to 11-C-01) control valve's block valve and bypass valve. Maintain column pressure by using column bottom stripping steam and place 11-PV-409A in service (auto mode). Maintain column pressure in the range of 1-1.5 kg/cm².
- 11) Depressurize flushing oil header, install blind at battery limit and back up MP steam to header.



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Note:-

- While carrying out above operation, coordinate with YSF/TPH for usage of cutter stock & slop system.
- Keep CBD pump in Auto mode and pump out the stock to slop system.
- While carrying out flushing ensure that control loop bypass valves are in open position.
- Ensure all the FTs & PTs impulse lines are flushed thoroughly.

21.4.2 DESALTER STEAM OUT PROCEDURE

- Bypass the Desalter and get the transformers isolated electrically.
- Keep Desalter R/V bypass valve in open position and empty out crude from Desalter to CBD. Keep close watch on Desalter top pressure to avoid creation of vacuum in Desalter.
- Steam out Desalter by introducing LP steam (MP steam would damage Teflon bushings of transformers), and steam the R/V line to 11-C-01 through R/V bypass valve.
- Steam Desalter brine line up to MEROX OWS and blind at unit limit.
- Steam the stripped water to Desalter line from SWSU up to Desalter via 11-E-18 by opening check valve discharge flange (coordinate with MEROX unit).
- Carry out steaming till the sweet steam comes out.

21.4.3 STEAM OUT PROCEDURE PREHEAT TRAIN AND FRACTIONATOR

- Introduce MP steam at 11-PM-01A/B discharge and displace 11-E-01 to 11-E-07 cutter stock to CBD and carry out initial steaming. (Ensure 11-PM-10A/B discharge and warm up valves are in closed position)
- After backing up steam into cutter into cutter stock header, steam out the 11-E-08 to 11-E-16 and 12-E-01 to 12-E-06 and 11-E-40 A/B using individual cutter lines and carry out thorough steaming (both initial and final steaming). Co-ordinate with FCCU-2 for steaming circulating oil side.
- Open purge steam to 11-F-01 passes and steam common inlet line up to PFD manifold.
- Steam out each pass for a minimum of 1½ hour and keep all the passes under steam purge for thorough steaming.
- Pump out 11-C-01 bottom to slop tank.
- After completion of 11-F-01 steaming, isolate passes purge steam and depressurize 11-F-01 coils and 11-C-01 vents/drains respectively.



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- 7) Steam out 11-C-01 thoroughly by opening stripping steam to column bottom and strippers (monitor column pressure and maintain a pressure in the range 1-1.5 kg/cm²).
- 8) Steam out all connected pump suction through OWS funnels using steam from 11-C-01. Isolate pumps and depressurize by opening vents and drains.
- 9) Open over-flash loop bottom drain valve boiler maker and steam the loop thoroughly. Also steam the connected gauge glasses and level trolls and other instrument leads of 11-C-01 thoroughly.

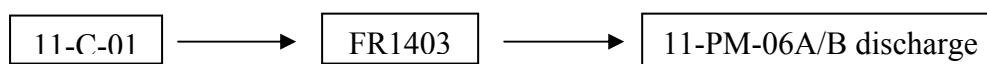
21.4.4 STEAMING OUT OF PRE-FLASH DRUM

- 1) Take cutter stock into PFD via cutter tie-in.
- 2) Displace this cutter stock at 11-PM-10A CBD/OWS and 11-PT-02B suction CBD/OWS by taking MP steam to PFD.
- 3) Steam the PFD and its manifold with existing MP SOP thoroughly. While steaming, ensure that PR1902 control valve bypass and PFD relief valve bypass are in open position.

21.4.5 STEAMING OUT OF CIRCULATING REFLUXES

1) Top Reflux:

- a) Isolate pump suction and discharge valves
- b) Steam out the 11-PM-06 A/B suction line to OWS by backing steam from 11-C-01.
- c) Steam the Top reflux line by backing steam from 11-C-01 up to 11-PM-06A/B discharge including FR1403 control valve and its bypass as shown in the following circuit:-



- d) Back up steam to pump casing and displace the hydrocarbon from pump casing by opening casing drain (Minimize steaming through pump to protect mechanical seals).

2) TPA

- a) Isolate pump suction and discharge valves
- b) Steam out the 11-PM-09A/B to TPA return line and steam upto pump discharge as shown in the circuit.



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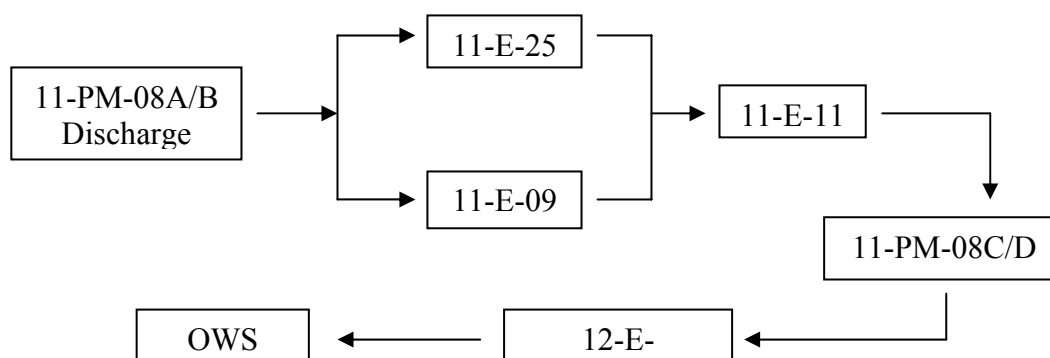


- c) Open warm up valve of the pump and steam the pump casing at slow rate and until hydrocarbon is displaced (Minimize steaming through pump to protect mechanical seals).

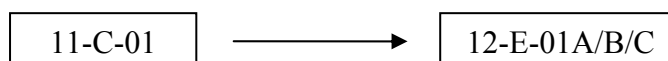
3) KERO CR

- Isolate the KERO CR pump suction and discharge valves
- Steam out the 11-PM-08A/B suction line to OWS by backing steam from 11-C-01.
- Steam out the KERO CR circuit (Both BH and PG circuit) SOP as shown in the circuit below.

i)



ii)

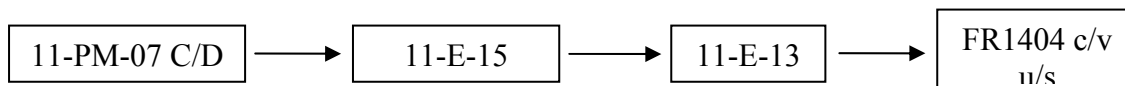


4) DSL CR

- Isolate the Diesel CR pumps suction and discharge valves
- Steam out the 11-PM-07 C/D suction lines to OWS by backing up steam from 11-C-01.
- Steam out the Diesel CR circuit (both BH and PG tie-ins) from 11-PM-07 C/D discharge with newly provided SOP as shown below.



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- d) Back up the steam from 11-C-01 to Diesel CR return upto FR1404.
- e) FR1404 bypass also to be steamed either from 11-C-01 or from circuit.

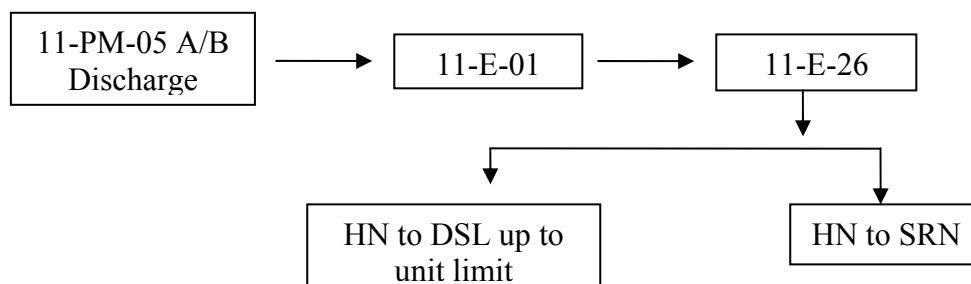
5) Overhead system

- a) Open boiler makers on salt water outlet of overhead condensers.
- b) Back up steam to 11-V-01 from 11-C-01.
- c) Displace 11-V-01 hydrocarbon to flare via PR1409 control valve bypass and newly provided 3" line at ground level.
- d) Open vents and drains of gauge glasses, LTs of 11-V-01.
- e) Steam the overhead condensers, 11-V-01 and its boot sour water to MEROX line thoroughly. Provide additional steam hose to 11-V-01 for effective steaming.

21.4.6 STEAMING OUT OF RUNDOWN CIRCUITS

1) HN Product

- a) Isolate the HN pumps suction and discharge valves.
- b) Steam out the 11-PM-05 A/B suction line to OWS by backing up steam from 11-C-01.
- c) Open the warm up valve of the pump and steam the pump casing at slow rate and till hydrocarbon is displaced.
- d) Steam out HN rundown lines both HN to SRN and HN to diesel using cutter tie-ins to 11-E-01 and 11-E-26 as shown in the circuit (cutter stock header is with steam).



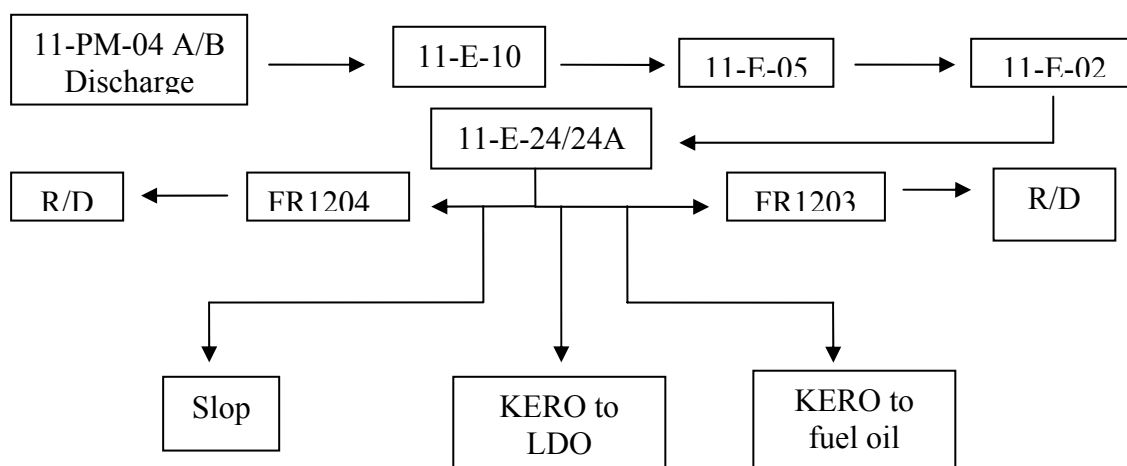
Note:- 11-E-26 salt water side boiler makers to be opened after isolation on cooling water side.



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2) KERO Product

- Steam out the 11-PM-04 A/B suction lines to OWS by backing up steam from 11-C-01.
- Steam out the KERO rundown lines i.e. KERO to Diesel, KERO to KERO, KERO to fuel oil, and KERO to LDO by using the newly provided MP SOP to 11-PM-04 A/B discharge as shown in the circuit.



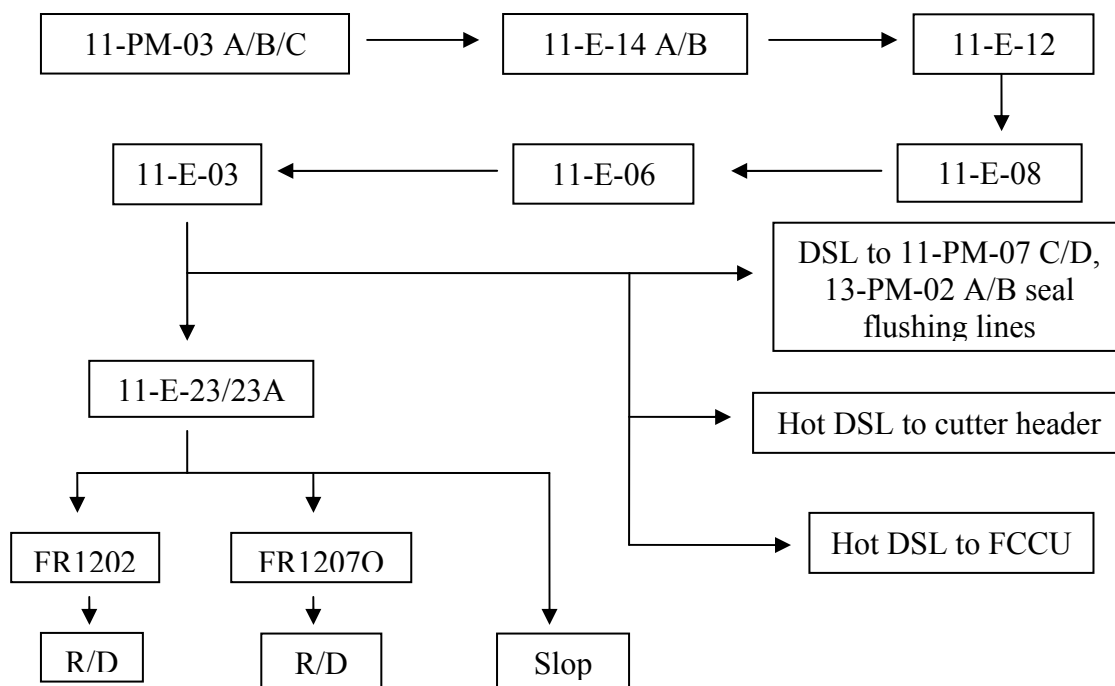
Note: - Isolate salt water to 11-E-24,24A and remove boiler makers.

3) DSL Product

- Steam out the 11-PM-03A/B/C suction lines to OWS by backing up steam from 11-C-01.
- Steam out the Diesel rundown lines i.e. Diesel to Diesel, Diesel to LDO and diesel to slop as shown in the circuit below.



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Note: - Isolate salt water to 11-E-23A and remove boiler makers.

4) RCO Circuit

- Displace the cutter stock of 11-PM-10 A/B suction lines to OWS and steam out those lines by backing up steam from 11-C-01.
- Displace the cutter stock of SR circuit to respective exchangers to OWS/CBD and steam the above circuit by newly provided MP SOP to 11-PM-10 A/B discharge.
- Displace the cutter stock and steam the quench line. After dropping TR2201 control valve steam quench line from both 12-C-01 and 12-E-03 thoroughly.

21.4.7 STEAMING OUT OF AUXILLIARY SYSTEMS

1) FUEL OIL SYSTEMS

- Close fuel oil supply and return lines unit limit block valves.
- Open the Cutter stock to fuel oil supply line at unit limit and flush it thoroughly to slop header.
- After flushing with cutter stock, disconnect individual fuel oil hoses of 11-F-01 & 12-F-01 and drain the line stock into empty drums and also drain the line stock of SDVs.



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- d) Steam out the whole fuel oil supply and return lines by backing up steam from 11-F-01 & 12-F-01 atomizing lines.
- e) After initial steaming, install spread blinds on fuel oil supply and return lines.
- f) Carry out final steaming till sweet steam comes out and then make up the blind.

2) FUEL GAS SYSTEM:

- a) Close fuel gas header block valve at unit limit.
- b) Depressurize header stock to flare via 11-V-01 (Ensure that before starting the steaming of 11-V-01, fuel gas depressurizing is to be completed).
- c) Open the SOP to fuel gas at the unit limit, steam out the 11-F-01, 12-F-01 fuel gas lines, 11-V-01 fuel gas make up line and 12-C-01 vacuum break up fuel gas line.
- d) After initial steaming install spread blinds at unit limit and other dead ends.
- e) Carry out final steaming and make up the blinds.

3) FLARE HEADER

- a) Steam the flare header by backing steam from PR1409 bypass and 3" line at ground level.
- b) Block unit limit block valves.
- c) After initial steaming install spread blinds at upstream of block valves.
- d) Carry out the final steaming and make up the blind at unit limit.

4) CHEMICAL INJECTION SYSTEM

- a) Close chemical injection block valves on the 11-C-01 side and break open the flanges.
- b) Hook up the steam hoses to ammonia, Kontol, DMF and caustic lines and steam out the drums and connected lines.
- c) Steam out fresh caustic and spent caustic lines which are coming from MEROX by hooking steam hoses at unit limit.

Note: - Carry out thorough water washing of all the chemical pots connected piping before introducing steam

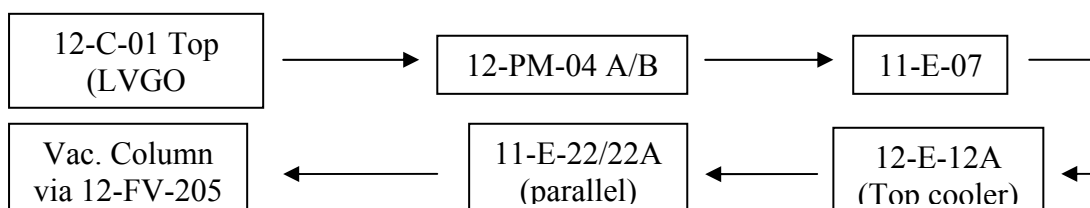


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21.4.8 VACUUM SECTION CUTETR FLUSHING PROCEDURE

1) LVGO Circuit:

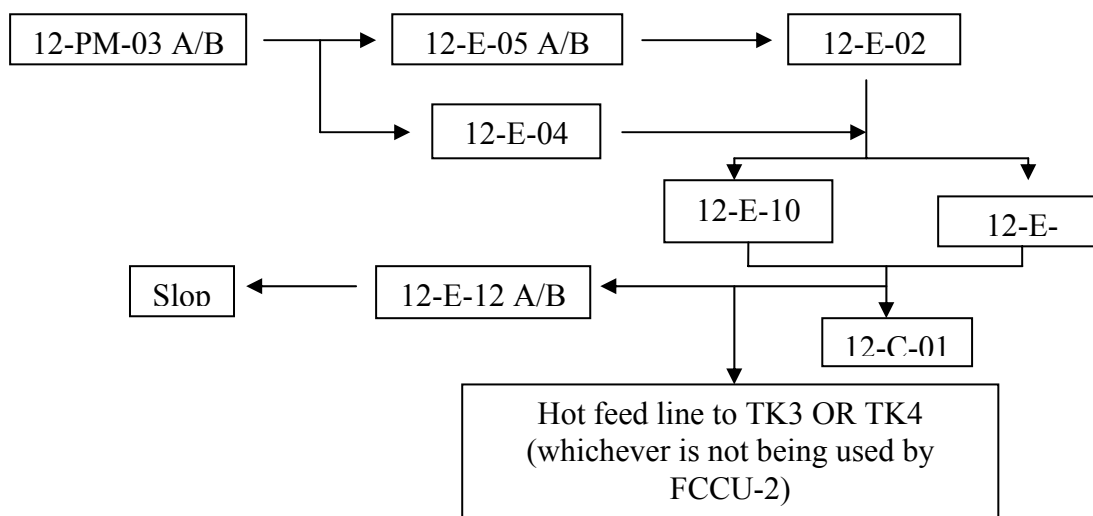
Take cutter stock to 12-C-01 top section via cutter stock tie-in to cold LVGO reflux line. Build up level in the LVGO section and start circulating in the following manner for stream No. 1:



Stream No. 2 of LVGO to HVGO is to be flushed via 12-E-11 under LI2205. The third stream of LVGO from pump (12-PM-04 A/B). Discharge to be flushed into HVGO section of 12-C-01 via 12-FV-201 to build up HVGO level.

2) HVGO CIRCUIT:

Take cutter stock into HVGO section via. 12-FV-201 (through hot LVGO reflux to cold HVGO reflux line using 12-PM-04 A/B). After building up level, start circulating in the following manner: -



Another stream of hot HVGO reflux is to be flushed to 12-C-01 via FR2202. Continue routing cutter stock via FR2202 to build up slop cut level.

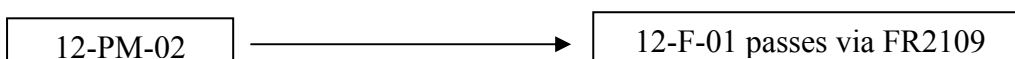


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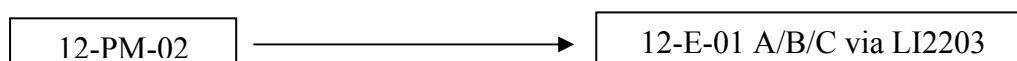
3) SLOP CUT CIRCUIT:

After building up slop cut level, start flushing as described in the following manner:

i) Circuit 1 : -

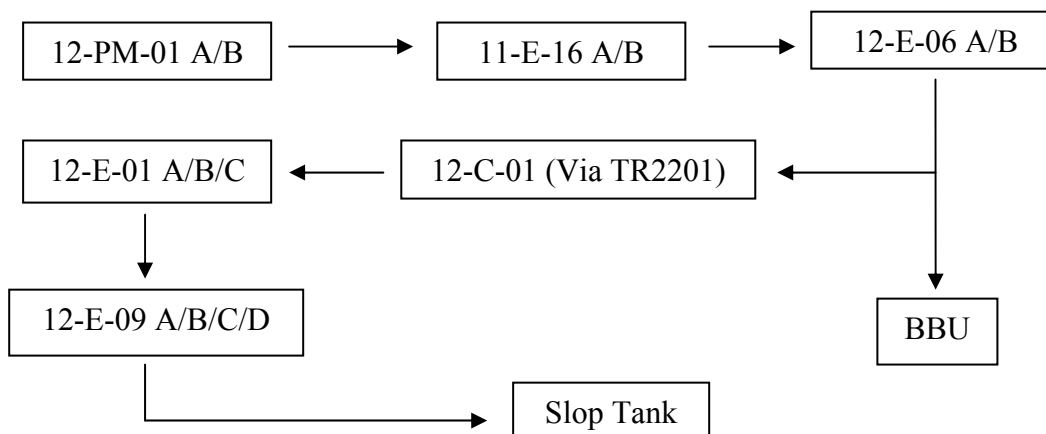


ii) Circuit 2 : -



4) SR CIRCUIT:

Take cutter into 12-C-01 bottom via 12-PM-01 A/B suction and start building up level. After up the desired level, start flushing as shown in the circuit given below.



Continue cutter stock flushing of all the above given circuits till we start observing clear cutter stock at SR sample point.

21.4.2.2 VACUUM SECTION STEAM OUT PROCEDURE

- Steam the 12-F-01 coils with passes purge steam for a minimum of two hours each pass keeping 12-C-01 under vacuum.
- After 12-F-01 coils are steamed, reduce coil steam to bare minimum and break vacuum by isolating the ejectors of 1st stage, 2nd stage and finally 3rd stage. Get the column vent blind removed. Stop surface condensers cooling water and drain salt



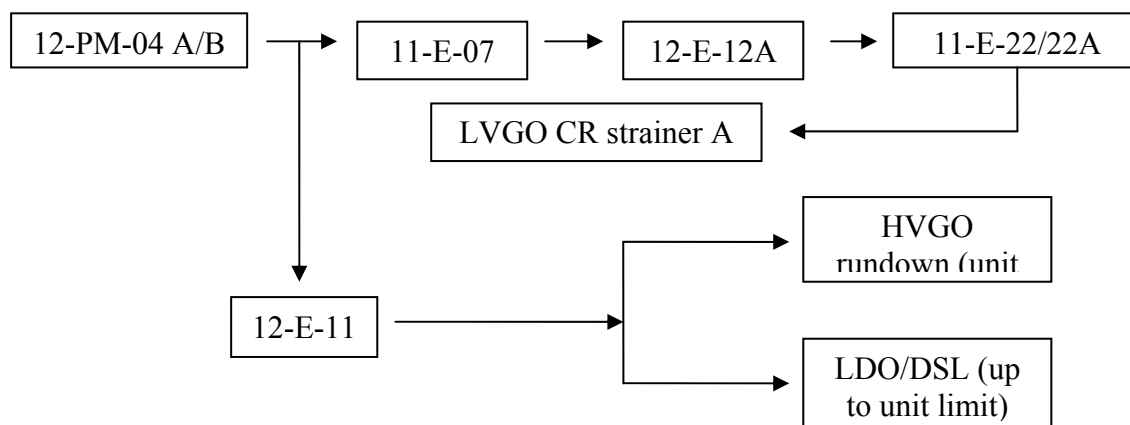
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water by opening boiler makers. Make sure that the column pressure is maintained at 0.5-1.0 kg/cm² maximum (Install one PG 0-5 kg/cm² range at 12-C-01 flash zone).

- After breaking vacuum, steam displace cutter from LVGO, HVGO, slop & SR pump suction lines and SR to HVGO pump suction to respective CBD/OWS.
- Isolate LVGO, HVGO, Slop & SR pumps and depressurize by opening vents and drains. Before backing up MP steam ensure that vent line block valve at 12-C-01 shell is closed. Now back up steam into vacuum section pumps venting line and displace the line content to respective pumps.

1) LVGO CIRCUIT

- a) Open SOP (pump vent line) to 12-PM-04 A/B discharge line. Displace the cutter stock and steam the LVGO circuit thoroughly as shown below.



Note:- a) 12-E-11, 12-E-12A & 11-E-22/22A coolers salt water side to be isolated and remove boiler makers.

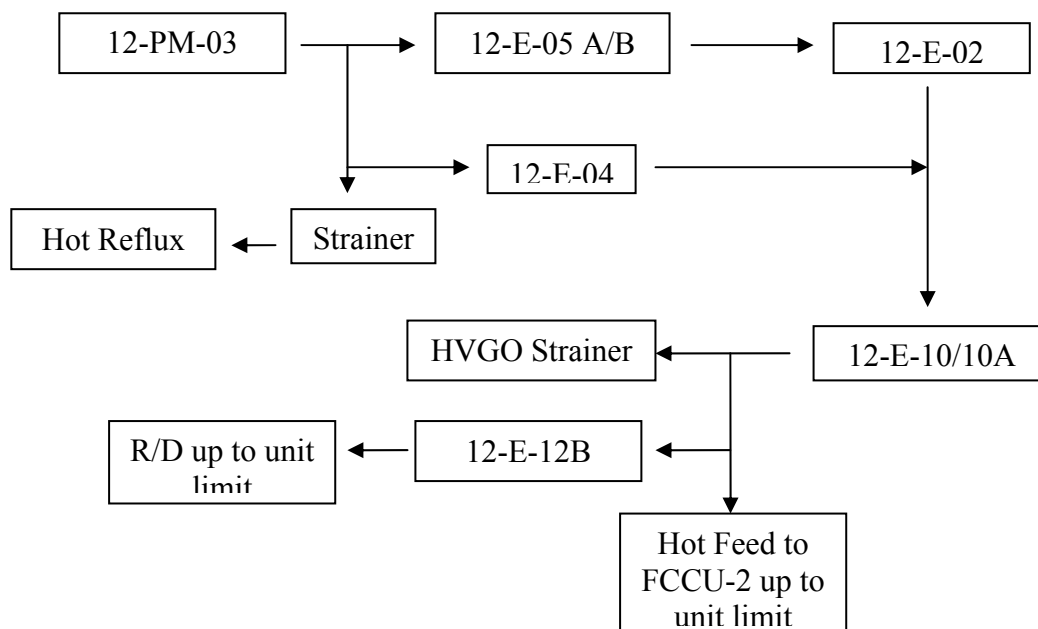
- b) If required open strainer drain valve.

2) HVGO CIRCUIT:

- a) Open SOP (pump vent line) to 12-PM-03 A/B discharge line. Displace the cutter stock and steam the HVGO circuit thoroughly in the following manner.



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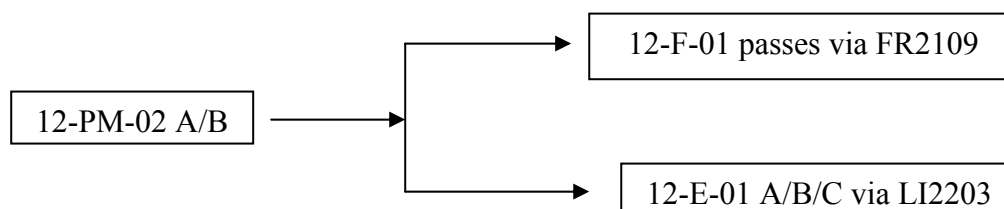
Note : - a) Isolate S/W to 12-E-12B and remove boiler makers.

b) Isolate BFW to 12-E-10/10A & open vents on steam side.

c) If required open strainers drain valve.

3) SLOP CUT CIRCUIT :

- a) Open SOP (pump vent line) to 12-PM-02 A/B discharge. Displace the cutter stock and steam the slop cut circuit as shown below.



- b) Slop cut to CDU-1 cooler box line cutter stock displacement and steaming also to be done.

Note: - Carry out steaming of 12-PM-02 A/B discharge to 12-F-01 passes via FR2109 after completion of 12-F-01 individual passes.



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4) SR CIRCUIT :

- Open the SOP (pump vent line) to 12-PM-01 A/B discharge and displace cutter stock at SR manifold.
- Steam the SR circuit thoroughly.
- Steam the tempered water drum before handing it over to Maintenance.

Note: - Isolate tempered water to 12-E-09 A/B/C/D and open boiler makers.

5) OVERHEAD SYSTEM:

- Remove the blind on LP steam to 12-C-01.
- Open Surface condensers salt water outlet line boiler makers.
- Steam 12-C-01 and connected systems with LP steam for initial steaming.
- Open block valves bypass valve of top pressure control valve and steam fuel gas make up line.
- Maintain column pressure at 0.5-1.0 kg/cm² maximum.
- Steam hot well oil/water lines up to unit limit by connecting steam hose at pump discharge line.
- Open the 12-C-01 vent boiler maker and then carryout final steaming.

After completion of final steaming, take water to 12-C-01 via 12-PM-04 A/B suction for water washing the column and then drain water at 12-C-01 bottom and continue till it gets clear water.



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TEMPORARY OPERATIONS		

TEMPORARY OPERATIONS

Temporary operations are those which we use to overcome the operation problems by adapting a method so that till the completion of job there will be no disturbance to plant operating conditions. The job can be either online or shut down or T&I nature. Till the completion of job the temporary operation will be in place. Generally in the following cases temporary operations are required:

1. If any control valve is having problem then as a temporary operation we will bypass the control valve and operate the plant till the rectification of the same.
2. If any line plugged then if at all any bypass line is available then as a temporary operation we will use the bypass line till the rectification of the same.
3. If any exchanger or cooler or APH is having tube leak then as a temporary operation we will bypass the same and operate the plant till the rectification of the same.

22.1 CDU-II UNIT TEMPORARY OPERATIONS STATUS:

S.NO	Temporary operation details	Reason for temp Operation	Date from which temp operation is in place	Temporary operation job completion (online/ Opportunity S/D / T&I)	Remarks
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					



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PROCESS SAFETY INFORMATION		

PROCESS SAFETY INFORMATION

The following Process Safety Information is to be included in the manual.

- Information on deviation from the design limits of major equipment and minimum Consequence- PSM/FR/2.6
- Information of plant relief system- PSM/FR/2.7
- List of Process System and Interlocks- PSM/FR/2.10
- List of Enclosed facilities- PSM/FR/2.8
- Information on Plant Holdups- PSM/FR/2.5
- Design Codes and Standards Employed.

As per TSD-PAD, preparation of documents is under consideration.



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SAMPLING REQUIREMENT AND SAMPLING PROCEDURES		

SAMPLING REQUIRMENT AND SAMPLING PROCEDURES

Product timely representative sampling, immediate lab analysis, feed back to plant personnel and necessary corrections in plant parameters, wherever required is an important loop, which should be continuously completed. Correct representative collection is an important aspect in the entire chain of sample collection. The results given by lab are utilized by unit supervisors for proper monitoring and improving the operation of the units. These results are being used by planning and technical personnel to fine tune and optimize the operation of the unit.

24.1 SAMPLE COLLECTION:

All regular samples should be collected as per LAB sampling schedule in cleaned bottles/grabbers/tin with the required quantities as mentioned by lab. These samples should be tagged correctly writing the origin of the sample, name of the stream, tests to be performed and date/time. The tagged bottles should be kept at the designated place 15 minutes ahead of sampling schedule. In case, sample is not collected in time due to any problem, the technician concerned should highlight to supervisor. The plant technician, during routine rounds, should observe that the samples are taken by laboratory and inform shift supervisor and laboratory if the samples are not collected in time. If sample collection is further delayed then shift supervisor should remind laboratory once again and inform Shift Foreman. The overall responsibility of ensuring availability of the sample at the collection point rests with the unit shift supervisor.

Shift supervisor should ensure that sufficient number of cleaned bottles/grabbers and sample tags are available in the unit for sample collection. The quantity of samples should vary as per the volume requirement list given by laboratory.

24.2 SAMPLING PROCEDURES:

Samples are to be collected in clean and dry bottles. Sometimes samples are taken to find out the affect of certain changes brought about in plant conditions. The samples are to be taken with great care so that they represent the actual plant conditions of the collection time. The line content is to be properly drained to ensure representative sample is collected and the bottles are to be thoroughly rinsed with the product before collection of sample. The products, specifically those which are prone to oxidation in contact with air, are to be filled to top of the bottle. The sample points are suitably located to provide the representative samples. There are various precautions that need to be taken during sample collection and also



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different samples have different precautions. Draining excessively for sample collection results in wastage of finished products and reduces overall through-put.

24.2.1 Liquid sampling procedure (non-flashing type):

- The person taking sample should wear proper safety clothing like face-shield, aprons, rubber-gloves etc, to protect face, hands etc.
- Whenever hot samples are taken check for proper cooling water flow in sample cooler.
- Sample points usually have two valves in series, one gate valve for isolation and second globe valve for regulating the flow. Open gate valve first and then globe valve slowly, after placing sample container. After the sample collection, close globe valve first and then gate valve. Then re-open globe valve to drain the content for congealing liquids. In case of bitumen sampling two gate valves are used. The purpose of two gate valves is to facilitate poking in case the line gets choked.
- Sample valve should be slowly opened, first slightly to check for plugging. If the plugging gets released suddenly the liquid comes-out at a dangerously uncontrolled rate. Never tap the line to clear plugging, instead ask maintenance for assistance. For avoiding plugging of the line in case of congealing type samples, sample points are to be equipped with proper copper coil type steam tracer. It should be ensured that steam tracing is functioning properly.
- The operator taking the sample should be careful to stand in a position such that the liquid does not spill on him and he has unobstructed way out from the sample point in case of an accident. While collecting dangerous toxic material's samples, proper gas mask is to be used. It is advisable to stand opposite to wind direction in case of volatile toxic liquid sampling.
- Sample should be collected in clean, dry and capped bottle. In case of congealing samples use clean and dry grabber/tin.
- Rinsing of bottles is to be done thoroughly before collecting the sample.
- Before collecting ensure that the content has been drained and fresh sample is collected.
- Gradually warm-up the sample bottle/metallic tin by repeatedly rinsing it before collecting the sample
- Cap the bottle immediately after collecting the sample
- Attach a tag to the bottle indicating the concerned details.



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- A few products suffer deterioration with time. For example, the color of the heavier distillates slowly deteriorates with time. So these samples should be sent to laboratory at the earliest after collection.
- The samples after collection should be kept away from any source of ignition to minimize fire hazard.
- Volatile samples (naphtha) should be collected in bottles and kept in ice particularly for some critical tests like RVP. Moreover sample should not be collected fully to maintain vapor phase.

24.2.2 High pressure hydrocarbon liquid samples (flashing type):

- The person taking sample should wear personal protection equipments like apron, gas mask and hand gloves to protect him.
- Ensure that sample bomb is empty, clean & dry.
- Connect the sample bomb inlet valve to the sample point with a flexible hose.
- Open the inlet valve of the sample bomb. Hold the sample bomb. Hold the sample bomb outlet away from person. Keep face away from hydrocarbon vapor and stand in such a way that prevalent wind should blow hydrocarbon vapor away. Open the gate valve of sample point slowly till it is fully open. Then slowly crack open the regulating valve. One should be careful at the time of draining because chances of icing are there. As a result the formation of solid hydrates is a continuing process which leads to the plugging of valves.
- When all the air in the hose and bomb are displaced as seen by the hydrocarbon vapor rising from the outlet of sample bomb, close the sample outlet valve. Allow a little quantity of liquid to spill to make sure that the bomb is receiving liquid. Frosting will be an indication of liquid spillage.
- Allow liquid hydrocarbon to fill the bomb. When the bomb is full, close the valves on sample point. Close inlet on the sample point. Carefully disconnect the hose from the sample bomb. To allow for some vapor space in the bomb for thermal expansion in case of overfilling, crack open the outlet valve of bomb and displace a small part of liquid and then close outlet valve.
- Closed sampling facilities are provided at some locations where it is not desirable to waste the costly product or if the material is toxic. For filling the sample bomb, pressure drop across a control is usually utilized or across pump suction and discharge air is expelled from the bomb after it is connected to the upstream of control valve. The bomb is then connected to the downstream of control valve or pump discharge



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side. The sample is then collected and bomb is detached after closing valves on the both sides. Send sample bomb to Laboratory for analysis.

24.2.3 Gas sample:

- For collection of gas sample which are not under high pressure and temperature, rubber bladders are used. For the operation under vacuum or low pressure, aspirator is used. For representative sample, purge the bladder 3 to 4 times with the gas and then take the final sample. Use of three-way valve with bladder/aspirator will facilitate purging & sampling.
- Sample bombs are to be used for taking gas samples from high pressure and high temperature.

Note: DM water Samples should be collected in clean dry plastic bottles to avoid silica leaching in glass bottles

24.3 RESULT ASSESSMENT AND CORRECTIVE ACTION:

When the testing is completed, laboratory will enter the results in LIMS. If the results are not entered in time, the shift supervisor should enquire the status of testing from the laboratory. In case of any problem, YSF should be kept informed. After checking the results in LIMS, the shift supervisor should compare them with planner's requirements in the unit regulation sheet and take appropriate corrective actions. This includes adjusting the unit parameters if required, informing the persons concerned, i.e., YSF, unit Manager, operations planner and TPH personnel (if any stream is to be re-routed). If any parameter becomes off even though unit conditions are same, the technician on the advice of unit supervisor should collect a check sample indicating on the tag the required parameter to be tested.

24.3.1 Significance of Individual Parameters in Product Testing:

Each individual test is designed to represent one particular property of the product. Every parameter is critical in itself. If the sample collected is not representative, the results will be erroneous and misleading. If the product with off spec parameters is routed to product tank, the product in the tank becomes off spec. Correction of the same subsequently becomes difficult and time consuming. Sometimes this leads to down grading of the product or sloping it back to the crude for reprocessing, which has a negative impact on operation



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expenditure. Also each test on each product consumes precious time of lab personnel. Hence proper knowledge of sampling procedure is essential.

Petroleum Products Determination of Color:

(ASTM SCALE)

Applicable to products: HSD, LDO, JBO

Significance: This method is used to determine color of black oils. Color is an indication of overall purity of the sample. When the color is deteriorated it indicates:

- Thermal decomposition, may be due to high temperatures
- Entrainment or trace contamination due to presence of heavier products

This scale ranges from 1 to 7, in increments of 0.5, 7 being the darkest and 1 being lightest.

(SAYBOLT CHROMOMETER)

Applicable to products: MTO, SKO, ATF, NAPHTHA

SAYBOLT Color: It is the number related to the depth of a column of material, the Color of which is compared with specified glass standards. The range of numbers is +30 (lightest color) to -16 (darkest color)

Significance: White oils are used in the manufacturing of paints, fertilizers, cosmetics, pharmaceuticals, agriculture, food and industrial usage. Color is an important factor to show that the product is not contaminated by the heavier streams while refining. When the color of the sample is $> +25$ then it is considered to be better than the color of the pure water whose saybolt color is +24.

Determination of Flash Point by:

(ABEL CLOSED CUP METHOD)

Applicable to products: MS, MTO, ATF, SKO, HSD

Significance: This method is for determination of flash point of petroleum products and other liquids ranging between -30°C and 70°C inclusive. Lowest temperature corrected to a barometric pressure of 101.3 kPa at which application of a test flame causes the vapor to ignite is the flash point. This method is suitable for products whose flash ranges between -30°C to 70°C . Flash point is used in shipping, storage and handling and safety regulations as a classification property to define 'flammable' and combustible materials. It also indicates



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possible contamination of a sample by highly volatile material in a relatively non-volatile or non-flammable material.

(PENSKY–MARTENS CLOSED CUP METHOD)

Applicable to products: FO, LDO, LSHS, Heavy Diesel, Bitumen

Measuring Range: This test method is applicable to petroleum products and other liquids, which are having flash points ranging between 40°C and 360°C.

Significance: It is one of the properties which must be considered to assess the overall flammability hazard of the material. Flash point is a response to heat and an ignition source under specified conditions in the laboratory but should not be used to appraise the fire hazard or fire risk of a material. It may be used as an element pertinent to an assessment of the fire hazard of a particular end use.

Determination of Kinematic and Calculation of Dynamic viscosity:

Applicable to products: ATF, JBO, LDO, HSD, FO, LSHS

Kinematic viscosity (γ) is the resistance to flow of a fluid under gravity.

Dynamic viscosity (η) is the ratio between the applied shear stress & rate of shear of a liquid. It is also called coefficient of dynamic viscosity or simply viscosity. Thus dynamic viscosity is a measure of the resistance to flow or deformation of a liquid.

Significance and use: Viscosity is one of the most important heating oil characteristics. It is indicative of the rate at which the oil flows in fuel systems and the ease with which it can be atomized in a given type of burner. With improper viscosity at the burner tip, carbon deposition on the walls of the fire box or other conditions, leading to poor combustion will occur. Difficulties in moving the oil in fuel oil systems will be encountered below the temperature at which the viscosity is reached.

Determination of Salt Content in Crude Petroleum:

Applicable to products: Crude

Significance: Excessive salt in the crude oil will result in higher corrosion in the process of refining. Knowledge of it is useful to decide desalting requirements.



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Determination of Pour Point:

Applicable to products: JBO, LDO, HSD, FO, LSHS

Pour point: It is the lowest temperature at which a sample of petroleum product will continue to flow when it is cooled under specified conditions. Pour point of a product indicates its ease of handling at low temperatures.

Determination of Carbon Residue by RAMSBOTTOM METHOD:

Applicable to products: LDO, HSD, FO, LSHS

Carbon residue: The residue formed by evaporation and thermal degradation of a carbon containing material.

Significance: The carbon residue value of a fuel oil is a rough approximation of its tendency to form deposits on the burners. In case of motor oils the value was once regarded as direct indication of its tendency to form carbon deposits in the combustion chamber of the engine. Additives may increase the value of carbon residue but they will decrease the tendency to form the carbon deposits. In case of gas oils it is a guide in the manufacturing of gas.

Distillation Characteristics:

Applicable to products: Naphtha, MS, MTO, ATF, SKO, HSD, JBO

The distillation (volatility) characteristics of hydrocarbons have an important effect on their safety in handling and performance, especially in case of fuels and solvents. The boiling range gives important information on composition and behavior during storage and use. Limiting values are applied in the end –use performance and to regulate the formation of vapors which may form explosive mixtures with air or otherwise escape into the atmosphere as emissions.

Definitions:

I.B.P: It is the temperature (corrected) observed at the instant that the first drop of condensate falls from the lower end of condenser tube.

E.P/F.B.P.: It is the maximum thermometer reading (corrected) obtained during the test. This usually occurs after the evaporation of all liquid from the bottom of the flask.

Corrosiveness of Copper Strip:

Applicable to products: LPG, MS, SKO, ATF, MTO, HSD

Significance: Hydrogen sulfide is one of the sulfur compounds present, in the refined products, having highly corrosive nature. Copper corrosion is an indication of the nature of



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the product to corrode the storage containers and other metal tubes during various operations and handling of petroleum products.

It is reported as 1A, 1B, 2A, 2B. Up to 4D

Conradson Carbon Residue:

Applicable to products: Heavy petroleum products

Significance: R.C.R. as discussed is preferred method for the samples that are mobile below 90°C and for other samples like heavy residue fuels, Coker feed stocks etc., C.C.R. is preferred.

Determination of Needle Penetration:

Applicable to products: Bitumen

Penetration of a bituminous material is its consistency expressed as the distance in tenths of a millimeter that a standard needle penetrates vertically into a specimen of the material under specified conditions of temperature, load and duration of loading.

The usual temperature is 25 °C, applied load is 100gm and the time traversed by the needle is 5 seconds. It is used as a measure of consistency. Higher values of penetration indicate softer material.

Determination of Softening Point:

Applicable to products: Bitumen

Softening point is the temperature at which the substance attains a particular degree of softening under specified conditions.

Significance: Softening is a measure of consistency or the hardness of the bitumen sample. Bituminous materials do not change from the solid state to the liquid state at any definite temperature but gradually become softer and less viscous as the temperature rises. For this reason determination of the softening point must be made by a fixed arbitrary and closely defined method.

WATER:

Determination of pH:

It is the logarithm of reciprocal of hydrogen ion concentration.

Significance: Values below 7 and approaching 0 are increasingly acidic. Similarly values higher than 7 and approaching 14 are increasingly basic. pH is a measure of relative acidity



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or alkalinity of a sample. pH indicates corrosive tendency of a sample. Low pH causes corrosion of equipment and pH higher than 8 causes sludge formation. pH values are to be maintained as per design values.

Determination of Alkalinity:

Alkalinity is due to presence of bicarbonates, carbonate and hydroxide ion. In lab, P & M alkalinity are measured.

Determination of Conductivity:

This helps in detection of ionized impurities dissolved in condensed steam, BFW etc. This is a simple blow down control parameter.

Determination of Chlorides:

The chlorides of Ca, Na and Mg are very soluble in water. Hence during usage in boilers etc their precipitation will not take place.

Determination of Iron:

Iron will get precipitated out of solution. Scale formation may be due to deposition from water or washing away of metal surface.

Determination of Silica:

Presence of silica is objectionable as it forms hard dense scales which have high resistance to heat transfer. In addition to scale formation, it also gets deposited on turbine blades, resulting in lowered turbine efficiency. As silica is relatively inert, it can not be precipitated like other inorganic materials. This can be removed by basic anion exchange.

Total Hardness:

Hardness of water is the amount of Ca & Mg salts present. Hardness of water is undesirable due to its scale forming nature. Also in evaporation surfaces, hardness leads to sludge formation.



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Total Dissolved Solids (TDS):

This indicates the concentration of dissolved impurities in water sample. This parameter helps in controlling the blow down.

Oil & Grease:

Oil contamination can occur in treated water due to pump seal leaks etc. As oil floats it can not be removed by blow down. This impurity if gets deposited on heating surface leads to reduction of heat transfer efficiency. This may lead to localized heating and tube rupture. Floating oil on the surface causes entrainment.

NOTE: For the tests required for every product and their significance, refer to the HOD guidelines Vol. 1.

LAB SAMPLING

Testing Details:

SRN

RVP ----- < 7 psi
H2S ----- Absent
Color ----- + 30 min

KERO

Saybolt Color ----- +20 min
Flash ----- 37 °C min
End Point ----- 290 °C

MTO/

Saybolt Color ----- +25 min
Flash ----- +32 min °C
IBP ----- 130 min °C
FBP ----- 230 max °C

ATF

Saybolt Color ----- +15 min
Flash ----- 38 min °C
EP ----- 250 max °C



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DSL

ASTM COL -----	< 2
Flash -----	> 100 °C
POUR -----	< +15
Recovery 85/95 % -----	350/370 °C (BS-II HSD regulation)
95 % -----	360 °C (EURO – III HSD regulation)
Density -----	0.82 – 0.86 (BS-II HSD regulation)
-----	0.82 – 0.845 (EURO – III HSD regulation)

BITUMEN

PEN (mm) -----	80-100
-----	60 -70

SR

Flash -----	180 min °C
-------------	------------

FRESH CRUDE (MON/WED/FRI/SUN)

SALT -----	140 ppm max
S&W -----	% vol 0.15 max

Desalted crude

SALT -----	5 ppm max
S&W -----	% vol 0.05

ATMOS WATER

pH -----	5.5-6.5
Cl -----	<20 ppm
Fe -----	0.3 wt% max

HOTWELL WATER

pH -----	5.5-6.5
Cl -----	<20 ppm
Fe -----	0.3 max wt%

WASH WATER

pH -----	5.0-8.0
Cl -----	100 ppm max



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EFFLUENT WATER

pH -----	> 7.0
OIL (ppm) -----	100 max
Si -----	<0.5 ppm
TDS -----	<1500 ppm

Note:

Time Period:

Regular samples: 07:00 hrs /19:00 hrs

Special Samples: 07:00 hrs / 11:00 hrs / 15:00 hrs / 19:00 hrs / 23:00 hrs / 03:00 hrs.



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25.1 HEAT EXCHANGERS: SHELL SIDE DATA:

EQPT NO.	DUTY MMKCal /hr	FLUID	QTY Kg/hr *1000	SP.GR @15 C	DG.PR Kg/cm ² g	OP.PR Kg/cm ² g	OP. TEMP c IN/OUT	D.P Kg/c m ² g	AREA m ² /shl
11-E-01	1.10	CRUDE	367.6	0.848	30.7	23.8	30/35	0.7	119
11-E-02	2.12	CRUDE	367.6	0.848	30.7	23.1	35/44	0.7	119
11-E-03	3.39	CRUDE	367.6	0.848	30.7	22.4	044/060	0.7	200
11-E-04A/B	7.40	CRUDE	367.6	0.848	30.7	21.7	060/095	1.4	298
11-E-05	3.43	CRUDE	367.6	0.848	30.7	20.7	095/112	0.7	200
11-E-06	1.46	CRUDE	367.6	0.848	30.7	20.0	112/118	0.7	200
11-E-07	1.44	CRUDE	367.6	0.848	30.7	19.3	118/125	0.7	119
11-E-08	0.58	CRUDE	183.3	0.848	40.0	35.0	120/126	0.7	48
11-E-09	3.89	KERO CR	333.4	0.810	13.4	09.7	208/191	0.7	200
11-E-10	0.88	CRUDE	183.8	0.848	40.0	33.6	161/168	0.7	48
11-E-11	3.89	KERO CR	190.2	0.810	13.4	10.7	241/210	0.7	335
11-E-12	1.52	DIESEL	066.1	0.853	15.0	10.6	250/218	0.7	331
11-E-13	1.13	DIESEL CR	135.1	0.853	11.2	08.6	277/257	0.7	156
11-E-14A/B	1.78	DIESEL	066.1	0.853	15.0	11.3	288/250	1.4	156
11-E-15A/B	3.00	DIESEL CR	135.1	0.853	11.2	09.6	307/277	1.4	156
11-E-16	1.44	CRUDE	183.8	0.848	40.0	28.8	278/290	0.7	156
11-E-17A-H	25.3	NAPTHA	114.6	0.670	04.8	02.3	128/45	0.4	230
11-E-18	1.37	DES. WATER	125.0	1.00	21.6	16.1	070/125	0.7	36
11-E-19A/B	2.36	STAB.FEED	057.8	0.667	24.9	18.6	045/110	0.7	82
11-E-20A-D	2.89	LPG	033.7	0.555	14.0	07.6	060/040	0.3	186
11-E-21	1.13	NAPTHA	049.2	0.684	14.2	08.2	082/043	0.7	167
11-E-22	1.19	LVGO CR	029.2	0.891	12.0	09.2	184/070	0.7	85
11-E-22A	1.19	LVGO CR	029.2	0.891	12.0	09.2	184/070	0.7	223
11-E-23	1.67	DIESEL	066.1	0.853	15.0	09.2	080/043	0.7	335



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11-E-23A	1.67	DIESEL	066.1	0.853	15.0	09.2	080/043	0.7	335
11-E-24	1.01	KEROSENE	067.5	0.810	15.0	07.8	070/043	0.7	167
11-E-24A	2.05	KEROSENE	060.0	0.810	15.0	09.2	085/043	0.24	283
11-E-25	3.21	STAB.BOT	140.89	0.684	14.0	07.9	155/160	0.1	243
11-E-26	0.72	HN	018.7	0.772	10.7	07.0	060/043	0.7	36
11-E-40A/B	4.90	CIRC. OIL	065.0	0.930	15.2	11.5	395/285	0.54	242
12-E-02	2.20	CRUDE	183.3	0.848	40.0	33.5	164/184	0.7	178
12-E-03	3.60	SR	112.0	1.020	28.7		300/250	0.7	311
12-E-04	3.19	CRUDE	183.8	0.848	40.0	32.1	213/241	0.7	311
12-E-05A/B	3.48	HVGO CR	126.4	0.921	16.0		315/276	0.7	114
12-E-06A/B	2.24	CRUDE	183.8	0.848	40.0	30.7	271/290	1.0	242
12-E-07A	3.90	HC+STEAM	8.39	-	1.5	71mm	250/65/3 5	3.2m m	963
12-E-07B	3.00	HC+STEAM	5.975	-	1.5	240mm	218/60/4 5	2.0m m	120
12-E-07C	2.78	HC+STEAM	5.409	-	1.5	812mm	247/60/4 5	2.0m m	120
12-E-08A	2.14	TEMP.WATE R	179.4	1	06.2		072/060	-	122
12-E-08B	8.24	TEMP.WATE R	154.8	1	06.2		090/050	-	330
12-E-09A-D	2.63	SR	083.5	1.020	28.7		154/40	1.0	242
12-E-10	4.42	STEAM	007.0		19.0	11.5	160/189		120
12-E-10A	4.42	STEAM	007.0		19.0	11.5	160/189		120
12-E-11	0.96	LVGO	11.5	0.891	12.5		213/070		43
12-E-12A/B	6.46	HVGO	63.0	0.920	16.0		242/070		167



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TUBE SIDE DATA:

EQPT NO.	FLUID Kg/hr	QTY Kg/hr *1000	SP.G R @15 C	DG.P R Kg/cm ² g	OP.PR Kg/cm ² g	OP. TEMP C IN/OUT	PASS	D.P Kg/c m ² g	TUBES
11-E-01	HY NAPTHA	18.7	0.772	10.7	08.6	141/43	08	0.7	320
11-E-02	KEROSENE	67.3	0.810	13.5	09.9	121/70	04	0.7	320
11-E-03	DIESEL	66.1	0.853	15.0	08.5	167/80	06	0.7	540
11-E-04A/B	TPA	333.6	0.694	10.2	08.3	131/91	02	1.4	800
11-E-05	KEROSENE	0.7	0.810	13.5	10.6	201/121	06	0.7	540
11-E-06	DIESEL	66.1	0.853	15.0	09.2	204/167	06	0.7	540
11-E-07	LVGO CR	29.2	0.891	12.6	08.9	213/136	06	0.7	320
11-E-08	DIESEL	66.1	0.853	15.0	09.9	218/204	02	0.7	128
11-E-09	CRUDE	183.8	0.848	40.0	34.3	126/161	04	0.7	540
11-E-10	KERO	67.4	0.810	13.5	11.3	219/201	02	0.7	128
11-E-11	CRUDE	183.8	0.848	40.0	32.9	168/205	04	0.7	898
11-E-12	CRUDE	183.8	0.848	40.0	32.2	205/218	04	0.7	850
11-E-13	CRUDE	183.8	0.848	40.0	31.5	218/235	02	0.7	420
11-E-14A/B	CRUDE	183.8	0.848	40.0	30.5	235/252	02	1.4	420
11-E-15A/B	CRUDE	183.8	0.848	40.0	29.5	252/278	02	1.4	420
11-E-16	SR	111.9	1.020	28.7	16.4	350/331	04	0.7	420
11-E-17A-H	SEA WATER	146.	0.677	05.0	02.0	032/045	02	0.7	619
11-E-18	LP STEAM	2.7	-	07.0	03.5	147/147	02	0.7	052
11-E-19A/B	SEA WATER	49.2	0.683	14.0	07.9	155/82	02	0.7	220
11-E-20A-D	SEA WATER	361.7	1.00	05.0	02.0	32/40	02	0.7	500
11-E-21	SEA WATER	102.9	1.00	05.0	03.0	32/40	04	0.7	450
11-E-22	SEA WATER	91.2	1.00	05.0	03.0	32/45	02	0.7	228
11-E-22A	SEA WATER	91.2	1.00	05.0	03.0	32/45	02	0.7	600
11-E-23	SEA WATER	108.0	1.00	05.0	03.0	32/45	06	0.7	900
11-E-23A	SEA WATER	108.0	1.00	05.0	03.0	32/45	06	0.7	900
11-E-24	SEA WATER	91.9	1.00	05.0	03.0	32/45	04	0.7	450
11-E-24A	SEA WATER	115.0	1.00	05.0	02.0	32/45	04	0.4	750
11-E-25	KERO CR	143.2	0.810	14.0	11.4	241/207	02	0.7	450
11-E-26	SEA WATER	16.1	1.00	05.0	03.0	32/43	06	0.5	382
11-E-40A/B	CRUDE	397.0	0.828	40.0	19.0	260/271	02	0.9	650
12-E-01A-C	CRUDE	183.8	0.848	40.0	34.0	120/164	02	1.5	850
12-E-02	HVGO CR	126.4	0.921	16.0	-	276/253	04	0.7	210
12-E-03	CRUDE	183.8	0.848	40.0	32.8	181/213	04	0.7	850
12-E-04	HVGO	63.1	0.921	16.0	-	315/242	06	0.7	850



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12-E-05A/B	CRUDE	183.8	0.848	40.0	31.4	241/271	02	0.7	470
12-E-06A/B	SR	112.0	1.020	28.7	-	331/300	04	1.0	650
12-E-07A	SEA WATER	1000.0	1000.0	05.0	02.0	32.1/36.1	03	0.3	2600
12-E-07B	SEA WATER	750.0	750.0	05.0	01.5	36.1/40.1	02	0.3	1050
12-E-07C	SEA WATER	250.0	250.0	05.0	01.5	36.1/47.2	02	0.3	390
12-E-08A	SEA WATER	356.6	-	05.0	03.0	032/045	02	0.7	
12-E-08B	SEA WATER	329.0	-	05.0	03.0	032/045	02	0.7	
12-E-09A-D	TEM. WATER	65.0	1.00	05.0	-	060/085	04	1.0	650
12-E-10	HVGO CR	126.3	0.92	16.0	-	253/230	04	0.7	260
12-E-10A	HVGO CR	126.3	0.92	16.0	-	253/230	04	0.7	260
12-E-11	SEA WATER	73.4	-	05.0	03.0	032/045	02	0.7	116
12-E-12A/B	SEA WATER	99.5	-	05.0	03.0	032/045	02	0.7	450

25.2 PUMPS: (motors/turbines)

Pump Number	Service	FLOW (m ³ /hr)			Discharge Pressure (kg.f/cm ²)	Avail NPSH (mts.)	Operating Temp (°C)	FLC (amp)/ (RPM)	KW
		Max	Min	Normal					
10-PM-01A/B	CAUSTIC CR	30.0	12.0	25.0	7.30	5.0	40	14.0	7.5
10-PM-03A/B	WATER	30.0	12.5	25.0	7.30	5	40	25.2	7.5
11-FM-01	11-F-01 FD FAN	-	-	-	-	-	Ambient	160	90
11-FM-02	11-F-02 ID FAN	-	-	-	-	-		34	150
11-PM-01A	CRUDE CHARGE	482.0	219.0	438.0	23.45	6.0	30	114	550
11-PT-01B	Feed turbine	482.0	219.0	438.0	23.45	6.0	30	5522	
11-PM-02A	CRUDE BOOSTER	515.0	234.0	468.0	33.60	6.0	120	114	500
11-PT-02B	PFD TURBINE	515.0	234.0	468.0	33.60	6.0	120	3120	
11-PM-03A/B	DIESEL PROD.	137.0	51.0	102.0	11.30	3.8	228	97	55
11-PM-04A/B	KERO. PROD.	128.0	52.0	105.0	10.40	6.0	219	97	55
11-PM-05A/B	HN	36.0	14.0	28.0	8.60	6.0	141	27	15
11-PM-06A	TOP REFLUX	250.0	89.0	178.0	18.5	3.5	45	43	200
11-PT-06B	Top reflux turbine	250	89	178	18.5	3.5	45	2970	



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11-PM-07C/D	DIESEL CR		280.0	425.0	10.10	7.1	307	218	132
11-PM-08A	KERO. CR	643.0	268.0	536.0	11.40	10.5	241	43.5	200
11-PT-08B	KERO CR turbine	643.0	268.0	536.0	11.40	10.5	241	2950	
11-PM-08C/D	KERO. CR BOOSTER	----	----	625.0	12.5	AMPLE	190	236	145
11-PM-09A/B	TPA	699.0	291.0	583.0	9.00	7.6	131	224	125
11-PM-10A/B	RCO	232.0	83.0	200.0	11.00	3.4	343	152	90
11-PM-11A/B	LPG/ REFLUX	80.0	21.0	66.0	14.60	3.2	40	63	55
11-PM-12A/B	DES. WATER	25.0	6.0	15.0	17.10	6.0	30	63	37
11-PM-13A/B/C	CAUSTIC INJECT.	75*	----	-	15.00	3.0	30	1.1	0.28
11-PM-14A/B/C	AMMON. INJECT.	20*	----	-	15.00	3.0	30	1.1	0.38
11-PM-15A/B	CORR. INHIB.	10*	----	-	14.00	3.0	30	1.1	0.38
11-PM-16A/B	DMF	12*	----	-	10.00	----	45	1.3	0.18
11-PM-18A	CBD	----	----	40.0	6.00	FLOOD	80	62	37
12-FM-01	12-F-01 FD FAN	-	-	-	-	-	ambient	53	45
12-FM-02	12-F-01HD FAN	-	-	-	-	-	-	80	45
12-PM-01A/B	SR	174.0	30.0	133.0	22.30	4.1	350	49.7	200
12-PM-02A/B	SLOP CUT	32.0	12.0	24.0	17.00	>6.0	385	97	55
12-PM-03A/B	HVGO	400.0	211.0	333.0	11.22	9.0	315.0	43.5	200
12-PM-04A/B	LVGO	109.0	42.0	85.0	8.90	10.0	213	137	55
12-PM-05A/B	HOTWELL OIL	3.0	----	3.0	7.0	4.0	45	18	9.3
12-PM-06A/B	HOTWELL WATER	15.0	----	6.0	7.0	4.0	45	78	15
12-PM-07A/B	TEMP. WATER	200.0	87.5	175.0	4.80	5.5	60	62	17
12-PM-08A/B	Vac Neutralizer	25.0	0	----	----	----	----		

Note: * =Chemical Injection Pumps flow rate is in litres/hr.



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Details of Lube oils used in the pumps:

Pump Number	Service	Pump side lube oil	Turbine side lube oil	Remarks
10-PM-01A/B	CAUSTIC CR	T-68	-	
10-PM-03A/B	WATER	T-68	-	
11-FM-01	11-F-01 FD FAN	T-68	-	
11-FM-02	11-F-02 ID FAN	T-68	-	
11-PM-01A	CRUDE CHARGE	T-68	-	
11-PT-01B	Feed turbine	T-68	T-77	
11-PM-02A	CRUDE BOOSTER	T-68	-	
11-PT-02B	PFD TURBINE	T-68	T-77	
11-PM-03A/B	DIESEL PROD.	T-68	-	
11-PM-04A/B	KERO. PROD.	T-68	-	
11-PM-05A/B	HN	T-68	-	
11-PM-06A	TOP REFLUX	T-68	-	
11-PT-06B	Top reflux turbine	T-68	T-77	
11-PM-07C/D	DIESEL CR	T-68	-	
11-PM-08A	KERO. CR	T-68	-	
11-PT-08B	KERO CR turbine	T-68	T-77	
11-PM-08C/D	KERO. CR BOOSTER	T-68	-	
11-PM-09A/B	TPA	T-68	-	
11-PM-10A/B	RCO	T-68	-	
11-PM-11A/B	LPG/ REFLUX	T-68	-	
11-PM-12A/B	DES. WATER	T-68	-	
11-PM-13A/B/C	CAUSTIC INJECT.	T-68	-	
11-PM-14A/B/C	AMMON. INJECT.	T-68	-	
11-PM-15A/B	CORR. INHIB.	T-68	-	
11-PM-16A/B	DMF	T-68	-	
11-PM-18	CBD	T-68	-	
12-FM-01	12-F-01 FD FAN	T-68	-	
12-FM-02	12-F-01ID FAN	T-68	-	



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12-PM-01A/B	SR	T-68	-	
12-PM-02A/B	SLOP CUT	T-68	-	
12-PM-03A/B	HVGO	T-68	-	
12-PM-04A/B	LVGO	T-68	-	
12-PM-05A/B	HOTWELL OIL	T-68	-	
12-PM-06A/B	HOTWELL WATER	T-68	-	
12-PM-07A/B	TEMP. WATER	T-68	-	
12-PM-08A/B	Vac Neutralizer	T-68	-	

25.3 COLUMNS:

11-C-01	Atmospheric Column
11-C-02	Heavy Naphtha Stripper
11-C-03	Kerosene Stripper
11-C-04	Diesel Stripper
11-C-05	Naphtha Stabiliser
12-C-01	Vacuum Column

25.4 VESSELS:

Tag No	Description	Operating Temp (C)		Operating Pressure (Kg/cm2g)	
		PG	BH	PG	BH
10-V-01	Naphtha Caustic Wash	40	40	6.5	6.5
10-V-02	Naphtha Water Wash	40	40	5.5	5.5
11-V-01	Overhead Naphtha Accumulator	44	44	2.0	2.0
11-V-02	Crude Desalter	129	136.5	14.0	14.0
11-V-03	Stabilizer Reflux Drum	40	40	8.0	8.0
11-V-04	Desalting Water Vessel	50-70	50-70	Atmospheric	Atmospheric
11-V-05	Decoking Drum	100	100	Atmospheric	Atmospheric
11-V-06A/B	Ammonia Vessels	Ambient	Ambient	Atmospheric	Atmospheric
11-V-07A/B	Caustic Drums	Ambient	Ambient	Atmospheric	Atmospheric
11-V-08	CBD Drum	40-150	40-150	Atmospheric	Atmospheric
11-V-10	PFD	250-275	250-280	8-15	3.5-9
12-V-01	Hot Well Drum	44	44	Atmospheric	Atmospheric



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12-V-02	Tempered water drum	60-80	60-80	Atmospheric	Atmospheric
12-V-03	Steam Blow Down Drum	100	100	Atmospheric	Atmospheric
	Demulsifier drum	Ambient	Ambient	Atmospheric	Atmospheric
	Corrosion Inhibitor	Ambient	Ambient	Atmospheric	Atmospheric

CDU-II PSV LIST

S NO	Tag Number	Description	Set Pressure (Kg/cm ² g)
1.	10-PSV-101	10-V-01 SRN CAUSTIC WASH	14.00
2.	10-PSV-102	10-V-02 SRN CAUSTIC WASH	14.00
3.	11-PSV-101	11-V-02 DESALTER	12.50
4.	11-PSV-301	11-F-01 SUPERHEATED STEAM OUTLET	14.00
5.	11-PSV-302	11-F-01 SUPERHEATED STEAM OUTLET	14.00
6.	11-PSV-401A	11-C-01 ATMOS DISTILLATION COLUMN	4.80
7.	11-PSV-401B	11-C-01 ATMOS DISTILLATION COLUMN	4.80
8.	11-PSV-401C	11-C-01 ATMOS DISTILLATION COLUMN	4.80
9.	11-PSV-403	11-V-01 ATMOS. ACCUMULATOR	5.30
10.	11-PSV-501	11-C-05 NAPHTHA STABILIZER	14.00
11.	11-PSV-502	11-V-03 STABILIZER REFLUX DRUM	14.00
12.	11-PSV-601	11-V-06 AMMONIA TANK	2.50
13.	11-PSV-603	11-P-13A CAUSTIC SOLUTION PUMP	19.00
14.	11-PSV-604	11-P-13B CAUSTIC SOLUTION PUMP	19.00
15.	11-PSV-606	11-P-14 AMMONIA SOLUTION PUMP	19.00
16.	11-PSV-608	11-P-15 CORROSION INHIBITOR PUMP	19.00
17.	11-PSV-609	11-P-16 DE EMULSIFIER PUMP	11.30
18.	11-PSV-701	11-P-17A SEA WATER BOOSTER	12.47
19.	11-PSV-702	11-P-17B SEA WATER BOOSTER	12.47
20.	120-PSV-1201A	PFD RV	25.5
21.	120-PSV-1201B	PFD RV	25.5
22.	12-PSV-202	12-C-01 VACUUM DISTILLATION COLUMN	3.50
23.	12-PSV-202B	12-C-01 VACUUM DISTILLATION COLUMN	3.50
24.	12-PSV-203	12-E-07A EJECTOR CONDENSER	1.50
25.	12-PSV-204	12-E-07A EJECTOR CONDENSER	1.50
26.	12-PSV-205	12-E-07A EJECTOR CONDENSER	1.50
27.	12-PSV-206	12-E-07B EJECTOR CONDENSER	1.50
28.	12-PSV-207	12-E-07B EJECTOR CONDENSER	1.50
29.	12-PSV-208	12-E-07B EJECTOR CONDENSER	1.50
30.	12-PSV-209	12-E-07C EJECTOR CONDENSER	1.50



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31.	12-PSV-210	12-E-07C EJECTOR CONDENSER	1.50
32.	12-PSV-211	12-E-07C EJECTOR CONDENSER	1.50
33.	12-PSV-301	12-E-10 HVGO CR STEAM GENERATOR	13.50
34.	12-PSV-301A	12-E-10A STEAM GENERATOR	13.50
35.	12-PSV-301B	12-E-10A STEAM GENERATOR	13.50
36.	12-PSV-302	12-E-10 HVGO CR STEAM GENERATOR	13.50

25.5 CDU-II TSV LIST

S NO	Tag Number	Description	Set Pressure (Kg/cm ² g)
37.	11-PSV-703	LP STEAM TO PUMP SEAL	1.48
38.	11-STV-001	11-PT-01 CRUDE CHARGE PUMP	4.87
39.	11-STV-002	11-PT-02 CRUDE BOOSTER PUMP	4.87
40.	11-STV-003	11-PT-06 ATMOS. COLUMN REFLUX PUMP	4.87
41.	11-STV-004	11-PT-08 KERO CR PUMP	4.87
42.	11-TSV-101	11-E-22 COOLER SW OUTLET	4.40
43.	11-TSV-102	11-E-26 HN COOLER SW OUTLET	4.40
44.	11-TSV-103	11-E-07 SHELL OUTLET	12.10
45.	11-TSV-201	11-E-21 SRN COOLER SW OUTLET	4.40
46.	11-TSV-202	11-E-23 DIESEL COOLER SW OUTLET	4.40
47.	11-TSV-203	11-E-24 KERO COOLER SW OUTLET	4.40
48.	11-TSV-204	11-E-16 CRUDE/SR SHELL OUTLET	35.10
49.	11-TSV-303	12-E-01 CRUDE/SR CRUDE OUTLET	35.20
50.	11-TSV-401	ATMOS. O/H CONDENSER SW OUTLET	4.40
51.	11-TSV-402	ATMOS. O/H CONDENSER SW OUTLET	4.40
52.	11-TSV-403	ATMOS. O/H CONDENSER SW OUTLET	4.40
53.	11-TSV-501	11-E-20A SW OUTLET	4.40
54.	11-TSV-502	11-E-20C SW OUTLET	4.40
55.	12-TSV-201	Vac. o/h condensers SW outlet 12-E-07B	4.4
56.	12-TSV-202	Vac. o/h condensers SW outlet 12-E-07A	4.4
57.	12-TSV-304	12-E-02 CRUDE/HVGO CRUDE OUTLET	35.20
58.	12-TSV-305	12-E-03 CRUDE/SR CRUDE OUTLET	35.20
59.	12-TSV-306	12-E-04 CRUDE/HVGO SHELL OUTLET	35.20
60.	12-TSV-308	12-E-05 CRUDE/HVGO SHELL OUTLET	35.20
61.	12-TSV-403	12-E-08A TEMP. WATER COOLER	4.40
62.	12-TSV-404	12-E-08B TEMP. WATER COOLER	4.40
63.	12-TSV-404	12-E-11 LVGO COOLER SW OUTLET	4.40
64.	12-TSV-405	12-E-12A HVGO COOLER SW OUTLET	4.40
65.	12-TSV-407	12-E-12B HVGO COOLER SW OUTLET	4.40



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26.1 CHEMICALS WITHDRAWAL MANAGEMENT (FROM WARE HOUSE) AND MINIMUM QUANTITY:

The monthly requirement for CDU-II for various chemicals is shown below (no chemical is used in BBU). An inventory corresponding to 30 days chemical consumption (buffer stock) is to be always maintained in the field at any point of time. The details are as follows:

SL. No.	Generic Name	Approved Chemical Name	Considered quantity for withdrawal from Stores every month		30 days inventory needs to be maintained at field (Kg) approx.
			Kg/ Month	No. of drums (approx.)	
1.	Demulsifier	D 9004	2177	11	2200
		EC2040A	1512	8	1600
		DM 101	2722	14	2800
		Embreak 2W157			
		- HS Case	1240	6	1200
2.	Filmer (Atmos)	- LS Case	696	3	600
		Nalco EC1021A	411	2	360
		ESK50	1512	8	1440
		Philmplus 5K7	423	2	360
3.	Neutralizer	Thermosol V	472	3	540
		Ammonia	605	12	600
4.	Caustic	Caustic	1512*	-	1800*

* Caustic is obtained from MEROX Plant as 15 % (20 °Be) solution which is directly used for naphtha-caustic wash and for dosing in crude after dilution with water to 3 % (5 °Be) solution.

Note:

1. Information to be given by Operations to PAD in case any chemical consumption increases above or below the recommended quantities.
2. Information to be given by Materials to PAD in case of change in any proprietary chemical for usage in CDUs.



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26.2 MAXIMUM ALLOWABLE STORAGE OF CHEMICALS IN THE REFINERY:

CHEMICAL	STORAGE ALLOWABLE (MT)
Anhydrous Ammonia	9.099
Caustic 100%	327
Corrosion Inhibitor (EC1020A)	0.39
Corrosion Inhibitor (Eloguard)	5.4
Corrosion Inhibitor (NALCO 5186)	7.344
Demulsifier (EC2040A)	10.656

26.3 STORAGE PRECAUTIONS:

26.3.1 Storage in Tanks:

- The tanks should be located so as not to pose safety problems due to leakage and reaction with other chemicals stored nearby. The environment at the location must be compatible with the chemical stored.
- The tanks should be fitted with vents/relief systems of adequate capacity discharging to a safe height. Capacity requirement in case of high vapor evolution resulting from heating due to fire of adjacent tank also should be considered. Wherever atmospheric release of the vapor is not advisable, the relief should be routed to a neutralizing system.
- The vents of atmospheric storage tanks should contain flame arrestors for Class A chemicals (flash point less than 23°C). The location of flame arrestor must be easily accessible for a periodic inspection to ensure that flame arrestor is free of any choking.

26.3.2 Storage of Chemicals in drums/Other Containers:

The chemical drum storage of hazardous chemicals (liquids) pose potential hazard compared to tank/bulk storage due to the following:

- The drums normally being less resistance to fire would collapse faster escalating and spreading of fire to other drums of the stack.
- Each drum of a cluster of drums can become a source of leakage.

Basis the above, ensure the following while handling chemical drum/other containers:



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- Periodic site inspection should be carried out to ensure that there is no leakage from any of the drums.
- Where combustible solid materials are stored, the dust content of the air must be kept below the lower explosive limit. Efficient dust collection system and good house keeping should be followed.

26.3.3 Storage of Compressed Gas Cylinders:

The nature of chemical gas apart, the high pressure of storage amplifies the hazard of the storage gas cylinders. The following guidelines shall be ensured for storage of cylinders.

- Cylinders containing flammable gases and toxic gases shall be kept separated from each other and from cylinders containing other types of gases by an adequate distance or by a suitable partition wall.
- For inflammable gases, the cylinder storage area should be made of non-combustible materials and flameproof fittings should be used.
- Cylinder nozzle should be suitably protected against damage. Oil or similar lubricant should not be used on valves or other fittings of cylinder containing gas.
- In case of liquefied gas like liquid nitrogen, the cylinders should be stacked vertical so that in case of defective regulator only gas would escape.
- The floor level at storage area should be sufficiently above ground level to prevent water logging and corrosion. Flooring of the storage areas should not be wet or muddy.
- Periodic inspection of cylinders for ascertaining leakage should be made. Automatic gas detectors where applicable should be provided at suitable points so that in case of leakage, alarm is hooted at control room.
- Emergency kit, safety protective equipment and clothing should be available in the close vicinity to storage area. Adequate and suitable fire extinguishers must be available at site.
- Some of the other storage precautions for agents/chemicals used in refinery as follows:
 - a) **Oxidizing Agents:** Oxidizing substances must be stored away from all flammable materials even if they are only slightly flammable. Oxidizing material must be kept away from substances which are reducing agents.



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- b) **Water Sensitive Substances:** These substances react with water or steam to produce flammable or explosive gases and evolve heat. For example, conc. acids like sulfuric acid and conc. Alkali like sodium hydroxide react with water later to evolve heat. Such materials must not be stored in areas where water flooding from pipe leakages or leaky roofs can happen.
 - c) **Toxic Materials:** These substances must be stored in well ventilated areas, preferably cool and certainly away from direct sun rays. Periodic checks should be organized for all parts of storage areas.
 - d) **Incompatible Chemicals:** Incompatible chemicals should not be stored near each other.
 - e) **Corrosive Chemicals:** The flooring of the area where corrosive chemicals are stored shall be impervious and made of corrosion resistant materials.
- The inventory of all hazardous chemicals shall be stored as minimum as possible. Attempt should be made to find suitable less hazardous alternate chemicals, to replace hazardous chemical.

26.4 LOADING AND UNLOADING PROCEDURES:

- The tank truck should be in good condition with proper seal for contents.
- Driver should have all the relevant papers.
- CISF should have given the clearance for the vehicle.
- Permit is issued to driver and he is wearing proper PPE.
- MSDS and chemical handling procedures are available in field room.
- Area is barricaded and caution boards are displayed.
- Condition of hoses and other hardware is good.
- Receipt sample is collected and lad report is verified with vendor specification.
- Valves to be opened slowly and in case any leak is observed, abort the operation immediately.
- Safety shower in the surrounding area has to be in working condition.
- In case of flammable chemicals, the prime mover (engine) should be kept off. The tanker should be properly blocked from movement before connections are made for unloading hazardous chemicals. The unloading point should be located at a safe distance outside the storage dyke.
- Pressurizing with air/inert gas for unloading should be avoided. It is recommended to use pumps/vacuum systems for unloading. Pumps should preferably be of seal less



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type and valves should be of glandless type. Solid chemicals in bulk should be handled with lifting machines and conveyors.

- Coupling used for connecting hose to tanker must be leak proof. Flange connections are preferred. Where threaded connections are used, the threaded portion should be properly preserved against corrosion/wearing of threads and thoroughly inspected before connecting are made.
- The unloading hose should be devoid of cracks & blisters and should be capable of withstanding whatever pressure developed during unloading operation. The hose should be hydro-tested at a frequency guided by experience. Proper records of hydro-test should be maintained.
- Provision of sample quantity of water/ neutralizing medium to take care of leakage/spillage must be made. Also steam and inert gas hose stations must be available at unloading point.
- The unloading systems should have facility to vent/drain the remaining chemical in the hose to a suitable safe point. The hose should be kept blinded when hot in use. The thermal safety valve discharging to safe disposal or handling facility should be provided.
- Suitable protective clothing should be used while handling drums/containers and the operator should position himself such that he is I the upwind direction so that even in case of accidental release of chemical, he is safe.
- Manual handling of drums/containers/cylinders should be minimized. It is preferable fork-lifters and suitable cradles are used to handle drums.
- Carboys/cylinders containing hazardous chemicals should not be subjected to impact.

26.5 EMPTY CONTAINER DISPOSAL:

- There should be a designated place for all empty drums and cylinders with proper name plates indicating name and how many drums to be stored there. It has to be ensured that all the empty drums/cylinders are reaching to designated place (scrap yard).
- The empty drums are to be timely removed and they are made free of their contents before being kept in safe place.
- The consumed cylinder valve should be fully closed first and then the process valve has to be closed. Then the connection to be removed and put an “empty” label on the cylinder and then stored in designated area.



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26.6 HANDLING PRECAUTIONS:

- The proper precautions are to be taken while loading and unloading, as discussed in this chapter separately.
- Before handling any chemical, its MSDS should be thoroughly read and understood. Also the emergency system to be followed in case of any leaks should be known.
- Proper PPE should be used before handling any chemical and everyone should be forced to follow the same.
- Every chemical injection pump must have a “drawdown cylinder” to facilitate periodic dosing rate measurement and pump stroke adjustment accordingly.
- Every chemical storage tank must have necessary instruments to monitor its level, pressure and temperature.
- Every tank should have a dyke of a suitable material (compatible with the chemical) of volume equal to the volume of the larger storage tank. The dyke should have facility to:
 - a) Drain off rain water into storm water channel.
 - b) Route high volume spillage/leakage to suitable neutralizing pit nearby.
 - c) Discharge safe effluent to oily water system, as applicable. The isolation valves on dyke drains should be located outside the dyke.
 - d) Connections and openings to tank should be as less as possible so that the possibility of leakage and maintenance hazards is minimized.
- Suitable breathing canisters and first aid box should be available at site for use in case of emergency.
- The fire access roads provided to storage are should not be blocked and storage tanks/area should have adequate fire protection and fire fighting facility.
- Adequate communication facility like PA system is available for interaction with control room, fire station, medical unit and communication system is in working condition.
- No loose cables or temporary electrical fittings should be used to prevent risk of open spark.
- All the drums in use should be kept in a proper rack.
- Before charging any drum, it has to be ensured that sufficient ullage is there before the filling is started.
- Drums should never be filled full with the liquid chemical and sufficient ullage to take care of thermal expansion and provision to collect accidental spills for safe disposal should be available.



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- As soon as the chemical is received from the stores, it should be ensured that it is the right chemical and check for proper labeling, authenticity, etc.
- Due date should be checked for cylinder testing and if the date is over, the cylinder should not be accepted.
- Periodic check should be done to ensure that there is no leakage from cylinder top or body. The cylinder should not be dropped /bumped against any other cylinder or hard object.

26.7 DESCRIPTION OF CHEMICAL DOSING SYSTEM:

Four process Chemicals are used in CDU-II. The name of the chemicals and their functions are as below:

SL. No.	Chemical	Use of Chemical
1	Demulsifier	To facilitate proper separation of water and crude in desalter by breaking emulsion formed between crude and desalter wash water
2	Atmos and Vacuum O/H Neutralizer	To maintain pH of column overhead liquid and minimize corrosion.
3	Atmos and Vacuum O/H Filmer (Corrosion Inhibitor)	To prevent overhead corrosion by forming a protective film on the overhead pipe internal surface and prevent direct contact of corrosive materials with the pipeline metal.
4	Caustic	1. To neutralize Calcium and Magnesium chlorides carried over in desalted crude. 2. For SRN Caustic wash to remove H ₂ S from SRN.

26.7.1 Approved Chemicals for usage:

Following are the presently approved Process Chemicals/ brands for usage in CDU-II:

SL No	Generic Name	Brand Name
1.	Demulsifier	SM DYECHEM, D 9004
		NALCO, EC 2040A
		ATSUAN, DM 101
		EMBREAK 2W157
2.	Atmos Overhead Neutralizer	Ammonia
3.	Atmos Overhead Filmer	ESK 50
		NALCO, EC1021A



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SL No	Generic Name	Brand Name
		PHILM PLUS 5K7
		THERMOSOL V
4.	Vacuum overhead Neutralizer	Ammonia
5.	Caustic	Conventional Chemical

26.7.2 Location of Injection for Process Chemicals:

Following are the chemical injection locations.

SL No	Process Chemical	Injection Point
a	Demulsifier ¹	1. To crude line upstream of desalter (after 11-E-07) 2. To crude line at crude feed pump (11-PM-01 A/B) suction
b	Atmos and Vacuum O/H Neutralizer ²	1. To Column Top vapor line 2. To Column Atmos Column Top Reflux return line
c	Atmos and Vacuum O/H Filmer ³	1. To Column Top vapor line 2. To Column Atmos Column Top Reflux return line
d	Caustic ⁴	1. To desalted crude line downstream of desalter (at the suction of 11-PM-02 A/B) 2. To crude line at crude feed pump (11-PM-01 A/B) suction

¹ Provision for DMF (and wash water) injection in crude line at feed pump suction is to be used only if salt content of fresh crude is > 50 ppm. Normal dosing shall be in crude line upstream of desalter (after 11-E-07).

² Provision for injection neutralizer or filmer in atmos column top reflux line (for 11-C-01) to be used only if overhead corrosion is consistently high in spite of maintaining overhead filmer as per norm. Normal dosing shall be in column overhead line.

³ Provision for caustic injection at crude feed pump suction to be used only if desalter effluent pH is < 8.0 . Normal caustic dosing in crude shall be at crude line downstream of desalter.

4 Caustic in subsequent sections refers only to caustic injection in crude.



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26.7.3 Chemical Diluents and their Ratios:

Usage details required for the chemicals are as follows:

SL No	Generic Name	Approved Chemical Name	Diluent Medium	Dilution Ratio (chem:diluent)
1.	Demulsifier	SM DYECHEM, D 9004	Water	1:3
		NALCO, EC 2040A	Water	1:3
		ATSUAN, DM 101	Water	1:3
		Embreak 2W157	KERO	1:4
2.	Atmos Overhead Neutralizer	Ammonia	Water	4 %
3.	Atmos Overhead Filmer	ESK 50	KERO	1:3
		NALCO, EC1021A		1:3
		PHILM PLUS 5K7		1:3
		THERMOSOL V		1:3
4.	Vacc overhead Neutralizer	Ammonia	Water	4 %
5.	Caustic	Conventional Chemical	Water	3 % (5 °Be)

i. Norm for Chemical Injection:

Consumption norm for the above mentioned process chemicals are as follows:

SL No	Generic Name	Approved Chemical Name	Base Norm of injection w.r.t crude feed rate (wt-ppm)	Quantity of neat chemical to be injected (Kg/Hr)	Quantity of neat chemical to be injected (Kg/Day)	Quantity of neat chemical to be injected (Kg/Month)
1.	Demulsifier	D 9004	7.2	3.0	73	2177
		EC2040A	5	2.1	50	1512
		DM 101	9	3.8	91	2722
		Embreak 2W157	4.1	1.7	41	1240
		- HS Case	2.3	1.0	23	696
2.	Atmos Vacuum Overhead Neutralizer	Ammonia	2	0.8	20	605
3.	Atmos	Nalco EC1021A	1.36	0.6	14	411



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SL No	Generic Name	Approved Chemical Name	Base Norm of injection w.r.t crude feed rate (wt-ppm)	Quantity of neat chemical to be injected (Kg/Hr)	Quantity of neat chemical to be injected (Kg/Day)	Quantity of neat chemical to be injected (Kg/Month)
	Overhead Filmer	ESK50	5.0	2.1	50	1512
		Philplus 5K7	1.4	0.6	14	423
		Thermosol V	1.56	0.7	16	472
4.	Caustic	Caustic	5*	2.1	50	1512

* Caustic in crude to be dosed as per requirement only. Caustic Dosage Rates for caustic dosing guidelines is based on overhead chloride. 5 ppm is the max limit for caustic injection irrespective of overhead chlorides content.

Note:

- Unit is considered to be in service for all the days in the month.
- Actual dosage to be recorded and monitored daily based on the daily consumption of the chemical.
- Consumption figures are arrived at considering 420 TPH (500 m³/hr) of Crude throughput. The same will vary based on actual crude throughput of the unit.

ii. Dosage Rates:

Dosing rates (along with diluents) for the above mentioned chemicals are as follows:

SL No	Chemical	Dosage Rate (Lt/Hr)	Pump Number & Capacity (Lt/Hr)	Pump stroke %
1.	Demulsifier			
	D 9004	12.1	11 PM 16 A/B (12 Lt/Hr)	100
	EC2040A	8.4		70
	DM 101	15.1		100, 25
	Embreak 2W157			
	- HS Case	8.6		70
- LS Case	4.8		40	
2.	Atmos Overhead Filmer			
	EC1021A	2.3	11 PM 15A/B (10 Lt/Hr)	20
	ESK50	8.4		80
	Philmplus 5K7	2.4		20



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	Thermosol V	2.6		25
3.	Ammonia	21	11PM14A/B/C & 12PM 08A/B (20 Lt/Hr)	50 50

Caustic Dosage Rates (3 % caustic solution) in Crude

Overhead Chlorides (ppm)	Sweet Crude Case			Sour Crude Case		
	Neat Caustic (ppm)	Diluted Caustic (LPH)	Pump Stroke (%)	Neat Caustic (ppm)	Diluted Caustic (LPH)	Pump Stroke (%)
25	0	0	0	0	0	0
50	1	14	20	2	28	40
75	1.5	21	30	3.5	49	65
100	1.7	24	35	4.5	63	85

Note:

- Chemical dosing rates to be measured every shift using draw down cylinders and pump stroke to be adjusted to meet the required dosing rate.
- It is assumed that the pump stroke of the pump follows a linear relationship with flow
- Unit is considered to be in service for all the days in the month.
- Actual dosage to be monitored and consumption of the chemicals to be recorded on daily basis.
- Dosages rates are arrived at considering 420 TPH (500 m³/hr) of Crude throughput. The same is required to be varied based on actual crude throughput of the unit and monitoring parameters (as mentioned in section 26.7.3.iii)
- After every batch of 3 % caustic solution preparation (for dosing in crude), caustic strength is to be confirmed by sending sample to lab for checking strength.



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iii. Monitoring Requirements:

Target monitoring parameters are as follows:

SL No.	Chemical	Monitoring Parameter	Allowable Parameter Limit	Testing Schedule
1	Demulsifier	Oil content in Desalter effluent Water	< 100 ppm	Daily Thrice (once in a shift)
2	Atmos Filmer	Atmos Sour Water Iron	< 0.3 ppm	Once in 2 days
3	Atmos Neutralizer	Atmos sour water pH	5.5-6.5	Daily Thrice (once in a shift)
4	Vacuum Filmer	Hot well Water Iron	< 0.3 ppm	Once in 2 days
5	Vacuum Neutralizer	Hot well water pH	5.5-6.5	Daily Thrice (once in a shift)
6	Caustic (to crude)	Atmos Overhead Chlorides	<30 ppm	Once in 2 days
7	Caustic (for SRN wash)	H ₂ S content in SRN	0	Twice Daily

Note:

- Filmer dosing to be increased if iron content in overhead water is higher than allowable limit. If iron content in overhead water is within limit, filmer injection rate to be maintained as per recommended dosage rate.
- Demulsifier dosing rate to be increased if oil content in desalter effluent water is higher than the allowable limit. If oil in desalter effluent water is consistently within limit then demulsifier dosing can be reduced and may even be stopped, if necessary.
- Neutralizer injection to be varied in order to maintain overhead water pH content within 5.5–6.5. In case of pH consistently higher than 6.5, neutralizer injection can be stopped.
- In case of any monitoring parameter persists to be beyond the limit consistently, the same is to be referred to PAD for any further analysis and recommendations.
- Caustic in crude to be injected as per guidelines provided under section 3.5.2
- Caustic to SRN caustic wash drum to be checked for strength (% availability) every 15 days. However, if SRN is reported to contain H₂S, caustic strength to be checked immediately and fresh batch to be loaded, if % availability of caustic in circulation is < 25 % (strength <3.5 % or 5 °Be). Irrespective of H₂S levels in SRN, caustic to be dumped and fresh batch to be loaded once caustic strength falls below 3.5 % (5 °Be).



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iv. Preparation for Chemical Dosing:

- The concerned plant technician should be familiar with procedure, physical and chemical properties and respective MSDS of the chemical to be prepared for dosing into the system, before starting the preparation of chemical.
- Proper PPE should be used and required safety precautions pertaining to that chemical as mentioned in PPE safety manual and MSDS register should be followed.
- The chemical should be prepared as per the standard procedure given in manager instructions. After preparation of the chemical, proper data should be maintained in the chemical inventory book and also the TOB. Any problems faced should also be recorded in TOB.
- The records should always be maintained as soon as the preparation is done so that any confusion is avoided.
- It has to be ensured that there is no wastage or spillage of chemical like pump leak, flange leak, hose leak, etc., in the plant area. If any leak is observed then immediately standby pump has to be placed in service.
- The dosing rates have to be measured /checked by draw down cylinder.
- Periodical cleaning of pumps strainer has to be done. Also for viscous chemicals, suction lines may get plugged often and need to be unplugged.

Standing Instructions for Empty oil & chemical drums collection for washing(SI 23):

The following activity to be carried out on continuous basis as per the procedure given below

S.No	Activity	Record	Responsibility
1.	Maintain a record of full lube oil/chemical drums received	Enter in Chemical inventory register	Field Technicians
2.	Maintain a record of empty drums after loading. Ensure that drums are capped.	1.Enter in unit waste log book 2.Raise job card for cleaning of drums	Field supervisor
3.	Empty drums are to be labeled as given below Empty oil based drums- Label type 1 Empty water based drums- Label type 2	Label the Empty drums with respective stickers.	Field Technicians



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4.	Mobilize and store the empty drums at the designated area in field	Segregate (Oil or water based) the drums at the storage area	Field Technicians
5.	Arrange for lifting and transporting the empty drums to washing area(WWTP) as per routine schedule	Issue work permit to Maintenance	Field supervisor
6.	Maintain a record of empty drums dispatched for washing	Enter in dispatched empty drums register.	Field supervisor



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27.1 CHEMICAL HAZARDS (MSDS):

HPCL-VR is determined to ensure that all the chemicals used are handled in a safe manner and information on their toxicity and hazards are made available to the personnel involved in handling the chemical. It will enable adequate safety planning and prevent accidents. This manual termed as “MSDS Manual” contains the Material Safety Datasheets (MSDS) of all the chemicals used by the refinery & the products manufactured by the Refinery.

For detailed MSDS of all the chemicals, refer to the MSDS manual available in the field room or the same can be viewed or downloaded from VISAKH REFINERY portal. VR Portal > Department > Technical > Process Safety Management Portal > MSDS Manual > Chapter 2.

27.2 FIRST AID PROCEDURES:

27.2.1 Animal Bites:

In case of animal bites, the following action has to be taken:

- Calm the affected person
- Wash hands before attending to wound
- Wash wound with soap & running water
- Apply antibiotic ointment
- Dress using sterile bandage
- After first aid, medical treatment must be sought quickly
- Suturing may be required
- Tetanus booster / antibiotics required
- Treatment depends on type / location of wound

27.2.2 Back Pain:

In case of back pain, the following first aid has to be provided:

- Rest in a comfortable position.
- Apply ice pack to affected area
- Painkillers or relaxants may be used
- Avoid strenuous exercise



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- Avoid pillows
- Avoid sleeping on soft mattress

27.2.3 Burns:

Burns can be of first, second and third degree. Categorization depends on severity of tissue damage.

First Degree Burns:

Injuries are superficial / mild, swelling & redness of the injured area is observed. Pain develops but no blisters are seen. Burned area becomes white on touching. Follow the first aid mentioned below in this case:

- Remove patient from heat source
- Remove the burnt clothing
- Run cool water over burnt area
- Gently clean the injured area
- Gently dry
- Apply anti biotic such as silver sulpha-diazine
- Use a sterile bandage to cover burns
- Take tetanus vaccination, if required

Second-degree burns:

Burns extends to middle skin layer, dermis and swelling, redness and pain is observed. Burnt area may turn white on touching, blisters develop, that ooze a clear fluid. There are chances of scars development. Follow the first aid mentioned below in this case:

- Clean the affected area thoroughly
- Gently dry
- Apply antibiotic cream over affected area
- Make the patient lie down
- Keep burnt body part at a raised level
- Skin graft may be required
- Physical therapy may be essential to aid mobility
- Splints may be used to rest affected joints
- Hospitalization is essential



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Third-degree burns:

Damage occurs to all three skin layers and destroys adjacent hair follicles, sweat glands, nerve endings. There is lack of pain due to destroyed nerves and injured area does not turn white on touching. No blisters are observed and swelling occurs. The skin develops leathery texture and discoloration of skin observed. Follow the first aid mentioned below in this case:

- Requires immediate hospital care
- Dehydration treated through intravenous fluid supply
- Oxygen is administered
- Eschars are surgically opened
- Periodically run clean cool water over burns
- Nutritious diet helps to heal quickly
- Regular monitoring essential
- Mental Depression treated by anti-depressants

27.2.4 Cardio-pulmonary Resuscitation (CPR):

CPR is an emergency life-saving measure and is a combination of rescue breathing & chest compressions. It is done on unconscious/ non-breathing patient. It is done on people suffering from cardiac arrest and also for near-drowning/ asphyxiation/ trauma. CPR conducts defibrillation and supports heart pumping for short duration. It allows oxygen to reach brain, thus buying time till help arrives. It is more effective when done as early as possible

The Vital Steps:

- Clear the airway
- Assess if the person is conscious / breathing
- Lay the person on his back on a hard surface
- Using a head tilt -chin lift open his airway
- Check for breathing sound
- If not breathing, start mouth-to-mouth breathing

Mouth- to-mouth breathing:

- Pinch the person's nostril shut
- Seal his mouth with your own



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- Give the first breath, lasting one second
- Watch if chest rises
- If it rises, give second rescue breath
- If it does not rise, give a head tilt- chin lift
- Now give second rescue breath

Restore circulation through compression:

- Place heel of your palm on patient's chest
- Place your other hand above first
- Keep elbows straight
- Push down using upper body weight (compress)
- Push hard and fast
- After 30 compressions, clear airway
- Give two rescue breaths
- This is one cycle
- Give 100 compressions /minute
- Continue CPR till medical help arrives

27.2.5 Chemical Burns:

In case of burns, the following first aid has to be provided:

- Remove patient from accident site
- Wash injury with tepid water liberally
- Identify chemical for effective therapy
- Seek medical treatment
- IV fluids need to be administered
- Pain medications and antibiotics needed
- Wounds cleaned and bandaged
- Follow -up care compulsory
- Consultation with specialist is a must

27.2.6 Chemical Splash in the Eye:

In case of Chemical splash in the eye, the following first aid has to be provided:

- Lay the person on the floor
- Keep eye lids open forcibly



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- Use clean cold water to wash eyes gently
- Keep washing steadily for at least 20 minutes
- Rinse/wash hands thoroughly to remove chemical
- If wearing contact lens remove them
- Do not rub eyes
- Do not use eye drops until told
- Wear sunglasses to minimize irritation
- After these basic steps seek medical help
- Remember the name of chemical to tell doctor

27.2.7 Dislocation:

In case of dislocation, the following first aid has to be provided:

- Call medical help as soon as possible
- Do not move the joint or try to place it back
- Place ice to control swelling
- If skin is cut, clean gently and bandage with sterile gauze
- Sling or splint the injury in its original position
- If injury is serious, check for breathing
- If not breathing, provide **Cardio pulmonary resuscitation (CPR)**. Elevate the feet up to 12 inches
- Cover the patient with a blanket

27.2.8 Electric Shock:

In case of Electric shock, the following first aid has to be provided:

- Do not attempt to move the victim from current source
- First step is to switch off the current source
- Otherwise, move the source using a wooden stick
- Attend to the victim
- Check for breathing
- No breathing, do Cardio pulmonary resuscitation (CPR)
- Call emergency medical aid
- If breathing, do a physical examination
- Treat for minor burns
- Re-establish vital functions
- Excessive burns may require hospitalization/ surgery
- Supportive care must be provided



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27.2.9 Foreign Object in the Ear:

In case a foreign object falls in the ear, the following first aid has to be provided:

- If object is protruding, use tweezers to remove
- If object is small, shake head with ear facing downward
- If it is insect, turn head to place affected ear upward
- Place few drops of mineral oil/baby oil inside ear
- Flush the insect out using clean water
- Use oil only in case of insect otherwise it may lead to swelling

Steps to Avoid:

- Do not push your finger into the ear
- Do not strike the head to dislodge object
- Do not shake a child to remove object
- Do not try to remove object on your own
- Do not block any discharge from ear
- Do not try to clean the ears

27.2.10 Foreign Object in the Eye:

In case a foreign object falls in the eye, the following first aid has to be provided:

- Wash hands before helping the victim
- Seat the person in a lighted area
- Gently examine the eye
- Pull lower eyelid downward
- Ask the person to look upward
- Then hold upper eyelid while person looks down
- If object is floating try flushing it out
- Otherwise, touch the object with wet cotton bud
- Object should cling to the cotton bud
- If object is removed, flush eyes with saline/warm water
- If object cannot be removed, see a doctor
- If object is embedded, do not touch
- Cover the eyes with paper cups and tape it
- Consult doctor immediately



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Steps to Avoid:

- Avoid rubbing eyes
- Do not remove an embedded object
- Do not try to remove a large object

27.2.11 Foreign Object in the Nose:

In case a foreign object falls in the nose, the following first aid has to be provided:

- The person must be urged to breathe through mouth
- The person should avoid breathing with force
- Close the unaffected nostril
- Blow out gently through the affected nostril
- Get medical aid if this method fails

Steps to Avoid:

- Do not probe an object which is not seen
- Do not probe an object that is not easy to grasp
- Do not blow nose too hard
- Do not use sharp instruments to remove the object

27.2.12 Foreign Object in the Skin:

In case a foreign object falls in the skin, the following first aid has to be provided:

- Wash hands well
- Clean the affected area using soap, water
- If object is visible above skin, squeeze the area around
- When object pops out, remove using sterile tweezer
- If embedded under skin, use a sterile needle
- Sterilize needle by flaming/wiping with alcohol
- Use needle to break skin over affected area
- Lift tip of the object
- Use a small tweezer to pull it out
- Gently squeeze the area and let bleed



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- Clean the area with soap, water. Pat dry
- Apply an antibiotic

Steps to Avoid:

- Do not wet if the object is of wood
- Wet wooden objects swell- becomes difficult to remove

27.2.13 Fracture:

There are two types of fractures.

- Open fracture: Skin breaks causing open wound
- Closed fracture: Skin not broken

In case of both the fractures, the first aid differs and is followed as discussed below:

For open fractures:

- Control bleeding before treatment
- Rinse and dress the wound

For open / closed fractures:

- Check the breathing
- Calm the person
- Examine for other injuries
- Immobilize the broken wound
- Apply ice to reduce pain / swelling
- Consult a doctor

Do not:

- Massage the affected area
- Straighten the broken bone
- Move without support to broken bone
- Move joints above / below the fracture
- Give oral liquids / food



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27.2.14 Heart Attack:

In case of a heart attack, the following first aid has to be provided:

- Try to relax
- Loosen tight clothes
- Take medicines if any
- Pain subsides within 3 min of medicine intake
- If not, see a doctor.
- Give artificial respiration if required
- Give **Cardiopulmonary Resuscitation (CPR)**:
 - a) If no pulse is detected
 - b) By placing palm on chest to pump
- 15 pumps are followed by 2 artificial respiration
- Continue till ambulance / doctor arrives

27.2.15 Heat Stroke:

In case of a heat stroke, the following first aid has to be provided:

- Remove the person to a shady place
- Cool the person by sponging with wet towel
- Apply ice packs in armpits and groin
- Water with electrolyte, fruit / vegetable juice should be given
- Victim must be rested

27.2.16 Severe Bleeding:

In case of severe bleeding, the following first aid has to be provided:

- Wash hands well before administering to patient
- Wear synthetic gloves
- Make the victim lie down
- Slightly elevate the legs
- If possible keep the affected area elevated
- Remove any obvious debris/particle
- Apply direct pressure using clean cloth/bandage
- Use hand if cloth is not available
- Apply pressure continuously for at least 20 minutes
- Do not remove the cloth to check the bleeding



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- Hold the bandage in place using an adhesive tape
- If bleeding seeps through bandage, do not remove it
- Add extra bandage on top of the first one
- Apply direct pressure on the artery if necessary
- The pressure points for arm--below arm- pit/above elbow
- For leg--behind knee/near groin
- Squeeze the artery keeping finger flat
- Continue applying pressure on the wound
- Once bleeding stops immobilize the affected part
- See a doctor

27.2.17 Spinal Cord Injury:

In case of spinal cord injury, seek medical help or call for an ambulance as soon as possible. Meanwhile the following first aid has to be provided -

- Move the person, if surrounding is not safe
- Immobilize the head, neck and body on both sides
- Movements may dislocate vertebra and cause further injury
- If there is no sign of breathing, perform CPR
- Do not tilt head backward during CPR

27.2.18 Sprain:

In case of sprain, the following first aid has to be provided:

- Apply a cold compress to injured area for 20 min
- This may be done 4-8 times a day
- Use a plastic bag with crushed ice, wrapped in a towel
- Use compression bandages to reduce swelling
- Keep the injured leg elevated on a pillow
- Take anti inflammatory pills if necessary
- Take rest for the recommended period
- When pain/swelling is diminished, do recommended exercises

Steps To Avoid:

- Do not return to normal activities if not completely cured
- This could lead to the problem turning chronic



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The contents of the First Aid box in the refinery are as follows:

S. No.	First Aid Box Content	Required No.
1	Small Sterile Dressings	4
2	Medium Sterile Dressings	4
3	Large Sterile Dressing	4
4	Burn Dressings	4
5	Absorbent Cotton (25gm)	1 Roll
6	Dettol Bottle	1
7	Burnol	1 tube
8	Disprin Tablets	6
9	Tincture Iodine Bottle	1

27.3 PERSONAL PROTECTIVE EQUIPMENT (PPE):

PPE are the equipments used to protect the person from hazards associated with the work being performed. PPE acts as a barrier between the hazard and the person exposed. Each unit and section is provided with certain PPEs depending on the layout, potential hazards and access for fire fighting. Fire & Safety carries out monthly check of all the equipments and releases inspection report.

PPE are characterized into two categories:

- a) Respiratory devices
 - b) Non-respiratory devices
- All the personnel should always ensure that everyone in field is wearing required PPE. For regular jobs like collecting H₂S rich gas, heater checking, chemical handling, etc. specific and recommended PPE is a must in an addition to safety helmet and work gloves.
 - Breathing apparatus should be wore before attending/attempting to arrest leaks like H₂S, CO, LPG, hydrocarbon vapor leaks.
 - Breathing apparatus should be checked and ensured that accessories are maintained in good and ready to use condition like oxygen cylinder is filled with pressure specified and face mask is free of dust etc. Organic and ammonia canisters should not be kept after their expiry date.
 - PPE boxes in the control rooms should be painted with a display of all the required items in the unit. All the PPE items should be cleaned by the plant personnel twice in week and maintain in good condition.



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- The usage of any PPE in case of any incident in the unit is to be recorded in the TOB by the shift in charge with details of the persons.
- All the safety showers in the unit have to be checked and ensure that both eye washers and the shower are in working condition.
- In every shift, it has to be recorded in every shift that “the contents of the First Aid Box are intact”. If any item has been used in the shift, it should be recorded in shift TOB with proper reasons.

27.3.1 Commonly Used PPE and its functions:

Safety Helmet:

Helmet is one of the most important items of personal protective equipment used by the industrial workers for protection against head injuries. The injuries can be caused either by falling objects or by hitting inadvertently against the pipe lines.

Safety Shoes:

Safety shoes are also one of the most important items of PPE used in the industrial workers for protection against foot injuries. These injuries can be caused either by falling objects or hitting anywhere inadvertently.

Safety Belt:

It is used to protect persons from free falling while working at higher elevation. It is a must for working above a height of two meters.

Chemical Cartridges:

It will provide against low concentration of relatively non-toxic gases and vapors as given in the table.

Type of Chemical Cartridge	Suggested maximum use concentration in ppm
AMINES	100
AMINE DERIVATIVES	30
AMMONIA	300
CHLORINE	10
SULFUR DIOXIDE	50

Canisters:

It can provide protection against high levels of toxic air contaminants as high as 20,000 ppm. Canisters/respirators are classified according to the type of gas and their



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concentration. The most commonly used canisters are acid gas, ammonia, and hydrocarbon vapor canisters.

Safety Gloves:

Safety gloves are used to protect hands against common industrial hazard. Asbestos gloves are used in hot work environment. It will provide protection up to 120°C.

PVC gloves/apron/shoes/suit:

It provides protection against chemicals such as caustic /acid /amine, etc. It should not be used as a protection against hydrocarbon and hot work environment. The maximum temperature allowed is 50°C.

Ear Muffs and Ear Plugs:

They are used to protect ears from high decibel noises. They are to be used while moving in the field.

Face Shield:

Face shield is used to protect face from any thing falling while looking up. Generally for checking the heaters, it has to be used to protect from any oil dripping.

PPE available in CDU-III field room:

S. No.	PPE Name	Quantity Available
1	Acid Gas Canister	2
2	Ammonia Canisters	4
3	Asbestos Gloves	2
4	Ear Muff	4
5	Face Mask Canister	2
6	Face Shield	2
7	Organic Vapor Canisters	4
8	Pan View Goggles	4
9	PVC gloves	4
10	PVC hood	2
11	PVC pant and Coat	2
12	Water Gel Blanket	1
13	Safety Belt	1



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27.4 FIRE FIGHTING SYSTEM & EQUIPMENT:

Water is the most important medium available for fire protection. Water is used for fire extinguishing, fire control, cooling of equipments and protection of equipment and personnel from heat radiation. For these purposes water is used in various forms such as straight jet, water jog, water curtain, water spray, deluge, sprinkler for foam making etc. The system is designed in such a way that it can fight two fires requiring largest water demand at the same time.

Fire water system is designed for a minimum residual pressure of 7.0 kg/cm^2 at the hydraulically remotest point of application at the designed flow rate at that point. The fire network shall be kept pressurized at minimum 7.0 kg/cm^2 all the time.

27.4.1 Hydrants:

Hydrants are located keeping in view the fire hazards at different sections of the premises to be protected and to give most effective service. At least one hydrant post shall be provided for every 30 meters of the perimeter of unit battery limit. The hydrants should be located at a minimum distance of 15 meters from the periphery of hazardous equipment. Double headed hydrants with two separate landing valves on 4" stand post should be used. All hydrant outlets shall be situated at a workable height of about 1.20 meters above the ground level.

27.4.2 Monitors:

Monitors are located at strategic locations for protection of cluster of columns, heaters, etc. and where it may not be possible to approach the higher levels. Monitors shall be located to direct water on the object as well as to provide water shield to firemen approaching the fire. The monitors should not be installed less than 15 meters from the hazardous equipment and shall not exceed 45 meters from the hazard to be protected. The butterfly valves of the monitors should be periodically checked for free operation and if isolation valves are gate valves, it should be greased regularly and medium sized wheel spanner should be welded near it. Monitor rotation levers should be periodically checked for its function.

27.4.3 Fixed Water Spray System:

Fixed water spray system is a fixed pipe system connected to a reliable source of water supply and equipped with water spray nozzles for specific water discharge and



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distribution over the surface of area to be protected. This system should be provided in high hazard areas where immediate application of water is required.

27.4.4 Fixed Water Sprinkler System:

Fixed water sprinkler system is a fixed pipe tailor made system to which sprinklers with fusible bulbs are attached. Each sprinkler system includes a controlling valve and a device for actuating an alarm for the operation of the system. The system is usually activated by heat from a fire and discharges water over the fire area automatically.

27.4.5 Carbon Dioxide System:

Carbon dioxide is an odorless and colorless inert gas and it extinguishes fire by cutting off the oxygen and creating an inert atmosphere around the hazard. If the inert atmosphere is maintained for a reasonable time the possibility of flash back is also reduced. Carbon dioxide is best applied to fire hazards through the fixed system consisting of CO₂ storage, distribution and discharge nozzles.

27.4.6 Dry Chemical Powder (DCP)-10 kg capacity:

This is the most common fire extinguishing medium in the refinery. The number should be determined based on the maximum traveling distance of 15 meters in hazardous areas. At least one fire extinguisher should be provided for every 250 sq.mt. of hazardous operating area.

27.4.7 Dry Chemical Powder (DCP)-75 kg capacity:

It is provided in critical operating areas. At least one fire extinguisher should be provided for every 750sq.mt.of hazardous operating area.

27.4.8 Dry Chemical Powder (DCP)-5 kg capacity:

It is provided at elevated locations for easy handling.

Note: Any of these cylinders if used, should be immediately informed to Fire and Safety and get a replacement cylinder. Also the same has to be mentioned in the TOB and informed to unit manager/YSF.



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27.4.9 1" Rubber Hose Reel:

They are provided for immediate usage of fire water during fire emergency or push the spilled oil into OWS. They are of 30 meters length as per OISD guidelines and contain a nozzle at the open end. It should not be used for floor washing but can be used for pushing spilled oil into nearest storm sewer/OWS. In this condition, immediately car seal should be put back.

27.5 SPILL HANDLING:

27.5.1 Ammonia:

If a spill or release to the environment has occurred, fire department, emergency response, and/or hazardous materials spill personnel should be notified immediately. Cleanup should be attempted only by those trained in proper containment procedures. Contaminated soils should be removed for incineration and replaced with clean soil. If ammonia should contact the water table, aquifer, or navigable waterway, time is of the essence. It is readily miscible in water and total containing and remediation may not be entirely possible. When such spills occur, the local and/or state emergency response authorities must be notified. A comprehensive emergency response or disaster preparedness recovery plan should be in place prior to any operations involving the use, transportation, storage, or disposal of ammonia. If ammonia is spilled or leaked, the following specific steps are recommended:

- Restrict persons not wearing protective clothing from area of spill or leak until cleanup is complete and area can be opened for normal work.
- Remove all ignition sources.
- Ventilate area of spill or leak.
- If applicable, stop flow of leaking liquid or gas. If leak source is a cylinder and the leak cannot be stopped in place, remove leaking cylinder to a safe place in the open air, and repair or allow cylinder to empty.
- Keep ammonia out of a confined space, such as a sewer, because of the possibility of explosion (unless the sewer is designed to prevent the buildup of explosive concentrations).
- It may be necessary to dispose of ammonia as a hazardous waste. The responsible state agency or the regional office of the federal Environmental Protection Agency (EPA) should be contacted for specific recommendations.



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27.5.2 Caustic 100%:

If a spill or leak to the environment has occurred, fire department, emergency response and/or hazardous materials spill personnel should be notified immediately. Cleanup should be attempted only by those trained in proper spill containment procedures. If sodium hydroxide should contact the water table, aquifer, or navigable waterway, the local and/or state emergency response authorities must be notified. A comprehensive emergency response or disaster preparedness/recovery plan should be in place prior to any operations involving the use, transportation, storage, or disposal of sodium hydroxide. If sodium hydroxide is spilled or leaked, the following specific steps are recommended:

- Restrict persons not wearing protective clothing from area of spill or leak until cleanup is complete and area can be opened for normal work.
- Do not allow exposure to incompatible materials.
- Close or remove all ignition sources and ventilate area of spill.
- Collect powdered materials in the safest and most efficient manner possible. Place materials in a sealed drum. Do NOT dry sweep (generates dusts). Use a vacuum equipped with a high efficiency particulate air (HEPA) filter instead.
Do NOT use water or damp mop residues.
- It may be necessary to dispose of sodium hydroxide as a hazardous waste. Consult state, local, or federal environmental regulations.

27.5.3 EC 1020A:

- Small spills: Contain with absorbent material such as clay, soil or any commercially available absorbent into recovery or salvage drums for disposal.
- Large Liquid Spills: Dyke to prevent further movement and reclaim into recovery or salvage drums or tank truck for disposal.
- For large indoor spills, evacuate employees and ventilate the area. Those responsible for control and recovery should wear breathing apparatus, impermeable (PVC) gloves, boots apron and a face shield with chemical splash goggles.

27.5.4 ELOGUARD 86:

In case of any spillage or leak, the spill has to be flushed with plenty of water. Those responsible for flushing should wear breathing apparatus, impermeable (PVC) gloves, boots apron and a face shield with chemical splash goggles because of alkaline nature of the chemical.



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27.5.5 NALCO 5186:

Restrict the access to area till the clean up operations are complete and ventilate the area properly. Do not touch the spilled material by bare hands. Stop or reduce the source of leak if possible and remove source of ignition if any. Those responsible for control and recovery should wear breathing apparatus, impermeable (PVC) gloves, boots apron and a face shield with chemical splash goggles.

- For cleaning small spills, soak the spill with absorbent material and the residue should be kept in properly labeled and covered containers. The affected areas should be washed.
- For large spills soak the spill with absorbent material or by digging trenches or by making dykes. Reclaim it into salvage drums or tank trucks for proper disposal. Wash the spillage site thoroughly with water. Contact an approved water hauler for disposal of contaminated recovered material.

27.5.6 EC 2040A (DMF):

Restrict the access to area till the clean up operations are complete and ventilate the area properly. Do not touch the spilled material by bare hands. Stop or reduce the source of leak if possible and remove source of ignition if any. Those responsible for control and recovery should wear breathing apparatus, impermeable (PVC) gloves, boots apron and a face shield with chemical splash goggles.

- For cleaning small spills, soak the spill with absorbent material and the residue should be kept in properly labeled and covered containers. The affected areas should be washed.
- For large spills soak the spill with absorbent material or by digging trenches or by making dykes. Reclaim it into salvage drums or tank trucks for proper disposal. Wash the spillage site thoroughly with water. Contact an approved water hauler for disposal of contaminated recovered material.

Note: For any new chemical, the spillage handling procedure can be obtained form MSDS.



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All hydrocarbon stream contaminated with water, water contaminated with hydrocarbons or chemicals and such liquid streams that cannot be disposed off directly are collected and treated either for recovery of hydrocarbons or to render themselves harmless to receiving stream or for both the purposes. For these purposes CBD system and OWS system are provided. These systems are described sequentially in the following paragraphs.

28.1 CLOSED BLOW DOWN SYSTEM:

(Refer P&ID No. 10-1100-E-204 Rev.7 and 10-1100-E-205 Rev.7)

Those hydrocarbon streams that are either free of water or only slightly contaminated with water are received in closed blow down (CBD) vessel (11-V-08). Such hydrocarbon streams are generated especially during shutdown periods when equipment and systems are drained under gravity to clear oil hold up. The hydrocarbon in the CBD vessel should be received at a temperature well below flash point of lightest component present inside.

CBD network helps in reducing the amount of hydrocarbon loss. It helps in the avoiding hazardous conditions that may result in from draining of hot liquid from the equipment, by providing a collection point for such liquids. It also helps in reducing the oil going to the effluent treatment facility. Uncontaminated hydrocarbon from equipment draining is collected and routed to CBD vessel via CBD headers, covering the entire unit.

An 8" underground CBD header has been provided with ballast points at different locations on the CBD header for flushing the line after draining heavy or viscous oil into it. The header is finally routed to the CBD drum (11-V-08). It is an under ground vessel placed horizontally. The drum is provided with a coil through which cooling (salt water) water or LP steam can be passed. The drum caters to hydrocarbon blow down from CDU/VDU-II, BBU.

LI-1701 and TH-1702 on 11-V-08 indicate the level and temperature of the CBD drum. It is also provided with low and high level alarms for LI-1701 respectively. There is also an indication of the CBD inlet temperature TH-1701. It is provided with **low and high temperature alarm for (TI-2601)**. CBD pump (11-P-18) is interlocked with high / low level alarms for auto start and auto stop. Hydrocarbon is pumped out by CBD pumps to slop header in the unit.

Under normal conditions it is expected to remain at atmospheric pressure. A small purge of steam is provided at the event of the drum to guard against fire at the vent top due to lightening and static electricity hazard. Service water connection is given to the drum for cleaning purpose



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28.2 OILY WATER SEWER:

This system also called OWS is mainly to collect water contaminated with hydrocarbon oil. Such streams are generated during equipment draining and flushing during routine operation. Streams suitable for OWS also generated as a result of floor washing and cleaning or spilled oil etc. Drained streams received through OWS funnels etc. are routed to ETP-II through a combined header. Contaminated rain water from the unit paved area is also routed to OWS System. All equipments having CBD connections are normally provided with OWS connection also. No OWS points are provided near the heater for safety reasons. The OWS manholes are sealed and a vent pipe provided releases any hydrocarbon trapped at a higher elevation. The final outlet of the OWS is located at the south east corner of the unit. The OWS system is common for both CDU and BBU units. The effluent generated in the unit reaches ETP-II for treatment and oil recovery.

Care should be taken that at no time any spark reaches vapor space of OWS funnels. This may result in explosion immediately there or elsewhere in network wherever explosive mixture of air and hydrocarbon vapors is present in appropriate concentration for explosion.

28.3 SURFACE DRAIN SYSTEM:

The peripheral surface Drain with double valve arrangement to route it either to OWS or to Storm Sewer System depending upon its oil content. Normally, surface drain outlets are kept floating on OWS system. During rains, the outlet is changed over to Storm water sewer after one hour to reduce the load on ETP-II.

28.4 SLOP SYSTEM

Off-spec hydrocarbon streams are routed to an 8" slop header at unit battery limit which is then routed to slop tanks. The slop generated within CDU-II is primarily dry slop and is hooked up to slop headers from flare area. This header is routed to the dry slop tanks located in off-sites.

Dry slop can be recycled back to CDU for reprocessing either by direct injection of slop oil to crude feed line during Refinery crude tank (120-T-01 C/D/E/F) processing or via crude tanks as slop is sent to crude oil tanks. All product rundown lines have access to slop header to route them to slop tanks when their quality goes off-spec. (Mainly during start-up / shut down).



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Standing Instructions for Effluent monitoring: (ADM/OPRN/PRODN/SI/011):

The following guidelines needs to be followed for monitoring of the draining's of Oil/Water in to OWS,

- The normal operation liquid draining's in to OWS each shift wise to be recorded, starting with shift I personnel of the day.
- Each shift to ensure record the approximate quantity in the column with signature.
- A team has been formed, consisting of Operations and Technical members for surprise check of the system weekly.
- Objective of this implementation in operations is to achieve “MINAS” standards of ETPs outlets.

Standing Instruction on OWS system (SI:12):

Presently during wet weather, the collection of Rain water was exceptionally high and is more than what ETP's are actually to hold. This is resulting in over flowing of the surge ponds and oil breakthrough in to Refinery Effluent.

In View of the above, it is required to ensure that, the entire OWS system is properly maintained and operated in order, not to get excess/un wanted influent entry in to oily water sewer system.. The following actions need to be compiled in Process Units.

- a. Ensure that all the OWS catch basin covers are in position.
- b. During wet weather conditions, switch surroundings OWS to open drain after first hour of the rain.
- c. Ensure that no Fire water usage for Housekeeping/Floor washing.
- d. Ensure no draining of spent caustic and chemicals in to OWS.
- e. Ensure sour water routing to SWSU.
- f. If any oil carry over in Desalter Effluent water reported by ETP-II operator, CDU block shift in charge to respond immediately, if required suspend wash water injection to Desalter till it is corrected in consultation with YSF.

Ensure that spent caustic is routed to Merox unit for dumping in Chemical sewer.



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The Refinery got ISO 14001 certified for its EMS in May 2002 by SGS Yardley Certification Services of UK and since then it has been sustained till now with recertification in May 2006 complying with the revised ISO 14001:2004 standards. Refinery also submits environmental compliance reports to APPCB regularly. In the unit, the air emissions are generated mainly from the heater as combustion gases (flue gases). They contain a mixture of SO₂, CO, NO_x, SPM, etc. If proper care is not taken for these emissions, they can be fatal to living beings.

29.1 ENVIRONMENTAL POLLUTION CONTROL MANAGEMENT & FACILITIES:

Environmental pollution through air emissions, liquid effluents discharge and hazardous wastes is managed by :

- (i) Control
- (ii) Monitoring mechanisms as described below:

29.1.1 Air Emissions Control:

Emissions into ambient air from Refinery operations which significantly affect the environment are (i) Sulfur Dioxide (SO₂), (ii) Carbon Monoxide (CO), (iii) Oxides of Nitrogen (NO_x), (iv) Suspended Particulate Matter (SPM), (v) Hydrocarbon (HC), and (vi) Noise.

Sulfur Recovery Units (SRUs):

Refinery has installed two identical units of 16 LTPD capacity Liquid Oxidation based Catalytic units (LOCAT) licensed by erstwhile ARI Technologies, USA for desulfurization of the sour fuel gas (0.5 wt%-14wt% H₂S) to sweet fuel gas (less than 100 ppm H₂S) for use in heaters/ boilers in Sept. 1994 and Dec. 2000. Another unit (DHDS-SRU) with a capacity of 130 MTPD was installed along with DHDS block in the year 2000 to desulphurize Amine Acid Gas (AAG) and Sour Water Stripper Gas (SWSG). The SRUs will help in reduction of SO₂ emissions to atmosphere when the gases are burnt. Under VRCFP, an additional SRU of 65TPD was commissioned.



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Low sulfur content in the internal Fuel:

Sulfur in the internal fuel, that is, Low Sulfur Heavy Stock (LSHS) used in Boilers, Furnaces & Gas Turbines is maintained in the range of 0.5 to 0.6wt% by in-house blending so as to meet present SO₂ emission norm of 11.5 TPD stipulated by APPCB

CO Boiler in FCCUs:

CO boilers are installed at downstream of Fluid Catalytic Cracking Units (FCCUs) for converting highly harmful CO to CO₂, and in turn generating HP steam as it is an exothermic reaction.

Low NO_x burners in Furnaces:

Low NO_x burners have been provided in furnaces and boilers installed after the year 1999 to reduce NO_x emissions.

High efficiency cyclones in FCCU:

FCCUs utilize zeolite based catalyst in regenerator and reactor for catalytic cracking of heavy molecular weight complex hydrocarbons to low molecular weight simple hydrocarbons. In this process, part of the catalyst gets deformed and forms fines which tend to escape to the atmosphere along with the flue gas from the regenerator. The installation of cyclones within regenerator and reactor enables retention of the fines within the process system.

Smoke less Flare:

The refinery flare gases are safely burnt in a tall vertical flare stack with proper water seal and steam dispersion to avoid black smoke. The flare piping network is closed system interconnecting all the process units.

Floating roof tanks & mechanical seals:

To facilitate control of fugitive emissions from tanks and pumps, refinery has installed floating roof tanks for Crude oil tanks and light product storage tanks. All pumps in use are equipped with mechanical seals and for of LPG and lighter product pumps double mechanical seals are used.



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Low noise machines/ motors:

At the time of equipment purchase itself it will be specified that noise level to be below the statutory limit with required acoustic enclosures.

Tall Stacks:

Process furnaces, boilers and gas turbines are provided with tall stacks (about 60m) for better dispersion of flue gases and to get ground level concentration within allowable ambient air quality limits.

After installation of the pollution control facilities, it is to be ensured that the facilities are functioning properly so as to meet the emission norms consistently. Monitoring of air emissions is carried out as follows:

On-line stack monitors:

Refinery has all the stacks equipped with on-line stack analyzers to monitor SO₂, NO_x, SPM, HC and CO.

Continuous Ambient Air Monitoring Stations (CAAMS):

Based on predominant wind direction and neighboring population, three CAAMS have been installed to monitor ambient air quality with respect to SO₂, NO_x, HC, SPM, RSPM, CO and Noise. One of the CAAMS is also equipped with Weather Monitoring Station (WMS) to measure wind speed, direction, relative humidity, rainfall, temp, etc.

CCTV for flare:

A Closed Circuit color TV (CCTV) is installed at the centralized process unit control room for monitoring the flare flame and its quality. The unit personnel keep a watch on the CCTV and diagnose the problems within their units and check the condition of the relief valves for their passing based on the quality of the flare.



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Noise Monitoring:

Exposure to noise at high levels and for long duration can have adverse physiological and psychological effects. Periodic measurement of the noise pollution is carried out by Fire & Safety Department.

29.1.2 Liquid Effluent Management:

Liquid effluents which significantly affect the environment are (i) Sour Water (ii) Equipment/ Floor washings, contaminated with oil & chemicals, (iii) Spent Caustic and Chemicals from process & product treatment units, and (iv) Hydrocarbon draining from columns, vessels, heat exchangers etc. These pollutants are controlled with use of the following facilities:

Sour Water Stripping Units:

During the distillation process of the hydrocarbons in the column sour water, containing hydrogen sulfide, ammonia, phenols etc, gets collected in the overhead accumulator drum. The sour water is stripped off the pollutants in strippers with a steam reboiler. The off gases viz. hydrogen sulfide and ammonia are routed to SRUs while the stripped water is reused partly as desalter wash water and the balance stripped water is routed to Effluent Treatment Plant (ETPs) prior to discharge as refinery effluent to the sea.

OWS & Chemical Sewers:

Segregated sewer systems of Oily Water Sewer (OWS) and Chemical sewers are provided for better separation of oil from water and other spent chemicals. All the process units, product treatment units and tank farm areas are provided with underground OWS network. Spent chemicals like spent caustic (rich in sulphides) from Mercaptans Oxidation (Merox) units and Diesel Hydro-desulfurization (DHDS) unit have dedicated underground sewer system for routing the effluents separately to the ETP.

Closed Blow Down System:

All process and treatment units have Closed Blow down (CBD) system installed to facilitate draining of hydrocarbons from equipment/ heat exchangers prior to giving for maintenance & repairs. This drain facility ensures that the hydrocarbon is recycled back to the process system as slop oil and not let out to ETPs.



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Effluent Treatment Plants:

ETPs are equipped with physico-chemical, biological treatment and effluent polishing facilities to remove free oil, emulsified oil and reduction of Biological Oxygen Demand (BOD). There Tilted Plate Interceptors (TPIs), Dissolved Air Floatation (DAF) cells, ammonia stripping tower, trickling filter, activated sludge process reactor, Dual Media Filters (DMFs), Granulated Activated Carbon (GAC) filters etc. for effective stage-wise pollutant removal from the liquid effluents. The effluent quality is measured at the outlet through a composite auto-sampler.

The Refinery liquid effluent is tested daily to check its quality and take corrective actions. APPCB also monitors the liquid effluent discharge from the refinery premises by collecting regular legal samples for checking compliance with Minimal National Standards (MINAS). Following studies have been taken up for further improving the liquid effluent:

Zero effluent discharge:

To achieve zero liquid discharge from the Refinery by internally reusing and recycling the waste water, the study was taken up with assistance of EIL.

Dedicated sour water collection and reprocessing:

The study was taken up with assistance of EIL to provide dedicated sour water collection system in the refinery to safeguard ETP from shock loads during emergency or abnormal plant status.

Correction of excess oil ingress at ETP-I/II:

The study was taken up with assistance of EIL to reduce oil ingress into the ETP's. The study was completed and project is under implementation.

Performance evaluation of Bio-system at ETP-II:

The study was taken up with assistance of NEERI and the recommendations on improvement of plant performance are under implementation.



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29.1.3 Hazardous Waste Management:

Hazardous wastes are managed through minimization at source, recycle or reprocess, sale and explore usage or alternate disposal options and store it for safe disposal in future. In Refinery, oily sludge generation from storage tanks forms major portion of the hazardous wastes, other wastes being spent catalysts, charcoal, lead acid batteries, slop oil, etc. The following facilities are provided to effectively control the hazardous wastes:

Side entry mixers in tanks:

The sludge generation in crude tanks varies with type of crude oil stored and duration of usage of the tank. With usage of side entry mixers, settlement of the tank sludge to the bottom is avoided. All crude oil and heavy oil tanks are provided with side entry mixers and the service factors of the mixers are maintained high.

Proper Housekeeping in Process units:

Wastes generated in the units are segregated into ferrous and non-ferrous and kept stored at the earmarked unit peripheral limits for subsequent removal and proper disposal of the wastes.

Empty Drum Washing:

The empty chemical and oil drums (plastic and metallic) are labeled into oil based and water based and kept stored at the earmarked unit peripheral limits for subsequent removal to the empty drum washing area near WWTP. These empty drums are washed with suitable solvent depending on whether the drum was oil based or water based and using jet sprayer and tilted drum shaking equipment.

Centrifuges at ETPs:

ETPs are equipped with centrifuges for effective solid–liquid separation and the ETP sludge is then treated in bioremediation farm, through the process of which the quantum of sludge reduces considerably.

Monitoring and disposal involves an elaborate mechanism. The hazardous waste generation is monitored through Disposal Requisition Notice (DRN) system since the year 2000. The waste generating section/ unit raises DRN indicating the waste details (type of waste, source



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of generation, quantity etc.) and based on this management or disposal method for the waste shall be advised. Following actions have been taken for reprocessing or recovery and disposal of the hazardous wastes:

29.1.4 Ex-situ processing of high oily sludge:

The Refinery pioneered in processing high oily sludge generated from tank farms. About 34,000 m³ of the sludge was treated by M/s.Singapore Cleanseas Pte Limited, Singapore during Apr. 2001 thru Aug. 2003. Presently, M/s Balmer Lawrie & Co. Ltd, Kolkata, carried out the sludge processing till Nov 2008. Presently, sale of sludge is being taken up.

29.1.5 In-situ processing of high oily sludge:

- Chemical method:

Crude tank bottom high oily sludge from four Crude tanks was processed successfully to recover oil using a chemical (EC9007A) supplied by M/s Nalco Chemicals since the year 1998.

- Mechanical method:

Crude tank bottom sludge was also processed in 2005-06 using mechanical method of Blabo technology.

29.2 EFFLUENT, EMISSIONS AND SOLID WASTES:

The liquid effluent of CDU-II is the sour water which is removed from the column overhead drum, Desalter and hot well drum. This is sour water and needs to be treated in Sour water stripping unit. While routing this water to SWSU, it has to be ensured that the water contains very minute traces of oil (limit up to 100 ppm). If the water contains more oil, then it becomes difficult to dispose off or treat this water. The water should be oil free and the stripped water is used for desalting in the desalter.

The effluent sometimes is required to drain to OWS or local drains in case of some problem in Merox or due to any process or line constraints. This time of this operation has to be minimized as this is not an environment friendly practice. This can only be done as a temporary measure and cannot be continued for a longer period of time.



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The flue gas of heaters is the emissions from the unit. There are various methods of controlling these emissions. The environmental regulations are very stringent for emissions and already the actions taken in the refinery to control emissions have been discussed. All the stacks have online analyzers for SO_x, NO_x, CO, HC and SPM. They will give us an idea of the amount of these gases generated and which gives an idea of the quality of fuel that is getting incinerated in the heater. The fuel being used is generally sweet fuel (sweet fuel gas and LSHS) and thus it helps to keep the emissions inside the permissible limits. The technical department keeps a constant check on the values of these meters and the emissions from the unit so that there is no harm caused to environment from the refinery.

Solid wastes generated in the unit are either ferrous or non-ferrous. The wastes include insulation material, cotton wastes, insulation cladding material, replaced line portions, etc. They are kept in the designated scrap yard of the unit. The scrap yard is situated in south-west corner of the unit and separate yards are available to keep ferrous and non-ferrous materials.

The scrap generated from the unit has to be kept in these yards. Once the quantity of scrap is more in the yard, job-card is given to remove scrap from the unit and the same is carried out by the maintenance department. It is to be ensured that the scrap yard is always emptied once it becomes full and thus there is always sufficient dumping space for the scraps. Any waste generated has to be shifted to the scrap yard. If during maintenance, scrap is generated then it should be ensured that proper housekeeping has been carried and the scrap generated has to be put in the scrap yard. Always the ferrous and non-ferrous scraps have to be kept in their separate yards



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“THE BEST PROTECTION AGAINST ALL HAZARDS IS PREVENTION”

Hazard is defined as a chemical or physical condition that has the potential for causing damage to people, property or the environment. People working in the field must be well known to the actions that need to be taken in case of such situations. The unexpected hazards are discussed in the chapter but there are some hazards which may not be covered in the chapter. Accidental release of flammable or toxic vapor can result in severe consequences like BLEVE, vapor Cloud Explosion or Toxic Vapor Cloud. A BLEVE occurs when there is a sudden loss of containment of a pressure vessel containing a superheated liquid or liquefied gas. Delayed ignition of flammable vapors can result in blast overpressures covering large area. Toxic clouds may cover yet larger distances due to the lower concentration threshold value for potentially lethal situations in relation to those in case of explosive clouds.

In contrast, fires have localized consequences. The extent of damage to people depends on the heat flux and duration of exposure. Fires can be put out or contained in most cases, but there are few mitigating actions one can take once a vapor cloud gets released. Major accident hazards arise, therefore, consequent upon the release of flammable or toxic vapors.

Safety of personnel and equipment is very important. Ignorance of the details of the unit or the techniques of safe and efficient operation reduces the margin of safety of personnel and subjects the equipment to more hazardous conditions. All the operating and maintenance crew therefore must be fully familiar with the equipment and materials being handled in the unit and recognize the hazards involved in handling them and the measures taken to ensure safe operations.

Since the unit handles with one of the most potential source of fire and explosion, LPG and chemicals like caustic, adherence of safety rules should be given uphill importance.

30.1 SAFETY OF PERSONNEL:

General safety rules, which shall be practiced and enforced for all personnel who enter the unit, are summarized below:

- Safety helmets and boots shall be worn by all personnel at all times in the plant. They may be removed when inside rooms or buildings which do not have overhead or other hazards.
- Smoking is not permitted in side the refinery.
- Each employees assigned to work in the unit shall know where the safety and fire suppression equipment is located and how to operate equipment.



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- Safety glasses, goggles or face shields shall be worn while performing work, which could result in eye or face injury.
- Operations personnel golden rule
- **Do not open or close any valve without first determining the effect.**
- Maintenance personnel golden rule:
- **Treat each piece of equipment or piping as if it is under pressure.**

30.2 PRESENCE OF FLAMMABLE HYDROCARBONS AND SAFETY PRECAUTIONS:

30.2.1 Hazards Related to Handling of Liquefied Petroleum Gases Systems:

In the design of the LPG handling area (absorber section) major consideration has to be given to the Hazards of LPG which are listed in this subsection.

Liquefied Petroleum Gas is a mixture of commercial Butane and commercial propane both saturated and unsaturated hydrocarbon. LPG is inherently dangerous on account of fire, explosion and other hazards. This calls for special attention on the manner in which it is stored, transported or used. These hazards can have an impact on both the public. As a consequence of this special emphasis has been given on the safety regulations while designing this system.

- LPG at atmospheric pressure and temperature is a gas which is 1.5 to 2.0 times heavier than air. It is readily liquefied under moderate pressure. The density of liquid is approximately half that of water and ranges from 0.525 to 0.58 at 15 °C. Since LPG vapor is heavier than air, it would normally settle down at ground level / low lying places and accumulate in depressions. Under still conditions the dissipation of accumulated vapor is slow. This accumulation of LPG vapors gives rise to potential fire and explosion hazard.
- LPG has an explosive limit range of 1.8% to 9.5% volume of gas in air. This is considerably narrower and gives an indication of hazard of LPG vapors accumulated involving areas in the eventuality of leakage or spillage.
- The combustion reactions of LPG increase the volume of product in addition to the generation of heat. LPG requires 24 to 30 times its own volume of air for complete combustion and at the same time yield 3 or 4 times its own volume of carbon dioxide yielding approximately 10,900 k cal of heat per kg. Leaking liquid phase LPG will



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rapidly expand around 250 times its own volume therefore creating a greater risk than would occur with a similar size vapor leakage. When the pressure is reduced, LPG vaporizes rapidly reducing the surrounding temperature. This may lead to burns in contact of LPG with the skin.

- LPG liquid released forms a vapor which travels long distance. This explains the hazardous aspects of faster spreading of LPG fire. The vapors of LPG can give more violent conclusion in explosion as in case of Hydrogen due to higher flame propagation
- Liquid phase LPG expands considerably when its temperature increases, exceeding most of petroleum products. The coefficient of expansion is around 0.000237 per degree centigrade at 15 C. This high rate of expansion has been taken into account when specifying the maximum quantity of LPG permitted to be filled into any pressure vessel.
- LPG is colorless in both liquid and vapor phase.
- The vaporization of liquid cools the atmosphere and condenses the water vapor contained in them to form a whitish fog which makes it possible to see an escape of LPG.
- LPG has only faint smell and consequently it is necessary to add some odorant so that any escaping gas can easily be detected.
- LPG has low viscosity and can leak when other petroleum products normally cannot thus demand a very high integrity in the pressurized system to avoid leakage. It is also poor lubricant and leaks are therefore likely to occur at pump seals and glands. In view of this, special attention has been given e.g. to the selection of mechanical seals etc.
- The hazards associated with release of a boiling flammable liquid to atmosphere show themselves disproportionately when large quantities are involved - they can give rise to two phenomena which are highly hazardous the unconfined vapor cloud explosion and the boiling liquid expanding vapor explosion.

Vapor cloud can ignite and burn as fireballs causing lot of damage by radiation starting secondary fires at some distance. Vapor cloud ignites and explodes causing high overpressures and very heavy damage. The latter is termed as 'Percussive unconfined Vapor Cloud Explosion' i.e. PVUCE in short. Even though large quantities of LPG emission are necessary only a fraction of this contributes to percussive effect.



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The boiling liquid expanding vapor explosion (BLEVE) is due to holding pressurized flammable liquid above its boiling point. This may result from inadequate vapor space for temperature expansion of the content or high temperature due to radiation heat from the adjacent vessel under fire or due to a mechanical damage. BLEVE even though not as serious as PUVCE, the potential for spreading fire over a wide area makes them significant in terms of both life and property posing hazard to fire-fighters.

Ejection of boiling liquid produces reaction rupturing the vessel causing the rocket projecting of dish ends. The released liquid flashed and atomizes immediately often resulting in large fire ball. Although the fire ball lasts only a few seconds, its effects can be devastating due to flame contact and thermal radiation.

30.3 SAFETY PRECAUTIONS WHILE HANDLING FLAMMABLE HYDROCARBONS:

- i) LEL Detectors: LEL detectors are provided at various locations in the unit at critical locations. This will help in early identification of any hydrocarbon leaks so that immediate responsive action can be taken. Activation of LEL detector raises alarm in DCS.
- ii) The unit handles hydrocarbon liquids and gases which can cause fire and explosion if conditions for the same are favorable. Direct contact of hydrocarbons with any source of ignition of fire such as spark or open flame should be avoided. Sparks or open fire should not be allowed in the operating area while the unit is operating or has potential of hydrocarbon leakage. Even during shutdown. When an open flame work is absolutely essential, this should be allowed with the permission of the authorized person after suitable precautions have been taken. Approved safety & work permit systems should be strictly followed.
- iii) Propane gas is highly explosive in air mixtures containing 2.4 to 9.5 % by volume of Propane vapors. Its concentration in air should normally be not allowed to exceed 0.4%. It must be stored in a cool, isolated and ventilated area free of acute fire hazard. A concentration in air above 1% of Propane may have mild narcotic effect. Inhalation of excessive amounts causes progressive aesthesia. Liquid Propane in contact with skin will cause frostbite through rapid evaporation.
- iv) Improper earthing of equipment may produce static charges. All equipment must be provided earthing connections as per standards.
- v) The drain / vent valves and sample points on hot and heavy products and steam lines are likely to get plugged, if the lines are not in constant use. Such valves should be crack opened for a few minutes under personal supervision and only after the liquid flow has started the opening of the valves should be adjusted as required.



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vi) Leakage of hydrocarbons from equipment and pipe joints are undesirable and must be reported to the immediate supervisor without delay.

vii) The set pressure of relief valves should never be altered without prior permission of the authorized persons in writing.

Wherever block valves are provided for isolation of relief valves, these valves should be kept opened all the times as per stipulations in P&ID's.

viii) The closure of block or control valves which isolate exchangers or piping from relief protection devices can subject the equipment to possible overpressure due to thermal expansion of the blocked fluid. The expansion may be caused by heat from the atmosphere or adjacent hot equipment. In order to prevent such over pressurizing some suitable drains/vents or other valves connected with some closed system in the network should be left opened, to take care of such expansion.

ix) Poor house-keeping in the operating area is a potential cause of accidents. Spillage of oil etc. is very dangerous. Maintain a high standard of house-keeping in the work area.

x) Releasing equipment for maintenance without properly preparing it e.g. blinding / isolating & making it hydrocarbon free and electrical isolation etc. should be avoided as this can also result in major accidents.

30.4 CAUSTIC SODA:

Permissible Exposure Limits in Air:

The recommended limit on airborne work place sodium hydroxide concentration is 2.0 mg sodium hydroxide / m³. This is to protect against the irritation of the respiratory tract.

Harmful Effects and Symptoms:

This compound is extremely alkaline in nature and very corrosive to body tissues. Dermatitis may result from repeated exposure to dilute solutions in the form of liquids, dusts or mists. Local contact of sodium hydroxide with eyes, skin etc. can result in extensive damage of tissues, with resultant blindness and burns.

First Aid:

If this chemical gets into eyes, irrigate immediately. Washing should start within 10 seconds and continue for at least 15 minutes to prevent permanent injury. If this chemical contacts the skin, flush with water immediately to prevent slow healing chemical burns. If a person breathes in large amount of this chemical, move the exposed person to fresh air at once and



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perform artificial respiration. When this chemical has been swallowed, get medical attention. Give large quantities of water and do not induce vomiting.

Personal Protective Methods:

Wear appropriate clothing to prevent any possibility of skin contact. Wear eye protection to prevent any possibility of eye contact. Wash immediately when skin is wet or contaminated. Work clothing should be changed daily if it is possible that clothing is contaminated. Provide emergency showers and eyewash.

Recommended Respirator Selection:

100 mg/m ³	HiEPF/SAF/SCBAF
200 mg/m ³	PAP HiEF/SAF/PD.PP,CF
Escape	DMXSF/SCBAF

Disposal Method Suggested:

Discharge into tank containing water. Neutralize then flush to sewer with water.

30.5 HYDROGEN SULPHIDE

Hydrogen sulfide is a deadly poison, having a very quick action. Its hazard is compounded by the fact that the sense of smell is paralyzed once the concentration of H₂S becomes dangerous. Therefore, under no circumstances should appropriate safety measures be omitted whenever H₂S is present. When sampling or purging operations are being carried out on streams carrying H₂S, appropriate precautions must be taken.

Prevention of hydrogen sulfide poisoning:

The best method of preventing H₂S poisoning is to stay out of the plant areas known or suspected to contain it. Sense of smell is not an infallible guide to its presence; though the compound has the distinct and unpleasant odor of rotten eggs, it will in higher concentrations paralyze the olfactory nerves so the victim cannot smell it.

A good method of prevention is to check systematically the atmosphere of the suspected areas for the presence of H₂S by testing with portable H₂S detectors, moist lead acetate paper, or by Tutweiler H₂S determination.



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Gas masks suitable for use with hydrogen sulfide must be worn during work when exposure to it is likely to occur; masks must be frequently checked to ensure they are functional.

When working under conditions of likely exposure, the presence of a second operator is compulsory.

Acute hydrogen sulfide poisoning:

Breathing air or gas containing as little as 600-1000 ppm of H₂S for one minute can cause acute poisoning. One full breath of highly concentrated hydrogen sulfide gas will cause unconsciousness, and death may occur if the victim falls and remains in the presence of the gas.

Operations using processing gases containing H₂S are safe, provided precautions are taken. No work should be undertaken where there is danger of breathing H₂S, and no one should enter or remain in an area without wearing a suitable mask.

The symptoms of acute hydrogen sulfide poisoning are muscular spasms, irregular breathing, feeble pulse, odour of the breath, nausea. Loss of consciousness and suspended respiration quickly follow.

When acute poisoning has occurred, immediately remove the victim to fresh air. If breathing has not stopped, keep the victim in fresh air and keep him calm and warm. Call a physician and keep the patient quiet and under close observation for about 48 hours for possible edema of air passages on lungs.

If the victim is unconscious, breathing has stopped and there is no heartbeat (check on the main artery), then artificial respiration must be started at once. Quick action in starting respiration is essential. For correct procedure on artificial respiration, consult the first aid manual.

In all cases of acute H₂S poisoning, the use of oxygen containing 5% carbon dioxide is recommended as an inhalant, whether or not breathing has stopped. This mixture is a respiratory stimulant which must be administered by someone familiar with the use of respiratory apparatus.



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Sub-acute hydrogen sulfide poisoning:

Breathing air or gas containing 100-600 ppm of H₂S for an hour or more may cause sub-acute hydrogen sulfide poisoning. Symptoms of sub-acute poisoning are headache, inflammation of eyes and throat, dizziness, indigestion, excessive saliva and weariness.

Edema of the air passages on lungs may also occur. When sub-acute poisoning has occurred, keep the patient in the dark to reduce eye strain and have a physician treat inflamed eyes and throat. Check for possible edema.

H₂S leak handling procedure:

Actions to be taken up for the following cases :

- H₂S hooter alarm is activated
- H₂S smell is detected
- Portable detector indicates presence of H₂S
- DCS supervisor informs high H₂S indication in any of the detector

ACTIONS:

- The DCS supervisor will announce in PA 'H₂S leak' and 'vacate the unit' atleast six times in English and Telugu, once an alarm is activated, so that all persons present in the unit can go out to a safe place.
- Concerned Operator to wear BA set and carry spare BA set/cylinder. Spare BA set/cylinder will be left in the field in a safe location, where field supervisor is stationed.
- With portable detector, the Operator (with BA set 'on') moves to the location of leak and tries to ascertain the leak.
- All other operating personnel (technicians and supervisor) in the field will also wear BA sets and go to location along with all the spare BA sets & cylinders available in the field room, but will stand in a place where there is no H₂S is present (they also carry portable detectors).
- Field supervisor will station himself in a safe location where communication with DCS is available and H₂S is not present. All spare BA sets/cylinders will also be kept at the location. He shall co-ordinate all activities from this location.
- Stand by Operators will go around the unit to evacuate any person affected by H₂S leak and send out all other contract/maintenance persons of the unit towards upwind direction.



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7. The first Operator will immediately give preliminary condition of leak to Supervisor after coming to the safe location. The supervisor shall send additional operators with BA set 'ON' for follow-up actions to the location of leak, if required.

8. BA set shall not be removed for any reason (like communication), while the person is in the area of H₂S leak.

9. Field supervisor shall assess the situation and inform YSF/F&S for additional help (if required) through DCS supervisor.

10. If the search takes more than 30 min or low pressure alarm is given by BA set, the operator shall come to the field supervisor and gives feed back regarding the areas checked.

11. Once leak is ascertained, field supervisor will decide on the course of action.

- i. isolation if possible
- ii. attending the leak if minor
- iii. Shutdown of the plant

30.6 FURNACE EXPLOSIONS:

Statistics have shown that 90% of the furnace explosions occur during start-up or low load operation. More than 50% of the furnace explosions happen during a start-up some years after the initial start-up. These observations pinpoint the need for additional care, instructions and training in start-up operations. Most explosions happen in gas-fired furnaces.

30.6.1 Explosion theory

Three conditions must be met to get an explosion:

- A combustible gas (vapor) must be present
- The combustible gas (vapor) is mixed with oxygen (air) in the proportions required to produce an explosive mixture.
- A source of ignition must be available to start the explosion (unless the temperature of the mixture is above the auto-ignition temperature).



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30.6.2 Fundamental operating procedures:

Heater flame-out

The questions of what action should be taken after a flame-out of a furnace is subject to controversial opinions resulting from different interpretations of the explosion theory. Take care of following operation:

- Shut off fuel and feed supply (this is automatically done in the case of shut-down)
- Close main hand valves
- Close individuals burner valves

30.6.3 Heater light-off:

Heater light-off after flame-out

When the cause of the flame-out (shut-down) is such that the furnace can be re-ignited again before it has been cooled off entirely, the recommended procedures are as follows:

- Check firebox with an explosion meter for hydrocarbons and hydrogen to make sure that there are no explosive mixtures in any part of the firebox.
- When negative hydrocarbon tests are obtained, start re-ignition in the normal way.

Heater light-off during normal start-up

- Make sure individual burner fuel valves and registers are closed
- Open air registers
- Check furnace firebox for hydrocarbons
- Follow start-up instructions

Note: During the light-off of the burner, check very carefully through the end-wall peepholes and/or burner peep sights to ascertain that the pertaining burners are really ignited and are not impinging on the heater tubes.

30.6.4 Fire in the firebox of the heater:

In case of an unintended fire in the firebox, for instance caused by the tube rupture, the following procedure is recommended :



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- Shut off fuel and feed
- Close all openings (air registers) and keep the peepholes closed to prevent air introduction
- Stop forced draft fan
- Open snuffing steam to firebox

30.7 EMERGENCY HANDLING OF TOXIC / CORROSIVE CHEMICALS:

DO'S:

- In case of leak move in upwind direction away from the contaminated area.
- If respiratory equipment's are not available then use wet handkerchief to cover your nose.
- In case of skin contact, wash the affected area with large quantity of water. In case of eye contact, flush the eye with large quantity of water atleast for 15 minutes, keeping eyelids open.
- In case of inhalation, go to open fresh air area and take plenty of water.
- Keep the victim warm.
- In case a feeling suffocated, then give artificial respiration.
- Consult doctor and act as per his advice.

DONT'S:

- Become panicky.
- Go close to the leaky cylinders without right PPE and respiratory equipment.
- Apply little water on to the leaky cylinder.
- Spread rumors.
- Apply ointment to the injured due to the chemicals.
- Take any medicine without consulting Doctor
- Induce vomiting, keep warm and quiet.
- Attempt to neutralize acid in contact with skin.

DO's:

- Assess the wind direction and approach from upwind side.
- Keep all persons out of vapor cloud area. Evacuate the area in the path of vapor cloud.
- Isolate the source of vapor if possible.



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- Eliminate any source of ignition in the surrounding areas (Spark arising out of operation of electrical switches or electrical equipment, operation of i/c engines and vehicles, or any hot job including welding and grinding).
- Stop the traffic on the road if the vapor cloud is moving towards the road.
- If the vapor cloud is moving out of the refinery, stop the traffic on the main road and alert the nearby residents to eliminate all source of ignition. Activate mutual aid if required and contact district authority for activating Offsite Emergency Preparedness plan. Operate water spray system for dispersing vapor cloud and operate water monitors for dispersing the cloud.
- Continue the water operation till the entire vapor cloud is safely dispersed into the atmosphere.
- In case of the escaping gas/ vapor in fire, immediately apply large quantity of water as quickly as possible to all surface exposed to heat.

DONT'S:

- Don't do any hot jobs in the close vicinity.
- Don't operate electrical switches or electrical equipment in nearby vicinity.
- Don't approach from downwind direction.
- In case of fire, do not extinguish unless leak is stopped, if the source can not be isolated after extinguishing the gas will continue to flow and spread over a wide area and re-ignition can result in more hazardous situation.

30.8 INCASE ANY ACCIDENT HAPPENS:

The probable areas where a person can get trapped during an emergency are:

- Switch Gear room
- Within a lift and elevator
- High rise structures
- Confined space like tanks / vessels under maintenance
- Buildings
- Near to the source of release of toxic/ flammable gas leak In the process unit / pipe tracks

Search and rescue operation to be carried out with the help of following equipments:

- Self contained Breathing Apparatus sets with life line.
- Rescue tools such as hydraulic spreader/ cutter, pneumatic lifting bags, fire man hook.



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- High voltage resistant safety shoes and PVC / rubber gloves
- Extension ladder
- Torch lights
- Stretchers

During the search and rescue following precautions to be taken:

- Carry out the rescue operation in a pair not as an individual.
- Proceed cautiously anticipating tripping hazard.
- Keep yourself as low as possible.
- Follow the instructions of the Rescue controller.
- Cut-off the water, electricity and gas supplies of rescue operation area.
- Collect and check all equipments before returning from site.
- In case of usage of Breathing Apparatus sets follow the BA set usage instructions and in no case the user to remain in the danger area after the sounding of warning whistle.
- Keep constant communication with rescue coordinator.

Important considerations for fighting petroleum liquid fires and Gas fires:

- Approach should be made from the upwind side.
- The vehicles should be parked in such a way that in case of danger to the vehicles or fire fighters, the vehicle can be removed immediately without any delay.
- Role clarity of all the personnel available at the emergency site is very important. The personnel whose presence is not warranted at the fire / emergency site should be evacuated from the site.
- Brief details of the fuel involved in fire to be collected from the Incident Controller / operation controller/ concerned Plant In-charge.
- Cutting off the fuel supply is the best method for putting off any petroleum fire. Before making any attempt for fire fighting it is preferable to cut off the fuel supply.
- Selection of extinguishing media plays a very important role in putting off the fire. Depending on the equipment and fuel involved in fire the extinguishing media is to be selected.

Selection of extinguishing media:

The following points are to be kept in mind during selection and use of different fire extinguishing media:

- Quantity of different extinguishing media required to be used in different type of fires to be pre-estimated so that sufficient quantity of extinguishing media can be made available without any delay.



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- In case water attack is to be initiated, minimum two parallel lines to be put into action.
- In case Fire is to be attacked with DCP or CO₂ in an enclosed chamber, proper ventilation, evacuation of the area and means of exit to be ensured before attack.
- If foam is to be applied in low lying areas or on ground, the stream has to be applied to the nearest smooth elevated surface, where from the foam generated will flow down and spread over the burning liquid uniformly.
- Before applying DCP through DCP tender the downwind area is to be evacuated after consulting the Incident Controller. DCP should be applied from the upwind direction and preferably from hose reel hose.
- Back up of fire fighting equipment and manpower is very important for fire fighting operations. While fighting with foam ensure that sufficient quantity of foam is reaching the site to prevent the chances of re-ignition. Further there should not be any shortage of water. After extinguishing the fire ensure the use of water/ Foam application for some time for preventing re-ignition. The frontline fire fighters should be given adequate protection.
- The mutual aid scheme help can be mobilized in case of requirement.

30.9 FIRE IN PROCESS UNITS:

Fire in process unit can be due to leak from flanges, mechanical seal leaks and leak from pipelines etc. It may be at ground level or at elevation. It can be a minor fire due to leak or a major fire due to rupture and formation of pool of oil. Before attacking the fire details of the source of fuel, isolation of the source and adjoining equipment to be essentially obtained.

- If the fire is minor, it can be attacked with DCP extinguisher or fine spray of water using spray nozzles. However DCP attack should be back-up with water spray to prevent re-ignition.
- If the fire is major, it is to be attacked with the help of DCP from DCP tender and with back up of foam for ground fires and back up of water for ground fire.
- Simultaneously isolation of the source of fuel to be carried out followed by isolation of adjacent equipment, stoppage of rotary equipment, (from substation), shutdown of fired equipment, shutdown of entire unit, isolation of power to unit etc. as per requirement.
- Water spray if provided to be operated and cooling of nearby vessels and vessels exposed to the fire to be carried out.

Water spray protection may be provided to the operation staff to approach and close the valve for isolation of fuel if isolation valve is on fire. The fire at a height can be fought with water spray monitor and DCP tender.



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SAFE WORK PRACTICES

Set of best industrial practices, standards and statutory requirements grouped together and compiled to act as a guide to the employees in for executing the assigned job in a safe way.

They act as guidelines helping the executors for the safe execution of the jobs in the refinery.

Safe work practices list out the essential requirements that are to be ensured for carrying out jobs safely in the refinery.

So far total of 9 safe work practices were developed, and released for implementation.

SWP – 01 : Specification of Personal Protective Equipment.

SWP – 02 : Minimum Personal Protective Equipment for various activities.

SWP – 03 : Examination of Eyesight of certain personnel.

SWP – 04 : Requirement of PPE availability in plants and other locations.

SWP – 05 : Control of loose materials, unfixed tools and equipment in plant areas.

SWP – 06 : Contractor job equipment safety assurance.

SWP – 07 : Operations safety huddle.

SWP – 08 : Control & monitoring of slippery areas in plants.

SWP – 10: Lock out and tag out procedure.

For updated SWP details refer safe work practices manual or in website link http://teaming.hpcl.co.in/WebApplications/VR/IS_VR/fs/Safe%20Work%20Practices/Forms/AllItems.aspx

31.1 WORK PERMIT PROCEDURES:

Work permit system in the refinery is for carrying out various types of works after ensuring the safety of people, equipment and material when performing any work within the refinery



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premises. The issuer and the receiver of the permit have their responsibilities and roles before issuing the permit. The details have been discussed in this chapter.

Based on the nature of hazards involved in the work to be carried out, the “Work Permits” are of following types:

Type No.	Type of Work Permit	Color of Permit
I	Cold Work Permit	Yellow
II	Hot Work Permit	Dark Pink
III	Electrical Isolation /Energisation permit	Dark Blue
IV	Excavation Permit	Light Green
V	Confined Space Entry Permit.	Brown
VI	Loading / Unloading of Hazardous Chemicals permit	Light Blue

Note: Only the employees undergone Training on “Revised Work Permit System” and with “Authorization Card for Work Permit” are eligible for permit issuing or Receiving based on their nature of job.

ASSET CUSTODIANS FOR ISSUE OF WORK PERMITS:

Sr. No.	ASSET TYPE	ASSET NAME	CUSTODIAN DEPARTMENT	PERMIT ISSUING AUTHORITIES
1	BUILDING	Block-A	Concerned Department	Nominated officer of Respective Dept for the respective premises.
		Block-B	Concerned Department	Nominated officer of Respective Dept for the respective premises.
		Block-C	HR	Nominated officer of the Dept.
		Dispensary	HR	Medical Officer
		Security Gates & Check Posts	HR	HR security officer
		Refinery Warehouse	Materials	Nominated warehouse officer.
		Project Warehouse	Project Materials	Nominated warehouse officer.
		F&S Building	Fire & Safety	F&S shift in charge



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		Sub-stations	Maintenance	Shift in-charge
		Rack Rooms, Instrumentation panel rooms etc.	Maintenance	Instrumentation shift in-charge or nominated Instrumentation officer.
2	PLANTS	All process units, P&U & OM&S facilities and CPP	Operations	Respective Plant shift in-charge
3	UTILITIES	Steam lines, water lines , Air lines, in offsite area, N2 lines	Operations	TPH shift in charge and counter signed by Manager of P&U and Merox respectively.
4	ROADS	Between plant areas	Operations	Respective plant supervisor and counter signed by other plant Manager
		Offsite areas around tank farms, along pipe tracks.	Operations	TPH shift Officer and counter signed by adjacent plant Manager
5	FABRICATION YARDS	North-west boundary & east of O/H foam tanks etc.	Respective Projects / Minor Project	Nominated officer.
6	FIRE GROUND, FIRE WATER LINES	Fire ground and fire water lines	Fire & Safety	F&S shift in charge
7	FLASH / VIDEO PHOTOGRAPHY	-	Operations	Area in charge on approval of GM-Operations (Designated Factory Manager)
9	STORAGE / SCRAP YARDS	-	Materials	Designated Materials Officer



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TYPES OF WORK PERMITS WITH DETAILS:

TYPE OF WORK PERMIT	DEFINITION	EXAMPLE SCOPE		APPLICABILITY
Cold Work Permit	Permit for an activity which does not produce sufficient heat or spark to ignite a flammable air-hydrocarbon mixture or a flammable substance.	Opening process machinery.	Blinding & Deblinding.	All areas under Refinery control.
		Tightening of flanges.	Hot bolting	
		Inspection.	Painting.	
		Scaffolding.	Grass Cutting.	
Hot Work Permit	Permit for an activity that can produce a spark or flame or other source of ignition having sufficient energy to cause ignition, where the potential for flammable vapors, gases, or dust exists.	Welding.	Grinding.	All areas under Refinery control.
		Gas Cutting.	Shot Blasting.	
		Burning.	Soldering.	
		Chipping.	Hammering (metal)	
		Open Fire.	Use of certain non-explosion proof equipment.	
		Use of Diesel Generator.	Stress relieving, preheating and induction heating.	
		Electric flash unit/ flash bulb for taking videography / photography.	Concrete chipping	
		Fire Water usage, Isolation of Fire Water network	Radiography.	
		Hot tapping.	Road Closure.	
		Stopping of steam leaks on line using pneumatic drill.	Use of ultrasonic instruments for thickness measurements.	



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Electrical Isolation & Energisation Permit	Permit to enable an equipment / facility electrically safe and that the electrical power is isolated to the extent necessary for the safe conduct of the authorized work.	Energizing & isolation of all electrical Equipment connections at sub-stations.		All areas under Refinery control.
Excavation Permit	Permit for an activity which involves breaking of any ground surface by person or machinery for any purpose, including penetrating the ground surface to drive steel or wooden pickets. Core drilling, drilling for fixings, vertical boring into decking, floor slabs etc. (Taking out layers of soil using any tool)	i. Road Cutting. ii. Tank Dyke wall breaking. iii. Digging for cable & pipeline laying		All areas under Refinery control.
Confined Space Entry permit	Permit for an activity within an enclosure with known or potential hazards and restricted means of entrance and exit which is not normally occupied by people and is usually not well ventilated.	Process Vessels (above ground & under ground)	Boilers.	All areas under Refinery control.
		Furnaces	Storage tanks.	
		Large Diameter pipes.	OWS manholes.	
		Effluent Switch pits.	Chemical storage room / shed.	
		Chemical / Process Pump Houses	Entering Excavation more than 1.2 meter deep	



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Loading /Unloading of hazardous chemicals Permit	Permit for an activity of loading / unloading of chemicals having potential of causing spillage / leak resulting into environmental / health / property damage.	Unloading of HCL, H ₂ SO ₄ , NaOH, NH ₃ , H ₂ O ₂ and such other chemicals from a tank or other containers	All areas under Refinery control
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Validity of Work Permit:

- Permit shall be issued for single shift. Where the work has to be continued, the same permit may be revalidated / extended shift wise for a period not exceeding Seven Calendar Days in the succeeding shifts by authorized person after satisfying the permit conditions.
- In instances like plant turn around, project jobs, Storage Tank T&I jobs work permit of extended duration may be issued with the approval of Division Head for a period not exceeding Thirty Calendar Days. In such cases Manager of Executing Department should receive the permit. An additional Extension of Work Permit sheet shall be used for this purpose.(attached as in Format PSM-FR-9.17)

DSI on Generation of Work requests for Plant Maintenance (DSI-0-6):

Work request is fundamentally an instrument for requesting repairs and in kind replacement of plant equipment through maintenance department.

Work request is also used for facilitating various plant shutdown and start-up operations like steaming, blinding, other isolation and sanitizing methods, inert methods like hose connection, drain & vent connection etc.

Types of jobs requiring work request: Job types for which Work Request is required to be placed are as below:

No	Job type
J-1	Requirement of Repair or in kind replacement of existing plant equipment and facilities
J-2	Requirements of isolation, blinding, de-blinding, de-pressuring, testing, inerting,



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	system cleaning, steam-water-air hose connections for start-up shut down operations.
J-3	Housekeeping, Cleaning, Dressing
J-4	Preventive, Predictive maintenance of existing plant equipment
J-5	Provision (part or full, modification) of external features like supports, access structures, protective guards or covers, instrument steam tracing, external fire fighting systems, Civil works like kerb walls, routing channels, paving, fire proofing, Insulation, painting, stencilling, new lighting, safety boards, external safety features, Field room facilities, personnel communication systems.

Approvals for different types of jobs are authorized as below:

Job Type	Approval by
J-1	Plant Shift In charge
J-2	Plant Shift In charge
J-3	Plant Shift In charge
J-4	Area Maintenance Manager
J-5	Plant Manager

In all the case Except J-4 , The work requests are generated by the respective plant shift in charge and they are routed to respective officials based on the job type and priority.

“Work Request” generation is done for the Jobs in following CIRCUMSTANCES :

NO.	DESCRIPTION
C-1	Observation by Operations Personnel in Operating Plant.
C-2	Receipt of Inspection Work List.
C-3	Receipt of On-stream Inspection Report, F&S Inspection Report.
C-4	Requirement of unplanned Plant shutdown jobs. (Note : Any shutdown of the Plant except the Turnaround is considered as Unplanned Shutdown)
C-5	Requirement of Plant Turnaround jobs.
C-6	Statutory Requirements, Audit Reports (ISO14001, ESA, SSA etc)

In each of the above circumstances, any type of Job can be there.



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All “Work Requests” are assigned a required “priority of attention” as below :

Priority No.	Definition
0	A job if not done has an imminent danger to the safety of the Plant and/or personnel
1	Job which is required to be done to handle a situation of Plant operational emergency which may cause loss of production or thuput in that plant or another plant.
2	A job is required to be done for a situation which though currently in control is in a gradually deteriorating mode and may become critical in a week.
3	Regular repair requirements which should be attended at the earliest possible anytime upto 3 weeks.
4	Job is essential but the equipment cannot be released and it is possible to carry out the job only in a part or total unit shutdown.
5	Job to be taken up in a planned shut down.

Job Type approval:

Priority No	Approval By
0	Division Head or above
1	Division Head or above
2	Plant Manager
3	Shift In charge
4	Plant Manager
5	Shift In charge

Quantitative limits for raising work requests:

Maximum work requests per shift					
	C-1	C-2	C-3	C-4	C-5
J-1	5	n.u.l	n.u.l	n.u.l	n.u.l
J-2	n.u.l	n.u.l	n.u.l	n.u.l	n.u.l
J-3	2	n.u.l	n.u.l	n.u.l	n.u.l
J-4	n.u.l	n.u.l	n.u.l	n.u.l	n.u.l
J-5	1	n.u.l	n.u.l	n.u.l	n.u.l

n.u.l : No upper limit



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- The quantitative limits for raising work requests is to avoid bunching of work requests, not intended that in every shift work request rising is necessary.
- In case of requirement crossing the quantitative limit, then work request should be routed through division head.
- At the time of placing work request an assessment of the existing area of the work environment to be carried out.
- Information on plant process conditions for non-routine work PSM-FR-9.7 is to be attached to every work request.
- It is ensured that for all the recommendations of the OSI, IWL, work request is compulsorily raised within 2 working days of its receipt.

GUIDELINE FOR EXPLANATORY NOTES TO WORK PERMIT FORMS

Sr. No	Checklist points	Explanatory notes
1	Exact Location (Area / Unit / Equip no)	Exact location of the area or the unit in which the work is to be carried out shall be written against it. For example: CDU-I, 2-F-01 Tank farm of 20-D-4, Tank 501 etc.
2	Description of work	Precise description of the work to be performed shall be written. Sketches should be attached wherever possible to avoid miscommunication.
3	Equipment / Area inspected	Equipment or area where work is to be conducted should be inspected to ensure that it is safe to carry out the work and assess other safety requirements / stipulations. In case of equipment / vessel box-up permit, it should be ensured that the work is complete, all personnel are out, no maintenance gear is left behind and debris removed.
4	Surrounding area checked / cleaned	Unsafe conditions for performance of work may arise from surrounding area. It should be cleaned up to remove flammable material such as oil, rags, grass etc.
5	Sewers, Manholes, Closed Blow Down (CBD) etc. and Hot Surfaces covered.	Flammable gases may be released from nearby sewers. Hot un-insulated surfaces (equipment / pipelines) may provide a source of ignition and therefore, these are to be properly covered to prevent fires.



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6	Hazard from other routine / non-routine operations considered and persons alerted:	Other activities (routine / non-routine) being carried out near-by, which can create conditions unsafe for performance of the permit work, should be taken into consideration and the concerned persons should be alerted accordingly.
7	Equipment electrically isolated and tagged	Before issuing a permit, it shall be ensured that electrical isolation has been done, switches are locked-out and cautionary tags duly signed with date and time are attached.
8	Running water hose / Portable extinguisher provided / Fire water system available	Running water hose and portable fire extinguisher are required to flush / dilute in case of release of any hazardous chemical or to quench sparks and to put out small fires immediately. In order to meet any contingency, it should be ensured that the fire water system including firewater pumps, storage, network etc. is checked and kept ready for immediate use.
9	Equipment blinded / disconnected / closed / isolated / wedge opened	Equipment, for which the work permit is being issued, should be isolated from the rest of the plant in order to ensure that there is no change in the work environment with respect to presence of toxic / flammable gases, liquids, hazardous chemicals etc. in the course of the work. Blinding is one of the most effective ways of isolation. Blinds should be installed as close to the equipment as possible. If lines cannot be blinded, these should be disconnected and the open ends should be made safe by installing pipe caps / plugs, blind flanges etc.
10	Equipment properly drained / depressurized	Equipment under pressure should be depressurized after isolation. This should be followed by draining / purging / water flushing etc. as the case may be. Equipment containing liquid hydrocarbons should be drained completely. There may be a possibility of overlooking of liquid collected in pockets or inaccessible areas such as level gauges, small nozzles / bleeders on vessels, laterals in pipe work etc. All low point drains should be in unplugged condition.
11	Equipment properly steamed / purged	Purging of equipment (tanks, vessels, pipelines etc.) is done to make them free of flammable hydrocarbon and toxic gases. Steam / Inert Gas is used for gas freeing of vessels and pipes in refineries and other locations. Other means of purging is by displacement with water and final traces of gas removed by properly earthed air eductor. All high point vents should be unplugged while purging. It should be done in a systematic manner to cover the entire equipment / plant and continued till the concentration of toxic / flammable gas is lowered to allowable level.



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12	Equipment water flushed	Water flushing is an effective means of cooling, cleaning and even gas freeing of equipment. It is also employed to remove traces of acids / chemicals. Equipment metallurgy must be considered before using sea / saline water. Sometimes, flushing with demineralized water would be necessary depending upon the metallurgy of the equipment.
13	Gas / Oxygen deficiency test done	Gas test includes measurement of: (i) Hydrocarbons, (ii) Oxygen Deficiency by Oxygen Meter, (iii) Toxic gases like Hydrogen Sulphide, Carbon Monoxide, Nickel Carbonyl, and Chlorine etc. by techniques like Indicator Tube method, Lead Acetate Paper etc. Measurement of lead in air is required for entering tanks (which stored leaded gasoline earlier) and personal protective equipment shall be worn before entering the tank
14	Shield against sparks provided	Hot works like welding, grinding etc generates sparks which can provide source of ignition to the surroundings. In order to protect operating area from the hazards of sparks generated, shields are to be provided to contain the sparks generated. The shield material should be non-flammable and should be kept wet with water.
15	Proper ventilation and lighting provided	Where natural ventilation is not available, fans / air eductors are provided. These are also required for Speedy dispersal of fumes generated by welding job. Only approved reduced voltage extension lights (not exceeding 24 volts) are to be allowed for work inside vessels from consideration of personal safety.
16	Proper means of exit provided	Proper means of exit is required in case of emergencies developed on account of the work or otherwise. Availability of an alternate route of escape should be considered.
17	Area cordoned off and caution boards provided	To prevent any unwarranted entry in the work area and also to caution other personnel taking actions which may endanger people working on the permit job, precautionary tags / boards are to be provided to display like "No Entry" sign on roads or "Caution - Men at Work Inside" on the manhole of a vessel, Danger - Radiography in progress" etc.
18	Portable equipment / Hose nozzles properly grounded	As a precaution against static electricity generation, portable equipment / hose nozzles e.g. nozzle of a shot blasting gun, are to be grounded. Use of hydrocarbon lines for earthing should be avoided. 30 Milli-amp circuit breaker shall be installed on electrical circuits / cables feeding power supply to portable equipment.
19	Standby person provided for entry to confined space	Whenever a person is entering in a confined space, minimum two designated persons shall be kept at the manhole or entry point. The designated person shall be in constant communication with the persons inside the confined space.



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20	Standby personnel provided for fire watch from Process / Maintenance / Contractor / Fire Department	Depending on the criticality of the job, work permit issuer shall decide the type of standby to be provided i.e. from which department, of which level, how many and also additional firefighting support facilities etc.
21	Iron Sulphide removed / kept wet	Pyrophoric substances may be present in operating area / equipment handling hydrocarbon. Iron Sulphide scale is the most common pyrophoric substance encountered. These should be either removed to safe locations or kept wet all the time to prevent their auto-ignition.
22	Clearance obtained for excavation / Road cutting / dyke cutting etc	For any excavation work which may affect underground sewers / telephone lines / cables / pipelines etc., clearance shall be obtained from all the concerned sections. Markers should be put around the area where excavation is to be done and the depth to be indicated in the work permit. Road cutting can hamper the movement of the fire vehicles; initial clearance should be obtained from Fire Department and final approval from the higher designated authorities. When the dyke is cut, any mishap in the tank farm can lead to a free flow of oil to outside the bund. A higher level authority should be designated for authorizing dyke cutting. Further, it should be ensured that dyke would be reconstructed in the shortest possible time.
23	Checked spark arrestor on mobile equipment	No vehicle / engine without spark arrestor shall be permitted in operating areas.
24	Checked for oil / gas trapped behind lining in equipment	Before undertaking hot jobs, a check should be done for oil / gas trapped behind lining in the equipment.

1. BASIC RULES OF WORK PERMIT SYSTEM:

- 1.1 It is the policy of the company to ensure the safety of people, equipment and materials when performing any work within the Refinery premises.
- 1.2 Jobs shall be performed only after obtaining the stipulated permit.



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- 1.3 Permit shall be issued / renewed and received only by personnel designated for the same and are to be signed by the issuing / renewing and receiving personnel.
- 1.4 All necessary checks and tests shall be conducted to ensure safety of the jobs prior to issue of permits. All precautionary measures required to be taken shall be clearly indicated on the permit and their compliance ensured before handing over the permit to receiver
- 1.5 Permits shall be worded so as to limit the work to the designated areas / equipment / facilities, agencies and time.
- 1.6 If any of the conditions vary during the work which may affect safety of human beings and equipment, the job is to be stopped, the permit to be withdrawn and given back to the issuer. Reason for withdrawing the permit to be mentioned on the permit
- 1.7 If a permit is withdrawn, the work should be restarted only after safe conditions are established back and clearance obtained from the issuing / renewing authority and Fire & Safety.
- 1.8 Work permits shall be made readily available at the work site for verification by concerned personnel. The permit shall be put in a plastic folder and same will be displayed at work site. Failure to present the permit on demand for verification shall result in stoppage of work.
- 1.9 Work permits are valid only for the time period specified in the permit. Normally the permits will be issued for the General Shift timings i.e. 08:00 Hrs to 16:30 Hrs.
- 1.10 If the job is to be continued beyond the time limit specified in the permit, it has to be renewed at the beginning of the next shift.
- 1.11 Permits up to maximum of 30 days can be issued for major shutdown / project jobs. However, this permit must be presented every day prior to starting of the work in each shift by the concerned Executing person to issuing authority for renewal.
- 1.12 While renewing a work permit that has already been issued by another Issuing authority of the concerned department, the renewing authority is no way relieved of the duties and responsibilities specified by the original issuing authority.
- 1.13 At the time of renewal of a permit or when the job is in progress, if the issuing or renewing authority as the case may be feel that some additional conditions or precautions are to be stipulated or any of the conditions or precautions already stipulated are not necessary based on the prevailing conditions, a new permit is to be issued incorporating the changes and the previous permit should be withdrawn.



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- 1.14 All work permits must be closed on completion of the job by the Supervisor of the executing department / section and the permit duly signed for closure is to be returned to the issuer. Work permits once closed cannot be renewed.
- 1.15 The work permit should be issued for one job one location only. Separate permits to be issued for performing separate work on same equipment.
- 1.16 No hot work shall be allowed to be included in the hot work permit already Issued for other jobs.
- 1.17 Whenever there is a job in pipe track which is in close vicinity (5 meters) of process unit then prior to issuing of the permit the clearance from the concerned unit in - charge should also be taken.
- 1.18 Each hot work permit shall have only a single job with scope written on it and the person who is taking the gas test must sign on the permit.
- 1.19 Always "Receiver" has to sign the permit last, after taking all the necessary approvals and understanding the necessary safety measures.
- 1.20 The job demarcation of each contractor shall be clearly defined on the permit beforehand, incase more than one contractor working in the same area.

2. SPECIFIC RULES FOR WORK PERMIT ISSUING / RENEWING PERSONNEL:

The issuing/renewing authority is primarily responsible for the safe execution of the job covered under the work permit and as such is to perform the following duties:

- 2.1 Analyze and determine the condition of the facility and the area involved and Stipulate necessary measures to be taken to ensure safe execution of the work.
- 2.2 Stipulate precautionary measures such as area preparation, blinding / isolation / tagging, provision of safety and fire protection equipment, standby personnel etc. in the permits and ensure the compliance.
- 2.3 Stipulate preparatory, preventive and precautionary measures to be carried out and ensure their compliance.
- 2.4 Coordinate with other departments / sections wherever precautions are necessary to be taken from their end also and counter signature to be obtained prior to issuing of the permit
- 2.5 Verify and certify installation and removal of blinds / isolation and their tagging.
- 2.6 Perform necessary gas and other tests and the person performing the gas test to sign on the permit.



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- 2.7 Ensure that the facilities and the area at and around the place where permit is issued / renewed are made safe prior to commencement and also during the progress of the job.
- 2.8 Check periodically the progress of the job permitted and stop the job if conditions become unsafe.
- 2.9 Ensure that while hot work is in progress, no other work conducive to hazards is permitted in the vicinity eg. repairs to Oil or Gas lines, bleeding or sampling of flammable materials etc.
- 2.10 Ensure that all personal protective equipment / clothing stipulated in the work permit are available at site before commencement of work and the same used by the personnel while performing the work.
- 2.11 If a hot work permit is issued or if Hot Work Permit continues beyond the normal office working hours (after 4.30 p.m.) then the permit is to be renewed as per permit renewal procedure.
- 2.12 If the job is being done through contractor and in case more than one contractor is working in the same area then the issuer shall clearly define beforehand the demarcation of each job per contractor to avoid over lapping of different jobs.
- 2.13 The issuer shall give clearance to the Job Engineer for carrying out the hot jobs and not directly to the contractor personnel.
- 2.14 In case of Hot Work Permit for cutting, welding, grinding etc. the issuer or his representative and the receiver must be present for the first cut / first weld / first grinding.
- 2.15 For unit housekeeping, mopping (regular contract) Operations Supervisor will give cold work permit to contractor Supervisor directly.
- 2.16 Ensure that "Receiver" returns back permit duly signed on completion of job and "Housekeeping" is maintained. Scrap is shifted to designated area. Returned closed permit are preserved for 30 days.

3. SPECIFIC RULES FOR WORK PERMIT RECEIVING / EXECUTING PERSONNEL:

Although a work permit has been issued for performing a certain job, no way it relieves the supervisor of the executing department or the person executing the work of their responsibility for safe execution and completion of the job. The receiving authority / executing personnel is / are responsible for the compliance of the precautions /



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conditions prescribed in the permit. Their duties and responsibilities include the following:

- 3.1 Carry out Job Hazard Analysis and ensure precautions as per that.
- 3.2 Secure necessary work permit from the concerned department before commencement of the job.
- 3.3 Provide for and comply with all necessary precautionary measures stipulated in the permit.
- 3.4 Ensure basic safety precautions like usage of Safety belts, provision of ELCB (of 30mA), Flash back arrestor to cutting torch, DCP extinguisher / charged fire hose are provided as per the need of the job. Ensure that DCP and fire hoses are in good operable condition and available at site prior to start of the permit
- 3.5 Take extra care when commencing the work, especially while breaking flange joints, cutting through lines, striking an arc for the first time etc. Permit receiver to be present at job site for first cut or first weld.
- 3.6 Ensure safe working practices throughout by continuous supervision (Either by executing personnel or contractor personnel)
- 3.7 Ensure the availability of all the Personal Protective Equipment / Firefighting equipment at site before commencement of work and also ensure that the same are used during the work.
- 3.8 Stop the job if conditions become unsafe and report to the issuing / renewing authority.
- 3.9 Ensure that the work permit is duly put in a plastic folder and displayed at the work site for verification.
- 3.10 Ensure that no blinds or other isolations specified for the work are removed or disturbed during the work.
- 3.11 Ensure usage of grit blasting hood while Grit Blasting with air supply line connected with vortex cooling system and job area is fully covered to contain dust particles.
- 3.12 Grit blasting hopper to be tested at 1.5 times of maximum working pressure. Contractor should maintain test certificate.
- 3.13 Ensure that 24 V (Ac) hand lamps are used in the confined space.
- 3.14 Ensure that grade area is barricaded with red & white colour tape to cover excavated area or in case of material handling at overhead.



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3.15 On completion of the job, measures and maintain normal conditions at the area of work.
Clear all scrap material and maintain good housekeeping.

3.16 On completion of job the receiver will sign for closure of the permit at space provided at the back of the permit and return the permit to the issuer.

3.17 Receiver to ensure that registration number is entered in the permit before commencing the job

3.18 Ensure that the Contractor has received a booklet on "Safety Regulations for Contractors".

31.1.1 Procedure for Cold Work Permit:

Issuing Authority:

Cold work permits can be issued/renewed by authorized personnel of the department to which the equipment or area belongs.

31.1.2 Procedure for Hot work Permit:

Responsibilities of Renewing/Issuing Authority:

- Hot work permits will be issued/ renewed by duly authorized supervisor of the owner department of the equipment or the area.
- Identify the hazards and note on the permit. Suggest additional safety measures to carry out the job.
- Permit is valid for one shift and may be extended for one subsequent shift if condition permits. Issuer to note the extension of permit on his copy and in TOB.
- Issuer and receiver of the permit should be company employees, will arrange to deposit Fire House copy of the permit at box provided at MOI control room by 10:00 hrs, 18:00 hrs and 02:00 hrs.
- After office hours the hot work to be carried out with a fire watch from Operations department/ executing department/ F&S, as decided by the permit issuing authority.
- If the permit is renewed, the issuer will inform fire house on telephone.



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31.1.3 Procedure for Electrical Work Permit:

Duties of Permit Issuer:

The operations department personnel/owner of the equipment will raise the perm in the prescribed format and fill the permit as required. On receipt of permit from electrical section duly filled and signed then only issuer can take further action.

Explanatory Notes for Hot Work Permit Form:

Excavation/ road cutting/ fire water usage/crane operation/ radiography/ vehicle entry into restricted areas:

- Clearance obtained for excavation from Electrical departments and telephone in charge: For any excavation which may affect underground electrical cables, electrical maintenance should give clearance. Markers would be put around the area where excavation can be done and depth should be indicated in the work permit. Clearance from telephone in charge to be taken to ensure that no telephone cables get damaged while excavation.
- Shoring Provided: If the depth of excavation is mote than 2 meters, shoring is to be provided to avoid the caving in of the earth.
- Equipment/ Area Inspected: To be checked to ensure it is safe to carry out work and access other safety requirements.
- NRV provided on fire inlet line: In case of fire water usage for filling water in the tank or flushing the pipelines, NRV is to be provided on fire water line to avoid the back up of tank content/ line content into fire water network.
- Overhead HT cables Isolated: In case of crane entry and movement, the issuer of the permit will inform the electrical department to isolate the overhead electrical cables if there is a chance of crane touching the cables.
- Flame arrestor on exhaust checked: Every vehicle must have a flame arrestor on the exhaust.
- Area Cordoned off: In order to prevent unauthorized entry of personnel and to avoid accidents due to excavated pits, work areas must be cordoned off by red and white tape.
- Danger Sign Displayed: To prevent unwanted entry into the work area and to caution the people, danger sign boards must be displayed.
- Hydrocarbon Gas testing Reading: Hydrocarbon presence to be tested by explosive meter to ensure that the area is free of hydrocarbon. The person carrying out the gas test has to write to value and put his signature at the designated place.



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- Clearance from F&S: In case of crane movement and road closure, the issuer of the permit will inform F&S department and take shift F&S officer's signature at the prescribed place in the permit.

Vessel Entry/ Vessel Box-up/ Hot work/ hot tapping:

- Equipment/ Area Inspected: To be checked to ensure it is safe to carry out work and access other safety requirements. In case of box-up permit, it should be ensured that the work is complete, all personnel are out, no maintenance gear is left behind and debris removed.
- Surrounding area checked/ cleaned: Areas should be clean and free of any oil, rags, grass, etc.
- Sewers, manholes, CBD etc and hot surfaces covered: The hydrocarbon vapors may be present in these areas, which may be ignited during hot jobs. Therefore such areas should be covered. Hot un-insulated lines/ surfaces may provide a source of ignition. Therefore these are to be covered properly to prevent fires.
- Consider hazard from routine/non-routine operations and concerned personnel alerted: Other activities being carried out nearby areas, which can create conditions unsafe for performance of the intended job. Therefore these activities should be taken into consideration and the concerned personnel should be alerted accordingly.
- Equipment electrically isolated and tagged: Before issuing the permit for starting mechanical/electrical job, it has to be ensured that the equipment is electrically isolated and switches are locked out, cautionary tags duly signed with date and time area attached.
- Running water hose, portable extinguishers provided and fire water system checked for readiness: They are used to flush/ dilute in case of release of any hazardous chemical or to quench sparks and to put out small fires.
- Equipment blinded/ isolated/ disconnected/ closed/ wedged open: Equipment on which work permit is being issued should be completely isolated from the rest of the plant with which it is connected during the normal operation, in order to ensure that there is no change in the work environment with respect to toxic/ flammable gases, liquids, hazardous chemicals, etc.
- Equipment properly drained and de-pressured: Equipment under pressure should be de-pressured after isolation. This will be followed by draining/purging, water flushing as the case may be.
- Equipment properly steam purged/ water flushed: It is done to make equipments free of flammable hydrocarbons and toxic gases.
- Gas/ oxygen deficiency test done and found ok: Area should be free of hydrocarbon and same has to be tested by explosive meter. In case of confined space entry, in addition to hydrocarbon, presence of oxygen and toxic gases to be checked. The results have to be as follows, only then conditions are considered to be safe for working.



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- Hydrocarbon = 0.
- Oxygen Content > 19.5%.
- Hydrogen sulfide < 2 ppm.
- CO < 10 ppm.

- Shield against spark provided: It is provided to protect the operating areas from the source of ignition generated during welding.
- Proper ventilation and lighting provided: Where natural ventilation is not available fan/ air educators are provided. Wherever sufficient light is not available, lighting has to be arranged.
- Proper means of exit provided: proper means of exit is required in case of emergencies developed on account work or otherwise. Availability of an alternate route should be considered.
- Precautionary tags/boards provided: To prevent any unwarranted entry and also to caution other personnel taking actions which may endanger people working on the permit job.
- Portable equipments/ hose nozzles properly grounded: As a precaution against static electricity generation, portable equipment/ hose nozzle is to be grounded.
- Check flame/spark arrestor on mobile equipment: No vehicle to be allowed without them.
- Welding machine checked for safe location: It should be in non-hazardous and ventilated areas. It should have proper supervision while welding job is in progress. It should be switched off immediately after completion of job.
- Checked for earthing/ return connection to the equipment being welded: All earthing connections should be given as per welding codes.
- Oxygen and acetylene cylinder kept outside the vessel/tank: While cutting/welding in confined spaces, oxygen and acetylene cylinders should be outside and their hoses should be in healthy condition.
- Standby person provided for vessel entry: Standby person has to be present at the man way or entry point holding the rope connected to safety belt of person working inside. In case of emergency inside or outside the vessel, the standby person can pull the person out.
- Standby person provided for fire watch from Process/ Maintenance/contractor/ F&S: Depending on the criticality of the job, the work permit issuer will decide the type of standby to be provided that is from which department and of what level, how many and also additional fire fighting support facilities.
- Iron sulphide removed and kept wet: Pyrophoric substances may be present in the operating area/ equipment handling hydrocarbon. Iron sulfide scale is the most common pyrophoric substance encountered. This should be either removed to safe location or kept wet all the time to prevent auto ignition.



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- Checked for oil/ gas trapped behind lining in the equipment: Many times oil/ gas trapped behind linings depicts itself in the form of swelling and can be confirmed by way of drilling holes.
- Area cordoned off: In order to prevent un-authorized entry of people and to avoid accidents during excavations work areas are to be cordoned off.
- Clearance obtained for dyke cutting: Applicable to offsite tank dyke cutting.
- Inspection Approval: While it is presumed that modification job will be undertaken always with the approval of designated authority, it is further to be noted that hot tapping should be undertaken after an approval from Inspection personnel. Continuous flow in the line is to be ensured. Gas and airlines are not permitted for hot tapping work.

Exempted Areas:

The following areas do not require a work permit to carry out a job.

- Project Building
- Machine Shop
- Fire House
- Garage
- Contractor Yard
- Dispensary
- Administration and Technical Service Building
- South and main gate

31.2 CONFINED SPACE ENTRY PROCEDURE:

Any place is termed as confined space if it meets the following criteria:

- a) It is large enough and configured such that an employee can bodily enter and perform the assigned task.
- b) And has limited or restricted means for entry or exit.
- c) And is not designed for continuous employee occupancy.
- d) And it may contain or produce dangerous contaminants.

The examples of confined space in refinery are vessels, tanks, furnaces, boilers, pits, manholes, sewers, heat exchanger shells (open from one end), excavation deeper than 1.2 meters, entry on floating roof tanks when roof is more than 3 meters below from the top of tank shell, AC ducting systems, etc.



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31.2.1 Hazards Involved:

- Oxygen Deficiency (less than 19.5% volume): This leads to asphyxiation.
- Presence of toxic, corrosive, hazardous materials: Maximum allowable are H_2S = 2ppm, HC =20% LEL, CO = 10 ppm, NH_3 =25ppm.
- Presence of flammable, combustible, explosive or pyrophoric materials: They lead to incidents like fire, etc.
- Restricted Access- Limited number of entry, exit points: This can lead to problem of evacuating the entrants immediately in case of an emergency.
- Restriction of freedom of movement inside confined space: This can cause unwanted emergencies due to space constraint problem.
- Falling/tripping hazard: Falling hazards can cause injuries, accidents, etc when any object falls from higher elevations. Uneven flooring/obstructions in the walk way etc. Can cause tripping hazard.
- Inadequate Illumination/visibility/communication: this can cause unwanted incidents and confusion in case of emergencies.
- High temperature and humidity: Surfaces of high temperature can cause injuries humidity/vapour can cause suffocation problems and breathing problems.
- Electrical, static or radioactive hazards: This can cause electric shocks to human life and may create sparks/fire, etc.
- Mechanical Hazards: Mechanical hazards can cause accidents, incidents, injuries to human life.

31.2.2 Preparatory work Before Issuing Confined Space entry Permit:

- All personnel involved with confined space entry should be aware of hazards and no person should enter the confined space unless entry permit has been obtained by the concerned supervisor.
- All the connected lines to the confined space have to blinded or disconnected by blind flanges. No entry can be given without positive isolation.
- Power driven internal equipments such as mixers shall be disconnected electrically by an authorized person.
- Mechanical ventilation equipment shall be properly grounded (earthed) to dissipate any static charges. Pneumatic air movers (educators) and exhaust fans are recommended for this purpose. Electrical powered fans if any shall be explosion proof types. Air intake of forced ventilation equipment shall be from an uncontaminated location.
- Utility or instrument air cannot be used as means of ventilation or as air supply to breathing apparatus. All entrants shall wear personal monitors as required (H_2S , CO , O_2).



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- Adequate illumination shall be provided using 24V or below explosion proof lamps. Use of any electrical equipment above 24V shall have 'earth leakage circuit breaker' and also F&S department approval.
- Wherever possible, the confined space shall be adequately ventilated to enable entry without requirement of respiratory protection equipment. At least two manways should be open in every chamber of vessel for proper ventilation.
- Care should be taken when ventilating vessels containing 'pyrophoric iron scales to avoid spontaneous ignition. In such situation entry with required safety precautions or handling pyrophoric iron in wet condition should be adopted.
- The confined area shall be made safe for entry by such methods as depressurizing, venting, draining, steaming, washing and ventilating. Radiation sources if any shall be removed.
- Shift in charge should be aware of possible behavioural effects of hazard exposure in confined space area before issuing the work permit at confined space area.

31.2.3 Gas Testing Procedure:

- Shift in charge has to ensure all the meters/instruments to be used for gas test are ready to use.
- Technicians have to ensure that the meter is in good condition and it has to be checked and logged in TOB at the starting of the shift. If any problem is found with the instruments, it has to be brought immediately to shift in charge's notice.
- The gas tests have to be carried out and the testing result values have to be recorded in the permit. F&S personnel can be called for entry into critical areas for witnessing gas test and providing guidance as required.
- First, the meter should be checked in fresh air (preferably in safe zone/field room). Batteries should not be changed or charged in hazardous area. Testing immediately after stopping the purge may give misleading results. Purge medium or air mover should be stopped temporarily for 15 minute or more depending on the size of space prior to measurement.
- Initially the test has to be done from outside the confined space, using a long probe. If the initial gas tests indicate a concentration above the permissible limits, further was freeing should be done until the gas concentration is within permissible limits.
- Test result should be representative of the entire confined space. Hence the need may arise to enter for gas testing at different locations inside large tanks or towers and complex vessels based upon number of man ways, toxic material handled, size of vessel, etc to get a representative result. Operation shift supervisor should decide at initial stage, the need for entering the confined space for gas testing.



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- The job executing supervisor should witness or satisfy himself, that actual gas test has been done before issuing and accepting the permit. Receiver can refuse the permit if proper gas test was not done. For critical entries, the receiver should accompany the gas tester up to the man way.
- At least 13% oxygen is required to obtain an accurate LEL reading from a combustible gas meter. Hence, these cannot give a proper reading in atmospheres such as vessel purged with steam/nitrogen. Hence LEL and O₂ shall be measured simultaneously.
- Gas meters should be calibrated once in a month and certified by HPCL instrumentation section. Record for each gas tester is maintained by instrumentation section. Next calibration due date should be marked on the tester.

31.2.4 Confined Space Entry Permit:

- The permit receiver should give “safety pep-talk” to entrants about the hazards and precautions prior to entry.
- The receiver has to provide means for easy exit and entry of personnel into and out of the confined space. Safety rope should be provided where required.
- Hot work permit has to be issued in case of confined space entry and should be completely filled. Non-relevant items should be struck off.
- The issuer has to make the receiver aware of the hazards involved in confined space entry and precautions to be taken.
- Where atmosphere within confined space is initially safe, but there is a reason to believe that it may become unsafe during the period for which entry is authorized (from emission of fumes from sludge or deposits contained in the space or welding fumes), continuous gas monitoring is required. The same has to be mentioned on the permit at the time of issuing.

31.2.5 Communication at Confined Area:

- Shift in charge should make effective communication visually or by voice with the entrants, effective means of communication should be available. Intrinsically safe hands-free communication sets should be preferred.
- In IDLH (immediately dangerous to life or health) atmosphere, a common communication link system should be provided and used by personnel who are entering, standby, and maintaining the life support system. IDLH values for toxic gases are H₂S =300 ppm and CO= 1500 ppm.
- Shift in charge should ensure that all the entrants should be evacuated if the communication is interrupted.



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- Shift in charge may enter a confined space to attempt a rescue with the help of F&S personnel and co-ordinate for rescue operations.
- Shift in charge should monitor all the activities inside a confined space to determine if it is safe for entrants to remain in the space and can order all the entrants to evacuate in case he feels the presence of any prohibited condition.
- F & S personnel have to be communicated as soon as the shift in charge determines that authorized entrants may need assistance to escape from confined space hazards.

31.2.6 Requirements for Entry with Respiratory Protection:

- Entrants should wear air supplied respirators in oxygen deficient atmospheres, when toxics are beyond TLV and where atmosphere within confined space is initially made safe, but there is a reason to believe that it may become unsafe during the period for which entry has been authorized.
- Under no circumstances chemical cartridge/canister type gas masks should be used for confined space entry. Particular respirators have to be used if required.
- Receivers have to ensure that air supplied respirators are in good condition, well maintained and inspected according to manufacturer's specifications.
- Breathing apparatus users should be medically certified and properly trained to use breathing apparatus.
- Entrants should wear harness attached to lifelines and be attended by an attendant who should have breathing apparatus and clothing ready for use outside the confined space.
- In IDLH atmospheres, entrants shall wear breathing apparatus or airline mask attached with standby escape set. Rescue arrangements should be made ready with the help of F&S.

31.2.7 Additional Requirements for Inert Entry:

Creating inert atmosphere is required where it is impossible to gas free below 20% LEL and/or presence of pyrophoric material. Inert atmosphere are IDLH due to oxygen deficiency.

- Oxygen concentration in the inert confined space should not exceed 5%. Hydrocarbon should be below 100% LEL at the manway. LEL reading inside is not required.
- Issuer has to ensure continuous oxygen monitoring and alerting the attendant to evacuate entrants if oxygen level is exceeded. Audio-visual alarm is recommended.
- The people not entering the confined area have to wear respiratory protection if the effluent from the confined space contaminates the air.
- Independent source of air for different people with low pressure alarm and escape cylinder attached to waist shall be provided.



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31.3 PROCEDURE FOR OPENING PROCESS EQUIPMENT AND PIPING:

In the refinery any maintenance job needs a preparatory work prior to conducting the job. When any jobs (cold or hot) has to be done in areas where flammable/hazardous mixtures are likely to be present, the isolation of equipment involved in the job, has to be done in a proper and scientific way for the safety of the equipment and personnel.

31.3.1 Purging/ Isolation of the Equipment:

Before any maintenance job is done in the plant, it is necessary to remove all flammable or toxic gases from the system. Inert gas, nitrogen or steam may be used for purging. Water is used for flushing after ensuring that the equipment can carry weight of water involved. The part of the plant or the system to be purged is to be isolated either by closing appropriate valves or by proper blinding. No valves should be considered leak proof and in any the cases of maintenance jobs. Isolation of equipment is to be done by blinding. A blind list should be prepared for isolation of equipment and check shall be done to comply with it.

If the line to be welded has contained flammable material it must be drained and section should be disconnected at both ends. The whole section or the isolated portion should be then gas freed and tested before welding commences. Where isolation and gas freeing are impracticable, the line after draining should be washed thoroughly with water and welding/hot job while charged with low-pressure steam or filled with water.

31.3.2 While opening any flange joint for installation of blind, the following procedure must be followed:

- Wear protective clothing and appropriate eye/face protection.
- Ensure adequate working platform is provided.
- Initially flange bolts to be slackened slowly by standing upwind direction and cross check for presence of flammable or toxic gas, bolts opening can be performed at a faster rate.
- Removing flange bolts leaving a minimum of two bolts, then loose last two bolts without completely removing the nuts. Spread the flanges to install blinds. Always open flange on the side away from the person so that any sudden release will be directed away from the personnel.
- The presence of concerned operation technician is a must for any emergency help, like further isolation etc.



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- Hot cutting for flange nut bolts will be permitted only when presence of flammable or toxic gases is ensured zero.

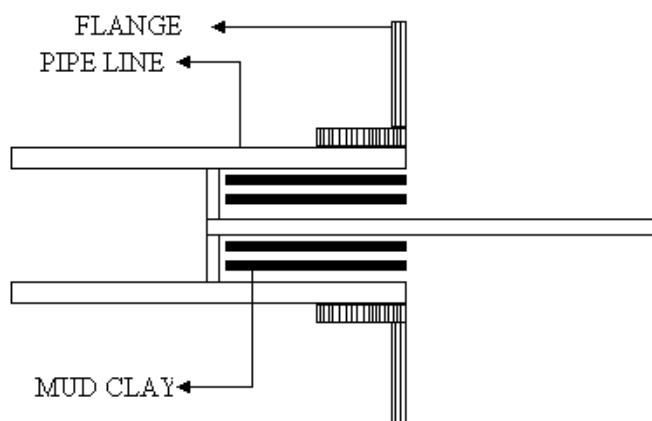
31.3.3 Mud Packing:

Most common method used for plugging the end of pipeline for hot work is “MUD PACKING”. The main objective of mud packing is to block the hydrocarbon vapour, which might be present in the pipeline.

Procedure for Mud Packing:

- Cold cut the line on which the flange is to be welded.
- Insert the plug into the pipe as shown in the sketch.
- The plug must have a hollow 1” pipe connected at the centre and pipe will be extended to a distance 5 feet away from the hot work area. The purpose of hollow pipe at the centre is to prevent any pressure build-up on the upstream. At frequent intervals the presence of toxic/flammable gases is to be checked and recorded so as to ensure that there is no abnormal situation at the site. The gap between the pipe and plug shall be nicely packed with clay mud as shown in the sketch.
- The gas coming out from the vent point to be continuously monitored.
- Once the welding is completed, the plug has to be taken out along with the mud by pulling.

Typical sketch of mud packing:





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31.3.4 Removal of Scaling:

- After draining and isolation of the vessel is completed, the material left, such as sludge has to be removed. A powerful jet of water applied through an open manhole can remove sludge. If this is not possible, use reagent to dissolve the sludge/scale. The liquid may be removed through the drain and the whole system may then be flushed with water.
- Where the scaling cannot be removed by liquid, it should be chipped away. Care in this case has to be taken that non-sparking tools are used for chipping purposes and the area is kept wet with water if there is even a possibility of generation of inflammable vapour.
- When process equipment is removed from a unit and must be sent elsewhere for repair, it is operations supervisor's responsibility to see that it is thoroughly cleaned and/or properly tagged if it has contained flammable, corrosive or toxic chemicals or iron sulfide.
- Equipment having contained 'sour' stock must be inspected by operation personnel for iron sulphide deposits. Arrangements must be made to remove them after wetting till such time such deposits should be kept wet to prevent spontaneous combustion.
- Personnel entering tank which are not declared 'free' of hydrocarbon vapour should wear breathing apparatus set. Personnel should enter with "Entry Permit" and keeping a person as standby.

31.3.5 Preparation for Hot work after cleaning the equipment:

- All cold work cleaning rules must be observed in preparing equipment for hot work.
- When normal cold work preparation has not satisfactorily removed rust, oil or lead sludge from internal surfaces of vessels they must be hand scrapped or if practicable, they must be sand blasted with the nozzle bonded to the equipment being sand blasted.
- Sewers openings in the area of hot work are to be sealed to ensure prevention of flammable atmosphere. The seal must be removed upon job completion. Sand should not be used for sewer sealing because vapors may escape through the porous material. Pit covers can be sealed by mud packing.
- Lines, vessels, etc which have contained flammable, toxic or otherwise injurious materials must be carefully checked and appropriate measures taken to ensure that liquids or gases are not entrapped between lines and vessel shell.

31.3.6 Preparation of Equipment for return to service:

- It should be ensured that all foreign materials/scales/rust etc are removed and equipment is properly cleaned prior to giving box-up permit.



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- It has to be ensured that water or test liquid has been drained properly.
- Carefully inspect all the equipment that has been repaired before returning it to service in order to be sure that all the guards and safety devices are placed back and they are in safe condition to operate.

31.4 PROCEDURES FOR CRITICAL EQUIPMENT HANDING OVER:

31.4.1 Pumps / Turbines seal repairs:

For Handing over to maintenance:

1. Isolate suction, discharge and warm up valves.
2. Allow the system to cool down and then flush it with cutter to CBD through casing after removing blinds on cutter and CBD of the concerned.
3. After ensuring the stock becomes light depressurize it to CBD. Open pump vent point valve for proper depressurizing.
4. Blinding is to be done for proper isolation on suction, discharge, warm-up, flushing oil, seal oil, seal steam and CBD lines and update blinds register.
5. Isolate power to the pump & tag.
6. Isolate steam to the turbine & depressurize and blind it on steam side.
7. Issue permit, decouple turbine /pump after ensuring the above.
8. Keep equipment under maintenance display board.

Note: disconnect instrument thermocouples as per rotary.

After the seal replacement on pump/ Turbine:

1. Check BCW, seal system, seal oil systems and remove any foreign materials in and around pump/turbine basements
2. Remove blinds on suction, discharge and warm up, CBD, seal systems, seal oil and flushing oil. Ensure proper gaskets are installed.
3. Close pump vent, open casing drain to CBD. Throttle pump discharge PG drain valve to remove air.
4. Open flushing oil slightly to fill casing and to remove air from downstream PG point.
5. Once oil is coming through downstream PG point, close it. Also close casing drain to CBD and check pump blind flanges with FLO pressure (9 Kg\Cm²) for leaks.
6. De-pressure the cutter to CBD and open pump warm up valve slightly for warm up condition (only for hot service pumps).
7. Allow rotary to do hot alignment.



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8. Allow maintenance to do hot bolting on blind flanges.
9. After hot alignment and coupling fixing, check for free movement of shaft.
10. Check no load and direction of motor rotation before coupling as per rotary/Electrical and isolate power.
11. After ensuring all the jobs are over power to motor is to be released. Steam blind to turbine is to be removed.
12. When all the above factors are ok place the pump /Turbine and see the performance.
13. Keep casing drain to CBD\OWS blind.
14. Ensure updating of gaskets and blind registers.
15. Remove equipment under maintenance display board.

31.4.2 Pump \ Turbine suction strainers cleaning:

For Handing over to maintenance:

1. Isolate suction, discharge and warm up valves.
2. Allow the system to cool down and then flush it with cutter to CBD through casing after removing blinds on cutter and CBD of the concerned.
3. After ensuring the stock becomes light depressurize it to CBD. Open pump vent point valve for proper depressurizing.
4. Blinding is to be done for proper isolation on suction, discharge, warm-up, flushing oil, seal oil, seal steam and CBD lines and update blinds register.
5. Isolate power to the pump & tag.
6. Isolate steam to the turbine & depressurize and blind it on steam side.
7. Issue permit, decouple turbine /pump after ensuring the above.
8. Keep equipment under maintenance display board.

After the seal replacement on pump/ Turbine:

1. Check BCW, seal system, seal oil systems and remove any foreign materials in and around pump/turbine basements
2. Close pump vent, open casing drain to CBD. Throttle pump discharge PG drain valve to remove air.
3. Open flushing oil slightly to fill casing and to remove air from downstream PG point.
4. Once oil is coming through downstream PG point, close it. Also close casing drain to CBD and check pump blind flanges with FLO pressure (9 Kg\cm²) for leaks.
5. De-pressure the cutter to CBD and open pump warm up valve slightly for warm up condition (only for hot service pumps).



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6. After ensuring all the jobs are over power to motor is to be released. Steam blind to turbine is to be removed.
7. When all the above factors are ok place the pump/Turbine and see the performance.
8. Keep casing drain to CBD\OWS blind.
9. Ensure updating of gaskets and blind registers.
10. Remove equipment under maintenance display board.

31.4.3 LPG/ Naphtha pumps seal repairs:

1. Isolate suction, discharge and warm up valves
2. Isolate self coolant lines.
3. Depressurize the content to flare through casing vent to flare.
4. Provide steam hose connection pump D\S PG point.
5. After ensuring gas is depressurized close casing vent to flare, Open casing drain to OWS and check if any gas is coming.
6. Observe passing of valves during depress ring to flare through _____ of lines.
7. Ensuring system is depressurized blind S\C,D\S ,and self coolant, casing CBD\OWS and flare systems, update blind register.
8. Isolate power to the pump and disconnect steam hose.
9. Issue permit to rotary to carry out the seal replacement job.
10. Keep equipment under maintenance display board.

Note: Inform instrument people to check and repair seal pot pressure and low level switches if any.

After the seal replacement on pump/ Turbine:

1. Provide steam hose at pump downstream PG point.
2. Check self coolant systems.
3. Check and remove all foreign materials in and around pump basement area.
4. Remove blinds on suction, downstream and self coolant systems.
5. Open steam and pressure test the pump for blinding flange leaks.
6. Close vents and bleeders.
7. During slight steam purge open suction valve and casing vent to flare for some time, and close steam completely.
8. Close casing vent to flare after some time.
9. Open casing drain to displace condensate if any.
10. Blind pump casing drain to OWS\CBD.



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11. Check for no load test before fixing coupling as per rotary \ electrical.
12. When all jobs are over release power to motor and check direction of pump.
13. Place the pump online slowly and check the performance.
14. Ensure updation of blind and gasket registers.
15. Remove equipment under maintenance display board.

31.4.4 Releasing and commissioning of coolers, condensers and exchangers for tube leaks:

Coolers/condensers:

1. Isolate inlet and outlet valves on shell and tube.
2. Depressurize the hydro carbon content to CBD after removing blind.
3. Cooling water is to be de-pressured to intermediate sewer.
4. Provide steam connection on shell side vent and start initial steaming.
5. After ensuring sweat steam and non passing of isolation valves stop steaming and blind both sides shell and tube inlet and outlet lines and update blind register.
6. Again carry out final steaming for 2 to 3 hrs then stop steam.
7. Again put blinds to CBD lines.
8. Allow maintenance to carry tube leak jobs.
9. Care should be taken while bundle is pulled out and when fixing and dropping of components.

Air fin coolers:

1. Isolate inlet and outlet valves on shell and tube. .
2. Depressurize the hydro carbon content to CBD after removing blind.
3. For attending the tube leak of one AFC, both the common inlet and outlet are to be isolated. Power to AFC fans is also to be isolated.
4. Provide steam connection on tube side and start initial steaming.
5. After ensuring sweat steam and non passing of isolation valves stop steaming and blind both sides shell and tube inlet and outlet lines and update blind register.
6. Carry out final steaming for 2 to 3 hrs then stop steam.
7. Put blinds to CBD lines.
8. Allow maintenance to carry tube leak jobs.
9. Care should be taken while bundle is pulled out and when fixing and dropping of components.
10. Other standby AFC can be placed on line as per procedure after properly isolating & blinding the AFC handed over to maintenance.



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For exchangers:

1. Isolate inlet, outlet, warm up valves, TSV's on shell and tube sides and open bypass.
2. Steam connections are to be given to vents and depressurize to CBD\OWS on both sides.
3. Ensuring sweat steam and non passing of isolation valves blind both shell and tube side inlet and outlet. Also carry out hot bolting of component bolts for dropping during steaming condition.
4. Release exchanger for dropping components after closing steam valve. Disconnect it only after final steaming for 2 to 3 hours.
5. Blind CBD lines and update blind register.

Commissioning of cooler/condenser:

1. After the maintenance jobs are over connect steam hoses on shell side vents.
2. Deblind the shell and tube sides.
3. Carry out steam test for blinding flanges at 5 to 6 Kg\cm².
4. When there is no leak, deblind FLO and CBD blinds.
5. Cutter flush to CBD.
6. Remove all steam hoses and cap off vents.
7. Ensuring water/cutter is displaced open inlet and outlet valves slightly keeping bypass in wide open condition. Slowly open inlet and outlet valves and slowly close bypass valve.
8. Establish cooling water flow through cooler/condenser and close intermediate sewer valve before establishing flow on shell side.
9. Observe pressure and temperature and slowly open valves on shell & tube.
10. Check for flange leaks if any.
11. Blind CBD\OWS and FLO lines.
12. Ensure blind and gasket registers are updated.

Commissioning of Air fin cooler:

1. After the maintenance jobs are over, connect steam hoses on tube side after isolating the common inlet and outlet valves and power to fans.
2. Deblind the tube side.
3. Carry out steam test for blinding flanges at 5 to 6 Kg\cm².
4. When there is no leak, deblind CBD blinds.
5. Flush to CBD.
6. Isolate all steam hoses and cap off vents.



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7. After filling with product on tube during slight steam purge by opening slightly inlet and outlet so as to displace water to CBD/OWS, remove all steam hoses.
8. After ensuring water is displaced, open inlet and outlet valves wide.
9. Restore power to the fans for starting.
10. Observe pressure and temperature and slowly open valves on tube.
11. Check for leaks for flanges if any.
12. Blind CBD\OWS lines.
13. Update gasket and blind register.

Commissioning of exchanger:

1. After the maintenance jobs are over connect steam hoses on shell and tube side vents.
2. Deblind the shell and tube sides.
3. Carry out steam test for blinding flanges at 5 to 6 Kg\cm².
4. When there is no leak deblind FLO and CBD blinds.
5. Cutter flush to CBD\OWS.
6. Remove all steam hoses and cap off vents.
7. After filling with cutter to displace water/condensate to CBD\OWS through top of the bundle on shell & tube, displace cutter. Open inlet and outlet valves slightly. Keep open wide seeing the unit conditions then close bypass valve.
8. Ensuring water is displaced open warm-up valve on crude from top to displace cutter. Slowly open inlet or outlet valves and slowly displace cutter to CBD on shell side. Ensuring cutter is displaced to CBD on crude side open the inlet and outlet valves, warm-up valves slightly to prevent vapour lock. This may lead to feed failure to furnace, hence slowly open them wide after checking unit conditions. Afterwards open inlet and outlet valves slightly. Ensuring the conditions keep valves wide open and close bypass valve.
9. Observe pressure and temperature on both sides.
10. Slowly keep the exchanger fully in service. Check for leaks on flanges and update blind register.

For steam generator:

1. Isolate inlet and outlet valves, TSV's on shell and tube sides and open bypass.
2. Isolate BFW. Depressurize 42V21 level at 42E19A/B drains and blow down valves. By opening 2" vent depressurize the system after closing PSV and warm-up valves on MP steam.
3. Steam connections are to be given to vents and depressurize to CBD\OWS on both sides.
4. Ensuring sweat steam and non passing of isolation valves blind both shell and tube side inlet and outlet.



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5. Release exchanger for dropping components after closing steam valve. Disconnect it only after final steaming for 2 to 3 hours.
6. Blind CBD lines and update blind register.

Commissioning of steam generators:

1. After the maintenance jobs are over connect steam hoses on shell and tube side vents.
2. Deblind the shell and tube sides.
3. Carry out steam test for blinding flanges at 5 to 6 Kg/Cm².
4. When there is no leak deblind FLO and CBD blinds.
5. Cutter flush to CBD.
6. Remove all steam hoses and cap off vents.
7. Build up levels in steam generators and 42V21 up to 30 %.
8. Keep steam generator vents open.
9. After filling with cutter displace cutter to CBD with SR through top of the bundle.
10. Ensuring cutter is displaced open inlet and outlet valves slightly keeping bypass in wide open condition. Slowly open inlet and outlet valves and slowly close bypass of one steam generator.
11. Keep 42V21 warm up open.
12. Observe steam generator pressure and temperature and slowly open steam outlet valves.
13. Slowly close SR side bypass and keep steam generators one by one online.
14. Close local vent and keep the steam pressure control on system finally. Blind CBD\OWS and FLO lines. Update gasket and blind register.

31.4.5 Process Equipment: Towers, Vessels etc.:

Before opening any equipment, it should be purged to render the internal atmosphere non-explosive and breathable. Operations to be carried out are:-

- Isolation with valves and blinds.
- Draining and depressurization.
- Replacement of vapors or gas by steam, water or inert gas.
- Take care about instrument tapping.
- Washing of towers and vessels with water.
- Ventilation of equipment.
- Opening of top manhole.
- Testing of inside atmosphere with explosive meter.
- Complete opening if inside atmosphere is satisfactory.



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- Analyze the atmosphere inside for O₂ content and any poisonous gas.

Note: Open a vent on the upper part of the vessel to allow gases to escape during filling and to allow air inside the vessel during draining. Ensure proper ventilation inside the vessel by opening all manholes. For hydrocarbon or other gases, pressurize the vessel with N₂ or gas and fill in the liquid and drain under pressure. This is to avoid hydrocarbon going to atmosphere.

Precautions before Handing Over Equipment:

Following items should be checked by a responsible operating supervisor before equipment is handed over for maintenance after it has been purged.

- Ascertain that equipment is isolated by proper valves and blinds.
- Ascertain that there is no pressure of hydrocarbons in the lines, vessels and equipment.
- Purge the system with N₂ first and later by air and check for O₂ content at vent and drain to ensure that the vessel is full of air.
- Check that steam injection lines and any inert line connections are disconnected or isolated from the equipment.
- Provide tags on the various blinds to avoid mistakes. Maintain a register for blinds.
- Check for pyrophoric iron and if existing. Keep this wet with water.
- Keep the surrounding area cleaned up.
- Get explosive meter test done in vessels, lines, equipment and surrounding areas.

If welding or hot work is to be done also:

- Keep fire fighting devices nearby, ready for usage.
- Close the neighboring surface drains with wet gunny bags,
- Keep water flowing in the neighboring area to cool down any spark bits etc.
- Keep steam lancers ready for use.

After the above operations have been made, a safety permit should be issued for carrying out the work. A responsible operating supervisor should be personally present at the place of hot work till the first torch is lighted. Hot work should be immediately suspended if instructed by the supervisor or on detecting any unsafe condition.

When people have to enter a vessel for inspection or other work, one person should stand outside near the manhole of the vessel of for any help needed by the persons working inside.



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The person entering the vessel should have tied on his waist a rope to enable pulling him out in case of urgency.

31.4.6 Procedure for dropping Hot well off gas burners:

- Inform to DCS supervisor before dropping burner
- During dropping and placing Hot well off gas fire, make sure stack damper is in wide position with DCS to prevent excessive pressure developing in the furnace.
- Open hot well vent.
- Remove Hot well off gas fires from the all the hot well burners for this isolate flame arrester inlet valve. Wait for some time till hot well gas in the line is completely burnt. After that remove gas fires and pilot fires.
- Start steaming hot well off gas header from the downstream of flame arrester. M.P. steam provision is available.
- Make sure that all the hot well burners are properly steamed out and then isolate flame arrester outlet valve.
- Blind the hot well off gas line, fuel gas line and pilot line (one which is required to drop).
- Close air registers.
- Use face shield and asbestos gloves during checking and dropping of Hot well off gas burner tips
- After dropping Hot well off gas burners with face shield and asbestos gloves check for tip condition and line plugging condition.
- Replace the tip if required and unplug the line by soaking in oil and then with steam
- Fix Hot well off gas burner with new gaskets.
- De-blind Fuel-gas/Hot well off gas lines & pilot gas lines.
- Keep open hot well off gas individual valves and flame arrester outlet valve and start steaming
- Ensure no leaks then stop steaming.
- Inform to DCS supervisor before placing any fire in heater.
- Ignite pilot first and put gas fire. Adjust air registers.
- Keep open flame arrester inlet valve.
- Keep one person at hot well vent valve, one person at “PA” for communication and one person at heater to observe hot well gas fire.
- Hot well vent to be closed slowly and hot well gas fire to be to be monitored.
- Advise maintenance to carry out house keeping.
- Finally, up-date burner spares register. Blind registers.



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31.4.7 Procedure for dropping Fuel gas burners:

- Inform to DCS supervisor before fuel gas raisers cleaning.
- During dropping and placing Fuel-gas make sure stack damper is in wide position to prevent excessive pressure developing in the furnace.
- Remove Fuel-gas fire& oil and pilot fires. Also isolate air registers.
- Purge oil gun with steam for some time to congealing fuel-oil flexible hose.
- Atomizing steam and purge steam valves are to be closed.
- Blind fuel gas line. One person to be stationed there till completion of blinding. While slackening the blinding flange make sure fuel gas line plug cock valve is not passing for which gas tester may be used while slackening blinding flange. If valve is not passing, fuel gas line to be blinded. If valve is passing suspend the job.
- Check burner in-side for oil accumulation through burner view ports
- Use face shield and asbestos gloves during checking and dropping of Fuel-gas burner tips
- Keep burner drain at assembly open to drain oil inside, if any. After removing all fires& closing air registers allow the burner to cool for 2to3hrs.
- After dropping Fuel-gas raisers with face shield and asbestos gloves soak it in the soaking pit after cooling
- Remove and replace damaged tips of the burners with new parts after checking condition if any, before assembling after cleaning.
- Fix Fuel-gas raisers.
- De-blind Fuel-gas line.
- Inform to DCS supervisor about completion of job and placing of fires.
- Ignite pilot.
- Adjust air registers.
- Place gas fire and then oil fire slowly.
- Check for any leaks with gas detector.
- Advise maintenance to carry to house keeping.
- Finally, up-date burner spares register, blind registers.

31.4.8 Procedure for burner's assembly dropping:

- Inform to DCS regarding the job and ensure stack damper is in wide open condition to prevent excessive pressure developing in the furnace
- Remove oil fire, gas fire and pilot fires and hot well off gas fire (if it is a hot well burner).
- Purge oil gun with steam for some time to prevent plugging with fuel-oil
- purge steam valves are to be closed



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- Close atomizing steam valves.
- Disconnect oil& steam flexible hoses.
- Drop oil gun, after cooling keep it sample trough for soaking and subsequently gun to be cleaned as per procedure.
- After removing all fires& closing air registers allow the burner to cool for 2 to 3hrs.
- Disconnect pilot burner electrical power.
- Check burner in-side for oil accumulation through burner view ports
- Use face shield and asbestos gloves during checking and dropping of burner assembly
- Keep burner drain at assembly open to drain oil inside, if any.
- Drop Fuel-gas/pilot gas burners/tips as per the procedure mentioned in fuel gas burner dropping; with face shield and asbestos gloves soak it in the soaking pit after cooling.
- Drop the burner assembly for repairs and fix a dummy plate at the bottom.
- Fix burner assembly after repairs and after removing plate at the bottom after taking clearances from INSPECTION/TSD.
- Fix Fuel-gas/Hot well off gas/pilot burners .connect oil burner flexible hoses and oil gun. Restore power to pilot burner.
- De-blind Fuel-gas/Hot well off gas lines and pilot gas lines.
- Lit up pilot burner and then keep Fuel-gas/Hot well off gas fires (if it is a hot well burner) as per procedure.
- Check for any leaks at pilot gas line, fuel gas line and hot well gas line flanges with gas detector.
- Place oil fire as per procedure and check for leaks if any after placing oil fire
Finally, up-date burner spares register, blind registers.

31.5 ELECTRICAL LOCKOUT/ TAGOUT PROCEDURES:

Multi-lock system is used to prevent injury by accidental energizing of any equipment while it is attended by different sections/agencies:

- The executing authority and the issuing authority will jointly decide the requirement.
- The issuing authority issues the work permit to competent electrical person to isolate the electrical equipment from sub-station.
- The competent electrical person and the executing authorities install their locks in the multi-lock pad as per colour coding.
- Electrical and other parties sign the isolation work permit.



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Colour coding of pad locks:

- a) Electrical maintenance- Brass Yellow
- b) Mechanical Maintenance- Black
- c) Others- Blue

- Each lock should be numbered and the key should be same number.
- After locking, the person who installs the lock is the responsible custodian of the key. The locks should be removed by individual craft after completion of their jobs.
- If the custodian of the key has to leave the site, responsibility has to be transferred to the next shift person.
- Electrical maintenance division shall be the last party to remove the lock only after receiving the necessary permit.

Use of “**DANGER-DO NOT OPERATE**” tag with isolation by local switch shall be limited to minor electrical jobs carried out by Electrical Maintenance division, such as re-lamping.

Electrical Isolation Procedure:

- Electrical isolation may be required before starting work on or near electrical equipment to avoid electric shock and other hazards. The extent of isolation required will depend on the nature of work.
- Requirement of isolation and extent of isolation shall be jointly decided by the issuing authority and executing authority by using electrical isolation permit.
- When multiple sections/ agencies are involved, the multi-lock system shall be followed. If only electrical maintenance is involved they may use single lock.
- Wherever the possibility of electric shock or injury is expected due to inadvertent starting, lock of power circuit at substation is mandatory. Locking the control circuit at local switch shall not be considered as adequate.

When an electrical circuit/equipment is fed from two different sources of power supply, both the source of power to be switched off to avoid back feeding.



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EQUIPMENT OPERATING PROCEDURES

32.1 CENTRIFUGAL PUMPS

32.1.1 INTRODUCTION

A centrifugal pump is a rotodynamic pump that uses a rotating impeller to increase the pressure of a fluid. Centrifugal pumps are commonly used to move liquids through a piping system. The fluid enters the pump impeller along or near to the rotating axis and is accelerated by the impeller, flowing radially outward into a diffuser or volute chamber (casing), from where it exits into the downstream piping system. Centrifugal pumps are used for large discharge through smaller heads.

General procedures for start-up/shutdown and trouble shooting of centrifugal pump are discussed in this section. For detail operating procedure refer the vendor operating procedure.

32.1.2 START UP

- i) Inspect and ensure all the mechanical jobs are completed before going for start-up of pump.
- ii) Carryout no load run of the motor without coupling the pump. Then, de-energize the electrical supply and couple the motor with pump.
- iii) Check bearing cooling water, Jacket cooling water for proper flow Open external seal flushing liquid to mech. seal in pumps having such facility.
- iv) Check oil level in the bearing housing; flushing may be necessary if oil is dirty or contains some foreign material.
- v) Rotate the shaft by hand to ensure that it is free and coupling is secured. Coupling guard should be in position and secured properly.
- vi) Testing of New pump with water: Ensure that pump motor is suitable for operation of increased load due to pumping of water. Also such operation may need approval of pump vendor/specialist rotary group engineer in certain cases. However, normally all pump's minimum continuous flow and motor ratings are checked with water as pumping medium.



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- vii) Open suction valve. Ensure that the casing is full of liquid. Bleed, if necessary, from the bleeder valve. Keep discharge valve closed.
- viii) Energize the motor. Start the pump and check the direction of rotation. Rectify the direction of rotation if not correct.
- ix) Check the discharge pressure.
- x) Open the discharge valve slowly. Keep watch on the current drawn by the motor, if ammeter is provided. In other cases check at motor control centre/MCC.
- xi) Check temperature of the bearings and if necessary adjust the cooling water flow (if provided).
- xii) Check the gland/seal and if necessary adjust gland tightness/flow of the coolant for the seal.
- xiii) In case of hot stand by pump: Ensure that casing attain pumping temperature by draining to suitable closed blow down system. This is to avoid vapour locking.

32.1.3 SHUT DOWN

- i) Close discharge valve fully if pump is single stage. If pump is multi-stage, having high tension electric motor, follow pump vendors instructions particularly regarding minimum continuous flow requirements.
- ii) Stop the pump.
- iii) If pump is going to remain as stand by and has provision for keeping the pump hot-cold proceed as follows:
 - Open the valve in the bypass line across the discharge valve and check valve.
 - The circulation rate should not be so high as to cause reverse rotation of idle pump or overloading of the running pump. Reverse rotation of pump may have adverse effect on thrust bearings as they are not designed for the same.
- iv) If pump is to be prepared for maintenance, proceed as follows:
 - Close suction and discharge valves.
 - Close valve on check valve by-pass line, if provided.
 - Close cooling water to bearing, if provided.



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- Close external flushing liquid to mechanical seals, if provided.
- Slowly open pump bleeder and drain liquid from pump. If the liquid is very hot or cold, allow sufficient time before draining is started. Ensure that there is no pressure in the pump. Also drain pump casing.
- Blind suction and discharge and check valve by-pass line.
- Cut off electrical supply to pump motor prior to handing over for maintenance.

32.1.4 TROUBLE SHOOTING

i) Pump not developing pressure

- Bleed/ vent pump casing.
- Check the lining up of the suction side.
- Check the suction strainer for plugging.
- Check the liquid level from where the pump is taking suction (physical verification).
- Check pump coupling and rotation
- Check the foot valve (in case of vertical lift pumps).
- Check the temperature of liquid. If it is higher than for what the pump has been designed, available NPSH may come down.
- Check for any air leakage in the pump suction line or pump casing. This may occur at various joints and packing boxes as the packing ages.

ii) Unusual Noise

- Check if coupling guard is touching coupling.
- Check for proper fixing of fan and fan cover of the motor.
- Check for pump cavitation.
- Get the pump checked by a Rotary EED group.

iii) Rise of bearing temperature

- Generally the bearing oil temperature up to 80°C or 50°C above ambient whichever is lower can be tolerated. Refer vendors' instruction manual for maximum tolerable pump bearing temperature.
- Arrange lubrication if bearing is running dry or oil level is low.



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- Adjust cooling water to the bearing housing if there is such provision.
- Stop the pump, if temperature is too high, call the Rotary EED group.

iv) Gland Overheating

- Adjust cooling water if facility exists.
- Slightly loosen the gland nut, if possible.
- Stop the pump and hand over to maintenance.
- Arrange external cooling if pump has to be run for some time.

v) Unusual Vibrations

- Check the foundation bolts.
- Check the fan cover for looseness.
- Stop the pump and hand over to maintenance for checking alignment.

vi) Leaky Gland

- Check the pump discharge pressure.
- Tighten the gland nut slowly, if possible.
- Prepare the pump for gland packing adjustment or replacement of mechanical seal as the case may be.

vii) Mechanical Seal Leak

- Stop and isolate the pump and hand over to maintenance.
- Refer to vendor's instruction for more details on trouble shooting of pumps.

32.2 POSITIVE DISPLACEMENT PUMPS

32.2.0 INTRODUCTION

A positive displacement pump causes a fluid to move by trapping a fixed amount of it then forcing (displacing) that trapped volume into the discharge pipe. A positive displacement pump has an expanding cavity on the suction side and a decreasing cavity on the discharge side. Liquid flows into the pump as the cavity on the suction side expands and the liquid flows out of the discharge as the cavity collapses. The volume is constant given each cycle of operation.

General procedure for start-up/shutdown and trouble shooting of the positive displacement pumps are discussed. Vendor's operating manual should be studied for



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further details, specific to pump under reference:

32.2.1 START UP

- i) Check for mechanical jobs are completed.
- ii) No load runs of the motor to be carried out before coupling with the pump. De-energize electrical supply. Couple the motor with the pump.
- iii) Flush and renew oil in pump gear box.
- iv) Check whether suction/discharge blinds are removed.
- v) Check whether suction strainer is installed and is clean.
- vi) Check for proper lining up including the pulsation dampener and pressure safety valve in the discharge. Open suction valve fully.
- vii) Check that the motor shaft is reasonably free and coupling secured. Coupling guard should be in position.
- viii) Energize motor. Open discharge valve. Start the motor and check direction of rotation. If wrong correct it. Never start the pump with discharge valve closed.
- ix) Adjust the pump stroke in case of reciprocating pumps and run the pump at desired settings.
- x) Watch discharge pressure and check the rate of pumping using the flow meter or by taking suction from the calibration pot. The valve on the recirculation line (provided in case of gear pump, screw pump etc.) shall be adjusted to obtain the required discharge pressure.
- xi) Care should be taken to avoid dry running of pump and back flow of liquid. Bleed if necessary to expel vapour/air.
- xii) Check for unusual noise, vibrations, rise of temperature of both motor and pump.

32.2.2 SHUT DOWN :

- i) Stop the pump.
- ii) Check the discharge pressure for gradual reduction.
- iii) Close the suction and discharge valves and flush the pump if required.
- iv) Drain the liquid if maintenance jobs are to be carried out on the pump.



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32.2.3 TROUBLE SHOOTING

i) Insufficient discharge pressure

- Check the line up in suction side.
- Checkup suction pressure.
- Check the functioning of the safety valve and pressure control valve on discharge to suction.
- Check the strainer on the suction side.
- Check for insufficient liquid level in the vessel from which pump is taking suction.
- Check pump's coupling and rotation.
- Get the pump checked by pump technician.

ii) High Discharge Pressure

- Check the line up on the discharge side.
- Check pressure control valve opening.

iii) Leaky Gland

- Check for normal pump discharge pressure.
- Tighten the gland nut slowly if possible.
- Handover the pump for replacing gland packing.

iv) Unusual Vibration

- Check the foundation bolts.
- Check motor fan cover for looseness.
- Stop the pump and hand over to maintenance.

32.3 CENTRIFUGAL COMPRESSORS

32.3.0 INTRODUCTION

Centrifugal compressors, sometimes referred to as radial compressors, are a sub-class of dynamic axisymmetric work-absorbing turbo machinery. In an idealized sense, the compressive dynamic turbo machine achieves a pressure rise by adding kinetic-



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energy/velocity to a continuous flow of fluid through the rotor or impeller. This kinetic energy is then converted to an increase in static pressure by slowing the flow through a diffuser. This may seem trivial, but it is actually quite complex to understand and precisely predict. This general class of turbo machinery includes pumps, fans, blowers, and compressors in axial, mixed-flow and radial/centrifugal configurations

General procedure for start-up, shut-down and troubleshooting are discussed in this section. For detail operating procedure refer the vendor operating procedure.

32.3.1 START UP

- i) Ensure all mechanical jobs have been completed on the compressor that is to be started including suction line passivation.
- ii) No load run of the motor to be completed before the compressor is coupled. After no load run, de-energize the electrical supply. Couple the motor and compressor.
- iii) Check oil level in lube oil tank and refill as required to bring the oil level to the high mark on the sight glass. Do not overfill.
- iv) Start the standby motor operated lube/seal oil pump keeping pressure regulator at a value given by the Vendor. Establish the pressure of lubricating oil in bearings and seal oil to seal.
- v) Bar the unit over once to be sure all moving parts are clear.
- vi) Open cooling water and ensure it is operative. Commission cooling water to all water coolers viz. oil cooler, gas coolers etc. as applicable.
- vii) Ensure that compressor casing has also been purged with inert gas, if applicable.
- viii) Line up the antisurge valves of the compressor.
- ix) Open suction valve/discharge valve and line up to the system. Remove any accumulated liquid from the casing by opening casing drain.
- x) Start the drive as per the vendors' recommendations and run the compressor. Adjust the minimum circulation valve, so that the drive does not consume excess power.
- xi) Allow the oil to warm up to 40o C. Watch compressor motor amperage. Listen for unusual noise during the warm up period. Auxiliary motor driven lube oil



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pump will be stopped when shaft driven pump develops normal oil pressure and kept ready for auto start.

- xii) When the unit is warmed up and determined to be operating satisfactorily, gradually start throttling the valve beyond surge requirements of the compressor.
- xiii) Load the compressor to the required capacity in the above mentioned manner.

NOTE : The compressor before trial run on the actual medium may be tried on air with suction and discharge open to atmosphere. While attempting to run the Compressor on air, Vendor's instructions and recommendations are to be followed. It is to be noted that scaling arrangement may be required to be changed to suit the machine running on air.

32.3.2 SHUT DOWN

- i) Stop the drive and close the antisurge valves.
- ii) Start the oil pump to lubricate the bearing till required.
- iii) Close the discharge valve.
- iv) Close the suction valve.
- v) Turn off cooling water to oil cooler as applicable.
- vi) If the compressor is to be given for maintenance, isolate, depressurise and purge with inert gas to make the compressor free of hydrocarbons.

32.3.3 NORMAL OPERATION

- i) Check oil level in lube oil tank and add oil as required to maintain the proper level as indicated on sight glass.
- ii) Log all temperatures, pressures, levels, flows and amperage.
- iii) Adjust cooling water flows to compensate for changes in inlet water temperature or ambient temperature.
- iv) Listen for any unusual noise while the machine is operating. These should be investigated immediately.
- v) Periodically drain from suction KOD/inter cooler etc.
- vi) Watch differential pressure across oil filter to check cleanliness; change over filter, if necessary and arrange cleaning of choked filter.



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- vii) Keep the exterior of the compressor and the compressor room floor clean.

32.3.4 TROUBLE SHOOTING

Follow vendor's recommendations. However, some general guide lines are given:

i) Surging

- Restricted flow due to plant operating at partial load or throttling at discharge.
- Blocked suction due to strainer choking or line choking if a permanent strainer is not provided.
- Liquid carry over from suction K.O. drum.

ii) Heavy vibrations in machine

- Misalignment
- Bent rotor
- Damaged rotor
- Imbalance
- Weak foundation
- Mechanical Loosening etc.
- Increase in gas temperature at suction along with drop in compression ratio
- Increased circulation of gas due to internal leakage as a result of 'O' ring of end cover/diaphragm damage.

iii) Oil carry over in the compressor unit in the product side

- Faulty operation of seal oil level control system.
- Damaged oil seals.
- H.P. seal oil drain collector levels not being maintained properly.
- At stand still condition, check the seal oil level controls, and check the drain oil quantities from HP seals near the compressor and compare with previous data.
- Check seal oil traps and drain collectors.



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- Check for any damage to the membranes in differential pressure indicators, and differential pressure control valves to avoid leakage of pressure oil into the reference gas chamber.
- Check that the balance gas pressure at the compressor is at least 200 mm WC higher than the respective reference gas pressure.
- Check oil seals.
- Avoid flooding of oil through reference gas lines from seal oil overhead tanks.

32.4 RECIPROCATING COMPRESSORS

32.4.0 INTRODUCTION

General procedures for start-up, shutdown and troubleshooting are discussed in this section. Vendor's operating manual should be studied for further details.

32.4.1 START UP

- i) Ensure all mechanical jobs have been completed on the compressor machine and auxiliary systems. Check if fine mesh strainer is installed as per vendor's recommendations.
- ii) No load run of the motor to be carried out before the compressor is coupled to the motor. De-energize electrical supply and couple the compressor with the motor.
- iii) Check oil level in frame sump and refill as required bringing the oil level to the high mark on the sight glass. Do not over fill.
- iv) Start the stand by motor operated frame oil pump keeping pressure regulator at a value given by vendor. Establish the required pressure of lubrication oil in bearings & crossheads.
- v) Bar the unit over once to be sure all moving parts are clear.
- vi) Open cooling water and ensure it is operative. Commission cooling water to frame oil cooler, cylinder jacket cooling and packing cooling as applicable. Remove the suction and discharge valve of the compressor if recommended by compressor vendor. Valve flanges should be covered to prevent entry of foreign material into casing. The machine has to be tried with the valves in removed condition.
- vii) Before running on no load, give a bump start. Time for which an unloaded machine continues to roll after driving power has been cut-off gives a fair indication of no load friction.



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viii) Start the motor and run the compressor with the suction and discharge valves in removed condition for about say 1 hour. After ensuring that the machine performance is okay, stop the compressor, Isolate power, and Reinstall the valves.

ix) Commission inert gas (IG) purge pressure controller to the packing and vent to safe height in atmosphere, as applicable (distance piece purging). It is assumed that compressor has been purged with IG while desiring of the system using IG. Compressor cylinder is to be purged with IG by temporary hose connection if for some reason compressors was under maintenance and cylinders are full of air. Steam purging of compressor should be avoided as it may lead to excessive rusting of parts afterwards.

N.B. The above is applicable when the gas handled is of explosive nature.

x) Unload the compressor manually.

xi) Open suction valve/discharge valve and line up to the system. Open bypass valves/purge valve as applicable.

xii) Before running on no load, give again a bump start. Time for which an unloaded machine continues to roll after driving power has been cut off gives a fair indication of no load friction.

xiii) Start the motor and run the compressor on no load for 10 minutes until the frame oil warms to about 35oC. Watch compressor motor amperage. Listen for unusual noise during the warm up period. Periodically check any heating of bearing and other moving parts. Operating satisfactorily gradually start loading the compressor by manually operating the suction unloader/clearance pocket.

xiv) Load the compressor to the required capacity in ascending sequence of steps.

xv) Observe the discharge pressure, temperature and bearing temperatures when the machine is loaded step by step.

32.4.2 SHUT DOWN

- Unload the compressor in descending sequence of steps and bring it to 0% capacity.
- Stop the motor.
- Close the discharge valve.
- Close IG purge to piston rod packing, if provided.
- Close suction valve.
- Turn off cooling water to cylinders, packing and frame oil cooler.



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- vi) If the compressor is to be given for maintenance, isolate, depressurise and purge with inert gas to make the compressor free of hydrocarbon gas, if handled.

32.4.3 NORMAL OPERATION

- Check oil level in frame sump and add oil as required to maintain the proper level as indicated on sight glass.
- Log all temperatures, pressures, levels and amperage.
- Adjust cooling water flows to compensate for changes in inlet water temperature or ambient temperature and for change in compressor loading.
- Listen for any unusual noise while the machine is operating. These should be investigated immediately and periodically.
- Periodically drain from volume bottles, distance piece etc.
- Watch differential pressure across oil filter to check cleanliness. Change over filter if necessary and arrange cleaning of choked filter.
- Keep the exterior of the compressor and the compressor room floor clean.

32.4.4 TROUBLE SHOOTING

Follow vendor's recommendation. Some guide lines are given below:

i) Low Lube Oil Pressure

- Low oil level-Plugged oil pump strainer
- Leaks in suction and discharge lines of the oil pump.
- Worn out bearings of the oil pumps.
- Defective oil pump.
- Dirt in oil filter check valve.
- Broken oil filter check valve spring
- PSV/ by pass of PSV passing.
- Defective pressure gauge.

ii) High Oil Pressure

- Plugged oil pressure lines.



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- b) Defective oil filters mechanism.
- c) Excessive spring tension in oil pressure adjusting mechanism.
- d) Defective pressure gauge.

iii) Overheated Cylinders

- a) Insufficient cooling water, scoured piston or cylinder.
- b) Broken valve and valve springs.
- c) Insufficient lubrication
- d) Packing too tight.
- e) Choked cooling water passage.

iv) High Intercooler Pressure

- a) Broken or leaking valves.
- b) Defective gauge.

v) Low Inter Cooler Pressure

- a) Broken or leaking valves.
- b) Leak in inter cooler.
- c) Piston rod packing leaking.
- d) Defective pressure gauge.

vi) Knocking Sound

- a) Scoured piston or cylinder.
- b) Defective lubrication.
- c) Foreign material in cylinder.
- d) Piston hitting cylinder head.
- e) Loose piston or piston pin.
- f) Loose main bearing.
- g) Scoured cross heads or crosshead guides.

vii) High Suction Temperature

- a) Broken or leaking suction valves.



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32.5 HEAT EXCHANGERS

32.5.0 INTRODUCTION

A heat exchanger is a piece of equipment built for efficient heat transfer from one medium to another. The media may be separated by a solid wall, so that they never mix, or they may be in direct contact.

Shell and Tube type heat exchangers can be broadly classified into following types:

- Water Coolers/condensers
- Steam heaters
- Chillers
- Exchangers

Start-up/shut down procedures for each unit shall vary slightly from case to case. However, general start-up/shut-down procedures are discussed in the following paragraphs.

32.5.1 START UP

After the heater exchanger has been pressure tested and all blinds removed, proceed as follows:

- Open cooling medium vent valve to displace non-condensable (air, fuel gas, inert gas etc.) from the system. Ensure the drain valves are capped. For high pressure system, drain valves should be flanged. This activity is not required if gas is the medium.
- Open cooling medium inlet valve. Close vent valve when liquid starts coming out through it, then open cold medium outlet valve and fully open the inlet valve also. Where cold medium is also hot, warming up of cold medium side gradually is also essential.
- Open hot medium side vent valve to displace non condensable (air, fuel inert gas etc.). Check that the drain is closed and capped. This activity is not required if gas is the medium.
- Crack open hot medium inlet valve. When liquid starts coming out from the vent valve, close it. Open hot medium inlet valve and then open the outlet valve fully. In case of steam heaters, initially the condensate shall be drained to sewer till pressure in the system builds up to a level where it can be lined up



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to the return condensate header.

- v) In case by passes are provided across shells and tube side, gradually close the bypass on the cold medium side and then the bypass across the hot medium side.
- vi) Check for normal inlet and outlet temperatures and pressures. Check that TSVs are not passing.
- vii) The opening of inlet and outlet valves should be done slowly ensuring that the exchangers are not subjected to thermal shock.
- viii) In case of coolers/condensers, adjust the water flow to maintain the required temperatures at the outlet. The return water temperature should not exceed 45°C.
- ix) For avoiding fouling, velocity of water should be at least 1 m/sec in a cooler/condenser.

32.5.2 SHUT DOWN

- i) Isolate the hot medium first. In case both hot and cold medium are from process streams, exchanger shall remain in service till the hot stream has cooled down enough.
- ii) Isolate the cold medium next.
- iii) Drain out the shell and tube sides to OWS/Sewer/Closed blow down system as applicable.
- iv) Depressurize the system to atmosphere/flare/blow down system as applicable.
- v) Purge/flush if required. This is particularly important in congealing services.
- vi) Blind inlet and outlet lines before handing over the equipment for maintenance.

AIR COOLERS

The air coolers/condensers comprise of a fin tube assembly running parallel between the inlet and outlet headers. These are of the forced draft type. The forced draft fans provided have auto variable pitch rotors in which the fan blades are adjustable in pitch during rotation. This allows variation in air flow as per the cooling requirements. These coolers are also provided with manually and/or automatically operating louvers for the control of the cooler outlet



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temperature.

Refer to vendors instructions for the detailed procedure of start-up, shut down and normal operation.

32.6 AGITATORS

32.6.0 INTRODUCTION

General procedures for start-up, shut down and troubleshooting are discussed. Vendor's operating manual should be studied for further details.

32.6.1 START UP

- i) Ensure that all mechanical & electrical jobs have been completed on the agitation assembly that is to be started. ii) Check lubrication of bearing housing, gear box etc. It is preferable to change to fresh lubrication material before starting.
- ii) Energize the motor. Start the motor & check the direction of rotation. Rectify the direction of rotation if necessary. Check the no load current.
- iii) Check cleanliness of the vessel.
- iv) Rotate the shaft by hand to ensure that it is free & coupling is secured. Coupling guard should be in position & secured properly.
- v) Before filling liquid check whether the vessel outlet valve is closed or not. If not, close the valve.
- vi) Fill liquid in the tank up to normal operating height. Generally water could be used for initial test. Commission level instrument if any.
- vii) Start the motor & check for any vibration/heating of gear box, any excessive vibration of the shaft etc. Measure load current drawn by the motor.
- viii) If any solid to be mixed, slowly open the solid charge. Hold & start mixing slowly. xi) Check for unusual noise, vibration, rise of temperature of both motor & gear.
- ix) If any heating arrangement is there slowly commission the system & ensure it is operative.

32.6.2 SHUT DOWN

- i) Stop the motor.



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- ii) If any heating arrangement is there, stop it. If any hot oil heating is there, slowly change over to cold oil from hot oil. But keep the circulation on.
- iii) If the liquid is sticky type, drain it as early as possible (in hot condition) & flush the system.
- iv) Motor to be de-energized

32.6.3 NORMAL OPERATION

- i) Check lubrication system of bearing housing & gear box.
- ii) Log all temperatures, level & amperage.
- iii) Listen for any unusual sounds while the m/c is operating. If any, it should be investigated immediately.

32.6.4 TROUBLE SHOOTING

Follow vendors' instructions. General guidelines are given below:

- i) Unusual vibration: Check for misalignment and improper facing of bracket.
- ii) Seal getting heated: Adjust cooling water flow, if provided.

32.7 EJECTORS

32.7.0 INTRODUCTION

An ejector is a pump-like device that uses the Venturi effect of a converging-diverging nozzle to convert the pressure energy of a motive fluid to velocity energy which creates a low pressure zone that draws in and entrains a suction fluid. After passing through the throat of the injector, the mixed fluid expands and the velocity is reduced which results in recompressing the mixed fluids by converting velocity energy back into pressure energy. The motive fluid may be a liquid, steam or any other gas. The entrained suction fluid may be a gas, a liquid, a slurry, or a dust-laden gas stream. [

General procedure for startup, Shut down & trouble shooting are discussed here in this section. Vendor's operating manual should be studied for specific details.

32.7.1 START UP



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- Ensure all mechanical jobs have been completed on the ejectors with all accessories.
- Check all the blinds have been removed or not.
- Check hot well is properly filled with water or not. If not, fill it up.
- Charge the steam header to the ejector. Drain condensate from low point drains. Ensure steam is dry before it is charged to the ejector.
- If there are pre-condensers, inter condensers & after condensers, open condensate drain valve provided on the pre condenser, inter condenser & after condenser.
- If there is an after condenser, be sure that the air vent on the hot well is open & free to discharge to the atmosphere.
- If there is no after condenser, be sure that the ejector discharge is open & free to discharge to the atmosphere or against a back pressure only equal to that for which it was designed.
- If there are pre-condensers, inter condensers & after condensers, start circulation of cooling water through' the tubes of inter, pre and after condensers. In case of barometric condensers, open water to the spray nozzles.
- Open all isolating valves on the first & subsequent stages of the ejectors.
- Open upstream & downstream isolating valves of the pressure controller for controlling vacuum.
- Before opening steam to the ejector, open the strainer bleeder and purge the strainer.
- Open steam valve slowly to the last stage (which discharges to the atmosphere or after condensers). Next open steam valve on preceding stage & so on until all stages are in operation. The vacuum on the vessel to be evacuated should then start to rise steadily. Observe maximum vacuum pulled. Adjust vacuum to operating level.
- When hot well level starts increasing, try to control level of the hot well. After level becomes steady put it on auto level control, if provided.
- In case of H/C service, once sufficient level is there on the H/C side, start H/C recovery pump, and control the level. When steady, put it on auto, if provided.

32.7.2 SHUT DOWN

- i) Close steam valve to first stage.



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- ii) Close steam valve to second & subsequent stages (if any) in their respective order.
- iii) Close isolating valves on all stages.
- iv) Close circulating water valve.
- v) Open water make up for hot well.

32.7.3 TROUBLE SHOOTING

If vacuum starts falling, the reason may be:

- i) Insufficient inlet steam pressure.
- ii) Inlet water temperature to the condensers higher than normal temperature.
- iii) Air leaks in the tail pipe of inter condensers.
- iv) Flooding of the inter condensers by excessive water flow (direct C.W. condensers).
- v) Starving of any inter-condenser by insufficient cooling water flow.
- vi) Plugging of water distribution system in the condenser.
- vii) Plugging of the tail pipe.
- viii) Plugging of the steam nozzle ejector and jets due to pipe scale.
- ix) Steam leak at nozzles throat. The defects mentioned above are to be ascertained & rectified for proper operation.

32.8 PRECOMMISSIONING

As the new projects nears mechanical completion of new equipment / units, operating personnel have to carry out preparatory works for ensuring safe and smooth start-up of the facility. These activities are termed as Pre-commissioning activities

- Some of the pre-commissioning works can be carried out simultaneously along with construction. But, care in carryout work is necessary so that it will not interfere with construction work. It is most important to plan schedule and record with checklists and test schedules all the preliminary operation and to co-ordinate the construction program.

Once mechanical contractor completes work, sections of the unit should be checked out by PMC, refinery and vendor personnel in those areas. Immediately punch lists that indicate the



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deviations from the design specifications, should be written following inspection of those areas, and distributed to the contractor. In this manner mistakes in construction can be found and corrected early.

Inspection of the plant can be basically divided into the following areas:

- Vessels
- Piping
- Heater
- Exchanger
- Pumps
- Compressors
- Instrumentation
- Catalyst/Chemicals Inventory

While detailed Format for checking is provided in Annexure-I, a discussion and lists of the major points that must be examined in the inspection of these areas follows:

32.9 VESSELS INSPECTION

The actual installations must be compared against the drawings to assure that the vessels will function as intended. The reactor internals must conform exactly to the design specifications if good distribution is to be attained and catalyst migration is to be avoided. Particular attention must be paid to the following details:

32.9.1 Specification Check

- i) Review design specifications with the vendor drawings to verify agreement on:
 - Pressure, temperature, and vacuum ratings.
 - Shell metallurgy, thickness, and corrosion allowance.
 - Nozzle size and orientation; flange rating, type and finish.
 - Type of lining, thickness and material.
 - Stress relieving and/or heat treatment.
 - Foundation design for full water weight.
- ii) Confirm that the vessel has been hydrostatically tested.
- iii) Verify that all code plate information on the vessel is correct.



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32.9.2 Internal Inspection

i) Reactors

- Inlet distributors, quench distributors: metallurgy, type, size, opening sizes, freedom to expand.
- Vapor/liquid collection and distribution trays: tightness, vertical positioning, liquid tightness of bubble caps and risers, metallurgy, dimensions, packing, supports, welding, levelness, cleanliness.
- Catalyst support grids: metallurgy, grid type and dimensions, screen type and size, supports, welding.
- Catalyst unloading nozzles: metallurgy, orientation, length.
- Thermowells: orientation, length, and metallurgy.
- It must be verified that sufficient quantities of bolts, washers and hold downs of the proper size and metallurgy are available to reassemble any disassembled portions of the reactor internals.

ii) Other Vessels

- Vessel trays: spacing; levelness, orientation and dimensions of weirs, downcomers, accumulators, draw off and trap trays, seal pans, distributors, baffles, nozzles, tray contact devices; metallurgy of trays, contact devices, clips, bolts, nuts and gaskets; freedom of movement of valve caps or other contact devices; number, size, and distribution of tray contact devices or perforated plate holes; proper fit of all internals and proper welding of support rings or other support devices; liquid tightness of draw off trays, seal pans and accumulators, all bolting and clips tightened.
- Mesh blankets and outlet screens: size, location, and levelness, suitability of fit (no bypassing allowed), metallurgy of blanket, support, tie wires, and grids.
- Vortex breakers: type, size, and orientation.
- Baffles: type, orientation, levelness.
- Instrument nozzles: location, orientation, cleanliness, thermowell length and metallurgy, baffle size and type.



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- Inlet distributors: type, size, orientation, levelness, freedom to expand.
- Non-fired reboilers: location, orientation, proper supports.
- Packing: type, size, support, installation.
- Internal ladders and other devices: location, size, orientation, properly secured.
- Lining and refractory:-
 - Hex-steel for concrete lining: clean and properly secured. Lumnite or other specified cement applied according to the specifications, with no holes or gaps in the applications.
 - Metal linings in good condition. Weld overlays have no gaps or holes in the application.
 - Lining is of the proper thickness and covers the required portion of the vessel.
 - Other refractory installed correctly with no gaps or holes in the application.

iii) General

The vessel should be clean (free from trash) and should not have excessive mill scale.

32.9.3 External Inspection

- i) Manways and nozzles: location, size, flange rating and finish, metallurgy, with proper gaskets, nuts and bolts.
- ii) Ladders and platforms: correctly positioned, secure and free to expand.
- iii) Insulation and steam tracing: provided as specified and has expansion joints as required.
- iv) Vessel grounded correctly.
- v) Correct vessel elevation.
- vi) Valves and instrumentation: easily accessible from grade or platform.
- vii) Piping:
 - Adequate supports and guides for all connecting lines.
 - Level and pressure instrument connections drain to a safe location.
 - Vents to atmosphere or blowdown provided as specified.
 - Relief valves have been bench tested.



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- Check valves exist on utility line connections where hydrocarbon backup could occur.
- Connections available for steaming/purging of the vessel.
- viii) Fireproofing of structure and supports is complete.
- ix) Instrumentation:
 - Level glass floats center positioned correctly with respect to vessel tangent line, and are readable from grade or platform.
 - Through-view level glasses have rear light for illumination.
 - Flange ratings, metallurgy, size, etc. are all correct.
 - Reactor skin thermocouples are located properly and installed so that they have good contact with the wall.

32.10 PIPING

The unit must be constructed in accordance with Piping and Instrumentation Diagrams (P&ID's), including all details, elevations, dimensions, arrangements, and other notes on the P&ID's.

Check the piping adequacy for carrying out normal operations of the unit as envisioned in the licensor design. Also, check whether piping is adequate for special procedures such as dry-out, special materials preparation, regeneration and/or alternative flow schemes incorporation in the unit design.

Check piping adequacy for receiving feed and sending products without contamination of these streams. Check all tankage interconnections to minimize the possibility of stream contamination outside of the battery limits.

Check that adequate means of measuring flows, pressures, and temperatures, and sampling of all process streams has been provided.

The following items must be checked to ensure conformity to the design specifications:

32.10.1 Flanges: rating, facing, and metallurgy; type (typically, 2" and smaller are socket weld, 2½" and larger are weld neck flanges).

32.10.2 Gaskets: type; metallurgy (materials of retainer, jackets, winding, filler, etc.); thickness, ring size, etc.



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- 32.10.3 Fittings, connections and couplings: rating and metallurgy.
- 32.10.4 Valves: rating and metallurgy (body, trim, seats, etc.); packing; seat inserts; bonnet gaskets; grease seals; socket-weld or flange type, rating and facing; installed in correct direction of flow; lubricant provisions; gear operators; extended bonnets; stops; ease of operation.
- 32.10.5 Bolting: stud or machine bolts; bolt and nut metallurgy; bolt size.
- 32.10.6 Pipe: metallurgy, thickness; seamed or seamless; lining.
- 32.10.7 Tubing: size and thickness; metallurgy; seamed or seamless.
- 32.10.8 Gauge glasses:**
- Through-view types should have rear-mounted lights.
 - Design pressure and temperature.
 - Special materials of construction.
 - Drains to safe location.
 - Visible from grade (or platform, if required).
- 32.10.9 Pressure relief valves:**
- Size and style.
 - Lever requirement.
 - Inlet/outlet flange material, facing and rating.
 - Set pressure - must be bench tested.
 - Metallurgy of nozzle, disc, spring, etc.
 - Type (pilot operated, balanced, etc.).
 - Inlet/outlet block valves car-sealed open; valve stems installed in horizontal or below.
- 32.10.10 General:**
- i) Utility systems within the battery limit should follow all relevant pipe class specifications in the same detail required for process lines.
 - ii) Package systems (modular units, etc.) shown on the P&ID should follow all relevant pipe class specifications in the same detail required for other process



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lines.

- iii) Expansion: review the physical installation to ensure that no expansion problems will occur when the unit gets hot and that:
 - Column overhead, reflux, feed and other lines are free to expand.
 - Rotating equipment will not be pulled out of alignment.
 - Sufficient expansion loops have been provided on long hot lines.
 - Pipe shoes are free to move in one direction, and are resting on supports of sufficient size that the shoe will not fall off the support.
- iv) High point vents and low point drains should be installed where necessary.
- v) Spectacle blinds should be provided where required.
- vi) Car-sealed valves should be locked in proper position.
- vii) Spring hangers should have locking pins removed (after hydrotesting) and necessary adjustments should be made for hot/cold position after startup.

32.11 HEATER

The heaters must be inspected to ensure that they can be operated in a safe and efficient manner and that the required heat duty needed for the process can be provided. After all, it is important that the possibility of a tube rupture or other heater mishap be minimized. In particular the following items must be checked:

32.11.1 Specification Check

All design specifications should be reviewed with vendor drawings to verify agreement on:

- i) Conformity to process requirements.
- ii) Heater type.
- iii) Tube arrangement, metallurgy, size, and thickness (note that tube metallurgy may be different for radiant, convection, and convection shield tubes).
- iv) Instrumentation connections.
- v) Tube supports and support metallurgy.
- vi) Refractory.
- vii) Access doors, observations ports, steam smothering connections, and explosion doors.



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viii) Stack arrangement.

32.11.2 Internal Inspection

i) Radiant Section

- Arrangement and symmetry of tube-coil with respect to heater wall, burner rings, and tube spacing.
- Vertical length of tube coil with respect to supports and guides.
- Fuel gas, fuel oil and pilot burner tips are clean and oriented properly. Burners are properly mounted with clearance for firing and removal. Castable refractory has not been used for burner blocks. Fuel oil tip sizing is adequate with respect to actual fuel oil viscosity.
- Tube skin thermocouples, if required, are located properly and installed so that they have good contact with the tube skin.
- Refractory is in good condition before and after refractory dry-out. No refractory is resting on tubes.
- Heater shell expansion joints are packed with designed insulation material and clean.
- Adequate space for tube expansion.
- Heater shell is sealed to prevent escape of hot gases and entrance of atmospheric moisture during shutdown.
- Smothering steam and instrumentation connections are not covered by refractory.
- Heater is clean and free from debris.
- Heater instrument connections are open - not filled or covered with refractory.

ii) Convection Section and Stack

- If extended surface elements are allowed, the bottom three or more rows of convection tubes must be bare.
- No refractory is on the tubes.
- Expansion provisions are adequate.
- Damper is free to move fully open and closed; its position indicator is correct both at the stack and at the damper control; damper is weighted to fail open; the damper, support pipe and bolting are all of the correct metallurgy.



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- e) Soot blowers, if specified, are provided with provision for inspecting the soot blowing operation.
- f) Other checks should be conducted as in the Radiant Section inspection.

32.11.3 External Inspection

- i) Location with respect to process equipment.
- ii) Platforms for access to all observation ports, instrumentation, sample connections, soot blowers, and damper connections.
- iii) Adequate number and arrangement of observation ports to permit visual inspection of the entire length of all wall, hip and shock/shield tubes, and the burner blocks.
- iv) Hand firing equipment located adjacent to an observation port from which that burner can be viewed.
- v) Explosion doors located such that heater gases will not flow towards process equipment and platforms.
- vi) Explosion doors located such that doors can open completely.
- vii) Symmetry of external piping and crossovers.
- viii) Instrumentation and sampling connections.
- ix) Damper position indicator visible from damper control; damper control functioning properly.
- x) Pocketed crossover connections have flanged drains.
- xi) Decoking connections as specified.
- xii) Sufficient smothering steam connections into heater firebox. Box valves on smothering steam are located remote from the heater, with drain valves and/or steam traps upstream of final block valve for condensate removal. Weep holes provided in smothering steam lines at low points.

32.11.4 Fuel Systems

- i) Fuel lines have battery limit block valves that are remote from the heater and easily accessible. Fuel oil piping and its steam tracing are arranged such that no dead legs or pockets are formed. Fuel lines to burners can be easily disconnected from burners for burner removal. All fuel lines have been leak tested.
- ii) Fuel oil lines at burner valves are correctly piped with steam crossovers. All



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steam lines have adequate traps and condensate drains.

- iii) Shutdown solenoids for fuel shutoff valves have been set properly.
- iv) Fuel oil circulating lines are provided.

32.11.5 Heater Instrumentation

- i) All draft gauge, pyrometer and analyzer connections are as specified.
- ii) All heater TRC's fail upscale during power failure or open circuit.

32.11.6 PROCEDURE FOR REFRACTORY DRYING

Purpose

Furnace dry-out procedure is utilized as a means of curing of furnace refractory before initial process operation and also whenever refractory is repaired / replaced. At the same time the dry-out period is also used as means of checking the operation of heater components such as burners and such control device as may be used during the dry-out period.

Duration

The drying time of the furnace refractory varies with the porosity of the refractory and the Atmosphere humidity. A 3-5 day dry-out period is generally required during which furnace temperature is gradually increased to the point at which refractory is completely dry. If linings are excessively wet, then only the pilot burners should be left alight for a day or two to dry off this excess moisture. During the dry-out cycle, moisture in heater close to the burners is evaporated within a very short time. However, the incipient moisture of the refractory needs gradual heating and evaporation. Generally over a three day cycle, moisture inside refractory is removed completely.

Tube Protection

During the dry out period it is necessary to pass steam through all the process tubes to prevent overheating of the tubes. Steam is introduced after warm-up to prevent undue condensation and water hammering. Steam will be vented out through the vents provided on preheat coil outlet, steam super heater outlet and the decoking stack. The tube metal temperature should be monitored and the same should not exceed the tube metal design temperature.



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Dry out

For the initial phase of dry-out, only fuel gas should be used to facilitate control and easy change over of burners. Carry out following steps for dry-out.

- i) Open the stack damper wide. Purge furnace firebox thoroughly with steam.
- ii) Open all burner air register and peep holes. Allow refractory to dry under natural draft for 24 hours.
- iii) Bypass cut off circuits on fuel gas lines and open fuel gas shutdown valves.
- iv) Remove blinds on fuel gas circuit.
- v) Light up the pilot burners as uniformly as possible as per ***Burner / Vendor's instruction.***
- vi) Raise the temperature with lighting sufficient main burners to bring arch temperature up to 120 oC at a rate of 25 oC/h.
- vii) Hold the temperature at 120 oC for 12 hours by rotating the burners for heat uniformity and then increase to 200 oC at a rate of 25 oC/h. Introduce steam through tubes at 200 oC after making sure that all condensate has been drained off. Check all burners, observe the condition inside the firebox and the inspect areas of expansion.
- viii) Hold the temperature at 200 oC for 12 hours. Increase the arch temperature to 400 oC again at 25 oC/h. Hold at this temperature for 24 hours and repeat checks as above. Increase the flow of steam if tube metal temperature is found exceeding design temperature.
- ix) The heat duty requirement during dry-out may not warrant firing of all burners. Sufficient number of burners, with good flame, may be fired and the operation of burners may be rotated every 2 hours to maintain uniformity.
- x) Before proceeding to the final stage of drying out, ensure that sample supply of steam is available to prevent over-heating of the tubes.
- xi) Increase the arch temperature at a rate of 25 oC/h to 500 oC and hold for 24 hours .Make a thorough check so the heater during this period and in particular, expansion should be noted.
- xii) During dry-out period it is advisable to try all instruments on automatic control and see that all alarms etc. are functioning properly.
- xiii) Check tube wall temperature frequently during process, to avoid possible overheating.
- xiv) At the end of 500 oC drying out period, reduce the arch temperature at a rate of 50 oC/h and accordingly start reducing the steam flow through the tubes.
- xv) When the arch temperature is around 200 oC, firing can be cut off together with steam flow through tubes. Allow the heater for natural cooling.
- xvi) When firebox is cool enough and the inside entry permits available, inspect refractory thoroughly for any damage and if required repair it.

32.11.7 OPERATING PROCEDURE



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The CDU/VDU furnace can be fired both on fuel oil and fuel gas. Each main burner is provided with a pilot gas burner. The furnace should be fired so that at no time flames impinge on the tube. The burners should be operated to provide fires, which are as uniform in length as can be obtained. It is normally desirable to fire all burners even at reduced operating capacities for a uniform heat flux distribution. Excess air for satisfactory combustion is 30% for fuel oil and 20% for fuel gas. The amount of excess air should be measured at the inlet to convection and at the base of the stack. The excess air in the furnace should be reduced till the flue gas analysis indicates traces of carbon monoxide. It must then be increased till no carbon monoxide exists. The pass outlet temperatures should be maintained equal. Inequality of flow among the passes should not go beyond 10% (maximum). Skin, box and intermediate pass temperature indicators have been provided. Refer to vendors instructions on operation of the burners.

32.11.8 START-UP IN NATURAL DRAFT MODE

a) Preliminary Checks:

- i) Ensure cleanliness inside the heater. Check and confirm that there are no flammable materials such as oil accumulation in the fire box.
- ii) Zero check all draft systems (gauges / meters). Stroke check the control valves of FO, FG and atomizing steam. Also check the action of FO and FG SDV's.
- iii) Ensure that the burner air registers are moving freely.
- iv) Check the free operation of the stack damper and all the DOD's.
- v) All safety valves including the superheating coil RV should be commissioned.
- vi) Commission the tempered water system and cooling water system.
- vii) Commission the atomizing steam. Disconnect steam flexible hoses and purge them till dry steam appears and then connect them back. This is to ensure that no condensate enters the fire box.

b) Commissioning the FO & FG system:

- i) Ensure that steam tracing to the FO lines is commissioned before circulation is established
- ii) After opening the SDV's, the return header has to be commissioned first. Establish fuel oil circulation in all the headers of the furnace. The rate of circulation should be such that the fuel oil returned should be equal to the fuel oil consumption in the furnace. Keep instruments on manual control and ensure that the burner valves and gaskets are not leaking.
- iii) Commission the fuel gas system after ensuring that the FG is free from any liquid by draining the liquid thoroughly at the FG KOD in gas plant.
- iv) After commissioning the FO and FG lines, check and rectify any leaks in the system.



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v) It is also to be ensured that the fire box is free from any flammable mixtures. For this, keep the stack damper and the DOD's of the furnace open and steam purge the furnace chamber at least for 30 minutes using **LP steam**. Atomising steam or FO purge steam can also be used for this purpose.

c) Before lighting the furnace,

- i) Keep the primary air register and secondary air register open.
- ii) Keep stack damper and all the DOD's wide open. Close all peepholes explosion doors.
- iii) Sufficient amount of feed flow is to be established in all the passes and each pass inlet pressure is to be maintained at least 14 kg/cm² g.
- iv) Cut the snuffing steam and light the pilot burner with the portable ignitor. If the pilot burner does not light up, shut off fuel gas and steam the furnace again and light the pilot. All the pilot burners are to be ignited in a similar way. While lighting the burner, it is advisable not to stand under the burner.

d) Lighting the furnace:

- i) Keep all burners valves closed. Bypass shut-down circuit.
- ii) For lighting oil burner, flush the fuel oil line by opening the cross over valve. Set the steam differential pressure controller so that steam pressure is about 1.8 – 2.0 kg/cm² higher than fuel oil pressure.
- iii) Crack open the atomising steam to the burner that is to be lighted. Insert a lighted torch and open oil valve for a small flame. Steam valve is to be adjusted for a clean bright flame.
- iv) When required number of burners have been lighted, open the FO and atomising steam valve full and control the flames by adjusting the fuel oil pressure.
- v) Maintain furnace draft about minus 1 to 2 mm water gauge at the inlet to convection zone.
- vi) Check excess air in the furnace and adjust the damper, primary secondary air registers to give the required excess air.
- vii) For lighting the fuel gas burners, operate the FG plug-cock valves and maintain 1.5 kg/cm² g. on fuel gas header pressure and follow the same lighting procedure as in the case of FO burner.
- viii) It is preferable to operate a burner either on fuel gas or fuel oil only at a time.
- ix) When the unit stabilizes, take all controls and commission the TRC control cascading either on FG or FO depending on the number of burners on FO or FG service. x) Commission the shutdown circuit of the furnace.

32.11.9 NATURAL DRAFT TO BALANCED DRAFT CHANGEOVER

The following steps have to be taken to change the furnace operation from natural draft to balanced draft mode



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- i) Ensure that the display of the annunciator lamps is normal. Check whether they are lit up during the lamp test. The trip switches of FD fan and ID fan are to be kept in manual bypass mode.
- ii) Start the FD fan with the outlet damper and the suction vanes in fully closed position.
- iii) Close the air bypass damper and open all the damper blinds in the air and flue gas lines.
- iv) Open the suction vanes to half open position. Open the outlet damper to allow about 20% of the maximum flow. Start closing the drop out doors (DOD's). There are 5 DOD's for 11-F-01 and 2 for 12-F-01.
- v) Close one DOD from the panel.
- vi) Open FD fan outlet damper to 40% and close the second DOD
- vii) Similarly, close the other DOD's one-after-the-other by opening the discharge damper in increments of 20%
- viii) While doing so, care should be taken so that the furnace pressure doesn't go to the positive side. If it does, the discharge damper should be closed till such extent where the furnace pressure becomes negative again.
- ix) After checking the firebox conditions for sufficient combustion air, the FD fan trip can be put in auto-interlock mode.
- x) Take the ID fan suction vanes in manual control and close them fully.
- xi) Start the ID fan (after obtaining clearance from CPP in case of 11-FM-02)
- xii) Increase the suction vanes opening slowly watching the furnace arch pressure.
- xiii) After the ID fan is fully stabilised, close the stack damper and put the ID fan suction vanes in auto control.
- xiv) The ID fan trip can be put in auto-interlock mode after ensuring normal operating conditions.

32.11.10 BALANCED DRAFT TO NATURAL DRAFT CHANGEOVER

- i) Place the ID fan trip in manual-bypass mode
- ii) Open the stack damper and check the lamp position. Then trip the ID fan
- iii) Allow the system to run on forced draft so that the APH system cools down.
- iv) Place the FD fan trip in manual-bypass mode.
- v) Open one DOD and decrease the FD fan discharge correspondingly so that the furnace pressure never to the positive side.
- vi) Open the other DOD's in the same procedure as mentioned above. The number of DOD's to be kept open is decided by checking the fire box conditions and the O2 analyser.
- vii) Trip the FD fan
- viii) Now the furnace is under natural draft.

32.11.11 SHUT DOWN THE HEATER



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Fuel to the heater should not be cut off abruptly unless an emergency shut down is called for. The product pass outlet temperatures should be slowly reduced. As burnerstend to become unstable when fired at low rate, reduce the number of burners as required. Follow the procedure outlined below for shutting off individual burners.

- i) First change the heater to natural draft operation from the balanced draft mode.
- ii) Close fuel oil valve and open steam purge valve on the fuel oil line to the burner.
- iii) Purge all the oil into the firebox and continue steaming for about 5 minutes.
- iv) Close atomising steam to the burner.
- v) Pull out oil gun in case the gun is not withdrawn; leave a little stem blowing through the tip to cool it. This may be done when a burner is to be taken out of service during normal operation.
- vi) For shutting down the furnace, cut the burners one by till all oil fires and gas fires are cut off. Bypass the shut down circuit of the furnace, so that fuel oil and fuel gas supply are not interrupted when the feed flow to the furnace is reduced.



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“Spill” means an event of coming out of a liquid of its container especially accidentally. In a refinery, this will mean oil coming out of any of the equipments whether it is storage, transport, pump, processing equipment, etc. Every time a spill occurs, it has a potential to lead to secondary events like fire, environmental damage, personnel injuries or injury/effects on public outside the refinery. The occurrence and extent of these events will depend on the size and type of oil spills. Best strategy is to prevent an oil spill. In order to achieve this, the design of the facilities needs to be in a manner that prevents oil spillage from entering any surface drain or a water body and the mechanical integrity of the equipment will have to be ensured all this time. This will also require proper management practices and day to day administration in a manner that the activities are under close control. However, residual risk always exists and in spite of all precautions and an emergency response plan is meant to tackle and manage the residual risk.

33.1 GENESIS OF AN OIL SPILL:

An oil spill can occur during any of the routine or non-routine operations associated with the operating facilities of the refinery. There are four broad reasons for the generation of an oil spill. It is possible that in a certain instances, inadequacy in one of them can get covered by the adequacy of the other. These reasons are:

33.1.1 Mechanical Integrity Failure:

The failure of structural/construction integrity of any civil or mechanical part of facility can be termed as mechanical integrity failure. Successful management of Mechanical integrity has many elements. The mechanical integrity of the facility can get violated due to inadequate/improper design, job execution inadequacies, gaps in preventive and turnaround inspection programs, operation outside safe operating envelope and many such gaps in management. A safe operating envelope again can get violated due to many reasons which may not be in control every time like instruments failure, etc. every time a mechanical integrity failure occurs in a hydrocarbon oil bearing system, the oil spill (small or large) will occur. The oil spill may be a surface oil spill or ingress of oil into soil if it is an underground system. The consequence of the oil spill will depend upon layout, surrounding etc of the area. Examples can be line rupture, gasket failures, equipment failures, etc.



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33.1.2 Operational Control Inadequacies:

These are related to the day to day administration and operation of an operational facility. Well laid out procedures, controls on activity, proper authorization, clarity in instructions, proper communication, training, etc. are some of the important elements in managing operational activities. Gaps in these areas can lead to abnormal situations and depending on the system involved and type of operation, oil spill can occur. Management practice should ensure proper execution of every workflow element associated with the operation of the facility. Examples can be delay in reporting of a flange leak, de-pressuring of a line in pipe track without ensuring complete flushing, overflow of a tank due to gaps in monitoring system-both management and hardware.

Operational co-ordination in the management of loading and unloading operations associated with ships and tankers has very high importance. Impact of lapses in operational activities associated with loading and unloading can be very high as there is a high possibility of the oil getting into the sea.

33.1.3 Collection System Design Philosophy Inadequacies:

A design philosophy of oil containment and secured handling, considering the type of operations occurring in the facility, is absolutely essential to make sure that the oil gets properly contained and is lead to the collection system without causing any oil spill beyond the boundary of that facility. If proper philosophy of segregation of oil handling area and non-oil handling area is not followed, oil spills can become a routine affair. Examples can be non-provision of OWS facilities at an area which involves routine operations of valve operation, blinding, etc.

33.1.4 Capsizing or Damage to Ships/Tankers:

These are one of the most serious type of oil spill scenarios since they may involve huge volumes and the spill area directly takes place over a turbulent water body. The impact area can be very vast and serious environmental damage can result. Many living species get affected and liabilities can be very heavy and long term. In order to protect the interests of the company, it is required to ensure that the chartered ships are having appropriate mechanisms to prevent oil spills. Suitable agreements for mutual inspections and adherence to prescribed standards are some of the tools which can be adopted.



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33.2 OIL SPILL MITIGATING FEATURES:

There are certain features incorporated in the refinery facilities which help wither in preventing an oil spill into a surface drain or a non oil handling area. These features can be preventive type or mitigating type. The mitigating type may mean that it provides some elapsed time before a response can actually be implemented. These are described below:

33.2.1 Design Philosophy of Collection System:

The oily water system collection system has certain features which are provided for the purpose of segregating an oil handling area from a non-oil handling area. The basic philosophy features of the system are described here. These are the key features which contribute to keeping oil contained in oil handling area or contain the oil to a restricted area and prevent it from going to surface drain:

- An area where a large number of routine operations are held and a number of oil handling equipments are located is a paved area. Paving is generally impervious and is done by RCC. This paved area is provided with OWS facility. Depending on the complexity and size, the area is also provided with a surround OWS channel. This surround OWS channel acts as a barrier between the outside unpaved area and the plant paved area and is thus a facility provided to contain an oil run-off.
- All storage tanks are provided with Dyke walls around it to contain the leakage from the tank in case of a failure of tank. The OWS system provided in the tank passes underground out of the Dyke wall with a valve at just outside the Dyke wall. Facility is provided to route the accidental oil collected in the Dyke area to OWS. Operating philosophy requires all the three valves at the Dyke wall outlet to be kept closed and open only when required for the specific purpose. This philosophy and Dyke wall provision ensures that no oil can pass to the external surface drain.

The above basic philosophy features play a key role in maintaining the surface drains free from oil in the Refinery. If the philosophy is followed and the other issues mentioned in this chapter are satisfied in a sustained manner, the risk of an oil spill can be minimized.

33.2.2 Certain Other Mitigating Features:

There can be many such provisions and features existing in the Refinery which help in increasing the response time available for handling an oil spill and some of them may contribute decisively in containing the oil spills to a restricted area or within the Refinery premises.



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Oil Catchers:

Existence of oil catchers at the outlet of all the Refinery surface drains before the drain leaves the Refinery premises can provide time to mobilize an oil evacuation device like gully sucker. Also in case of smaller oil spills, the oil catcher can fully prevent it from going out of the Refinery premises depending on the volume of the oil catcher and the spill volume.

Non-running Open drains:

If there is an open drain which is being used to cater to water flow requirement in normal operation of a facility in the Refinery, it can be big liability in case of an oil spill. As the drain is under water flow in normal times, it will take very less time for the oil to go out of the Refinery in case it enters such a drain. In wet weather, such a scenario will be valid for all the surface drains. Thus wet weather oil spills are to be considered more serious than dry weather oil spills in case of the Refinery. If there is no running drain in dry weather, it will enable total blocking of the drain by weirs/baffles to contain the oil spills and recover them. If there is water flow, total blockage is not possible and oil catcher type mechanism can only be provided which invariably leads to some oil breakthrough.

Kerb Wall Provisions:

Kerb wall provisions around the oil equipment facility in offsite areas help to contain the oil spill within the area of the facility. In oil wharf area a spill containment barrier is provided all around the loading/unloading manifold. This barrier provides protection against smaller spills and in most cases if the response during a hose rupture is rapid, these barriers can play a decisive role in preventing the oil from entering the sea.

Similarly in many offsite areas, pump bays, valve manifolds are provided with Kerb walls around them with OWS facility.

Provision of Floating Booms on water bodies:

Usage of barrier floating booms around the probable area of oil spill is a preventive measure against the spread of oil spill beyond a limit. If the oil spill occurs it can be recovered from the contained area and spread of oil spill can be avoided. Such provisions can be made while loading/unloading a ship, at the mouth of the intake canal, at the effluent canal etc. Nature of booms is different for still waters, turbulent waters and flowing waters.



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33.3 EFFECTS OF AN OIL SPILL

There can be many after effects of an oil spill. These can be listed as:

- Monetary loss.
- Loss of oil.
- Damage to environment- flora, fauna.
- Resultant fire and associated losses.
- Loss in organizations credibility.
- Public or statutory liability.
- Expenditure on cleanups.

Many effects may be secondary or tertiary nature also. In extreme cases where the damage to environment is severe, the liabilities may be huge. Marine environment comprising of Mangroves, coral reefs, fish, birds and other marine population are sensitive ecosystems and are the main sufferers during an oil spill in the sea. Sea waves make the oil to spread far and wide and increase the impacted area. The coastline can get awash with oil and result in loss of many areas of tourism for the populace. Such occurrences give wide publicity to the event and are detrimental to the existence and reputation of the industrial establishments like oil refineries.

Another complexity arises when there is an oil spill from a tanker which is chartered by the refinery and it spills oil in the far high seas. Since quick response is needed, logistics may be extremely cumbersome. And if the impacted area is a sensitive eco-system, the problems increase many fold. The refinery may have to depend on the services of very specialized agencies with huge costs. Also such events get wide media coverage and create almost a permanent negative image about the organization in the minds of public. Images of affected areas, marine species and birds provide motivation for penal action against organization.

33.4 CERTAIN MAJOR OIL SPILL POSSIBILITIES FROM REFINERY FACILITIES

- Rupture of any Crude Tank.
- Rupture of any product tank.
- Rupture of pipelines in any pipe-tracks.
- Rupture or leaks at loading/ unloading hoses from Ships.
- Leaks of the loading arm.
- Overflow of a HC storage Tank.
- Rupture of pipes between ATP and Refinery North wall.



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- Partial or full capsizing of the Ships chartered by the refinery.
- Overflow from the Tanker.
- Crude unloading line leak.

The associated risk in each case will depend upon the quantitative extent of the abnormality. Also the impact of the event will depend upon the environment around the event location. If the event is occurring at wharf, OSTT the impact is high due the escape of oil into the sea. The response time also can be low due to the rapidity of the event. Rupture of hoses, loading arms, joints leak, rupture of subsea pipeline, overflow of ship tanks are high impact scenarios. Oil can escape from the facilities in the refinery premises also due to any of the earlier mentioned reasons and reach the sea through the effluent canal and other drains.

Apart from the spill itself, the occurrence of secondary events like fire will depend upon the quality of the oil and situation can get complicated. Response strategies should be planned to suit the requirements of the event.

33.5 MOVEMENT OF OIL SPILL

It is important to predict the probable direction of movement of the oil spill. This helps in taking mitigating measures to control the impact of the oil spill. The movement and drift of spilled oil spill on a water body will depend upon spread of oil will invariably impact the shore as it will travel with the wave fronts.

In case of oil wharf area, since there is a continuous flow towards the LLPH intake canal, the oil spill is likely to move towards the intake area, though' there is a parallel impact of the wind direction. Areas towards Shipyard and ENC are likely to get washed with oil spill in case there is a larger oil spill. This again will depend upon the size of the spill and wind direction.

In case of OSTT, the movement of the spill will depend mainly upon the wind direction. The turbulence of the waves will result in movement of certain patches of spill in different directions also. In case of flowing streams, however, like refinery effluent canal, there is little impact of the wind direction. The flowing canals act as a contained transport medium for the oil to flow and reach the destination. At the final delivery point the oil spill will spread over water body and there the wind direction will affect the spill. It is important to know the wind direction at the time of oil spill. This helps in planning the response.



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33.6 FACILITIES & RESOURCES AT RISK

Different types of shorelines are affected in different ways. Thus cleanup is strategies should be suitable to the needs. The type of the affected area will dictate the cleanup strategy. The resources at risk will need to be protected with measures in order to minimize the impact. Environmentally and commercially sensitive areas which will get affected under different spill scenarios are presented below:

SPILL SCENARIO	SPILL MOVEMENT CHANNELS	AFFECTED AREAS	MECHANISM OF CLEANING
Spill from the main refinery area	Refinery Main Effluent Canal	Western Harbor Arm	Boat Mounted Skimmer
		Mangrove area of Western arm	Boat Mounted Skimmer
		Navy Jetty in the western arm	Boat mounted skimmer
	VPT Drain	Municipal drain on the south of refinery	Gully sucker/ Vacuum truck
		VPT Drain	Gully sucker/ Vacuum truck
		ENC/ Shipyard Area Bay/ Wharf	Boat mounted skimmer
	LLPH pipe track	ENC/ Shipyard Area Bay/ Wharf	Boat mounted skimmer/ Manual
Spill from Oil Wharf	Directly into the Sea	ENC/ Shipyard Area Bay/ Wharf	Boat mounted skimmer
		LLPH intake area/ Bay	Boat mounted skimmer/ Manual
Spill form OSTT area/ crude offloading line	Directly into the Sea	Fishing Harbor	Boat mounted skimmer
		Beaches and Coastline	Boat mounted skimmer and Manual. Drum Skimmers
		Main Harbor	Boat mounted skimmer and



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			Manual. Drum Skimmers
Ship Capsizing/ Grounding	Directly into the Sea	Beaches and Coastline	Boat mounted skimmer and Manual. Drum Skimmers
		Fishing Harbor	Boat mounted skimmer and Manual. Drum Skimmers
		Rushikonda Aqua farms	Boat mounted skimmer and Manual. Drum Skimmers
		Main Harbors	Boat mounted skimmer and Manual. Drum Skimmers

33.7 CHARACTERISTICS AND BEHAVIOUR OF OIL

The behavior of oil spill on a water body depends mainly on the characteristics of the Oil and the environment of the spill. Following characteristics of the oil are important for consideration and provide information about the possible behavior of the oil:

- Specific gravity of the oil provides a broad clue about the class of the oil as to whether it is heavy, middle or light etc. It will indicate whether oil will float or sink.
- Pour point will indicate congealing characteristic of the spilled oil. Oils like LSHS, VGO with high pour points will congeal on water bodies.
- Viscosity will determine the spread and flow of the spilled oil. The spilled oil film thickness will also depend on its viscosity. Low viscous 'oil spillage will spread more rapidly than a high viscous oil.
- Asphaltene content may determine the emulsion forming tendency of the spilled oil. Oil with Asphaltene content higher than 0.5% exhibit tendency to form oil-water emulsions. Turbulence in the water body increases the formation in emulsions. This then makes recovery, storage and handling of the oil difficult.
- Flash point of the spilled oil is an indicator of the fire hazard associated with the oil spill.
- Color of the spilled oil will provide a first hand visible conclusion of the area affected by the oil. Transparent clean oils spillage poses problems in judging the spread unless a detailed survey is made.

33.8 CATEGORISATION OF VARIOUS OIL SPILL SCENARIOS IN THE REFINERY

The oil spills occurring in a refinery may be of varied nature. The quantity-of the oil spilled will depend upon the type of failure, type of operation, operating condition of the system etc.



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The impact of the oil spill will depend upon the topography and the features of the area at which the oil spill has taken place. It will also depend upon the prevailing weather conditions.

An oil spill scenario can become a fire scenario if appropriate timely precautions are not taken. If there has been a breakthrough of oil from the refinery and there is a coupled fire scenario, the impact on the surrounding population and facilities of other establishments can occur. There can also be secondary fire scenarios due the fire originating from an oil spill of the refinery. In such cases the liability increases manifold. In some cases quantity of the spilled oil may be less but the danger of fire may be very high and thus requiring a more rapid response.

In case of an oil spill in oil handling process plant area, the issue may be more concerned with the immediate danger of fire rather than environmental damage. The process plant area is a paved area and has a barrier surround OWS system which will be instrumental in preventing oil escaping to the surface drains and unpaved area, though this is dependent on the quantity. In wet weather, however, the dynamics may be slightly different. The requirements of handling an oil spill event in off shore area are vastly different. The spread of oil in offshore can be very rapid and it immediately starts impacting the marine species. The wind direction, oil quality and turbulence/flow of the water body are the parameters which determine the spread of the spill.

While planning response to an oil spill scenario, all these factors need to be considered. While planning the infrastructure associated with the response elements, there always needs to be a lead over the severity of actual scenario so that adequacy is always ensured. It is also required that mechanism is in place for handling smallest to the largest spill.

Thus, based on the severity, effects and logistics involved, the OIL SPILLS are categorized as below in TABLE - 1 : (In the order of lower to higher environment impact):

CATEGORY	TYPE OF OIL SPILL
ZERO	An oil spill occurring in a paved area of a process plant or an operating facility which is provided with ZERO OWS facility provided the spill is of such magnitude that it is not entering any surface drain.
ONE	An oil spill excluding Naphtha, MS, Crude (or similar lighter oils with fire hazard), occurring in any area and the oil is entering surface drain. But there is no immediate danger of oil going out of the refinery.
TWO	An oil spill of Naphtha, MS or Crude (or similar light oils with fire hazard) occurring in any area and the oil is entering surface drain. But there is no immediate



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	danger of oil going out of the refinery.
THREE	An oil spill excluding Naphtha, MS or Crude (or similar lighter oils with fire hazard), occurring in any area and the oil is entering surface drain. But there is an imminent danger of oil going out of the refinery.
FOUR	- An oil spill of Naphtha, MS, Crude (or similar light oils with fire hazard) occurring in any area and the oil is entering surface drain. There is an imminent danger of oil going out of the refinery. - Catastrophic failure of any of the Storage Tanks. - If any of the oil spill is above 100 MT. - Oil Spill due to any abnormal occurrence in wharf and/or OSTT.
FIVE	Oil Tanker capsizing or running aground.

The requirement and level of response in each of the category will vary. Though theoretically there is a danger of fire in every oil spill, the danger is more in lighter oils and very high fire risk in case of Naphtha/MS oil spill. Situations where the fire risk is high will require simultaneous activation of Fire emergency Plan. Proper precautions will have to be taken to make sure that the personnel involved in managing such oil spills be aware of such dangers and are well protected and prepared. In case there is a risk of MS/Naphtha going out of the Refinery, the fire risk is much higher and the impact also will be very high.

The above category is not showing the fire risk ranking. The Categories are arrived at considering the oil breakthrough out of the Refinery and its impact outside the refinery.

NOTE:

- As stated in the preface of the plan, the oil spill response plan is a sub plan under the On-Site Emergency Plan. At any time during any oil spill event it can be declared as Onsite emergency by the incident controller with authorization from Head Refinery. Once it is declared as Onsite emergency all the requirements of response as per the Onsite emergency plan will have to be fulfilled from that time onwards apart from the requirements under NOSDCP when applicable.
- In the event of a fire resulting during the oil spill, the Fire emergency has to be given precedence over the oil spill emergency. But simultaneously resources must be directed towards carrying out containment, source arresting and recovery of the leaked out oil as it will reduce the severity of fire. And during such time the response organization will be as per the Fire emergency manual.



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33.9 GUIDELINES FOR IMPLEMENTING THE ELEMENTS OF OIL SPILL RESPONSE PLAN

No two oil spill events can be considered totally identical. Their origination, magnitude, after effects, resource requirements for response etc. can be vastly different. The dynamics involved in handling the on-shore oil spill are quite different from an Off-shore oil spill. The effectiveness in handling will depend upon many elements. Satisfying resource requirement, though very important, is not the only element in achieving effectiveness. Keeping abreast of the roles and practicing their execution at certain intervals enables improving the response capability and quality. Therefore it is important to conduct mock drills and exercise at certain intervals. These exercises not only help to understand the roles better but also are tools for further improving the Plan strategies itself based on the work-front experience.

Preparedness of each role player for playing his role to the fullest is dependent upon the psychological and resource adequacy. The resources have to be adequate and they have to be available in a manner that they can be rapidly deployed. Backup arrangement is also needed to mobilize additional resources in case required. The methodology adopted to deploy the resources nevertheless enhances the effectiveness of response. A well coordinated, rapid and on target response in the initial stages can make things easier to manage in subsequent stages.

The following guidelines for implementing each of the philosophy element of an Oil Spill Response will help in understanding the direction in which the O.S.R.A. has to act:

33.9.1 To identify and Stop the Source leak:

- If a leak is occurring in the Process Unit area where OWS facilities are provided, from oil spill point of view it is not of imminent danger (surely there is a risk of Fire). Such an event can become an oil spill scenario if it prolongs or if it is of a catastrophic nature. It is the rapidity of activities like isolating, de-pressuring and taking shutdown of the respective section which will be important in stopping source leak.
- A pipeline leak in an Offsite area where there are no OWS facilities and which is unpaved, theoretically the oil spill scenario starts from the moment the leak starts. For source leak stoppage, usage of clamps with rubber packing etc. should be resorted to. These Kits are available in Fire-House and with Maintenance Department. If leak does not stop, system will need to be isolated and de-pressured. A Tank Bottom Leak will require decision to pump-out the Tank contents.



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33.9.2 To Contain the oil spill:

- First Step in containment is to prevent the entry of the oil into the surface drain. This may not be fully possible every time but efforts have to be made to achieve this to the maximum. This will require deployment of Flexible Absorbent or containment Booms all around the oil pool/patch which has started developing due to the leak. Based on the surface contour and presence of other hurdles at the area, the boom invariably will leave some gaps. These gaps will have to be packed using absorbent pads, pillows or such other packing material.
- The adjacent Surface drains have to be blocked, towards upstream and downstream of the oil spill area, using baffle plates. Requisite grooves arrangement has been provided (or are being provided) at some locations in each of the Surface drain to enable such baffle insertion. It will be possible to insert such baffles and fully block the surface drain only in case of dry weather and when there is no water flow in the drain. In case of wet weather or water flow case, underflow-overflow baffle combination will have to be suitably inserted to allow water flow to go through and retain the oil. In such cases, traces of oil are expected to breakthrough and measures must include close monitoring of the situation.
- It will also help if protective block-baffles are provided at a distance after the first block-baffle in the surface drain this will act as secondary containment and thus will enable that much cushion in the response activity with respect to recovery of the spilled oil.
- Depending on the severity, boom and baffle should be deployed at the respective surface drain' outlet leaving the refinery.
- In case the above mentioned equipment is not available, methods like usage of sand bags, hay filters can -be used. But these have disadvantage that they cannot be re-used and end up as hazardous waste and their disposal becomes a problem.

33.9.3 To recover the oil spill:

- Recovery of spilt oil will require equipment which can be pressed into service at a short notice. These may be Gully sucker, Oil skimmers and in some cases manual bailout may be required.
- Usage of oil skimmers is generally not possible in shallow depths. In such cases gully suckers should be used.
- Gully sucker is versatile equipment and can be used in most cases.
- A floating suction mesh attachment helps in improving the effectiveness of a gully sucker.
- Recovered oil from a floating skimmer will need to be stored in a vessel close to the skimmer operation area. Gully Sucker is mobile truck and can transport the oil to a distance location for unloading the same.



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- Refinery has two gully suckers which can be used simultaneously to increase the recovery capacity.

33.9.4 To treat the breakthrough oil (which is going out of containment) spill:

- Only Treatment possible for break-through oil is application of dispersant. The dispersant can only be applied to relatively lesser quantities of oil-which is breaking through in the form of streaks.
- If there is a sheet of oil escaping, the application of dispersant has no use and in such cases, the containment area should be extended. .
- A dispersant does not have the capability to disperse the entire oil body to which it is applied. It has an efficiency ranging from 30 to 60 %.
- For applying dispersant, either a pump can be used or it can be sprayed using an eductor on the Fire water hose with nozzle.

33.9.5 For working area house keeping/cleaning:

- This is to enable smooth functioning of the personnel involved and equipment and maintain the ergonomics. A person should be nominated by the I.C. & E.M.C. to generally look after this aspect so that the frontline personnel involved in the actual activity of oil spill management have a manageable work-front available for them.
- The person who looks after the house keeping aspects must be in close contact with all the OSRAs and interact with them so that his activities are managed in a manner to suit their requirements.
- After the completion of the entire oil spill response activity a final mop up and area inspection to restore to original condition is necessary.

33.9.6 To ensure Safety during the entire activity:

- Emergency Fire and Safety Controller must nominate one of his representatives to look after this aspect.
- This representative should interact, prompt and advise the participating personnel in the safety aspects of their execution activity.
- The execution area may have lot of non standard accesses movement paths which the participating personnel will have to encounter. Proper prompting and interaction at the time of activity will help in avoiding injuries.



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- Road closure should be suitably done with respect to the roads in close proximity to the oil spill area.
- Personnel protective equipment should be provided.

33.9.7 To help in restoring the Damage to the Environment:

- This activity will generally mean the finishing touches in case of the area cleanups after the event has fully come in control.
In case any birds, animals are affected action shall be taken to treat them in the veterinary hospitals. Affected trees shall be cleaned



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34.1 UNIT MASTER BLIND LIST:

34.1.1 UNIT LIMIT BLIND LIST:

EAST BATTERY LIMIT BLIND LIST:

FIRST PLATFORM:

S no	Size	Specification	Description	rating
1.	6"	B7A	HVGO R/D	150#
2.	6"	A1A	BITUMEN TO HFO LINE	150#
3.	3"	A9A	AW TO MEROW	150#
4.	6"	B7A	HVGO HOT FEED TO FCCU-II	300#
5.	8"	A1A	HFO R/D	300#
6.	6"	A1A	OLD BITUMEN R/D	150#
7.	3"	A2A	LVGO TO DSL	150#
8.	6"	A1A	KERO R/D	150#
9.	6"	A1A	KERO R/D TO DSL	150#
10.	6"	A1A	DIESEL TO STORAGE	150#
11.	4"	A1A	DIESEL TO LDO	150#
12.	3"	A1A	KERO TO LDO	150#
13.	3"	A1A	KERO TO FO	150#
14.	6"	B1A	SR TO IFO	300#
15.	8"	A1A	SLOP Header	150#
16.	1 1/2"	A1A	HWW TO MEROX	150#
17.	4"	A1A	KERO TO FO	150#
18.	3"	B1A	FO RETURN LINE	300#
19.	2"	A9A	CAUSTIC LINE	150#
20.	6"	A1A	SRN TO MS	150#
21.	6"	A1A	SRN R/D TO STG	150#
22.	3"	B1A	FO SUPPLY	300#
23.	3"	B1A	CDU-I SR TO BBU	300#
24.	6"	A1A	OTN TO STORAGE	300#
25.	3"	A1A	NAPHTHA TO 11-C-05	300#
26.	3"	B4F	SLOPCUT TO PDU	300#
27.	6"	Redundant line	CTU RCO LINE	300#
28.	2"	B1A	HOT DIESEL TO FCCU-II	150#



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29.	3"	A9A	CAUSTIC TO 11-V-04	150#
30.	2"	B1A	LPG TO VAPORISER	300#
31.	4"	A3A	FCCU-II MAB COND. TO 11-V-04	150#
32.	3"	B1A	LPG R/D	300#
33.	2"	A9A	CAUSTIC TO 11-V-07	150#
34.	8"	B1A	RFO LINE	300#

SECOND PLATFORM:

S no	Size	Specification	Description	rating
1.	6"	B1A	SR FROM CDU-3	300#
2.	6"	A1A	BITUMEN NEW R/D LINE	300#
3.	3"	A1A	HN TO SRN	150#
4.	2"		DRINKING WATER LINE	
5.	3"	A1A	HN TO DSL	150#
6.	2"/3"	A3A	11-E-18 condensate to PP	150#
7.	3"/2"	A9A	CAUSTIC FROM MEROX	150#
8.	4"	B1A	SR TO LDO	300#
9.	8"	A1A	FUEL GAS LINE	150#
10.	3"	A1A	LVGO TO HVGO R/D	150#
11.	3"	A1A	LVGO TO LDO	150#
12.	4"	B1A	STRIPPED WATER FROM MEROX	300#
13.	3"	B1A	EFFLUENT WATER TO SWSU	300#
14.	4"	A1A	CUTTER	150#
15.	2"	J3A	INST. AIR	150#
16.	4"	A3A	YARD AIR	150#
17.	4"	A2A	SERVICE WATER	150#
18.	4"	A3A	BCW SUPPLY	150#
19.	4"	A3A	BCW RETURN	150#
20.	4"	A1A	11-V-03 GAS TO FCCU-II	150#
21.	3"	A1A	HWO TO TK 17	150#
22.	12"	A2A	TANK FARM LP STEAM	150#
23.	12"	A1A	OFF GAS	
24.	6"	B7A	FCCU HOT FEED	300#
25.	4"	A3A	DM WATER	150#
26.	4"	A3A	BFW	300#
27.	3"	A1A	HN TO TK 124	150#
28.	6"	B1A	SR TO VBU TK	150#
29.	10"	A2A	LP STEAM	150#
30.	8"	D2A	HP STAEM	600#
31.	12"	B2A	MP STEAM	300#



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32.	4"	A1A	MEROX FLARE KOD TO OTN	150#
33.	3"	A1A	SRN TO FCCU	300#
34.	3"	B4F	RCO TO FCCU-II	300#
35.	3"	A1A	11-V-01 TO FCCU-II	150#
36.	6"	A3A	BCW RETURN	150#
37.	4"	A1A	KERO TO FO (OM&S)	150#
38.	20"	A1A	FLARE	150#

THIRD PLATFORM:

S no	Size	Specification	Description	Rating
1.	14"	A1A	SR TO VBU TK	300#
2.	10"	A1A	SR TO CDU-III	300#
3.	8"	B3F	CIRCULATING OIL	300#
4.	8"	B3F	CIRCULATING OIL RETURN	300#
5.	6"	A1A	KERO TO ATP	150#
6.	3"	A1A	HN TO ATP	150#

FOURTH PLATFORM:

S no	Size	Specification	Description	Rating
1.	10"	B1A	HOT DIESEL TO DHDS	150#
2.	4"	A1A	PG HN TO STG	150#

WEST BATTERY LINES:

S no	Size	Specification	Description	Rating
1.	10"	A1A	FEED LINE FROM TANK FARM	150#
2.	12"	A2A	TANK FARM LP STEAM	150#
3.	4"	A3A	BCW SUPPLY LINE	150#
4.	4"	A3A	BCW RETURN LINE	150#
5.	10"	A1A	CBD INLET V/V	150#
6.	4"	A1A	CBD PUMP DISCH.	150#



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34.2 INDIVIDUAL EQUIPMENT BLIND LIST:

34.2.1 CDU-II PUMPS BLIND LIST:

S. No	PUMP	SERVICE	BLINDS	SIZE & RATING	SPEC
1	11-PM-01A/B	CRUDE	Suction Discharge Warm up	10" X 150 10" X 300 1"X300	A1A B1A B1A
2	11-PM-02A/B	CRUDE	Suction Discharge Warm up	12" X150 10" X300 1"X300	A1A B1A B1A
3	11-PM-03A/B/C	DIESEL	Suction Discharge Warm up	8" X150 6" X300 ¾"X300	A9A B7F B7F
4	11-PM-04A/B	KERO Product	Suction Discharge Warm up	8"X150 6"X300 ¾"X300	A9A B7F B7F
5	11-PM-05A/B	Heavy Naphtha	Suction Discharge Warm up	6"X150 3"X150 ¾"X150	A1A A1A A1A
6	11-PM-06A/B	Top Reflux	Suction Discharge	12"X150 8"X300	A1A B1A
7	11-PM-07C/D	Diesel CR	Suction Discharge Warm up	10"X150 8"X300 ¾"X300	A4F B4F B4F
8	11-PM-08A/B	KERO CR	Suction Discharge Warm up	16"X150 12"X300 ¾"X300	A9A B7A B7A
9	11-PM-08C/D	KERO CR booster	Suction Discharge Warm up	10"X 300 10"X300 ½"X300	B7A B7A B7A
10	11-PM-09A/B	Top pump around	Suction Discharge Warm up	16"X150 10"X300 ¾"X300	A1A A3A A3A
11	11-PM-10A/B	RCO	Suction Discharge Warm up	12"X150 8"X300 ½"X300	A4F B4F B4F
12	11-PM-11A/B	LPG	Suction Discharge	8"X150 6"X150	A1A A1A
13	11-PM-12A/B	Wash water	Suction Discharge	4" X 150 3"X300	A3A B3A



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S. No	PUMP	SERVICE	BLINDS	SIZE & RATING	SPEC
14	11-PM-13A/B/C	Caustic injection	Suction Discharge	1"X150 ¾"X300	A9A B7A
15	11-PM-14A/B/C	Ammonia injection	Suction Discharge	1"X150 1"X300	A9A B7A
16	11-PM-15A/B	Filmer	Suction Discharge	¾"X150 ¾"X300	A9A B7A
17	11-PM-16A/B	DMF	Suction Discharge	¾"X150 ¾"X300	A9A B7A
18	12-PM-01A/B	SR	Suction Discharge Warm up Vent	12" x 150 8" x 300 1" 2"X300	A4F B4F B4F B1A
19	12-PM-02A/B	Slop cut	Suction Discharge Warm up Vent	6" x 150# 4" x 300# 1"X300 2"X300	A4F B4F B4F B1A
20	12-PM-03A/B	HVGO	Suction Discharge Warm up Vent	16" x 150# 10" x 300# 1"X300 2"X300	A4F B4F B4F B1A
21	12-PM-04A/B	LVGO	Suction Discharge Warm up Vent	8" x 150# 6" x 300# 1"X300 2"X300	A1A B1A B1A B1A
22	12-PM-05A/B	HWO	Suction Discharge	2" x 150# 1 ½" x 150#	A1A A1A
23	12-PM-06A/B	HWW	Suction Discharge	3" x 150# 2" x 150#	A9A A9A
24	12-PM-07A/B	Tempered water	Suction Discharge	10" x 150# 8" x 150#	A3A A3A
25	12-PM-08A/B	Vac. neutralizer	Suction Discharge	¾" X150. ¾"X300	A9A B7A
26	10-PM-01A/B		Suction Discharge	4"X150 6"X150	A9A A9A
27	10-PM-03A/B		Suction Discharge	2"X150 2"X150	A3A A3A



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34.2.2 EXCHANGERS BLIND LIST:

S. No	Exchanger	Service	blinds	size & rating	spec
1.	11-E-01	Crude/ HN	i) Crude inlet valve downstream flange ii) Crude outlet valve upstream flange iii) HN inlet valve downstream flange iv) HN outlet valve upstream flange v) Crude inlet cutter line vi) Crude to CBD vii) HN to CBD	10" X 300 10" X 300 3" X 150 3" X 150 1" X 150 1 ½ " X 150 1" X 150	B1A B1A A1A A1A A9A A1A A1A
2.	11-E-02	Crude/ KERO	i) Crude inlet valve downstream flange ii) Crude outlet valve upstream flange iii) KERO inlet valve downstream flange iv) KERO outlet valve upstream flange v) Crude inlet cutter line vi) Crude to CBD vii) KERO to CBD	10" X 300 10" X 300 6" X 300 6" X 300 1" X 150 1 ½ " X 150 1" X 150	B1A B1A B1A B1A A9A A1A A1A
3.	11-E-03	Crude/ Diesel	i) Crude inlet valve downstream flange ii) Crude outlet valve upstream flange iii) Diesel inlet valve downstream flange iv) Diesel outlet valve upstream flange v) Diesel to 11-E-23A upstream flange vi) Crude inlet cutter line vii) Crude to CBD viii) Diesel to CBD	10" X 300 10" X 300 6" X 300 6" X 300 6" X 300 1" X 150 1 ½ " X 150 1" X 150	B1A B1A B1A B1A B1A A9A A1A A1A
4.	11-E-04A/B	Crude/ TPA	i) Crude inlet valve downstream flange ii) Crude outlet valve upstream flange iii) TPA inlet valve downstream flange iv) TPA outlet valve upstream flange v) Crude inlet cutter line vi) Crude to CBD vii) TPA to CBD	10" X 300 10" X 300 10" X 150 12" X 150 1" X 150 1 ½ " X 150 1" X 150	B1A B1A A1A A1A A9A A1A A1A
5.	11-E-05	Crude/ KERO	i) Crude inlet valve downstream flange ii) Crude outlet valve upstream flange iii) KERO inlet valve downstream flange iv) KERO outlet valve upstream flange v) Crude inlet cutter line vi) Crude to CBD vii) KERO to CBD	10" X 300 10" X 300 6" X 300 6" X 300 1" X 150 1 ½ " X 150 1" X 150	B1A B1A B1A B1A A9A A1A A1A



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S. No	Exchanger	Service	blinds	size & rating	spec
6.	11-E-06	Crude/ Diesel	i) Crude inlet valve downstream flange ii) Crude outlet valve upstream flange iii) Diesel inlet valve downstream flange iv) Diesel outlet valve upstream flange v) Crude inlet cutter line vi) Crude to CBD vii) Diesel to CBD	10" X 300 10" X 300 6" X 300 6" X 300 1" X 150 1 ½ " X 150 1" X 150	B1A B1A B1A B1A A9A A1A A1A
7.	11-E-07	Crude/ LVGO	i) Crude from 11-E-06 valve downstream flange ii) Crude from 11-PM-02A valve downstream flange iii) Crude to Desalter valve upstream flange iv) Crude to 11-E-08 valve upstream flange v) LVGO inlet valve downstream flange vi) LVGO outlet valve downstream flange vii) Crude inlet cutter line viii) Crude to CBD ix) LVGO to CBD	10" X 300 10" X 300 10" X 300 10" X 300 4" X 300 4" X 300 1" X 150 1 ½ " X 150 1" X 150	B1A B1A B1A B1A B1A B1A A9A A1A A1A
8.	11-E-08	Crude/ Diesel product	i) Crude inlet valve downstream flange ii) Crude outlet valve upstream flange iii) Diesel product inlet valve downstream flange iv) Diesel product outlet valve upstream flange v) CBD inlet crude inlet vi) Diesel product to CBD	8" X 300# 8" X 300# 6" X 300# 6" X 300# 1 ½ " X 150# 1" X 150#	B1A B1A B1A B1A A1A A1A



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S. No	Exchanger	Service	blinds	size & rating	spec
9.	11-E-09	Crude / KERO CR	i) Crude inlet valve downstream flange ii) Crude outlet valve upstream flange iii) Crude inlet from 11-E-11 at 11-E-11 valve downstream flange iv) KERO CR inlet valve downstream flange PG mode v) KERO CR inlet valve downstream flange BH mode vi) KERO CR outlet valve upstream to control valve flange vii) KERO CR outlet to 11-C-01 valve upstream flange viii) Crude inlet to CBD ix) KERO CR inlet to CBD.	8"X 300# 8" X 300# 8" X 300# 12" X 300# 12" X 300# 12" X 300# 12" X 300# 1 ½ " X150 # 1 ½ " X150 #	B1A B1A B1A B7A B7A B7A B7A A1A A1A
10.	11-E-10	Crude/ KERO product	i) KERO product inlet valve downstream flange ii) KERO product outlet valve upstream flange iii) Crude inlet valve from 11-E-11 downstream flange iv) Crude outlet valve from 11-E-09 downstream flange v) Crude outlet shell flange vi) CBD drain	6" X300 # 6" X300 # 8" X300 # 8" X300 # 8" X300 # 8" X300 # 1 ½ " X300 #	B7A B1A B1A B1A B1A B1A A1A
11.	11-E-11	Crude/ KERO CR	i) Crude inlet valve from 11-E-10 bypass downstream flange ii) Crude inlet valve from 11-E-08 downstream flange iii) Crude outlet valve upstream flange iv) KERO CR inlet from 11-E-09 and 11-E-25 inlet valve downstream flange v) KERO CR inlet valve downstream flange vi) KERO CR outlet valve upstream flange vii) KERO CR outlet valve upstream flange	8" X 300# 8" X 300# 8" X 300# 12" X 300# 10" X 300# 10" X 300# 12" X 300#	B1A B1A B1A B7A B7A B7A B7A



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S. No	Exchanger	Service	blinds	size & rating	spec
12.	11-E-12	Crude/ Diesel	i) Crude inlet valve downstream flange ii) Crude outlet valve upstream flange iii) Diesel product inlet valve downstream flange iv) Diesel product outlet valve upstream flange v) Diesel CR from 11-E-13 valve downstream flange vi) Diesel inlet to CBD vii) Crude inlet to CBD	8" X 300# 8" X 300# 6" X 300# 6" X 300# 8" X 300# 1 ½" X 150# 1 ½" X 150#	B1A B1A B7A B1A B7A A1A A1A
13.	11-E-13A/B	Crude/ Diesel CR	i) Crude inlet valve downstream flange ii) 11-E-13 inlet bypass valve downstream iii) Crude outlet valve at 11-E-14 upstream iv) At 11-E-14 and 11-E-15 bypass valve upstream flange v) Diesel CR inlet downstream flange vi) Diesel product inlet downstream flange vii) Diesel CR outlet valve upstream flange viii) Diesel product outlet valve upstream ix) Crude to CBD x) Diesel CR to CBD	8" X 300# 8" X 300# 8" X 300# 8" X 300# 8" X 300# 6" X 300# 8" X 300# 6" X 300# 1 ½" X 150 1 ½" X 150	B1A B1A B1A B1A B7A B7A B7A B7A A1A A1A
14.	11-E-14A/B	Crude / Diesel	i) Crude inlet valve downstream flange ii) Crude outlet valve upstream flange iii) 11-E-13 crude bypass valve downstream flange iv) Diesel inlet valve downstream flange v) Diesel outlet valve upstream flange vi) Diesel CR inlet valve downstream flange vii) Diesel CR outlet valve upstream flange viii) Crude inlet and outlet to CBD ix) Diesel inlet to CBD	8" X 300# 8" X 300# 8" X 300# 6" X 300# 6" X 300# 8" X 300# 8" X 300# 1 ½" X 150# 1 ½" X 150#	B7A B7A B7A B7A B7A B7A B7A A1A A1A
15.	11-E-15A/B	Crude / Diesel CR	i) Crude inlet valve downstream flange ii) Crude outlet valve upstream flange iii) Diesel CR inlet valve downstream iv) Diesel CR outlet valve upstream flange v) Diesel CR inlet to CBD vi) Crude inlet and outlet to CBD	8" X 300# 8" X 300# 8" X 300# 8" X 300# 1 ½" X 150# 1 ½" X 150#	B7F D7F B4F B7A A1A A1A



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16.	11-E-16	Crude / SR	i) Crude inlet valve downstream flange ii) Crude outlet valve upstream flange iii) SR inlet valve downstream flange iv) SR outlet valve upstream flange v) Crude inlet to CBD vi) Crude outlet to CBD vii) SR inlet to CBD viii) SR outlet to CBD	8" X 600# 8" X 600# 8" X 300# 8" X 300# 1 ½" X 150# 1 ½" X 150# 1 ½" X 150# 1 ½" X 150#	D7A D7A B4F B4F A1A A1A A1A A1A
17.	11-E-18	Wash water / LP	i) Water inlet flange ii) Wash water outlet flange iii) LP steam inlet flange iv) LP outlet (condensate) v) Wash water to CBD	4" X 300# 4" X 300# 2" X 150# 2" X 150# 1 ½" X 150#	B1A B1A A3A A3A B1A
18.	11-E-19 No isolation valves	Stabilizer feed / SRN	i) Stabilizer feed inlet to 11-E-19 ii) Unstabilized naphtha to stabilizer from 11-E-19 iii) SRN inlet iv) SRN outlet v) SRN to CBD vi) Unstabilized naphtha to CBD	6" X 300# 6" X 300# 6" X 300# 6" X 300# 2" X 150# 1" X 150#	B1A B1A B1A B1A A1A A1A
19.	11-E-25	Stabilizer reboiler	i) Naphtha to reboiler flange (2 no's) ii) Naphtha from reboiler flange (2 no's) iii) KERO CR to reboiler inlet flange iv) KERO CR outlet flange from reboiler v) KERO CR to CBD vi) Naphtha to CBD	10" X 300# 10" X 300# 10" X 300# 10" X 300# 2" X 150# 1 ½" X 150#	B1A B1A B7A B7A A1A A1A
20.	11-E-40A/B	Crude/ circulating oil	i) Crude inlet valve downstream flange ii) Crude outlet valve upstream flange iii) Circulating oil inlet valve flange iv) Circulating oil outlet valve upstream flange v) Crude in let to CBD vi) Cutter inlet to crude line vii) Circulating oil to CBD viii) Cutter to Circulating oil	8" X 300# 8" X 300# 8" X 300# 8" X 300# 1 ½" X 150 1" X 150 1 ½" X 150 1" X 150	D7A D7A B4F B4F A1A A1A A1A A1A



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S. No	Exchanger	Service	blinds	size & rating	spec
21.	12-E-01A/B/C	Crude/ KERO CR	i) Crude inlet valve downstream flange ii) Crude outlet valve upstream flange iii) KERO CR inlet valve flange iv) KERO CR outlet valve upstream flange v) SR to shell inlet vi) SR from shell outlet line vii) Cutter to crude viii) Crude to CBD 2 nos ix) Cutter to KERO CR x) KERO CR to CBD	8"X300# 8"X300# 8"X300# 8"X300# 10"X300# 10"X300# 1"X150 1"X150 1"X150 1 ½"X150	B1A B1A B7A B1A B4F B4F A1A A1A A1A A1A
22.	12-E-02	Crude/ HVGO	i) Crude inlet valve downstream flange ii) Crude outlet valve upstream flange iii) HVGO inlet valve downstream flange iv) HVGO outlet valve upstream flange v) Cutter to crude vi) Crude to CBD 2 nos vii) Cutter to HVGO viii) HVGO to CBD 2 nos	6"X300# 6"X300# 6"X300# 6"X300# 1"X150 1 ½"X150 1"X150 1 ½"X150	B1A B1A B7A B7A A1A A1A A1A A1A
23.	12-E-03	Crude/ SR	i) Crude inlet valve downstream flange ii) Crude outlet valve upstream flange iii) SR inlet valve downstream flange iv) SR outlet valve upstream flange v) Cutter to crude vi) Crude to CBD 2 nos vii) Cutter to SR viii) SR to CBD 2 nos	8"X300# 8"X300# 8"X300# 8"X300# 1"X150 1 ½"X150 1"X150 1 ½"X150	B1A B1A B4F B7A A1A A1A A1A A1A
24.	12-E-04	Crude/ HVGO	i) Crude inlet valve downstream flange ii) Crude outlet valve upstream flange iii) HVGO inlet valve downstream flange iv) HVGO outlet valve upstream flange v) HVGO vertical line valve vi) Cutter to crude vii) Crude to CBD 2 nos viii) Cutter to HVGO ix) HVGO to CBD 2 nos	8"X300# 8"X300# 6"X300# 6"X300# 6"X300# 1"X150 1 ½"X150 1"X150 1 ½"X150	B7A B7A B7A B7A B7A A1A A1A A1A A1A



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S. No	Exchanger	Service	blinds	size & rating	spec
25.	12-E-05A/B	Crude/ HVGO	i) Crude inlet valve downstream flange ii) Crude outlet valve upstream flange iii) HVGO inlet valve downstream flange iv) HVGO outlet valve upstream flange v) Cutter to crude vi) Crude to CBD 2 nos vii) Cutter to HVGO viii) HVGO to CBD 2 nos	8"X300# 8"X300# 6"X300# 6"X300# 1 "X150 1 "X150 1 "X150 1 "X150	B7A D7A B7A B7A A1A A1A A1A A1A
26.	12-E-06A/B	Crude/ SR	i) Crude inlet valve downstream flange ii) Crude outlet valve upstream flange iii) SR inlet valve downstream flange iv) SR outlet valve upstream flange v) RCO inlet valve downstream flange vi) RCO outlet valve upstream flange vii) cutter to Crude side viii) crude to CBD ix) cutter to shell side x) SR to CBD	8"X300# 8"X300# 8"X300# 8"X300# 8"X300# 8"X300# 1 "X150 1 ½"X150 1 "X150 1 ½"X150	D7A D7A B4F B4F A1A A1A A1A A1A A1A A1A
27.	12-E-10	HVGO steam gen	i) HVGO inlet ii) HVGO outlet iii) MP steam outlet 2 nos iv) BFW inlet v) Cutter to HVGO vi) HVGO to CBD	6"X300 6"X300 2"X300 2"X300 2"X 150 1 ½ " X150	B7A B7A B2A B2A A1A A1A
28.	12-E-10A	HVGO steam gen	i) HVGO inlet ii) HVGO outlet iii) MP steam outlet 2 nos iv) BFW inlet v) Cutter to HVGO vi) HVGO to CBD	6"X300 6"X300 2"X300 2"X300 1"X 150 1 " X150	B7A B7A B2A B2A A1A A1A



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COOLERS AND CONDENSERS:

S. No	COOLER/ CONDESERS	SERVICE	BLINDS	SIZE & RATING	SPEC
1.	11-E-17 A/B	Atmos O/h vapors	i) Vapor inlet valve down stream flange ii) 2" service water line iii) Condenser outlet valve to drum upstream flange iv) To CBD v) Gas make up flanges 2 nos vi) Salt water inlet valve vii) Salt water outlet flange viii) Salt water reverse inlet valve ix) Salt water reverse outlet flange	16"X 150 2"X150 8"x150 1 ½" X150 2"X 150 10" 10" 10" 10"	A3A A9A A9A A1A A9A J5A J5A J5A J5A
2.	11-E-17 C/D	Atmos O/h vapors	i) Vapor inlet valve down stream flange ii) 2" service water line iii) Condenser outlet valve to drum upstream flange iv) To CBD v) Gas make up flanges 2 nos vi) Salt water inlet valve vii) Salt water outlet flange viii) Salt water reverse inlet valve ix) Salt water reverse outlet flange	16"X 150 2"X150 8"x150 1 ½" X150 2"X 150 10" 10" 10" 10"	A3A A9A A9A A1A A9A J5A J5A J5A J5A
3.	11-E-17 E/F	Atmos O/h vapors	i) Vapor inlet valve down stream flange ii) 2" service water line iii) Condenser outlet valve to drum upstream flange iv) To CBD v) Gas make up flanges 2 nos vi) Salt water inlet valve vii) Salt water outlet flange viii) Salt water reverse inlet valve ix) Salt water reverse outlet flange	16"X 150 2"X150 8"X150 1 ½" X150 2"X 150 10" 10" 10" 10"	A3A A9A A9A A1A A9A J5A J5A J5A J5A



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S. No	COOLER/ CONDESERS	SERVICE	BLINDS	SIZE & RATING	SPEC
4.	11-E-17 G/H	Atmos O/h vapors	i) Vapor inlet valve down stream flange ii) 2" service water line iii) Condenser outlet valve to drum upstream flange iv) To CBD v) Gas make up flanges 2 nos vi) Salt water inlet valve vii) Salt water outlet flange viii) Salt water reverse inlet valve ix) Salt water reverse outlet flange	16"X 150 2"X150 8"X150 1 ½" X150 2"X 150 10" 10" 10" 10"	A3A A9A A9A A1A A9A J5A J5A J5A J5A
5.	11-E-20 A/B	stabilizer O/h vapors	i) Stabilizer vapors to condenser inlet ii) LPG from condensers iii) to flare iv) make up or depressuring line v) Salt water inlet valve downstream flange vi) Salt water outlet valve upstream flange vii) Salt water reverse inlet valve viii) Salt water reverse outlet flange	6"X150 4"X150 1"X150 1 ½"X 150 8" 8" 8" 8"	A1A A1A A1A A1A J5A J5A J5A J5A
6.	11-E-20 C/D	stabilizer O/h vapors	i) Stabilizer vapors to condenser inlet ii) LPG from condensers iii) to flare iv) make up or depressuring line v) Salt water inlet valve downstream flange vi) Salt water outlet valve upstream flange vii) Salt water reverse inlet valve viii) Salt water reverse outlet flange	6"X150 4"X150 1"X150 1 ½"X 150 8" 8" 8" 8"	A1A A1A A1A A1A J5A J5A J5A J5A
7.	11-E-21	SRN	i) SRN inlet valve downstream flange ii) SRN outlet valve upstream flange iii) Take off to FCCU-II iv) SRN to CBD v) Salt water inlet valve vi) Salt water outlet flange vii) Salt water reverse inlet valve viii) Salt water reverse outlet flange	6"X300 6"300 3"X300 1 ½ "X150 6" 6" 6" 6"	B1A B1A B1A A1A J5A J5A J5A J5A



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S. No	COOLER/ CONDESERS	SERVICE	BLINDS	SIZE & RATING	SPEC
8.	11-E-22	LVGO	i) LVGO inlet valve downstream flange ii) LVGO outlet valve upstream flange iii) LVGO to CBD iv) Cutter to LVGO v) Salt water inlet valve vi) Salt water outlet valve	4"X300 4"X300 2"X150 1"X150 6" 6"	B1A A1A A1A A1A J5A J5A
9.	11-E-22A	LVGO	i) LVGO inlet valve downstream flange ii) LVGO outlet valve upstream flange iii) LVGO to CBD iv) Cutter to LVGO v) Salt water inlet valve vi) Salt water outlet valve	4"X300 4"X300 1 ½"X150 1"X150 6" 6"	B1A A1A A1A A1A J5A J5A
10.	11-E-23	Diesel	i) DSL from 11-E-03 or KERO from 11-E-02 inlet valve downstream flange ii) DSL/KERO to CBD iii) Cutter to DSL line iv) t/o to 11-E-24 v) to DSL storage vi) salt water inlet valve downstream flange vii) salt water outlet valve upstream flange	6"X300 1 ½"X150 1"X150 6"X300 6"150 6" 6"	B1A A1A A1A B1A A1A J5A J5A
11.	11-E-23A	Diesel	i) Dsl from 11-E-03 inlet valve ii) t/o to FCCU-II iii) DSL to CBD iv) Cutter to DSL v) DSL to storage vi) Salt water inlet valve vii) Salt water outlet valve	6"X300 2"X150 1 ½"X150 1"X150 6"X150 6" 6"	B1A A1A A1A A1A A1A J5A J5A
12.	11-E-24	KERO	KERO from 11-E-02 Cutter to KERO Kero from 11-E-24 outlet KERO to CBD Salt water inlet valve Salt water outlet valve	6"X300 1"X150 6"X300 1"X150 6" 6"	B1A A1A B1A A1A J5A J5A



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S. No	COOLER/ CONDESERS	SERVICE	BLINDS	SIZE & RATING	SPEC
13.	11-E-24A	KERO	KERO from 11-E-02 Kero from 11-E-24 outlet KERO to CBD Salt water inlet valve Salt water outlet valve	6"X300 6"X300 1"X150 6" 6"	B1A B1A A1A J5A J5A
14.	11-E-26	Heavy Naphtha	HN from 11-E-01 inlet valve HN to CBD HN from 11-E-26 Salt water inlet valve Salt water outlet valve	3"X150 1"150 3"X150 3" 3"	A1A A1A A1A J5A J5A
15.	12-E-08 A	Tempered water	i) tempered water from 12-E-09A-D-inlet ii) tempered water outlet valve iii) salt water inlet valve iv) salt water outlet valve v) salt water reverse inlet valve vi) salt water reverse outlet valve	8"X150 8"X150 10" 10" 10" 10"	A3A A3A J5A J5A J5A J5A
16.	12-E-08 B	Tempered water	i) tempered water from 12-E-09A-D-inlet ii) tempered water outlet valve iii) salt water inlet valve iv) salt water outlet valve v) salt water reverse inlet valve vi) salt water reverse outlet valve	8"X150 8"X150 12" 12" 12" 12"	A3A A3A J5A J5A J5A J5A
17.	12-E-09A/B	SR/ Tempered water	i) SR from 12-E-01 A/B/C inlet valve downstream flange ii) SR to run down. iii) SR to CBD 2 nos iv) Tempered water inlet valve v) Tempered water out let valve vi) Tempered water to OWS	6"X300 6"X300 1 ½"X150 6"X150 6"X150 ¾"X 150	B1A B1A A1A A3A A3A A3A
18.	12-E-09C/D	SR/ Tempered water	i) SR from 12-E-01 A/B/C inlet valve downstream flange ii) SR to run down. iii) SR to CBD iv) Tempered water inlet valve v) Tempered water out let valve vi) Tempered water to OWS	6"X300 6"X300 1 ½"X150 6"X150 6"X150 ¾"X 150	B1A B1A A1A A3A A3A A3A



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S. No	COOLER/ CONDESERS	SERVICE	BLINDS	SIZE & RATING	SPEC
19.	12-E-11	LVGO	i) LVGO inlet valve downstream flange ii) LVGO from 11-E-22/22A inlet iii) LVGO outlet iv) LVGO to CBD v) Cutter to LVGO line vi) Salt water inlet flange vii) Salt water outlet flange	4"X 300 4"X300 4"X300 1 ½"X150 1"X 150 6" 6"	B1A B1A B1A A1A A1A J5A J5A
20.	12-E-12A	HVGO	i) HVGO inlet from 12-E-04 ii) HVGO outlet r/d iii) Bypass valve from 12-E-12B iv) HVGO from 12-E-10/10A v) VGO to CBD vi) Cutter to VGO line vii) Salt water inlet flange viii) Salt water outlet flange	6"X300 6"X300 6"X300 6"X300 1 ½"X150 1"X150 6" 6"	B7A B7A B7A B7A A1A A1A J5A J5A
21.	12-E-12B	HVGO	i) LVGO inlet from 11-E-07 valve downstream flange ii) LVGO outlet iii) HVGO inlet from 12-E-04 iv) HVGO outlet r/d v) Bypass valve from 12-E-12A vi) VGO to CBD vii) Cutter to VGO line viii) Salt water inlet flange Salt water outlet flange	4"X 300 4"X300 6"X300 6"X300 6"X300 1 ½"X150 1"X 150 6" 6"	B7A B7A B7A B7A B7A A1A A1A J5A J5A

34.2.3 Heaters Blind List:

11F01 ENTRY BLIND LIST:

S. No	DESCRIPTION	SIZE	RATING	SPEC
1.	To install b/m on FO to individual burners(12nos)	¾ "	300	B1A
2.	FG c/v 1 st block valve downstream flange and bypass valve upstream flange	6 "	150	A1A
3.	4" yard air to decoking at pillar no 26	4 "	150	A3A
4.	Soot blowing steam valve at down stream flange	4 "	300	B2A
5.	LP smothering steam to take valve down stream flange	4 "	150	A2A
6.	Di -sulphide gas line valve at up stream valve down stream	¾ "	300	



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	flange			
7.	Hot well off gas line	3 "	150	A1A
8.	Atomizing steam take off valve down stream flange	3 "	300	B2A
9.	Super heater steam out let flange	6"	300	B2A
10.	Super heater steam in let flange	10"	300	B2A
11.	Pass A control valves upstream block valve downstream flange & its bypass valve upstream flange	6"	300	B7A
12.	Pass B control valves upstream block valve downstream flange & its bypass valve upstream flange	6"	300	B7A
13.	Pass C control valves upstream block valve downstream flange & its bypass valve upstream flange	6"	300	B7A
14.	Pass D control valves upstream block valve downstream flange & its bypass valve upstream flange	6"	300	B7A
15.	Pass A out let flange at heater	6"	300	B7A
16.	Pass B out let flange at heater	6"	300	B7A
17.	Pass C out let flange at heater	6"	300	B7A
18.	Pass D out let flange at heater	6"	300	B7A
19.	Purge steam (for each pass flow) 4 no's	2"	300	B2A

12F01 ENTRY BLIND LIST:

S. No	DESCRIPTION	SIZE	RATING	SPEC
1.	Fuel gas line take off valve flange (2 no's)	3"	150	A1A
2.	Atomizing steam take off valve downstream flange	3"	300	B2A
3.	Soot blowing steam take off valve down stream flange	4"	300	B2A
4.	Smothering steam take off valve down stream flange	6"	150	A2A
5.	FG line at EBL valve down stream flange	6"	150	A1A
6.	FO supply mass flow meter Downstream valve and bypass valve downstream (2nos)	3"	300	B1A
7.	FO return downstream of SDV	3"	150	B1A
8.	Turbulising steam (4 no's)	2"	300	B2A
9.	Purge steam (4no's)	2"	300	B2A
10.	Pass –A control valves upstream block valve downstream flange & its bypass valve upstream flange	4"	300	B4F
11.	Pass –B control valves upstream block valve downstream flange & its bypass valve upstream flange	4"	300	B4F
12.	Pass –C control valves upstream block valve downstream flange & its bypass valve upstream flange	4"	300	B4F
13.	Pass –D control valves upstream block valve downstream flange & its bypass valve upstream flange	4"	300	B4F



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14.	Radiation out let (4 no's)	8"	300	B2G
15.	4" yard air to decoking	4"	150	A3A
16.	Service water and steam combined line to APH	3"	150	A1A

Columns Blind List:

11C01 BLIND LIST:

S. No.	DESCRIPTION	SIZE& RATING	SPEC
1.	TOP OUTLET	24"X150	A3A
2.	VENT	3"X150	A9A
3.	REFLUX	6"X150	A1A
4.	SAFETY VALVE TAKE-OFF (BOILER MAKER)	14"X150	A1A
5.	TPA RETURN	12" X150	A1A
6.	TPA TAKE OFF (2 no's)	14"X150	A1A
7.	HEAVY NAPHTHA VAPOR RETURN	8"X150	A1A
8.	SIDE DRAW (HEAVY NAPHTHA)	6"X150	A1A
9.	KERO VAPOR RETURN	12"X150	A9A
10.	KERO CR RETURN	12"X150	A1A
11.	SIDE DRAW (KEROSENE)	18"X150	A9A
12.	DIESEL VAPOR RETURN	14"x150	A4F
13.	DIESEL CR RETURN	14"x150	A4F
14.	PFD VAPOR TO COLUMN	10"X300	B1A
15.	SIDE DRAWOFF (DIESEL)	14"X150	A4F
16.	DESALTER AND PFD RV OUTLETS	10"X150	A1A
17.	OVERFLASH DRAW	6"X150	B4F
18.	OVERFLASH RETURN	6"X150	B4F
19.	FEED WITH COMPANION FLANGE	24"X300	B4F
20.	BOTTOM STEAM	8"X300	B2A
21.	BOTTOM OUTLET	12"X300	A4F
22.	UTILITY CONNECTION	1 ½" X150	B4F

11C02 BLIND LIST:

S. No.	DESCRIPTION	SIZE& RATING	SPEC
1	FEED HEAVY NAPHTHA	6"X150	A1A
2	BOTTOM OUTLET OF HEAVY NAPHTHA	6"X150	A1A
3	TOP OUTLET HEAVY NAPHTHA	8"X150	A1A
4	VENT	2"X150	A9A
5	UTILITY CONNECTION	1 ½"X150	A9A
6	STRIPPING STEAM HEAVY NAPHTHA	4"X300	B2A



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11C03 BLIND LIST:

S. No.	DESCRIPTION	SIZE& RATING	SPEC
1	FEED KEROSENE	12"X150	A9A
2	BOTTOM OUTLET KEROSENE	8"X150	A9A
3	TOP OUTLET KEROSENE	12"X150	A9A
4	VENT	2"X150	A9A
5	UTILITY CONNECTION	1 ½ "X150	A9A
6	STRIPPING STEAM KEROSENE	6"X300	B2A

11C04 BLIND LIST:

S. No.	DESCRIPTION	SIZE& RATING	SPEC
1	FEED DIESEL	12"X150	A4F
2	BOTTOM OUTLET DIESEL	8"X150	A9A
3	TOP OUTLET DIESEL	14"X150	A4F
4	VENT	2"X150	A9A
5	UTILITY CONNECTION	1 ½ "X150	A9A
6	STRIPPING STEAM DIESEL	6"X300	B2A

11C05 BLIND LIST:

S. No.	DESCRIPTION	SIZE& RATING	SPEC
1.	VAPOR TAKEOFF	10"X150	A1A
2.	RV TAKE OFF	6"X300	A1A
3.	VENT	2"X300	A1A
4.	REFLUX FLOW	4"X300	A1A
5.	FEED ENTRY	6"X300	B1A
6.	REBOILER RETURN	18"X 300	B1A
7.	REBOILER DRAW OFF	16"X300	B1A
8.	SRN DRAWOFF	6"X300	B1A
9.	MP STEAM CONNECTION	1 ½ "X 300	A2A



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12C01 BLIND LIST:

S. No.	DESCRIPTION	SIZE& RATING	SPEC
1.	TOP OUTLET	36"X150	A1A
2.	VENT	4"X150	A1A
3.	SAFEY VALVE	10"X150	A1A
4.	LVGO REFLUX	4"X150	A1A
5.	LVGO DRAWOFF	10"X150	A1A
6.	HVGO CR+LVGO IR	8"X150	A3A
7.	HVGO DRAWOFF	18"X150	A4F
8.	HVGO IR	6"X150	A4F
9.	SLOPCUT DRAWOFF	8"X300	A4F
10.	PUMP OUT VENT	3"X300	B4F
11.	FEED	52"X300	B6A(C*)
12.	LP STEAM	3"X300	A2A
13.	QUENCH RETURN	6"X300	B7A
14.	BOTTOM OUTLET	12"X300	A4F
15.	UTILITY CONNECTION	1 ½"X300	A4F

32.4.5 Vessels blind list:

11-V-01 BLIND LIST:

S. No.	DESCRIPTION	SIZE& RATING	SPEC
1	OVERHEAD NAPHTHA FROM 11-E-17A-H	12"X 150	A9A
2	RV TO FLARE	4"X 150	A9A
3	TAKE OFF TO FLARE OR MAKEUP	6"X 150	A9A
4	VENT	2"X 150	A9A
5	REFLUX DRAW OFF	12"X150	A1A
6	BOOT TAKEOFF	3"X150	A9A
7	DRAIN	1 ½"X150	A9A

11-V-02 BLIND LIST: DESALTER:

S. No.	DESCRIPTION	SIZE& RATING	SPEC
1.	FEED INLET	10"X300	B1A
2.	FEED OUTLET	10"X300	B1A
3.	FEED OUTLET BOTTOM	8"X300	B1A
4.	RELIEF VAVLE TAKE OFF	10"X300	B1A



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5.	DESLUDGING WATER LINE UP (3 NOS)	2"X300	B1A
6.	VENT	3"X300	B1A
7.	EFFLUENT WATER TAKE OFF (2 NOS)	2"X300	B1A
8.	LP STEAM	2"X300	A2A
9.	BOTTOM DRAIN	¾"X 300	B1A
10.	DRAIN	6"X300	A1A
11.	TRI-COCKS	¾" X300	B1A

11-V-03 BLIND LIST: **STABILIZER OVER HEAD DRUM**

S NO	DESCRIPTION	SIZE& RATING	SPEC
1.	VAPORS FROM 11-E-20A-D	8"X150	A1A
2.	RV TAKE OFF	1 ½"X150	A1A
3.	STABILISER OFF GAS TO FUEL GAS	4"X150	A1A
4.	BYPASS LINE TO CONTROL VALVE	3"X150	A1A
5.	SPLIT CONTROL VALVE RETURN	3"X150	A1A
6.	VENT	2"X150	A9A
7.	UTILITY CONNECTION	1 ½"X150	A1A
8.	BOOT TAKE OFF	2"X150	A1A
9.	LPG PUMP SUCTION	8"X150	A1A

11-V-04 BLIND LIST: **DESALTER WATER TANK**

S. No.	DESCRIPTION	SIZE& RATING	SPEC
1	DM WATER MAEK UP	3"X150	A3A
2	DRAIN	3"X150	A3A
3	VENT	2"X150	A3A
4	SUCTION TO WASH WATER PUMPS	4"X150	A3A
5	MAB CONDENSATE	4"X150	A3A
6	TOP DRAIN	3"X150	A3A

11-V-05 BLIND LIST: **DECOKING POT:**

S. No.	DESCRIPTION	SIZE& RATING	SPEC
1	SERVICE WATER INLET	3"X150	A3A
2	VENT	20"X150	A9A
3	INLET	18"X300	B7A
4	DRAIN	6"X150	A9A



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11-V-06A/B BLIND LIST: AMMONIA SOLUTION TANKS

S. No.	DESCRIPTION	SIZE& RATING	SPEC
1	AMMONIA FEED	1 ½"X150	A3A
2	SERVICE WATER	2"X150	A3A
3	OVERFOLW	1"X150	A9A
4	DRAIN	¾"X150	A9A
5	SOLUTION TO 11-PM-14A/B/C	1"X150	A9A
6	COMMON CONNECTION TO DRUMS	1"X150	A9A

11-V-07A/B BLIND LIST: CAUSTIC TANKS:

S. No.	DESCRIPTION	SIZE& RATING	SPEC
1	CAUSTIC INLET	3"X150	A9A
2	SERVICE WATER MAKE UP	1 ½"X150	A3A
3	VENT	2"X150	A9A
4	AIR INLET	2"X150	A3A
5	CAUSTIC MAKE UP	2"X150	A1A
6	VENT	1"X150	A9A
7	TAKE OFF TO PUMP SUCTION	1"X150	A9A
8	DRAIN	¾"X150	A9A
9	OVERFLOW	2"X150	A9A

11-V-08 BLIND LIST: CBD DRUM

S. No.	DESCRIPTION	SIZE& RATING	SPEC
1	CBD INLET	8"X150	A1A
2	PUMP SUCTION MANWAY		
3	SERVICE WATER INLET	4"X150	A3A
4	COOLING WATER INLET	3"	J5A
5	COOLING WATER OUTLET	3"	J5A
6	VENT		
7	LP STEAM INLET	2"X150	A2A

11-V-10 BLIND LIST:

S. No.	DESCRIPTION	SIZE& RATING	SPEC
1	CRUDE BOTTOM DRAW OFF	14"X 300	B7A
2	CRUDE INLET	10"X300	D7A



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3	PFD VAPOR OUTLET TO 11-C-01	10"X300	B1A
4	RELIEF VALVE	8"X300	B1A
5	VENT	3"X300	B1A
6	DRAIN	6"X300	B1A
7	CUTTER CONNECTION	4"X300	A1A
8	DRAIN TO OWS	3"X300	B1A

12-V-01 BLIND LIST:

S. No.	DESCRIPTION	SIZE& RATING	SPEC
1.	1 ST STAGE EJECTOR CONDENSATE	4"X150	A1A
2.	2 ND STAGE EJECTOR CONDENSATE	4"X150	A1A
3.	3 RD STAGE EJECTOR CONDENSATE	3"X150	A1A
4.	UN-CONDENSABLE GAS	8"X150	A1A
5.	OFF GAS TAKE OFF	8"X150	A1A
6.	VENT	2"X150	A1A
7.	HOT WELL WATER TAKEOFF (2 NOS)	2"X150	A9A
8.	HOT WELL OIL TAKE OFF	2"X150	A1A
9.	WATER MAKE UP	3"X150	A1A
10.	UTILITY CONNECTION	1 ½ "X150	A1A
11.	DRAIN	2"X150	A9A

12-V-02 BLIND LIST:

S. No.	DESCRIPTION	SIZE& RATING	SPEC
1	TEMPERED WATER INLET	10"X150	A3A
2	TEMPERED WATER OUTLET TO 12-PM-07A/B	12"X150	A3A
3	OVERFLOW	3"X150	A3A
4	DM WATER MAKEUP	2"X150	A3A
5	TEMPERED WATER PUMP RETURN	3"X150	A3A
6	LP STEAM TO DRUM	1 ½ "X150	A2A

12-V-3 BLIND LIST:

S. No.	DESCRIPTION	SIZE& RATING	SPEC
1	1 ST INLET FEED	2"X150	A3A
2	2 ND INLET	2"X150	A3A
3	VENT	4"X150	A3A
4	SYPHON LOOP	2"X150	A3A
5	STEAM CONDENSATE	2"X150	A1A



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10-V-01 BLIND LIST:

S. No.	DESCRIPTION	SIZE& RATING	SPEC
1	SRN TO 10-V-01	3"X150	A9A
2	SRN OUTLET TO 10-V-02	6"X150	A1A
3	RELIEF VALVE	3"X300	A1A
4	VENT	2"X300	B1A
5	CAUSTIC TAKEOFF	6"X150	A9A
6	UTILITY CONNECTION	½"X150	A9A

10-V-02 BLIND LIST: NAPHTHA WASH WATER DRUM

S. No.	DESCRIPTION	SIZE& RATING	SPEC
1	SRN FROM 10-V-01	6"X150	A1A
2	RELIEF VALVE	3"X300	A1A
3	VENT	2"X300	B1A
4	SRN TAKE OFF	6"X 150	A1A
5	WATER MAKE UP	6"X150	A1A
6	UTILITY CONNECTION	½"X150	A9A

34.3 LIST OF VENDOR MANUALS:

The following vendor manuals are available in CDU II.

- i) Desalter manual
- ii) EIL design Package
- iii) ZEECO burner Manual
- iv) DCS Operating Manual
- v) APC manual



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34.4 START-UP & SHUTDOWN CHECK LISTS:

34.4.1 CDU-II START-UP CHECK LIST:

Pre start up check list:

S. No	Activity	Responsibility	Sign	Time	Remarks
1.	Information given to P&U for consumption of BCW, STEAM(MP/LP),BFW,DMW,FO AND UNBOOSTED COOLING WATER	Senior supervisor			
2.	Information given to FCCU-2/FCCU-R for consumption of FG	Senior supervisor			
3.	Ensured portable pilot igniter in charged condition.	PO			
4.	information given to MEROX for consumption of FLUSHING OIL	Senior supervisor			
5.	information given to CPP for consumption of power	Senior supervisor			
6.	Information given to PSS-20 for readiness of all motors.	Senior supervisor			
7.	information given to TPH for readiness of SLOP TANK and FEED BOOSTER	Senior supervisor			
8.	good housekeeping done	Senior supervisor			
9.	availability of instrument air	Senior supervisor			
10.	ensured availability of all critical pumps	Senior supervisor			
11.	readiness of FIRE WATER	atmos "A/B"			
12.	readiness of utilities (BCW,STEAM,DMW,BFW)	atmos "A/B"			
13.	check for readiness of FG and effective FO circulation for atmos heater	atmos "A/B"			
14.	check for readiness of FG and effective FO circulation for vac heater	vac "A/B"			
15.	readiness of feed PH neutralization dosing system, feed de-emulsifier system, atmos O/H PH neutralization dosing and atmos O/H filmer injection.	atmos "A/B"			



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16.	readiness of SRN caustic and water wash system	atmos "A/B"			
17.	readiness of vac O/H PH neutralization dosing	vac "A/B"			
18.	readiness of CBD pump	vac "A/B"			
19.	ensured CBD level is under control	vac "A/B"			
20.	ensured U/L feed B/V in closed condition	vac "A/B"			
21.	ensured cooling water to O/H condenser	atmos "A/B"			
22.	ensured stripping steams B/V's are closed	atmos "A/B"			
23.	ensured atmos heater stack damper in open condition	atmos "A/B"			
24.	ensured atmos heater is in natural draft	atmos "A/B"			
25.	ensured vac heater stack damper in open condition	vac "A/B"			
26.	Ensured vac heater is in natural draft.	vac "A/B"			
27.	ensured BFW to steam gens are isolated	vac "A/B"			
28.	ensured availability of pilot flames in atmos heater	atmos "A/B"			
29.	ensured availability of pilot flames in vac heater	vac "A/B"			
30.	proved clear SR and HVGO R/D's	vac "A/B"			
31.	ensured atmos column bottom level is under control	atmos "A/B"			
32.	ensured atmos column O/H accumulator level in under control	atmos "A/B"			
33.	Ensured all DCS software switches are in start up mode.	paner officer			
34.	ensured vac column bottom level is under control	vac "A/B"			
35.	ensured steam traps on atmos stripping steam header are working condition	atmos "A/B"			
36.	atmos heater pass FCV's and atmos column O/H PCV stroke checked	atmos "A/B"			
37.	vac heater pass FCV's stroke checked	vac "A/B"			
38.	ensured atmos stripper levels are pumped out	atmos "A/B"/PO			
39.	ensured vacuum is broken with FG	vac "A/B"			
40.	ensured vac. Column vent in closed position.(for breaking vacuum)	vac "A/B"			
41.	PPE box checking and found ok	vac "A/B"			



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Atmos start up check list:

S. No	Activity	Responsibility	Sign	Time	Remarks
1.	Ensured tempered water coolers are in service	Vac "A/B"			
2.	Maintain atmos column pr at 0.5 kg/cm ²	PO			
3.	Checked line up of feed circuit	Atmos "A/B"			
4.	Checked line up of RCO circuit for feed circulation	Atmos "A/B"			
5.	Checked for any leaks on flanges of feed and RCO circuits	Atmos "A/B"			
6.	1st phase condensate drained and from exchangers and columns (after one hour circulation and half an hour settlement)	Atmos "A/B"			
7.	Ensured atmos column bottom level gauge glass indication tallying with the level transmitter	Atmos "A/B"/ PO			
8.	Ensured all instruments are working on feed and RCO circuits	Atmos "A/B"			
9.	2nd phase condensates drained from exchangers and columns (after one hour circulation and half an hour settlement)	Atmos "A/B"			
10.	Line up of cooling water to overhead condensers	atmos "A/B"			
11.	ensured atmos heater in natural draft	Atmos "A/B"			
12.	Ensured heater FG line deblinded	Atmos "A/B"			
13.	Checked pilot flame healthiness in atmos heater	Atmos "A/B"			
14.	Checked main gas flame healthiness in atmos heater	Atmos "A/B"			
15.	Checked for any leaks on flanges of feed passes of atmos heater	Atmos "A/B"			
16.	Allowed COT to raise at the rate of 40 ⁰ c/hr	PO			
17.	Allowed atmos column Flash Zone temp at 120 ⁰ c for about two hours to remove condensate	PO			
18.	Ensured tempered water temp operates in between 60 ⁰ c and 80 ⁰ c	Vac "A/B"/ PO			



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19.	Ensured feed pump suction temp operates below 55 ⁰ c	Atmos "A/B"/PO			
20.	Checked for any leaks on flanges of feed and RCO circuits at 250 ⁰ c of atmos heater COT (hot bolting if required) and hold for 2hrs for temp profile of the column	Atmos "A/B"			
21.	Checked for any leaks on flanges of atmos heater inlet and outlet at 250 ⁰ c COT (hot bolting if required)	Atmos "A/B"			
22.	Checked atmos column bottom level, overhead accumulator H/C and interface levels are tallying with corresponding level transmitters	Atmos "A/B"/PO			
23.	Ensured that after each makeup of atmos column bottom level feed B/V at U/L is in isolated condition	Atmos "A/B"			
24.	Ensure all circulating reflux pump suction lines are free off condensate	Atmos "A/B"			
25.	Lined up circulating reflux circuits to column	Atmos "A/B"			
26.	Lined up product circuits to slop	Atmos "A/B"			
27.	Lined up atmos water to MEROX	Atmos "A/B"			
28.	Ensured atmos heater pass O/L' s are below 260 ⁰ c with 5 ⁰ c imbalance	Atmos "A/B"			
29.	Ensured atmos heater skin temp's are below 500 ⁰ C and arch temp is below 800 ⁰ C	Atmos "A/B"			
30.	place overhead condensers in service (if condensers inlet temp >65 ⁰ C)	Atmos "A/B"/PO			
31.	FEED CUT IN TO ATMOS SECTION	Atmos "A/B"/PO			
32.	Maintained pass flows above 50 m3/hr	Atmos "A/B"			
33.	Increased atmos heater cot to 350 ⁰ c	Atmos "A/B"			
34.	Increased atmos column pressure to 2.0 kg/cm2	Atmos "A/B"			
35.	Start up mode switches kept back to normal and RCO routed to vac heater after introduction of bottom stripping steam	Atmos "A/B"/PO			
36.	Atmos O/H temp controlled below 135 ⁰ c by using top reflux and routed un-	Atmos "A/B"/PO			



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	stabilized naphtha to slop				
37.	Started KERO CR bypassing stab reboiler after KERO stripper senses level	Atmos "A/B"/PO			
38.	Route Un-stabilizer Naphtha stabilizer and ensure bottom level is below 70%.				
39.	Route KCR thru reboiler and stab bottoms lined up to R/D and maintain stabilizer pressure @ 8.5Kg/Cm2g.	Atmos "A/B"			
40.	Maintained stab vapor line temp below 60° c by using stab reflux and LPG lined up to R/D	Atmos "A/B"/PO			
41.	Started DSL CR	Atmos "A/B"/PO			
42.	Started TPA	Atmos "A/B"/PO			
43.	Started HN product to slop	Atmos "A/B"			
44.	Started KERO product to slop	Atmos "A/B"			
45.	Started DIESEL product to slop	Atmos "A/B"			
46.	Adjusted all other process parameters suited to the type of feed	PO			
47.	Introduced stripping steams to strippers after removal of condensate	PO			
48.	HN routed to diesel R/D	Atmos "A/B"			
49.	KERO routed to diesel R/D	Atmos "A/B"			
50.	Diesel routed to diesel R/D (ensured diesel color is good)	Atmos "A/B"			
51.	Routed product samples to LAB	Atmos "A/B"			
52.	Feed booster placed I/S	Atmos "A/B"/PO			
53.	Desalter pressured up (ensured that pass flows are not Cascaded with total flow) at 10 kg/cm2	Atmos "A/B"/PO			
54.	Power to Desalter energized	Atmos "A/B"			
55.	Desalter effluent water lined up to MEROX	Atmos "A/B"			
56.	Wash water (service water/DMW) injection to Desalter started	Atmos "A/B"			
57.	Interface level counter checked with LT	Atmos "A/B"/PO			
58.	Ensured Desalter PT tallying with local PG	Atmos "A/B"/PO			
59.	Demulsifier injection started	Atmos "A/B"			
60.	Atmos O/H neutralizer injection started	Atmos "A/B"			
61.	Atmos O/H filter injection started	Atmos "A/B"			



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62.	HN to SRN/DIESEL R/D	Atmos "A/B"/PO			
63.	KERO to KERO R/D	Atmos "A/B"/PO			
64.	Commission PFD as per procedure.	Atmos "A/B"/PO			

Vac. Section start-up check list:

S.No	Activity	Responsibility	Sign	Time	Remarks
1.	Lined up vac bottom circuit for circulation	Vac "A/B"			
2.	Lined up BFW to HVGO steam generators	Vac "A/B"			
3.	ensure vac. Heater in natural draft	Vac "A/B"/PO			
4.	Checked pilot flame healthiness in vac heater	Vac "A/B"			
5.	Checked main gas flame healthiness in vac heater	Vac "A/B"			
6.	Allowed Cot to raise at rate of 40 ⁰ C/hr	PO			
7.	Checked for any leaks on flanges of vac column at 250 ⁰ C	Vac "A/B"			
8.	Lined up cooling water to surface condensers	Vac "A/B"			
9.	Checked for any leaks on flanges of feed passes of vac heater	Vac "A/B"			
10.	Checked for any leaks on flanges of VAC heater inlet and outlet at 275 ⁰ C of COT	Vac "A/B"			
11.	Checked for any leaks on flanges of VAC column at 275 ⁰ C of COT	Vac "A/B"			
12.	Ensured HW level tallying with LT	Vac "A/B"/PO			
13.	Ensured all instruments are in working condition on vac column and bottom circuit	Vac "A/B"/PO			
14.	Ensured all pump suction lines are free off condensate	Vac "A/B"			
15.	Lined up LVGO and HVGO R/Ds to HVGO R/D tank	Vac "A/B"			
16.	Lined up HWO to slop	Vac "A/B"			
17.	Lined up HWW to MEROX	Vac "A/B"			
18.	Ensured vac heater pass O/L's are	Vac "A/B"			



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	below 275 ⁰ C with 5 ⁰ C imbalance				
19.	Ensured vac heater skin temp's are below 570 ⁰ C and arch temp is below 776 ⁰ C	Vac "A/B"			
20.	Ejector lined up (1A/B/C, 2A/B/C,3A/B/C)	Vac "A/B"			
21.	FEED CUT IN TO VAC SECTION	Vac "A/B"/ PO			
22.	Maintained pass flows above 18 m3/hr	PO			
23.	SR routed to RFO/HFO	Vac "A/B"			
24.	Vacuum pulled	Vac "A/B"/ PO			
25.	HWW routed to MEROX	Vac "A/B"			
26.	HWO routed to slop	Vac "A/B"			
27.	Increased vac heater COT to 385 ⁰ C	PO			
28.	HVGO product routed to SLOP	Vac "A/B"			
29.	Slop cut routed to SR	Vac "A/B"			
30.	Adjusted all other process parameters suited to type of feed	PO			
31.	Vac O/H neutralizer injection started	Vac "A/B"			
32.	Vac O/H filmer injection started	Vac "A/B"			
33.	Routed product samples to LAB	Vac "A/B"			
34.	LVGO product routed to HVGO R/D TK	Vac "A/B"			
35.	HVGO product routed to HVGO R/D TK	Vac "A/B"			
36.	HW off gases routed to 11-F-01 and vent closed	Vac "A/B"/ PO			



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34.4.2 CDU-II PLANNED SHUTDOWN CHECKLIST:

S no	Activity	required / not required	done/ not done	Sign	Time	Remarks
1.	Clearance to be taken from manager & YSF					
2.	Information to be given to CPP					
3.	Information to be given to SS20					
4.	Information to be given to P& U for the expected reduction of BCW, Steam(MP/LP), BFW, DMW, FO & Cooling water consumption					
5.	Information to be given to TPH for readiness of slop tank					
6.	Information to be given to FCCU-II, VRCFP & DHDS before the removal of hot feed					
7.	Information to be given to FCCU's for the expected reduction of FG consumption.					
8.	Information to be given to MEROX for the expected reduction of stripped water flow consumption.					
9.	Ensured readiness of CBD pump					
10.	Ensured all hot work permits stopped.					
11.	Ensured readiness of Fire water					
12.	Reduce feed rate gradually from 500 m ³ /hr to 430 m ³ /hr. and accordingly adjust other parameters like Heater firing, stripper levels, Reflux drum levels etc					
13.	Slowly increased the PFD pressure to ensure no vaporization.(PFD pressure C/V closed fully)					
14.	After PFD level reduced to about 20%, PFD by-pass valve to be opened					
15.	11-PT-02B turbine speed reduced slowly (informed power plant), Level C/V to be closed fully.					
16.	11-C-01 tray 14 shell V/V from PFD vapor line to be closed					
17.	11-E-40 crude side to be bypassed					



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18.	Reduce feed rate from 430 M3/Hr to 350 M3/Hr slowly.					
19.	Simultaneously product R/D flows & CR flows to be reduced (first TPA, DSL CR, then KERO CR)					
20.	Hot feed to FCCU-II to be taken out					
21.	12-E-12 top cooler changed over to HVGO mode from LVGO CR mode.					
22.	12-E-06 A/B changed over to SR mode from RCO mode.					
23.	Disulphide off gas to 11-F-01 taken out & line blinded					
24.	Desalter water injection reduced slowly & finally taken out					
25.	Desalter to be taken out					
26.	R/D flows & CR flows further to be reduced in atmos. & vac. column					
27.	12-F-01 COT slowly reduced.FO & FG trip bypass taken in manual					
28.	Hot well off gas to 11-F-01 no. 17, 18, 19 and 20 burners to be taken out & vented to atmosphere.					
29.	Off gas line to burners to be blinded					
30.	When the Vac. furnace COT reduced to 300 deg C, furnace is taken out from balanced draft & placed in natural draft. First stack damper opened, then ID fan stopped, then DODs opened & simultaneously air flow to furnace reduced & finally FD fan stopped					
31.	Slop cut to be taken out from CDU-I cooler box (tk.4) & line cutter flushed					
32.	When Vac. furnace COT reduced below 300 deg C, all the fires to be taken out.					
33.	FG & FO line to furnace blinded					
34.	BFO kept under circulation					
35.	take out vac. Feed					
36.	ejector steam flow to be taken out & lines blinded					
37.	Vac. column product pumps (LVGO & HVGO) to be stopped after the levels came down. Pump d/s & R/D v/v closed					



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38.	Slop cut to be recycled back to 12-F-01 until the level came down. Then slop cut pump to be stopped.					
39.	Vac. section kept under circulation. 12-C-01 bottom --12-PM-01 -- 12-F-01—12-C-01					
40.	remove FG blind to vac. Column					
41.	FG introduced in vac. column to maintain positive pressure					
42.	Feed slowly reduced from 350 m ³ /hr to 250 m ³ /hr. At this flow rate feed booster pump 11-PM-02B stopped					
43.	Diesel to DHDS taken out					
44.	All the chemical injections taken out					
45.	When atom. Furnace COT came down to 270-280 deg C furnace to be taken out from balanced draft to natural draft. First stack damper opened, then ID fan stopped, then DODs opened & simultaneously air flow to furnace reduced & finally FD fan stopped					
46.	As the KERO CR flow came down to 200 m ³ /hr, KERO CR booster pump stooped.					
47.	All fires to be taken out from atmos. Furnace and pilot burners					
48.	All the product pumps & CR pumps stopped except top reflux pump					
49.	Pump d/s V/Vs & R/D V/Vs closed.					
50.	Reflux pump stopped when reflux drum level came down. Pump d/s V/Vs closed					
51.	Atmos. Column bottom & strippers stripping steams stopped & shell B/Vs closed.					
52.	Stabilizer feed C/V to be closed					
53.	CDU-I unstabilized naphtha to be taken out and v/v to be closed					
54.	reduce stabilizer bottom level to minimum SRN C/V to be closed					
55.	When stabilizer reflux drum level came down LPG pump to be stopped. Pump d/s & R/D V/Vs closed.					
56.	Atmos. Section kept under circulation.					



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57.	unit limit feed v/v to be closed					
58.	11-C-01 Bottom—11-PM-10 B—SR circuit—11-PT-01B—11-F-01 (by-passing Desalter & PFD) –11-C-01					
59.	Cutter to be taken at the top of vac. column & LVGO level build up, pumped out by LVGO pump & again put back into vac. Column					
60.	Build up HVGO level and pump out to slop & the whole HVGO circuit to be cutter flushed & again put back to vac. column.					
61.	Build up slop cut level with cutter and pump out the level to SR After that pumps to be stopped. D/S & R/D V/Vs closed					

34.5 LEL DETECTORS STATUS:

S No.	TAG. No.	LOCATION	TYPE	REMARKS
1	HCD1001	Stabilizer reflux pumps (11-PM-11A/B south side)	HC	
2	HCD1002	DL1503 boot drain point/ OWS	HC	
3	HCD1003	11-F-01 FD fan suction	HC	
4	HCD1004	12-F-01 FD fan suction	HC	
5	HCD1005	South east of Atmos Over head drum	HC	
6	HCD1006	North east of Atmos Over head drum	HC	
7	HCD1007	South east of over head condensers	HC	
8	HCD1008	North east of over head condensers	HC	
9	HCD1009	Top of 11-V-04 wash water drum	HC	
10	HCD1010	South west of 11-V-04 wash water drum bottom	HC	



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34.6 INSTRUMENT AIR FAIL TO OPEN CONTROL VALVES IN CDU-2

S. No.	CONTROL VALVE NUMBER	DESCRIPTION
1.	11DPV301	Atomizing Steam to FO DP
2.	11FV101	Vac. split control valve
3.	11FV204	Kero to diesel
4.	11FV301	11F01 Pass1
5.	11FV302	11F01 Pass2
6.	11FV303	11F01 Pass3
7.	11FV304	11F01 Pass4
8.	11FV403	Reflux
9.	11FV404	DSL C/R
10.	11FV405	Kero C/R
11.	11FV406	Top Pump Around CR
12.	11FV501	Stab. RX.
13.	11LV101	Desalter interface level
14.	11LV902	PFD level
15.	11PV101	PFD bypass SDV
16.	11PV105	Desalter pressure
17.	11PV409B	Makeup control valve
18.	11PV501B	Stab makeup.
19.	11PV902	PFD pressure
20.	12DPV103	12F01 Atm. steam & FO
21.	12FV101	12F01 Pass1
22.	12FV102	12F01 Pass2
23.	12FV103	12F01 Pass3
24.	12FV104	12F01 Pass4
25.	12FV109	Slop cut Recycle



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26.	12FV201	Hot LVGO RX Flow
27.	12FV202	Hot HVGO Rx
28.	12FV203	Cold HVGO Rx.
29.	12TR201	SR Quench
30.	12FV205	LVGO top reflux
31.	12LV301	12E10 level
32.	12LV302	12E10A level
33.	12PV207	Ejector pressure
34.	12TV103	Cold SR to VBU
35.	13FV106	Air to BBU reactor

NOTE: All shutdown valves (SDVs), will close and all STACK DAMPERS and DOD's will open during instrument air failure.

34.7 DCP CYLINDERS, FIRE HOSE REELS AND SAFETY SHOWERS LOCATIONS:

DCP BOX NO.	LOCATION
1.	West of CSS - 20
2.	North West of MOI
3.	East of 11 - F - 1
4.	South of 11 -F - 1
5.	East of 11 - F - 1 (FD Fan)
6.	South of 11 - E - 21
7.	South of 11 - E - 14 AB
8.	West of De Salter
9.	West of Unit
10.	East of Field Room
11.	North of Vac Heater



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12.	East of 12 - PM - 01A
13.	East of 11 - PM - 09A
14.	East of 12 - PM - 04A
15.	East of 13 - PM - 02B
16.	Near 11 - PM - 12B
17.	West of 11 - PM - 02A
18.	South of 1 st Floor (11 - V - 01)
19.	South of 1 st Floor (11 - V - 01)
20.	South of Column (2 nd Floor)
21.	South of 2 nd Floor(11-E -17E/F)
22.	South of 2 nd Floor(11-E-20A/B)
23.	North Floor near 12 - E - 07A
24.	3 rd Floor Vac Column
25.	Top Floor Vac Column
26.	Near 12 - E - 7C
27.	Atmos Heater
28.	Vac Heater
29.	Vac Heater (12 - F -1)
30.	Atmos Heater Top
31.	Atmos Column (8 th Landing)
32.	Extreme Top on 11 - C -01
33.	One CO ₂ extinguisher at field room

FIRST AID FIRE HOSE REELS

HR. No	LOCATION
1.	South of 11 - F - 1 (ID Fan)
2.	South of 11 - E - 12
3.	East of De Salter Drum



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4.	South West Corner
5.	South West of 11 - PM - 02A
6.	North of 11 - PM - 02A
7.	Near 12 - E - 01 A/ B
8.	Near 12 - PM - 04B
9.	East of 13 - KM - 01B
10.	North of Vac Column
11.	North of Vac Heater
12.	East of Vac Heater
13.	West of Vac Heater
14.	West of Vac Column
15.	O/H Condenser 1 st Floor South of 11 - V - 03
16.	O/H condenser 2 nd Floor S/E of 11 - E - 17 E/F
17.	Near O/H Condenser North of 12 - E - 07A
18.	South of 11-F-01 (ID FAN)
19.	South of 11-E-16

SAFETY SHOWERS

Safety shower number	Location
1	12-F-01 Heater Area - East Side
2	Chemicals Pump Area- North Side
3	10-PM-01A/B Caustic Pump Area – South Side
4	11-F-01 Heater Area - East Side
5	10-V-01 & 10-V-02 Area - North Side
6	BBU Compressor Area- Towards North side



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34.8 AUTO IGNITION TEMPERATURES:

The following are the auto ignition temperatures of the products handling:

S no	Liquid	Auto ignition temperature
1	Crude	500-850 °F
2	LPG	466.1 °C
3	Naphtha	232 °C
4	Motor spirit	280-456 °C
5	Kerosene	280 to 456 °C
6	ATF	230 °C
7	Diesel	257 °C
8	LDO	257 °C
9	JBO/MTO	253 °C
10	Fuel Oil 360 cst	487 °C
11	Fuel Oil 180 cst	500 °F
12	Bitumen	>300 °C
13	Sulphur	232 °C
14	H ₂ S	260 °C
15	Methane	537 °C
16	Ethane	515 °C
17	Ethylene	543 °C
18	Butane	420 °C
19	Propane	450 °C
20	Propylene	472 °C
21	Carbon monoxide	700 °C

34.9.1 A. Provision of Corrosion Coupons, Corrosion Probes

1. Corrosion Probes

S. No.	Tag	Loop No.	Line Description	Location	Service	Line size	Piping Class
1	11-CP-301	13	10"-P-11-319-D7A-Ih	Crude to Atmos Furnace	Crude	10"	D9A/CS
2	11-CP-401	52(B)	14"-P-11-410-A4F-Ih	Diesel CR + Product drawoff line	Diesel	14"	A4F/P5(5 Cr)
3	12-CP-202	72	8"-P-12-222-	LVGO Pump 12-P-04 A/B	LVGO	8"	A1A/CS



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			A1A-It	Suction Line			
4	12-CP-301	76(E)	6"- P-12-318-B7A-It	HVGO exchanger 12-E-05A/B Outlet	HVGO	6"	B7A/CS
5	12-CP-203	73(A)	16"-P-12-214-A4F-It	HVGO pump 12-P-03A/B suction line	HVGO	10"	A4F/P5(5 Cr)

2. Corrosion Coupons

1	11-CC-301	13	10"-P-11-319-D7A-Ih	Crude to Atmos Furnace	Crude	10"	D9A/CS
2	11-CC-401	52(B)	14"-P-11-410-A4F-Ih	Diesel CR + Product drawoff line	Diesel	14"	A4F/P5(5 Cr)
3	12-CC-202	72	8"-P-12-222-A1A-It	LVGO Pump 12-P-04 A/B Suction Line	LVGO	8"	A1A/CS
4	12-CC-301	76(E)	6"- P-12-318-B7A-It	HVGO exchanger 12-E-05A/B Outlet	HVGO	6"	B7A/CS
5	12-CC-203	73(A)	16"-P-12-214-A4F-It	HVGO pump 12-P-03A/B suction line	HVGO	10"	A4F/P5(5 Cr)
6	12-CC-201	83	6"-P-12-208-B4F-It	Slop cut pump 12-P-02A/B Suction	Slop cut	6"	A4F/P5(5 Cr)
7	12-CC-204	73(A)	12"-P-12-201-A4F-It	Vac Residue pump 12-P-01A/B pump section	Vacuum Residue	12"	A4F/P5(5 Cr)

B. VR improvement schemes

1. ER Probes

1	CPI 1401	26	8"-P-11-437-2-A9A	O/L of 11-E-17E/F	Light Naptha	8"	A9A
2	CPI 1402	26	12"-P-11-437-A9A	I/L of 11-V-01	Light Naptha	12"	A9A
3	CPI 2201	68	4"-P-12-247-1-A1A	O/L of 12-E-7A	Vac O/H condensate	4"	A1A

2. PH Meter

1	PH 2201	88	1"-A9A	O/L hotwell drum near 12-LV-201	Sour Water	1"	A9A
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34.9.2 FSM: PROCESS /LOCATION DETAILS FOR CDU2

S l N o	Location	Service	State	Line Size	Parameters				Piping classes / Metallurgy	Max flow T/Hr	Probe metallurgy	Corroive constituent	Bend
					Operating conditions		Design conditions						
					Pressure Kg/cm 2g	Temperature Deg.C	Pressure Kg/cm 2g	Temperature Deg. C					
FSM (FOR MONITORING NAPHTHENIC ACID CORROSION)													
1	Atmos furnace 11-F-01 Radiation outlet first elbow 6"-P-11-314-B4F-Ih	Crude	Vapor + Liquid	6"	3.5	350-378	4.8	400	B4F / P5 (5Cr)	112.6	Not applicable	Sulphur, H2S (ppm), CO2 (ppm), Naph Acid	On elbow



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2	Atmospheric Transfer line 24"-P-11-318-B4F-Ih	Crude	Vapor + Liquid	24"	3.5	350-378	4.8	400	B4F / P5 (5Cr)	450		Sulphur, H ₂ S (ppm), CO ₂ (ppm), Naph Acid	-
3	Vacuum furnace 12-F-01 Radiation outlet first elbow 8"-P-12-115-B2G-Ih	Reduced Crude Oil	Vapor + Liquid	8"	50 mm Hg abs	410	3.5 / Full Vacuum	425	B2G / P9 (9Cr)	62		Sulphur, H ₂ S (ppm), CO ₂ (ppm), Naph Acid	On elbow

Standing instructions No 32

Procedure for monitoring online ER probes, PIN matrixes and Corrosion coupons for High Acid crudes in CDU-II

Objective: To monitor the system corrosion and keep it in the limited values while processing HAC in the unit.

Background: CDU-II was designed for BH Crude. Later it was planned to process HAC in CDU-II for getting better margins. During the 2010 T&I, projects had installed corrosion probes, corrosion coupons, PH meter and Pin Matrix in CDU-II for monitoring the corrosion factor when processing HAC. Monitoring the above parameters and injecting the chemicals with proper dosage will reduce the corrosion factor of lines and equipment's.

Scope: This standing instructions is generated for monitoring the probes and injecting the chemicals with proper dosage for reducing the corrosion factor when processing the HAC in CDU-II.

Responsibility: The overall responsibility to implement these guidelines rests with the unit shift in charge.

Standing Instructions:

The following are to be observed while processing HAC crudes in CDU-II :

- PIN matrix device: In CDU-II ROXAR group had installed pin matrix devices in 11F01 pass B inlet and outlet lines and 12F01 pass B outlet lines for monitoring the corrosion factor. Dedicated PC is provided in MOI for monitoring the corrosion rates. Actual corrosion rate will display and there are calculated in time bound period. Maximum allowable corrosion factor is 5mils/year.**



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2. ER probes: These were installed in the following locations for monitoring the corrosion rate during processing HAC. Below are the locations where corrosion probes are installed in CDU-II.

S.No	Tag	Location	Service
1	CI301	Crude to Atmos furnace	Crude
2	CI400	Disel CR+Product drawoff line	Diesel
3	C2202	LVGO pump 12-P-04 A/B suction line	LVGO
4	C2301	HVGO exchanger 12-E-5 A/B outlet	HVGO
5	C2203	HVGO pump 12-P-3 A/B suction line	HVGO
6	C1401	11E17E/F O/L	Naphtha
7	C1402	11V01 I/L	Naphtha
8	C2201	12E07O/L	VAc. O/H
9	PH2201	Hotwell water	Sour water

In every shift the readings are to be noted (0400hrs, 1200hrs and 2000hrs) and corrosion rate to be calculated.

- a) The following calculations are to be done once in the shift for measuring the corrosion rate (mils/yr):

For Example: value of C1402 at 31st Dec at 0400hrs :736076mil.

Value of C1402 at 31st Dec at 1200hrs: 7.6114mil

Corrosion rate = (present value –previous value)*365*24/(diff in hrs)

= (7.6114-7.6076)mil*365*24/8= 4.161mil/yr.



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When the value of ER probe reaches 9.5mil(ER probe range 0-10 mil), it has to be informed to Tech-PAD for the replacement of the probe.

If the corrosion rate is more than 5mils/yr, the chemical dosing should be increased. Corrosion rate and pH of the overhead system should be controlled by increasing corrosion inhibitor and caustic injection to Desalter. HAC chemical dosing should be increased to control the corrosion rate of side stream products. The HAC chemical dosage rates will be provided by Tech-PAD.

The following tags are to be monitored and rate at which the value decreases implies corrosion rate and proper action to be taken. To reduce the corrosion rate the chemical injections are to be introduced and the following are the locations where chemicals are introduced.

Injection point	Diluent	Source of diluent	Diluent flow tag
Suction of Crude booster pumps 11P02A	Crude	Crude from common discharge of 11P01A/B	F1604
Diesel CR and Product common draw off line from 11C01	Diesel	11-E-03 (Dsl/crd) down stream	F1605
Suction of RCO pumps 11P10 A/B.	Diesel	11-E-03 (Dsl/crd) down stream	F1606
Suction of LVGO pumps 12P04 A/B	HVGO	12-E-10/10A to 12-C-01	F2607
Suction of HVGO pumps 12P03 A/B	HVGO	12-E-10/10A to 12-C-01	F2608
Suction of SR pumps 12P01 A/B	HVGO	12-E-10/10A to 12-C-01	F2609

If the dosing rate reaches maximum limit but the corrosion rate does not come down, it has to be informed to Tech-PAD for further advice.

- b) Atmos water and hot well water pH are to be maintained between 5.5 and 6.5. Online monitoring devices are installed to check HWW pH (PH2201) and high and low values alarms were provided at 5.5 and 6.5. Corrective actions need to be taken in case the value deviate from the given range. The main aim to maintain the pH between 5.5 and 6.5 is when pH is acidic more corrosion takes place and leads to depletion of pipelines and equipment. If pH is above 7, then salts will form and plug the Overhead Condensers.



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3. Corrosion coupons: indicates the corrosion in a particular stream. These are monitored in field. The coupons should be removed once in a month with the help of maintenance (raise respective job card) and send to lab for weighment for pre-determined period. The difference in weight of the corrosion coupon will be monitored by Technical (PAD and inspection). The corrosion coupons are inserted at the following locations:

S.No	Tag	Location	Service
1	11-CC-301	Crude to Atmos furnace	Crude
2	11-CC-401	Disel CR+Product drawoff line	Diesel
3	12-CC-202	LVGO pump 12-P-04 A/B suction line	LVGO
4	12-CC-301	HVGO exchanger 12-E-5 A/B outlet	HVGO
5	12-CC-203	HVGO pump 12-P-3 A/B suction line	HVGO
6	12-CC-201	Slop cut 12-P-02 A/B suction	Slop cut
7	12-CC-204	VR pump 12-P-01 A/B pump section	Vac residue

4. Corrective actions:

- Prior to changing over to HS(1hr before spiking/ switch), the following chemical injection rates are to be maintained:

S.No	Chemical	Dosage rate w.r.t crude feed rate	Lt/hr	Pump max flow	Pump stroke
1.	DMF	5PPM	10	12	80%
2.	Ammonia	2PPM	24	20	50% with 2 pumps
3.	Filmer	3PPM	5.5	10	50%
4.	Caustic	5PPM	58	75	80%



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Caustic should be introduced downstream of desalter and injection should start along with the crude changeover activity.

The above dosage rates are to maintained to ensure minimal corrosion in the system. However the dosage rates can be optimized based on the actual corrosion rate.

- In addition to the above, wash water rate of 5 to 6% on crude rate should be maintained and mix valve DP to be maximized (target value: 05 Kg.cm2)
- If pH is less or more 6.5, ammonia dosage is to be increased or decreased respectively.

While processing HAC, the HAC chemical (EC1245A) has to be introduced at the target locations mentioned in the table.



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Introduction to Utility System

The supply of utilities is common to CDU / VDU and BBU. The Utility System consists of

- Instrument Air (IA)
- Plant Air (PA)
- once through cooling (Sea) Water
- Bearing Cooling Water
- Service Water (WS)
- DM Water (DMW)
- Boiler Feed Water (BFW)
- Tempered Water (TW)
- LP Steam
- MP Steam
- HP Steam
- Fuel Oil (FO)
- Fuel Gas (FG)
- Flushing Oil (FLO).
- Flare system

35.1.1 INSTRUMENT AIR SYSTEM

(Refer P&ID 10-1100-E-203 Rev.5):

The existing compressed air system at MEROX plant has an installed capacity of 15,000 m³/hr and comprises three big Elliott air compressors, each of capacity 5000 m³/hr. A 2" Instrument Air header supplies IA to CDU/VDU-II and BBU. The header is provided with an isolation valve and a spectacle blind. IA is used as motive force for pneumatically operated control valves and for the operation of 11-F-01 and 12-F-01 soot blowers. A pressure gauge 11-PG-709, a temperature gauge 11-TG-708, a low pressure switch 11-PAL-703 and a flow integrator 11-FQ-705 are provided on the header.

35.1.2 PLANT AIR SYSTEM

(Refer P&ID 10-1100-E-203 Rev.5):

A 4" plant air header supplies plant air to the entire unit. The header is provided with an isolation valve and a spectacle blind for positive isolation at the battery limit. A number of utility points are provided from this header. Dedicated 4" & 2" tapings are taken from main header for decoking of Atmospheric Furnace & Vacuum Furnace, respectively. A 2" tapping



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is provided to 11-V-07 A/B (Caustic solution tanks) for agitation. There is a provision to route Plant Air to the Instrument Air header at the unit limit. There is also a provision to route Plant Air to BBU reactor through 13-PIC-102. A pressure gauge 11-PG-710, a temperature gauge 11-TG-709, and a flow integrator 11-FQ-708 are provided on the header.

35.1.3 ONCE-THROUGH COOLING (SEA) WATER SYSTEM

(Refer P&ID 10- 1100-E-201-1 Rev.6):

The cooling water requirement is met by the Once-through Cooling Water System in which sea water is used. Sea water is supplied in two headers to the unit. One header of 30" size, known as the un-boosted water header, caters to the cooling water requirement at the ground level and its supply pressure is 2.5 kg/cm² g. The other header is a 24" one, known as the boosted water header, caters to the cooling water requirement to the overhead condensers and its supply pressure is 3.5 kg/cm² g. The connected salt water booster pumps (11-P-17 A/B) are there in FCCU-II.

Each of the supply headers is further bifurcated as follows

30" header into two headers of 26" and 18"

26" header into two headers of 18" and 26"

Both the return headers join one 36" header and go out of the unit to WWTP. There is battery limit isolation with blinding facility for both the supply and return headers. A pressure gauge 11-PG-701 and a temperature gauge 11-TG-702 are provided on the return header.

The following coolers have been provided with supply and return headers:

10" line to / from 11-E-17A to H

8" line to / from 11- E- 20A to D

6" line to / from 11-E-21

6" line to / from 11-E-23/23A

6" line to / from 11-E-21A

6" line to / from 11-E- 24/24A

6" line to / from 11-E- 22/22A

3" line to / from 11-E-26

18" line to / from 12-E-07A/B/C

10" line to / from 12-E-12B

6" line to / from 12-E-12A

6" line to / from 12-E-11

12" line to / from 12-E-08 A/B



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35.1.4 BEARING COOLING WATER SYSTEM

(Refer P&ID 10-1100-E-201-2 Rev.6):

Circulating bearing cooling water is used for pump gland and bearing cooling, turbine oil coolers etc., where use of sea water is prohibited. The BCW system for VREP-I caters to the requirement in CDU/VDU-II, FCCU-II, PP-II, MEROX, air compressors, off sites and PRU.

A bearing cooling water system with 2 cooling tower cells near the IFO system of the refinery complex meets the additional bearing cooling water requirement of units in VREP-II. A 4" BCW supply line and 4" and 6" return headers cater to the entire load of the unit. There is a battery limit isolation valve with a spectacle blind. Pressure gauges 11-PG-720/721, temperature gauges 11-TG-716/717 are provided on the supply and return headers respectively.

35.1.5 SERVICE WATER SYSTEM

(Refer P&ID 10-1100-E-203 Rev.5):

Service water is required mainly for cleaning and washing. A 4" service water header caters to the entire unit. It is provided with an isolation valve along with a spectacle blind at battery limit. A pressure gauge 11-PG-711, a temperature gauge 11-TG-710, and a flow integrator 11-FQ-709 are provided on the header at the battery limit.

The service water header supplies water to various hose stations in the plant, 11-V-05 (decoking pot), CBD drum (11-V-08), Caustic solution vessels (11-V-07 A/B) for dilution of concentrated Caustic solution, Ammonia solution tanks (11-V-06 A/B), LVGO product / CR pumps (12-P-04 A/B), Crude Charge pump (11-P-01 A/B) suction, to air pre-heaters (11-E-27, 12-AP-01) and to BBU off-gas quench drum (13-V-02). Service water is also provided for steam blow down drum (12-V-03), as makeup to Desalter water drum (11-V-04), to the suction of 11-P-06 A/B (for washing 11-V-01) and to other equipment within the unit.

35.1.6 DM WATER SYSTEM

(Refer P&ID 10-1100-E-203 Rev.5):

A 4" header caters to the DM water need for the entire unit. It is provided with an isolation valve and a spectacle blind at the battery limit. A pressure gauge 11-PG-714, a temperature gauge 11-TG-713, and a flow integrator 11-FQ-706 are provided on the header at the battery limit.



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DM water is used as Desalter wash water in (11-V-04) and in the tempered water drum (12-V-02) for level make-up.

35.1.7 BOILER FEED WATERSYSTEM

(Refer P&ID 10-1100-E-203 Rev.5):

A 3" BFW line caters to the entire load of the unit. It is provided with an isolation valve and a spectacle blind in the battery limit. A pressure gauge 11-PG-13, a temperature gauge 11-TG-712, and a flow integrator 11-FR/FQ-710 are provided on the header at the battery limit. It serves the following points

- 1" line for de-superheating the steam exit of Atmospheric furnace
- 2" line for HVGO CR steam generators 12-E-10/10A

The line then goes to the Bitumen plant to supply BFW to the steam generators. And splits into

- ✓ 2" line for BBU steam generators 13-E-01A/B, 13-E-03/03A.
- ✓ 1 ½" line to BBU reactor top.

35.1.8 LP STEAM SYSTEM

(Refer P&ID 10-1100-E-203 Rev.5):

A 6" LP Steam header caters to the requirement of the entire unit. The header is provided with isolation valves and **spectacle blind**. A pressure gauge 11-PG-712, a temperature gauge 11-TG-711, and a flow recorder / integrator 11-FR/FQ-707 are provided on the header at the B/L. A 4" steam line is provided on the superheated MP steam line from the Atmospheric Furnace. This line is to be used only when the LP steam header is low or when the superheated MP steam temperature is more than 375 °C.

The following connections are provided on the header.

- 4" line to 11-E-18
- 6" line as snuffing steam to 11-F-01
- 6" line as snuffing steam to 12-F-01
- 3" & 1 ½" line to 12-C-01
- 2" header for the pumps 11-P-09 A/B, 12-P-03 A/B, 11-P-05 A/B, 12-P-04 A/B (1" each)
- 4" line to 11-C-01
- 4" & 2" line to 11-V-08
- 2" line to biturox reactor.
- 1" line to bitumen pumps 13-PM-01A/B.
- 1" take off to the BBU sample point.



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Use of LP steam is mainly as follows:

1. Sample point purging
2. Heating medium for exchangers (for heating products with LP steam)
3. Body steam for steam out of columns
4. Quench steam for pump mechanical seal
5. Snuffing steam of fired heaters / furnace
6. Soot blowing of fan blades
7. Heat Tracing of process lines
8. Dilution of vented hydrocarbons from vent systems
9. Steam smothering and utility
10. Dead end of CBD headers

35.1.9 MP STEAM SYSTEM

(Refer P&ID 10-1100-E-202 Rev.7):

MP steam for CDU/VDU-II is supplied from Power Plant-II. A 12" medium pressure steam header caters to the entire unit. The header is provided with isolation valves and spectacle blind. A pressure gauge 11-PG-705, a temperature gauge 11-TG-705, a flow recorder / integrator 11-FR / FQ-702 and a low pressure alarm 11-PAL-701 are provided on the header at the battery limit.

MP steam serves mainly following purposes:

Atmos Heater (11-F-01)

a) 10" line for superheating

1. 4" line for Soot blowing Steam
2. 6" line for Decoking Steam
3. 4" line for Atomising Steam
4. 4" line for purging steam

b) Vacuum Heater (12-F-01)

1. 4" line for Soot blowing Steam
2. 4" line for Decoking Steam
3. 3" line for Atomising Steam
4. 4" line for purging steam
5. 4 nos. 2" lines for tubulising steam

c) 6" line as motive fluid in steam ejectors (12-J-01, 02 & 03 A/B/C)



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- d) As velocity steam to each pass in 12-F-01
- e) Utility Hose Stations (for use in equipment in hot services under shut down /maintenance)
- f) In BBU,
- ✓ 2" take off to Reactor
 - ✓ 2" take off to seal pot and quench drum
 - ✓ 1" take off to BBU product pumps 13-PM-03A/B
 - ✓ 2" take off to BBU product turbine 13-GT-01.
 - ✓ Utility Hose Stations (for use in equipment in hot services under shut down /maintenance)

From the 12" header, a 10" tapping has been taken along to which the MP steam produced by the HVGO CR steam generators 12-E-10/10A and Bitumen feed / product cooler 13-E-01A/B joins and gets superheated in the 11-F-01 convection bank. The total steam generated by the above steam generators is measured and integrated by 11-FQ-712.

A pressure gauge (11-PG-719) and a temperature gauge (11-TG-715) are provided on the generated header. The superheated steam temperature from the Atmospheric Furnace can be controlled by injecting BFW through 11-TCV-303, and is used to control the degree of superheating.

The temperature can also be controlled by allowing more MP steam into the superheating coil and the excess MP steam can be diverted to the LP header through a 4" line by manually operating a globe valve. At the time of refractory drying, superheated steam can be vented to the Atmosphere through a silencer. Safety valves are also provided on this header at 11-F-01 top.

This superheated steam is used as stripping steam for Heavy Naphtha, Kerosene and Diesel Strippers and also for the Atmospheric Column Bottom.

35.1.10 HP STEAM SYSTEM

(Refer P&ID 10-1100-E-202 Rev.7):

An 8" HP steam header caters to the need of the entire unit. The header is provided with isolation valves and spectacle blind. A pressure gauge 11-PG-708, a temperature gauge 11-TG-767, a flow recorder / integrator 11-FR/FQ-704 and a low pressure alarm 11-PAL-702 are provided on the header at the battery limit.



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The following connections are provided on the header.

1. 6" line to 11-P-01B turbine
2. 6" line to 11-P-02B turbine
3. 3" line to 11-P-06B turbine
4. 4" line to 11-P-08B turbine

35.1.11 FUEL GAS SYSTEM

(Refer P&ID 10-1100-E-202 Rev.7):

An 8" fuel gas header caters to the need of the entire unit. It is provided with a double block valve and a spectacle blind at the battery limit. A pressure gauge 11-PG-707, and a flow integrator 11-FR/FQ-703 are provided on the header at the battery limit.

The following connections are provided on the header

1. 6" line to 11-F-01 (with a separate flow recorder and integrator)
2. 3" line to 12-F-01 (with a separate flow recorder and integrator)
3. 3" line to the Atmospheric overhead reflux drum (11-V-01)
4. 3" line u/s of 1st stage vacuum ejectors

Fuel gas being produced in various units is routed to the Sulphur Recovery Units for removal of H₂S. The sweet fuel gas is then routed to the FG distribution network in FCCU-I (R), and FCCU-II. The FG produced within the CDU/VDU-II unit is routed to FCCU-II in a FG supply header of 4" size at the battery limit and is provided with double block valve and spectacle blind.

35.1.12 FUEL OIL SYSTEM

(Refer P&ID 10-1100-E-202 Rev.7):

A 3" fuel oil supply and return header caters to the entire need of the unit. It is provided with isolation valves and spectacle blinds at the battery limit. Pressure gauges 11-PG-706/703 and temperature gauges 11-TG-706/703 are provided on the supply and return headers respectively at the battery limit.

The following connections are provided on the header

- 3" supply to 11-F-01, with flow recorder and integrator 11-FR/FQ-305
- 2" supply to 12-F-01, with flow recorder and integrator 12-FR/FQ-105



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The return headers have a globe valve, an SDV and a flow recorder and integrator 11-FR/FQ-306 and 12-FR/FQ-106. Within the CDU/VDU unit, FO is supplied to the Atmospheric and vacuum Furnaces through 3" & 2" headers respectively with circulating lines back to the return headers. Constant circulation is necessary in the FO lines to prevent congealing in the headers which may occur during idle conditions.

35.1.13 FLUSHING OIL SYSTEM (FLO)

(Refer P&ID 10-1100-E-204 Rev.7 & 10-1100-E-205 Rev.7):

Flushing oil (FLO) is normally with boiling range and properties comparing well with gas oil. It is used as a flushing medium for displacing heavy congealing and viscous material from equipment & piping during unit shut-down or other maintenance jobs requiring cleaning of equipment from such hydrocarbons. It is also used as a shaft lubricant in 13/3-GT-01.

35.1.14 TEMPERED WATER SYSTEM

(Refer P&ID 10-1200-E-104 Rev.5):

The tempered water system serves the SR coolers (12-E-09A/B and 12-E-09C/D) and the bitumen product coolers (13-E-02A/B and 13-E-02C). The purpose is to make sure that SR and bitumen product rundowns are maintained above their pour points, else there are chances of plugging and congealing of the lines. A 1" take off from the discharge of tempered water pump (12-PM-07A/B) serves the SR pumps (12-PM-01A/B) for cooling the pump bearings. The tempered water system is a closed loop, wherein, the spent tempered water from the coolers exchanges the excess heat with the sea water in tempered water coolers (12-E-08A/B), and then it is pumped back into 12-V-02. In order to account for evaporation losses in the drum, a 1" DM water make up line has been provided from DM water tank (11-V-04).

35.1.15 FLARE SYSTEM:

(Refer P&ID 10-1100-E-202 Rev.7):

In the event of abnormal operating conditions or emergencies, the hydrocarbon operating system may get pressurised. In order to prevent this from shooting up and crossing design limit of respective system / equipment and causing accident and / or equipment damage, it may become necessary to relieve same amount of non condensable hydrocarbon vapours to a system that renders them harmless. For this header is provided for collection of relieved vapours in the unit to which all relevant equipment are connected. PSV's of the vessels of hydrocarbon service are all connected to 24" flare header. Flare lines are designed



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for a pressure of 3.5 kg/cm² g. and temperature of about 200 °C. Flare lines to be tested pneumatically because of line support considerations. Entry of steam and condensate in flare header to be avoided as it may lead to extinguishing of main flare flame.

The following from CDU/VDU-II are routed to flare header.

1. Atmospheric overhead accumulator (11-V-01) PSV
2. Atmospheric Column top pressure controller (11-PV-409B)
3. Stabiliser column PSV
4. Stabiliser reflux drum (11-V-03) PSV
5. Stabiliser reflux / LPG product pump seal pot
6. LPG pumps and Stabiliser reflux pump casing vents
7. Overhead Naphtha accumulator pressure control vent
8. Stabiliser overhead condenser shell vents.

There is a provision to measure the mass flow rate of the flare gases, which is indicated by FX-801 on the DCS panel. There is also a SOP provision at the dead-end of the flare header in the unit.



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CDU-II INSTRUMENT TAGS DESCRIPTION:

S no	Tag no	Description
1.	LI-1101	Desalter water interface level control
2.	LI-1404	HN stripper level control (11-C-02)
3.	DL-0101	SRN water wash drum interface level (10-V-02)
4.	LI-2202	Column bottom level control
5.	LI-3102	13-E-01 BFW level control
6.	LI-3108	13E-03 BFW level control
7.	LI-3402	13-E-03A Steam gen.
8.	DL-1406	11-V-01 Reflux drum boot interface control
9.	DL-2201	Hot-well level water control
10.	LI-1502	Stabilizer reflux drum level control
11.	LI-2203	Slop cut level control
12.	LI-2204	HVGO level control
13.	LI-2205	LVGO level control
14.	LI1701	CBD Level
15.	L2206	Hot-well oil level control
16.	LR-1902	PFD level control
17.	LR-1401	11-C-01 bottom level control
18.	LI-2201	12-C-01 bottom level
19.	LI-1103	Desalter wash water tank level control
20.	LI-1403	KERO stripper level control (11-C-03)
21.	LI-1402	Diesel stripper level control (11-C-04)
22.	LI-1405	11-V-01 Reflux drum level control
23.	LI-1501	Stabilizer bottom level control
24.	LI-2301	12-E-10 BFW level control (MP Steam gen.)
25.	LI-2302	12-E-10A BFW level control
26.	LI-3107	Quench drum level control
27.	DL-1503	11-V-03 boot water level (LPG Drum)
28.	LI-2401	Temperature water drum level
29.	LI1101A	Agar probe-
30.	LI1101B	Agar probe-
31.	LI1101C	Agar probe-
32.	DP1301	11F01 Atomizing steam DP control
33.	DP2103	12F01 Atomizing steam DP control
34.	L2201	12C01 Bottom Level
35.	LI1902	PFD Level
36.	LR1401	Atmos Column bottom level
37.	LI-3102	13-E-01 BFW level control



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38.	LI-3108	13E-03 BFW level control
39.	LI-3402	13-E-03A Steam gen.
40.	LI-3107	Quench drum level control
41.	AI3401	Capillary DPT of Bit. Viscosity meter

Level switches

42.	LL1401	Atmos Column bottom level low
43.	LH1401	Atmos Column bottom level high
44.	LL2201	Vac bottom low switch
45.	LH2201	Vac. Column bottom high
46.	LL2203	Vac. Column HVGO level low
47.	LL2204	Vac. Column LVGO level low
48.	LL1501	Stabilizer bottom level low
49.	LH1501	Stabilizer bottom level high
50.	LL1902	PFD level low
51.	LH1902	PFD level high
52.	LL2202	Vac. Slop cut level low
53.	LL2401	Tempered water low
54.	LH2401	Tempered water high
55.	LL11P11A	LPG pump seal level switch A
56.	LL11P11B	LPG pump seal level switch B
57.	LSL1101	Desalter level switch
58.	LL3001	Bit agitator Seal pot LS
59.	LL3401	BBU reactor level high
60.	LH3401	BBU reactor level low

Flow transmitters:

S no	Tag no	Description	Location
1.	F1402R	Over flash Flow	On 11-C-01,O/F Loop KERO LCV Platform(P/F)
2.	FI2100	RCO to 12-F-01	East of 11PM 10A
3.	FR2101	RCO to Pass A	South of Pass A C/v
4.	FR2102	RCO to Pass B	South of Pass B C/v
5.	FR2103	RCO to Pass C	South of Pass C C/v
6.	FR2104	RCO to Pass D	South of Pass D C/v
7.	FR2109	Slop cut Recycle	East of 12PM01A
8.	FR2202	Hot HVGO CR to 12C01	12PM01B Suction to CBD,OWS Line.
9.	FR2203	Cold HVGO CR to 12C01	Opposite to12PM05A



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10.	FR2301	SR to BBU feed flow	East of Bitumen K.O. Pot near 13PM03B
11.	FR2304	Slop cut to SR	South of 11PM05B near slop cut PDU manifold
12.	F2401R	SR R/D Flow	South of 12PM 04 A/B
13.	F2402R	HVGO R/D to Tank	West of 11PM 05B behind HVGO LCV
14.	FR2402	HVGO Hot feed to FCCU II	Opp. to LPG Reflux C/V
15.	F2406R	SR to HFO	Near HFO/RFO 3 way C/V
16.	F2408Q	HVGO to LDO	East of 12E11
17.	FR1101	Total crude flow -Vac section	Between 12-E-01A/B/C Bell Cover Side
18.	F1102	Desalter Wash Water Flow	11-E-18 West side
19.	FR1103	HN to DSL R/D	11-E-26 West
20.	F1104R	Total Crude Flow to Unit	South of 11PM01A
21.	F1105R	Crude split Flow to Atoms section	West of 11PM 11B
22.	F1106R	LP Steam to 11-E-18	East of 11-E-18
23.	FR1107	HN to SRN R/D	East of 11-E-26
24.	F1201R	SRN R/D Flow	East of SRN Cooler
25.	FR1202	DSL R/D to Tank	Between 11PM 8C/D
26.	FR1203	KERO R/D to Tank	11PM 8D discharge V/V
27.	FR1204	KERO R/D to DSL	KERO R/D to DSL C/V
28.	F1205Q	KERO R/D to LDO	South of KERO to LDO V/V
29.	F1206Q	KERO to FO	South of KERO to FO Loop
30.	F1207Q	DSL to LDO	East of 11PM 8C
31.	FR1301	Crude to Pass A	North of PassA C/V
32.	FR1302	Crude to Pass B	North of Pass B C/V
33.	FR1303	Crude to Pass C	North of Pass C C/V
34.	FR1304	Crude to Pass D	North of Pass D C/V
35.	F1307R	Atomizing Steam flow	Atomizing steam C/V
36.	FR1308	F.G. Supply to 11F01	North of FO Return SDV 5th from East
37.	FR1401	11-C-01 Bottom S/Steam	South of 11PM 4A
38.	FR1403	Top Reflux	South of 11PM 4A
39.	FR1404	DSL CR to 11-C-01	South of DSL CR C/V
40.	FR1405	KERO CR to 11-C-01	Desalter RV
41.	FR1406	TPA to 11-C-01	South of TPA C/V
42.	FR1407	DSL Stripping Steam	Behind 11-C-01 RV steam T/O, V/V
43.	FR1408	KERO Stripping Steam	Behind Top Reflux C/V
44.	F11409	HN Stripping Steam	Behind 11-C-01 RV Steam T/o, V/V
45.	F1450R	Atmos Water to 11V04	North of 11V04, Out of two East one
46.	FR1501	Stabilizer Reflux flow	West of 12PM7B
47.	FR1502	LPG R/D to MEROX	East of 11PM11A
48.	FR1503	Stabilizer Feed flow	Behind DL1406 C/V



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49.	FR1504	KERO CR to 11E25	Between 11PM8A&8B
50.	F1505	Cdu-1 Naphtha to 11C05	South of 11PM8A
51.	F1701Q	Salter water flow to CDU	Located at FCCU-2
52.	F1702R	MP steam to unit	North of 11F01 F.O.C/v 1st from east
53.	F1703R	F.G. Flow to Unit	Near 12F1 F.O. C/V 3rd from East
54.	F1704R	HP steam flow to unit	Near 11F1 F.O. C/V 3rd from East
55.	F1705Q	Inst. Air flow	12 F 1 F.O C/V 5th from East
56.	F1706Q	D.M Water	North of 11F1 F.O,C/V 4th from East
57.	F1707R	L.P steam to Unit	North of 11F 1` F.O,C/V 2nd from East
58.	F1708Q	Plant Air to Unit	South of 12F1 F.O,C/V 2nd from East
59.	F1709Q	Service Water to Unit	South of 12F1 F.O ,C/V 4th from East
60.	F1710Q	B.F.W to Unit	South of 12F1 F.O,C/V 2nd from East
61.	F1711Q	Salt water to FCCU	Located at FCCU-2
62.	F1712R	Total MP steam generated	West of 11F01 Pass' C ' C/V
63.	FR1801	AGO	South of 12PM02A
64.	FR1805	NGL	North of SR Quench C/v
65.	FR1820	Air to 11F01	On APH 3rd Platform from landing
66.	F1902R	PFD Crude to 11-F-01	North of 11-V-01
67.	FR3501	DSL to DHDS	West of TPA , C/V
68.	F4218Q	Tank farm steam	East of 11PT2B
69.	11FI9301	KERO to FO (OM&S)	West of MOI
70.	F2107R	12F01FG Volumetric Flow	South of 12F01 F.O. Supply C/V
71.	FR2107	F.G.to 12F01	South of F.O. Supply C/v
72.	F2108R	Atomizing steam to 12F01	South of 12F01 Atomizing Steam C/V
73.	FR2201	Hot LVGO CR to 12C01	Below 12C01, Opp. to Hot LVGO CR C/V
74.	FR2204	Quench to 12C01	West of 11PM004B behind Quench C/V
75.	F2207R	MP steam to 12C01 Ejector	Behind ejector steam C/V East of 11PM09A
76.	F2302R	Steam generated from 12E10	West of blow down
77.	F2302A	Steam generated from 12E 10A	North of 11V04 out of two East one
78.	F2303R	SR From 12E03	O/H of 11PM05A/B
79.	F2305B	BFW to 12E 10A	North of 11V4 out of two West one
80.	F2403R	LVGO R/D	North of 12PM4 A/B
81.	FR2404	LVGO R/D to DSL/LDO	West of 12E11
82.	FR2405	SR to RFO	U/L 1st stage P.F south corner
83.	F2407Q	SR to LDO	West of U/L Slop manifold
84.	FR2520	12F01Combustion air	Between DOD no1&no2
85.	F2650R	Hot well water to 11V04	North of 11V04 manifold, Out of two West one
86.	FR2205	Cold LVGO CR to 12C1	North of DSL CR C/V
87.	F2206	Hot well oil R/D	Near slop manifold EBL
88.	12FT23	12 E 10 A BFW FT	
89.	FT0104		
90.	FR2404A	Hot LVGO to FCCU-II	Behind 12-PM-04A/B



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91.	FR1101	Total crude flow -Vac section	Between 12-E-01A/B/C Bell Cover Side
92.	FI1102	Desalter Wash Water Flow	11-E-18 West side
93.	FR1103	HN to DSL R/D	11-E-26 West
94.	F1104R	Total Crude Flow to Unit	South of 11PM01A

Pressure transmitters:

S no	Tag no	Description	Location
1.	PI2101	F.O. Supply Pressure Control	South of 12F01FOsupply C/v
2.	PI1301	FO Supply to 11F01,Pr,	North of 11F01 FO Supply C/V
3.	P1101	11PM 01A/B discharge Pressure	East of 11PM 01A
4.	P1410R	11-C-01Flash Zone Pressure	On 11-C-01,Desalter RV
5.	P1801	11F01 FD Discharge Pressure	Behind 11F01 Safety Shower out of two south
6.	P1802	APH Air Outlet Pressure	Behind 11F01 Safety Shower out of two north one
7.	P1803	Flue gas APH I/L Pressure	11F01 APH 4th P/F, from landing
8.	P1804	Flue gas APH O/L, Pressure	West of 11F01, Opp. D0d 5,1st from North
9.	P1901R	PFD I/L, Pressure	PFD , LCV U/s, B/V
10.	P2202R	12C01, Flash zone Pressure	East of Surface condensers P/F
11.	P2501	12F01 FD Discharge Pressure	Opposite to DOD no.1 4th from south
12.	P2502	APH Air outlet Pressure	Opposite to DOD No1 ,3rd from south
13.	P2503	Flue gas APH I/L, Pressure	On 12F01 APH 4th Platform from landing
14.	P2504	Flue gas APH O/L	Opposite to DOD no.1 5th from south
15.	PH1807	11F01 ID Fan Suction Pressure Indication	North of 11F01 ID Fan
16.	PH1807A	11F01IDFan Suction Pressure high	Adjacent toPH1807South
17.	PH2507	ID fan suction Pressure high	Opposite to DOD no.1, 6th and 7th from south
18.	PH2507A	ID fan suction Pressure high	Opposite to DOD no.1, 6th and 7th from south
19.	PI1105	Desalter Pressure Control	South of 10PM 3B
20.	PI1409	11C01 Top Pressure	11C01 Top
21.	PI1501	Stabilizer bottom Pressure	11C5 2nd P/F from Bottom
22.	PL1806	11F01FDFan discharge Pressure Low	West of 11F01, Opp. DOD 5
23.	PL2506	FD fan Discharge Pressure Low	Opposite to DOD no1,1st from south
24.	PR1409A	Atmos O/H gas to flare	11 C 01 Top
25.	PR1409B	FG make up to Atmos column.	11 C 01 Top
26.	PR1501A	Stabilizer Pr Controller	North of 11E20A to D
27.	PR1501B	Stabilizer Pr Controller	North of 11E20A to D
28.	PR1808	11F01 Pressure Control	11F01SB Air V/v P/f west
29.	PR1809	11F01Pressure Control(switch)	11F01SB Air V/v P/f west
30.	PR1902	PFD Pressure Controller	PFD PCV
31.	PR2202R	12C01 Flash zone pressure	12C01 Vac Column
32.	PR2206	12C01TopPressure Control	12C01 RV P/F
33.	PR2207	12C01 Ejector Pressure	East of 11PM09A
34.	PR2508	12F1 Pressure Control	North of 12F1 soot blowing P/F



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35.	PO2509	12F1 Arch Pr trip	
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Temperature indications:

S no	Tag no	Description	Location
1.	T1101	Desalter I/L, Temp	Mix valve D/s near shell flange
2.	T1102	Desalter O/L Temp	Crude O/L D/s shell flange
3.	T1103	11 E 07 Crude O/L Temp	Near 11 E 23 A Bell cover
4.	T1104	HN R/D Temp	On O/L, of 11 E 26
5.	T1201	SRN R/D Temp	East of 11 E 26
6.	T1202	DSL R/D Temp	At the d/s of DSL r/d c/v
7.	T1203	KERO R/D Temp	O/H 11PM08D
8.	T1205	11 E 09 KERO CR return	South of 11E09,U/S of I/L Shell flange
9.	T1306	PASS A Radiation O/L	Just below SB P/F, On it's O/L to Common Line 1st from east
10.	T1307	Pass B Convection O/L	Soot blowing P/F north side 2nd from east
11.	T1311	Pass B Radiation O/L	Just below SB P/F on it's O/L to common line 2nd from east
12.	T1312	Pass C convection O/L	Soot blowing P/F north side 4th from east
13.	T1316	Pass C Radiation O/L	Just below SB P/F, on it's O/L to common line,4th from east
14.	T1317	Pass D Convection O/L	Soot blowing P/F north side 3rd from east
15.	T1321	Pass D Radiation O/L	Just below SB PF, on it's O/L to common line,3rd from east
16.	T1322	Heater O/L, Temp	On Transfer Line to 11C01 Shell flange
17.	T1324	MP Super heated steam temp	On Super heat steam O/L, Soot blowing P/F
18.	T1332	11E16 Crude O/L, Temp	11E16 Crude O/L,V/V
19.	T1334	Pass A Decoking I/L Temp	Near burner platform 1st from east
20.	T1335	Pass B Decoking I/L Temp	Near burner platform 2nd from east
21.	T1336	Pass C Decoking I/L Temp	3rd from east
22.	T1337	Pass D decoking I/L Temp	4th from east
23.	T1338	Pass A Decoking O/L	Just near elbow swing portion before coil o/l
24.	T1339	Pass B Decoking O/L	Same as above 2nd from east
25.	T1340	Pass C Decoking O/L	Same as above 1st from west
26.	T1341	Pass D Decoking O/L	Same as above 2nd from west
27.	T1401R	11 C 01 Flash Zone Temp	On 11 C 01, Near Desalter Rv
28.	T1402A	RCO D/O Temp	RCO D/O Shell flange D/s,
29.	T1402R	DSL D/O Temp	Near D/O, Shell flange
30.	T1403R	KERO d/o Temperature	U/S of KERO LCV bypass valve
31.	T1404R	HN d/o TEMP	At the d/s o/f d/o shell flange
32.	T1406	11C01 Top temperature	On 11C01 Vapor line
33.	T1407	DSL stripper d/o temp.	O/H of 11PM10B
34.	T1408	KERO stripper d/o Temp.	D/O shell flange
35.	T1409	HN stripper d/o Temp.	D/O shell flange
36.	T1410	11-V-01 drum Temp.	On HC i/l to 11v01 u/s of shell flange
37.	T1411	TPA d/o Temperature	On TPA d/o south side at 11c01
38.	T1412	DSL CR return Temp.	DSL CR shell flange
39.	T1413	KERO CR Return Temp	KERO CR Shell flange, u/s



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40.	T1414	TPA Return Temp	TPA Return. Line Shell flange, u/s west side
41.	T1501	Stabilizer Feed Temp.	11C5 Feed I/L shell flange
42.	T1502	11 C 05 Flash zone Temperature	Above 11c5 SOP Shell flange
43.	T1503	Stabilizer Bottom Temp control	Adjacent to T1501
44.	T1505	Reboiler Naphtha O/L temp	Both T/O, of 11E25 Join & going to 11C05
45.	T1506	Reboiler Naphtha I/L, Temp	Before splitting to Two streams & entering to 11E25
46.	T1507	11E25 Kero CRO/L Temp	Reboiler C/V D/S
47.	T1509	Stabilizer REFLUX drum Temp	On 11pm 11A/B Suction line, before splitting to pumps
48.	T1510	Stabilizer Top temp control	On 11C05 Vapor line
49.	T1901R	PFD I/L Temp.	U/S of PFD LCV
50.	T1902R	PFD O/L Temperature	11PT2B Discharge D/s V/V
51.	T1C01	11E1 Crude I/L Temp	D/S of 11E01 Crude I/L V/V
52.	T1H010	11E01 HN I/L	O/H of 11E01 at the U/S V/V of HN
53.	T1H100		
54.	T1H11	11E11 KERO CR I/L	At the U/S of 11E11 KERO I/L Shell flange
55.	T1H12	11E12 DSL O/L	At the D/S of 11E12 DSL O/L B/V
56.	T1H140	11E14A/B DSL O/L	
57.	T1I01	11E1 Crude O/L, Temp	D/S of 11E01 Crude O/L V/V
58.	T1I02	11E2 Crude O/L, Temp	D/S, of O/L, V/V
59.	T1I03	11E3, Crude O/L, Temp	D/S, of O/L V/V
60.	T1I04	11E4, Crude O/L, Temp	U/S, of I/L to 11E5
61.	T1I05	11E5, Crude O/L Temp	O/L, of D/S B/V
62.	T1I06	11E6, Crude O/L, Temp	U/S, of O/L, B/V
63.	T1I07	11 E 07 CRUDE O/L	D/S of Crude O/L B/V of 11E07 to Desalter
64.	T1I08	11E08 Crude O/L	At the D/S of 11E08 Crude O/L V/v
65.	T1I09	11E09 Crude O/L	Located at 11E12 Crude I/L Shell flange U/S
66.	T1I10	11E10 Crude O/L	Located near 11E09 Crude I/L Shell flange U/S
67.	T1I11	11E11 Crude O/L	Located near 11E10 Crude I/L Shell flange U/S
68.	T1I12	11E12 Crude O/L	At the U/S of 11E12 Crude O/L V/v
69.	T1I13	11E13 Crude O/L	Located at 11E14A/B, on 11E13 Crude O/L to 11E14A/B Crude I/L valve U/S
70.	T1I14	11E14A/B Crude O/L	Located at 11E15A/B, on 11E14 Crude O/L to 11E15A/B Crude I/L valve U/S
71.	T1I15	11E15A/B Crude O/L	At the D/S of 11E15 A/B Crude O/L V/V
72.	T1R01	11E01 HN O/L	At the D/S of 11E01 HN O/L V/V
73.	T1R02	11E02 KERO O/L	Located near 11E24, At the U/S of KERO I/L V/V
74.	T1R03	11E03 DSL O/L	Located near 11E23, At the U/S of DSL I/L V/V
75.	T1R05	11E05 KERO O/L	At the U/S of 11E02 KERO I/L Shell flange
76.	T1R07	11E07 LVGO O/L	At the D/S of 11E07 LVGO O/L V/V
77.	T1R08	11E08 DSL O/L	At the D/S of 11E08 DSL O/L V/V
78.	T1R10	11E10 KERO O/L	At the D/S of 11E10 KERO O/L V/V
79.	T1R11B	11E11 KERO CR O/L	Near West of 11E12 On 11E11 O/L to 11PM8C/D Suction
80.	T1R13	11E13 DSL O/L	At the D/S of 11E13 DSL O/L V/V
81.	T1R14	11E14A/B DSL O/L	At the D/S of 11E14 DSL O/L V/V
82.	T1R15	11E15A/B DSL CR O/L	At the D/S of 11E15 DSL CR O/L V/V



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83.	T1R16	11E16SR I/L	At the U/S of 11E16 SR I/L V/V
84.	T2101	RCO to 12F1 I/L Temp	O/H of RCO, Manifold
85.	T2102	Pass A ,convection O/L	Convection O/L,P/F
86.	T2106	Pass A Radiation O/L	Decoking elbow P/F
87.	T2107	Pass B Convection O/L	At Convection O/L P/F
88.	T2111	Pass B Radiation O/L Temp	Decoking elbow platform
89.	T2112	Pass C Convection O/L	At convection O/L, platform
90.	T2116	Pass C Radiation O/L	Near Decoking elbow P/F
91.	T2117	Pass D Convection O/L	Convection O/L,P/F
92.	T2121	Pass D Radiation O/L	Decoking elbow P/F
93.	T2122	Vac. Furnace O/L Temp.	Transfer line Ext. Portion
94.	T2134	Pass A Decoking I/L	On Decoking elbow swing P/F, 1st from west
95.	T2135	Pass B Decoking I/L	On Decoking elbow swing P/F, 2nd from west
96.	T2136	Pass C Decoking I/L	On Decoking elbow swing P/F, 3rd from west
97.	T2137	Pass D, Decoking I/L	On Decoking elbow swing P/F, 4th from west
98.	T2138	Pass B Decoking O/L	On decoking elbow swing west side 1st from bottom
99.	T2139	Pass A Decoking O/L	On Decoking elbow swing west side 2nd from bottom
100.	T2140	Pass C Decoking O/L	On Decoking elbow swing East side 2nd from bottom
101.	T2141	Pass D, Decoking O/L	On Decoking elbow swing Eastside 1st from bottom
102.	T2202	SR Quench Temp.	SR Quench to 12C01 Shell flange
103.	T2203R	12C01, Flash zone Temp.	Surface condensers P/F
104.	T2204	Slop cut D/O	O/H of 12PM 02A/B
105.	T2205	HVGO D/o	O/H of 12PM02A/B
106.	T2206	Hot HVGO CR Temp.	D/S of Strainer
107.	T2207	LVGO D/O	LVGO D/O Shell flange
108.	T2208	Cold HVGO CR Temp.	D/S of Strainer
109.	T2209	12C01 Top Temp.	On Vapor line-RV P/F
110.	T2210	Cold LVGO CR Temp.	D/S of LVGO CR Strainer
111.	T2301	MP Steam Temp. from 12E10	O/H of 12E10
112.	T2302	MP Steam from 12E10A	O/h of 12E10A, D/s of Check V/v
113.	T2303	SR R/d to Bitumen	
114.	T2305	12E06A/B, Crude O/L Temp.	12E06A/B Crude O/L V/v D/s
115.	T2401	SR R/D Temp.	12E09A/B,C/D,SR O/L near HFO,RFO C/V U/S
116.	T2402	12E11 LVGO O/L Temp.	On 12E11O/LLVGO to R/D,O/h of Sample Point
117.	T2403	HVGO O/L from 12E12	On HVGO R/d near 12E12A/B O/L
118.	T2404	HVGO R/D Temp.	On HVGO R/D near 12E12A/B O/L
119.	T2405	Temp. Water Drum Temp	12E08A/B O/L to drum near G/G P/F
120.	T2406	SR R/d to IFO	Adjacent to 12PM07B on SR to IFO line
121.	T2501	Comb. air APH O/L,	On APH air O/L
122.	T2506		
123.	T2C010	12E01 Crude I/L	At the U/S of 12E01A/B Crude I/L V/V
124.	T2H01		
125.	T2H040		
126.	T2H050		
127.	T2H060	11E16 SR O/L	At the D/S of 11E16 SR O/L V/V
128.	T2I01B	1201A/B/C Crude O/L	2, tags at 12E01A/B,C O/L



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129.	T2I02	12E02 Crude O/L	At the U/S of 12E03 Crude I/L V/V
130.	T2I03	12E03 Crude O/L	At the U/S of 12E04 Crude I/L V/V
131.	T2I04	12E04 Crude O/L	At the U/S of 12E05 Crude I/L V/V
132.	T2I05	12E05 Crude O/L	At the U/S of 12E06 Crude I/L V/V
133.	T2R01	12E01 SR O/L	
134.	T2R02	12E02 HVGO CR O/L	At the D/S of 12E02 HVGO O/L V/V
135.	T2R03	12E03 SR O/L	At the D/S of 12E03 SR O/L V/V
136.	T2R04	12E04 HVGO O/L	At the D/S of 12E04 HVGO O/L V/v
137.	T2R05	12E05 HVGO O/L	At the d/sof12E05HvgoO/L V/v
138.	T2R06	12E06A/B SR O/L	At the U/S of 12E03 SR I/L V/V
139.	TH1701	CBD I/L, Temp High	Upstream of CBD I/L, V/V
140.	TH1702	CBD Drum Temp	On CBD Drum
141.	TH1801	Comb air O/L, Temp high	Same as above
142.	TH1802	Furnace ID Fan flue gas temp. high	Near ID fan suction
143.	TH1804	Flue gas APH O/L Temp	Near I.D fan suction
144.	TI1402	Over flash Liquid Temp	On O/F Loop , S/Steam Shell V/V Platform
145.	TR1301	Furnace COT controller	On 11F01 Passes O/L Common Header
146.	TR1403	11C01 Top Temp. Control	On 11C01 Vapor line shell flange D/S
147.	TR2133	Furnace COT controller	On Transfer line Ext. Portion
148.	TR2201	12C01 Bottom Temp.	O/H of cold LVGO FT or 12C01 SW corner pillar
149.	TR2302	12E10HvgoTemp.control	12E10 HVGO C/v
150.	TX1302	Flue gas convection O/L Temp.	Below SD P/F Adjacent to T1805, Out of two north is T1805 & South is TX1302
151.	T1301R	Below convection Temp.	
152.	T1325	Arch Temp	Just below soot blower NO.3
153.	T1326	Arch Temp	West of 11F01 soot blowing P/F
154.	T1327	Fire Box Temp	Near Burner No4 T/o V/V
155.	T1328	Fire Box Temp	Between burner No 11 T/O, V/V
156.	T1329	Fire Box Temp	Between No.7&8 Burner
157.	T1330	Fire Box Temp	Near No7 Burner peep hole
158.	T1331	Flue gas convention O/L	South of T1325
159.	T1333	Below stack damper Temp	
160.	T1801	Comb air O/L Temp high from APH	APH 2nd platform
161.	T1804	Flue gas APH O/L Temp	Flue gas APH O/L Temp
162.	T1805	Flue gas APH I/L Temp	Below damper P.F North one
163.	T1806	Stack Temp	Above damper
164.	T2123	Arch Temp	Below soot blower no.7
165.	T2124	Arch Temp	Blow Soot blower no.2
166.	T2125	Fire box	From south ---> north second peephole
167.	T2126	Fire box	From south--> north 7th peephole
168.	T2127	Fire box	Behind no3 burner v/v
169.	T2128	Fire box	Behind no10 burner v/v
170.	T2129	Ex. Convection Temp.	Just above SD West side, In between FG T/O to APH&FG from ID fan discharge to stack



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171.	T2132R	Flue gas Convection I/L	Between SBno.7&8 1st from north
172.	T2501	On APH air O/L	Combustion air APH O/L,
173.	T2504	Flue gas APH O/L Temp	
174.	T2505	Flue gas APH I/L Temp	
175.	T2506	Stack Temp	
176.	TH1801	Combustion air O/L, Temp high	Combustion air O/L, Temp high
177.	TH1802	Furnace ID Fan flue gas temp. high	Furnace ID Fan flue gas temp. high
178.	TH2501	Flue gas APH Temp. high	Combustion air O/L, Temp high
179.	TH2502	Furnace ID Fan flue gas temp. high	
180.	TL1803	Furnace ID Fan flue gas temp. low	
181.	TL2503	Furnace ID Fan flue gas temp. low	
182.	TX1302	Flue gas convection O/L Temp.	Below SD P/F, Adjacent to T1805, Out of two north is T1805 & South is TX1302
183.	T2131R	Flue gas Convection I/L	Between Soot Blower no.1&2
184.	T1303	Pass A Skin Temp.	Below 1st stage soot blowing P/F South side
185.	T1304	Pass A skin Temp.	N.E.corner,below No1 SB P/F
186.	T1305	Pass A skin Temp.	East of Burner No 12
187.	T1308	Pass B Skin Temp.	East side, below No1 SB P/F
188.	T1309	Pass B Skin Temp.	SE corner,below No1 SB P/F
189.	T1310	Pass B Skin Temp.	Between Burner No12 & No1 T/O V/v
190.	T1313	Pass C Skin Temp.	SW corner below No1 SB P/F
191.	T1314	Pass C Skin Temp.	NW corner below No1 SB P/F
192.	T1315	Pass C Skin Temp.	West of burner No1 V/v
193.	T1318	Pass D skin Temp	South side below 1st stage soot blowing P/F
194.	T1319	Pass D skin Temp	S/w corner below 1st stage soot blower
195.	T1320	Pass D skin Temp	Between burner no 12 & 1 V/V, adjacent no1 v/v
196.	T2103	Pass A Skin Temp.	12F01 West, O/H of 4th peep hole (from south), out of two south one
197.	T2104	Pass A Skin Temp.	12F01 West, O/H of no4 & no5 Peep holes 1st P/F from Top
198.	T2105	Pass A Skin Temp.	Same Location, 3rd P/F from Top
199.	T2108	Pass B Skin Temp.	12F01 West, O/H of 4th peep hole (from south), out of two north one
200.	T2109	Pass B Skin Temp	O/H of 4&5 peephole on west 2nd from Top
201.	T2110	Pass B Skin Temp	Same location 4th from Top
202.	T2113	Pass C Skin Temp	South of SB. No 5 2nd from south
203.	T2114	Pass C Skin Temp	O/H of no.7 burner operating v/v 1st from Top
204.	T2115	Pass D Skin Temp	Same location 3rd from Top
205.	T2118	Pass D Skin Temp	South of S.B no.5 1st from south
206.	T2119	Pass D Skin Temp	O/H of no7 burner operating v/v 2nd from top
207.	T2120	Pass D skin Temp	Same location 4th from Top
208.	T3108	Bitumen Compressor Discharge	On Bitumen Compressor Discharge header



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		Temp.	
209.	T3109	Bitumen R/D temperature	At the O/L of 13E02A/B to 13E02C
210.	T3402	Reactor Temp.	West of Reactor 2nd P/F, Out of two, Top one
211.	T3403	Reactor temperature middle.	West of Reactor 2nd P/F, out of two, 2nd one
212.	T3404	Reactor Bitumen Temperature	West of Reactor 1st P/F, O/H of LP Steam to Reactor V/V
213.	T3405	Reactor Waste gas Temp.	On Reactor top, south side, near Pr. Gauge

Trip instruments:

S no	Tag	Description	type
1.	11PH807A	11F01IDFan Suction Pressure high	Transmitter
2.	11PL806	11F01FDFan discharge Pressure Low	Transmitter
3.	12PL506	FD fan Discharge Pressure Low	Transmitter
4.	12PO509	12F1 Arch Pr trip	Transmitter
5.	11PO809	11F01Pressure Control(switch)	Transmitter
6.	11-PAL-301	11F1 fuel oil pressure low	Switch
7.	11-PAL-303	11F1 fuel gas pressure low	Switch
8.	12-PAL-106	12F1 fuel oil pressure low	Switch
9.	12-PAL-108	12F1 fuel gas pressure low	Switch
10.	11FR301	Crude to Pass A	Transmitter
11.	11FR302	Crude to Pass B	Transmitter
12.	11FR303	Crude to Pass C	Transmitter
13.	11FR304	Crude to Pass D	Transmitter
14.	12FR101	RCO to Pass A	Capillary
15.	12FR102	RCO to Pass B	Capillary
16.	12FR103	RCO to Pass C	Capillary
17.	12FR104	RCO to Pass D	Capillary
18.	11PH807A	11F01IDFan Suction Pressure high	Transmitter
19.	11PL806	11F01FDFan discharge Pressure Low	Transmitter



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Control valves:

S no	Tag no	Description
1.	10DL101	Wash water drum interface
2.	10FV104	SRN to MS
3.	11DL406	11V01 interface
4.	11DPV301	Atomizing Steam to FO DP
5.	11FV101	Vac. split control valve
6.	11FV102	Desalter water
7.	11FV103	HAN to Diesel
8.	11FV107	HAN to SRN
9.	11FV202	DSL R/D
10.	11FV203	KERO to MEROX
11.	11FV204	KERO to diesel
12.	11FV301	11F01 Pass1
13.	11FV302	11F01 Pass2
14.	11FV303	11F01 Pass3
15.	11FV304	11F01 Pass4
16.	11FV308	11F01 FG flow
17.	11FV3501	DSL to DHDS
18.	11FV401	11C1 Bottom stripping steam
19.	11FV403	Reflux
20.	11FV404	DSL C/R
21.	11FV405	KERO C/R
22.	11FV406	Top Pump Around CR
23.	11FV407	Diesel striping steam
24.	11FV408	KERO stripping
25.	11FV409	HN Striping steam
26.	11FV501	Stabilizer REFLUX
27.	11FV502	LPG to MEROX
28.	11FV503	Stabilizer feed
29.	11FV504	KERO C/R to Reboiler
30.	11FV805	Natural Gas to 11C01
31.	11LV101	Desalter interface level (located at MEROX)
32.	11LV103	11V04 level
33.	11LV402	DSL stripper level
34.	11LV403	KERO stripper level
35.	11LV404	HN stripper level
36.	11LV501	Stabilizer level
37.	11LV902	PFD level



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38.	11PV101	PFD bypass SDV
39.	11PV103	Steam
40.	11PV105	Desalter pressure
41.	11PV301	11F01 FO Pressure
42.	11PV402	Turbine Auto cut Steam
43.	11PV408	Steam
44.	11PV409A	flare control valve
45.	11PV409B	Makeup control valve
46.	11PV501A	Stabilizer pressure to FCCU
47.	11PV501B	Stabilizer makeup.
48.	11PV902	PFD pressure
49.	11SDV301A	SDV-11F1 Fuel oil supply
50.	11SDV301B	SDV-11F1 Fuel oil return
51.	11SDV303	11F01 Fuel Gas
52.	11TV303	BFW to super heated MP steam coil
53.	12DL201	Hot water level
54.	12DPV103	12F01 Atm. steam & FO
55.	12FV101	12F01 Pass1
56.	12FV102	12F01 Pass2
57.	12FV103	12F01 Pass3
58.	12FV104	12F01 Pass4
59.	12FV107	12F01 FG flow
60.	12FV109	Slop cut Recycle
61.	12FV201	Hot LVGO REFLUX Flow
62.	12FV202	Hot HVGO Rx
63.	12FV203	Cold HVGO Reflux
64.	12FV204	MC1 Quench
65.	12FV205	LVGO top reflux
66.	12FV301	SR to 12E3 (CDU-I SR & BBU)
67.	12FV402	FCCU Hot feed
68.	12FV404	LVGO to DSL
69.	12FV404A	Hot feed to FCC
70.	12FV405	SR+ Slop
71.	12LV202	12C01 Bot. level
72.	12LV203	Slop cut level
73.	12LV204	HVGO level
74.	12LV205	LVGO level
75.	12LV301	BFW
76.	12LV302	12E10A level
77.	12PV101	12F01 FO Pressure



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78.	12PV206	12C01 Top pressure
79.	12PV207	Ejector pressure
80.	12SDV102A	12F01 FO Supply
81.	12SDV102B	12F01 FO Return
82.	12SDV105	12F01 Fuel Gas
83.	12SDV900	HW Off gas to vent
84.	12SDV901	HW Off gas to 12F01
85.	12TV102	Tempered water CV
86.	12TV103	Cold SR to VBU
87.	12TV302	HVGO 12E10
88.	12TV303	HVGO, 12E10, Temperature